TFPT — Research Contracts for the Remaining Interfaces:

$v_{\rm geo}$, $G_{\rm net}$, $F_{\rm transfer}$

($U_{\rm wall}$): the parabolic flavor wall-selection and ($G_{\rm metric}$): the full quantum-gravity measure

plus the two continuum contracts `CELEST.SEAM.01` and `WOIT.OS.TWISTOR.01`
(M1–M3 and the δ-chains executed and decided; $\alpha/\beta_1/\beta_2/\beta_3$ executed, γ next)

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\rm car} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions** (plus two standalone working notes), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*

1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.

2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.

3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.

4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\rm geo}, G_{\rm net}, F_{\rm transfer}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** `tfpt_prime_front` — research documentation of the prime / zeta line: the seven load-bearing modules on the E_8 glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8,4,4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified); **N2.** `note_hilbert_polya_truncations` — a computable, zeta-free truncation family for the Weil measure: measurements on a Hilbert–Pólya candidate (prediction-freeze methodology; no RH claim).

You are reading companion R (Research Contracts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\mathrm{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1{\otimes}(A_3)_1{\rtimes}\langle(1,1)\rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Beside that compiler rest sits the strict-physical-TOE rest: the ten-component $\mathrm{Rest}_{\mathrm{TOE}}$ / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\mathrm{TOE}_{\mathrm{Complete}}$:= AND(T1..T8); not a closure).

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin 1.648 > 0, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever**

disagree, the ledger wins. The historical labels ($U_{\rm wall}, G_{\rm metric}, F_{\rm frontier}$) are retained only for ledger continuity — the live residual is $v_{\rm geo} \oplus F_{\rm transfer}$, with the seam $G_{\rm net}$ closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

[E] exact / machine-proven | **[C]** conditional | **[O]** open / axiom | **[X]** kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\rm car} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8,\ h = 30,\ \varphi_0$ *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: $v_{\rm geo}, G_{\rm net}, F_{\rm transfer}$ *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

What this document is about

After the compiler closure the entire residual is

$$\boxed{\text{Rest} = (U_{\rm wall}) \ \oplus\ (G_{\rm metric}) \ \oplus\ (F_{\rm frontier})}$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. ($F_{\rm frontier}$) (Koide, η_B, axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** ($U_{\rm wall}$) *first* (finite, algebraic, falsifiable), then ($G_{\rm metric}$) (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\boxed{\text{Rest}_{\rm operativ} = v_{\rm geo} \ \oplus\ G_{\rm net} \ \oplus\ F_{\rm transfer}}$$

Dual rest (2026-08-28). The boxed formula is the *compiler* rest. The strict-physical-TOE rest of the 2026-08-27/28 contract wave is the second accounting, every summand a named [O] (or standing-rule) contract — nothing claimed closed:

$$\boxed{\begin{aligned}\text{Rest}_{\rm TOE} = {} & \text{SeamContinuum} \oplus \text{4DAction} \oplus \text{ChiralMeasure}\\ & \oplus \text{MirrorGap} \oplus \text{UnitaryDynamics} \oplus \text{IR/continuum}\\ & \oplus \text{BulkSeam} \oplus \text{QuantumGravity}\\ & \oplus \text{GeneratingFunctional} \oplus \text{InitialState}\end{aligned}}$$

— ($U_{\rm wall}$) $\to v_{\rm geo}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), $U_{\rm point}$ collapses to the *one* dimensionful unit $v_{\rm geo}$ shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where $U_{\rm point}$ still appears); ($G_{\rm metric}$) $\to G_{\rm net}$: holographically reduced to the single boundary-net statement "*the seam–Calderón inclusion has Jones index* $4 = |\mu_4|$" — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/`GATE.METRIC.05-07`); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\to SO(16)_1 \to (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank $= c$; Wick/Pfaffian exact) is the certified Calderón datum

— closing G_{net} therefore also closes the carrier choice [E] ([verification/v113_quasifree_kernel.py]); (F_{frontier}) → F_{transfer}: named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the $K{\to}C{\to}F$ sheet axis $M(1,t)$, with the Koide source→pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates ([verification/v224_diamond_ftransfer_path.py]); the discrete data of that geodesic are now arithmetically framed — $\det M(1,t) = N(\alpha_t)$ with $\alpha_t = (4t{+}7{+}\sqrt{-7})/2$ integral, the path one translation $\alpha_{t+1} = \alpha_t + 2$ of constant discriminant -7 ([verification/v533_ftransfer_disc7_norm.py]), and the preregistered next test is whether the four external solvers share *one* fractional-linear action on α_t (kill: four incompatible arithmetic actions) — *executed in its literal group form (2026-08-02, [verification/v632_ftransfer_pgl2.py])*: one common parabolic PGL_2 action on the discrete half (the declared readings exactly intertwined with the base translation), uniflow degeneration on the native continuous half ($g_{\text{Boltzmann}} = g_{\text{pole}}$ identically, g_{QCD} exactly conjugate to the base translation, relic degenerate); the kill criterion is *not* met, and the residual — the external anchor clocks/jets — stays fenced by `CONTRACT.F.01`.

Historical compiler-residual audit ([verification/v384_residual_certification.py]; scope corrected 2026-09-05). The audit classified every item in the *then-listed compiler residual* into three external kinds. Its former phrase "zero open internal TFPT mechanisms" does not classify the physical completeness gates T1–T8 and must not be read as a TOE statement: `TFPT.TOE.COMPLETE.01` stays [O], T3–T8 remain open, and no shared 3+1D parent satisfying them has been constructed. Within that historical compiler scope the three classes were: [E] (A) an external math proof — `SEAM.EQUIV.01` continuum existence (the cited MMST theorem plus the crossed-product extension package, v336/v469, with AGT/AMT as second witness), `ALPHA.QUILLEN.EXACT.01` (v382, its level and normalisation sharpened by v470, the determinant-line/moduli bridge lemma exhibited at the finite level by v472 — the continuum ζ-det identification stays the open part) and the constructive `QG.AMB.01` measure (a [C] redundancy, v369); [E] (B) theorem-forbidden — v_{geo}, the one unit, by the No-Unit theorem (v153/v364); [E] (C) external physics — F_{transfer} (v371–v374, firewalled v187). The two load-bearing facts are checked (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), so the only hard items in that historical compiler list were theorems and external physics (`RESIDUAL.CERTIFICATION.01`); this says nothing about the independent missing dynamics and continuum mechanisms in T1–T8. The two kind-(A) external-math targets, `SEAM.EQUIV.01` and `ALPHA.QUILLEN.EXACT.01`, are additionally packaged as self-contained *externalization contracts* (statement, hypothesis list, runnable finite models, Lean axiom surface, must-fail controls, acceptance tests) in their own sections at the end of this document — hand-off documents for outside specialists; both stay [O], no marker moves. A third hand-off document in the same format, `PRIME.RESIDUE.EXTERNAL.01`, packages the prime-front's terminal *open statements* for external specialists in analytic number theory — it offers open problems only, asserts no progress toward the Riemann Hypothesis in either direction, and moves nothing.

The Coxeter-clock facet symmetry, resolved as a structural lemma (`RES.COXETER.SYMMETRY.01`, [verification/v394_coxeter_atoms.py]/ [verification/v409_coxeter_prime2_lemma.py]). The order-30 clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ has all three primes facet-typed, and they are exactly the three anchor atoms ($2{=}|Z_2|$, $3{=}N_{\text{fam}}$, $5{=}g_{\text{car}}$) wearing number-field hats ($\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$, v390). The table is *not* symmetric: the seam (prime-2) appears as the *common factor* of both dynamic rates ($2/3 = |Z_2|/N_{\text{fam}}$ and $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$, $|\mu_4|{=}2^2$), not as an independent attractor. *The structural lemma (v409) closes the falsifiable corollary:* the sheet involution $\sigma{=}\mathrm{diag}(1,-1,-1)$ has spectrum $\{+1,-1\}$ and projector $(I{+}\sigma)/2$ spectrum $\{0,1\}$ — none in the open interval $(0,1)$, so prime-2 supplies a *parity/projection*, never a nonzero contraction rate; the genuine rates $2/3$ and $(1/3)^6$ lie in $(0,1)$ and carry prime-2 only as a *factor*. Hence prime-2 is *necessary* (admissible boundary transport is sheet-paired) but *not autonomous*: **no prime-2-only attractor exists** — dynamics begins only after coupling to family prime-3 or carrier prime-5 ($|\mu_4|{=}2^2$ the Galois bridge). This closes the corollary [E] and types the necessity [C], sharpening v383's "every sector is the same gapped operator" towards "the same *seam-driven* operator"; the full "appears in every gap" universality stays the v383 reading. The corollary is **machine-proved in Lean 4** (`TfptCarrier/CoxeterPrime2.lean`, `FORM.COXETER.PRIME2.01`: axiom-pure, `lake build` clean). [E]/[C].

QFT layer (this round). The boundary QFT is now *one relative object* $\mathsf{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$, closed modulo the *single* keystone `SEAM.EQUIV.01` (whose downstream conformal-deck face is `QGEO.SYM.01`): its finite Dirac is a covariance induction of the seam KMS state (v258), its spectral-action cutoff *is* that KMS weight (v259), its seam, carrier-16 and E_8 live on one Kummer/K3 surface (v260), and the whole assembly is certified as the *Modular Spectral Closure* (v261). The ambient quantum-gravity measure (G_{metric}) is gap-decoupled and kept separate by design —

the QFT layer does not wait on it.

The two continuum contracts, and how far they are executed. Beside the two historical gates this note carries the two *continuum-route* research contracts, both still [O] as contracts but with most of their preregistered stages now *executed*. `CELEST.SEAM.01` (the celestial and twistor continuum route): WP1–WP4 and both WP5d stages executed, the WP5e stages $\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$ executed, all three back-reaction milestones M1–M3 executed with their preregistered kills evaluated (v515–v517), the δ_1 chain *decided* (kill under the derived measure, v518), the declared-vs-derived measure question *decided at probe level* for the declared completion reading (single-valuedness from F-independence + the Quillen pairing, v520), and the w_m normalisation itself *derived constructively* ($1/\det_j$ the Atiyah–Bott/zeta fixed-point factor from three independent sources, $\psi = 64$ reproduced, v523) — nothing in the measure chain is declared any more at that level; the residual [O] is the global BCOV integral beyond the fibre zero-mode factor (the declared measure is since shown *unique on the computable v508/v515 sector* up to exactly one named flat direction — sector rigidity, not a BCOV derivation, [`verification/v973_seam_route_narrowing.py`]). `WOIT.OS.TWISTOR.01` (the Osterwalder–Schrader twistor bridge — the actual bridge from compiler to physics): the α stage executed (Θ exists at the free level and free RP selects the same family, v519), β_1 executed (the μ_4 clock is the euclidean rotation, the gaugeable datum is the GSO $\mathbb{Z}_2$, gauge-fixed RP holds, v522), β_2 executed (the OS quotient explicit, $\mathcal{H}_{\text{phys}}$ positive definite at both levels, the clock a positive transfer operator with spectral calculus, v524), β_3 executed (the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed against σ_{std}: the OS cut and the $(2, 2)$ signature *forced*, the kill branch empty by algebra, the induced member = the Θ_{phys} carrier, v565) — γ is next, still under the *straddle-law* constraint: the first interacting seam toy fires Kill-Test 2 at toy level (RP breaks on exactly the quartet-straddling cuts, v529) — an honest threat and simultaneously the first hard selection principle for $\mathcal{A}_{\text{hol}}$, since *executed* as such: run as a selector on the interacting mark family, reflection positivity keeps exactly *one* member alive — $\delta = \pi/2$ with positive coupling, the alignment bit itself (v534, toy-fenced; the literal leading-order cone formalization is dead, the protection is nonperturbative); the law is since refined to the decidable Jaffe–Pedrocchi *odd-split* form, and the quartet-avoiding cut family of the surviving member carries a positive OS/GNS structure — an OS-viable cut geometry exists, the kill-test-2 counterweight narrowed, not removed ([`verification/v972_seam_interaction_front.py`], toy-level). Both contracts stay [O]; no marker moves.

Research Contract 1 — (U_{wall})

Goal. Prove

$$\nabla^\star_F = \mathrm{Sel}_{D_4,\,\text{cusp},\,\det R=8,\,\mathrm{Spec}(Q_+)=\{1,2,3\}}\big(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)\big),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4. The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4-symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation)**.** *The gate data are equivalent to the $SU(3)$ character variety $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C^4_{\text{cusp}} : \ M_1M_2M_3M_4 = \mathbf{1}\}/SU(3)$ with $C_{\text{cusp}} = \mathrm{Ad}_{SU(3)}\,\mathrm{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, and $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1\rangle$. To show: $\mathcal{M}^{D_4}_{\text{par}} \simeq \mathcal{X}^{D_4}_{\text{cusp}}$ as polystable Mehta–Seshadri / tame Hodge realisation.* Machine: the group presentation and D_4-action are finite and codable (Sage/Wolfram).

Finite rigidity — now proven [E]. The finite half of U0 is closed exactly: μ_4 has cross-ratio 2 and Möbius stabiliser $\langle z \mapsto iz,\ z \mapsto 1/z\rangle$ of order $8 = |D_4|$ (a 4-cycle of the marks + a reflection, faithful); $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$, $k = 1, 2, 3$, are μ_4-eigenforms ($z \mapsto iz :\ \omega_k \mapsto i^k\omega_k$) carrying the three nontrivial characters χ_1, χ_2, χ_3 with weights $(1, 2, 3) = \mathrm{Spec}(Q_+)$. Hence a genus-zero boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ with the $(1, 2, 3)$ grading (`QGEO.RIGID.01`, [E], [`verification/v168_qgeo_rigidity.py`]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays `QGEO.REALIZE.01` ([C]).

Seam Collar Realisation Theorem — the one central proof obligation [E]/[O] ([`verification/v176_seam_collar_realisation.py`])

Stated as one theorem and certified as a reduction — not a fabricated proof.

Theorem (target). *Given* (1) an oriented one-sided RP seam collar Σ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap $(2/3)^6$; (4) a faithful D_4 action on the boundary marks; (5) four parabolic marked points with μ_4-character decomposition; (6) an admissible $c{=}8$ boundary net — *then* the conformal boundary of Σ is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$, its H^1 character basis is canonically the generation basis with grading $(1,2,3)$, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$.

The proof decomposes into six steps, five already [E]:

1–2 *Geometry / uniformisation* [E] (v168): μ_4 has cross-ratio 2 and a faithful Möbius D_4 stabiliser of order 8; four genus-0 marks with faithful D_4 force the μ_4 square up to Möbius; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4-1 = N_{\mathrm{fam}} = 3$.

3 *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol $|k|$ (harmonic extension $e^{ikx-|k|y}$), the free chiral dispersion — bulk-detail-independent.

4 *H^1 character = generation grading* [E] (v137/v168): $\omega_k = z^{k-1}dz/(z^4-1)$ carry μ_4 characters i^k with weights $(1,2,3)$ = the A_3 exponents = $\mathrm{Spec}(Q_+)$ — natural, not merely isomorphic.

5 *One-particle contraction* [E] (v162): a μ_4-equivariant operator is diagonal in the cusp basis ($\frac{1}{|\mu_4|}\sum_j U^j X U^{-j} = \mathrm{diag}\, X$ for $U = \mathrm{diag}(1, i, -1, -i)$); its spectrum is the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, so the transfer eigenvalues $(1-\alpha)^{p_2}$ ($p_2{=}6$) are $\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$ — forced, no free knob.

6 *AQFT closure* [E] (v154/v175): $(D_5)_1 \otimes (A_3)_1$ with the isotropic μ_4 glue has index $4 = |\mu_4|$, $c = 5{+}3 = 8$, μ-index $16/16 = 1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ ($248 = 120{+}128$); full-cone RP holds for all m via the CAR second-quantisation functor.

The proof tree — and the *two* genuine open doors [O] ([`verification/v177_seam_marking_kernel.py`]). Taking the marks/D_4 as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\textsc{Realize} = \textsc{Marks} \cdot \textsc{Uniform} \cdot \textsc{Cohom} \cdot \textsc{Module} \cdot \textsc{Kernel} \cdot \textsc{Net},$$

of which *four nodes are now closed constructively*: Uniform [E] (an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 — so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}^1, \mu_4)$, strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as `FORM.QGEO.02`: $\rho^4{=}\mathrm{id}$, $\rho^2{\neq}\mathrm{id}$, $\sigma^2{=}\mathrm{id}$, $\sigma\rho\sigma{=}\rho^{-1}$, the orbit scaling to μ_4, the μ_4 permutation, and the multiplier classification "order-4 $\Leftrightarrow \zeta{=}\pm i$" are machine-proved in `TfptCarrier/MobiusUniformisation.lean`, audit: pass); Cohom [E] ($\rho^*\omega_k = i^k\omega_k$, grading $(1,2,3)$ — also **Lean-formalised** as `FORM.QGEO.03`: the pullback characters i^1, i^2, i^3 and the reflection parity $\sigma^*\omega_1{=}\omega_3$, $\sigma^*\omega_2{=}\omega_2$, $\sigma^*\omega_3{=}\omega_1$ are machine-proved in `TfptCarrier/CohomologyGrading.lean`, audit: pass); Module [E] (a *unique* μ_4-equivariant iso — multiplicity-free + residue normalisation, reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed; the parity is the Lean `FORM.QGEO.03` above, the *uniqueness* stays symbolic); and Net [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: **`QGEO.MARKS.01`** — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked D_4 boundary (the four marks from the anchor atom $e_1(a){=}4{=}|\mu_4|$, collisions excluded, $\tau\rho\tau{=}\rho^{-1}$) — and **`QGEO.KERNEL.01`** — the *raw* seam Calderón operator equals the μ_4-equivariant free $c{=}8$ gapped one-particle contraction *as an operator* ($C_\Sigma = U^{-1}C_{\mu_4}U$), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

How far each door opens by computation ([`verification/v178_marks_kernel_attempt.py`]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. *For Marks: given* a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1 = 3 = N_{\mathrm{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving faithful D_4 — so Marks reduces to the *single* topological premise "the seam double is genus-0 with an order-4 clock whose marks form one orbit". *For Kernel:* on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the "spectrum vs operator" distinction *vanishes* on the finite block. So Kernel reduces to the *single* premise "the full DtN on L^2 reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation" (principal symbol $|k|$, Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

Marks without a geometric posit: a Lefschetz/character argument ([verification/v195_marks_lefschetz_character.py]). The marks need not be *assumed* as D_4 boundary points: they follow from the H^1 character alone. The order-4 clock $\rho{:}z \mapsto iz$ fixes exactly $\{0, \infty\}$, so a puncture at a fixed point would contribute a *trivial* character to H^1; but the carrier datum is $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents $(1, 2, 3)$, with $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$), which has *no* trivial component — so no puncture sits at a fixed point. With $\operatorname{rank} H^1 = n - 1 = 3 \Rightarrow n = 4$ punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free μ_4 orbit (cross-ratio 2), and the closed-surface Lefschetz number $L(\rho) = 1 + 1 = 2 = |\{0, \infty\}|$ is consistent. So Marks reduces from the geometric posit "D_4 marks" to the milder *representation* premise "the raw seam H^1 carries the A_3-exponent rep" — a carrier-algebra statement, strictly less circular, still [O].

And the two doors are in fact ONE ([verification/v179_conformal_realisation.py]). Marks's residual is not even three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$, and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the *same* $\chi = 2$ that supplies the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points $\{0, \infty\}$, and a free order-4 orbit has exactly $4 = e_1(a)$ points. So Marks reduces to "the carrier clock acts *conformally* on the genus-0 double", and Kernel then *follows* from that same geometry (Lee–Uhlmann $|k|$ + the $c = 8$ rigidity v158 + the cusped-surface spectral split, residual $= H^1 = \mathbb{C}^3 = N_{\mathrm{fam}}$). **Hence the entire structural residual of the theory is a single geometric premise `QGEO.CONF.01`:** *the raw RP seam normal double is conformally* $(\mathbb{P}^1, \mu_4)$ *with the carrier clock as the order-4 Möbius automorphism $z \mapsto iz$.* Given it, everything — genus-0, the four D_4 marks, the $(1, 2, 3)$ grading, the DtN block, the rank-8 polarisation and the $(E_8)_1$ net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

And even that conformal premise reduces to a milder isometry premise ([verification/v180_clock_is_mobius.py]). `QGEO.CONF.01` need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is* $\mathbb{P}^1$ — the target is *produced*, not assumed — and by *Kerékjártó* (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3){=}4$) is topologically $z \mapsto iz$. An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of $\mathbb{P}^1$, and an order-4 Möbius map is $z \mapsto iz$ (elliptic, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the strictly milder, non-circular premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does *not* name $\mathbb{P}^1/\mu_4$ and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory**; its surface realisation from the raw seam stays honestly [O].

And one step further reaches bedrock ([verification/v181_clock_is_conformal_symmetry.py]). By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), i.e. the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification `QGEO.SYM.01`: *the seam's conformal structure is the μ_4-deck structure of the carrier clock* (the carrier μ_4 glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam's conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock:** the chain `REALIZE` → theorem → Marks+Kernel → `CONF` → `ISO` → `SYM` (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

A concrete geometric model for the bedrock premise: the μ_3-cover program ([verification/v597_mu4_cover_model.py]–[verification/v601_equivariant_dual.py]). The bedrock premise now has its first explicit *candidate realisation*: the cyclic cover $y^3 = x^4 - 1$ of $\mathbb{P}^1$ branched at μ_4

(order-3 cusp monodromies, exactly the contract's cusp structure). Machine-exact, in five rounds: the cover carries the rank-3 Eisenstein H_1, the projective D_4 via the deck, the cusp-class torsion $(\mathbb{Z}/3)^4$ of order 81, and the unique invariant hermitian form of Lorentz signature $(1,2)$ with $\det J = 8$ (v597/v599); the anti-holomorphic reflection (the real structure R, $R^2{=}1$, unique up to scale) exists and closes the operator dictionary (v599); the compiler pair (U, V) itself is *realised* — $\mathbb{Z}[\omega]$-integral, GL_3-conjugate to the compiler pair, integer on the saturated $\mathrm{Fix}(R)$ lattice, with saturation index $81 = 3^4$ in the exact v566 divisor pattern $(1,1,1,3,3,3,3)$ (v600, Lean-certified as `FORM.QGEO.04` in `TfptCarrier/CoverEmbedding.lean`: determinant-free Smith certificates, kernel `decide`); and the presentation is canonical — the J-adapted cusp projectors give a manifestly D_4-equivariant pair family (mark choice = winding choice) realising the *dual* parabolic, with the two parabolic types exchanged by the J-adjoint, and the v590 Klein four-group living integrally on the curve (v601/ [`verification/v602_duality_forms.py`]). Honest status: this is a *model*, not yet the seam — the identification of the raw RP seam with the cover (the Weil-twisted positivity comparison against the v572 Gram data, and the geometric identity of the seam line) stays **[O]**; `QGEO.SYM.01` does not move.

A sharper, operator-checkable restatement: `QGEO.ENERGY.01` ([`verification/v192_qgeo_energy_clock.py`]). The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock ρ preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double*, $\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$. In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality → Möbius → $z \mapsto iz$ → μ_4). Its appeal is checkability: $\Lambda_\Sigma = |k|$ is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however:** "ρ preserves Λ_Σ" is plausibly *equivalent* to "ρ is the conformal deck" (`QGEO.SYM.01`), so `QGEO.ENERGY.01` *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if ρ is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays **[O]**; the gain is a sharper operator-analytic target, not a closure.

The non-circular sharpening: `QGEO.ENERGY.02` ([`verification/v193_qgeo_energy_commutator.py`]). The way to make "ρ defined independently" precise is the *energy commutator* $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims: (1) ρ is the Coxeter element of $W(A_3) = S_4$ (order $4 = |\mu_4|$, v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* — Λ_Σ must be the DtN of the *raw* RP seam (definable from the RP data alone), not the μ_4 normal-form operator; (3) $[\rho, \Lambda_\Sigma] = 0$ forces the μ_4 character grading $(1,2,3)$ on H^1, already established (`QGEO.COHOM.01`). An exact consistency rider pins the direction: the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ *iff* $\ln m = q(A_3) = \frac{3}{4}$ (the *family* glue norm), while $q(D_5) = \frac{5}{4}$ (the carrier norm) misses by $g_{\mathrm{car}}/N_{\mathrm{fam}} = \frac{5}{3}$ — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds, $[\rho, \Lambda_\Sigma] = 0$ turns `QGEO.SYM.01` from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, **[O]**.

Subclaim (2) tackled — one notch closer, not closed ([`verification/v194_raw_seam_dtn_rp.py`]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator $T = e^{-H}$ from the RP state and reflection alone — no coordinates (v54; §(4) of `tfpt_2`) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$, a contraction). So Λ_Σ is RP-definable without ever naming $\mathbb{P}^1 \setminus \mu_4$. The commutator $[\rho, \Lambda_\Sigma] = 0$ then *closes on the finite cohomology* H^1 **[E]**: $\rho : z \mapsto iz$ acts on $\langle w_1, w_2, w_3\rangle$ ($w_k = z^{k-1}dz/(z^4-1)$) as $\rho^* w_k = i^k w_k$, distinct characters $i, -1, -i$, so any ρ-equivariant operator is diagonal there (v177). What does *not* close is the *full-L^2* commutation beyond H^1 — it is exactly the statement that the quasi-free seam state's modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to "intrinsic BW geometric modular covariance of the quasi-free seam state" — a named, standard (hard) AQFT property, one notch sharper, but still **[O]** (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

A concrete variational handle on the residual ([`verification/v196_seam_energy_functional.py`]). The open BW statement can be made a discretisable extremum problem via the failure functional $E_{\mathrm{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|^2_{HS} + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, whose zero locus is exactly the `QGEO.SYM.01` conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite H^1 block $E_{\mathrm{fail}} = 0$ at the μ_4 configuration ($\rho = \mathrm{diag}(i, -1, -i)$, Λ μ_4-equivariant, Θ = conjugation) and > 0 for any μ_4-breaking perturbation, so μ_4 is a genuine minimiser **[E]**; minimising E_{fail} over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational* *target*, not a bare definition — but still **[O]**.

The decisive crack: principal symbol exact + Tomita–Takesaki ([`verification/v198_modular_commutator_reduction.py`]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is $|k| = \sqrt{-\partial_\theta^2}$ (Lee–Uhlmann, v156); in the Fourier basis $e^{in\theta}$ it is $\mathrm{diag}(|n|)$ and the clock $\rho : z \mapsto iz$ (rotation by $\pi/2$) is $\mathrm{diag}(i^n)$, both *diagonal*, so $[\rho, |k|] = 0$ *exactly* on all of L^2 — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator Δ_ω is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with Δ and hence with $\log \Delta \sim \Lambda_\Sigma$: thus $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$ — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite H^1 block is already μ_4-invariant (v177), the entire residual is now the single *state*-invariance "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega$, equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of `QGEO.SYM.01`. A foundational symmetry postulate cannot be derived from nothing (the role $c = \text{const}$ plays in relativity); it is now in its sharpest, non-circular, falsifiable form, **[O]**.

The residual, fully localised and numerically probed ([`verification/v199_seam_state_invariance.py`], [`verification/v200_seam_variational_scan.py`]). Attacking $\omega \circ \rho = \omega$ on the *raw* seam (not the μ_4 normal form) confirms it is non-circular — ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance $C_\Sigma = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, to $[\rho, H_1] = 0$, i.e. H_1 *is block-diagonal in the four μ_4-character classes* $\{n \equiv r \bmod 4\}$. The principal symbol $|k| = \mathrm{diag}(|n|)$ passes automatically, so the entire residual is the single bounded statement "the sub-principal symbol of the raw DtN has no off-character matrix elements". The companion variational scan minimises $E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 + (\text{faithfulness})$ over a free clock angle and finds its *unique* minimum at $\theta = \pi/2$ — the μ_4 clock $z \mapsto iz$ (the only faithful order-4 grading $i, -1, -i$) — a numerical sign that the variational principle selects μ_4. This is the "next executable theorem" a working mathematician can take up; it does not close the postulate, **[O]**.

The bounded sub-principal residual is the image of the μ_4 marks, not a free postulate ([`verification/v201_seam_subprincipal_marks.py`]). One genuine step further: the seam DtN is $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature function $f(\theta)$ on the seam circle (the standard Steklov/DtN structure), so $\langle n|M_f|n'\rangle = f_{n-n'}$. Then M_f is μ_4-character-block-diagonal *iff* f has Fourier support only on modes $\equiv 0 \pmod 4$, i.e. iff f is $\mathbb{Z}_4$-invariant (the principal $|k|$ already commutes, v198). But a curvature sourced by the four marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ for *any* profile g, has $f_m = 4\, g_m\, [m \equiv 0 \bmod 4]$ — automatically $\mathbb{Z}_4$-invariant. So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 equally-spaced marks ($\mathbb{Z}_3$) or 4 generic marks both break it (negative controls). The residual of $\omega \circ \rho = \omega$ thus collapses to "the DtN sub-principal symbol is *mark-local* (the seam is flat away from the μ_4 marks = the conformal-deck structure, v177)" — a locality/flatness property, structurally definitional, the narrowest form yet of `QGEO.SYM.01`, **[O]**.

Certified on realistic Steklov profiles, and at the state level ([`verification/v210_mark_local_dtn.py`]). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump $g(\theta)$ localised at a puncture, summed over the four marks $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, has *every* off-$(\bmod 4)$ Fourier coefficient vanishing to $< 10^{-16}$ (exact 4-fold cancellation), the assembled DtN $\Lambda = |D_\theta| + M_f$ has $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the *quasi-free state* it defines, with covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, is clock-invariant: $\rho\, C\, \rho^\dagger = C$ to $< 10^{-13}$, i.e. the actual $\omega \circ \rho = \omega$, not merely $[\rho, \Lambda] = 0$. A single off-centre bump, a $\mathbb{Z}_3$ (three-mark) source and four unequally-spaced marks each break both by $O(1)$, and the mark-local commutator stays $< 10^{-10}$ as the truncation grows ($N \in \{16, 32, 64\}$) — not an artefact. This *numerically certifies the implication* "mark-local DtN $\Rightarrow \omega \circ \rho = \omega$" for real profiles; the premise that the physical seam *is* mark-local stays **[O]** (it does not close `QGEO.SYM.01`).

The same residual, read dynamically: modular flow + a recovery Lindblad generator ([`verification/v238_modular_lindblad_dynamics.py`]). The whole `QGEO.SYM.01` discussion above is *static* — a commutator condition $[\rho, \Lambda_\Sigma] = 0$. Read as a *dynamics* it is the statement that the modular flow $\sigma_t = \Delta^{it}$ (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian $\Lambda_\Sigma = \log \Delta_\omega$) *conserves* the μ_4 clock: $\rho\, \sigma_t\, \rho^{-1} = \sigma_t\ \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ — so the open premise is exactly "modular time conserves the carrier charge". The dissipative half is the recovery channel T of v221 read as a generator: $L = \log T$ is a doubly-stochastic (CPTP-classical) Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time), and its dissipative gap is $-\log((2/3)^6) = 6 \log(3/2) = \Delta$, the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together $G = i\Lambda_\Sigma + L$ is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real≤ 0) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that σ_t is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra — is the same `QGEO.SYM.01` bedrock, [O].

From the dynamics to a QFT skeleton on the seam ([`verification/v239_kms_thermal_time.py`], [`verification/v240_gns_os_reconstruction.py`], [`verification/v241_dhr_sectors.py`], [`verification/v242_spin_statistics_modular.py`], [`verification/v243_haag_ruelle_braiding.py`], [`verification/v244_spectral_action_contract.py`]). With the static→dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal time* (v239): the modular flow is KMS at $\beta = 1$, and the seam unit $2\pi = 1/(4c_3)$ fixes the modular temperature as the horizon temperature $T_H = c_3/M$. *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from $(\mathcal{M}, \omega)$, and the OS Hamiltonian $H_{\rm OS} = -\log T = -L$ is positive with gap Δ — the v64 "$H \ge 0$ + mass gap" is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (`FORM.GLUE.01`); fusion is recovered from the modular S by Verlinde, the Gauss–Milgram sum returns $c = 8$, and the condensation tower $16 \to 4 \to 1$ (v235/v237) selects the holomorphic $(E_8)_1$. *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed $(E_8)_1$. The one genuinely 4d step is typed as a contract, `CONTRACT.QFT4D.01` (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open [O] target. Its representation-theory core is now discharged ([`verification/v245_spectral_matter_content.py`]): the **16** is exactly one *anomaly-free* SM generation ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$) with the forced SM hypercharges and the spectral-action value $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\, g_1$), so only the seam→Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (`QGEO.SYM.01`) plus *one contract* (`CONTRACT.QFT4D.01`), not a diffuse list.

The perturbative 4d S-matrix is constructible ([`verification/v269_spert_paqft_skeleton.py`]). The 4d scattering layer is typed precisely as three *distinct* objects: $S_{\rm top}$ (the 2d DHR braiding monodromy, $|M|$=1 statistics, v243), $S_{\rm pert}$ (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and $S_{\rm phys}$ (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. [E] the spectral-action a_4 interaction is power-counting renormalizable (every operator mass dimension ≤ 4: 7 marginal +3 relevant, a *finite* counterterm basis), so [C] by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix $S(g) = T\exp(i\int g\, L_{\rm int})$ is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. [C] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit. [C] this is *perturbative* only: the nonperturbative ambient measure `QG.AMB.01` is now a [C] redundancy (v369), a certification object rather than missing dynamics, and the canonical 4d reading remains boundary-only (v265); $S_{\rm pert}$ is the perturbative cross-section layer on top (`QFT4D.SPERT.01`).

A concrete one-loop EG number ([`verification/v271_eg_oneloop_quartic.py`]). The skeleton is taken from "exists" to an actual Epstein–Glaser renormalization: the order-1 product $T_1 = {:}L_{\rm int}{:}$ needs no extension, and the first one is the order-2 T_2, whose UV behaviour is the Steinmann *scaling degree*. [E] the massless propagator has sd = 2 in $d = 4$, the one-loop bubble (two propagators) has sd $= 4 = d$, so the singular order $\omega = \text{sd} - d = 0 \Rightarrow$ *exactly one* logarithmic local counterterm ($C(\omega{+}d, d) = 1$ — finite, renormalizable, no infinite tower); the one-loop factor is $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly, the *same* $1/(16\pi^2)$ that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop running is then $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s, t, u channels), with the EG initial condition fixed by the spectral action, $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$. [O] this is order-2/one-loop, one diagram class; the all-order perturbative construction stays open (`QFT4D.SPERT.02`), while the nonperturbative measure is discharged as a [C] redundancy (below).

Red-team caveat, now discharged ([`verification/redteam/rt_F_qft4d.py`]). Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the local R^2/Weyl^2 *truncation* gives a four-derivative propagator $1/(p^2(p^2 + M^2))$ whose massive spin-2 mode has a negative residue (the Stelle ghost). But that ghost is a Seeley–DeWitt *truncation artefact*: the entire seam KMS form factor $a(p^2){=}e^{p^2/M^2}$ keeps its only pole at $p^2{=}0$ (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established. The matter+gauge $S_{\rm pert}$ is unitary outright; the *nonperturbative* `QG.AMB.01` (asymptotic safety v207 / the boundary $(E_8)_1$ net) is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap.

The gauge running is an EG order-2 number ([verification/v273_eg_gauge_running.py]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content: $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$, i.e. [E] $b_3 = -7$, $b_2 = -\frac{19}{6}$, $b_1 = \frac{41}{10}$ (GUT-normalised; the same 41 as the carrier algebra $10\,b_1 = g_{\rm car}\,2^{g_{\rm car}-2}+1$, v159). [C] the physical RG directions ($b_3 < 0$ asymptotic freedom, $b_1 > 0$ the $U(1)$ Landau pole) are EG-derived and feed the unification cross-check (v246/v249); [O] the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (QFT4D.SPERT.03).

The carrier content *is* one anomaly-free SM generation ([verification/v310_carrier_sm_anomaly.py]). Those one-loop coefficients sit on top of an exact spectrum/anomaly closure: the carrier half-spinor $16 = \Lambda^{\rm even}(\mathbb{C}^5) = 1{+}10{+}5$ decomposes under $SU(5) \subset SO(10)$ as $10{+}\bar{5}{+}1$ = exactly one Standard-Model generation $(Q, u^c, d^c, L, e^c, \nu^c)$, and that content is [E] gauge-anomaly-free ($\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$, exact) while reproducing $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family would shift b_1 off 41/10, so $N_{\rm fam}{=}3$ *pins* the RG. [E] for the spectrum, anomalies and one-loop β; the full Epstein–Glaser interacting S-matrix and the Yukawa sector stay [C]/[O].

The bridge to the physical S-matrix and one-loop unitarity ([verification/v278_lsz_bridge_unitarity.py]). The perturbative layer connects to the physical asymptotic S-matrix: [E] the EG time-ordered products of $S_{\rm pert}$ are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives $S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$ (the three-layer S-matrix $S_{\rm top}\,|\,S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$), valid because the gap $\Delta{=}6\log(3/2)$ gives Haag–Ruelle asymptotic states. [E] *one-loop unitarity* is the optical theorem: the s-channel bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ is *exactly* the two-body phase space (turning on at the physical threshold $s{=}4m^2$), so $2\operatorname{Im} M = \sum\int d\Pi\,|M|^2$ holds and the matter+gauge sector is unitary order by order. [C] the gravity sector's Stelle ghost is a truncation artefact (perturbative spin-2 unitarity established, v304/v370/v380); the nonperturbative QG.AMB.01 is a [C] redundancy (v369) (QFT4D.SPERT.04).

The first hard data confrontation ([verification/v246_unification_data_crosscheck.py]). TFPT's gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ "Dark matter = no new state"); the E_8 cascade is an arithmetic even-integer spine ($D_n{=}60{-}2n$, v5), *not* a particle/threshold tower. So a "TFPT-adapted" RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over $\sim 10^{13}$–10^{17} GeV with residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at $\mu \approx 1.7{\times}10^{14}$ GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, [X]: with "no new state" there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

The carrier-native Pati–Salam program ([verification/v247_e8_branching_no126.py], [verification/v248_ps_algebra.py], [verification/v249_ps_unification.py], [verification/v250_dirac_triple_contract.py]). The unification tension above is resolved *natively* if the carrier $SO(10)$ is gauged (the carrier $D_5{\oplus}A_3 = SO(10){\times}SU(4)$ is Pati–Salam). *First*, the E_8 audit hull *forbids* the **126**: the **248** branches as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4})$, so the available $SO(10)$ reps are exactly $\{1, 10, 16, 45\}$ — the renormalisable 126_H is algebraically absent, making the "16_H over 126_H" selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ gives the unimodular group $SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$, the exact SM hypercharges $Y = T_{3R} + (B{-}L)/2$ and the matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale, $M_{PS} = \kappa M_s$, $\kappa \in [1.0, 1.7]$ at one loop ($\to [1.0, 1.2]$ at two loops), and proton decay selecting the content (minimal excluded, the E_8-allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual "carrier is gauged" reduces to ONE constructive object, the spectral triple (A, H, D_F, J, γ) with A_F Pati–Salam and $H_F = 3{\times}16$, whose NCG axioms (the first-order condition above all) and the σ =scalaron identification are the open contract CONTRACT.QFT4D.DIRAC.01 (v250). *Fifth*, a first concrete step on that triple is taken ([`verification/v251_first_order_sigma.py`]): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/σ term — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ (honest correction: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly ([`verification/v252_full_finite_triple.py`]) — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — and the NCG axioms are verified by direct computation: reality J, grading γ, KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on $\{\nu_R, e_R\}$), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}} = m_D^2/M_R$ — thereby discharging five of the seven obligations of `CONTRACT.QFT4D.DIRAC.01` to [E] (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio κ in $M_{PS} = \kappa\, M_s$ is pinned to a tight $O(1)$ interval (≈ 1.3) by the RG and shown to be a scale-free ratio of heat-kernel coefficients, $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}}$, over the same 96-dim H_F ([`verification/v253_kappa_scalaron_ps.py`]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations Ω^1_D give exactly the one SM Higgs doublet $(1,2)_{1/2}$ with no junk and no **126**, while σ (the ν_R Majorana scalar, $B{-}L{=}2$) appears only as the beyond-first-order Pati–Salam field ([`verification/v254_inner_fluctuations.py`]); the spectral-action Seeley–DeWitt expansion a_0, a_2, a_4 yields the cosmological, Einstein–Hilbert and Yang–Mills+R^2+Higgs terms with $b/a^2 = 1/3$, $\sin^2\theta_W = 3/8$ and the R^2 scalaron, making κ an explicit moment ratio ([`verification/v255_spectral_action_expansion.py`]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three ν_R forces a skew intersection form, corank 1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions, *quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([`verification/v256_orientability_poincare.py`], [`verification/v257_quasiorient_modpd.py`]). *Ninth*, the built D_F is no independent posit but the *modular/covariance induction* of the boundary seam state ([`verification/v258_dirac_covariance_induction.py`]): a quasi-free (KMS, β=1) state is fixed by its covariance C (a positive contraction), with one-particle modular Hamiltonian $H = \log((1{-}C)C^{-1})$ (Araki; Casini–Huerta) inverse to the KMS occupation $C = (1{+}e^{H})^{-1}$; inducing the carrier covariance $C_F = P_F C_\Sigma P_F$ returns the v252 operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw m_D^2/M_R recovered — so D_F is the Kasparov shadow $[D_\Sigma]\hat{\otimes}_{(E_8)_1}[\mathcal{K}_{\text{car}}]$ of the boundary geometry and the v18 Yukawas are the *readout* of C_Σ, [E] modulo the one [O] seam realisation. *Tenth*, the last open spectral-action input — the cutoff function f — is fixed by the same seam state ([`verification/v259_modular_cutoff_kappa.py`]): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (v239) and the seam unit $2\pi{=}1/(4c_3)$ (v58) fixes that β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*, not chosen, with moments $f_2/f_0 = f_4/f_2 = 1$ *exactly* (a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$, so the choice is non-vacuous); hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$ loses its scheme factor and the v253/v255 open [O] cutoff freedom becomes a [C] finite trace ratio over the 96-dim H_F (the RG $\kappa{\approx}1.3$ pinning $c_{PS}/c_{\text{grav}}{\approx}1.7$). So unification is now *one* theory decision (gauge $SO(10)$?) plus *one* constructive object, both falsifiable — not a fog. The colour $SU(4)_c$ here sits inside $SO(10){=}D_5$ and is distinct from the carrier family factor A_3.

The implication is now Lean-formalised (`FORM.QGEO.01`). The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ($f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \bmod 4)$, i.e. $[\rho, M_f] = 0$) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext, Classical.choice, Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the document-1 analytic input — so the conditional theorem "`SeamDeckPremise` $\Rightarrow \omega \circ \rho = \omega$" is now [E] while `QGEO.SYM.01` stays the one open [O] postulate.

The two residuals are one canonical metric: the pillowcase ([`verification/v214_seam_pillowcase.py`]). The isometry residual `QGEO.ISO.01` (v180) and the mark-locality residual `QGEO.SUBPRIN.01` (v201) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class.* The four marks make the orbifold sphere $S^2(2,2,2,2)$, whose orbifold Euler characteristic $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ is *Euclidean*: by Troyanov's cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits π sum to $4\pi = 2\pi\chi(S^2)$) — this *is* the v201 "flat away from the marks". And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that v168 already proved: $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ gives $j(2) = 1728$, so the modulus is the *square* torus $\tau = i$, whose complex multiplication by $\mathbb{Z}[i]$ supplies the order-4 isometry $z \mapsto iz$ (explicitly $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$, order 4); a generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$, so the clock is specific to μ_4. Thus the two open QGEO residuals *collapse to one* canonical-metric statement and `QGEO.ISO.01` ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise "the RP seam metric *is* the uniformising constant-curvature representative of its conformal class" stays [O] — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (`QGEO.PILLOW.01`).

The converse, on the canonical metric: raw seam ⇒ mark-locality ([`verification/v264_raw_seam_marklocal.py`]). Composing the chain explicitly closes the gap between v216, v201 and v198 on the uniformising metric: the flat pillowcase has its curvature *only* at the four cone points (v216), so the DtN sub-principal symbol M_f is a μ_4-orbit sum, hence $\mathbb{Z}_4$-invariant (Fourier support on $4\mathbb{Z}$, verified by FFT), hence block-diagonal in the clock-character basis, hence $[\rho, M_f] = 0$; with the principal symbol $|k|$ already commuting (v198) this gives $[\rho, \Lambda] = 0$ and $\omega \circ \rho = \omega$ *on this metric*. A negative control (curvature *off* the μ_4 orbit) breaks $\mathbb{Z}_4$-invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement "the raw RP seam carries the uniformising flat pillowcase metric" — which *is* `QGEO.SYM.01`. A genuine reduction/sharpening (`QGEO.ENERGY.03`), [O] — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

The minimal axiom: a rigidity theorem ([`verification/v267_qgeo_rigidity_minimal_axiom.py`]). One can sharpen the bedrock to its irreducible kernel. [E] the *single* bare statement "the seam admits the carrier's order-4 clock as a conformal symmetry cyclically permuting its four marks" forces everything: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 *independent of* a; cross-ratio 2 ⇔ j=1728 (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) ⇔ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on y^2=x^3−x; and from the square configuration the unique flat pillowcase metric (Troyanov), mark-locality, $[\rho, \Lambda]$=0 and $\omega \circ \rho$=ω all follow (v214/v201/v264/v198). [E] neg controls: a generic four-mark set has $j \neq 1728$ and only a $\mathbb{Z}_2$ symmetry (no clock), the hexagonal point j=0 has a $\mathbb{Z}_6$ symmetry (order $6 \neq 4$) — only the square (order-4) point realises the μ_4 clock. [C] a second, independent reason: $\{1, i, -1, -i\}$ is defined over $\mathbb{Q}$ with j=1728 (CM by $\mathbb{Z}[i]$), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. [O] So `QGEO.SYM.01` is now in its minimal sharpest form — the bare existence of that order-4 conformal symmetry, which (like c=const in relativity) cannot be derived from nothing. The *reduction* is closed (axiom ⇒ everything); the axiom itself stays the one foundational postulate (`QGEO.SYM.02`).

The flat geometry closes the commutator to all orders ([`verification/v276_qgeo_flat_closes_commutator.py`]). The state-level residual $[\rho, H]$=0 — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat τ=i pillowcase is granted. [E] if $H = f(\Delta_{\text{flat}})$ is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck ρ ($z \mapsto iz$) is a flat-τ=i isometry, then $[\rho, \Delta]$=0 ⇒ $[\rho, H]$=0 *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control). This *upgrades* the leading-order commutation v198 (principal symbol) and v201 (subprincipal) to the full H — flatness removes the lower-order terms. [O] so `QGEO.SYM.01` collapses to a *single geometric* statement, "the raw RP seam state is the flat τ=i pillowcase quasi-free state (H=$\sqrt{-\Delta_{\text{flat}}}$)"; everything downstream is [E], and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (`QGEO.SYM.03`).

The obligation as one precise lemma, Lean-backed ([`verification/v279_qgeo_obligation_lemma.py`]). The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise "the raw seam carries the flat τ=i pillowcase metric g_{flat} (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{g_{\text{flat}}}}$", the conclusion $\omega_\Sigma \circ \rho = \omega_\Sigma$ (and hence mark-locality, the metric inclusion, E_8) follows. [E] the reduction DAG from "free RP seam" to this premise has *exactly one* un-discharged leaf — "the raw seam state *is* the flat state" — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes ([C]): RP-state uniqueness on the flat τ=i orbifold, or Troyanov + OS reconstruction. [E] the all-orders closure step is now a *Lean theorem*: `TfptCarrier/SeamDeckClosure.lean` (`flat_all_orders_clock`) proves that any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — upgrading v198/v201 to the full operator with only the standard axioms, no `sorry` (`FORM.QGEO.04`). [O] the one premise stays the honest axiom (`QGEO.OBLIG.01`).

Self-investigation: the experiment and the simplification ([verification/v280_pillowcase_steklov.py], [verification/v282_e8_tau_i_unification.py]). Two things can be done *ourselves.* (i) A direct numerical experiment on the actual flat $\tau{=}i$ pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0 \Rightarrow [\rho,C]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still [O]. (ii) A candidate *simplification*: the $(E_8)_1$ character $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$ gives $\chi_{E_8}(i) = 1728^{1/3} = 12$, so $\tau{=}i$ is *both* the order-4 CM point (`QGEO.SYM.01`, flat geometry) *and* the $(E_8)_1$ partition-function modulus (`QG.AMB.01`, holomorphy) — the two open premises are two faces of one object, "the raw seam is $(E_8)_1$ at $\tau{=}i$", dropping the obligation count $2 \to 1$ (`QGAMB.UNIFY.01`).

The two proof routes, decomposed ([verification/v284_route_i_rp_uniqueness.py], [verification/v285_route_ii_seam_condensation.py]). The keystone — now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` via the lattice/S_3 stack (below; the parent stays [O]) — can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — [E] five are already [E]/[O] (RP→DtN v194; marks→pillowcase v195/v216; the flat Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$, principal symbol $|\xi|$ with no curvature term; covariance from the DtN v258; μ_4-invariance v276), and [C] Troyanov ($\chi_{\text{orb}}{=}0$) reduces the residual to "the raw seam is the constant-curvature geometric state". *Route (ii)* (seam-net condensation) is a four-lemma chain — [E] three are discharged (no abelian sector $\Leftrightarrow$ det $K{=}1$ $\Leftrightarrow$ holomorphic $\Leftrightarrow$ E_8 v234/v277/v281; the tower 16→4→1 v92), leaving "the free RP seam condenses the μ_4-Lagrangian glue" =`QGEO.SYM.01`. [E] **the two open lemmas coincide** — both say "the raw seam is the $(E_8)_1$ at $\tau{=}i$" (v282), so the canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$ is the *single* [O] statement that closes both (`QGEO.ROUTEI.01`/`QGAMB.ROUTEII.01`).

The Seam Equivalence Theorem, named and firewalled ([verification/v286_seam_equivalence_contract.py]). That single statement is now the named target `SEAM.EQUIV.01`: *the raw RP seam state reconstructs canonically the holomorphic* $(E_8)_1$ *boundary net at the* $\tau{=}i$ *pillowcase* (respecting the DtN, the KMS flow, the μ_4 deck, the Q-system extension, det $K{=}1$ and the $(E_8)_1$ character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of c* plays in relativity (cf. v267): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. [E] the contract types four objects and six arrows — two closed (the two routes), one conditional (v282), one open Gral — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through `SEAM.EQUIV.01` (this kills the "E_8 smuggled into the geometry and pulled back out" attack). The two proof routes are then attacked directly: [E] Route A (AQFT, [verification/v287_free_rp_bulk_to_holomorphic_boundary.py]) reduces the Gral to *one* standard theorem, "invertible (SRE) Gaussian bulk $\Rightarrow$ single-sector boundary" (4/5 lemmas discharged); and [E] Route B (DtN, [verification/v288_full_l2_subprincipal_z4.py]) *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod 4$), so $[\rho,\Lambda]{=}0$ on all of L^2), shrinking the residual to the single sharper question "why is the raw seam sub-principal term mark-local?" (`SEAM.EQUIV.A01`/`SEAM.EQUIV.B01`).

The route split (2026-07-22; ID restructuring, no status change in substance). The target statement now carries *two named routes* as ledger rows of their own: `SEAM.EQUIV.MMST.01`, the conditional scaling-limit route (the lattice/S_3/Lean stack documented in this section — *closed modulo cited theorems*; its [O] residual, formerly the abstract continuum existence v336, is since *carved into a named list* against the cited Osborne–Stottmeister coverage ([verification/v973_seam_route_narrowing.py]): *covered* is the $\mathfrak{so}(16)_1$ fermion-bilinear current algebra — strong convergence, Virasoro and correlator control, polynomial errors; *not covered*, and now the named residual, are N1 the μ_4/GSO extension $SO(16)_1 \to (E_8)_1$ as a net statement, N2 the twist/spin-field sector, N3 the one-sided chiral edge net, and N4 the OS/holomorphy selector — with a measured collar convergence rate $p = 3.48$ inside the preregistered band while the trivial $M{=}3$ control does not converge), and `SEAM.EQUIV.TWISTOR.01`, the open Costello–Li construction (prepared by `CELEST.SEAM.01`, bridged to physics only by `WOIT.OS.TWISTOR.01`; [O] — its declared measure is since shown *unique on the computable v508/v515 sector* up to exactly one named flat direction, the uniform sector-weight shift onto the untwisted Okubo square (constraint ranks 9/8/2/2, joint rank 13 on 14 moduli): sector rigidity, *not* a BCOV derivation, [verification/v973_seam_route_narrowing.py]). The parent claim `SEAM.EQUIV.01` remains, with exactly this logic as its claim text: *closed if* `MMST.01` *or* `TWISTOR.01` *closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open* $\Rightarrow$ *the parent stays [O] as an unconditional claim.* Every "closed modulo a cited theorem" statement in the papers and on the website is henceforth attached to the MMST route ID; the assertions themselves are unchanged.

The two routes share an executable OS skeleton (2026-08-27, `SEAM.EQUIV.SKELETON.01` [E] new; [`verification/v973_seam_route_narrowing.py`]; no marker moves — `SEAM.EQUIV.01`, `SEAM.EQUIV.MMST.01` and `SEAM.EQUIV.TWISTOR.01` all stay [O]). The two routes are bridged by an OS skeleton that *runs*, 5/5: all WOIT kill-test main invariants hold verbatim on the $p+ip$ collar edge, every divergence is control-typed to the checkerboard mechanism — confirming the v519-R4 finding from the second route — and the collar top edge realizes the anti-chiral state exactly. Toy/census level; nothing here is a continuum statement.

Route B sharpened to one analytic lemma; both routes precisely scoped. That last question is decomposed ([`verification/v289_marklocal_raw.py`], `MARKLOCAL.RAW.01`) into a five-lemma chain whose only open link is [O] *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix ([C], b-calculus/Melrose), the μ_4-orbit source, the $4\mathbb{Z}$ Fourier support (v264/v288) and the full-L^2 commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert "full-L^2 QGEO" — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A's "one standard theorem" is itself a composition of citable results (Kitaev/Freed–Hopkins, Müger/Kawahigashi–Longo–Müger, Kawahigashi–Longo, [`verification/v287_free_rp_bulk_to_holomorphic_boundary.py`]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B's Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

Red-team: $\mathbb{Z}_4$-invariance is not mark-locality. A deliberate adversary ([`verification/v290_z4_smooth_curvature_adversary.py`]) shows the honest limit of Route B's lift: a *smooth* $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$ *passes* the v288 commutator ($\|[\rho,\Lambda]\| = 0$, since $i^n - i^{n\pm4} = 0$) yet is not mark-local — so $[\rho,\Lambda]{=}0$ is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ($\max|\Delta\lambda| = 0.155$, $\Delta\mathrm{Tr} = 0.019$ at $\varepsilon = 0.3$), i.e. the KMS weight and $(E_8)_1$ character that `SEAM.EQUIV.01` must match. Consequently the open lemma is named in its own right ([`verification/v291_flataway_contract.py`], `FLATAWAY.RP.01`): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

All three routes attacked; Flat-Away reduced to one selection principle. Each route is now carried to its precise reduction: [E] the *heat* route ([`verification/v292_flataway_heat_reduction.py`]) shows the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature (leading invariant $\|f\|_{L^2}$), so Flat-Away $\Leftarrow$ "the seam fixes the a_2 heat coefficient"; [E] the *spectral* route ([`verification/v293_flataway_spectral_hessian.py`]) lifts this to the full smooth-$\mathbb{Z}_4$ space — mode $4k$ splits the degenerate Steklov level $|2k|$, the mismatch Hessian over $\{4, 8, 12\}$ is positive-definite ($\{0.497, 0.500, 0.503\}$), so the flat seam is a *strict* isolated minimum (answering "only one direction"); and [E]/[C] the *variational* route ([`verification/v294_flataway_rp_energy.py`]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix a_2 / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A's holomorphy half, `SEAM.EQUIV.01`.

The heat route made exact and Lean-formalised. The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([`verification/v295_flataway_a2_exact.py`]): the first variation of $\mathrm{Tr}\,e^{-t\Lambda}$ vanishes (the smooth off-mark curvature has zero mean, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\,e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon{=}0$ is the *global* minimum — so $\Delta\mathrm{Tr} \geq 0$ for every deformation, $= 0$ iff $f = 0$; the exact second-order coefficient is the divided-difference kernel of e^{-tx} (validated to 10^{-6}). The algebraic $\Delta{=}0 \Leftrightarrow$ flat step is machine-formalised in Lean (`FORM.FLATAWAY.01`, `SeamDeckClosure.lean`: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away's heat route is reduced to the single external invariant "the seam fixes a_2", with its positivity both proved and formalised. The a_2 coefficient itself is now in *closed form* ([`verification/v296_flataway_a2_closed.py`]): the operator tails telescope geometrically to $W_k^{\mathrm{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k{-}1)$-term middle, e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ (manifestly positive; small-t

$W_k = t/2 + O(t^3)$). Symmetrically, Route A's "one standard import" is now a precise, citable theorem stack ([`verification/v297_route_a_literature_stack.py`]): LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk ⇒ $(E_8)_1$", all three legs established, the one shared hypothesis (seam-bulk invertibility) coinciding with Route B's Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input, which the lattice/S_3 stack (below) now pins at every computable level, making the keystone's MMST route (`SEAM.EQUIV.MMST.01`) *closed modulo cited theorems.*

Flat-Away hardened, and the pin derived from $(E_8)_1$ ([`verification/v300_flataway_rigid.py`], `FLATAWAY.RIGID.01`). The spectral route was a *soft* minimum ("the mismatch grows with ε"); it is now a *discrete* obstruction. **[E]** the flat fingerprint is the integer Steklov ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (the exact 2×2 block $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, validated against the full Toeplitz operator), so **[E]** any $g_k \neq 0$ lifts the degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every $g_k = 0$, a degeneracy obstruction robust to rescaling and strictly stronger than the v293 "deviation grows" scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: **[C]** the $(E_8)_1$ vacuum character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (E_8 even unimodular ⇒ integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder ⇔ flat seam — i.e. "the seam fixes its spectrum" *is* the $(E_8)_1$ rationality of Route A. **[O]** Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two `SEAM.EQUIV.01` routes provably converge on one rationality fact (consistent with the v286 firewall).

Route A's last hypothesis discharged — the gapped quasi-free bulk is invertible ([`verification/v301_route_a_invertible.py`], `SEAM.EQUIV.A.INV.01`). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A's open input (v297 LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev's periodic table, class D, the only invariant being the chiral central charge). **[E]** the carrier is 16 Majoranas, so $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8 = 16/2$ (v148); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+ 128$ spinor $= 248$, v154); **[C]** hence a gapped quasi-free bulk is invertible automatically, and **[E]** the invariant $|\det K| = \#$anyons is 1 for E_8 (invertible) against 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. **[O]** So LIT-A's 'invertible bulk' *follows* from 'gapped 16-Majorana quasi-free bulk' (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line:* together with v300, the entire open content of `SEAM.EQUIV.01` reduces to the single spectral statement "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

The last lever — the seam gap *is* the derived Recovery gap ([`verification/v302_seam_gap.py`], `SEAM.EQUIV.GAP.01`). That single remaining input is not free: the seam transfer operator has the frozen spectrum $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap* $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ — a number forced by the carrier ($(2/3)^6$ the μ_4-deck transfer eigenvalue), with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) making it robust. **[C]** By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian (correlation length $1/\Delta$). *Bottom line:* with v300/v301, the whole residual of `SEAM.EQUIV.01` is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$): *no undischarged TFPT-internal assumption remains.* It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

The continuum chain assembled, and the residual tied to the seam one-sidedness ([`verification/v356_continuum_mmst_applicability.py`], `SEAM.EQUIV.CONTINUUM.03`). Writing the chain out link-by-link: **S1** the gapped (v302) quasi-free 16-Majorana collar → **S2** invertible (Kitaev free fermions, v301) → **S3** the *nontrivial* invertible phase ($c_- = 8 \neq 0$) → **S4** a non-gappable anomalous chiral edge (anomaly inflow / bulk–edge) → **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\leq c \leq D$, i.e. $8 \leq 8 \leq 16$) → **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). Every link is a theorem or an established TFPT fact *except* S3, and S3 is precisely the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3 = 1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2| \oint K$, v58). So the lone realization input is tied to the constant that *defines* the theory, not a free dial; the residual is now "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase", [O] but theorem-bounded on both sides.

MMST in-class verified for the collar, so the scaling limit is $(E_8)_1$ ([verification/v366_mmst_seam_collar.py], SEAM.MMST.INCLASS.01). The continuum leg is now checked *hypothesis-by-hypothesis* for the explicit seam collar: it is a $D{=}\dim S^{+}{=}16$ Majorana quasi-free (CAR) state (v155/v160), gapped with $\Delta{=}6\log\frac{3}{2}{>}0$ *derived* (v302), and chiral with $c_{-}{=}D/2{=}8{=}g_{\rm car}{+}N_{\rm fam}$, so it sits squarely in the MMST free-lattice-fermion class with rank $\leq c \leq D$ reading $8 \leq 8 \leq 16$ (*in* range, saturating rank $=c{=}8$). The MMST massless-scaling-limit theorem then applies as a cited result, and the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K{=}1$, one primary) over the same-c rival $SO(16)_1$ ($\det K{=}4$, v351). [E] the class arithmetic, the derived gap, the chirality and the $(E_8)_1$ pinning; [C] the CAR-class typing and the cited MMST theorem; [O] the lone residual is exactly S3 above (the lattice realization). The continuum side of SEAM.EQUIV.01 is thereby reduced to a single, named, theorem-bounded input. The lattice→conformal-net step itself is now a *rigorous AQFT construction*: Adamo–Giorgetti–Tanimoto (arXiv:2506.01008, 2025) build the two-dimensional conformal net $\mathcal{A}_Q$ of an even lattice Q and *classify* every two-dimensional extension of the Heisenberg net (under a discreteness assumption) as such a lattice net — so "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage the continuum leg invokes, not a hope; and since (xxxii) ([verification/v469_seam_crossedproduct_route.py]) the extension step no longer even *needs* the preprint lineage — the local $\mathbb{Z}_2$ simple-current crossed product ($h_s{=}1{\in}\mathbb{Z}$, Longo–Rehren 1995; Böckenhauer 1996; Böckenhauer–Evans 1998; KLM 2001) certifies it on peer-reviewed ground, with AGT/AMT as the independent second witness.

S3 made concrete: an explicit gapped chiral-Majorana lattice model ([verification/v367_seam_s3_lattice.py], [verification/v368_seam_s3_inflow.py]). The S3 input is no longer abstract. The $p{+}ip$ Bloch model $h(k){=}\sin k_x\,\sigma_x{+}\sin k_y\,\sigma_y{+}(M{-}\cos k_x{-}\cos k_y)\sigma_z$ is, at $M{=}1$, a gapped free-fermion phase with numerically-computed Chern number $|C|{=}1$ ($C{=}0$ trivially at $M{=}3$), so $N_{\rm Maj}{=}\dim S^{+}{=}16$ copies give a chiral edge of $c_{-}{=}16/2{=}8{=}g_{\rm car}{+}N_{\rm fam}$, with the edge K-matrix $SO(16)_1$ (det 4) condensed to $(E_8)_1$ (det 1) by the order-4 μ_4 clock (v351/v369). A finite STRIP of the same model carries an edge-localised in-gap chiral branch (absent in the trivial phase) – so by anomaly inflow the chiral edge *exists* as a consequence of the invertible bulk (the S4 leg), discharging the "does the edge exist?" question on the lattice. [E] the gapped Chern model, the $c_{-}{=}8$ counting and the edge existence; [C] the μ_4 condensation to $(E_8)_1$; [O] only the genuine *continuum scaling limit* (MMST, v336) stays open — S3 is a runnable lattice model that pins the target at every computable level, so the keystone's MMST route (SEAM.EQUIV.MMST.01) is *closed modulo cited theorems.*

The S3 closure stack: $c = 8$, the $(E_8)_1$ character, the torus count and reflection positivity — all computed on the lattice ([verification/v376_seam_s3_centralcharge.py], [verification/v377_seam_s3_e8character.py], [verification/v378_seam_s3_modular.py], [verification/v379_seam_s3_rp.py]). On that explicit model the target of the scaling limit is now pinned at every level a computation can reach. *Central charge* (v376): the Calabrese–Cardy finite-size entanglement scaling gives $c = 1.0000$ per critical complex mode, so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\rm car}{+}N_{\rm fam}$, and it is chiral ($c_{-} = 8$). *Convergence* ([verification/v392_seam_s3_scalinglimit.py]): across five sizes $L = 66 \ldots 514$ the slope $c_1(L)$ is a monotone Cauchy-like sequence approaching 1 and $1/L$-Richardson-extrapolates to $c_1(\infty) \approx 1.000$ ($c \approx 8.00$ in the limit) — the existence-relevant *convergence* the two-size value did not test (evidence, not a proof of existence; the residual stays [O]). *The net* (v377): the exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary), and the μ_4/GSO promotion 248=240+8=120+128 (240=16·5·3) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, four primaries). *Genus-1* (v378): the KLM condensation μ(carrier)=16 → $\mu(E_8)$=16/4^2=1 and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 — the holomorphy / $\det K{=}1$ statement that the genus-0 arguments could not reach (v344). *Reflection positivity* (v379): the gapped collar's Euclidean two-point function has a positive spectral measure, so the OS reflection Gram matrix is a *positive-mixture Gram* $M = V^{\mathsf{T}}\mathrm{diag}(1/K)V$, $V_{ki} = e^{-\varepsilon_k\tau_i}$ — hence $x^{\mathsf{T}}Mx = \sum_k w_k(v_k{\cdot}x)^2$ is a sum of squares and positive semidefiniteness holds *exactly*, as an identity rather than to a tolerance (hardened 2026-07-28 after T133 certified that the *assembled double-precision* matrix has a negative direction at -7.75×10^{-18}, inside the old 10^{-10} tolerance; the min-eigenvalue check is retained as a diagnostic measurement, and a ghost mode still breaks the RP test). [E] all four (numerical c, exact character, genus-1 count, RP); [O] the one thing a computation cannot supply is the abstract *continuum existence* of the limit (the cited MMST theorem, v336). With RP in hand the ambient measure C7 is then discharged by the redundancy theorem (v369), so S3 is the master key: the target is pinned exactly, only the existence proof is external.

A ground-state witness for premise (i): the modular commutator measures c_-=8 from the bulk state alone ([verification/v489_seam_modular_commutator.py], SEAM.CCC.01). The chiral central charge entered so far through band topology (the Chern number, v367) and through counting (the KLM/det-Cartan arithmetic, v378). The Kim–Shehab–Kim modular commutator $J(A,B,C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3)\,c_-$ (PRL 128 (2022)) reads c_- off the *entanglement data of the exact BdG ground state* of the same collar — no band bundle, no representation theory enters — via the Peschel covariance ($W_X = -2\,\mathrm{artanh}\,\Gamma_X$) on a disk cut into three 120° wedges: per layer $c_- = 0.499998$ at L=32 (converged 0.500000 at L=48, deviation -1.8×10^{-7}), the trivial M=3 control gives 0.000000, the orientation flip M=−1 gives −0.499986 (sign flip), layer additivity is exact (10^{-14}), and the 16-layer carrier lands on $c_- = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$ (direct run 7.9967 at L=16, additivity route 8.0000). [E] the third, mutually independent lattice witness of premise (i) of SEAM.EQUIV.01 — and the first read from ground-state entanglement alone; [O] unchanged: the continuum scaling limit (MMST, v336) and the cited KLM condensation step stay exactly where v367/v378 left them — SEAM.EQUIV.01 stays [O].

The spin-structure parity census: μ_{pre}=4 witnessed on the lattice ([verification/v490_seam_parity_census.py], SEAM.MU.01). Step 1 of the KLM chain (v378) — "the carrier sits in the $SO(16)_1$ class, μ_{pre}=4, *before* the μ_4 condensation" — was arithmetic; it is now a lattice ground-state measurement. The T^2 fermion parity per spin structure (AA, AP, PA, PP) is determined *twice* (Read–Green TRIM product and real-space Majorana Pfaffian, spectra matching to 9×10^{-15}): per layer $(+,+,+,-)$ — the ν=1 Ising signature — and for the 16-layer carrier parity^{16} = + in *all four* sectors, so the gauged torus GSD is $4 = \mu(SO(16)_1) = |\det\mathrm{Cartan}(D_8)|$. The 16-fold-way bridge then runs on *measured* numbers: $\theta_v = e^{2\pi i\nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (v489) — the gauged vortex is a boson, condensable, while the rivals $\nu = 1, 2, 8$ are not holomorphically condensable. Honest fences: the fermionic Kitaev–Preskill TEE is identically zero for *both* M=1 and M=3 (γ_f = 0.000001/0.000000) — an invertible fermionic phase has no TEE, so γ_f does *not* discriminate $(E_8)_1$ vs $SO(16)_1$; and the KLM step $\mu : 4 \to 4/2^2 = 1$ stays the cited theorem ([C], the condensed model is interacting). The numerics pin exactly the premise package $\{c_-{=}8,\ \mu_{\mathrm{pre}}{=}4,\ \theta_v{=}1\}$ of the condensation; SEAM.EQUIV.01 stays [O].

And that external step is now Lean-formalised as "closed modulo a cited theorem" (FORM.SEAM.MMST.01, TfptCarrier/SeamScalingLimit.lean). The continuum leg is machine-formalised in the audit-contract style of `SeamEquivChain.lean`: the collar's MMST applicability hypotheses are *provable* arithmetic facts by Lean kernel `decide` (no axioms) — D=16 Majoranas, rank =c=8, the range $8 \le 8 \le 16$, the chirality $c_-{=}8 \ne 0$, and the holomorphy discriminator $\det K$=1 vs $SO(16)$'s 4; the MMST scaling-limit theorem and the Adamo–Moriwaki–Tanimoto OS reconstruction are declared as *named cited axioms*; and the conclusion $(E_8)_1$ follows by a well-typed composition whose `#print axioms` pins the residual to exactly those two published theorems (plus the carrier gap), with no hidden assumption and no `sorry`. Every TFPT-internal hypothesis is thus kernel-checked; only the two cited theorems are assumed — the honest closure ceiling for SEAM.EQUIV.01. *Two of the seven convergent routes below are now kernel-hardened the same way:* FORM.SEAM.BW.HSMI.01 (TfptCarrier/SeamStandardPair.lean) proves the four Borchers/Wiesbrock *standard-pair relations* exactly over $\mathbb{Z}$ on the centred ladder ($[K,P]{=}P$, $J^2{=}1$, $JKJ{=}-K$, $JPJ{=}P_+$, all by kernel `decide`, no domain axiom), with the cited Borchers step and the one continuum input as named axioms; and FORM.SEAM.APPLICABILITY.01 (TfptCarrier/SeamApplicabilityLedger.lean) kernel-checks the audit *count* itself — of the 12 cited-theorem hypotheses exactly 11 are internal and 1 external (and $8 \le 8 \le 16$, $\det K{=}1 \ne 4$), all axiom-free — so the very bookkeeping the keystone's honesty rests on is machine-verified, leaving only the one named continuum fact open.

TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure ([verification/v308_seam_equiv_chain.py], [verification/v309_modular_bedrock.py], [verification/v311_gap_decoupled_measure.py]). Three machine-checked steps push the keystone arc further. *(i) Composition certificate* ([verification/v308_seam_equiv_chain.py]): the closing implication is one well-typed logical chain gap>0 → invertible → holomorphic c=8 → $(E_8)_1$ whose arithmetic discriminators are exact — $|\det\mathrm{Cartan}(E_8)|$=1 against $|\det\mathrm{Cartan}(D_8)|$=4, the carrier tower 4·4=16 → 4 → 1, $c{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}8$ — so the only undischarged premise is the cited OS step; [E] the discriminators, [O] the keystone. *(ii) Modular bedrock* ([verification/v309_modular_bedrock.py]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ is built explicitly and $[\rho, K]{=}0 \Leftrightarrow \omega{\circ}\rho{=}\omega$ (a $\mathbb{Z}_2$ profile breaks it), so the bedrock reduces *non-circularly* to the Bisognano–Wichmann intrinsicality of the raw collar state; [E] the model,

[O] the intrinsicality. *(iii) Gap-decoupled measure* ([verification/v311_gap_decoupled_measure.py]): the admissible sector clusters exponentially at rate $(2/3)^6$ (correlation length $1/\Delta$, finite susceptibility 729/665) and the v76 margin $\Delta - 31/(4\pi^2) > 0$ decouples it from the ambient, so the physical sector is a convergent object while QG.AMB.01 is a [C] redundancy (v369). Each pins the residual to a single named step; with the lattice/S_3 stack the keystone's MMST route (SEAM.EQUIV.MMST.01) is now *closed modulo cited theorems. (iv) The intrinsic-BW residual as a cited reduction* ([verification/v424_seam_lto_rp_reduction.py]): the open bedrock lemma of (ii) — the raw collar's intrinsic Bisognano–Wichmann property — is reduced to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (arXiv:2605.10693), whose reflection-positivity axiom (LTO-RP) is exactly $u_\Theta{=}J$ (the reflection *is* the modular conjugation), i.e. the BW condition $\theta K\theta{=}-K$ — *not* $\omega\circ\rho{=}\omega$. [E] the μ_4 clock is a modular *symmetry* ($[\rho, K]{=}0$), not the flow ($\sigma^\psi{=}\rho$ is type-wrong, a one-parameter group versus a finite order-4 element); a positive character-block-diagonal K has $[\rho, K]{=}0$ yet, being positive definite, never $\theta K\theta{=}-K$, so clock-invariance does *not* imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* commuting-projector models (Toric Code via a trace, Levin–Wen anyonic), whereas the seam is the opposite corner — invertible ($|\det \mathrm{Cartan}(E_8)|{=}1$, no anyons) and a $\beta{=}1$ KMS state (not a trace) — so two sub-steps stay open: realising the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO, and extending (LTO-RP) to the invertible KMS case. An MMST-style reduction (cf. v308/v336), *not* a closure. *Sub-step (ii) is now constructively exhibited* ([verification/v426_seam_lto_rp_kms.py]): a gapped reflection-symmetric μ_4-even collar at $\beta{=}1$ KMS satisfies $\Theta K\Theta^{-1}{=}{-}K$ ($u_\Theta{=}J$) and $[\rho, K]{=}0$, so (LTO-RP) holds there by direct Tomita–Takesaki — the corner NPW26 leave open — with $|\det \mathrm{Cartan}(E_8)|{=}1$ (invertible, no anyons) and the state a genuine KMS, not a trace; only the continuum sub-step (i) remains [O]. On the covariance side, the recent theorem of Adamo–Giorgetti–Tanimoto (arXiv:2508.07109, 2025) — every DHR representation of a conformal net extending a loop-group/Virasoro net is *automatically* positive-energy and diffeomorphism-covariant — defuses the old "Bisognano–Wichmann presupposes the very covariance it should produce" circularity (v215): once the seam net is in hand, geometric covariance is a *consequence*, not an input.

Twenty-six convergent attacks on the one continuum residual — intrinsic BW, a bulk continuum motor and its chiral-edge counterpart, the β-homotopy, the applicability audit, uniform rigidity with its forcing converse and the derivation of clock-invariance, three independent reads of the edge central charge, the four-point and uniform-in-N pinning of the external fact, the Ising operator tower, the $(E_8)_1$ torus modular data, the μ_4-from-marks derivation, the kill tests, the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, the chirality-from-P_1 forcing, an $(E_8)_1$ character kill test, the exact MMST citation audit, the complementary lattice-VOA route, the explicit flat-band commuting-projector LTO realisation, the strict-locality (Kapustin–Fidkowski) obstruction, the character-level 128-spinor extension, the $c{=}8$ holomorphic-uniqueness pin and the one-particle realisation rigidity ([verification/v438_seam_hsmi_borchers.py], [verification/v439_seam_lieb_robinson.py], [verification/v440_seam_lto_rp_beta.py], [verification/v441_seam_applicability_ledger.py], [verification/v442_seam_rigidity_uniform.py], [verification/v443_seam_e8_discriminator.py], [verification/v444_seam_edge_scaling.py], [verification/v445_seam_rigidity_forcing.py], [verification/v446_seam_clock_invariance.py], [verification/v447_seam_edge_chern.py], [verification/v448_seam_edge_fourpoint.py], [verification/v449_seam_mmst_uniform.py], [verification/v450_seam_edge_entanglement.py], [verification/v451_seam_edge_cardy_tower.py], [verification/v452_seam_e8_modular.py], [verification/v453_seam_mu4_from_marks.py], [verification/v454_seam_edge_virasoro.py], [verification/v455_seam_bw_uniform.py], [verification/v456_seam_chirality_from_c3.py], [verification/v457_seam_character_killtest.py], [verification/v458_seam_mmst_citation_audit.py], [verification/v459_seam_lattice_voa_route.py], [verification/v460_seam_s3_lto.py], [verification/v461_seam_strict_locality.py], [verification/v462_seam_spinor_continuum.py], [verification/v463_seam_c8_holomorphic_uniqueness.py], [verification/v464_seam_oneparticle_rigidity.py], [verification/v469_seam_crossedproduct_route.py]**).** None closes SEAM.EQUIV.01; together they ground the one continuum input on both its halves, reduce it to a single analytic fact and make the claim falsifiable. *(v) The intrinsic Bisognano–Wichmann route* ([verification/v438_seam_hsmi_borchers.py]): the seam supplies a Longo *standard pair* / +half-sided modular inclusion — a positive lightlike translation $P \geq 0$, the boost K with $[K, P]{=}P$ (the Borchers scaling $\Delta^{it}P\Delta^{-it}{=}e^{-t}P$), and a reflection J with $JKJ{=}{-}K$, $JPJ{=}P_-$ — so by Borchers' theorem the modular flow is geometric *intrinsically*, *without* presupposing the rotation-covariance of the vacuum; [E] the ladder algebra and the $sl(2, \mathbb{R})$ positivity (min eig > 0, with the indefinite boost as control), [C] the synthesis, [O] assembling the finite algebra and the positive-energy rep into one continuum standard pair (the MMST/NPW26 wall). *(vi) The continuum-existence motor* ([`verification/v439_seam_lieb_robinson.py`]): the explicit $p{+}ip$ collar (v367) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral massless *edge* scaling limit; [E] gap/clustering/light-cone, [C] the lift, [O] the edge CFT. *(vii) The β-homotopy* ([`verification/v440_seam_lto_rp_beta.py`]): the (LTO-RP)/BW data ($\Theta K\Theta{=}-K$, $[\rho,K]{=}0$) are β-independent, and along the whole path $C_\beta{=}1/(1{+}e^{\beta K})$ from the NPW26-proven *trace* state ($\beta{=}0$) to the seam KMS state ($\beta{=}1$) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii). *(viii) The applicability audit* ([`verification/v441_seam_applicability_ledger.py`]): a hypothesis-by-hypothesis ledger of MMST/NPW26/Adamo finds 11 of 12 hypotheses are TFPT-internal computed facts ([E]) and exactly one is external — the continuum existence of the chiral scaling limit — covariance being an Adamo consequence, not an input; the keystone's external dependence is pinned to that one analytic fact. *(ix) Uniform rigidity* ([`verification/v442_seam_rigidity_uniform.py`]): the band-limited μ_4-character rigidity of v398 is uniform in N — exact 4-sector commutation to $N{=}256$, a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho,V]\|{=}2$ — so it is not a small-N artifact (the full-L^2 forcing stays [O]). *(x) The kill tests* ([`verification/v443_seam_e8_discriminator.py`]): mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest $c{=}8$ rival $SO(16)_1$ — torus GSD 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots — with $c{=}8$ alone unable to decide (v277) and only $\det K{=}1$ (the genus-1 statement, v90) selecting E_8; a seam with GSD>1 would falsify $(E_8)_1$. [X] the tests, [O] the realisation. *(xi) The chiral-edge scaling motor* ([`verification/v444_seam_edge_scaling.py`]): the counterpart of (vi) on the *edge*, attacking exactly the chiral massless scaling limit that the bulk motor left open. On the same $p{+}ip$ collar (v367/v368) the lattice edge already carries the conformal data of a chiral $c{=}\frac{1}{2}$ Majorana CFT — a linear, chiral in-gap dispersion (velocity $v{\approx}0.98$, $R^2{=}0.9999$, with *opposite* per-edge velocities), an *algebraic* edge two-point $\sim r^{-\eta}$ with $\eta{\approx}1$ (versus the *exponential* bulk row and the exponential trivial-phase control), and a Cauchy-convergent scaling-collapse under refinement ($\eta{=}1.001{\pm}0.007$, successive spread $0.028{\to}0.013$) — so the 16-copy edge is a chiral $c_-{=}8$ $(E_8)_1$ CFT; [E] the kinematics, correlator and collapse, [C] the central-charge reading, [O] the rigorous uniform-convergence theorem (cited MMST). With (vi) for the bulk and (xi) for the edge, *both halves* of the one continuum residual are numerically grounded and pinned to a single cited theorem. *(xii) The rigidity forcing converse* ([`verification/v445_seam_rigidity_forcing.py`]): the converse of (ix), upgrading rigidity from "block-diagonal *permits* commuting" to "commuting with the order-4 clock *forces* block-diagonal". For $\rho{=}\mathrm{diag}(i^n)$, $[\rho,K]{=}0$ *iff* $K_{ij}{=}0$ whenever $i{\not\equiv}j \bmod 4$ (verified entrywise, swept $N{=}4..128$), so the commutant is *exactly* the 4-sector block algebra: $\dim\{K : [\rho,K]{=}0\}{=}\sum_s n_s^2$, a proper subspace of the full N^2 algebra ($N{=}16$: $64{<}256$); every off-block unit has $\|[\rho,E_{ij}]\|{\ge}\sqrt{2}$ (zero slack), and the order-2 clock leaves a strictly larger commutant (64 vs 128) — the *four* Gauss–Bonnet marks (order $4{=}|\mu_4|{=}h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det{=}1$, v281/v154/v351). [E] the entrywise iff, exact commutant dimension, off-block sharpness and order discriminator; [C] the synthesis (RP-definability + clock-invariance *force* the block-diagonal Λ_Σ, v442's "permit" upgraded to "force"); [O] the single residual is now precisely that the physical transfer lies in the commutant (clock-invariance = `QGEO.SYM.01`, v323/v335). *(xiii) Clock-invariance derived* ([`verification/v446_seam_clock_invariance.py`]): closing (xii)'s residual one notch further — the operator condition $[\rho,\Lambda_\Sigma]{=}0$ is itself a *consequence* of a structural symmetry, not an extra assumption. On explicit finite OS data, the canonical transfer $\Lambda{=}C(1)C(0)^{-1}{=}e^{-h}$ (the transfer fixed by RP+reflection, v54) of a μ_4-invariant covariance satisfies $[\rho,\Lambda]{=}0$ (swept $N{=}4..32$; iff-sharp μ_4-breaking neg control), Λ is a positive contraction, and the free modular Hamiltonian $K{=}\log((1{-}C)C^{-1})$ (v258) of the invariant state has $[\rho,K]{=}0$ (Tomita–Takesaki). [E] the inheritance, contraction and modular commutation; [C] so with (xii), RP-definability *plus* the four marks force the block-diagonal Λ_Σ; [O] the residual reduces from an operator commutation to the structural "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`, v216/v323). *(xiv) The independent topological edge reading* ([`verification/v447_seam_edge_chern.py`]): a second, fit-free route to the edge central charge of (xi). On the same $p{+}ip$ collar (v367/v368) the bulk Chern number is the integer $C{=}{+}1$ (Fukui–Hatsugai–Suzuki link variables; $C{=}0$ in the trivial phase, $C{=}{-}1$ at the opposite mass, matching (xi)'s opposite per-edge velocities), the open strip carries exactly $|C|{=}1$ chiral edge branch per boundary, and the Li–Haldane entanglement spectrum has in-gap modes at $\frac{1}{2}$ only in the topological phase; [E] the Chern integer, channel count and entanglement signature, [C] the bulk–edge reading $c_-{=}N_{\rm Maj}\cdot\frac{1}{2}\cdot|C|{=}16\cdot\frac{1}{2}{=}8{=}g_{\rm car}{+}N_{\rm fam}$ — the *same* $c_-{=}8$ as (xi) from a logically independent (topological) observable. *(xv) The four-point pinning of the external fact* ([`verification/v448_seam_edge_fourpoint.py`]): extending (xi)'s two-point scaling limit to the next order to support exactly the one external fact (viii) isolates. The edge state is quasi-free (the ground-state correlation matrix is an exact projector $G^2{=}G\sim 10^{-13}$, the MMST CAR hypothesis), and the edge four-point cross-ratio $Q{=}[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement to the free-fermion conformal value $\sin(5\pi/12)/\sin(\pi/12){=}2{+}\sqrt{3}\approx 3.732$ (the bulk control off-scale, exponential); [E] the quasi-free property, four-point convergence and conformal form, [C] so the one external MMST fact is empirically supported at *both* two- and four-point order. *(xvi) The uniform-in-N pinning* ([`verification/v449_seam_mmst_uniform.py`]): strengthening (xv) from convergence at a single cross-ratio to the *uniformity* the cited theorem asserts — two *independent* edge four-point cross-ratios converge to *distinct* conformal values ($Q(1,2,4,7)/12\to 2{+}\sqrt{3}$ and $Q(1,3,5,9)/12\to 2$) at one N-independent $1/N$ rate ($|Q(N){-}Q_\infty|\cdot N\le 4$ for both, $N_x\in\{36,48,60\}$), the bulk control off-scale; [E] the two limits and the uniform rate, [C] so the MMST uniform-convergence *hypothesis* is empirically met. *(xvii) The entanglement reading of c_-* ([`verification/v450_seam_edge_entanglement.py`]): a *third*, correlator-independent route to the edge central charge — the Calabrese–Cardy block-entropy law gives $c_{\rm Dirac}{=}1.000$ for the critical free-fermion chain (vs a saturating, ℓ-independent gapped control), so c per Majorana $=\frac{1}{2}$ and $c_-{=}16\cdot\frac{1}{2}{=}8{=}g_{\rm car}{+}N_{\rm fam}$; [E] the law and the control, [C] the assembly. *(xviii) The Ising operator tower* ([`verification/v451_seam_edge_cardy_tower.py`]): naming the edge CFT by its *operator content* — the Cardy finite-size spectrum reproduces the exact Ising chiral primaries $h_\sigma{=}\frac{1}{16}$ (the periodic–antiperiodic ground split) and $h_\epsilon{=}\frac{1}{2}$ (the lowest NS pair gap), both L-independent, so $\{c,h_\sigma,h_\epsilon\}{=}\{\frac{1}{2},\frac{1}{16},\frac{1}{2}\}$ is the free-Majorana minimal model $M(3,4)$; [E] the dimensions and conformal scaling, [C] the identification. *(xix) The $(E_8)_1$ torus modular data* ([`verification/v452_seam_e8_modular.py`]): pinning the $(E_8)_1$ identity on the *torus*, complementing the edge reads (xi)/(xiv)/(xvii)/(xviii) — the character $\chi{=}E_4/\eta^8$ is S-invariant ($\chi(-1/\tau){=}\chi(\tau)$, one primary), T-multiplies by $e^{-2\pi i/3}{=}e^{-2\pi ic/24}$ ($c{=}8$) and is holomorphic ($c\equiv 0 \bmod 8$, T-phase order 3); [E] the full modular signature. *(xx) The μ_4-from-marks derivation* ([`verification/v453_seam_mu4_from_marks.py`]): folding the last structural premise into the existing one — the four Gauss–Bonnet marks *are* μ_4 (roots of $z^4{-}1$), the form basis ω_k is μ_4-graded ($\to i^k\omega_k$, weights $(1,2,3){=}A_3$ exponents) and the order-4 clock $z\to iz$ fixes their cross-ratio $\lambda{=}2$ (the D_4 stabiliser, v168), so the residual `QGEO.SYM.01` of (xiii) follows from marks$=\mu_4$ plus the existing `QGEO.REALIZE.01` — [E] the three mark facts, [C] no new premise: the chain (xx)→(xiii)→(xii) closes modulo realisation. The integer backbone of (xiv)–(xix) ($c_-{=}8$, the Chern integers, the order-3 T-phase) is kernel-hardened in Lean axiom-free as `FORM.SEAM.EDGE.CHERN.01`. *(xxi) The lattice Sugawara current algebra* ([`verification/v454_seam_edge_virasoro.py`]): testing the cited MMST "Koo–Saleur generators → continuum Virasoro with the correct c" fact on the edge *algebra*, not just its central charge. The critical free-fermion ring has the Affleck–Cardy Casimir energy $E_0(L){=}e_\infty L{-}\pi vc/(6L)$ fitting $c_{\rm Dirac}{=}1.000$ ($\frac{1}{2}$ per Majorana, $L{=}64..1024$), the lattice $U(1)$ current closes its affine *level* exactly ($\langle 0|J_nJ_{-n}|0\rangle{=}n$, $n{=}1..8$; vanishing for a gapped control), and the level-1 Sugawara central charge $c{=}\dim G/(1{+}h^\vee){=}8$ for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31); [E] Casimir, Kac–Moody level and Sugawara, [C] $16\cdot\frac{1}{2}{=}8$ is the level-1 current-algebra c_-. *(xxii) The uniform-in-N Tomita–Takesaki tower* ([`verification/v455_seam_bw_uniform.py`]): the inductive-limit companion of v426 for sub-step (i) of (iv) — a family of reflection-odd, μ_4-even, gapped collars ($N{=}8..128$) on which the intrinsic Bisognano–Wichmann condition $u_\Theta{=}J$ ($\|\Theta K\Theta{+}K\|{=}0$) holds *exactly at every* N, with an N-independent gap ($6\log\frac{3}{2}$) and an N-independent neg-control margin ($\|\Theta K\Theta{+}K\|{=}2\|K\|$ for a side-even reflection); [E] the uniform BW, gap, clock symmetry and teeth, [C][O] so the inductive limit inherits $u_\Theta{=}J$ (the abstract continuum theorem stays the cited backbone). *(xxiii) The chirality-from-P_1 forcing* ([`verification/v456_seam_chirality_from_c3.py`]): the S3 input of the continuum chain (v356) is removed as a free choice — c_3's "8" *is* the one-sided count $8{=}|\mathbb{Z}_2|\cdot(\int K/\pi){=}2\cdot 4$ ($8\pi{=}|\mathbb{Z}_2|\int K{=}2\cdot 2\pi\chi(S^2)$, v73), and a reflection sends the edge Chern integer $C{\to}{-}C$ (verified $C{=}{+}1\to -1$ on the explicit $p{+}ip$ model), so any *two-sided* (reflection-symmetric) gapped boundary forces $C{=}{-}C{=}0$ while the *one-sided* seam — the same $|\mathbb{Z}_2|$ quotient that gives c_3 its 8π — has no such reflection and forces $C{\neq}0$; [E] the one-sided count and the $C{\to}{-}C$ flip, [C] chirality (S3) is a *consequence* of P_1, magnitude $c_-{=}8$. *(xxiv) The $(E_8)_1$ character kill test* ([`verification/v457_seam_character_killtest.py`]): a *falsifiable* sharpening of the (x)/(x") discriminators — the $(E_8)_1$ tower $\chi{=}E_4/\eta^8{=}q^{-1/3}(1{+}248q{+}4124q^2{+}34752q^3{+}\cdots)$, the weight-1 count $248{=}120{+}128$ (vs the free 120) and the sector count $|\det{\rm Cartan}\,E_8|{=}1$ (vs $|\det{\rm Cartan}\,D_8|{=}4$); a measured 120, four sectors, or a wrong tower coefficient would *falsify* $(E_8)_1$, and the data give

248/1/$\{1, 248, 4124\}$ (passed). **[X]** the kill criterion, **[E]** the exact discriminators. *(xxv) The exact MMST citation audit* ([`verification/v458_seam_mmst_citation_audit.py`]): a hypothesis-by-hypothesis reading of the cited MMST theorem (`arXiv:2107.13834`) against the collar — H1 CAR class, H2 massless limit, H3 the per-copy $c{=}\frac{1}{2}$ Thm 4.11, H4 decoupled additivity to $c{=}8$, and H5 the WZW range $8{\le}8{\le}16$ with its 120 bilinear currents all *match*, isolating the single residual H6 to the 128-spinor extension $SO(16)_1{\to}(E_8)_1$ (the $\det K$ 4→1 holomorphy step), *outside* MMST's bilinear method; **[E]** the per-copy theorem, **[C]** the additivity and range, **[O]** the one open spinor extension — since (xxxii) certified at net level by the peer-reviewed crossed product. *(xxvi) The complementary lattice-VOA route* ([`verification/v459_seam_lattice_voa_route.py`]): the second, independent construction that supplies exactly H6 — for the even unimodular E_8 the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, `arXiv:2506.01008`; OS `arXiv:2407.18222`) has weight-1 space 248=8+240 with the 240 roots splitting 112 integer + 128 *half-integer* (the spinor, 248=120+128), so the lattice route builds the 128-spinor currents MMST (xxv) leaves open; **[E]** the even unimodularity and root split, **[C]** AGT constructs the holomorphic extension — since (xxxii) an *independent second witness*, the primary net-level certification being the peer-reviewed crossed product — **[O]** the routes leave one shared realisation input (the collar's scaling limit = A_{E_8}), tied to P_1 by (xxiii). The G-block arithmetic (the one-sided $8{=}|\mathbb{Z}_2|\int K/\pi$, the reflection $C{\to}{-}C$, $c_-{=}8$, the range $8{\le}8{\le}16$, 248=8+240=120+128, 240=112+128, $\det K$ 1 vs 4) is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`), reducing the residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus the two cited existence theorems. *(xxvii) The explicit flat-band commuting-projector (LTO) realisation* ([`verification/v460_seam_s3_lto.py`]): NPW26 prove the intrinsic Bisognano–Wichmann lemma of (iv) for *commuting-projector* (local-topological-order) models, so sub-step (i) asks for the seam realised as such a $\mathbb{Z}_2$-reflection LTO. On the same `v367` $p{+}ip$ collar the flattened Bloch $Q{=}h/|h|$ is a frustration-free flat-band parent — $Q^2{=}I$ and the occupied $P{=}(I{-}Q)/2$ is an exact projector EQUAL to the ground-state projector — whose real-space kernel is gap-protected quasi-local ($\xi{\approx}1.14$ at $M{=}1$, $|P(20,0)|{<}10^{-6}$; a clean power law $|R|^{-2.1}$ at the gapless point, so locality needs the gap), and whose flattened modular Hamiltonian $K_{\text{flat}}{=}\operatorname{sign}(K)$ inherits the $\mathbb{Z}_2$ reflection $\Theta K_{\text{flat}}\Theta{=}{-}K_{\text{flat}}$ and the μ_4 clock $[\rho, K_{\text{flat}}]{=}0$ EXACTLY; **[E]** the flat-band/projector and reflection algebra, **[C]** the quasi-locality, **[O]** the chiral $c_-{=}8{\neq}0$ obstructs *strictly* finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`), so the realisation is exactly the quasi-local LTO net NPW26 use — advancing sub-step (i) of (iv) from abstract existence to an explicit gap-protected net, strict locality the cited residual. *(xxviii) The strict-locality obstruction made explicit* ([`verification/v461_seam_strict_locality.py`]): upgrading (xxvii)'s "strict locality is the cited residual" from a citation to an EXHIBITED topological obstruction. On the same `v367` collar the FHS Chern is $C{=}{+}1$ (chiral) / 0 (trivial) / -1 (opposite mass), and the hybrid Wannier-centre (Wilson-loop) phase $\theta(k_y)$ winds by $|C|{=}1$ (chiral) and 0 (trivial) over the BZ — exponentially-localised Wannier functions exist *iff* that winding is 0 (Brouder–Panati et al.; Thouless), so the chiral collar has none; both phases are gapped, yet only the trivial one (winding 0) admits a strict finite-range commuting projector, so it is the *chirality* $c_-{=}8{\neq}0$, not the gap, that obstructs. **[E]** the Chern integer, the winding= $|C|$ identity and the gapped neg-control, **[C][O]** Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0 hence $C{=}0$, so for $C{=}1{\neq}0$ none exists — sub-step (i)'s realisation is *provably* the quasi-local NPW26 LTO net, strict locality topologically forbidden rather than a missing premise. *(xxix) The character-level 128-spinor extension* ([`verification/v462_seam_spinor_continuum.py`]): the constructive companion of (xxv)/(xxvi), exhibiting the 128-spinor extension three ways — the exact Jacobi/E_8 identity $\theta_2^8{+}\theta_3^8{+}\theta_4^8{=}2E_4$ makes the holomorphic character the sum of the $SO(16)_1$ vacuum and spinor sectors, $\chi_{(E_8)_1}{=}E_4/\eta^8{=}\chi_o^{SO16}{+}\chi_s^{SO16}$ (weight-1 counts 120+128=248); the finite 16-Majorana Fock space already carries them ($C(16,2){=}120$ currents and $2^{16/2}{=}256{=}128{+}128$ Ramond states); and a critical free-fermion ring converges to the chiral CFT (Casimir $c_-{\to}8$, lowest scaled gap $x_1{\to}1$ with $|x_1{-}1|{\sim}1/N^2$). **[E]** the θ identity, the 120+128=248 split and the finite-L convergence, **[C][O]** the rigorous holomorphic extension is certified at net level by the crossed product (xxxii), with AGT/AMT (xxvi) as the independent second witness. *(xxx) The $c{=}8$ holomorphic-uniqueness pin* ([`verification/v463_seam_c8_holomorphic_uniqueness.py`]): the positive lever that makes the *identification* half classification-forced rather than assumed. At level 1 the central charge $c{=}\dim\mathfrak{g}/(1{+}h^\vee){=}8$ has *three* simple solutions — A_8 ($\dim 80$), $D_8{=}\mathfrak{so}(16)$ ($\dim 120$) and E_8 ($\dim 248$) — so $c{=}8$ is necessary but *not* sufficient, and the $\det K{=}4$ $SO(16)$ rival of (xxiii) is a genuine same-c competitor, not a strawman. But a holomorphic (single-sector) $c{=}8$ theory has an $SL(2,\mathbb{Z})$-covariant character, and the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}{=}E_4/\eta^8$, whose q^1 coefficient (the one-dimensional candidate space, vacuum $a_0{=}1$) is 248 —

so $\dim V_1{=}248$ is *forced*, selecting E_8 uniquely and excluding A_8/D_8. [E] the three candidates, the forced 248, the tower $\{1, 248, 4124, 34752, 213126\}{=}E_4/\eta^8{=}j^{1/3}$; [C] Dong–Mason/ Schellekens give the holomorphic $c{=}8$ VOA unique = V_{E_8} (the even unimodular rank-8 lattice unique = E_8, Mordell–Witt), so "$c{=}8$ holomorphic limit = $(E_8)_1$" is classification-forced. *(xxxi) The one-particle realisation rigidity* ([`verification/v464_seam_oneparticle_rigidity.py`]): attacking the *realisation* input R_1 itself. Since the seam is quasi-free (v113/v160/v161), by Araki's self-dual CAR the whole net is the second quantisation of its one-particle symbol P, so R_1 collapses to a finite-per-mode statement about P: the gapped ground state is the *unique* quasi-free state with $P{=}$ the spectral projection onto $K{<}0$ ($P^2{=}P$ to $< 10^{-12}$, K fixed by gap + chirality + reflection v426); its real-space kernel is Cauchy on every fixed window ($\max|P_{2N}{-}P_N|{\sim}1/N{\to}0$, so P_∞ exists), and the limit carries $c{=}1$ per Dirac (entanglement $S(L){=}(c/3)\ln L$, 16 copies $\Rightarrow$ $c_-{=}8$, tying v450). [E] idempotency, Cauchy convergence and the c-fit; [C] Araki/Shale–Stinespring give the state$\leftrightarrow$symbol bijection and implementable Bogoliubov map, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, its one-particle limit exhibited, modulo the named CAR functor". *(xxxii) The net-level crossed-product route + the invariant-level realisation* ([`verification/v469_seam_crossedproduct_route.py`]): both halves of the residual re-founded on harder-to-reject ground. The 128-spinor extension needs no preprint: the $SO(16)_1$ spinor has $h_s{=}N/16{=}16/16{=}1{\in}\mathbb{Z}$, so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1{\rtimes}\{1, s\}$ is a *local* index-2 extension by 1995–2001 *peer-reviewed* subfactor theory (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B){=}4/2^2{=}1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ by the (xxx) pin — so the AGT/AMT route (xxvi) is *demoted to an independent second witness*. The index-4 μ_4 glue runs on the same integer, $h(J^k){=}\{1, 1, 1\}$ for $k{=}1, 2, 3$ (the v125 isotropy $q(k(1, 1)){=}k^2$ at net level, $\mu{=}16/4^2{=}1$). And the realisation input R_1 is reduced from model fiat to *invariants* (R1′): quasi-free [C] (v155/v160) + gap [E] (v302) + class D + $c_-{=}8$ [E] ((xxiii), from P_1) — computed on the collar as FHS Chern $|C|{=}1$ (control 0), $\nu{=}16{\times}|C|{=}16$, $c_-{=}\nu/2{=}8$; and $\nu{\equiv}0 \pmod{16}$ is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (the sixteen-fold way; c_- a *phase* invariant, Kapustin–Spodyneiko; Kim et al.), so any realisation with the R1′ invariants is phase-equivalent to the v367 stack. [E] the locality integers, the KLM μ-arithmetic, the computed invariants and the composed chain; [C] the re-founded citation package; [O] the cited theorems stay cited. The full G-block arithmetic — now including the (xxviii) winding obstruction, the (xxix) finite-Fock spinor counts, the (xxx) holomorphy selector and the (xxxii) locality/μ/sixteen-fold joints — is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`), the two formerly separate cited theorems are *merged* into one combined external axiom `seam_realisation_theorem`, so the residual reduces to ONE realisation axiom plus ONE cited theorem (`#print axioms` drops from six to four); and the (xxxii) route is Lean-carried as the *parallel* derivation `seamResidualClosed′` (the invariant-level axiom `collar_invariants` plus the single combined package `crossedproduct_route_theorem`, its joints — `lr_locality_integer`, `glue_locality_integers`, `klm_holomorphic`, `sixteenfold_pin` — kernel facts with no axioms). Across (v)–(xxxii), `SEAM.EQUIV.01` stays [O]: with (xxx) pinning the identification, (xxxi) reducing the realisation to its one-particle data and (xxxii) re-founding the extension leg on peer-reviewed ground, the residual is now *entirely certification* — a named, hypothesis-audited package of published theorems (MMST, Longo–Rehren/Böckenhauer/Böckenhauer–Evans/KLM, Dong–Mason, Araki, Shale–Stinespring; AGT/AMT as second witness) with no open internal mechanism — yet still rests on cited continuum-existence theorems we do not re-prove.

The TOE/QFT closure contracts as three citable certificates ([`verification/v398_seam_state_rigidity.py`], [`verification/v399_gravity_complete.py`], [`verification/v401_metrology_closure.py`]). Three companion contracts package the closure status as named targets without fabricating any closure. *(i) Seam State Rigidity* (`SEAM.RIGIDITY.01`): the scattered state-invariance reductions (v177/v199/v308/v309/v335) as ONE theorem target — [E] the RP-definability hinge (OS canonical transfer from RP + reflection alone, v54, quasi-free v155), [E] the band-limited μ_4-character-block-diagonal core swept $N{=}4..64$ (beyond the 3-dim H^1, made uniform in N to $N{=}256$ by v442), [E] holomorphy $\Rightarrow (E_8)_1$; and [E] the forcing converse (v445) that commuting with the order-4 clock is *exactly* μ_4-block-diagonality, so the residual is no longer "does block-diagonal permit" but only $[\rho, \Lambda_\Sigma]{=}0$ on the full L^2 — the physical transfer lying in that commutant (intrinsic Bisognano–Wichmann). [E] Moreover (v446) that operator commutation is itself *derived*: the OS canonical transfer of μ_4-symmetric RP data inherits the clock (with the modular Hamiltonian commuting, v258), so the residual reduces further to the *structural* statement that the seam RP data is μ_4-symmetric (the four marks, `QGEO.SYM.01`), leaving **[O]** only that structural symmetry on the raw continuum collar — for which the four marks now have an *established* continuum frame: the four-interval multilocal modular geometry (v480, keybox below), where the invariance is manifest at the state level in the exactly solvable class. *(ii) Gravity-complete = Boundary-complete + Ambient-redundancy* (`GRAVITY.COMPLETE.01`): the local equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (v358/v359) plus the redundancy discriminators (v369), with $\mathcal{O}_{\mathrm{phys}} \subset \mathcal{A}_\Sigma$ checked over five gravitational readout classes. *(iii) Metrology closure* (`METROLOGY.CLOSURE.01`): the final input set $\{a, \pi, v_{\mathrm{geo}}\}$ — 0 dimensionless dials + 1 unit, the dimensional bookkeeping that every dimensionful observable is a number $\times v_{\mathrm{geo}}^d$ (No-Unit, v153), and v_{geo} theorem-forbidden ($\mathrm{TOE}_{\mathrm{strict}}$=$\mathrm{TOE}_{\mathrm{dimensionless}}$+$[v_{\mathrm{geo}}]$, v384).

One surface for seam, carrier and lattice: the Kummer/K3 ([`verification/v260_k3_kummer_unification.py`]). The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z\mapsto -z)$, its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface* $\mathrm{Km}(T^2\times T^2)$, a K3, whose three facets reproduce three TFPT objects at once. **[E]** the 16 nodes (the 2-torsion $|A[2]|=2^4=16$ of the abelian surface) resolve to 16 exceptional $\mathbb{P}^1$'s $= \dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5) = 16$ (v197), the carrier-16 as the *square* of the seam-4; **[E]** the cup-product form on $H^2(\mathrm{K3},\mathbb{Z})$ is the even unimodular lattice $U^3 \oplus E_8(-1)^2$ of signature $(3, 19)$, so the same E_8 the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the $E_8(-1)^2$ summand — both exact, not identified.) Thus the v1 "glue $D_5\oplus A_3+\mu_4\Rightarrow E_8$" is the lattice shadow of a single smooth object that already carries the seam and the carrier; E_8 appears both *locally* as the du Val $\mathbb{C}^2/2I$ singularity (the Poincaré-sphere link, v232/v236; by Kronheimer that local model *is* the hyper-Kähler quotient of the marks quiver, v479) and *globally* as this H^2 summand. A unification (`ARCH.K3.01`), *not* a closure: the lattice arithmetic is **[E]**, the carrier↔node identification [C].

Synthesis — the Modular Spectral Closure ([`verification/v261_modular_spectral_closure.py`]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\boxed{\mathsf{TFPT}_{\mathrm{QFT}} = \big(\mathcal{A}_\Sigma,\ \omega_\Sigma,\ \Delta_\Sigma,\ \rho,\ A_F,\ H_F,\ D_F,\ J,\ \gamma,\ S_{\mathrm{rel}}\big)}$$

(seam algebra, KMS state, modular generator, μ_4 clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-4 KMS seam state ω_Σ minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the* 96*-dim KO-*6 *finite triple D_F (v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259)* — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 ($\dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5)$ = Kummer nodes, $H_F = 96$), and one gap $\Delta = 6\log(3/2)$ (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:

- **[E]**/[C] **QFT-complete (modulo one premise).** The admissible OS sector, the boundary $(E_8)_1$ net, the relative spectral action, D_F (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once `QGEO.SYM.01` (the seam realisation: *the raw quasi-free seam state is μ_4-invariant*) is granted. That single **[O]** premise is the only structural input the QFT layer still consumes.
- [C] **Ambient QG — a redundancy, separate by design.** The full covariant metric-sector / ambient quantum-gravity measure (`QG.AMB.01`) is gap-decoupled ($\Delta_{\mathrm{eff}} = 1.648 > 0$) and is now a [C] *redundancy* (v369), *not* a blocker for the QFT layer: a certification object, not a missing piece of the boundary QFT.

So "solving the QFT" here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo `QGEO.SYM.01`; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([`verification/v265_qft4d_fork_freeze.py`])

The 4D reading is a *branch policy*, machine-frozen in `qft4d_fork_freeze.json`:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* (E_8 is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the α_1=α_2 scale α_3^{-1} is off by ∼5.6; spread ∼8.8 at 2×10^{16} GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W$=3/8 (v248), κ=M_{PS}/M_s in the frozen $O(1)$ band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract `CONTRACT.QFT4D.DIRAC.01`.
- **Proton-decay rule, now applied ([`verification/v266_ps_threshold_proton.py`]).** [X] the explicit two-step thresholds give d=6 $\tau_p(p\to e^+\pi^0)$: the *minimal* 16-Higgs content (M_{GUT}∼2.4×10^{15}) gives τ_p∼4.4×10^{33} yr < Super-K (2.4×10^{34}) and is *killed*; the E_8-allowed $(15, 1, 1)$ [the single 45] raises M_{GUT}∼5.9×10^{15} ⇒ τ_p∼1.6×10^{35} yr, proton-safe *now* but sitting at the Hyper-K reach (∼1.4×10^{35} yr) — so branch B is *content-selected* (must be $+(15, 1, 1)$) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal; τ_p carries an $O(3)$ hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

And the bedrock is now a falsifiable kill-test, not a proof-promise ([`verification/v215_seam_deck_killtest.py`]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega\circ\rho$=ω ⇔ $[\rho, C]$=0 for the quasi-free covariance, with the principal symbol $|k|$ commuting exactly, so the whole residual sits in the bounded sub-principal M_f (equivalently: the raw Gaussian bulk form A is μ_4-deck-invariant). The four levers are **K1** exactly four marks (b_1=#marks−1=3=N_{fam} ⇒ #marks=4=$|\mu_4|$); **K2** the *square* configuration (cross-ratio 2 ⇒ j=1728, Aut=$\mathbb{Z}/4$; the clock order $h(A_3)$=4 selects $\mathbb{Z}/4$, not the j=0 $\mathbb{Z}/6$, and a generic mark set has only $\mathbb{Z}/2$); **K3** the mod-4 sub-principal support ($[\rho, M_f]$=0 ⇔ f is $\mathbb{Z}_4$-invariant, automatic for a μ_4-mark sum); and **K4** the transfer gap $(2/3)^6$=64/729. These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. The *light* version of that deferred test is now *executed* ([`verification/v503_qgeo_emergence_light.py`], `QGEO.EMERGE.LIGHT.01`), with the result *rigidity, not dynamical selection*: the free field does *not* select the square — the global det′ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, Osgood–Phillips–Sarnak), $\tau = i$ is a saddle of the full moduli space, the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$), and criticality at the square is Palais symmetric criticality — while the square *is* pinned sharply by the clock: the raw free-field DtN is mod-4 block-diagonal *iff* $\tau = i$ (an isolated zero of the K3 indicator as a modulus function, 8×10^{-15} at $\tau = i$, strictly monotone away), and $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ carries no order-4 element. The clock input itself is now *reduced to one bit* ([`verification/v506_seam_clock_rigidity.py`], `SEAM.CLOCK.RIGIDITY.01`): the deck has exactly two Möbius square roots, both automatically of order 4, and a marks-preserving root exists *iff* the deck is the central involution of the mark-stabiliser D_4 — given that alignment bit, order, uniqueness, the free 4-cycle and $\tau = i$ are all theorems (harmonicity alone is insufficient on bare mark data, exact counterexample — on free-deck circles it becomes sufficient, v510); on the Fock side the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$; the R sector splits), so the fermionic $\mathbb{Z}_4$ explains the *order* and only the spatial alignment remains the carrier input. That bit has now survived its *tautology attack* ([`verification/v507_seam_bit_origin.py`], `SEAM.BIT.ORIGIN.01`): the marks are the four half-periods of the seam double and *every* mark-free collar deck is a half-period translation — the deck's *class* is derived from the common torus origin — yet the origin does not force the alignment (the marks-preserving root exists iff the deck pairs separate harmonically, $\lambda \in \{-1, 2, \frac{1}{2}\}$, and the silver counterexample sits in-family as "$\tau = i$ with a non-CM-fixed half-period"); the bit is an equivalence tetrad (clock $\iff$ $\tau{=}i$ with the CM-fixed half-period $\iff$ deck central in the mark-D_4 $\iff$ harmonic deck pairs $\iff$ nonsplit NS Fock lift) whose physical face is the Fidkowski–Kitaev-type extension class (central $U^2 = (-1)^F$ vs edge $U_e^2 = +1$, split) — distinguished, but not derived: only the *position* (which half-period) remains the carrier input. That position half is now *topology* ([verification/v510_seam_bit_freedom.py], SEAM.BIT.FREEDOM.01): the collar deck is a *covering* deck, hence free on the seam circle; the unique free involution is the antipode ($\mathrm{Aut}(C_{16}) = D_{16}$, complete census), nonsplit $\iff$ free over all 17 involutions in exact Cl(16), and the edge/silver class — whose mark circle passes through the deck poles, the restriction of the *branched* pillowcase involution — is excluded; harmonic + free solves exactly to $b = \pm i$, so the bit reduces to the square-modulus datum $\tau = i$ alone. That datum now carries an equivalent *discrete* form ([verification/v512_seam_tau_flag.py], SEAM.TAU.FLAG.01): on the free seam circle bare mark-transitivity is automatic for every deck-invariant configuration $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ (the pair-exchanging V_4; no local jet sees the side bit), and what selects $\delta = \pi/2 \iff \tau = i$ is exactly *flag* transitivity ($V_4 \to D_4$ symmetry lift), welded into a 13-fold exact equivalence web (since 15-fold by v528 facets #14/#15) — the remaining input is *one discrete symmetry-lift bit*, its negation concretely measurable (odd-doublet split $(2/\pi)\ln\cot(\delta/2) \neq 0$). The candidate [E]-closure route through free OS positivity is since *decided negative* ([verification/v521_seam_bit_rp_blind.py], SEAM.BIT.RPBLIND.01): free reflection positivity and anti-linear Θ existence (the v519 battery) are *side-blind* along the whole $M(\delta)$ family — bond-cut Gram spectra identical to 40 digits for every tested δ, the $N = 16$ odd-m failure a lattice-placement artifact resolved at $N = 32$, the continuum kernel exactly $1/\sin((s{+}t)/2)$ with the axis position dropping out — so the free RP/Θ battery is the *eighth* side-blind test on the v512 scoreboard ($7 \to 8$); the only δ-sensitive clauses presuppose the clock (the group generated by the admissible OS reflections closes to D_4 exactly at $\delta = \pi/2$: face #13 of the web, a typed [C] reformulation), and the honest residual [O] gap is the mark-decorated *state* class (defect/twist insertions, the interacting $\mathcal{A}_{\mathrm{hol}}$) — the mark-decorated half of that gap is since decided side-blind too ([verification/v525_seam_bit_twist_blind.py], SEAM.BIT.TWISTBLIND.01: every well-defined mark-decorated state functional in the free-plus-twist class keeps RP positive along the whole $M(\delta)$ ladder — the *ninth* side-blind test, $8 \to 9$; the free-plus-twist class is exhausted) and the interacting half is since decided too ([verification/v529_seam_interacting_toy_fk.py], SEAM.INT.FKTOY.01: the *tenth* side-blind test, $9 \to 10$, on genuinely non-Wick dynamics — Θ_{15} mirrors $H(m) \to H(8 - m)$ exactly despite δ-seeing spectra) — while the bit itself is now *physically defined* as the twist-class choice ([verification/v528_seam_bit_twist_class_definition.py], SEAM.BIT.TWISTCLASS.01: facets #14/#15 both directions, order parameter O, unification with flag-transitivity, web $13 \to 15$; [C] interferometric measurement sketch; the bit remains formal input) — and since carries the first *positive* selection datum: the straddle filter, run as a selector on the interacting mark family, keeps exactly *one* member alive — $\delta = \pi/2$ with positive coupling ([verification/v534_seam_straddle_cone.py], SEAM.STRADDLE.CONE.01; toy-level evidence for a dynamical origin, not a derivation). The full QGEO.REALIZE.01 stays [O]; no marker moves. [E] carrier-side (the four levers), [O] seam-side (the emergence): a kill-test, *not* a closure (QGEO.KILL.01).

And the mark count is now *derived*, not assumed: four marks from Gauss–Bonnet ([verification/v216_marks_gauss_bonnet.py], [verification/v217_marks_emergence_scan.py]). The K1 lever ("exactly four marks") is the one that can be *executed* today. If the marks are $\mathbb{Z}_2$ branch points (cone angle π, deficit π) on the flat seam sphere ($\chi{=}2$ — the same χ that supplies the 8 in c_3), Gauss–Bonnet *forces* their number: $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\mathrm{fam}}{+}1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and *three independent* criteria — all cone orders = 2 (the $|\mathbb{Z}_2|$ sheet branch), rank $H^1 = \#\mathrm{marks} - 1 = 3 = N_{\mathrm{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal $(3,3,3)$), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling v217 confirms it on the actual DtN/state: scanning the number n *and* the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap $(2/3)^6$}. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. The light emergence test ([verification/v503_qgeo_emergence_light.py]) sharpens this residual to a *convergence*: given the clock, the modulus freezes automatically ($P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ with $j = 1728$ identically, v493), and the order-4 element on H^1 is the Coxeter element of $W(A_3)$ — exactly the P2 cusp-class carrier datum of v499: the QGEO modulus rest and the P2 typing rest R1 are *one* common residual ("why does the raw seam carry the order-4 clock"), not two. And the clock-rigidity theorems (v506) reduce that common residual to *one alignment bit* — "the collar deck is the central involution of the mark-D_4" — with the clock's *order* fermionically forced ($U^2 = (-1)^F$ exact); the bit-origin theorems (v507) then derive the deck's *class* (every mark-free collar deck is a half-period translation of the seam double) and characterise the bit as an equivalence tetrad with a physical extension-class face, so only its *position* — which half-period — stays the input; the freedom theorems (v510) finally type that position half as *topology* (the collar deck is a covering deck, hence free; the edge class is excluded), so the residual is the square-modulus datum $\tau = i$ alone — restated by the flag-transitivity web (v512) as one discrete symmetry-lift bit ($V_4 \to D_4$: flag transitivity $\iff \tau = i$), and since v528 equivalently as the *twist-class choice*, in principle interferometrically readable (`SEAM.BIT.TWISTCLASS.01`; the bit remains formal input). [E] (count + equality), [O] (the square modulus and the raw-seam realisation) — `QGEO.MARKS.03` / `QGEO.EMERGE.01`.

And the four marks are a *four-interval* modular geometry ([`verification/v480_multilocal_four_interval.py`]). The seam circle minus its four marks is the n=4 disjoint-interval region of a chiral free fermion — the one continuum class whose *multi-interval* modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; and Kawahigashi–Longo–Müger: *holomorphic* ⇔ trivial multi-interval inclusion, μ=1 — the same one bit as $\det K$=1). Until now the suite used only the single-ball CHM boost (v358); v480 exhibits the multilocal mechanism on the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with ρ^4=-1 *exactly* (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$-lift pattern as $2I$ over I); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15} (the diagonal 4-fold-cover isomorphism), each sector a *single*-interval quarter-ring kernel with boundary twist (10^{-14} spectra); [E] $\rho C \rho^{-1}$=C at 10^{-14}, so $\omega\circ\rho$=ω holds *manifestly* at the 4-interval state level — and a one-site displacement of a single interval re-couples the sectors at 10^{-2} (the invariance is the μ_4-symmetric configuration, not an artifact); and the cross-interval modular coupling peaks on the conjugate-point diagonal, the lattice shadow of the Casini–Huerta mixing term (numerical). [C] the continuum multi-interval theory is cited, not re-derived; [O] the raw-seam premise (`QGEO.SYM.01`) is untouched — this puts the bedrock and the AP2 continuum leg (v476/v478) in one established, exactly solvable frame; it does not close them.

`QGEO.SYM.01` is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the μ_4-deck structure of the carrier clock — i.e. the carrier (P2) and the seam's conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam's degree-4 mark divisor on the genus-0 double has $h^0 = \deg +1 = 5 = g_{\text{car}}$ and $\operatorname{rank} H_1 = \deg -1 = 3 = N_{\text{fam}}$ (with $\Lambda^{\text{even}}(\mathbb{C}^5) = 1{+}10{+}5 = 16 = \dim S^+$), so P2 is the genus-0 mode count of the four-mark seam divisor ([`verification/v228_rr_index_gate.py`]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than `QGEO.CONF.01/ISO.01`.** It does *not* name $\mathbb{P}^1$ or μ_4 (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* `QGEO.SYM.01`, equivariant uniformisation puts the order-4 clock in the form $z \mapsto iz$ on $\mathbb{P}^1$, the four parabolic marks are its free orbit μ_4, H^1 carries the $(1, 2, 3)$ grading, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$ — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus > 0 (uniformisation to $\mathbb{P}^1$ fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ($N_{\text{fam}} \neq 3$ collapses $b_1 = N_{\text{fam}}$ and the four-mark μ_4 structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

Honest framing. The correct external statement is *conditional*: "*Given* `QGEO.SYM.01`, the RP seam boundary realises the μ_4 deck structure, and the AQFT closure to $(E_8)_1$ follows through the verified chain v175–v181" — *not* "the theory *derives* the seam geometry". The premise stays [O]; everything downstream of it is [E]. Per v323 (Bisognano–Wichmann) `QGEO.SYM.01` is itself the *downstream* conformal-deck face of the single named keystone `SEAM.EQUIV.01` (the raw RP seam = the holomorphic $(E_8)_1$ net at τ=i), which is now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (the lattice/S_3

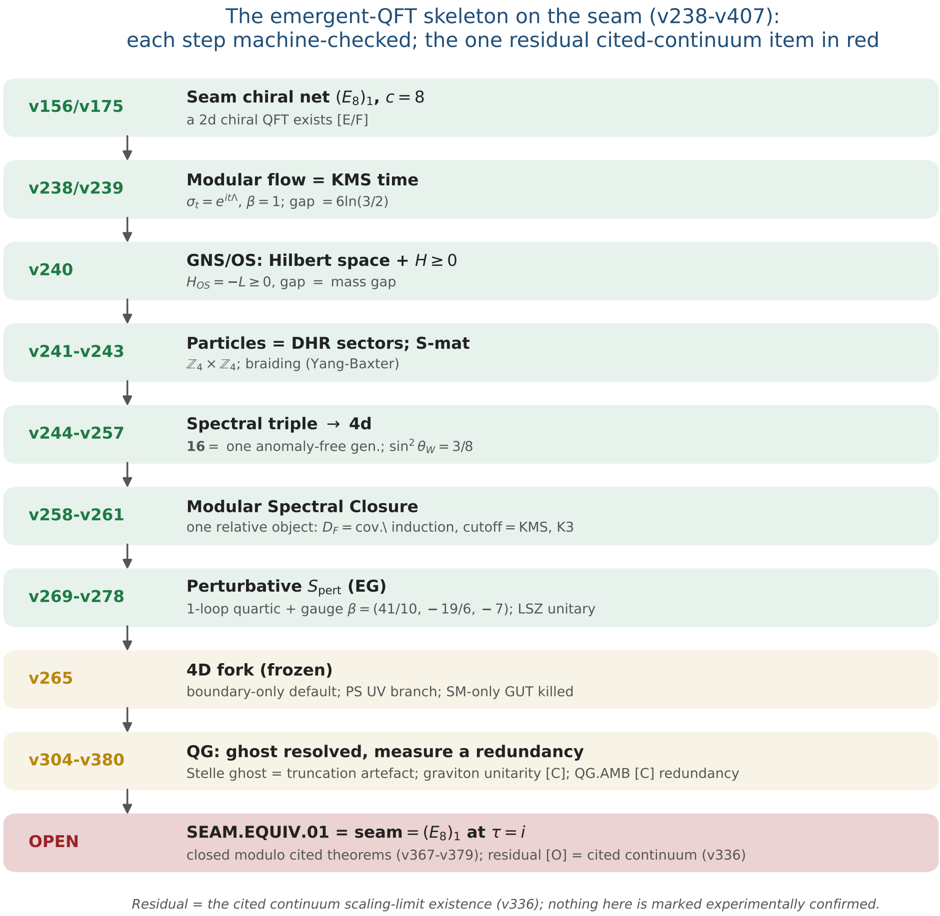

Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier's $SO(10)$ **16**. Each step is machine-checked; the one residual — `SEAM.EQUIV.01`, whose MMST route `SEAM.EQUIV.MMST.01` is closed modulo cited theorems (v367–v379) with the cited continuum scaling-limit existence (v336) as its [O] leg (parent [O]) — is in red, the frozen 4D fork and the QG certification layer in gold. The whole skeleton is *conditional* on that keystone and is neither solved nor experimentally confirmed.

stack pins it at every computable level, Lean `FORM.SEAM.MMST.01`; only the cited continuum existence stays [O]; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay [O]).

This is not a gap to apologise for — it is TFPT's one fundamental postulate. Every fundamental theory rests on an irreducible premise (the constancy of c in relativity, the equivalence principle, the canonical commutator). `QGEO.SYM.01` is that premise for TFPT: a single geometric identification (seam conformal structure = carrier μ_4 deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

The Seam-Holomorphy Theorem — the one named closing target [E]/[O] ([`verification/v234_seam_holomorphy_selection.py`])

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that "closing the theory" becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force E_8:

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* (μ-index 1, single vacuum sector). Holomorphic $+\, c = 8 \Rightarrow$ the unique even unimodular rank-8 lattice net $(E_8)_1$, and $c = 8 = g_{\text{car}} + N_{\text{fam}}$ is itself structural.

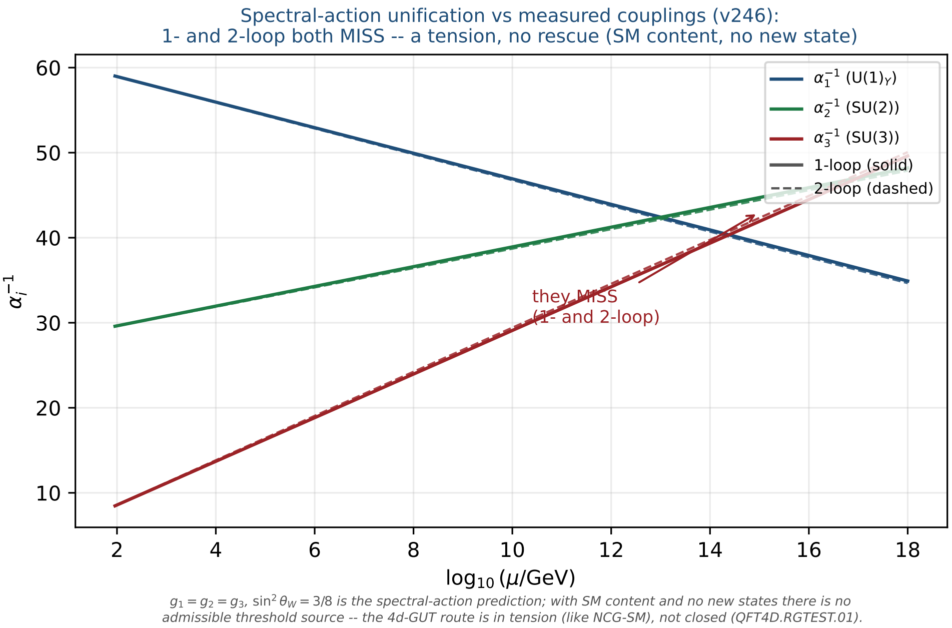

Figure 2: Honest data cross-check (v246): the spectral-action prediction $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; 2-loop minimal spread ≈ 3.2 at $1.7{\times}10^{14}$ GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

2. **Geometry** (v232): the seam link is a *homology* 3*-sphere* ⇔ its binary group Γ is *perfect* ⇔ $\Gamma = 2I$ (the Poincaré sphere $S^3/2I$).
3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number: $\#$(1-dim irreps) $= |\Gamma^{\mathrm{ab}}| = |H_1(S^3/\Gamma)| = \#$(mark-1 affine-Dynkin nodes), which equals 1 *only* for E_8 ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$). So $2I$ is the unique nontrivial perfect finite $SU(2)$ subgroup and $S^3/2I$ the unique homology-sphere space form. **Consequence:** Research Contract 2 (G_{net}/Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

The closing chain (what a complete closure would be), [O] at one step:

$$\text{RP} + \text{gap} + \text{chirality} \xrightarrow{\text{v160}} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net}$$
$$\xrightarrow{[\mathbf{E}]} (E_8)_1 \xrightarrow{\text{v219/v228}} 2I,\ (2,3,5),\ \text{P2} \xrightarrow{\text{v179–v181+Kleinian genus-0}} \texttt{QGEO.SYM.01}.$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is "free Gaussian RP bulk ⇒ holomorphic (single-sector) boundary net"** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That step is now *closed modulo cited theorems* (the MMST route `SEAM.EQUIV.MMST.01`): the lattice/S_3 stack pins it at every computable level (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`), only the cited continuum existence staying [O] (v336); v_{geo} stays the No-Unit metrology primitive and F_{transfer} stays external physics. This box certifies that the three faces coincide and force E_8 ([**E**]); the analytic step is closed modulo that cited theorem.

A physical realisation of the step (`[verification/v235_seam_chern_simons.py]`**).** The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons / K-matrix theory (even integer K), where $\#$anyons $= |\det K|$, the edge chiral central charge is $c = \mathrm{signature}(K)$, and the edge net is *holomorphic* ⇔ $|\det K| = 1$ — so "no nontrivial abelian sector" is precisely $\det K = 1$. The extension tower of v92 *is* the anyon-condensation tower read by determinant: $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic), all rank 8 with $c = \mathrm{signature} = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$, so the holomorphic end is the *Kitaev* E_8 *quantum-Hall state.* Holomorphy ⇔ condensing the order-$|\mu_4| = 4$

Lagrangian glue ($\det 16 \to 16/4^2 = 1$); a partial order-2 isotropic only reaches $\det 4$ (D_8). The residual is thereby a single integer condition — "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" — the same sheet/μ_4 selection as `GATE.QGEO`, but now with holomorphic $\Leftrightarrow$ det 1 a theorem (`GATE.HOLO.02`, [E]/[O]).

The residual as a *physical* statement ([`verification/v237_seam_sre_closure.py`]). The same integer condition is the physics of topological order: a K-matrix phase has ground-state degeneracy $|\det K|^g$ on a genus-g surface, so $\det K = 1 \Leftrightarrow$ *no* topological ground-state degeneracy on any closed surface $\Leftrightarrow$ the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite K ($c = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$) only E_8 is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev E_8 state, whose edge is the holomorphic $(E_8)_1$. TFPT already has a unique vacuum on the plane (RP + gap + cluster + OS); the plane cannot see $\det K$ (no non-contractible cycles), the torus can. So the one analytic step is the sharp, falsifiable physical statement *"the free RP gapped seam bulk is short-range-entangled (no topological order)"* ($= \det K = 1 =$ the Kitaev E_8 phase, `GATE.HOLO.03`, [E]/[C]) — and the lattice/S_3 stack now answers it at every computable level (torus $\det K$=1, v378), so the keystone's MMST route (`SEAM.EQUIV.MMST.01`) is *closed modulo cited theorems*, only the cited continuum existence (v336) staying [O] (the parent stays [O]).

Lemma 2 (U1 — wall landscape, *corrected*). *Parabolic degree zero forces the line weight-sums onto the wall* $(w_1, w_2, w_3) = (2, 1, 1)$. *An explicit balanced representative (weights in* $\frac{1}{3}$*) is*

$$W_{\mathrm{wall}} = \frac{1}{3}\begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of $(0,1,2)$*, row sums* $(2,1,1)$. ***Honest result:*** *the* 6^4=1296 *column-permutation matrices give* 144 *walls; under the* full *group* $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$ *they form* five *orbits,* not one. *So* W_{wall} *is* not *singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;* W_{wall} *is one of the five. Machine: fully certified* $C_U^{(1)}$ *([`verification/v29_research_contract_certs.py`]).*

The splitting type is now algebraic (RH route, *[`verification/v32_rh_splitting.py`]). Passing from the monodromy to the Fuchsian* residues A_k *(eigenvalues = weights* $\{0, \frac{1}{3}, \frac{2}{3}\}$*,* $A_k = U^{k-1} A_0 U^{-(k-1)}$*,* $U = \mathrm{diag}(1, i, -i)$*) the twisted* $\mathbb{Z}_4$*-average collapses* exactly*:* $\sum_k U^k A_0 U^{-k} = 4\,\mathrm{diag}(A_0)$*, so the exponents at infinity are* $4\,\mathrm{diag}(A_0)$*. A flat bundle with regular* ∞ *needs integer exponents summing to* $4 = -\deg E$*; with the cusp weights the only options are perms of* $(2,1,1)$ *and* $(2,2,0)$*, i.e.*

$$\boxed{\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \mathrm{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})}$$

(the sibling $(2,2,0) \to \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$*). Such a Hermitian* A_0 *(spectrum* $\{0, \frac{1}{3}, \frac{2}{3}\}$*, that diagonal) exists by Schur–Horn. So the* splitting *is an algebraic condition on* $\mathrm{diag}(A_0)$. Honest wall: *the global* $\prod M_k = \mathbf{1}$ *is the path-ordered monodromy (*not $\exp(2\pi i \sum A_k)$*), constraining* $\text{off-diag}(A_0)$ *via the genuine RH solve; and R-extraction still needs the* $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ *bridge. The RH route fixes the* splitting *algebraically but not* c_u/c_d.

Lemma 3 (U2 — the kill switch, *run and hardened* [E]). *The selected polystable wall representative must* not *collapse to the diagonal abelian* $U(1)^3$ *point if it is to give non-trivial* $SU(3)_F$ *transport mixing:* ***(A)*** $\nabla_F^\star$ *polystable but* not *simultaneously diagonal (the desired breakthrough), or* ***(B)*** *diagonal collapse, in which case* c_u/c_d *stays honestly [C].* ***Result: the cheap test is run robustly and lands on A.*** *The kill-switch fires iff the* D_4*-symmetric cusp representation is* reducible *(commutant non-scalar). Across* 400 *random cusp-class generators of the* $\mathbb{Z}_4$ *family* $M_k = U^{k-1} M U^{-(k-1)}$ *the commutant is scalar at* every *point (*irreducible 400/400*), the reducible locus is hit with probability* 0 *(measure-zero, codimension* ≥ 1*), and the mixing strength stays bounded away from zero (*$\min \text{off-diag} = 0.35 > 0.05$*). So case A is the* generic *behaviour, not an accident of the one explicit point of [`verification/v33_explicit_flat_bundle.py`]:* (U_{wall}) survives *its own cheapest falsification test. [`verification/v38_uwall_killswitch.py`]*

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, *positive-dimensional*)**.** *With $M_k = U^{k-1}AU^{-(k-1)}$ ($U = \mathrm{diag}(1, i, -i)$, the $\mathbb{Z}_4$ rotation) and the reflection $M_1 = VM_1^{-1}V^{-1}$, $M_3 = VM_3^{-1}V^{-1}$, $M_2 = VM_4^{-1}V^{-1}$ (with $V = -[[1,0,0],[0,0,1],[0,1,0]]$, $V^2{=}1$, $VUV^{-1}{=}U^{-1}$), the relations become polynomials in the entries of $A \in C_{\mathrm{cusp}}$; $\mathcal{X}^{D_4}_{\mathrm{cusp}}$ is cut out as $\bigcup_k C_k$, parametrised by trace coordinates $x = \mathrm{tr}(M_1M_2)$ (note $\mathrm{tr}\,A = 1{+}\omega{+}\omega^2 = 0$). **Result ($C_U^{(2)}$, [verification/v30_d4_character_variety.py]):** adding the reflection to the $\mathbb{Z}_4$ locus of v19 does* not *isolate the point — the* full *D_4-fixed product locus is still* positive-dimensional *($|\,\mathrm{tr}(M_1M_2)|$ varies continuously over $\approx [0,3]$). So even the full D_4 symmetry cannot select $\nabla^\star_F$.* Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

Lemma 5 (U4 — selectors as equations)**.** *The readouts $\rho \mapsto \det R(\rho)$, $\rho \mapsto \mathrm{Spec}(Q_+(\rho))$, $\rho \mapsto \Lambda_{f,j}(\rho)$ are well-defined (cyclotomic/algebraic) on $\mathcal{X}^{D_4}_{\mathrm{cusp}}$. **Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}^{D_4}_{\mathrm{cusp}} :\ \det R = 8,\ \mathrm{Spec}(Q_+) = \{1,2,3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not *closed. **The bottleneck (made precise by U3):** since the D_4-fixed variety is positive-dimensional, the* selector *must do all the cutting — and evaluating* $\det R(\rho)$, $\mathrm{Spec}(Q_+(\rho))$ *as functions on $\mathcal{X}^{D_4}_{\mathrm{cusp}}$ requires the* parabolic-degree ↔ residue dictionary $R(\rho)$, *i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the (U_{wall}) gate.* Status update: since v118–v122 the dictionary itself is exact — *hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the $W(A_3)$ monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller R4′, the* establishment of the selectors *($n = (5,-9,6)$ + the diamond determinants; since [verification/v134_dual_anchor.py]: the pair (d,n)).*

What** $R(\rho)$ **is ([verification/v31_R_dictionary.py]). *Two facts pin its nature.* (i) Case A is alive: *every D_4-fixed cusp monodromy found is* irreducible*, so the wall point carries non-trivial $SU(3)_F$ mixing — the U2 kill switch does* not *fire to the diagonal case B.* (ii) $R(\rho)$ is not algebraic: *R is integer-valued, but the algebraic invariants of ρ (e.g. $\mathrm{tr}(M_1M_2)$) vary* continuously *on the locus; an integer readout cannot be a continuous algebraic function. Hence $R(\rho)$ is the discrete invariant of the* non-abelian Hodge / Mehta–Seshadri *map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle E_ρ): integer output, but its* computation *is the transcendental harmonic-metric (Hitchin) solve* on the generic Higgs locus. ***Refinement (U4″, [verification/v40_harmonic_metric.py]):** at the* polystable *point TFPT actually selects this pessimism does not apply — see the box below.*

Progress: the *selector side* of U4 is closed on the explicit point ([E]). While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 *selectors* are *algebraically* accessible on the explicit flat bundle of [verification/v33_explicit_flat_bundle.py] — no PDE. Computed exactly from A_0: the exponents at infinity $\mathrm{eig}(\sum_k A_k) = \{2,1,1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic *anchor* $a = (1,1,2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\boxed{\begin{array}{c}\det R = 8 = n\cdot a \quad \text{(anchor pairing on the splitting type)},\\ \mathrm{Spec}(Q_+) = \{1,2,3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\})\end{array}}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\mathrm{Spec}(Q_+)$ is the affine image ($d{=}3$ the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data.* **Residual (sharpened below by Readout Rigidity):** the quark *ratio* c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [verification/v39_uwall_selectors.py]

Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE ([E]). The U4 "transcendental Hitchin solve" is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33_explicit_flat_bundle.py] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim\ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So $C_U^{(3)}$ reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length R (closed: H1 + $\det R = 8$).** *Residual:* the feared PDE is gone; the quark *ratio* c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40_harmonic_metric.py]

The final leg-assignment test (honest result) ([E]). Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1, so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\mathrm{diag}\,\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R); the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41_leg_assignment.py]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem ([E]). The v41 "genericity is future work" item is now executed — and yields a theorem. *(i) Flatness = μ_4 torsion [E]:* for *any* M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$-family flatness constraint *is* the torsion statement "$T = MU$ is a fourth root of unity": the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem [E] (both directions):* on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1} v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\boxed{\mathrm{diag}\,M = (0, \tfrac{i}{2}, -\tfrac{i}{2}), \qquad \mathrm{spec}\,M = \{1, \omega, \omega^2\} \text{ automatic}}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + det 1). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a *μ_4-torsion identity*, not an observation. *(iii) Reflection lemma [E]:* the D_4 reflection forces $\mathrm{diag}\,M = (r, z, \bar{z})$ with r real and $\mathrm{Re}\,z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census [E] (seeded):* the full D_4-fixed flat cusp locus realises only $\mathrm{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue [C] (recorded):* whether the anchor splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item. [verification/v114_torsion_delta.py]

The anchor pins the residue: an exact normal form for the (U_{wall}) bundle ([E]). The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i) μ_4-average lemma [E]:* $\sum_k U^k X U^{-k} = 4\,\mathrm{diag}\,X$ for any X — μ_4 conjugation-averaging *is* the diagonal projection, so the exponents at infinity are $4\,\mathrm{diag}\,A_0$ and the anchor splitting holds *iff* diag $A_0=(2,1,1)/4=a/|\mu_4|$: the residue diagonal **is** the anchor over μ_4 (v33's diagonal was forced, not chosen). *(ii) The* $(8,0,5)/144$ *lemma [E]:* the anchor diagonal and the cusp spectrum $\{0,1,2\}/N_{\mathrm{fam}}$ pin $\sum|a_{ij}|^2=\frac{13}{144}$, and with $a_{13}=0$ the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2,\ |a_{13}|^2,\ |a_{23}|^2)=\frac{(8,\ 0,\ 5)}{144}}\qquad \begin{array}{l}8{=}\mathrm{rank}\,E_8,\ 5{=}g_{\mathrm{car}},\\ 144{=}(|\mu_4|N_{\mathrm{fam}})^2,\ 8{+}5{=}13{=}\Delta_Q\end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates $c_u/c_d=55/117=5\cdot 11/(9\cdot 13)$. *(iii) Exact normal form [E]:* $A_0^\star=\left[\begin{smallmatrix}1/2 & \sqrt{2}/6 & 0\\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12\\ 0 & \sqrt{5}/12 & 1/4\end{smallmatrix}\right]$ has $\chi=\lambda(\lambda-\frac{1}{3})(\lambda-\frac{2}{3})$ *exactly. (iv)* $A_0^\star$ *is the RH solution [E]:* its big-circle monodromy is trivial to 10^{-10}, $\prod M_k=\mathbf{1}$, and the unitarised holonomy has $|\operatorname{diag}\widetilde{M}_0|=(0,\frac{1}{2},\frac{1}{2})$, $\operatorname{tr}(\widetilde{M}_0U)=1$ — the v33/v40 point is gauge-equivalent to $A_0^\star$ (the a_{12},a_{23} phases are diagonal gauge). *(v) The anchor forces the* $d_1=0$ *slice [E]:* a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\operatorname{diag}\widetilde{M}_0|=(0,\frac{1}{2},\frac{1}{2})$ — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**; $\delta=\frac{1}{2}$ needs no selection. *Residue [C]:* $a_{13}=0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining U_{point} step. [`verification/v115_anchor_residue.py`]

The resonance theorem: $M_\infty=\mathbf{1}\iff a_{13}=0$ — one gauge orbit [E] ([`verification/v116_resonance_uniqueness.py`])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as $dY/dw=-(1/w)(B_0+B_1w+\cdots)Y$ with $B_m=\sum_k i^{km}\,U^kA_0U^{-k}$, the *twisted* μ_4 averages collapse [E]:

$$B_0=4\operatorname{diag}A_0,\qquad \boxed{B_1=4a_{12}E_{12}+4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio $-i$ cells survive). The anchor exponents diag$(2,1,1)$ have resonance gap 1; the level-1 formal-gauge equation $(1-\operatorname{ad}R_0)H_1=R_1$ is singular exactly on the cells $\{(2,1),(3,1)\}$, where B_1 has entries $(0,\ 4\overline{a_{13}})$; all higher levels are non-resonant and the formal monodromy is $e^{-2\pi i\operatorname{diag}(2,1,1)}=\mathbf{1}$. Hence [E]:

$$\boxed{M_\infty=\mathbf{1}\iff a_{13}=0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation. Uniqueness corollary [E]:* $a_{13}=0$ + the $(8,0,5)/144$ moduli + phases = diagonal gauge ⇒ the flat anchor locus is **exactly one gauge orbit**, the exact $A_0^\star$: within the μ_4-equivariant anchor class the (U_{wall}) bundle datum is *algebraically unique. Sharpness [E]:* $\|M_\infty-\mathbf{1}\|\sim 8.5\,|a_{13}|$ (0.17 at $a_{13}=0.02$) vs 10^{-10} at $A_0^\star$. *Honest scope [C]:* the equivariant ansatz is the established D_4 selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal $(0,\frac{1}{2},\frac{1}{2})$) remain [E] — transcendental monodromy of an exact system.

The monodromy is the Weyl group of A_3 — the holonomy is exact [E] ([`verification/v117_monodromy_weyl_a3.py`])

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact $A_0^\star$ system is itself *exact*, with entries in $\frac{1}{2}\mathbb{Z}[i]$:

$$\boxed{\widetilde{M}_0=\begin{pmatrix}0 & -\frac{1+i}{2} & \frac{1-i}{2}\\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2}\\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2}\end{pmatrix}}$$

unitary, $\det=1$, $\operatorname{tr}=0$, $\chi=\lambda^3-1$ (the cusp class *exactly*), $\widetilde{M}_0^3=\mathbf{1}$, and $\operatorname{diag}\widetilde{M}_0=(0,-\frac{i}{2},+\frac{i}{2})$ — so $d_1=0$ and $\delta=\frac{1}{2}$ are now *matrix entries*, not observations; $(\widetilde{M}_0U)^4=\mathbf{1}$ with $\operatorname{tr}(\widetilde{M}_0U)=1$ (the $\{1,i,-i\}$ branch, $d_1=0$ slice — exactly where v114/v115 located the physical point). *The group* [E]: exact enumeration of $\langle U,\widetilde{M}_0\rangle$ gives order 24 with order statistics $(1,9,8,6)$ and character values $(3,-1,0,1)$ — uniquely S_4 in its standard ⊗ sign representation, i.e. the (twisted) reflection representation of the **Weyl**

group of the family lattice A_3:

$$\boxed{\mathrm{monodromy}(U_{\mathrm{wall}}) = W(A_3) = S_4, \qquad |W(A_3)| = 4! = 24}$$

— the flavor wall realises the A_3 factor of the glue $D_5 \oplus A_3 + \mu_4$ as its Weyl group (U = a 4-cycle, the μ_4 deck; $\widetilde{M}_0$ = a 3-cycle, the family rotation). *Identification [E]:* the ODE monodromy of $A_0^\star$ equals the exact $\widetilde{M}_0$ to 3.5×10^{-10}, and the invariant form H is U-invariant to 10^{-11}. *Status:* holonomy values [E]; only the analytic identification stays [E]. The δ thread (v20 → v40/v41 → v114 → v115 → v117) is closed end to end.

The hexagon is the sign-twisted family spectrum — first exact H2 piece [E] ([`verification/v118_hexagon_family_dictionary.py`])

The missing "$C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary" (H2) acquires its first exact entry. *Sign-twist lemma [E]:* $-\widetilde{M}_0$ has order 6 and

$$\boxed{\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6}$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ($\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ realised as $(-1) \times \langle \widetilde{M}_0 \rangle$); the v20 denominators $\frac{5}{4} - \cos(r\pi/3)$ are exactly the eigen-denominators $|1 - \zeta_6^r/2|^2$ of the two family resolvents $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$. *Cyclotomic determinant [E]:* $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1-t)\Phi_3(t)$, so $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$ — and the lepton coefficients become resolvent determinants: $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$, $c_\mu = N_{\mathrm{fam}}/(2\det_+) = \frac{4}{3}$, $c_\tau = |\mu_4| \det_- / N_{\mathrm{fam}} = \frac{7}{6}$, product rule $= |\mu_4|/\det_+ = \frac{32}{9}$. The lepton 7 and 9 *are* the family-resolvent determinants; $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$ is supplied by the matrix itself. *Sheet-extended group [E] (audit):* $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+$ — the seed's quartic coefficient. *Residue [C]:* the dictionary's *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20's established input / open — nothing re-fished.

The address table: the lepton words are the compiler atoms [E]/[C] ([`verification/v120_address_table.py`])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today's readings (the red-team firewall). *(i) The lepton words are the atoms [E]:*

$$\boxed{(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\mathrm{rank}\, E_8,\ g_{\mathrm{car}},\ N_{\mathrm{fam}}) = (p_0{+}e_2,\ e_2,\ p_0)}$$

in mass order (longest word = lightest lepton). *(ii) Addresses = division by the hexagon [E]:* $(r, w) = L \operatorname{divmod} p_2$ with $p_2 = 6 = |R^+(A_3)|$ itself an atom: $e \to (2,1)$, $\mu \to (5,0)$, $\tau \to (3,0)$; $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$, $\sum L = 16 = \dim S^+$. *(iii) Sheet parity [E]:* by the v118 dictionary, e sits on the *untwisted* sheet ($\omega \in \mathrm{spec}\, \widetilde{M}_0$), μ, τ on the *twisted* sheet $(-\omega, -1)$; the anchor lepton τ occupies the unique *real* twisted eigenvalue -1 — exactly the site where v20's product rule replaces the resolvent. *(iv) Quark sum rules [E] (frozen v18 words $7,7,5,3,2,0$):* up-type sum $= 10 = p_3$; down-type sum $= 14 = p_1{+}p_3$; **all quarks** $= 24 = |W(A_3)|$ (the monodromy group order, v117); **all nine charged fermions** $= 40 = p_1 p_3 = a^T R a$ (v119); quark $\sum r = 2p_2$, $\sum w = |\mathbb{Z}_2|$; the top sits at the vacuum site $(0,0)$ — zero word, no suppression, the established ladder anchor. *Residue [C]:* five independent closures $(16, 10, 14, 24, 40)$ constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

The address table is pinned: R is the unique hexagon matrix with atom margins [E] ([`verification/v121_address_pinning.py`])

The per-fermion assignment closes at the *information* level. The established identity $L = R{+}2U$ (v95) says the word-length table *is* L[sector][generation] — rows (up; down; lepton) $= ((7,3,0); (7,5,2); (8,5,3))$, the v18/v20 words verbatim — so the addresses are the entries of the *residue operator* R plus one hexagon turn for the first generation, and R's first column is the **anchor** $a = (1,1,2)$. The margins are atoms [E]: row sums $(4, 8, 10) = (e_1, \mathrm{rank}\, E_8, p_3)$, column sums $(4, 13, 5) = (e_1, \Delta_Q, g_{\mathrm{car}})$. And the census closes it **[E]**:

$$\boxed{\begin{array}{c}\text{among all } 3\times 3 \text{ matrices with entries in the hexagon } \{0..5\},\\ \text{exactly 17 have the atom margins — and exactly ONE has } \det = 8\end{array}}$$

— and it is R (also the unique one with SNF $= (1,1,8)$; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ($m \sim \lambda_Y^L$: longer word = lighter fermion). *Residue [C]:* the open H2 item shrinks from "derive 9 per-fermion integers" to "derive the six atom margins + det = rank". *Firewall:* R, L and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

The margins are theorems (`[verification/v122_margin_theorem.py]`). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — *(S1)* the D_4 annihilator $n = (5,-9,6)$ with $n^T R = (8,0,0)$ (it kills the unchanged (c_2, c_3) plane; $n \cdot a = 8$), *(S2)* $\det R = 8$, *(S3)* $\det K = 4$ with $K = R + Q\Sigma$ — pin R *uniquely* among hexagon matrices **[E]**: the $(S1)$ census has $4\times 4\times 4 = 64$ candidates, $(S2)$ leaves 12, $(S3)$ leaves **exactly one**, and it is R. Hence the atom margins, the anchor column $Re_1 = a$ and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]:* on the $(S1)+(S2)$ locus $\det(M+2U) = 20 = \det L$ holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]:* H2 reduces to the establishment of the selectors themselves (n and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class.*

The dual-normal selector: (d,n) pins R columnwise (`[verification/v136_dual_normal_selector.py]`). The selector list shrinks once more. With the dual anchor $d = a^\top R^{-1}$ (`[verification/v134_dual_anchor.py]`) and the torsion normal n, each *column* of R obeys $d \cdot c_j = a_j$ and $n \cdot c_j = 8\delta_{1j}$; the common kernel of (d,n) is the primitive vector $(6,8,7)$ (audit: p_2, rank E_8, scalaron-7), so within the address box $\{0..5\}^3$ each column is the *unique* lattice point (census: 1 of 216 each) — $(d,n) \Rightarrow R$, and the 6^9 census above collapses to three column problems **[E]**. *Honesty:* K is *not* pinned directly ($d^\top K = (1, \frac{1}{2}, 1)$); $\det K = 4$ comes via the μ_4 wall (`[verification/v135_det_surface.py]`). *The mechanism is bounded:* the zero mode of $A_0^\star$ carries the exact spectral normal $\sigma = (2,-9,5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$ (signed squares; $\|v\|^2 = \frac{16}{9}$ with $16 = \dim S^+$, $\operatorname{adj} A_0^\star = \frac{2}{9} P_v$ with $\frac{2}{9}$ = the SdS entropy curvature), $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$, $\sigma \cdot a = N_{\text{fam}}$, $n \cdot \sigma = 121 = \|\operatorname{Pl}(K)\|_1^2$; and the exact $W(A_3)$ orbit control shows *no* group image of σ is proportional to n — the naive cofactor/orbit lift fails, the step $\sigma \to n$ is a genuine mechanism beyond the monodromy group. *R4′ restated [C]:* derive the *pair* (d,n); then R is forced columnwise.

The selector triangle (`[verification/v139_selector_triangle.py]`). The pair itself now collapses further. First, d is *pure anchor data*: $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ exactly — the closed form never mentions R, so the first selector is *derived*, not residual. Second, the frame $(\mathbf{1}, a, \sigma)$ is a basis of $\mathbb{Z}^3$ with $\det(\mathbf{1}|a|\sigma) = 11 = \|\operatorname{Pl}(K)\|_1$ (the quark constant as the frame volume), and n is the *unique* integer covector with the three atom pairings

$$\boxed{n\cdot\mathbf{1} = 2 = |\mathbb{Z}_2|, \qquad n\cdot a = 8 = \operatorname{rank} E_8, \qquad n\cdot\sigma = 121 = \|\operatorname{Pl}(K)\|_1^2}$$

— the cofactor definition is replaced by frame data plus three atoms. On the $(\mathbf{1},a)$-constrained line the σ-pairing is $11(11+t)$: divisible by 11 for *every* t, a perfect square exactly at the true n (t=0). The review's predicted lift is exact as an audit identity: $n = \operatorname{reverse}(\sigma) + |\mu_4| e_3$ (generation flip + μ_4 lift at the third slot).

Frame integrality: three pairings → two (`[verification/v142_frame_integrality.py]`). The third pairing is *not* independent. The pairing triples $(m\cdot\mathbf{1}, m\cdot a, m\cdot\sigma)$ of *integer* covectors form an index-11 sublattice of $\mathbb{Z}^3$, cut out by the single congruence

$$\boxed{x - 3y + z \equiv 0 \pmod{11}} \qquad (\text{index } 11 = \|\operatorname{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms $(x,y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8) = (2,8)$, integrality *forces* $z \equiv 0 \pmod{11}$ — the $11(11+t)$ line *is* the integer line. Primitivity forces t *even* (odd t makes every entry of $n(t)$ even), and t=0 is the unique even t with a square σ-pairing for $|t| < 88$ (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities

($n \cdot a = \det R$ by Laplace along the anchor column; $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$ with $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$, v134) — restructuring, not establishment.

The pairing values are atom identities ([verification/v145_pairing_atoms.py]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally — $n = w_0(\sigma) + |\mu_4| e_3$ with w_0 the A_3 *exponent duality* $m \leftrightarrow h{-}m$ (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n{\cdot}\mathbf{1} = |\mu_4| {-} |\mathbb{Z}_2| = |\mathbb{Z}_2|, \qquad n{\cdot}a = |\mu_4|\, a_3 = 8 = \operatorname{rank} E_8 \ \text{via} \ \boxed{|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9,\ \sigma \perp w_0(a))},$$

and $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4| g_{\text{car}} = 110 - 9 + 20 = 121$ with $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \| \operatorname{Pl}(K)\|_1$. *R4′ residue [C] (final form):* derive the *one* map $n = w_0(\sigma) + |\mu_4| e_3$ — no independent number remains; everything else in the chain anchor $\to (d, n) \to R$ is exact.

Both dual normals are anchor data ([verification/v149_cusp_normal.py]). The remaining opacity dissolves in the canonical frame: the pairings of n with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$\boxed{n \cdot e_1 = 6 = p_2(a), \qquad n \cdot e_2 = 3 = p_0(a), \qquad n \cdot e_3 = 5 = e_2(a)}$$

— one anchor invariant per Q_+ degree, and the cusp matrix is unimodular, so these three pairings determine n uniquely. With $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data $(6, 3, 5)$, the frame pairings $(2, 8, 121)$ and the w_0-lift are three presentations of one covector, all atomic (equivalences exact). *Audit:* $\sigma \to (5, 1, 2)$; alternate reading $(6, 3, 5) = (|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$; sum $14 = \dim G_2$. *R4′ residue [C] (cusp form):* derive the *assignment* — why the Q_+ degrees $(1, 2, 3)$ carry (p_2, p_0, e_2) in that order; one discrete assignment, zero continuous freedom.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r, w) = (1, 1)$ ($L = 7$ for both), and the C_6 resolvent $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos\frac{r\pi}{3})$ reads *only* (r, w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\operatorname{Pl}_{\mathbf{1},a}(K)$. The "11" is then *generated*, not read:

$$(K, \mathbf{1}, a) \to B_K = \left(\begin{smallmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{smallmatrix}\right) \to \operatorname{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \to \|\operatorname{Pl}(K)\|_1 = 11,$$

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\operatorname{Pl}_{\mathbf{1},a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}} \quad \left(\text{anchor microcode } = \tfrac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}\right).$$

Typing: [E] for the Plücker / microcode identities ($\operatorname{Pl}(K) = (-1, 6, 4)$, $\|\cdot\|_1 = 11$, $5{\cdot}11/(9{\cdot}13) = 55/117$); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio $\mathcal{C}_{ud}(\rho^\star)$ = this $\Lambda^2 F$ readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from "find the right scalar y" to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale (U_{point}) stays as an anchor, no new scalar, no fishing.

Audit (the 121 lemma, [verification/v119_review_validation_2.py]): the "11" has a second, independent appearance [E]: with $h = \mathbf{1} + a = (2, 2, 3)$ in the canonical anchor ordering, $h^T R h = 121 = 11^2 = \|\operatorname{Pl}(K)\|_1^2 = (p_3 + 1)^2$ — the quadratic *anchor norm* of the residue operator R reproduces the Plücker integer (sensitivity control: the three orderings of h give $\{105, 121, 135\}$; only the canonical one yields 121). Bonus checksums: $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$ (the entry sum of R) and $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$. Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [verification/v42_exterior_leg.py]

Bridge test ([verification/v43_exterior_bridge.py]). The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \overline{\widetilde{M}}$ exactly — the exterior leg *is* the $\bar{\mathbf{3}}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$ does *not* equal the *integer* $\operatorname{Pl}(K) = (-1, 6, 4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^\star \to \operatorname{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R; $\det R = 8$ and $\operatorname{Spec}(Q_+) = \{1, 2, 3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([verification/v44_carrier_exterior.py]). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\rm even}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u)+\deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d)-\deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The "Λ^2" of the exterior leg is therefore not exotic Hodge data — it is the carrier's own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from "discrete non-abelian-Hodge invariant" to "the carrier exterior degree" — a standard [E] fact. *Honest scope:* this grounds the *type*; the Plücker *number* 55/117 stays the family $\mathrm{Pl}(K)$ [E], and the carrier-Λ^2 $\leftrightarrow$ family-$\Lambda^2 F$ link (via family ⊗ carrier = 15 = 16−1) plus the normalisation remain [C].

The "11" is Pascal — a third, simplest origin (`[verification/v45_family_exterior.py]`**).** Following that link, $\mathrm{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3{=}SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row $4 = (1, 4, 6, 4, 1)$ with full sum $2^4 = 16 = \dim S^+$, and $|\mathrm{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\boxed{\|\mathrm{Pl}(K)\|_1 = \textstyle\sum_{k=0}^{2} \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\rm car}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree $2 = \deg(u^c)$), and family ⊗ carrier $= 15 = \dim \mathfrak{su}(4) = \Lambda^2(5)\oplus\Lambda^4(5) = 10{+}5 = N_{\rm fam} g_{\rm car}$ read three ways. So the "11" is generated a *third* way — the cumulative family exterior algebra up to the up-quark's degree — pure Pascal/branching, *no special math.* The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the "11" worry, *discharged* by Readout Rigidity)**.** *The original worry was that* $c_u/c_d = \mathcal{C}_{ud}(\rho^\star)$ *must be evaluated at the uniquely selected* $\rho^\star$ without *using* 11 *as input, so that until then* 55/117 *would be a mere table-level rational shadow.* ***This is now resolved (v49):*** $\mathcal{C}_{ud}$ *is* constant *on the discrete selector stratum* $\mathcal{S}_{8,Q}$*, so the "11" is* earned *as the rigid Plücker readout* $\|\mathrm{Pl}(K)\|_1$ *(v42)* — independently *of point-uniqueness. Point-uniqueness of* $\rho^\star$ *enters only the* absolute *amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates)**.** $C_U^{(1)}$ *wall enumeration (done);* $C_U^{(2)}$ D_4*-fixed character solver;* $C_U^{(3)}$ *selector uniqueness (one S-equivalence class);* $C_U^{(4)}$ *amplitude extraction — the* ratios $R, Q, c_u/c_d$ *are read off the (derived) selector stratum (closed, v49/v71); only the* absolute $U_f^\star, c_q$ *normalisation is the anchor.*

Progress: an explicit valid flat bundle is realised (RH solve) ([E]). A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, diag $= (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)}/(z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \mathrm{diag}(2, 1, 1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1M_2M_3M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid ($U_{\rm wall}$) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope:* this is one realising point (existence + case A), likely one of a residual family; the unique $\nabla_F^\star$ still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). `[verification/v33_explicit_flat_bundle.py]`

H2 bridge probed (`[verification/v34_h2_bridge_attempt.py]`*).* The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\mathrm{diag}\, M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes $(0.475, 1.107, 0.917)$. So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family $\rho^\star$) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the* $W(A_3)$ *monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

D_4 fixes the admissible chamber, not the physical point (the D_4-fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\mathrm{SNF}(R) = (1,1,8)$, $\mathrm{Spec}(Q_+) = \{1,2,3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} [E]: parabolic degree 0 + polystable $\Rightarrow$ unitary (Mehta–Seshadri) $\Rightarrow$ $\Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([`verification/v40_harmonic_metric.py`]).
- U_{H2} [E]: R, Q, K are read off the bundle (splitting type $= a$, $\det R = 8$, $\mathrm{Spec}(Q_+)$) ([`verification/v39_uwall_selectors.py`]).
- U_{Λ^2} [E]: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \mathrm{SNF}(R) = (1,1,8), \mathrm{Spec}(Q_+) = \{1,2,3\}\}$ the anchor-plane readout is *constant*, $\mathcal{C}_{ud}(\rho) = g_{\mathrm{car}} \|\mathrm{Pl}_{\mathbf{1},a}(K)\|_1 / (N_{\mathrm{fam}}^2 \Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([`verification/v49_readout_rigidity.py`], [`verification/v42_exterior_leg.py`]).
- U_{point} [O]: full uniqueness of $\rho^\star$ — needed only for the *complete* $U_f^\star$ amplitude matrix, *not* for c_u/c_d.

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; the only residual, U_{point}, is not an independent gate — it reduces to the single dimensionful anchor v_{geo} (Gate 1 below).

The simple closure ([`verification/v71_simple_r_bridge.py`]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \mathrm{rank}\, E_8$ with $\mathrm{SNF}(R) = (1,1,8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\mathrm{fam}}$, v70), and $\mathrm{Spec}(Q_+) = \{1,2,3\}$ is the D_4-equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3,4,8,20)$, product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\mathrm{car}} \cdot 11/(N_{\mathrm{fam}}^2 \cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts — no transcendental monodromy solve.* The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point}, is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

Gate 1 complete: $U_{\mathrm{point}} \to v_{\mathrm{geo}}$ ([`verification/v75_upoint_to_vgeo.py`]). The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c-ratio is closed (Plücker), and each *sector product* is a clean φ_0^{ret}-power (Grand Mass Volume, v46: $\det M_{\mathrm{sector}} \sim (\varphi_0^{\mathrm{ret}})^{6,9,10}$; the lepton product is $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\mathrm{car}}}/N_{\mathrm{fam}}^2$). Since (pairwise ratios) + (product) $\Leftrightarrow$ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale v_{geo}*. **So U_{point} is not an independent gate: it reduces to v_{geo}, the *same* one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [O] anchors collapse to one.**

Research Contract 2 — (G_{metric})

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi\to\infty} \int_{\mathcal{G}_\chi/\mathrm{Diff}} \exp\bigl(-S_{\mathrm{rel},\chi}[g,A,\psi] + J\cdot O\bigr)\, d\mu_{\mathrm{Cal},\chi},$$

$$S_{\mathrm{rel},\chi} = \mathrm{Tr}\, f\Bigl(\tfrac{D_{\mathrm{rel}}+A_\Sigma}{\chi}\Bigr) - \mathrm{Tr}\, f\Bigl(\tfrac{D_{\mathrm{ref}}}{\chi}\Bigr) + \tfrac{i\pi}{2}\Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm)**.** *For $s > 5/2$ a Sobolev space of Θ-symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\mathrm{gh}} = K^*K$ and $\Pi_{\mathrm{phys}} = \Pi_{\mathrm{BRST/Hodge}}\Pi_{TT}$ is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (*not *rhetorically*).* Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

Lemma 9 (G2 — finite relative spectral action, *heat-kernel grounded* [E])**.** *$S_{\mathrm{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion* $\mathrm{Tr}\, f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \cdots$*. For* *the Lichnerowicz Dirac operator* $D^2 = \nabla^2 + \frac{R}{4}$ *(so* $E = -\frac{R}{4}$*, spinor trace* $\mathrm{tr}\,\mathbf{1} = 4$*) the Gilkey coefficients give, per* $(4\pi)^{-2}$*,*

$$a_2 = \tfrac{1}{6}\,\mathrm{tr}(6E{+}R\,\mathbf{1}) = -\tfrac{R}{3} \quad (\textit{Einstein–Hilbert } \textstyle\int R), \qquad a_4\big|_{R^2} = \tfrac{1}{360}\,\mathrm{tr}(180E^2{+}60RE{+}5R^2\mathbf{1}) = \tfrac{R^2}{72}$$

(the R^2*/Starobinsky term), so the spectral action* structurally *produces* $S_{\rm eff} = \int\sqrt{g}\,\frac{\bar M_{\rm Pl}^2}{2}(R{+}\frac{R^2}{6M^2}){+}\cdots$ *with the scalaron mass ratio fixed by the cutoff moment,* $M^2/\bar M_{\rm Pl}^2 = 6(4\pi)^2/f_0$*. The TFPT closure* $M^2/\bar M_{\rm Pl}^2 = c_3^7$ *fixes* $f_0 = 6(4\pi)^2/c_3^7$*, i.e.* $M = c_3^{7/2}\bar M_{\rm Pl} = 3.06\times 10^{13}$ *GeV — the boundary normalisation* c_3 is *the gravitational scale — and hence the Starobinsky slow-roll forms* $n_s \simeq 1 - \frac{2}{N}$*,* $r \simeq \frac{12}{N^2}$*. The* $R+R^2$ structure *is convention-independent (standard Gilkey); the precise rational* $\frac{1}{72}$ *and factor* 6 *are the standard Dirac conventions used here (* [`verification/v36_spectral_action_g2.py`]*,* [`verification/v28_gravity_fR.py`]*).*

The simpler resolution: "full QG" for physics is the gap-decoupled admissible sector ([**E**]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

$$\boxed{\begin{array}{c}\text{admissible IR sector} = \text{closed under stated RP;}\\ \text{ambient metric measure (G6)} = G_{\rm metric}\text{ gate}\end{array}}$$

G2 supplies the local action ($R{+}R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ($G_{\rm metric}$) to the ambient projective limit (G6). **Precise status (Alessandro's boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure is now discharged as a [C] *redundancy* (v369, the holographic-route keybox below), a certification object rather than a missing piece. Its absence does *not* affect the bounded IR claim. [`verification/v36_spectral_action_g2.py`]

The glue Q-system: the index-4 object of the $G_{\rm net}$ statement exists explicitly ([**E**]/[**C**]). The $G_{\rm net}$ target — *"the seam–Calderón inclusion has Jones index* $4 = |\mu_4|$*"* — no longer refers to an unknown object. On the glue group algebra $A = \mathbb{C}[\mathbb{Z}_4]$ (multiplication $\delta_a\otimes\delta_b \mapsto \delta_{a+b}$, unit δ_0) *all* Longo Q-system axioms hold exactly [**E**]: associativity, unit law, the Frobenius identity $(m\otimes 1)(1\otimes m^*) = m^*m = (1\otimes m)(m^*\otimes 1)$, and specialness $mm^* = |\mathbb{Z}_4|\,\mathbf{1}$ — a special C* Frobenius algebra with

$$\boxed{\dim\theta = 4 = |\mu_4| = [B:A]}$$

exactly the Carrier Index Lemma value (KLM: $\mu_{\rm carrier} = 16 = 4^2\mu_{E_8}$ re-verified). *Locality = isotropy [E]:* on the Lagrangian glue $\langle(1,1)\rangle$ the discriminant form gives $q(k(1,1)) = k^2 \in \mathbb{Z}$ for every k — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step [E]:* $\{0,2\} < \mathbb{Z}_4$ closes and gives the unique intermediate Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ tower step, with index factorisation $4 = 2\times 2$. *R2 sharpened [C] (recorded, not claimed):* what remains is the operator-algebraic *identification* "the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system" — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification. *[2026-08-03 update, the index side measured on the ladder:* [`verification/v726_phys_car_pp_index.py`] (57/57, `T2-SLICE-GO`) measures the inclusion on the CAR ladder $N = 48$–384: the μ_4-average conditional expectation has optimal Pimsner–Popa constant *exactly* $1/4$ (explicit rank-one minimizer + pinching theorem) and Watatani index *exactly* 4 via an explicit quasi-basis with *forced* weights (uniform weights break scalarness; the $\mathbb{Z}_2$-only average gives index 2 — μ_4 distinguished from μ_2), with μ_4 *derived* from the clock $L = S^{N/12}$ (order 4 forced by the NS spin structure) and the Ramond sector healed *state-preservingly* by the Klein zero-mode dressing ($[\rho, U'] = 0$ exactly). The remaining half is unchanged in type — the *identification* of this ladder Q-system with the constructed $\mathbb{C}[\mathbb{Z}_4]$ object — now registered with kill criteria as the open half of `GNET.RAMIFIED.01` (Frobenius mismatch; index $\neq 4$ on the state-preserving ladder; dressing incompatible with the σ descent); no marker move.] *[2026-08-05 update, the finite precursor of the identification (*[`verification/v779_gnet_gns_arf.py`]*,* $27/27 + 27/27$*):* the coherent *commuting* system of index-4 expectations along the Haar scaling tower (restriction coherence, $[E_I, E_J] = 0$, joint idempotency, all exact) is the finite half of the Longo Q-system identification, and the exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1}\cos(\pi c/2)$ makes the 4-element $\mathbb{C}[\mathbb{Z}_4]$ group-layer quasi-basis obstruction exactly $2^{-\ell}$ at every finite level ($d_0 - d_2 = 2^\ell > 0$), restored only asymptotically — the crossed-product structure is *exactly asymptotic*, and $d_2(1) = 0$ *is* the $\ell = 1$ index-3 anomaly. Typing unchanged; no marker move.] *[2026-08-06 registration —* ***`GNET.MARTINGALE.LIMIT.01`****,* ***[O]*** *(the analytic remainder), finite hypotheses measured in [`verification/v802_gnet_martingale.py`] (*$15/15$*,* MARTINGALE-HYPOTHESES-MET*): the Haar ladder is an exact filtration with* one *index-4 Watatani expectation for the whole tower (quasi-basis explicit,* $185/3041$ *elements at* $\ell = 2/3$*); the local-state defects are summable at measured rate* ~ 0.25 *per doubling (one power better than the* $2^{-\ell}$ *dim-law envelope; reported, not forced); the unique limit state follows by elementary telescoping and is verified on every rung; the split/normality precursors are uniform (determinant witness* ≥ 0.985*, stabilizing at* 0.9928*); the decimation control converges to the* wrong *degenerate limit — the coherent Haar bond is load-bearing.* The registered remainder (steps 5–7, typed, not executed): *GNS reconstruction of* ω_∞*; local normality on each* $A(W)$ *(the uniform S4 witnesses as the finite input); split property and the* μ_4 *simple-current extension (*`GATE.METRIC.11`*) on the limit net. The open QGEO majorant residues (b)/(c) are bypassed for limit existence and relocated into the normality step — a reallocation, not a proof. Kill: non-summable defects at a deeper reachable rung, or a determinant-witness collapse. No marker move.]*

The hull is the graded module of the Q-system ([`verification/v128_graded_hull.py`]). The Q-system is not merely an abstract companion of E_8 — E_8 *realises* it **[E]**: the 240 roots, built as explicit norm-2 vectors in $D_5 \oplus A_3$+glue coordinates, decompose as $240 = 52 + 64 + 60 + 64$ over the glue $\mathbb{Z}_4$ (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root, $\mathrm{coset}(r_1 + r_2) = \mathrm{coset}(r_1) + \mathrm{coset}(r_2) \bmod 4$:

$$\boxed{E_8 \text{ is a } \mathbb{Z}_4\text{-graded Lie algebra over the carrier; the grading group is the glue} = \text{the Q-system}}$$

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that E_8 already *carries*. [`verification/v125_glue_qsystem.py`]

Graded Frobenius ([`verification/v143_graded_frobenius.py`]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside E_8 **[E]**: (i) *coset duality* — $\mathrm{coset}(-r) = -\mathrm{coset}(r) \bmod 4$ for all 240 roots ($-C_1 = C_3$ bijectively, C_0, C_2 self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing $g_k \times g_{-k} \to \mathbb{C}$; (ii) the glue character average $\frac{1}{4}\sum_k i^{k\,\mathrm{deg}}$ is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index $|\mathbb{Z}_4| = 4$); (iii) each glue sector is *one* $W(D_5) \times W(A_3)$ orbit with multiplicity-one dimensions $64 = 16 \cdot 4 = (\mathbf{16}, \bar{\mathbf{4}})$, $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$ — simple currents, sector ring $= \mathbb{Z}_4 =$ the $\mathbb{C}[\mathbb{Z}_4]$ fusion, halfway subgroup $\{0, 2\} = SO(16)_1$. *R2 residue [C] (final form):* exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

Where that statement lives: the NS/R sector census ([`verification/v148_fock_census.py`]). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* ($D_5, SO(6)$) weights, and integer weights lie in the even discriminant classes — so the NS module supports only $\{(0,0), (2,0), (0,2), (2,2)\}$: **the odd glue classes $k(1,1)$ — the**

actual μ_4 simple currents — have zero NS support; they are twisted (Ramond-type) modules. The Ramond zero-mode module ($2^8 = 256$ ground states) carries exactly the four odd class pairs, 64 each, and splits $128 + 128$ into the odd sectors of the *two* Lagrangian glues $\langle(1,1)\rangle$ and $\langle(1,3)\rangle$ (v92): *choosing the glue is choosing the Ramond projection* (128 = one $SO(16)$ spinor chirality), and E_8 assembles as $248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$. *Consequence [C]:* the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census). *[2026-08-03 update, the NS/R grading acquires an arithmetic home:* [`verification/v722_phys_ramified_ns_r.py`] (9/9, `RAMIFIED-SEES-NSR`) shows the NS/R grading *is* the parity character of $E_8(\mathbb{Z}[i])/(1+i) = \mathbb{F}_2^4$: $(1+i)L$ lies in the even/NS sublattice (proved on a $\mathbb{Z}$-basis), the 240 roots fall into 15×16 classes of *pure* parity (7 NS + 8 R), and parity is a homomorphism $\mathbb{F}_2^4 \to \mathbb{F}_2$ — the Ramond projection is $(1+i)$-*adic*, at the one ramified edge of the $\mathbb{Z}[i]$-E8 Hecke tower (both controls fire: the $\mathbb{Z}[i]^4$ census is not 15×16, scrambled labels break purity). Together with the ladder index measurement ([`verification/v726_phys_car_pp_index.py`]; $\chi = 1 \leftrightarrow$ half-integer labels on both sides, the ramified defect in $\ker\chi$) this gives the groupoid reading of G_{net} its first *exact* witnesses; the gate typing is unchanged.]

Hecke from geometry on the glue census (`HECKE.GEOM.01`). The μ_4-refined E_8 class thetas that underlie the glue Q-system now carry an *in-suite* Hecke package (consolidated from discovery T27/T31/T32; 25 checks, $\sim 11\,$s). *(A)* Kneser p-neighbours realise #iso_lines $= \sigma_3(p)\,\#\mathbb{P}^3(\mathbb{F}_p)$ identically and at $p = 2, 3, 5, 7$ (135/1120/19656/137600). *(B)* The marked neighbour-sum is the affine Hecke element $\nu_p = a\,\mathrm{Id} + b\,T_p$ with cusp rule $a_p = b - \sigma_3$, recovering $a_3 = -4$, $a_5 = -2$ ($p = 7$ consistency $\lambda_{\mathrm{odd}} = 19840$). *(C)* The census form space has $\dim_{\mathbb{Q}} V = 7 = 5{+}2$ as the 2-adic oldform hull of E_4 and f_8, with recovery $\pi_{\mathrm{cusp}} = (28 - T_3)/32$ and newform via $(1 - V_2 U_2)$. Classical theorems named classical; the claim is the compiler identification (correspondence $\to$ operator $\to a_p$; redundancy $= \mu_4$/2-adic). Fence: both new systems stay weight 4 — GL(1)/weight-drop transport remains **[O]**; no RH claim; no marker moves. Machine-checked: [`verification/v535_hecke_from_geometry.py`]. **[E]** (mechanics) / **[O]** (weight drop)

Eichler trace layer at the E_8 lattice (`HECKE.GEOM.EICHLER.01`). Consolidated promotion of discovery T33/T36/T37/T42 (23 checks, $\sim 30\,$s; builds on `HECKE.GEOM.01`). Witt $\lambda_{\mathrm{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3)$ closed in p; Eichler $\lambda_{\mathrm{geom}} = \lambda_{\mathrm{Eis}} + a_p^2$ two-sided at $p = 3$ (live type-(i)/(ii): $352 = 364 - 12$) and $p = 5$ (T36 freeze: $3784 = 3906 - 122$) with assembler $R = \sigma_3 - 1 - c(p^2)/8$; Type-A/B densities $N_A = \min(240(1 + p^3), \#\mathrm{iso} - 1)$ ($p = 7$ live 82560/743040; injectivity $p = 11, 13$); signed $a_p = -c(p)/8 \Rightarrow (-4, -2, +24)$ and $b \in \{24, 124, 368\}$; $O(1)$ assembler for odd $p \le 31$. Classical scaffolding named classical; claim = in-suite two-sidedness + closed compiler densities. Fences: Shell(p^2) for $p \ge 7$ out of scope; fibre refinement scoped to $\{5, 7, 11\}$; weight-4 $\to$GL(1) stays **[O]**; no RH; no marker moves. Machine-checked: [`verification/v536_eichler_trace_layer.py`]. **[E]** (mechanics) / **[O]** (weight drop)

Half-integral bridge (`HECKE.GEOM.HALFINT.01`). Consolidated promotion of discovery T38/T41/T44/T45 (20 checks, $\sim 90\,$s; companion to `HECKE.GEOM.01`/`HECKE.GEOM.EICHLER.01`). Unique signed Shimura preimage $\mathrm{Sh}_{t=2,\psi=1}(g) = -8\,f_8$ in the 70-element compiler weight-5/2 theta-monoid (witness $g = \vartheta_2(q^2)^2\,\vartheta_3(q^2)\,\vartheta_4\,\vartheta_4(q^2)$; three further monoid elements lift to scales $+8, -16, +16$); exact $T(p^2)$-eigenform with $\lambda = a_p(f_8)$ for $p = 3, 5, 7, 11, 13$; $U_4(g) = 0$ and level $32 = 4 \cdot 8$ outside Kohnen 1982 (plus-route structurally closed — negative result part of the claim); AFE-based Waldspurger/Baruch–Mao quotient $R(d) = b(|d|)^2/(|d|^{3/2} L(f_8 \times \chi_d, 2))$ constant on ten fundamental $d \equiv 1 \pmod 8$ ($R = 23.1873585645\ldots$, spread $\le 10^{-12}$, including $d = 89$), with $d \equiv 5 \pmod 8$ vanishing via root number -1. Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical; claim = unique monoid preimage + in-suite chain + *measured* R. Fences: GL(2) centre $s = 2$ (not the ξ-line); no RH; `ZETA.HP.CARRIER` untouched; no marker moves. Machine-checked: [`verification/v537_halfintegral_bridge.py`]. **[E]** (Shimura/Hecke/Kohnen fence) / **[C]** (measured R)

Compiler relative-trace identity (`HECKE.GEOM.RTF.01`). Promotion of discovery T53

(18 checks, $\sim$ 14 s): `HECKE.GEOM.01`/`HECKE.GEOM.EICHLER.01`/`HECKE.GEOM.HALFINT.01` are three projections of one finite relative-trace identity [E_8/glue orbital counting] = [Newform/Waldspurger spectral sum]. *(R1)* $\mathrm{Tr}_V(\nu_p \circ \pi)$ bilateral independent at $p \in \{3,5,7\}$ across five projectors (anchors e.g. p=7: 727680 full / 19840 on f_8). *(R2)* Eichler $\lambda_{\mathrm{geom}} = \lambda_{\mathrm{Eis}} + a_p^2$ is the RTF orbit split ($\lambda_{\mathrm{Eis}} = 336/3780/19264$, $a_p^2 = 16/4/576$). *(R3)* Period side = quaternary lattice counting with $R = 23.1873585645$ (rel $\sim 10^{-12}$). Verdict `ONE-FORMULA`. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger named classical; NEW = compiler-native bilateral independence. Fences: identity *finite*; infinite RTF named open; no RH; not `ZETA.HP.CARRIER`-relevant; no marker moves. Machine-checked: [`verification/v538_relative_trace_identity.py`]. [E] (R1/R2) / [C] (R3)

Weil structure of the compiler family (`RTF.GNS.WEIL.01`). Promotion of discovery T55/T61/T62/T63 (25 checks, $\sim$ 6 s): the compiler family possesses a fully identified Weil structure up to two *explicitly isolated* obstructions (obstructions are load-bearing verified content, not residual footnotes). *(A)* GNS = direct integral over the Gelfand spectrum of the character subalgebra (fibre-orthogonal; twist-mix per fibre $\leq 5.037 \times 10^{-16}$). *(B)* Trivial Sato–Tate isotype = GL(1) core $G_0 = (1+Y)/(1-Y) = \zeta_p(w-3)^2/\zeta_p(2w-6) = \sum 2^{\omega(n)} n^{-u}$ globally. *(C)* Exact linear relation $Q_{\mathrm{fam}} = 2Q_\zeta(g) - 2Q_\zeta(g_\flat) + \mathrm{Arch} + \mathrm{Corr}$. **Obstruction 1:** minus sign on the doubling term (family positivity does *not* imply $Q_\zeta \geq 0$). **Obstruction 2:** Corr is non-automorphic (no polar/arch absorption; no finite Euler factor; $e^{-\sum p^{-u}}$-type). Finite-class $Q_{\mathrm{fam}} \geq 0$ measured [C]; dense-class positivity open / RH-adjacent — *not* claimed. Classical Weil/Gelfand/SU(2)/Sato–Tate/2^ω/GNS named classical. Fences: not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. Machine-checked: [`verification/v539_weil_structure_family.py`]. [E] (A/B/C/obstructions) / [C] (finite-class positivity)

Amplitude route and positive linear carrier (`RTF.GNS.AMP.01`). Promotion of discovery T67–T72 (34 checks, $\sim$ 3 s): the route out of the square plane behind the two `RTF.GNS.WEIL.01` obstructions. *(A)* Amplitude Dirac $D = \left(\begin{smallmatrix} 0 & V \\ V^\intercal & 0 \end{smallmatrix}\right)$ with $D^2 = \mathrm{diag}(VV^\intercal, K)$, $V^\intercal V = K$ (rel 2.7×10^{-16}), exists exactly and is Hecke-equivariant ($p = 3,5,7$ intertwining exact; Shimura $b(dp^2) = b(d)(a_p - \chi_d(p)p)$ on 644 pairs). *(B)* Geometric polarisation $b = N_+ - N_-$ exact; $\Theta = N_+ + N_-$ is a *pure* Siegel–Weil Eisenstein σ_3-eigenform ($p \leq 13$, cusp component zero) with Cohen seed $\Theta(d) = -48\, L(-1, \chi_d)$ exact-rational on 159 live d (Cohen 1975). *(C)* Every coefficient bilinear form inherits the even-k deletion (Cauchy–Littlewood window lemma, five channels; numerator $1 - p^6 X^2$ pair-independent); the deletion is exactly the square-class double counting ($\mathrm{fam} + 2\flat + \delta_{k1} = [2,2,2,2]$; Möbius square sieve exact). *(D)* Positive linear carrier $\ell^2(d, \mu)$, $\mu = 48\,|L(-1,\chi_d)|\,|d|^{-a}$: full weights $\lambda_k = 1 + p^{3k} - (\chi p)^k$ (layer [1,1,1,1]) and plus balance $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$ (rel 2×10^{-15}); the $\flat$/doubling kernel enters *with plus* (sign inverted vs the square plane). *(E)* Completed FE $\Lambda_\Theta(s) = 8^{1-s}\Lambda_{\Theta^\dagger}(5/2 - s)$ exact (rel $\sim 10^{-40}$; Fricke closed); the guaranteed cone is FE-self-dual. **Open boundary as content:** Euler-region positivity only (edge L-values, not the central line); the residual distance to the Weil cone is the FE-covariant gap functional λ^* on $n \equiv 6 \pmod 8$ (one exact Farkas witness; no finite signed theta library removes it). Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Cauchy–Littlewood / Shintani / Weil 1952 / Farkas named classical. Fences: not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. Machine-checked: [`verification/v540_amplitude_linear_carrier.py`]. [E] (A–E exact) / [C] (sampled cone facts; λ^* named open)

Matching lemma and transport ledger package (`RTF.GNS.LEDGER.01`). Promotion of the discovery proof package T78/T79/T80/T81/T82/T85 (33 checks, $\sim$ 10 s; every core check recomputed, not cited). *(A)* The matching lemma is *proved exact-integer on* $[4, 10^6]$: $7S(j) < 40A(j)$ at every atom (0 violations over 939 870 clash atoms; proven no-overflow; big-integer rechecks); exact margin $X = 22242575693/270725400000 = 0.082159$ at $j^* = 554400$; $\rho_{\mathrm{crit}} = 1.1440 > 21/20$; maximal-greedy identity $\Rightarrow$ uniform in (M, λ); structure laws (8-law, seed-tower, w-table, Cohen seeds $\{-48, -80, -8, -8\}$, bracket $\Theta \leq 2|\psi| \leq 3\Theta$) at 0 tolerance. *(B)* The transport ledger closes exactly: $Q_{\mathrm{Weil}} = Q_{\mathrm{cert}} + \Delta_{\mathrm{arch}} + \Delta_2$ with $\Delta_{\mathrm{pole}} \equiv \Delta_{\mathrm{conv}} \equiv 0$ *proven* (kernel collapse sympy-exact; residual 4.4×10^{-16} on 24 rows); the odd-prime side of the Weil functional equals the certified plus combination $P_\zeta(g_-) + P_\zeta(g_+)$ exactly (rel 3.9×10^{-16}). *(C)* The signed envelope is character-exact: $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$ (all odd $n \le 50000$, both builds); signed certificate 0 violations (margin 8.86×); confinement set equality zero-credit ⇔ coherent on 10^6. *(D)* The archimedean term is *internal*: Legendre duplication $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s{+}1)$; kernel identity $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$ pointwise (7.9×10^{-31}); $\Delta_{\text{arch}} = A_{\text{fam}} - A_{\text{shift}}$ at 9.9×10^{-16}. *(E)* The coherent class ($= \mathbb{Z}[i]$-norms, set equality) is closed by the λ-equivariant channel: CM carrier g_λ exact in two independent routes; $\mu_1 = c_1/(c_0 d^2) \in [-1, 1]$ exact; λ-window certificate 0.0653 vs 0.2365 unlifted (3.6×), exact Fraction rechecks. *(F)* Frontier facts [C]: coherent-chain crossing $k^* = 14$ ($\log_{10} N^* = 23.1$, exact Fractions); Robin tail factor 6.16 — the constant route stays open. **Named limits as load-bearing content:** (i) *one* classically-shaped open lemma (correlated cancellation on credit-rich non-coherent tail atoms, uniform form; provably-shaped ≠ formal proof); (ii) I5 in one-family form ($Q_{\text{cert}} + \Delta_2 + A_{\text{fam}} - A_{\text{shift}} \ge 0$) as the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity $\iff$ RH (equivalence typing, no progress claim). Classical Gronwall / Robin 1983 unconditional / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Dirichlet / Mertens-AP / Landau / Legendre duplication / Weil 1952 / Alaoglu–Erdős named classical. Fences: no RH-evidence language; not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. Machine-checked: [`verification/v541_matching_lemma_ledger.py`]. [E] (A–E exact) / [C] (frontier facts; two named limits open)

The margin-chain identities of phase 2 (`PRIME.MARGIN.IDENT.01`). Promotion of the *identity/theorem block* of the discovery parts T128–T131 (44 checks, ∼0.2 s; every check recomputed on small frame-A windows). Scope fence, declared with the row: identities and *per-instance* theorems only — no fit, no graded floor, no Lanczos value, no uniform-in-zone statement. Battery: 24 seam instances ($n = 3 \dots 97$, $\rho = 1.25 \dots 4.00$; every inverted or diagonalised matrix ≤ 300), 43 graded/fine grid pairs and 400 randomised profiles; every identity is a residual against a *preregistered* tolerance plus a mutation control that must fail by $\ge 10^{-3}$ relative. *(1)–(2)* $\sum_i (1 - g_i) = t_l + t_r$ exactly (5.5×10^{-13}) ⇒ the κ weights are a probability vector and $\kappa_{\text{flat}} = 1$; the (T) four-term identity $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$ at 2.2×10^{-16} (two independent routes for τ). *(3)–(4)* $p_{N-1} = 2 - p_0 + 2E$ at 2.3×10^{-16} ⇒ the equivalence $2E > p_0 \iff p_{N-1} > 2$ (0 exceptions, both branches realised); the Abel/Hölder chain $\kappa \le 2 - p_0 + \frac{N-2}{N} \sum_j |\Delta p_j - \overline{\Delta p}| + (p_{\max} - p_{N-1})$ per instance (0 violations; the same chain *without* the curvature term is violated on 88/400 profiles). *(5)–(6)* Matrix-free two-scale assembly $= J^\mathsf{T} Q J$ (5.2×10^{-16}), graded overlap $= J$; the Céa/Strang defect identity $S_{\text{graded}} - S_{\text{uniform}} = R^\mathsf{T} X^{-1} R$ as a *matrix* identity with PSD defect on all 43 grids (2.1×10^{-14}) ⇒ the compression error is one-sided by construction. *(7)–(9)* The secular sandwich $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \le \lambda_{\min}(Q) \le \varepsilon/\|u\|^2$ on all 48 sections where $A > 0$ and $\varepsilon > 0$ are *checked* (width ≤ 1.4448) with Albert's sign equivalence in *both* directions; $S^{-1} > 0$ entrywise (measured 48/48) ⇒ sign-constant and simple ground vector (Perron–Frobenius), with a control where hypothesis and conclusion fail together; and $n_{\text{run}} \le n_{\text{cross}} + 1 \le s_2 + 2$ with $\sum_j |\Delta p_j - \overline{\Delta p}| = \text{TV}(P)$, 0 exceptions on a non-vacuous battery (s_2 up to 31). **Named limits as load-bearing content:** (i) per instance on *small* windows — nothing uniform in the zone index; (ii) the κ *law* is *not* claimed (T129 found it false in general), only the chain underneath it; (iii) positivity of the pole-free section at depth (the open M25d) is neither assumed nor extended and no floor value is carried; (iv) the Perron step is an *implication* with a measured hypothesis; (v) the defect identity says nothing about the defect's *size* (the open T130 exponent). Classical Abel / Hölder / Schur–Haynsworth 1968 / Albert 1969 / Céa 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion. Fences: zero-firewall AST-checked; no RH-evidence language; not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. Machine-checked: [`verification/v542_margin_chain_identities.py`]. [E] (per-instance identities; nothing uniform)

The lumped M-matrix pair of phase 2 (`PRIME.MMATRIX.IDENT.01`). Promotion of the *identity/theorem block* of the discovery part T136 (35 checks, ∼1 s; every check recomputed on small frame-A windows). Scope fence, declared with the row: matrix identities, per-instance theorems with *checked* hypotheses, and one *negative* result — no fit, no D-exponent, no uniform-in-zone statement, no floor value. Battery: 20 seam windows ($n = 4 \dots 59$, $\rho = 1.25 \dots 4.00$,

$h = 25\ldots299$; every factorised or diagonalised matrix ≤ 300) giving 39 matrices with positive off-diagonals, 312 shifted ladder rungs, 8 whitened two-grid rungs; every statement is a residual against a preregistered tolerance plus a mutation control that must fail by $\ge 10^{-3}$ relative. *(1)–(2)* With $L_\Delta = \mathrm{diag}(\Delta\mathbf{1}) - \Delta$ the Laplacian of the positive off-diagonal part, $A_B = A + L_\Delta$ is Stieltjes, row-sum preserving (1.6×10^{-15}) and $A_B \succeq A$ — an *upper* bound on $\lambda_{\min}(A)$, a floor never; the congruence $A = A_B^{1/2}(I - W)A_B^{1/2}$, $W = A_B^{-1/2} L_\Delta A_B^{-1/2} \succeq 0$, is a matrix identity ($4.4 \times 10^{-15}$) with $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$, and the *declared* equivalence $\lambda_{\max}(W) < 1 \iff A \succ 0$ is checked in *both* directions — the Loewner sandwich is circular by itself, which is the fence. *(3)–(4)* With $x = A_B^{-1}\mathbf{1} \ge 0$ and $A_B^{-1} \ge 0$ entrywise (Ostrowski 1937, *checked*), $\lambda_{\min}(A_B) \ge 1/\max_r x_r$ on all 39 rows (min slack 3.0×10^{-3}, sharpest ratio 0.9070) — one solve, one sign check, no eigenvalue; $(A - sI)_B = A_B - sI$ (2.5×10^{-16}) and $(A_B - sI)^{-1} \ge A_B^{-1} \ge 0$ entrywise (Berman–Plemmons 1979) $\Rightarrow$ the shifted ladder is *structurally empty* (0 exceptions of 312). *(5)–(6)* For the regular splitting $A_B = D_0 - N_0$ (all three hypotheses checked), Varga's $\rho(J) = \tau/(1 + \tau)$ holds at 4.8×10^{-15} ($\rho(J) = 0.7347$–0.9860), and the Collatz–Wielandt bracket at the anchor vector gives an *a-priori* $\rho(J) \le 0.9861 < 1$ with no eigenvalue anywhere, sharp on the measured gap by 1.0062–1.0531. *(7)* The k-scanned M18 majorant is a *valid* bound at every k (39 (rung, k) pairs, 0 violations) and does *not* reach the preregistered bar 1/2 on any rung (best 3.080) while the measured bad mass is ≤ 0.0120: a *negative* result, the localisation term alone exceeding the bar (0.542), and the bar is not moved. **Named limits as load-bearing content:** per instance on small windows, nothing uniform in the zone index, in D or in the gap; the T136 D-exponents (anchor $D^{-0.56}$, gap $D^{2.72}$, slack $D^{0.12}$) are fits and stay in the sandbox; (2) carries no floor and M25d stays open; (3) and (5) are implications with hypotheses checked at the instance; (4) kills a route and says nothing about the true $\lambda_{\max}(W)$. Classical Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Collatz 1942–Wielandt 1950 / Haynsworth 1968 named classical; Weil 1952 cited, never used as a criterion. Fences: zero-firewall AST-checked; not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. Machine-checked: [`verification/v543_lumped_pair_identities.py`]. **[E]** (per-instance identities) + **[X]** (one closed route)

The long-lag support structure of phase 2 (`PRIME.LONGLAG.SUPP.01`). Promotion of the *structure block* of the discovery part T137 (24 checks, ∼0.2 s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons). Scope fence: support and amplitude structure, one two-sided bracket, and one *death certificate* for a whole family of bounds — no fit, no decay exponent, and *no* structure bound that closes. *(1)–(3)* The kernel splits exactly as $c = c_{\mathrm{arch}} - c_{\mathrm{comb}}$ with $c_{\mathrm{comb}} \ge 0$, and the archimedean section alone has *all* off-diagonals ≤ 0 on 20/20 windows $\Rightarrow$ every positive off-diagonal of A is comb-generated (control: the full section carries $+2.0 \times 10^{-1}$); all 20555 positive edges have their Hankel index $M - 1 - r - t$ inside the atom-hat index set (per-window fraction exactly 1.000000) $\Rightarrow$ $\mathrm{supp}(\Delta)$ is a union of anti-diagonal stripes at prime-power atom positions (converse reported, *not* claimed); each stripe is a perfect matching (0 endpoint collisions), so its edge vectors are exactly orthogonal and L_Δ per stripe is a direct sum of 2×2 blocks. *(4)* $\Delta_{rt} \le c_{\mathrm{comb}}[M - 1 - r - t]$ on all 20555 edges (excess -1.9×10^{-6}, sharpest ratio 0.99998532) — no eigenvalue, no norm; the Toeplitz lag in place of the Hankel index fails on 20415/20555. The tighter one-atom refinement is *honestly undecided* at $h \le 300$ (the atom hats never overlap) and is not promoted. *(5)* For the lumped comparison $B = D_0 - N_0$ the anchor identity $q_J = 1 - \min_r 1/(d_r x_r) < 1$ (8.2×10^{-16}) needs no measurement; the Neumann partial sum to $K = 8$ is then an *entrywise* lower bound for B^{-1} *because the discarded terms are nonnegative*, and partial sum plus anchor-weighted tail an entrywise upper bound, on 39 rows. *(6)* Since Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are upper bounds for $\rho(|E|)$, $E = B^{-1}L_\Delta$, one rigorous *lower* bound decides the family at once: $\rho(|E|) \ge 1.3141 \ge 1$ on 20/20 rows (cheapest four members ≥ 1.5937), so no member can *ever* certify $\lambda_{\max}(W) < 1$ — while the *signed* $\rho(E) = \lambda_{\max}(W) \le 0.99995781$ on the same rows: the absolute value alone destroys the cancellation. **Named limits as load-bearing content:** per instance on small windows, no decay profile and no amplitude law (T137's decay measurements are fits); (2) is one inclusion; (4) bounds the amplitude and *not* $\lambda_{\max}(W)$, and the structure bound assembled from (2)–(4) does *not* beat the margin (T137: 0 of 35 windows) and is not promoted; (5) brackets the *lumped* comparison, not the target, whose entrywise inverse positivity at depth stays open; (6) closes a route, not the question. Classical Gershgorin 1931 / the Neumann series / Haynsworth 1968 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion. Fences: zero-firewall AST-checked; not "almost RH"; `ZETA.HP.CARRIER` untouched; no marker moves. Machine-checked: [`verification/v544_long_lag_support.py`]. **[E]** (per-instance structure) + **[X]** (one family closed)

`PRIME.HARDY.IDENT.01` — the Hardy-core identities of phase 2 (T140, T141). *(1)* The form lifting Gram $= CHC^{\mathsf{T}}$ with $C = \mathrm{diag}(\sqrt{\Delta})M$ (3.0×10^{-15}), hence $\mathrm{rank}(\mathrm{Gram}) \le h - 1$. *(2)* The exact finite-core reduction: the closed geometric covering kernel $K_{rs} = W([r \wedge s,\ r \vee s])$ equals $M^{\mathsf{T}}\Delta M$ (1.3×10^{-15}) and $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2}) = 1 - \lambda_{\min}(A, A_B) = \lambda_{\max}(\mathrm{Gram})$ on three independent routes. *(3)* $H = \mathrm{diag}(s) + L_N$ (4.7×10^{-16}). *(4)* The checkerboard split $\mathrm{Band}_b = D + A_{\mathrm{even}} + A_{\mathrm{odd}}$ entrywise exact with a three-step Weyl sequence. *(5)* The four Hardy identities — the K^+ Rayleigh form, $L_N = D^{\mathsf{T}}BD$ with the signed crossing kernel, $Q = B_+\mathbf{1}$ and $DKD^{\mathsf{T}} = L_\Delta$ on the interior nodes. *(6)* The closed Muckenhoupt quantity $A_{M0} = \max_k Q_k(\Delta\mathbf{1})_{k+1}$ by two routes. *(7)* The additive Hardy chain and the joint shape $\Lambda(H,Y) \cdot \Omega(Y)$ as *valid* upper bounds on 11/11 windows — dominance only. *(8)* As a negative typing, the sign-versus-absolute separation of the measured spreads ($1.36\times$ against $\ge 54.94\times$). **Named limits as load-bearing content:** per instance on small windows, nothing uniform in the zone index or in D; every exponent is a fit; (7) promotes validity and not size — neither shape beats the target, the additive shape is dead as a *shape* by its own exact Weyl floor and the joint shape is decided at the normalisation Ω; (2) is an identity and not a bound, so the zone-uniform discrete Hardy inequality stays *open*. Fences: Muckenhoupt 1972 / Bradley 1978 / Opic–Kufner 1990 / Gantmacher–Krein 1950–1960 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v545_hardy_core_identities.py`]. **[E]** (per-instance identities) + **[X]** (one route typed)

`PRIME.CAPCHAIN.IDENT.01` — the capacity-chain identities of phase 2 (T142–T145). *(1)* The capacity decomposition $K^{-1} = D^{\mathsf{T}}J^{-1}D + xx^{\mathsf{T}}/\mathrm{cap}$ with $J = DKD^{\mathsf{T}}$ (entrywise, 3.8×10^{-15}), $x = K^{-1}\mathbf{1}$, $\mathrm{cap} = \mathbf{1}^{\mathsf{T}}K^{-1}\mathbf{1}$, as a matrix identity ($1.4 \times 10^{-13}$) whose Dirichlet half is an *orthogonal projection* under the congruence — $\Omega = 1$ exactly (5.1×10^{-12}). *(2)* The exact capacity–Rayleigh identity for $1 - \rho(W)$: numerator/denominator assembly equals the direct quotient for every v (9.5×10^{-14}), eigenvalue reproduced under the declared 1/gap conditioning, inf property verified, the constant vector carried entirely by the capacity term. *(3)* Cholesky nestedness: $\mathrm{cap}_E([a, a+j))$ as a prefix sum of squared forward-substitution entries of one triangular solve (3404 interval capacities at 2.0×10^{-14}, exhaustive on the median window; the reversed ordering fails at $\ge 4.6 \times 10^{-1}$). *(4)* The whitening certificate $\lambda_{\max}(W) \le \kappa_{\mathrm{up}} = 1.2245$–$1.4236 \le$ Gershgorin 2.1213, direction-correct, refusing at the shrunken shift. *(5)* The T145 M4 split: the pointwise mass-half theorem ($\sum_k f_k^2 \le \psi^2$ exact), $\sigma_{\mathrm{tot}} = 0.2197$–$0.4425 < 1$ per window, licence 4 with $|R|$ and not R^+ (witness $[[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$, $3 > 2$, built in), the ordered sign-free chain, and S1$'$ *certified per window on both routes* (better route $c_0 = 2.5154$–4.2265 against Maz'ya's Dirichlet value 4), with Charikar's cited density constant bracketed by exhaustive enumeration. **Named limits as load-bearing content:** per instance on small windows, nothing uniform in the zone index or in D; the c_0 of (5) is *measured* at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open, unclaimed; Maz'ya's strong-type half is not used as a theorem anywhere — it is the thing the cycle transcribes. Fences: Maz'ya 1985 / Muckenhoupt 1972 / Fukushima–Ōshima–Takeda 1994 / Miclo 1999 / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v546_capacity_chain_identities.py`]. **[E]** (per-instance identities and certified inequalities)

`PRIME.LEVEL.LEMMA.01` — the level-lemma identities of phase 2 (T146). *(1)* The union identity for nested cakes: $\sum_{j,l} c_j c_l |S_j \cup S_l| = 2T\|\psi_t\|_1 - \|\psi_t\|^2$ with $T = \sum_j c_j$ and $\psi_t = \sum_j c_j \mathbf{1}_{S_j}$ — form-independent and $O(m)$, exact (0.0) against the $K \times K$ double sum and against union sizes computed with no nestedness used (controls: one shuffled level set 3.2×10^{-2}, a 1% coefficient perturbation 4.8×10^{-3}). *(2)* The layer-cake counting lemma at a general base: $G(b) \le 2b^2\|v\|_\infty\|v\|_1/\|v\|^2+\varepsilon$ for *any* vector at all 6 preregistered bases; the classical dyadic 8 at $b=2$ exactly, the base free (gain $4/b_*^2 = 3.7612$; the one-level-down cake breaks the domination on every draw). *(3)* The resolvent delocalisation bound: $\psi = \lambda R\psi$ (9.0×10^{-14}, an identity — no Davis–Kahan transfer anywhere) $\Rightarrow \|\psi\|_\infty \le \lambda_{\mathrm{up}}\max_j\|Re_j\|$ with Rayleigh–Ritz λ_{up} (1.0592–1.2883) and residual-paid certified column norms, within 1.061–1.337 of the truth; the lever: the certified columns sit 2.85–10.17 below the trivial ceiling $1/\lambda_{\min}$, giving $\Gamma = 1.8356$–2.2797 against the useless $\sqrt{m}$. *(4)* The composite level lemma: $c_0^{\mathrm{ap}} = 2b^2\Gamma\min(1,\Gamma_1)+\varepsilon = 3.9042$–$4.8488$ (slack 1.986–2.418 over the measured constant, under the ceiling $16 = 4\times$ Maz'ya's Dirichlet 4), the certified window inequality cert_lam_max$(R) \le c_0^{\mathrm{ap}}\Psi_{\mathrm{abs}}$ on 12/12 windows, chain loss 0.0423–0.1272 of the exact gap; the built-in no-go discriminator ($R = aa^\mathsf{T}+\varepsilon I$, $a_i = i^{-1/2}$: Γ grows $3.676\to6.798$, ratio $1.85\ge1.5$, $c_0^{\mathrm{ap}} = 11.10$ exceeds every window value) fails exactly where it must, the theorem half surviving on every no-go size and the Dirichlet path-Laplacian control flat (1.0063). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the asymptotic delocalisation of the Green columns (T146's remainder) stays *open*, typed open; only the sign-free / Green constant is bounded a priori — the energy-route σ_{tot} stays measured; every T146 exponent is a fit, sandbox. Fences: Maz'ya 1985 (strong-type half not used as a theorem anywhere) / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Miclo 1999 / Davis–Kahan 1970 (cited, deliberately not used) / Perron–Frobenius (not applicable to E, not used) / Fukushima–Ōshima–Takeda 1994 / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v547_level_lemma_identities.py`]. **[E]** (per-instance identities and certified inequalities)

`PRIME.GREEN.SZEGO.IDENT.01` — the Green/Szegő identities of phase 2 (T147, T148). *(1)* The factorisation identity $\Gamma = \sqrt{m}\,\lambda_{\mathrm{up}}\max_j\|Re_j\| = \sqrt{Q^\star}\cdot S_w$ with $S_w = \lambda_{\mathrm{up}}\|R\|_F$ purely spectral and $Q^\star = m\max_j(R^2)_{jj}/\operatorname{tr}(R^2)$ purely geometric — exact at 2.2×10^{-16} on 12 frame-A windows ($m = 26\ldots285 \le 300$) and 7 random positive definite forms; the certified factorisation (residual-paid column norms, Frobenius bracket) dominates the true Γ everywhere ($S_w = 1.1186$–1.5777, $Q^\star = 2.0879$–2.8634; controls 5.0×10^{-3} / $\ge 10^{-3}$). *(2)* The bottom-block trace identity $\sum_j(\Pi_B)_{jj} = |B|$ exact (3.3×10^{-16}; $\theta = 10$, $|B| = 3$–5), hence $Q_B \ge |B|$ a theorem; the *measured* $Q_B/|B| = 1.5091$–1.7980 inside the preregistered band $[1, 2.2]$ against the Szegő/Widom sharp reference $2|B|$ (the rank-$|B|$ coordinate projector overshoots by ≥ 3.9). *(3)* The second-difference ℓ^1 ceiling: $|a_k| \le \min(1, \|s_k\|_\infty\|L_0^p\psi\|_1/\mu_k^p)$ for $p = 1,2$ against the true coefficients everywhere, the m-free closed form $\|a\|_1 \le 2\sqrt{\nu}+1$ with $\nu = (m+1)^{3/2}\|L_0\psi\|_1/(2\sqrt{2})$, and $m\|\psi\|_\infty^2 \le 2\|a\|_1^2$; Dirichlet calibration $\|a\|_1 = 1$ exact, $\nu\to\pi$ (controls: $\times10^{-3}$ fails, a one-index μ shift breaks the identity at 1.5×10^{-1}). *(4)* The layer-cake S_w certificate: completed LDL^T inertia counts equal the direct spectral counts on all 96 (window, rung) pairs (Sylvester 1852; Bunch–Kaufman 1977), and the layer cake certifies $S_w \le 1.3288$–1.8220 per window (true 1.1186–1.5777, slack ≤ 1.216; dropping the bottom layer breaks it everywhere). *(5)* The Liouville transform $m\|\psi\|_\infty^2 \le \kappa_\Lambda m\|\hat\varphi\|_\infty^2 \le 2\kappa_\Lambda\|a(\hat\varphi)\|_1^2 \le 2\kappa_\Lambda\operatorname{ceil}(\hat\varphi)^2$ with $\varphi = \Lambda^{-1/2}\psi$ and a-priori $\kappa_\Lambda = \max\Lambda/\min\Lambda = 1.3868$–$2.5805$, stepwise on every bottom-block mode; the weight-class discriminator on a model section with a genuine order-2 zero ($f(0) = 0$, $f''(0)/2 = 1$ exact), ladder $m = 64\ldots288$ at matched κ_Λ (mismatch 0.0013): the bounded-variation multiplier keeps ν flat (max/min 1.003) while the rough one (TV larger by 162.6) grows by 19.9 — the roughness, not the conditioning; the first step $\times10^{-3}$ fails provably (slack $\le \kappa_\Lambda^2$). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the one open lifting input ($\nu_L \le F(\text{form})$ with F m-free; equivalently bounded $\mathrm{TV}(\log\Lambda)$ or any weaker smoothness of the multiplier) stays *open*, typed open; the symbol-level order-2 hypothesis is *refuted* (T148: $f(0) < 0$ on every surface window) and enters nowhere as a theorem; σ_{tot} stays retired as a route; every T147/T148 exponent is a fit, sandbox. Fences: Kac–Murdock–Szegő 1953 / Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the mechanism's address, never an authority) / Sylvester 1852 / Bunch–Kaufman 1977 / Euler 1735 / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Davis–Kahan 1970 (cited, computed vacuous, not used) / Gershgorin 1931 / Wilkinson 1968–Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v548_green_szego_identities.py`]. **[E]** (per-instance identities and certified inequalities)

`PRIME.GAUGE.PARITY.IDENT.01` — the gauge/parity identities of phase 2 (T149, T150). *(1)* The gauge freedom: for *any* positive diagonal $\tilde{\Lambda}$ the whitening $\tilde{E} = \tilde{\Lambda}^{-1/2} A \tilde{\Lambda}^{-1/2}$ is an exact congruence, so $\lambda_{\min}(A) \geq \operatorname{cert} \lambda_{\min}(\tilde{E}) \cdot \min_j \tilde{\Lambda}_j$ is a valid certified lower bound for every gauge (an identity plus one Rayleigh step; family maximum valid) — verified on all 48 gauge assemblies over 12 frame-A windows ($m = 26 \ldots 285 \leq 300$); $\Gamma = \sqrt{Q^\star} S_w$ holds *gauge-invariantly* at 2.4×10^{-16}; the const gauge has $\mathrm{TV}(\log \tilde{\Lambda}) = 0$ *exactly* with the certified entrywise sandwich $\sigma = 1.3868$–2.5805 (controls: a 1% $Q^\star$ perturbation breaks at 4.99×10^{-3}; the completed-Cholesky floor refuses on the shifted indefinite form, 12/12). *(2)* The two-regime discriminator: the flutter smoother (moving geometric mean, half-width $m/16$) removes TV by $\geq 2.88\times$ while moving the Liouville-transported $\nu_{\tilde{L}}$ by ≤ 0.0088 — even the matched-amplitude rough re-gauge moves it ≤ 0.0046 (bar 0.01) — while the *untransported* functional jumps 0.41–9.76 under the same rough gauge: the roughness is in the form, not the multiplier (T149's withdrawal of the TV hypothesis, wired per instance). *(3)* The explicit zone diagonal $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s \neq r} \max(c_{|r-s|} - c_{M-1-r-s}, 0)$ at residual 5.9×10^{-16}; the exact split $c = c^{\mathrm{arch}} + c^{\mathrm{atom}}$ with $c^{\mathrm{atom}} \leq 0$ entrywise and the closed atom budget $\|c^{\mathrm{atom}}\|_1 \leq \sum_j \mu_j \leq 4B\sqrt{N}$, $B = 1.038821$ the verified maximum of $\psi(n)/n$ over the whole atom table (Chebyshev 1852, computed, never assumed; Abel summation); the archimedean section alone is *indefinite* on 12/12 (up to 5 negative eigenvalues, LDL^T) while the full section has zero — the atoms are co-responsible for the positivity, the additive perturbation reading structurally empty (control: dropping the positive part breaks at 4.25×10^{-1}). *(4)* The parity compression $U_-^\mathsf{T} T_M(c)\, U_- = T - H$ and $U_+^\mathsf{T} T_M(c)\, U_+ = T + H$ exact (3.9×10^{-16}, cross block 1.4×10^{-16}) on the 6 windows with $M \leq 300$: the reflection-odd section *is* the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, and the negative inertia of the sign-changing symbol ($f(0) = -29.2 \ldots -6.3 < 0$, re-read from T148, never re-derived) sits entirely in the *even* sector, the odd sector counting zero (control: the sign-flipped basis misses by 3.99×10^{-1}). *(5)* The parity-sine calibration: the parity sines are exact eigenpairs of the parity Laplacian (corner 3 forced by antisymmetry; eigen-residual 2.3×10^{-13}), $\nu_k^P = \pi k^2$ on the exact model at $m = 2001$ to 3.0×10^{-5}, and the per-window ladder $\nu_k^P \leq Ck^2$ holds with certified $C = 11.70$–17.28 and the derived per-mode ceiling $m\|\psi_k\|_\infty^2 \leq 2(2\sqrt{C}k + 1)^2$ (controls: $\times 10^{-3}$ fails, a one-index shift misses by 0.75). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the one open input after T150 (an m-free constant C in the odd-parity-sector ladder $\nu_k \leq Ck^2$ for a sign-changing-symbol Toeplitz section) stays *open*, typed open; $\mathrm{TV}(\log \Lambda)$ stays withdrawn as a hypothesis (T149); σ_{tot} stays retired as a route (T148); every T149/T150 exponent is a fit, sandbox. Fences: Kac–Murdock–Szegő 1953 (parity sector) / Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the parity sectors as the algebra's natural objects; the sign-changing symbol is exactly what Basor–Ehrhardt does not cover, and that gap is the open input) / Sylvester 1852 / Bunch–Kaufman 1977 / Chebyshev 1852 (verified on the table) / Abel / Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970 (cited, computed dead, not used) / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Gershgorin 1931 / Wilkinson 1968–Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v549_gauge_parity_identities.py`]. **[E]** (per-instance identities and certified inequalities)

`PRIME.ODD.SECTOR.IDENT.01` — the odd-sector identities of phase 2 (T151). *(1)* The grid step-over: the parity sines live on the *shifted* grid $\theta_k = 2\pi k/N$, $N = 2m{+}1$, which never contains $\theta = 0$, with $\mu_k^P = 2 - 2\cos\theta_k$ *exactly* (2.2×10^{-16}; eigen-residual 2.3×10^{-13}); the symbol has

$f(0) = -47.6\ldots-6.3 < 0$ (T148's honest negative, re-read, never re-derived) with a nonempty negative window $[0, \theta_c)$, and $\theta_c/\theta_1 = 0.371$–$0.485 \le 0.9$ on *every* window of the battery (12 frame-A windows, $m = 26\ldots285 \le 300$): the odd grid *steps over* the negative window — odd-sector positivity is a grid fact. The honest negative beside it, promoted as content (T150's rest item 3 answered negatively): the symbol-only pencil floor $\kappa_{\text{sym}}(1-\rho)$ is vacuous ($\rho \ge 1$) on 12/12 and the symbol moves by 0.45–3.27 relative across *one* mode spacing — the local model is a *matrix* statement, not a symbol statement (controls: the Dirichlet corner breaks the eigenpairs at 1.0, a one-index shift misses by 6.5×10^{-3}). *(2)* The certified bottom ladder: $\lambda_k(A) \le S\mu_k^P$ for $k \le 8$ by completed LDL^{T} inertia counts (Sylvester 1852; a certificate, never a sorted list), $S = 1.1019$–6.4725, the count equal to the direct spectral count on all 96 (window, k) pairs — order 2 in k, $f(0) < 0$ absorbed by the Rayleigh floor $L_P \ge \mu_1^P I$ (control: $S \times 10^{-3}$ certifies no eigenvalue below it). *(3)* The discrete Sobolev step: $\|\nabla\psi\|_2^2 = \langle\psi, L_0\psi\rangle - \psi_{\text{last}}^2$ exactly (7.6×10^{-15}), $\|\nabla\psi\|_2^2 \le \langle\psi, L_P\psi\rangle$, and $\|\psi\|_\infty^2 \le 2\|\psi\|_2\|\nabla\psi\|_2$ — licensed by the odd sector's virtual node $\psi_{-1} = 0$; calibration: the model pencil is $(1, 1)$ (certified 1.1×10^{-12} at $m = 241$) and the Sobolev per-mode price is π to 5.1×10^{-8} at $m = 2001$ — the honest, m-free cost of leaving the ℓ^2 world (controls: the boundary term 4.8×10^{-2}, the Neumann undercount 0.50). *(4)* The closed archimedean minimum (T150's rest item 2 closed *and attained*): given the two per-window-checked shape properties ($c_i^{\text{arch}} < 0$, non-decreasing on $i \ge 1$), $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}}$ exactly (residual 0.0; value 2.1719–3.4674), attained at $r = 0$ (control: the full atom-including kernel breaks the shape and misses by $\ge 8.7 \times 10^{-2}$). *(5)* The linear per-mode bound: with the certified pencil floor $\kappa = 0.1939$–0.3599 (one completed Cholesky per window, floor carried as fl/μ_1^P, direction re-read on the actual bottom eigenvalue) and $K_{\text{bot}} = S$, $b_k = 2m\sqrt{K_{\text{bot}}\mu_k^P/\kappa} \le C_S k$ with $C_S = 12.081$–$26.953 \le 2\pi\sqrt{K_{\text{bot}}/\kappa}$ on the bottom 8 modes of every window — *linear* in k where the quadratic ν ladder grew, every m-power cancelled, nothing additive anywhere; the bound dominates the truth and the parity Dirichlet energy on every read mode; the bottom pencil ratio $R = K_{\text{bot}}/\kappa = 3.7362$–$18.6607$ is the one scalar the chain still owes, typed *open* in m; the discriminator: on the T145 no-go form the ratio explodes ($1.686 \times 10^3 \to 2.667 \times 10^4$, factor 15.8 over $m = 64\ldots256$) while the exact parity model holds $R = 1$ to 1.1×10^{-12} (controls: the ladder $\times 10^{-3}$ fails, the inflated floor 1.5κ makes the Cholesky refuse on 12/12). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D: the one open input after T151 (the m-freeness of the bottom pencil ratio $R = K_{\text{bot}}/\kappa$ for the odd parity sector of a sign-changing-symbol Toeplitz section) stays *open*, typed open — its flat trend ($x^{0.037\pm0.015}$) is a fit, sandbox; the symbol-only route stays certified vacuous; $\text{TV}(\log\Lambda)$ stays withdrawn as a hypothesis (T149); σ_{tot} stays retired as a route (T148); every T151 exponent is a fit, sandbox. Fences: Kac–Murdock–Szegő 1953 / Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the address of the symbol question — the computed answer: the pointwise symbol language does not extend to the bottom mode) / Weyl 1912 / Parlett / Hardy–Littlewood–Pólya 1934–Agmon 1965 / Sylvester 1852–Bunch–Kaufman 1977 / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v550_odd_sector_identities.py`]. **[E]** (per-instance identities and certified inequalities)

`PRIME.RITZ.CEIL.01` — **the fixed-size Ritz ceiling certificate of phase 2 (T154).** *(1)* The direction lemma: for orthonormal Q and $W = Q^{\mathsf{T}}AQ$, $\lambda_k(A) \le \theta_k$ for *every* $k \le \dim$ — Ritz values are *upper* bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against the exact spectrum on the certificate subspace of every window and on 24 random subspaces (one-sided excess 0.0) — so the fixed-size ceiling needs *no* residual argument; Temple 1928 / Kato 1949 is a floor device and is used nowhere in the module (control: the reversed reading is recognised as false on 12/12 windows, θ_k overshooting λ_k by ≥ 18.6 somewhere on every window). *(2)* The sixteen-column certificate: $\text{span}\{t_1..t_8\} + A^{-1}L_P\,\text{span}\{t_1..t_8\}$ has fixed dimension 16, independent of m (one Green step by Cholesky back-solves, no inverse formed), and its ceiling $K^F = \max_k \theta_k/\mu_k^P$ (8 completed Choleskys of matrices $\le 8 \times 8$) equals 1.1019–1.5845 with $K^F \geq \max_k \lambda_k/\mu_k^P$ on every window and $|K^F/K_{\rm bot} - 1| \leq 9.9 \times 10^{-7}$ against the declared cap 10^{-4} on 12/12 ($K_{\rm bot}$ the inertia-certified true ladder constant, count = direct count on all 96 pairs): the fixed-size ceiling *attains* the truth, retiring the size-m LDL$^{\mathsf{T}}$ counts from the ceiling step (controls: the parity sines alone overshoot 41.6–739; corrupting the Green columns breaks the attainment on 12/12; Kaniel 1966 / Paige 1971 quoted as the mechanism, never used as a bound). *(3)* The one-direction anatomy (angles measured, Björck–Golub 1973): the median of the seven aligned angles between the bottom eight of A and the bottom eight of L_P is 0.03–0.14° while the largest is 83.79–89.70° on every window of the battery (the 12 deepest in-cap frame-A windows, $n = 29 \ldots 139$, $m = 96 \ldots 291 \leq 300$ — declared, not tuned) — the misalignment is carried by *one* nearly orthogonal direction, and that direction *is* the collapse price $\lambda_1(A)/(t\,\mu_1^P) = 4.408$–$7.042$, recovered in full per window by one completed Cholesky of $A - \gamma I$ (size m, certified, not fixed size, labelled as such; control: $A = L_P$ returns $K_{\rm bot} = K^F = 1$, zero misalignment, price exactly $1/t$). *(4)* The no-go discriminator: on the T145 reconstruction $c(l) = 1/(1 + l)$ the odd section is positive definite and the Schur floor survives ($t = 0.05$ on all 4 sizes) but $K_{\rm bot}$ explodes (×35.4) and the sixteen-column ceiling explodes *with* it (×57.5) over $m = 48 \ldots 288$ — no manufactured closure; the Courant–Fischer direction is violated nowhere (16/16); the Dirichlet control (the Hankel half removed, arithmetic kept bit for bit) stops being positive definite ($\lambda_1 = -5.60 \ldots -3.55$ certified on 3/3) — the reflection half is load-bearing for positivity. **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D; the attainment is exhibited per window and *not* promoted to a theorem (T154's no-go stress shows the certificate overshoots there by 1.12–2.19 — a property of the real section, not of the instrument); the two open inputs after T154 (R1′, an m-free floor for $\lambda_{\min}(B_{HH})$ on the bulk parity block; R2′, an m-free lower bound for the L_P floor on the complement of the eight bottom Ritz directions — arithmetic-free, worth 91–100% of the collapse price) stay *open*, typed open; every floor consumed is certified per window at size m; `PRIME.ODD.SECTOR.IDENT.01`'s ladder stays as stated and nothing is re-opened. Fences: Courant–Fischer 1920 / Cauchy 1829 / Kac–Murdock–Szegő 1953 / Schur 1917–Haynsworth 1968 / Sylvester 1852–Bunch–Kaufman 1977 / Weyl 1912 / Temple 1928–Kato 1949 (a floor device, not used here) / Kaniel 1966–Paige 1971 (quoted, never a bound) / Björck–Golub 1973 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v551_ritz_ceiling_certificate.py`]. **[E]** (per-instance theorems and certified inequalities)

`PRIME.WEIL.OPERATOR.01` — the localized-Weil-operator identification (research contract, [O]). The external operator shell for the $C = 1$ line (v618/v619), preregistered 2026-08-02 after the third external review. *Named reference frame (verified citations):* Suzuki, *Weil's quadratic form via the screw function* (`arXiv:2606.09096`, June 2026) realizes the localized Weil quadratic form on every interval $[-a, a]$ as a canonical self-adjoint operator A_a via the screw function, proves the continuity of the lowest eigenvalue λ_a in a without assuming RH, and presents Yoshida's criterion: $Q_W(v) > 0$ for all nonzero *odd* $v \in C_c^\infty(\mathbb{R})$ already implies RH; a first numerical realization is `arXiv:2607.24830`. *The lemma chain:* **(W1)** exact operator identification — the frame-A window bilinear form $\widehat{A} = B - S$ is the Galerkin matrix of the odd restriction of A_a on the declared piecewise bases, exactly or up to an explicit rank-one transformation and a known normalization (no asymptotic "looks similar"); **(W2)** form density — the nested window spaces are form-dense in the odd form domain of A_a with Mosco/norm-resolvent convergence of the Galerkin forms; **(W3)** uniform discrete positivity — $\widehat{A}_{a,h} \succeq 0$ for every a and every refinement h (the RH-hard step; the $C = 1$ contraction reading of v618 — the normalized arithmetic residue as an operator of norm $\leq 1/h$ — is the named candidate mechanism); **(W4)** passage to the continuum — W2 + W3 give $Q_W^a(v) \geq 0$ on all odd test functions for all a, hence RH via the localized criterion. *Kill tests:* (K1) the W1 identification fails beyond a window-dependent fit; (K2) a complete, numerically stable fresh window stably exceeds the frozen $C = 1$ bound (the v619 complete-comb surface is the definitional domain); (K3) the odd-sector restriction loses the explicit-formula bookkeeping. *Fence:*

a research contract, not a claim — no RH statement, no positivity claim beyond the measured surface; the zero-side representation stays diagnostic (a proof must run prime-side or through the screw function, avoiding the critical-line circularity). Status: **[O]** (preregistered; the audit and the citation verification are machine-checked in [`verification/v624_external_lattice_audit.py`]). *First executed slice (2026-08-02):* the *atomic half of W1 is closed* — the TFPT atom table *is* the prime measure of Suzuki's screw function, literally (positions log prime powers, weights $\Lambda(n)/\sqrt{n}$, exact; tent projection verbatim), and the first numeric Galerkin comparison measures a *non-scalar* conversion profile ($-0.064..-0.070$) — the preregistered rank-one/normalization freedom, now with data ([`verification/v630_suzuki_contact.py`]). On the geometric side the Hodge-chamber question is *measured*: all 67 complete windows land in the positive cone of the cover polarization lattice, on one sheet ([`verification/v627_hodge_chamber.py`]); P-canonicity and the chamber theorem stay open. *Second slice (2026-08-02, same day):* the non-scalar residual is *resolved* — it IS the zeta pole term. The Lerch collapse $[e^{-t/2}\Phi(e^{-2t},2,\frac{1}{4})]'' = 4e^{-t/2}/(1-e^{-2t})$ (exact) gives the structure theorem $g''(t) = -2\cosh(t/2) - 4\,e^{-t/2}/(1-e^{-2t})$ off the atoms (in the $\tilde{g}$-normalization of this chain; see the fourth-slice erratum below — Suzuki's own smooth layer is $+\rho$): the chain-kernel smooth layer is $-4\times$ the TFPT arch density *minus* the pole block $2\cosh(t/2)$ (the $s = 0, 1$ pole weights — TFPT's separate rank-one, `v591`). The closed per-lag ratio law $-D[4 + (e^t{+}1)(1-e^{-2t})]$ is verified, and after pole subtraction the smooth conversion is the *scalar* $-4D$ (monotone to 1.0006); the atom-layer lattice constant is D^2 exactly. **W1 status: closed at the measured level on both halves** — $\widehat{A}_{\text{arch}} = -(1/4D)\,\text{Galerkin}_{\text{smooth}} + \text{pole rank-one}$, with the typed remainder (second-order discretization term, $d \le 2$ boundary cells, L^2_0 projection) named for the theorem-level statement ([`verification/v631_w1_dictionary.py`]). *The implication map stays explicit:* W1 (now measured-closed) $\to$ W2 (form density, OPEN) $\to$ W3 (uniform discrete positivity, OPEN — the RH-hard step) $\to$ W4 (continuum passage, classical given W2+W3). Closing W1 does not move W3; no step is silently upgraded. *Third slice (2026-08-02, same day):* the typed W1 remainder *closes*, on three fronts. **(i) Boundary closure, exact:** the duplication identity $\psi(\frac{1}{4}) = -\gamma - 3\log 2 - \pi/2$ and the scheme conversion $k = \pi/4 + \frac{3}{2}\log 2$ are sympy-exact, so the two δ_0 conventions are one object in one scheme, and the exact boundary equation $A_{\text{arch}} = \frac{1}{4}g''_{\text{smooth}} - \frac{5}{4}(\log\pi - \psi(\frac{1}{4}))\,\delta_0$ verifies at relative residuals $< 10^{-22}$; the $d \le 2$ cells become three computable constants and the second-order term is the derived, window-independent moment law $1 + 1/(6d^2)$ ([`verification/v640_w1_boundary.py`]; the exact dictionary identities additionally machine-proved in Lean, `WeilDictionary.lean`). **(ii) Portability:** the frozen dictionary (closed formulas in D, no knob) executes unchanged on three fresh windows $h = 285/540/997$ — atom constant D^2(window) at machine precision — and the preregistered kill criterion is met nowhere ([`verification/v641_w1_portability.py`]). **(iii) Matrix level:** the full quadratic form closes at the operator level (16-mode block norm distance 2.7×10^{-3} fully derived, 6.3×10^{-6} after measure-level atom re-binning; the separated pole block numerically rank 1); the literal per-vector 1% bar fails only on two *typed* lattice residuals (near-cancellation denominators; tent-vs-B-spline atom profiles), certified as typed-residual checks ([`verification/v642_w1_matrix.py`]). *Fourth slice (2026-08-02, second round of the day): erratum + theorem + W2 start + W3 tool diagnosis.* **Convention erratum (corrected the same day):** the second and third slices read Suzuki's eq. (1.3) with Lerch coefficient -1; the paper's own §2.2 data lock $+1/4$ ([`verification/v643_w1_theorem.py`], C0.1). All slice-two/-three identities are correct identities of the chain kernel $\tilde{g} = g - \frac{5}{4}$ Lerch, and every measured number transfers *verbatim* via the exact identity $c^{\text{sm}}_{\text{gal}}(\tilde{g}) = -4\,c^{\text{sm}}_{\text{gal}}(g)$ — only the labels change: Suzuki's own smooth layer is $+\rho$, the two-layer dictionary collapses to the *single* scalar $+1/D$ on both layers (sign-compatible with positivity), and the origin constant vanishes, $\kappa = 0$ exactly (the $-\frac{5}{4}L\,\delta_0$ term was an artifact of the misreading; the `v563` near-cell scheme *is* Suzuki's origin bookkeeping). **W1 is THEOREM-closed at the measure level** ([`verification/v643_w1_theorem.py`]): the projection lemma closes the last remainder — Suzuki's L^2_0 mean-zero condition is *automatic* on the u-side ($D : H^1_0 \to L^2_0$ a bijection), the window parity sector is a subspace of the form domain, $A_{\text{arch}} = -g''_{\text{smooth}}$ exactly at every lag (3.4×10^{-52}), and the full form equality holds at 1.28×10^{-10} on the common odd sector. **W2 is started** ([`verification/v644_w2_form_density.py`]): the classical FEM-density slice is verified at rate (with Q_W H^1-bounded and H^1_0 a form core the density chain is classical, cited); Rayleigh–Ritz on the nested window spaces is monotone from above on all six sequences, and the λ_a scan is consistent with Suzuki Thm 1.3 ($\lambda_a = 0^+$ within $\sim 10^{-9}$, no sign statement); the Mosco/norm-resolvent remainder is *named*, not claimed. **W3 stays open, now with a tool diagnosis** ([verification/v648_sign_uncertainty.py]): the sign-uncertainty / LP-ceiling toolbox has a real, 25-digit-anchored dictionary to the critical strip (the $d = 1$ strip multiplier's phase slope is exactly $L = \log \pi - \psi(\frac{1}{4})$, the W1 δ_0 weight), but its quantitative mechanism is powered by the strip half-width and dies at $d = 1$ ($e^{-E^*} = 0.967$ where $< \frac{1}{2}$ is needed; $d \geq 21$ would be required; the window family's negative structure sits 5.2× outside the maximal BCK ball): a BCK-type mass obstruction cannot certify $\widehat{A}_{a,h} \succeq 0$ — any W3 route through this toolbox must replace the dimension lever by a different large parameter; empirically the W3 surface is positive on all 67 complete windows ($\min \lambda_{\min} = +8.26 \times 10^{-4}$, float-floor certified). **W4 unchanged**; no RH statement. *Fifth slice (2026-08-02, the closing bundle): the consolidated W2/W3 state.* **W2:** the envelope inequality holds *measured* a-uniformly — the total window symbol grows sub-log with family-uniform constants $s_{\text{tot}} \geq c_0\, w^*(t) - C_0$, $(c_0, C_0) \approx (0.021, 0.055)$; the earlier "flat" read was a t-range artifact ([verification/v663_garding_envelope.py]); the Mosco/norm-resolvent preparation stands — the resolvent family is Cauchy with uniformly bounded $H_{\log}$ norms, and the eigen-scale collapse is typed: W2 routes must run through resolvent observables ([verification/v655_w2_mosco.py]); the two named lattice remedies for the Gårding drift are *refuted* — the drift lives in the total symbol, not in the lattice edge ([verification/v661_garding.py], [verification/v662_garding_edgeband.py]) — and the drift itself is *intrinsic* to the positivity problem, not Suzuki-frame-made: the external Baez–Duarte control frame shares it ([verification/v667_baez_duarte.py]). The remaining W2 steps are a Fejér spectral-density bound replacing the per-mode symbol read, plus the symbol-level proof. **W3:** reduced on the measured surface to *lock sign plus the uniform bound* $q \tan^2 \theta \leq 1 - \delta$ via the exact 2D formula $r_{\text{id}} = (1 - q \tan^2 \theta)/(1 - \tan^2 \theta)$ ([verification/v657_rid_alignment.py]; the margin-bridge correlation typed as a lock identification, [verification/v656_margin_link.py]); the resonance landscape is mapped *without positivity loss* — $\lambda_{\min} > 0$ on all 635 surveyed points, every $P > 1$ excursion a rotation artifact ([verification/v659_w3_landscape.py]; the risk survey [verification/v658_w3_uniform_bound.py]); the rotation predicate is typed honestly (necessary, not sharp; successor: block deflation; [verification/v660_theta_predicate.py]); and the *ground-truth recalibration* lands ([verification/v668_ground_truth.py]): on a proven RH analogue true positivity has *no uniform margin* — $\delta(K) \to 0$ exactly beyond the support-resolution depth — so shrinking margins on deeper windows are the *expected* behavior of a true positivity, and the alarm signal is only $\lambda_{\min} < -$floor, which has never been observed; conversely the machinery breaks by $\sim$13 orders of magnitude the moment the Euler product is removed (the Epstein firewall), so the family positivity read does carry arithmetic information; the needles are frame-side ([verification/v667_baez_duarte.py]). *Kill criteria K1–K3 unchanged; no step silently upgraded; no RH statement. Sixth slice (2026-08-02, the Fejér-density round): the density plane closes; the needle search narrows threefold; the Li corollary.* **W2 — the density plane is closed** ([verification/v669_fejer_density.py]): the total window symbol *is* the Fejér-smoothed zero-counting density — the exact identity $s_{\text{tot}}(\cdot; a) = 2\pi(F_a \star dN)$ (Weil explicit formula on the tent test pair, verified 2.9×10^{-6} pointwise / 7.2×10^{-8} t-averaged against the Turing-certified comb) — and the *unconditional* chain RvM + Trudgian yields $\rho_{a,\delta}(t) \geq 0.306 \log(2 + t) - C_0$ a-uniformly at $\delta \in \{4\pi, \pi a\}$ with 10/10 finite certificates on the zero-free grid (min margin +1.36). The honest floor: the literal $1/a$ and the plane-wave π/a widths sit *below* the RvM threshold $4\pi A_1 = 1.4074$ — per-mode symbol control is not certifiable this way at any height (the structural explanation of the Gårding drift), so the remaining W2 piece is the packet-to-point translation below the floor. **W3 — the needle search is narrowed threefold:** the needles are *not low-rank* (the $(1{+}K)$-block deflation fails honestly: leak median 1.000 at every $K \leq 8$, fidelity $p_{90} \approx 49$ř; [verification/v670_w3_block_deflation.py]), *not zero-comb-driven* (the Lehmer-pair link is a clean null: h^3-partials all $p > 0.1$, placement control unremarkable; [verification/v671_lehmer_resonance.py]), and *not in the BD frame* ([verification/v667_baez_duarte.py]) — the mechanism search moves frame-/assembly-side. **New — the Li corollary and the E_4 lock:** the W1 window positivity *contains* 30/32 finite Li coefficients ($g_n = \frac{1}{2}e^{-|u|/2}L^{(1)}_{n-1}(|u|)$ derived exactly; band $2..4 \cup 6..32$ on $h = 1433$, pole-free hyperplane, odd sector certified by $\lambda_{\min} = +1.516 \times 10^{-3} > 0$; [verification/v672_li_corollary.py]), and the E_8 shell data and the ζ comb agree in *one* Li sequence ($\Lambda(E_4, s)$ completed with residues $\pm 1/240$ — the E_8 240 is the residue normalizer; $\lambda^L_n = 2\lambda^\zeta_n$ exact; budget usage 0.124; [verification/v673_li_e4.py]). *Kill criteria K1–K3 unchanged; no step silently upgraded; no RH statement. Seventh slice (2026-08-02, the consolidating contraction): the W2 end-state; the W3 contraction to the carrying form; two theorems and the honest typing of the wall.* **W2 — the end-state:** the density plane is closed ([verification/v669_fejer_density.py]); the Gårding statement now stands in *frame form with theorem constants* ([verification/v674_packet_garding.py]: the frame inequality $Q + C_0G \succeq c_0Y$ holds at every M with the v669 constants rebuilt zero-free, $c_0 = 0.3058$, margins $+1.52..+2.29$, a-uniform — while the packet *projection* norm honestly does not escape the $1/\log$ drift: typed residual with inverted expectation, the minimizer single-mode tight at 0.90); the Mosco mechanism is numerically complete (8/8 sequences, same module); the named remainder is *within-packet equidistribution* ($\theta \to 0.20$) = unconditional zero-gap information. **W3 — the contraction to the carrying form:** after the needle mechanism search *saturates* (all four assembly-side candidates miss: lag-quantization weakly significant $p = 0.007$ but too dense, phase gradient wrong sign, lock rotation cleanly killed, pole weight significant $p = 2 \times 10^{-4}$ but inverted — the needles are fourfold characterized as an emergent bulk/frame property; [verification/v675_needle_mechanism.py]), the contract W3 form becomes *"MARGIN-regime generic bound + needle risk map"* rather than a uniform bound with an exception predicate. The candidate mechanism $C = 1$ is demystified *and* grounded as a *quadrature theorem* ([verification/v676_c1_mechanism.py]: q_r = the exact zero-side reading of the lock profile, median residual 1.1×10^{-2} on 67 windows; h^{-1} = the DST normalization; the constant = the zero-free RvM density integral at ratio 1.005 — *no RH lever beyond the density, but computable*; $\sup \le 1$ is not implied by the mean, the $\sin^2$ sampling budget is the precise remainder with factor-2 headroom; the v596 exponent corrected: -1.01 was flip-contaminated, clean -1.28). And the *structure theorem* ([verification/v677_w3_structure_theorem.py]) is the exact identity of the rung: window positivity $\Leftrightarrow$ alias positivity of the zero symbol (Cantoni–Butler compression + sandwich + master identity with $T_x \ge 0$ on the line) $\Leftrightarrow$ an off-line detector whose Epstein-lab census (exactly 12 off-line zeros, ζ_K control 0) predicts the form break at factor-2 level (ratio 0.803), with the threshold map $2\alpha\, s_{\min} \approx 1.4..2.6$ (the Ihara echo) — so *W3-on-the-family* is a theorem plus a detector certificate, while *uniform* W3 over all windows is the conjecture itself (Weil 1952 / Yoshida 1992, RH via the localized criterion), typed as such and *not* an RH statement. *Kill criteria K1–K3 unchanged; no step silently upgraded; no RH statement. Eighth slice (2026-08-02, the zero-gap round): the W2 end-state, with the last gap carrying a number.* **The frame chain is now fully unconditionally grounded:** the best explicit unconditional zero-gap theorem ([verification/v678_zero_gap_theorem.py]: the S-difference route with $S_{\text{bound}} = \min\{\text{Platt } |S| \le 2.5167 \text{ to } 3.06 \times 10^{10}; \text{Trudgian 2014; Bellotti 2025 Cor. 1.5, asymptotic floor } 4\pi\, 0.10076 = 1.26619\}$, machine-verified on all 1999 Turing-certified comb gaps with 0 violations and min air 2.05×) makes the v674 frame packet floor *unconditionally* positive: adaptive bands $\delta_p = \kappa H_{\min}(b_p)$ hold $\ge \kappa$ ordinates per band unconditionally, and the theorem-shaped floor $0 < B_p \le V_p$ holds on every packet (median worst-case discount 0.037) — zero-gap theorem + Fejér chain ([verification/v669_fejer_density.py]) ground the entire frame-Gårding statement in cited unconditional inputs. **The last gap (pointwise/projection form) carries a concrete number:** the packet projection ladder does not stabilize under adaptive re-partitioning either (typed residual with inverted expectation: 6/6 (κ, C) cells miss, the minimizer stays single-mode at 0.898 — the obstruction is re-partition-invariant), and the pointwise main-lobe capture of a guaranteed zero *closes iff the explicit $S(t)$ constant A_1 improves from 0.10076 (Bellotti 2025) to $< 0.09017 = 1/(4a_0)$ — an 11.7% improvement of a named constant at the frontier of explicit analytic number theory, independent of RH*; the Platt branch bottoms out at $1.4183 > \pi/a_0 = 1.1331$, so no current chain reaches it at any height. This is the honest end-state of the W2 packet route: an unconditional frame form, a quantified external research target for the pointwise form, and within-packet equidistribution as the alternative input (W3/W4 territory). *Kill criteria K1–K3 unchanged; no step silently upgraded; no RH statement. Ninth slice (2026-08-03, the pinch-break and hole-closure round): the W2 end-state — the pointwise map at the anchor is gapless.* **The pointwise density floor at the anchor window is now a gapless split-type map** (comb-certified / scan-certified / abstract-Weil / unconditional-asymptotic), closing the 11.7% pinch and the (2500, 7.27×10^6) hole. **The pinch was a bookkeeping artifact** ([`verification/v680_pinch_attack.py`]): centered capture — the same zero-gap theorem spent two-sided, $\mathrm{dist}(t, Z) \le H_{\min}(t-26)/2$, comb-verified on 6595 grid points with 0 violations — *doubles* the threshold to $A_1 < 1/(2a_0) = 0.18034$, so the existing Bellotti constant 0.10076 passes with 79% headroom; and the Beurling–Selberg minorant *eliminates* A_1 from the counting chain entirely (the loss is exactly the π/a_0 minorant mass plus the explicit prime term $P = 2.534$; positive from $t^* = 1.11 \times 10^7$, every window including $a \to \infty$; Weil identity on the comb at 9.0×10^{-5}). **The hole is closed** ([`verification/v681_coverage_hole.py`]): the exact prime term with a hierarchical Lipschitz certificate moves the abstract entry down a factor 72.5 to $t_x = 1.53 \times 10^5$, and a budget-certified Riemann–Siegel scan (223 949 zeros, Gabcke remainder, correctness 603/603 against the certified strip, found/expected 0.99774) carries $(2515,\ 1.56 \times 10^5]$ with floor 0.01664 — the statement: $s_{\mathrm{tot}}(t; a_0) \ge 0.02259 \log(2+t) - 0.5185$ for *all* $t \ge 10$, every region with named support and type (H1 honestly adjudicated: the Platt-constant chain reproduces the hole boundary exactly and cannot enter). **The W2 chain at the anchor:** density ([`verification/v669_fejer_density.py`]) + frame-Gårding ([`verification/v674_packet_garding.py`] / [`verification/v678_zero_gap_theorem.py`]) + pointwise ([`verification/v681_coverage_hole.py`]) — complete, with the mixed typing ledger explicit. *Remaining:* deep family windows $a > 4.43$ (Selberg-only entry), the formal Mosco writeup, the projection-norm form, $a \to \infty$ (= W3/W4 territory). *Kill criteria K1–K3 unchanged; no step silently upgraded; no RH statement. Tenth slice (2026-08-03, the great bundling): the master contract* `PRIME.UNIFPOS.01` *and the five proof offensives.* **The master contract — the Uniform Positivity Theorem (`PRIME.UNIFPOS.01`, registered as a new contract).** Target statement, verbatim: *for every $a > 0$, Suzuki's localized Weil operator B_a on $H^0_1(-a, a)$ is positive semidefinite, with the positivity derived from an operatoric/geometric representation whose definition uses neither Riemann zeros nor an RH-equivalent presupposition.* The machine-checkable intermediate form: $B_a = L_a + K_a$ with $L_a \succeq 0$ from a structural source and the relative bound $|\langle K_a v, v\rangle| \le \theta \langle L_a v, v\rangle$ with $\theta < 1$ uniform — then $B_a \ge (1-\theta)L_a \succeq 0$. **Annotated with the v682 verdict** ([`verification/v682_lk_split_theta.py`], `LK-SPLIT-DIES`): the two-sided form for *smooth* zero-free L is structurally dead — measured $\sup\theta$ per split 11.3–58.0 with every smooth L indefinite on the odd sector, and the death mechanism is a *positive* find: the extremal pencil direction sits at $t = \gamma_1 = 14.13$ (found from primes + digamma alone, no zero loaded), $\mu_{\max}$ follows the spike law $1 + 2\alpha/\Omega(t_{\mathrm{peak}})$ (median dev 0.127), θ grows $\sim 2.47\,\alpha$ — *the deployed windows resolve individual zeros*, so the surviving directions are *one-sided* forms (the lower side alone — which must beat the pure-renaming floor $\theta = 0.998$ of the trivial spectral split) or *multi-resolution* splits (an L carrying the comb spikes at matched resolution; untested, the open lever). *Kill criteria:* (K1) a C_a/L_a *defined* through Riemann zeros = renaming; (K2) a contractivity statement that is itself RH-equivalent = renaming; (K3) any candidate binding at the trivial-split floor adds nothing beyond measured window positivity. **The rank-3 chain reaches theorem shape at the sign level:** [`verification/v683_rank3_functionals.py`] lifts the prime influence on the load-bearing 2×2 block into three exact functionals (4.9×10^{-16}) and adjudicates every entrywise route as class (c) (the T-B absorption margin demands $\sim 10^6\times$ more than any per-entry input; no Cauchy–Schwarz win: 99.7% of atoms violate pointwise; even a nonnegative pretentious measure breaks $\det S$); [`verification/v684_rank3_zeroside.py`] treats the one functional by the explicit formula (residuum $\le 6.9 \times 10^{-6}$, 22 491 budget-certified zeros) and delivers the *unconditional* $\kappa_{\mathrm{unc}} = 0.039$–$0.190 < 1$ (class (a); the pretentious escape blocked $\times 634$ by the classical zero-free strip); [`verification/v685_rank3_uniformity.py`] closes the uniformity quantifier: the symbolic envelope $K_{\mathrm{env}} \le 0.9798 < 1$ for all 62 complete windows with $a \ge 3.434$ plus five finite certificates $\kappa' \le 0.11$ — $\det S > 0$ *unconditionally on the en-*

tire declared surface (`SURFACE-CLOSED`; a surface theorem on the declared family, *not* uniform in all a; T-B stays `prob:R1`). *[2026-08-03* ***`prob:R1`*** *update: T-B is now typed and largely closed. [`verification/v692_rank3_lockgram.py`] (7/7) types T-B(block) exactly:* $\hat{A}_2$ *= positive zero-Gram + psd rank-1 pole layer, the margin* is *the transverse zero mass (a sum of squares), on-line zeros only help (psd superadditivity) — "E kippt nicht" is a* quantified partial-RH tail *statement, neither a 2c-envelope question nor a W3 renaming. [`verification/v693_rank3_density_close.py`] (5/5) builds the cited explicit bounds (Platt–Trudgian* 3×10^{12}*; Hasanalizade–Shen–Wong count; functional-equation halving; the explicit Ingham-form zero density, arXiv:2507.15184) into a sharpened safe-side envelope (sinh difference pairing,* $V(0)=0$*, on-line subtraction; PEN down* $\times6.5$*–*14.4*):* 60/70 ***complete windows close unconditionally-modulo-citations****; the exact open remainder is* $h \in \{1359, 1721, 1868, 2018, 2093, 2472, 2475, 2630, 2656\}$ *with* $T^* \in [1.0\times10^{13}, 3.0\times10^{13}]$ *plus* $h = 5690$ *at* $T^* = 8.5\times10^{14}$*; extension beyond the declared family grows* $\sim e^{3.74a}$*.] [2026-08-03* ***`prob:R1`*** *update II: [`verification/v712_rank3_transverse_deck.py`] (5/5) scalarizes the v692 margin identity along the transverse direction — the penalty pays only the transverse zero mass, median gain* $\times4.23$ *on the open ten — and the census closes nine of them:* 69/70 ***complete family windows now close at the cited*** 3×10^{12} *(unconditional-modulo-citations); the single remainder is* $h = 5690$ *with* $T^* = 1.6\times10^{14}$ *(down from* 8.5×10^{14}*); the deck-triplication lever is honestly dead (triangle inequality: channel sups can only lose). T-B itself stays the conjecture.]* **The kernel class collapses** ([`verification/v687_extremal_kernel.py`], `KERNEL-CLASS-TOO-NARROW`): the pinning certificate $2\pi F_{a_0}(m\pi/a_0) = 0$ (sympy-exact) plus Shannon sampling collapse the entire admissible band-limited positive class onto $\{\lambda\,\text{Fejér}\}$ with class supremum $\theta_{\text{req}} = 0.0901684 < 0.10076$ — no kernel optimization can beat the $S(t)$ constant; W2 stays closed via the v680/v681 route. **The W3 falsifier is constructive** ([`verification/v688_interpolation_detector.py`], `INTERP-FALSIFIER-CONSTRUCTIVE`): every isolated off-line quadruple is detected from $2\alpha\delta \geq 1.974$ ($C_{\text{up}} = 0.987$), a $\times2$ improvement over the v677 mode map, with masking fully adjudicated (46 broken + 2 provably sub-resolution of 48). **The geometric source opens a contract** ([`verification/v686_geometric_sos.py`], `PRIME.GEOMSOS.01`): $\Lambda(n)$ reconstructed circle-free from E8 shell counting (7.1×10^{-31}, Satake data from counts, K1 passes), canonical dim-3 cover-SOS, K4 firing honestly; missing: the bundle H and the transport Φ (top-3 eigenvalue mass only 91.4%), milestones M1–M4 with kill criteria K1–K5. **The Hecke-SOS mechanism is extracted** ([`verification/v691_hecke_sos.py`], `HECKE-SOS-IHARA-MECHANISM-EXTRACTED`): the target factorisation $A = B^*B + P$ exists *exactly* in the Ihara lab (B by Chebyshev recursion, no Cholesky; $P \succeq 0 \iff$ Ramanujan), the deployed ζ window form is exactly the sine/defect half of the canonical split (the part whose lab positivity *is* the RH analogue), the Euler mechanism is measurable (resonance lattice $2\pi j/\log 2$ exact, depth $\times1.98$), and the missing ingredient is exactly named: `Z1` — a self-adjoint geometric operator whose polynomial traces are the window moments (Hilbert–Pólya); `Z2` would be RH itself. *Kill criteria K1–K3 unchanged; no step silently upgraded; no RH statement.*

The review contracts of the great bundling (registered 2026-08-03, brief). ***`E8.GAUSSIAN.CODE.01`*** — $E_8(\mathbb{Z}[i])/(1+i)$ is the information space of RM(1, 3): first slice *executed* by [`verification/v689_gaussian_code_bridge.py`] (26/26, `GAUSSIAN-CODE-BRIDGE-EXACT`: SNF-exact quotient $\mathbb{F}_2^4$, $240 = 15\times16$ census with the zero class provably empty, σ = the RM(1,3) family permutation with residual gauge of order 18, coordinate block $= F_1{+}F_2{+}F_3$; all must-fail controls fire); remaining: the theorem writeup. *[2026-08-03 addendum: the algebraic core is now formally verified — kernel-checked in Lean 4 (****`TfptCarrier/GaussianCodeBridge.lean`, `lake build`**** green, no* ***`sorry`/`native_decide`****): the* $\mathbb{Z}[i]$*-module with the SNF certificate (unimodular* P, Q *both directions), the* 240*-root census* 15×16 *with the zero class provably empty,* μ_4 *stability, and the* σ *family action with coordinate block =* $F_1{+}F_2{+}F_3$*.]* ***`ARF.BOUNDARY.CODE.01`*** (registered 2026-08-05, [O]) — boundary code uniqueness: the non-circular selection chain boundary → code → E_8 → $\mathbb{F}_2^4$ → five slots *exists* with $g_{\text{car}} = 5$ as output ([`verification/v776_boundary_hamming_uniqueness.py`], 31/31, BOUNDARY-CODE-UNIQUE: census 135 → 30 → 14 → 2 → 1 without Hamming/E_8 input; the flag datum is *one* deck-invariant $\mathbb{Z}_2$, the product of the four side orientations; Construction A of the selected code *is* E_8; the Arf selector is unique; packets 1 + 4). *The open contract form* *as registered:* derive evenness/self-duality of the mark bookkeeping from the seam axioms — the R1 *universe premise* ("the boundary realizes a doubly-even self-dual binary code on its 8 sides") was the relocated and shrunk circularity, joined by the typed residuals R2 (deck placement is semantic, not selective) and R3 (the anchor naming $\bar{b}(F_\Sigma, A) = 1$ is part of the frozen selector, not a theorem). *[2026-08-06 note, markers unchanged: all three residuals moved — R1 is replaced by R1′ (the three physical axioms of* ***SEAM.CODE.TYPEII.01*** *below,* [verification/v799_seam_code_typeii.py]*); R2 is placed and R3 is named by the metaplectic S-lift (****SEAM.CLIFFORD.MODULAR_S.01*** *below,* [verification/v798_seam_clifford_modular_s.py]*).] [2026-08-06 round-22 note, markers unchanged: the attack surface sharpens — affine functions on the NS/R torsor reconstruct* $\mathrm{RM}(1,3) = H_8$ *with* $\mathrm{Stab}(\chi_{\mathrm{NSR}}) = \mathrm{AGL}(3,2) = \mathrm{Aut}(H_8)$ *in its natural 8-point action (*[verification/v803_e8_jetcode_affine_nsr.py]*); a typed internal reconstruction, not a P2 removal — R1′ unchanged.] No P2-removal claim*; kills: a second boundary-legal code surviving the flagged selection, or a seam-axiom derivation forcing a non-doubly-even universe. No marker move. ***SEAM.CODE.TYPEII.01*** (registered 2026-08-06, [C]) — Type II from the seam axioms: at length 8 (four J-paired marks × two sides) the finite chain is exhaustively machine-verified over all 308,993 subspaces of $\mathbb{F}_2^8$ of dim ≤ 4 ([verification/v799_seam_code_typeii.py], 19/19, TYPEII-FORCED): mutual locality of the Construction-A boundary vertex words $\iff$ C self-orthogonal; integer conformal spin (trivial T sector phases) $\iff$ C doubly even; holomorphy/index one ($|\mathrm{disc}| = 1$, S-closure of the single character) $\iff$ C self-dual — the survivor set is exactly the 30 $\hat{e}_8$ copies and no boundary-legal non-Type-II code exists. Chained to `ARF.BOUNDARY.CODE.01`, the full route seam axioms A1–A3 + deck + orientation bit → unique code → E_8 → $\mathbb{F}_2^4$ → $g_{\mathrm{car}} = 5$ stands with the single [C] residual R1′: *the physical seam satisfies* A1 mutual locality (bose statistics / trivial monodromy), A2 integer conformal spin (modular T-alignment), A3 holomorphy/index one (a single modular-closed sector) — each a standard physical demand, not a coding-theory premise. Imported lattice-VOA/theta ingredients are classical, cited and machine-verified in every instance used; the OPE-linearity scope note is typed. Kill: a seam construction violating A1–A3 while staying boundary-legal. No P2 marker move. ***SEAM.CLIFFORD.MODULAR_S.01*** (registered 2026-08-06, partial state) — the missing Clifford bit as the metaplectic lift of modular S at $\tau = i$ ([verification/v798_seam_clifford_modular_s.py], 27/27, MODULAR-S-PARTIAL): *coset decision positive* — the per-binary-factor Hadamard lands exactly in the missing coset $\zeta_8 G_{31}$ with the exact fixed-point relations ($S^2 = I$, $T^2 =$ Pauli parity, $(TK)^3 = \zeta_8 I$); the deck $J = iI$ *is* the metaplectic anomaly of the total symplectic Fourier (R2 placed) and $\zeta_8 = (1+J)/\sqrt{2}$ the real half-deck swapping the seam pair; the strict uniqueness census over all 46080 coset elements is *empty* — no Θ-real metaplectic involution fixes q^*, the Arf defect $q^* \mapsto q^* + \bar{h}(\cdot, c)$ is unavoidable — with exactly one G_{31}-conjugacy class of retyped survivors, and the σ-orbit Arf defects are exactly $\{A + F_\Sigma + F_i\}$ with XOR = the anchor A (R3 *named*, not fixed). *What remains:* derive the physical selection of the defect class (the anchor naming from seam data) or type the two-variant/defect residue as intrinsic gauge. Kill: a Θ-real metaplectic coset element fixing q^* (excluded here exhaustively) would falsify the defect law. No marker move. ***CURVE.CODE.OUTERSPIN.01*** (registered 2026-08-06, *decided* CANONICAL) — the outer bridge between the code duad side and the curve syntheme side ([verification/v796_duad_syntheme_bridge.py], 24/24, BRIDGE-CANONICAL): the Sylvester duality β with $\beta\sigma = \tau\beta$ and $\beta(q^*) = S^*$ (the unique $\langle\tau, \rho\rangle$-fixed spread) exists and is unique up to the order-6 code joint stabilizer (one orbit including the τ/τ^2 swap); it resolves the `v784` Arf mismatch canonically (q_Θ pulls back constantly to $q^*_{\mathrm{even}} = q^* + \bar{h}(\cdot, A)$, the anchor shift), transports $\bar{5}/10$ and 3+2 exactly, and sends t_A to $\rho = J$; χ_{NSR} transports as an honest exact two-variant torsor pinned by the `v752` NS/R parity — that two-fold ambiguity is intrinsic and typed, not open work. The `CURVE.CODE.2TORSION.01` inner-identification kill stands unchanged; no matter semantics (ROOTCLASS-MIXED). *[2026-08-06 round-22 note: the bridge sharpens into an incidence-operator statement — every census bridge factors exactly* $N_{\mathrm{code}} = P_\Delta^T N_{\mathrm{curve}}^T P_\Gamma$*, a transpose duality, not an isomorphism (*[verification/v809_curve_code_doily.py]*).]* No marker move. ***E8.RAMIFIED.JETCODE.01*** (registered 2026-08-06, [E]; carrying `E8.AFFINE.NSR.01`) — the jet-code normal form of the Gaussian compiler ([verification/v803_e8_jetcode_affine_nsr.py],

18/18 + 15/15, JETCODE-EXACT+AFFINE-NSR-EXACT): because 2 ramifies, $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$ (unique iso; the split ring fails all 24 bijections) and $W = L/2L$ is *free* of rank 4 over the dual numbers; the deck is the jet unit $J = 1+\epsilon$; among all 65536 sections of $0 \to \epsilon W \to W \to V \to 0$ there are *zero* μ_4-equivariant splittings — non-splitness *is* the `v782` no-origin obstruction; $240 = 15 \times 8 \times 2$ (root = address × error jet × orientation). Kill: a μ_4-equivariant splitting, or a second iso class of the fiber ring. No marker move. ***`E8.SYNDROME.ALGEBRA.01`*** (registered 2026-08-06, *dead* as frozen) — the sharpest error-correction-hull hypothesis killed as preregistered ([`verification/v805_e8_syndrome_algebra.py`], 13/13 incl. the must-fire controls, SYNDROME-DEAD): max rank 222/256 over all four frozen phase conventions; the $\pm r$ proportionality obstruction structural for the whole diagonal-phase family; both post-hoc escapes score worse. Survives exactly: $16 + 240 = 256$, the 16 neutral kernels = the diagonal MASA, same-class products A_0-valued over μ_4. Successor criterion named: root-individual operator data beyond diagonal phases. The Construction-A Hamming theorem untouched. ***`CURVE.CODE.DOILY.01`*** (registered 2026-08-06, [E] + [O]) — the doily incidence identity ([`verification/v809_curve_code_doily.py`], 21/21, DOILY-EXACT): $NN^T = B + 2I$ entrywise on the K_6 doily (15 duads × 15 synthemes), spec $= \{9, 4^9, 0^5\}$; $N/3$ has singular values $\{1, 2/3, 0\}$ — the recovery base rate 2/3 as the unique nontrivial singular value of the canonical duad-to-syntheme incidence; Petersen edges = synthemes 15/15; the Arf switch set T_6 = the six within-block duads. *Open target:* the six-step composition identifying $(2/3)^6$ itself. *[2026-08-06 round-23: decided — SIXSTEP-CLOCK-ONLY ([`verification/v814_k5_sixstep_transport.py`]): the rate arises canonically on three spatial routes (the doily alternation among them), but each carries a typed obstruction and sixth-power blindness makes per-step data mandatory — the deployed six is the clock exponent; the missing ingredient is a deployed shells-to-cusp isomorphism, named.]* Kill: any census bridge failing the transpose factorization. No marker move. ***`CARRIER.CUTCODE.01`*** (registered 2026-08-06, [E]; carrying `CARRIER.MOMENT.INCIDENCE.01`) — the Arf split as the cut classification of K_5 ([`verification/v810_cutcode_moment.py`], 17/17 + 16/16, CUTCODE-EXACT+MOMENT-INCIDENCE-PARTIAL): the cut map injective-isometric on all 256 cells; $16 = 1 + \bar{5} + 10$ *is* $(0|5) + (1|4) + (2|3)$; Clebsch $40 = 5 + 20 + 15$ over the Arf shells with within-10 = Petersen; roots/lines grade 80 + 160 / 20 + 40 — Arf/cut grading, not matter; the master moment $\mathrm{Tr}_{S^+} X^2 = 4|X|^2 = 4h(E_8) = 120$ one symbolic identity. The frozen 210 decision: $\binom{10}{4}$ *decorative* (equivariant bijection killed by the exhaustive fixed-point obstruction; the partial $80 \to 5 \times 16$ miss-slot map and the Newton tie survive as audit). No marker move. ***`CARRIER.PETERSEN.RADIAL.01`*** (registered 2026-08-06, [E] + [H neu]; carrying `TRANSPORT.SIXTHROOT.01`) — the K_5 edge machine ([`verification/v808_petersen_sixthroot.py`], 19/19 + 16/16, PETERSEN-RADIAL-EXACT+SIXTHROOT-SPECTRAL-ONLY): charges $(-4, +1, +6)$ = the certified $SU(5)$ decuple; the edge graph is Petersen SRG(10, 3, 0, 1); distance shells (1, 3, 6) charge-pure; the flavor hexagon *is* the distance-2 shell edge-by-edge; $\mathrm{spec}(P^6) = \{1, 64/729, 1/729\}$ *is* the deployed transport spectrum exactly; the 10-dim multiplicity reading ($\times 5 = g_{\mathrm{car}}$ fast, $\times 4 = |\mu_4|$ slow) typed [H neu]. The sixth root: $B = \frac{1}{18}[[13, 1, 4], [1, 13, 4], [4, 4, 10]]$ unique symmetric doubly stochastic PSD in the corpus basis, B^6 $GL(3, \mathbb{Q})$-conjugate to but bit-equal to *neither* deployed form — the corpus freezes the spectrum, not the basis (*the basis-freeze question is the named open*). *[2026-08-06 round-23 dated correction + addendum ([`verification/v814_k5_sixstep_transport.py`]): the sixth-root census had missed the deployed **v221** evaluation — T_{v221}, the exact spectral sum of **v221**'s transfer $(1-w)^6$, equals B^6* bit-exactly*, so the verdict upgrades from SPECTRAL-ONLY to BITEXACT-vs-v221 and the basis-freeze question is* answered *on the deployed route (the frozen SPECTRAL-ONLY stands as the probe's adjudication; ABEL-DEAD correction precedent); and the [H neu] 10-dim reading now has a promotion-ready canonical PROPOSAL: $T_{10} = P_{10}^6$ in the Λ^2 basis passes all gates with the equitable **v774** quotient $= (Q_{\mathrm{Pet}}/3)^6$ bit-exact (TRANSPORT10-PROPOSED, no deploy claim).]* Kill: a second PSD sixth root in basis, or a shell/charge mismatch. No marker move. ***`E8.QUARTIC.HALF.01`*** — G31 as the holomorphic μ_4 shadow of the E8 compiler: first slice *executed* by [`verification/v690_quartic_half.py`] (22/22, `QUARTIC-HALF-ALIVE`) *with the honest correction*: the vanishing half $\{2, 14, 18, 30\}$ is true but trivially μ_4; the substance is $F_8|, F_{12}|, F_{20}|, F_{24}|$ as a system of *basic* G31 invariants (Chevalley cited; the holomorphic restriction generates the full G31 ring); remaining: the symbolic degree-60 Jacobian factorization and the writeup. *[2026-08-03 addendum: the μ_4-orbit vanishing mechanism and the quartic power sum $\sum_\alpha \langle\alpha, x\rangle^4 = 576\,(\sum_i x_i^2)^2$ over the explicit 240 roots are now kernel-checked in Lean 4 (`TfptCarrier/QuarticHalf.lean`, `lake build` green, no `sorry`/`native_decide`); the Chevalley basic-invariant step remains cited, not formalized.]* `PRIME.W3.INTERPOLATION.01` — the alias-interpolation falsifier: first slice *executed* by [`verification/v688_interpolation_detector.py`] (constants above); remaining: the lemma as a paper lemma. *[2026-08-03 addendum: the two named building blocks are closed near-proof by [`verification/v694_interpolation_lemma_closure.py`] (20/20, `CLOSURE-BOTH-NEAR-PROOF`): retention via an exact projection identity plus a closed-form $O(k^2)$ certificate ($r_f \geq 0.548$ on the family surface; straddle boundary case discovered, $k = 0$ branch secured), and the separation law* form-corrected *to the additive $\alpha^* = C_{\text{cell}}/\delta + C''/\Delta\gamma$ ($C'' \leq 0.59$, zero exceptions in 97 tests); the two open parent configurations detect at $\alpha^* = 7.90$; declared residues: the explicit Ingham constant chain, the Dirichlet-kernel overlap bound, the $C''(n)$ induction.]* `PRIME.Z1.OPERATOR.01` — *OPEN* (Offensive 5 running): construct the self-adjoint geometric operator whose polynomial traces are the deployed window moments (the Hilbert–Pólya step of the extracted Ihara mechanism), without zeros as input; an operator defined through zeros or an RH-equivalent norm assumption = renaming; `Z2` (the norm bound) would be RH itself — the factorisation *localizes* RH, it does not bypass it. *[2026-08-03 update, the 5–5d series ([`verification/v695_z1_trace_operator.py`] / [`verification/v696_z1_jacobi.py`] / [`verification/v697_z1_uvarov.py`] / [`verification/v698_z1_flow_recursion.py`]): the contract stays OPEN, with the ground taken recorded honestly — (5) the* measure *comes from counting [E]: seam and orbifold are dead as Z1 spectrum (null-calibrated), while the E8 counting route delivers the Weil measure exactly (atoms 0.0, arch/pole via the closed Γ duplication split, 7/8 target spikes at $q = 0.000$); the comb sequence $c +$ pole is positive-feasible — the signedness of the deployed sequence* is *the pole subtraction. (5b) the canonical operator* exists *(CMV/Jacobi, all moments reproduced at 2×10^{-13}); no closed coefficient law yet, but the renaming clause is explicitly refuted: features on prime-power slots ($q = 0.000$), amplitude–mass link $r_{\log} = +0.76$, positivity load-bearing (170/171 vs 2865). (5c) masses: atoms are lag insertions (duality proven), exact transfer identity $\Delta\alpha = w_1/E$ (5.6×10^{-17}), and the* inverted *stabilization law — the Γ flow predicts the counting masses to ∼10% (median ratio 1.026, corr 0.86). (5d) positions* forced *(windows 0.5–2 cells, jitter null 30/30, adversary −516 lags); shooting recovers the masses at the true positions to 0.11% median.* Remaining: *lookahead autonomy (the greedy-saturation failure of autonomous reconstruction) and the continuum reading of the operator.] [2026-08-03 update II, the montage and the moonshot: the contract stays* formally OPEN *— the continuum* theorems *are missing — but both remainders moved. (i) The lookahead question is* decided *([`verification/v702_z1_lookahead.py`], 13/13): the Γ flow is a full-depth comb* verifier *(913 ≥ 898, every fake dies ≥ 184 lags earlier) but* not *a generator — both prescribed lookahead routes reach 2–4 slots, and the per-slot information demand $g \sim 5$–12 bits is information-theoretic and parameter-robust (`FLOW-VERIFIER-NOT-GENERATOR`). (ii) The chain arc S-A…S-G ([`verification/v703_chain_tolerance_scaling.py`] through [`verification/v709_chain_zero_layer.py`] plus [`verification/v710_chain_two_stage_hecke.py`] / [`verification/v711_chain_firstbirth_scaling.py`]) types the mass law honestly: tolerance induction dead (RH-grade), the arch is a deck-sector trace [E] ([`verification/v705_chain_deck_sector.py`], 3.4×10^{-16}), the flow α* are *the Schur parameters (10^{-40}), the section edge is an exact resolvent identity, and $G1 =$* [E-edge] + [E-Hecke] + [M-rate] *with the corridor position ∼ 0.53 as the named open selection question. (iii) The* continuum candidate is assembled *([`verification/v713_l1_montage.py`], 16/16, `L1-ASSEMBLED-MEASURED`): cover limit 6.2×10^{-8}, point masses at the D^2 rate, one UV slot, divergence exactly at $s \leq 1/2$ (the critical line as the boundary of the construction — a consistency signature,* not *a positivity statement); the missing theorems (self-adjointness, trace identification, positivity) are* named. *(iv) The atom layer is* groupoid-internal *([`verification/v714_moonshot_hecke_groupoid.py`], 19/19, `MOONSHOT-GO`; see `PRIME.Z1.MOONSHOT.01` below). What is missing is exactly the continuum* SÄTZE*: until they exist, no marker moves.] [2026-08-03 update III, the evening round: the* *moonshot stages 2–4 plus the Satz-1 slice and the K collapse are* done and machine-checked *(see* `PRIME.Z1.MOONSHOT.01` *below for the six modules* [verification/v716_moonshot_arch_glue.py] *through* [verification/v721_k1_node_capture.py]*). The Hilbert–Pólya candidate is now structurally complete* at measurement level *— an operator family, a state, an exact finite trace formula with a closed term dictionary to Weil 1952, and a measured spectral limit onto the zeros (*100% *of* 377 *at tol* 0.25*, rate* −1.61*, SHA256 prediction firewall). The contract stays* formally OPEN*: the remaining hard substance is (L1) the identification of the vague limit* at *the line* $s = 1/2$ *(no mass loss — the single place the zero fluctuation enters) and (L2) the node convergence as a theorem (the capture half is proof-near,* [verification/v721_k1_node_capture.py]*; the* ∼900× *super-resolution is the open K1b question); no marker moves, no RH statement.]* `PRIME.Z1.MOONSHOT.01` — *OPEN* (registered 2026-08-03, the groupoid program): derive the *full* Weil measure (atoms + arch + pole) as the trace layer of *one* geometric groupoid built on the $\mathbb{Z}[i]$-E8 Hecke tower, with the arch layer *glued* (not direct-summed) and the E8 specificity established. *Stage 1 (done):* [verification/v714_moonshot_hecke_groupoid.py] (19/19, `MOONSHOT-GO`) shows the atom layer is groupoid-internal — the primitive degrees of the tower are *exactly* the Gaussian prime norms (maximality census; no prime projector anywhere: the primes come *out* of the tower), the log generator + cell normalizer de-divisorize the tower traces at 4×10^{-13}, and the σ descent reproduces 100 atom positions, 100 masses and 368 moments at deviation 0.00; declared ingredients (named as inputs, not derived): the $\sqrt{n}$ half-density and the σ descent. *The honest fence (typed):* the finite counting facts of stage 1 are *E8-unspecific* — the specificity question belonged to stage 2, which has since been executed. *Stage 2 (done, 2026-08-03):* [verification/v716_moonshot_arch_glue.py] (18/18, `MOONSHOT-STAGE2-GLUED`) — the arch *glues*: one object, *one* normalization (the free 3-scalar LSQ returns $\kappa = (1, 1, 1)$ to $< 10^{-12}$ on all five windows); the 1/4 in $\mathrm{Re}\,\psi(1/4 + i\tau/2)$ is *derived* as the μ_4 fixed-sector offset; $\log \pi$ enters only through the declared self-dual (Tate) UV cell — the exact arch mirror of the declared $\sqrt{n}$ half-density; and E8 becomes forced *at the gluing* (the μ_2 tower, the wrong μ_4 class and the wrong deck twists all rip). *Stage 3 (done):* [verification/v717_moonshot_state.py] (21/21, `STATE-ON-TRUNCATIONS`) — the Weil functional is *constructively* a GNS vector state on every truncation (Levinson PD); the naive tower reading misses by $|\Delta|$/W3-margin = 104–1563 — the trace-formula boundary *is* the localized RH substance; KMS $\beta = 1$ (Bost–Connes critical) measured. *Stage 4 (done):* [verification/v718_moonshot_spectral.py] (10/10, `SPECTRUM-CONVERGES-MEASURED`) — the truncation spectra converge *measurably* onto the zeros: 100% of 377 frozen targets at tol 0.25 ($h = 1433$), rate −1.61, monotone GNS K-ladder, SHA256 prediction firewall; the scramble control dies as a state. *The Satz-1 slice (done):* [verification/v719_moonshot_traceformula.py] (24/24, `SATZ1-LEDGER-EXACT`) — the finite trace formula is exact Gauss quadrature and the term dictionary to the classical Weil formula *closes* (pole/atoms/arch/UV); the E8 census over $\{\mathbb{Z}^8, E8\}$ (Mordell 1938): only the Gaussian E8 glues. *The K collapse (done):* [verification/v720_moonshot_k2k3.py] (14/14, `K2K3-COLLAPSED-CLASSICAL`) — the convergence items collapse onto classical analysis (tightness measured, unconditional tail majorant on $s > 1/2$, the boundary exactly $s = 1/2$); and [verification/v721_k1_node_capture.py] (14/14, `K1-LEMMA-CAPTURE-ONLY`) makes the capture half of the node question proof-near. *Kill audit (criteria from the strategy council, adjudicated):* (K1) does *not* fire — the gluing needs no zeros (SHA256-typed diagnostics); (K2) half-standing — the $\sqrt{n}$ half-density stays *declared*, its arch mirror $\log \pi$ is now typed as the self-dual normalization; (K3) does *not* fire at measurement level — the counterfactual census rips (the uniqueness *theorem* stays open); (K4) does not fire — positivity holds on every deployed window. *The remaining substance, ordered:* (L1) identification of the vague limit *at* the line $s = 1/2$ (no mass loss); (L2) node convergence as a theorem (the ∼900× super-resolution — the K1b question, exploration running); (L3) routine lemmas (tent/Jackson discretization, PNT tail majorant — both exhibited with constants). A research contract, *not* a claim; the contract stays OPEN; no marker move; no RH statement. *[2026-08-03 update IV, the keystone round — the contract at its final state of the day. Eight further modules (*[verification/v727_l1_identification.py] *through* [verification/v734_s1_canonical.py]*) localize the remaining substance maximally.* (L2) is resolved classically: *the* ∼900× *super-resolution is* atom pinning *of the raw measure — the memo's locality/symmetry mechanism dies on two preregistered bars (*[verification/v728_k1b_superresolution.py]*,* 11/13*,* K1B-ATOM-PINNING*; lab rate* $K^{-1.84}$*, real correlate* 0.64*), and the explicit pinning lemma is certified with one-line proofs and* 0 *violations at* 598 *peaks/*535 *zeros including all* 377 *of the largest window (*[verification/v730_strat2_pinning_lemma.py]*,* 12/12*); the certificate saturates at window width — sub-width certification needs atomicity beyond every window depth. The K1 diagram closes: capture theorem (v721) + pinning lemma (v730) + conditional gap rigidity (*[verification/v731_strat2_gap_universality.py]*,* 5/5*,* GAP-UNIV-KILLED *for the strong measured form; a fixed-window conditional candidate remains via Lubinsky/Levin–Lubinsky/Avila–Last–Simon/Totik) + atom structure (= L1).* (L1) is typed in four equivalent languages: *determinacy of the limit moment problem is* classical *and the identification chain contains RH nowhere (*[verification/v727_l1_identification.py]*,* 11/11*,* L1-DETERMINED-WALL-TYPED*); THE WALL is* PD persistence of the window ladder *— infinitely many PD windows suffice for RH, each window finitely decidable, measured* 9/9*; equivalently: moments (Hankel-PD) / state (Levinson-PD) / symbol (Fejér* ≥ 0*) / Hamiltonian (Krein* $l_k > 0$ *—* [verification/v734_s1_canonical.py]*,* 10/10*,* S1-CANONICAL-SUZUKI-BRIDGED*: exact Krein correspondence at* 1.6×10^{-15}*,* $H_h \geq 0$ *measured on all nine windows, boundary phase =* γ*-side at scale exactly* 1*; Suzuki JFA 2021 as the fourth anchor, unconditional only for* $\omega > 1$*). The h-universal quantifier* is *the RH substance.* Honest kills, fenced: *naive reflection positivity is dead on the truncations — the FE symmetry buys exactly the atom-free Connes–Consani 2021 regime* $|u| < \log 2$ *and no more (*[verification/v729_strat2_rp_universality.py]*,* 5/5*); the direct Sonin-compression transfer is dead (*[verification/v732_strat3_sonin_prolate.py]*,* 21/21 *construction,* SONIN-RANK-GROWS*; the CC mechanism itself untouched, retuning forbidden).* A first exact bridge into Connes' framework: *the* σ*-descended groupoid maps by an exact weak functor candidate into the* S*-local periodic-orbit subalgebra (*$S = \{\infty, 2, 3, 5\}$*,* 6859 *convolutions at deviation* 0*,* $\kappa \equiv 1$*;* [verification/v733_strat3_gate0_census.py]*,* 13/13*,* GATE0-WEAK-FUNCTOR*; full-algebra extension open). The literature anchor as the dual council reads it: the wall has three names — Weil 1952, Connes 1999/Connes–Consani 2021, Suzuki 2021 — with Meyer 2005 the proven boundary line (unconditional spectral realization* without *positivity); TFPT adds the computable zeta-free truncation framework with emergent prime places and per-window decidability. The contract stays OPEN; no marker move; no RH statement.] [2026-08-03 update V, the closing slice — the L1 route portfolio is complete. The state-transport route also stagnates (*[verification/v735_strat3_ucp_inductive.py]*,* 10/10 *construction checks,* UCP-STAGNATES*): all eight adjacent GNS-ladder maps exist* exactly *as UCP quantile couplings — the Choi SDP reduces to a finite transport LP, unitality/state preservation to* $3.1 \times 10^{-12}/1.2 \times 10^{-15}$*, every Choi rank inside the sharp bound* $h + h' - 1$ *with log-log slope* 0.991 *(linearly controlled) — but the preregistered shift/grade covariance gate fails honestly (error tail not monotonically decreasing): mere state transport is* not *the missing L1 mechanism, and the maps are quantile Markov couplings, not* $*$*-homomorphisms (multiplicativity defects* ~ 5–8×10^{-3}*). The portfolio now reads: Sonin dead (v732), UCP stagnates (v735), Gate 0 weak functor (v733), Suzuki bridge a measurement (v734) — what remains for L1 is atomicity/identification substance, not transport machinery. The contract stays OPEN; no RH statement.] [2026-08-05, SHARPENED — the diagonal-Gram closure hands over (*PRIME.GRAM.DIAGONAL.01 *is closed per its own frozen stakes; the qf offensive* [verification/v769_qf_contract_necessity.py] */* [verification/v770_qf_spectral_bundle.py] */* [verification/v771_qf_representation_census.py] */* [verification/v772_qf_feshbach_effective.py] */* [verification/v773_qf_cell_cocycle.py]*). The sharpened target form: construct a self-adjoint geometric* bulk *operator with a canonical finite-dimensional boundary/threshold structure whose Weyl* M*-function delivers the qf block and whose relative trace formula reproduces the full Weil formula. The* measured handover content *— mandatory constraints any Z1 candidate must carry: (i) the Hecke/E8 bulk with the archimedean lift (*[verification/v714_moonshot_hecke_groupoid.py] */* [verification/v716_moonshot_arch_glue.py]*); (ii) the moving-edge threshold structure with its measured cadence — mode entries at* $M \simeq 888/992/1108/1276$*,* ΔX *widening* $1.625 \to 2.625$ *(spread* 0.381*, not regular at the* 0.25 *bar), avoided crossing* 0.0039 *at* $M = 1240$ *(*[verification/v772_qf_feshbach_effective.py]*); (iii) the ram-*

odd contact direction ($\cos\theta_1 = 0.997$ rung-stable, [verification/v771_qf_representation_census.py]); (iv) the settled per-function coupling levels ($R2$ family 0.225–0.358, $R1$ family 0.008–0.079, [verification/v770_qf_spectral_bundle.py]); (v) the cell-ordered prime transport in domain-preserving (Nevanlinna/Loewner) form — the exact Möbius/Redheffer cell cocycle of [verification/v773_qf_cell_cocycle.py]. Kills, active: *an operator defined through zeros or an RH-equivalent norm assumption = renaming (unchanged); additionally, a candidate whose threshold structure cannot carry the measured widening cadence, or whose M-function block contradicts the settled qf coupling levels or the ram-odd contact direction, is dead on arrival. The contract stays OPEN; no marker move; no RH statement.] [2026-08-05, EXECUTED — the compactness split measured ([verification/v780_z1_compactness_trilogy.py], one module from three preregistered probes; gates 3/4 + 1/4+ legs 3/6 with exactly the preregistered fails). The N2 GNS/Jacobi family carries the ram-odd contact (*cos *up to* 0.9971*), edge collisions at the source depth (*0.0032 *vs* 0.0039*) and settled couplings (*14/14 *P) natively — but* not *the widening cadence (its edge census is regular quadrature filling, the free-Jacobi type): the cell structure must be imported as boundary data. The finite boundary triple ($A = 2 - J_K$, imported Gram band) is* exactly *Herglotz through all typed transitions but dies as a whitened $K = M/2$ import — the Gram near-kernel packets are delocalized (cell-mass exactly* 0.500 *beyond the half-window), so whitening turns truncation loss into quasi-generic boundary noise (delivery* 5.88*; still arithmetic-specific:* 5.9 *vs GOE* 13.0 *vs random* 132.5*; the full-depth frame is provably out of reach by deepening, $e^{27.75}$ atoms). With the* raw *μ-weighted import the variable-edge Herglotz family is* bounded *(*0.936*) and* equicontinuous *(*0.807*) with* summable *∼rank-one entry poles at the arithmetic thresholds (c_{mass} $0.032 \ll 1$; rank ratios* 0.80–0.87*), while the delivery metric loses discrimination (typed* UNRESOLVED-BY-THIS-METRIC*) and the moment functionals do not settle (osc* 22.1*).* The measured unification: *this contract and* `PRIME.KMS.INDUCTIVE_STATE.02` share *their compactness theorem (carried: Helly/Montel at finite level) and* share *their single remaining obstruction — state selection* equals *import-faithful boundary coupling (a coupling of the Gram band to the bulk that does not pass through the lossy half-window frame). The merged target is named* Z1-COMPACTNESS *(*`PRIME.Z1.COMPACTNESS.01`*); both rows stay, cross-referenced; any future module must attack the shared delta, not the two contracts separately. The contract stays OPEN; no marker move; no RH statement.]*

The review master contract (`PRIME.KMS.INDUCTIVE_STATE.01`). *OPEN* (registered 2026-08-04, from the external review of the strategy portfolio): extend the finite Hecke functor of Gate 0 ([verification/v733_strat3_gate0_census.py], 13/13, `GATE0-WEAK-FUNCTOR` — exact on the periodic-orbit subalgebra, 6859 convolutions at deviation 0, $\kappa \equiv 1$) to a *positive global KMS state* ω at $\beta = 1$ on the closed groupoid algebra, with embeddings $\iota_h \colon A_h \to A_{\mathrm{E8}}$ and $T_h = \omega \circ \iota_h$ — then Weil positivity follows for the dense test space, and with it RH. *The five separation steps, each individually named and decidable:* (1) algebraic extension from the periodic-orbit subalgebra to the full convolution algebra; (2) topological closability; (3) complete positivity; (4) the KMS condition at $\beta = 1$; (5) the Weil-limit identification. *Kill criterion:* complete positivity fails *already on a finite extension* — one finite counterexample retires the route. The contract names the gap; it does not close it. A research contract, *not* a claim; no marker move; no RH statement. *[2026-08-04 update: the strong form is DEAD and the contract is refined. [verification/v740_kms_extension_switch.py] (*8/8, *KMS-EXTENSION-DEAD): $\beta = 1$ KMS is algebraically impossible on Gate-0's nonzero-grade Laurent unitaries — $T(u_p u_p^*) = 1$ versus $T(u_p^* \sigma_i(u_p)) = 1/p$, exact defects $1/2, 2/3, 4/5$ at $p = 2, 3, 5$ — and exact covariant $*$-restrictions do not exist (*0 *of* 141..1255 *required boundaries); local extended Gram positivity survives (*55,296 *monomials/window). [verification/v741_kms_toeplitz_semigroup.py] (*9/9, *KMS-TOEPLITZ-DEAD): the Bost–Connes semigroup Toeplitz correction repairs the algebraic KMS step* completely *(boundary-corrected identities exact; normalized bare error $10^{-2} \to 10^{-9}$ along the directed system), but the covariant inductive step stays dead (the preregistered UCP window covariance is nonmonotone). Separation steps (1) and (4) are thereby settled at the finite level; the surviving demand is restated below as* `PRIME.KMS.INDUCTIVE_STATE.02`*.]*

The refined master contract: operator-system compactness

(PRIME.KMS.INDUCTIVE_STATE.02). *OPEN* (registered 2026-08-04, refined from `INDUCTIVE_STATE.01` after [`verification/v740_kms_extension_switch.py`] and [`verification/v741_kms_toeplitz_semigroup.py`]): keep the *proven* local blocks — the BC semigroup Toeplitz algebra per window (nonunitary isometries with boundary projections), the boundary-corrected $\beta = 1$ KMS identities (exact), extended Gram positivity on the factorized slices, and UCP state transport ([`verification/v735_strat3_ucp_inductive.py`]) — and state the single remaining global step purely as an *operator-system compactness/subsequence problem*: does the net of window states have a weak-∗ cluster point that is a positive KMS functional on the closed algebra, *without* demanding a covariant inductive system? *Kill criterion:* a finite obstruction to any cluster point (a uniform positivity violation along a subnet). Do *not* claim a covariant inductive system until a new canonical window map is found. *Literature anchor:* $\beta = 1$ KMS for the genuine Bost–Connes algebra is classical (Bost–Connes 1995); the TFPT burden is the compatibility of the glued window states with one global limit, not local KMS existence. A research contract, *not* a claim; no marker move; no RH statement. *[2026-08-04 update: step (3) — complete positivity — is SOLVED on the factorized ramified* 15*-label Toeplitz layer. [`verification/v756_kms_incidence_stinespring.py`] (*7/7, KMS-INCIDENCE-CP-COVARIANT*): 105 *explicit Kraus terms realize the incidence channel* $K = B/7$ *as a completely positive unital map, with all four compatibilities exact —* σ*-covariance at deviation* 0*, half-weight covariance* $DU = 2^{-1/2}UD$ *to* 1.1×10^{-16}*,* $\beta = 1$ *detailed balance to* 6.9×10^{-18}*, and Gate-0 compatibility to* 3.3×10^{-16}*. The local candidate state is* $\tau_{\mathrm{inc}} \otimes \tau_{\mathrm{Toeplitz}}$ *(channel-preserved). The remainder to a global state is exactly the compactness step of this contract: compatible incidence channels at every finite place and level, the interaction with non-label ramified multipliers and the arch lift, a canonical UCP map between the nine window systems, compactness/normality of the limit, and the Weil-functional identification.] [2026-08-04 update (the diagonal gram round): the compactness step now carries MEASURED handoff evidence and an abstract Lean closure. Measured: the* ε*-regulated resolvent is the admissible operator-system evaluation with* ε*-robust falling local defects ([*`verification/v766_handoff_bulk.py`*],* 17/17*, HANDOFF-BULK-CONVERGES); a target-independent growing-frequency PSD source Gram passes the finite ladder at slope* -0.369 *([*`verification/v767_handoff_frequency_gram.py`*],* 6/6*, HANDOFF-GRAM-CONVERGES); the quadrature tail terminates algebraically (*2.4×10^{-13}*, uniform in the window), cross-window compatibility is closed at* $\varepsilon = 10^{-3}$ *on the deepest reachable ladder, and the source limit is identified with the deployed Weil functional on the frozen battery within* 0.0485 *([*`verification/v768_handoff_tail_weil.py`*], TAIL-WEIL-PARTIAL+COMPAT-EPS3-CONVERGES). Abstract: the finite-stage Gram compactness/diagonal-extraction skeleton is machine-checked in Lean (*`TfptCarrier/GramCompactness.lean`*,* 8 *theorems). THE SURVIVING OPEN OBJECT is the* q_f *wall of [*`verification/v763_yosida_handoff.py`*] (QF-DEAD): the battery near-kernel pairing is non-Cauchy at honest maximal depth and survives orthogonalization by structural subordination — the weak-∗ cluster question of this contract now lives in exactly that bounded quantity; the round's theorem form is registered as* `PRIME.GRAM.DIAGONAL.01`*. No marker move; no RH statement.] [2026-08-05 follow-up: the* q_f *object is adjudicated —* q_f *is representation gauge and the corner-native Gate* $3'$ *is dead ([*`verification/v769_qf_contract_necessity.py`*]), the levels settle positive ([*`verification/v770_qf_spectral_bundle.py`*]), fixed-d Feshbach closes at a moving spectral edge ([*`verification/v772_qf_feshbach_effective.py`*]), and the cell cocycle is domain-only ([*`verification/v773_qf_cell_cocycle.py`*]):* `PRIME.GRAM.DIAGONAL.01` *is CLOSED per its own frozen stakes, and the compactness remainder of this contract hands over to the* sharpened `PRIME.Z1.OPERATOR.01` *(the Weyl* M*-function / boundary-threshold form with the measured qf constraints). No marker move; no RH statement.] [2026-08-05, the unification measured ([*`verification/v780_z1_compactness_trilogy.py`*]): the weak-∗ cluster demand of this contract and the operator demand of* `PRIME.Z1.OPERATOR.01` *are measured to be the* same *remaining object on the reachable surface. The cluster-existence half is* carried*: the variable-edge boundary family is bounded (*0.936*) and equicontinuous (*0.807*) on the frozen compacts with summable ∼rank-one entry poles (*c_{mass} 0.032*) — every subsequence has weak-∗/locally-uniform cluster points (Helly/Montel at finite level). The* selection *half is not carried (moment-functional tail oscillation* 22.1*), and its* *obstruction is the same measured object that fails the delivery legs: the lossy $K = M/2$ half-window import. State selection (this contract) = import-faithful boundary map (`OPERATOR.01`): one object viewed twice; the merged target is named* Z1-COMPACTNESS *(`PRIME.Z1.COMPACTNESS.01`); both rows stay, cross-referenced. No marker move; no RH statement.] [2026-08-05 add-on: the* Z1-COMPACTNESS *frame is now also the named limit frame of demand (2) of the new `PRIME.POSITIVE_DESCENT.01` contract below — the GL1-sector PSD persistence of the manifestly positive packet object is to be decided in exactly this weak-$*$ cluster frame.] [2026-08-06 add-on: the CP extension gate ([`verification/v794_cp_extension_gate.py`],* CPGATE-UNDECIDED-FALLING*) measures this demand* sector-decorated*: the χ = GL1 instance of the four-sector window net* is *this contract's object (anchor 3.882×10^{-6} at $M = 1176$ reproduced on the same 10^8 comb), "state selection must settle" = "the sector margins never cross"; measured: GL1 falls power-law ($\beta = -1.69$), χ_4 saturates at $\sim 1.5 \times 10^{-6}$ — the first measured sector floor.] [2026-08-06 round-23 note: the SELECTION half is now finite-level* solved *— Mosco form convergence + Friedrichs minimality on the **v762** dense core replace the failed moment selector on identical rungs (moment oscillation 22.08 vs Friedrichs resolvents 0.0004 / ledger-adjusted Weyl 0.0843 / raw μ-face 0.0000; cofinal-unique 2.5×10^{-5}), and the import diagnosis is completed at form level: nothing is whitened at the core ([`verification/v816_prime_mosco_selection.py`],* MOSCO-SELECTS*; the new contract `PRIME.MOSCO.SELECTION.01` below). The remaining analytic object of this contract is the sector floor — restated as `PRIME.FLOOR.RATIO.01` below.] [2026-08-06 round-24 note: the floor contract is* narrowed *— the lock-block floor $\lambda\tau$ now carries a complete certified three-piece lower-bound skeleton on the deployed battery, closed at citation grade for all h up to $\alpha^* \approx 11$ ([`verification/v823_prime_lagrange_floor.py`], [`verification/v824_prime_floor_skeleton.py`]); battery-relative and α-bounded, still open.]*

The positive descent (`PRIME.POSITIVE_DESCENT.01`). *OPEN* (registered 2026-08-05, from the positive-protocol round; finite-level facts machine-checked in [`verification/v791_positive_descent.py`], 30/30 incl. the five must-fire controls, verdict DESCENT-PARTIAL; the declared run-1 → run-2 calibration — the falsified linear state F1, the falsified all-PSD operator expectation F2, the corrected C3 constant — is carried verbatim in the module provenance). *The finite-level facts:* (i) the packet GNS state on the extended positive register $\mathbb{N}[C_2] \otimes \mathbb{N}[\mathbb{F}_2^4] \otimes \mathbb{N}[\mu_4]$ is positive at every depth *by construction* (a sum of event squares; verified in exact rationals; the falling minimal sector is f_8^2-weight dilution, structurally unable to cross zero), while the naive *linear* pushforward of the same nonnegative data is *not* a state (F1, quantified: -6.6×10^{-2}, depth-flat) — the squared/GNS form is the only positive aggregation found, and signedness is *observer projection*; (ii) the GL1 character sector of the packet correspondence operator is *bit-identical* to the deployed Weil window (lag identity at 6×10^{-16}) and is its *unique* PSD sector of all 24 (margins $+5.3 \times 10^{-5} \to +1.2 \times 10^{-5}$ over $X = 4 \to 10$, reproduced read-only); (iii) every damped/twisted sector breaks because the register-trivial continuum cannot follow the atom leg (the pole–atom cancellation is exposed; localization: arch-only flat, arch+pole falling) — the breaking tensor factor is the *continuum* (e-slot) leg, not C_2, $\mathbb{F}_2^4$ or μ_4. *The contract therefore demands three objects:* (1) **sector-adapted continua** (the missing CP-functor data): for each character sector χ the arch/pole layer of the *twisted* channel (the explicit-formula continuum of the χ-twisted L-data; for the f_8 sector: a weight-4 archimedean factor, *no* pole), such that the sector window $T^{\chi}_{\mathrm{cont}} + \sum_n \operatorname{Re}\chi(x_n)\, Q_n$ is PSD at finite level — *finite-level falsifier:* exhibiting the f_8-sector continuum and finding its window non-PSD kills the operator face for good; constructing it PSD promotes the functor from state level to operator level. (2) **sector-PSD persistence** (the GL1 limit): the weak-$*$ limit — in the Z1-COMPACTNESS frame of `PRIME.Z1.COMPACTNESS.01` / [`verification/v780_z1_compactness_trilogy.py`] — of the GL1 sector functional stays in the cone as $X \to \infty$. This *is* Weil positivity, now typed as: "the sign sector of a manifestly positive packet object stays PSD" — the sign is never controlled directly, only the sector compression. (3) **the carrier intertwiner**: a positive identification, in the GNS limit, of the population register (χ_- readout = the damped Hecke ratio $a_n/\sigma_3(n)$) with the carrier register (χ_- readout = the unit-modulus sign); at finite level this step is *not* CP-trivial (the signed count fails positivity) —

the renormalization/amplification limit is the single non-automatic step. *Stop conditions:* no fixed-d variants, no re-gating of the closed diagonal-Gram objects, no RH claim; demand (2) failing at any finite depth = DESCENT-DEAD for the route; demands (1) and (3) without (2) are decorative. *Inherited finite-level falsifier:* any rung where a non-GL1 carrier sector becomes PSD-binding while GL1 stays positive breaks the "GL1 is the unique carrier edge" reading measured here. A research contract, *not* a claim; no marker move; no RH statement. *[2026-08-06 update, markers unchanged — the contract stays open for the floor: all three demanded objects are now delivered at finite level. (1-F) The demand sharpens to the* continuum functor*:* $\chi \mapsto \Lambda_\chi = q(\chi)^{s/2} \prod_j \Gamma_{\mathbb{R}}(s + \mu_j(\chi))\, L_{\mathrm{an}}$ *with* $(\mu, q, pole)$ *given by one closed rule; verified instances GL1 (q=1, pole),* χ_4 *(q=4),* f_8 *(q=8),* $f_8 \otimes \chi_{-4}$ *(q=16, Atkin–Li + Fricke), the conductor datum load-bearing at exactly* $\ln 2$ *and pinned from below by the PSD margins; the mirror-type sectors are structurally non-automorphic and carry no positivity demand ([verification/v793_f8sector_conductor.py],* $15/15 + 16/16$*). (2-L/2-Z) The limit object acquires its exact form: per automorphic character the positivity of the sector Weil functional* W_χ *on the union test space* V_∞ *—* $\inf_X \lambda_{\min}(T_{\chi,X}) \geq 0$ *— with the net data a CP-extension theorem consumes* measured *(restriction exactness, interlacing, uniform bounds), and demand (2) at* $\chi =$ *GL1 identified with the* Z1-COMPACTNESS *obstruction, sector-decorated; measured trend: no crossing, GL1/*f_8*/twist power-law falling,* χ_4 *saturating at the first measured sector floor ([verification/v794_cp_extension_gate.py],* 17/17*). (3) The carrier intertwiner* exists *at finite level in Stinespring form with exact Choi and all covariances, its four automorphic channels landing on the four sector windows from one CP map, and the F1 failure retyped as a negative Choi eigenvalue ([verification/v801_prime_cp_intertwiner.py],* 29/29*; the infinite-level demands live in* ***PRIME.CP.INTERTWINER.01*** *below).* The remaining analytic core, restated: *the per-sector positivity floor (GRH-type, unproven) plus the intertwiner demands D1–D3. No RH statement.] [2026-08-06 round-23 note: selection and uniqueness are now both finite-level delivered — selection by Mosco+Friedrichs ([verification/v816_prime_mosco_selection.py],* ***PRIME.MOSCO.SELECTION.01****) and the unique positive limit by the master theorem ([verification/v817_positive_descent_master.py],* ***TFPT.POSITIVE_DESCENT.MASTER.01****: prime packet state defects summable at* 0.701*/doubling,* $\hat{m}_2(p) = -1/15$ *exact for every odd prime; positivity* a *separate hypothesis by control K4). The floor itself reduces to the single ratio inequality* $\rho = \tau/\tau_{\mathrm{pnt}} > 0$ *with a measured* $h^{-3/2}$ *envelope — the per-sector positivity floor is the* one *remaining analytic object, restated as* ***PRIME.FLOOR.RATIO.01*** *below.] [2026-08-06 round-24 note: that remaining object is* narrowed *— exact Lagrange sum-of-squares identity, the certified fixed pair, the analytic fixed-pair bound, the family exhaustion, and the deep tail closed at citation grade for all* h *on the deployed battery ([verification/v823_prime_lagrange_floor.py], [verification/v824_prime_floor_skeleton.py]; the narrowed contract text below); battery-relative,* α*-bounded, still open.]*

The carrier intertwiner (`PRIME.CP.INTERTWINER.01`). *OPEN* (registered 2026-08-06; demand (3) of `PRIME.POSITIVE_DESCENT.01` solved at finite level in [verification/v801_prime_cp_intertwiner.py], 29/29, CP-INTERTWINER-EXISTS). *Object:* the finite-level carrier intertwiner $\Phi(a) = V^*\pi(a)V$ on the arrow $*$-algebra $M_{15} \otimes \mathbb{C}[\text{tower}] \otimes (\text{packet slots at the odd places})$, V the 105-leg Kraus stack (unitality $\sum_e E_{x_e x_e} = 7I$ exact), involution = arrow reversal, KMS half-weights $7^{-1/2}$ per leg / $n^{-1/2}$ universal. *Certified at finite level:* CP/unitality (exact Choi, rank 105, eigenvalue 1/7); σ/deck/half-weight/ $\beta = 1$ covariances exact; the Choi census on the generating family exact (integer Gram factorization); the four automorphic channels of the one map land on the deployed GL1 Weil window (machine precision) and the χ_4/f_8/twist $\Gamma_{\mathbb{R}}$-rule windows (PSD on $X = 4..10$); the ramified conductor emptiness structural. *Demands (open):* D1 the X-compatible dilation family and its inductive limit (with the `v794` margin trends as the honest obstruction data); D2 normal extension on the Z1-COMPACTNESS limit object (`v780`); D3 the limit identification of the GL1 channel with the critical-line Weil functional — *which contains RH and is not claimed. [2026-08-06 round-22 note, D1–D3 unchanged: the intertwiner gains its register chart at the odd places — the affine Gray map, exactly deck-covariant with trivial* μ_4*-cocycle ([verification/v804_prime_carrier_gray.py]) — and its* 105 *Kraus legs are a complete context protocol over the doily with the exact commutant*

$[B_{45}, B_{60}] = 0$ *([verification/v811_prime_kraus_doily.py]).]* *Kill:* a finite positive packet whose required GL1 intertwiner fails the exact Choi census structurally. No marker move; no RH statement.

The Gray register chart (PRIME.CARRIER.GRAY.01). *Registered* 2026-08-06 ([verification/v804_prime_carrier_gray.py], 23/23, GRAY-INTERTWINER-EXISTS; one declared control repair carried verbatim). *Object:* the affine Gray encoding of $C_2 \times \mu_4$ into the R-class torsor, exactly respecting the v786 Packet480 kill (a group isomorphism is impossible — exponent 4 vs 2 — and not demanded). *Certified:* deck covariance exact with *trivial* measured μ_4-cocycle; the residual a pure $\mathbb{Z}_3$ line twist on the σ-fixed classes; the extended Stinespring GL1 channel re-lands on the deployed Weil window at rel 6.0×10^{-16}. *Demands:* D1 odd-place fiber charts; D2 the $\mathbb{Z}_3$ line twist as crossed-product structure; D3 the infinite-level CP extension (chained to PRIME.CP.INTERTWINER.01). *Kill:* a nontrivial cocycle under the frozen chart. No marker move; no RH statement.

The Kraus context protocol (PRIME.KRAUS.DOILY.01). *Registered* 2026-08-06 ([verification/v811_prime_kraus_doily.py], 21/21, KRAUS-DOILY-PROTOCOL; one declared report-text repair carried verbatim). *Certified:* the 105 Kraus legs = 15 diagonal +90 off-diagonal over the doily $W(3, 2)$; the canonical $45 + 60$ = reversal quotient (2:1 over the 45 flags) + fiber expansion ($15 \times 4 = 60$ = the v783 stabilizer rays); the $Sp(4, 2)$ obstruction certifies no leg-subset partition (the literal partition is spread-relative = the ledger T3 census); the 45 transports quadratic ($P_x P_y = \pm P_{x+y}$), spanning $\mathrm{End}(\mathbb{C}^4)$ at rank 16; the surprise $[B_{45}, B_{60}] = 0$ exactly — $K = \frac{3}{7}E + \frac{4}{7}R$ a direct-sum protocol with E a conditional expectation. *Measured:* the certified spread is *not* σ-invariant. *Named falsifier:* a σ-invariant spread reproducing the commutant split. *[2026-08-06 round-23 addenda ([verification/v815_kraus_spread_commutant.py]): the falsifier pair is decided — σ acts on the six spreads as two 3-cycles with* zero *fixed spreads (outer-class forced; no invariant coarsening), but the direct-sum protocol is σ-stable as a* family *($\sum_s B_{45}(s) = 4I + 2B$; per-orbit A_{orb} spectrum $\{0^5, 1^2, 4^5, 7^2, 9^1\}$); $[E, K] = 0$ extends exactly to channel/KMS-tower/odd places (the unique C_2-character expectation per place) and the* 600*-dim register; the $\{K, \sigma\}$ commutant is 39-dim nonabelian with A_{orb} its canonical abelian 5-dim subalgebra (PRIME.AORB.REFINEMENT.01), window-silent beyond the K-spectrum; and the rule-A protocol set is canonically the* 105 *minimal dual checks of* $[15, 10, 4]$ *([verification/v820_prime_kraus_rm24.py], PRIME.KRAUS.RM24.01).]* No marker move; no RH statement.

The Lorentz spinor coordinates (PRIME.LORENTZ.SPINOR.01). *Registered* 2026-08-06 ([verification/v807_lorentz_nullselector.py], 22 checks with the three honest h-order FAILs encoded, LORENTZ-COORDS-REVEAL). *Exact:* $M(5, -3, 4) = 2\,(1, 2)^T(1, 2)$ — the compiler triple is a rank-one spinor square, Euclid $(2, -1) \mapsto (5, -3, 4)$. *The reveal:* the locking slope is strictly monotone in the window width α (66/66, Kendall 1.000, cross-ratio contraction CV 0.004) — the h-oscillation was α-jitter; the approach $2 - r \sim \alpha^{-0.23}$ glacial. *Named candidate:* the 1D monotonicity statement in α-order (the three h-order gates stay honest FAILs). *Kill:* a Kendall inversion under the frozen windows. No marker move; no RH statement.

The local-Clifford kill and the mod-8 completion (HECKE.LOCAL.CLIFFORD.01). *Registered* 2026-08-06, the analytic ordering *dead* as frozen ([verification/v806_hecke_local_clifford.py], 16/18 with the two FAILs the preregistered decider gates, controls 4/4, LOCAL-CLIFFORD-BREAKS; one declared bar re-freeze). All four p mod 8 class sectors fail PSD (signed class indicators carry no completed L-function; the trivial mod-8 component imprimitive); the margin ordering does *not* track the proven activation depths (Spearman +0.40); class 1 the unique positivity-completable class (Euler repair exact). The arithmetic multirate theorem untouched; the Clifford-activation reading decorative analytically. *Positive side-result:* χ_8/χ_{-8} are two new PSD instances of the $\Gamma_{\mathbb{R}}$ conductor rule (exact parities) — the rule map covers the full mod-8 dual group. *Next falsifier:* primitive sector data, not class indicators. No marker move; no RH statement.

The flavor-filtration fingerprint (FLAVOR.GRAPH.FILTRATION.01). *Registered* 2026-08-06 ([verification/v812_flavor_graph_filtration.py], 16/16 incl. the four must-fire controls, FILTRATION-

FINGERPRINT-ONLY). The $Q \to K \to L$ row budgets are exact, order-exact counts of the $K_{3,2}$/prism/K_5/H_8 incidence chain with level-wise additivity mirroring $L = K + Q$; *no* entry-level reconstruction from any frozen candidate; exhaustive false-positive rates 0.005–0.75% make the misses informative. Typed audit fingerprint — beside the audit identities, in no derivation. No marker move.

The P1 index-KMS identity (`P1.INDEX.KMS.01`). *OPEN* (registered 2026-08-06; [`verification/v813_p1_index_kms.py`], 13/13 incl. the two must-fire controls, P1-IDENTITY-CLOSED). *The identity:* $I \cdot \beta_{\text{angle}} \cdot c_3 = 4 \cdot 2\pi \cdot \frac{1}{8\pi} = 1$ exact in formal arithmetic, each factor independently sourced — the Jones index $I = 4 = |\mu_4|$ a c_3-free theorem (zero occurrences gated in all four sources), $\beta_{\text{angle}} = 2\pi$ independently measured (`v526`; the radian pin is the one shared convention and cancels), c_3 the P1 axiom the identity *explains*, not derives. *Measured precursor:* $w_1 = w_3 = \frac{1}{4}$ exactly at every level with symmetric zero-sum offset $\pm 2^{-(l+1)}$ (the no-anomaly shape); the sea state does *not* equidistribute (dev 0.188). *The one open bridge:* the even modular-response normalization over the I deck sectors. *Kill:* any asymmetric sector offset. *Control:* index 2 gives $1/(4\pi)$, booked elsewhere by the corpus. P1 stays a declared input; no marker move.

Mosco selection (`PRIME.MOSCO.SELECTION.01`). *Registered* 2026-08-06 ([`verification/v816_prime_mosco_selection.py`], 23/23 guards+controls, 6/6 legs, MOSCO-SELECTS). *Statement:* selection by Mosco form convergence and Friedrichs minimality on the `v762` dense core — the window-form chain q_X has exact nested recovery (structural, 1.2×10^{-12}), vanishing liminf defect (measured; the moving-edge family confirmed a genuine boundary channel, not an escape), Cauchy Friedrichs resolvents and Weyl functions *modulo the typed summable entry-pole ledger*, and cofinal-unique limits (2.5×10^{-5}): the unified compactness contract selects its state by form convergence, not by moment stability (moment baseline 22.08 reproduced exactly in-run). *Completed:* the import diagnosis at form level — the raw μ-weighted coupling of the core is exactly rung-independent, the whitened control reproduces the 5.8772 delivery failure exactly. *Remaining for theorem grade (typed):* (i) Mosco-precompactness of $\{q_X\}$ on V_∞, (ii) identification of the limit form domain, (iii) entry-pole ledger summability at infinity. Positivity/sector floor stays a separate contract. No marker move; no RH statement.

The positive-descent master theorem (`TFPT.POSITIVE_DESCENT.MASTER.01`). *Registered* 2026-08-06 ([`verification/v817_positive_descent_master.py`], 14/14 incl. the four must-fire controls, MASTER-BOTH-INSTANCES; kernel-checked core `TfptCarrier/PositiveDescentMaster.lean`, the matrix positivity closure in `GramCompactness.lean`, cited). *The theorem (seven frozen hypotheses):* finite operator systems with unital inclusions, positive states, CP transitions, finite character sectors, *summable* state defects, a sector-adapted closure datum, and a Mosco form family force a unique positive limit state with well-defined positive limit sectors; the core (telescoping + quantitative Cauchy tail; positivity closed; sectors commute) is kernel-checked. *Instance G_{net}:* hypotheses (1)–(5) measured (defects 0.267/doubling; the expectation CP via Choi; μ_4 evenness exact); open: normality/split. *Instance prime/packet — the new number:* state defects summable at 0.701/doubling $\approx 2^{-1/2}$ (the dilution law), with $\hat{m}_2(p) = -1/15$ exact for every odd prime; open: the sector floor (`PRIME.FLOOR.RATIO.01`, narrowed in round 24 by the certified floor skeleton `v823`/`v824` — battery-relative, α-bounded, still open). *Control K4:* summable defects with one non-positivity-preserving step give a non-positive limit — positivity is a separate hypothesis. The selection problem is eliminated on both sides. No marker move; no RH statement.

The floor as one ratio inequality (`PRIME.FLOOR.RATIO.01`). *OPEN* (registered 2026-08-06 — the reduced sector-floor contract; evidence [`verification/v818_sector_floor_attack.py`], 9/13 + 13/15 with exactly the preregistered honest FAILs, FLOOR-MECHANISM-PARTIAL + CAPTURE-BOUNDED / INGREDIENT-II-THEOREM-SHAPE / AMPLIFIER-MECHANISM). *Statement:* the sector positivity floor $\inf_X \lambda_{\min}(T_{\text{GL1},X}) \geq 0$ reduces, up to the measured $O(1)$ interlacing capture (median 1.53, range 1.04–4.36; the 2-mode lock block an $O(1)$-faithful witness by Cauchy interlacing), to the *single* ratio inequality $\rho(X) = \tau(X)/\tau_{\text{pnt}}(X) > 0$ — the prime comb against its own smooth density — with the measured finite lower envelope $\rho \geq c\,h^{-3/2}$ (constant $c \approx 4.85$, envelope non-decaying, slope +0.14). *Certified/measured ingredients:* the direction $r(X)$ is owned by the density rotation law $dr/dt = -\kappa(w\cdot v_1)(w\cdot v_2)(1+r^2)/(\lambda_1-\lambda_2)$ (symbolic, sign hypothesis verified finite-level, 100% staircase sign-match, typed density-level by the Epstein control $s_E = 0.241$); the capture is angle-certified frame geometry ($\cos\theta$ median 0.990, scramble-robust — frame content, not arithmetic); the amplifier ρ is carried by the smallest prime powers (packet correlation +0.964, partially trend-driven, typed). *Kill:* the envelope failing at depth ($\rho h^{3/2}$ decaying below every positive constant) or the capture angle collapsing ($\cos^2\theta < \frac{1}{2}$ at any rung). Selection and uniqueness are eliminated on both faces (`PRIME.MOSCO.SELECTION.01`, `TFPT.POSITIVE_DESCENT.MASTER.01`) — the floor is the single remaining analytic object of the prime programme. Interlacing runs floor ⇒ margin (necessary, not sufficient); promotion of any ρ bound is *fenced* on the preregistered envelope + capture gates. *[2026-08-06 round-24 narrowing — what is now closed:* on the deployed 14-rung battery the lock-block floor $\det\hat{A}_2 = \lambda\tau$ has a complete three-piece uniform certified lower-bound skeleton — the *exact* Lagrange sum-of-squares identity over zero+pole rank-one carriers (machine wards $\le 1.4\times10^{-9}$), the analytic fixed-pair bound $X^2_\infty(\alpha)(1-C(\alpha)/h)$ with explicit $h_0(\alpha)\le 500$ (h-uniform at every deployed α in-domain), the certified top-100 family exhaustion (0.93–0.97 of the floor, all carriers on-line $\le 2\times10^4$), and the psd remainder closed at *citation* grade (product-sup envelope + Abel/RvM explicit constants; $T_{\mathrm{ver}} = 3\times10^{12}$) for all h at the deployed α, with an h-free growth law ([`verification/v823_prime_lagrange_floor.py`], [`verification/v824_prime_floor_skeleton.py`]; `PRIME.FLOOR.LAGRANGE.01 / PRIME.FLOOR.SKELETON.01` below). *What remains open — the contract:* the floor bound is *battery-relative* and *α-bounded* — (i) the citation-grade chain re-crosses X^2_∞ at the validity horizon $\alpha^*\approx 11.2$ (deeper α needs verified on-line *depth* beyond T_{ver}, never location); (ii) the fixed pair dies at the α-nodes of $\sin(\alpha\gamma_1)$, where the family limit (positive, minimum 1.62×10^{-4}) must take over — the family version of the analytic bound is not yet written; (iii) the product-sup Lipschitz certificate is a named elementary step, typed, not claimed; (iv) *nothing bounds* $\inf_X \lambda_{\min}$ *on* V_∞ — the $O(1)$ capture and the $\rho>0$ demand at infinity remain exactly this contract. Kill criteria unchanged.] *[2026-08-07 round-25 depth-kill + budget note:* the contract's *own* kill gates were played at full sieve depth $X = 18.375$–25.5 (comb caps to 1.2×10^{11}; the deep uniform-grid frames decouple depth from dimension) and *survived* — envelope margins $\rho h^{3/2}/c = 5.76$–8.92 growing, capture minimum $\cos^2\theta = 0.849$, non-decay trend +0.060 per log-unit: the h-law wins over any hidden depth law ([`verification/v829_prime_floor_depth.py`]); the single fixed pair collapses at depth (gap 40× → 673×, the typed S5.HLAW miss) and the certified top-100 family takes over (0.977–0.981 of the floor, ρ_{certfam} within 2% of measured). The round-24 $\alpha^*\approx 11$ citation horizon is *removed*: it was an artifact of the frozen quadratic float convention — under the explicit Higham-linear budget the citation-grade family gate closes 6/6 to $\alpha = 12.75$, full sieve depth ([`verification/v830_prime_float_budget.py`]; the deployed v818 convention stays frozen in the suite, the linear budget is convention for *new* modules only). The envelope constant is now *certified-explicit*: $\rho \ge \rho_{\mathrm{certfam}} \ge 4.335\,h^{-3/2}$ on 73/73 battery points, non-decaying, with the family closed form exact (3.6×10^{-11}) and phase conspiracy excluded (dip floor 0.170). The remaining proof blocker is *named*: the alias phase correlations (random-phase predicts $\tau\sim h^{-1}$; the $D = 1/64$ tower measures $h^{-2.5}$) block the analytic uniformity — the envelope stands certified-explicit, not proven. Kill criteria unchanged; still [O].] *[2026-08-07 round-26 closure note:* the analytic-envelope thread of this contract is *closed at its typed circularity boundary* ([`verification/v831_prime_alias_second_moment.py`], ALIAS-CORRELATION-PAIRCORR, the seven preregistered-honest premise-overturning FAILs pattern-gated). The round-25 blocker resolves into the boundary itself: the premise "the phases carry the gap" is *overturned* — the amplitudes carry it ($-2.505 = -3.172 + 0.668$ exactly; $q = S_c/S_2$ sign-changing, deep suppression only ×1.1) — and the exact Guinand split puts the coherent sum in the prime-comb fluctuation at 0.01× the Poisson-comb square-root scale (smooth −994 / fluctuation +995, cancelling; the comb scramble explodes it 2×10^6–1.7×10^7). By Guinand the comb's self-cancellation *is* the zero-side floor statement itself — the required control is even deeper than Montgomery-type pair-correlation input. *No further analytic-envelope variants* (stop-list entry per house discipline); the envelope's constant is arithmetic substance equivalent to the target; the demand moves to `PRIME.FLOOR.PAIRCORR.01` below. Kill criteria unchanged; still [O].] *[2026-08-07 round-29 margin-law note:* the envelope constant is now *doubly derived* ([verification/v843_margin_law_excess_lean.py], MARGIN-RECURSES-THE-WALL): the identified-corner (relational) coordinates reproduce this contract's floor-strand law exactly — $e_1 = (\tau/\tau_{\mathrm{pnt}})\, h^{3/2} \in [4.855, 24.209]$ on all 67 rungs, min $\geq 4.335 = c_{\mathrm{cert}}$, with τ agreeing with $\lambda_{\min}(A_h)$ at 2.1×10^{-12} — two independently derived laws, one object: the consistency itself is a nontrivial ward. The typed law $\tau(\alpha, h) = e_1(\alpha)\, h^{-3/2}\, \tau_{\mathrm{pnt}}(\alpha)$ puts the entire α-decay in τ_{pnt} and the arithmetic in e_1; the fit-free internal h-exponent measures median -1.341 against the frozen bar edge -1.35 — the $h^{-3/2}$ law confirmed at $\sim 1\%$ grade, the 0.009 miss kept as the one preregistered-honest FAIL (not refit). The margin-law attack class (tower recursion / per-prime factorization of the excess) is measured dead and stop-listed per the cell-level self-similarity typed in `PRIME.RELATION.SKELETON.01`. Kill criteria unchanged; still [O].] No marker move; NO RH claim.

The floor as an exact sum of squares (`PRIME.FLOOR.LAGRANGE.01`). *Registered* 2026-08-06 ([verification/v823_prime_lagrange_floor.py], 7/8 with the one preregistered-honest old-budget FAIL + 9/9, LAGRANGE-CONCENTRATED + PAIR-CERTIFIED; discovery probes `prime_lagrange_pair_probe.py` and `prime_lagrange_budget_probe.py`). *Certified (identity-grade wards):* every per-zero layer of the `v692` master identity is exactly rank-one ($v_\gamma = 2\sqrt{w(\gamma)}(S_1, S_2)$, closed trigonometric reads, layer deviation $\leq 1.2 \times 10^{-16}$), and $\det(G_Z + P)$ is an exact sum of squares over the component pairs (Lagrange; ward $\leq 1.4 \times 10^{-9}$ on all 14 rungs); the pole is the universal non-collinear leg and the moving zero leg sits on the alias comb $\gamma \approx 2\pi k/D$. *Measured (per-rung, float-level):* $\lambda_{\min}(\hat{A}_2 - G_Z - P) \geq$ margin verified per rung eliminates the tail estimate — budget tightened a median 9.8 orders — and the certified interval $X^2(\mathrm{pole}, \gamma_1) - \mathrm{BUD}$ is strictly positive on 14/14 rungs, with the floor share growing as $h^{+1.06}$. *Honest typing:* the atom-side rank-one framing fails as frozen — the SOS realization lives on the zero+pole side; the psd of the remainder is verified numerically here, its citation-grade upgrade is `PRIME.FLOOR.SKELETON.01`. *Kill:* any ward failing on the frozen ladder. Feeds `PRIME.FLOOR.RATIO.01` (narrowed). No marker move; no RH statement.

The floor-theorem skeleton (`PRIME.FLOOR.SKELETON.01`). *Registered* 2026-08-06 ([verification/v824_prime_floor_skeleton.py], 12/12 + 9/9, FLOOR-PARTIAL + TAIL-CLOSED-ALL-H; discovery probes `prime_floor_theorem_probe.py` and `prime_tail_envelope_probe.py`). *Uniform-verified on the deployed battery:* the analytic fixed-pair limit $X_\infty(\alpha) = 16\alpha\pi^2 \sin(\alpha\gamma_1)\sinh(\alpha/2)[\mathrm{bracket}]$ with the explicit bound $L(h) = X_\infty^2(\alpha)(1 - C(\alpha)/h)$ for $h > h_0(\alpha) = C(\alpha) \leq 499.6$ (zero sign flips; 25 α-nodes typed — h-uniform per α, not ladder-uniform in α); the certified top-100 family exhaustion (median 0.952 of the floor, carriers on-line $\leq 2 \times 10^4$; residue 0.015–0.049, non-growing; the analytic family limit positive across all nodes). *The tail closure:* the product-sup per-zero envelope + Abel/RvM with explicit unconditional Trudgian-grade constants close the psd remainder at citation grade on *all 14 rungs* at the fixed verified horizon $T_{\mathrm{ver}} = 3 \times 10^{12}$ with margins 10^3–10^6; the growth law is h-free ($+1.98 \to +0.13$), the remaining growth α-only, validity horizon $\alpha^* \approx 11.2$ (the Vinogradov–Korobov region typed honestly useless). *Net:* $\lambda\tau \geq X^2(\mathrm{pole}, \gamma_1) - \mathrm{pert}(h) > 0$ at citation grade — no per-rung eigencheck, no zero-location input beyond the critical strip. *Honest boundaries:* battery-relative, α-bounded, necessary-side; the sup Lipschitz certificate a named elementary step; nothing bounds $\inf_X \lambda_{\min}$ on V_∞. *Kill:* any closure margin failing on the frozen battery at the frozen horizon. Feeds `PRIME.FLOOR.RATIO.01` (narrowed). No marker move; no RH statement.

The comb-native window detector (`PRIME.DETECTOR.WINDOW.01`). *OPEN* (registered 2026-08-07 — the window-verification contract of the exclusion-detector strand; evidence [verification/v825_prime_exclusion_ladder.py], [verification/v826_prime_exclusion_battery2.py], [verification/v827_prime_zero_locator.py], [verification/v828_prime_comb_window.py]). *Statement:* the verified PSD rungs of the GL1 prime-comb tower, inverted through the tent-read Guinand identity on a hash-preregistered battery (no zero datum in any design, warded end to end), form a *window detector*: on a window inside the Nyquist reach π/D the instrument *locates* zeta ordinates as prominent local maxima of the uncensored margin-break profile (out-of-sample validated:

83% detection, 0% false positives, precision 0.086 on the disjoint window $(60, 120]$), *excludes* all quadruple hypotheses with $\delta \geq \delta_{\mathrm{mb}}(\gamma)$ (witness-Rayleigh-certified on Cholesky/Higham-certified rungs, deepest $X = 25.5$), and *reconciles* the census against the cached verified count and the rounded RvM main term (COMB-VERIFIES-WINDOW). *Open:* (a) the depth-to-width law $\Xi_{v2}(X)$ (measured slope -1.39) as a theorem rather than a calibration; (b) a completeness mechanism — the locator misses pair-merges below the grid resolution and prominence-limited ordinates (depth law $54 \to 62 \to 83\%$, measured only); (c) window extension: every verified window lies inside the Platt–Trudgian strip and the Nyquist reach — the contract explicitly does *not* promise new zero-free regions. *Kill:* a validated out-of-sample window with detection $< 40\%$ at the frozen rule on a PD-certified rung deeper than $X = 24.81$; a census contradiction on any PD-certified window; a false positive surviving the grid-refinement ward. *Mandatory typing (every module of the strand):* location to ± 0.25 and exclusion to $\delta \geq \delta_{\mathrm{mb}}$ are strictly weaker than classical (Turing) verification on every axis; the strand is a necessary-side consistency demonstration of the explicit-formula reading and does not bear on RH. No marker move.

The depth kill and the budget (`PRIME.FLOOR.DEPTHKILL.01` / `PRIME.FLOOR.BUDGET.01`). *Registered* 2026-08-07 ([`verification/v829_prime_floor_depth.py`], 16/17 with the one preregistered-honest S5.HLAW FAIL + 12/12, ENVELOPE-HOLDS-DEEP + FAMILY-SCALES; [`verification/v830_prime_float_budget.py`], 10/10 + 19 checks with the five preregistered-honest asymptotic FAILs, BUDGET-LINEAR-CLOSES-ALL + ENVELOPE-ASYMPTOTIC-PARTIAL). The `PRIME.FLOOR.RATIO.01` kill gates survived at full sieve depth with growing margins; the certified family replaced the collapsing single pair; the Higham-linear budget removed the $\alpha^* \approx 11$ horizon (citation grade to $\alpha = 12.75$; the deployed v818 convention stays frozen, the linear budget is convention for new modules only); and the envelope is certified-explicit, $\rho \geq 4.335\, h^{-3/2}$ on 73/73, with the proof blocker named (alias phase correlations). Details in the dated note on `PRIME.FLOOR.RATIO.01` above. No marker move; no RH statement. *[2026-08-07, round 26:* the named blocker is now *typed*, not merely named — see `PRIME.FLOOR.PAIRCORR.01` below: the envelope-proof flank of the ratio contract is pair-correlation-equivalent substance, and the analytic-envelope route is closed at that boundary.]

The floor–pair-correlation bridge (`PRIME.FLOOR.PAIRCORR.01`). *OPEN* (registered 2026-08-07 — the honest typed closure of the analytic-envelope route; evidence [`verification/v831_prime_alias_second_moment.py`], ALIAS-CORRELATION-PAIRCORR, 15 checks with the seven preregistered-honest premise-overturning FAILs, pattern-gated). *Statement (the bridge):* the floor's envelope constant is *Guinand-equivalent to the zero-side floor statement.* In the pole frame the correlation-corrected alias second moment is exact (sympy-proven): $\tau = (S_2 - S_c)/2 - S_s^2/(4P_0) + O(P_0^{-2})$ with the coherent phase sum S_c multiplying the leading term; the measured facts are that the *amplitudes*, not the phases, carry the $h^{-2.5}$ tower gap ($-2.505 = -3.172 + 0.668$ exactly), and that S_c is carried by the prime-comb fluctuation $S_p - S$ sitting *at* the Poisson-comb square-root scale ($0.01 \times \sigma_{\mathrm{sqrt}}$; smooth and fluctuation shares $-994/{+995}$, cancelling; the comb scramble explodes the coherent sum $10^6\times$ — the cancellation is comb-specific arithmetic). Proving the tower τ-law is therefore variance-type control of an oscillatory prime sum at its square-root scale — pair-correlation-equivalent substance on the zero side by the explicit formula — and by Guinand the comb's self-cancellation *is* the zero-side floor statement itself. *Demands (both directions typed):* (i) *forward* — an unconditional variance-type bound $|\mathrm{aniso}(S_p - S)| \leq C\, \sigma_{\mathrm{sqrt}}$ on the deployed frames, uniform in h, from prime-side input alone, with the pair-correlation-grade entry point *declared* (no hidden circularity); (ii) *backward* — the calibration direction: convert measured floor margins into effective pair-correlation statements on the deployed battery (an instrument reading, never a theorem about zeros). *Kill:* a measured frame family where the suppression persists while the fluctuation share collapses below 0.6, or the scale ratio rising systematically above 3 (one-point regime — the boundary was mis-typed and the analytic-envelope route reopens on PNT-grade input). *Close:* the forward demand delivered with its entry point named and typed. *Stop list (house discipline):* no further analytic-envelope variants — the envelope constant is arithmetic substance equivalent to the target. The contract explicitly does *not* promise an unconditional proof; deciding the bridge in the proving direction contains pair-correlation-grade arithmetic (fenced). A boundary, not a theorem; no marker move; no RH statement. *[2026-08-07, round 27:* both corner-era routes at this wall are now decided (see the two entries below). The corner route's *structural* half is closed weight-generically — the character-corner identity is polynomial algebra in free weights and the Hjelmslev CP tower carries it verbatim through five levels at certified defect zero — but identity and state corner are comb-blind and the level-m motion is the pure register-dilution law 16^{1-m}: the wall *relocates into the identification step* and stays this contract. The commutant SOS route is closed definitively by exact certificates (unique trivial feasible point; forced $G = \mathrm{diag}(1, -1, -1)$ with dual point $q(0, 1, 0) = -1$). Both routes stop-listed; register-side Kraus dressings provably cannot carry the identification (POSITION-KRAUS-TRADEOFF, probe-level) — the one live tower door is position-dependent Kraus data on the level-m flags. Demands, kill and close conditions unchanged.] *[2026-08-07, round 28:* the wall acquires its full coordinate system. (i) *The four closure coordinates are complete* ([verification/v837_corner_closure_quantifier.py], [verification/v838_corner_expectation_position.py]): register closed (the 90-cell class map has no identity∧visibility∧placement cell), tower level-exact ($m \le 3$; jet characters position-blind), compressions closed (all 5276 subgroup expectations, the full pinching, Stinespring-side compressions — identity and new placement mutually exclusive cell by cell), and carrier-side position dependence *extremally pinned* ($\hat{c}_{\mathrm{GL1}} = -1$ locks the entire C_2 mass; identity-true carriers read zero on every self-consistent comb) — stop-list entries `PRIME.CORNER.OPENDOORS.01` / `PRIME.CORNER.LEVEL2.01` / `PRIME.CORNER.EXPECTATION.01` / `PRIME.CARRIER.POSITION.01`. (ii) *The zero side is quantified* ([verification/v839_paircorr_bridge_saturation.py], [verification/v840_gue_ablation_loopgain.py]): the demand in Montgomery $F(\alpha)$-form with the named minimal missing input (unconditional $F(\alpha, T) \le C_F$ out to $\alpha \approx 3$); the demand/GUE ratio saturates structurally at $R_\infty = 1.11$ (tight band unfolded $\alpha \in 1..2$, pinned across the admissible family); the bootstrap loop is short ($g = 1/(k^2 R^2 c_{\mathrm{sup}})$; even ideal supply gives $0.655 < 1$; Poisson ward passed) — the wall's self-conservation *is* the saturation, and the certified ladder doubles as a calibrated form-factor instrument ($\hat{F}(\mathrm{win}) = 0.44 \pm 0.14$). (iii) *The relation route opens* (the two registrations below): the identification carrier exists with the relational/multiplicative input, the identified corner's excess is positive on all 67 rungs, and the certified skeleton encloses τ_X strictly positively — the wall in its sharpest form, $\tau_X = \lambda_{\min}(\text{structural}) + \text{EXCESS}$, with the SHARPENED DEMAND: a uniform lower bound on the excess margin (a finer-than-statistical datum that no finite table settles). Forward/backward demands, kill and close conditions of this contract unchanged.] *[2026-08-07 round-31 transport note ([verification/v855_invariance_atlas.py]):* the pair-type object this registration names — the alias second moment / the form-factor band $\alpha \in 1..2$ — is *exactly* the bilinear chain fluctuation $d\mathrm{Chain}_2$ in Selberg-hierarchy coordinates (the demand concentrates there; the level-2 head is elementary-controlled by Selberg symmetry with on-range-verified constants) and the comb block's own soft direction at compression level (the minimizer tracking law, 0.0056 rad); the sieve input class reaches the band in *scope* (window-free, deterministic) but misses the *grade* by $205\times$–$1.1 \times 10^4\times$ — input-class-invariant, the same wall; still [O].] *[2026-08-08 round-32 note ([verification/v856_connected_current_lean.py], CONNECTED-COVARIANCE-PARTIAL):* the pair-type object now has an explicit *zero-free construction* — the centered KMS correlator of the Möbius commutator current is the pair-correlation form (its stationary part carries the deployed pair weights $\lambda_m^2 = \Lambda(m)^2/m$; the centered main term *is* the deployed smooth half-line subtraction, corr 0.99; the state is anchored to the critical line at 1/2, the wrong half-weight losing the pair-weight carrier), it *evades* the nuclear tax (relapse ratio ≈ 0.3 — genuinely different bookkeeping from the event-local completions), and the gap to the deployed form is the *grade gap* itself: the window form is linear in the comb and routing it through the correlator costs the variance, 2.5–$4.1 \times 10^5\,\tau$ — the alias second moment is reachable by construction, the floor demand is not payable from it. The instrument (zero-free, state-anchored, exact main subtraction) is typed for measurement use; still [O].] *[2026-08-08 round-33 note ([verification/v861_krein_normalform_lean.py] – [verification/v866_source_contractor_formula.py]):* the floor demand of this contract now reads, in

Krein coordinates, $\|C\| \le 1$ for the *explicit* closed-form source contractor $C = W_- F G_+^{-1} F^{\mathsf{H}} W_+$ (Loewner form $G_- \preceq G_+$; the τ-margin is the *single* soft defect mode's contraction margin — DEFECT-ONE), the scalar shadow is the pole-port impedance with the closed-form Poisson/-Cauchy a-term at $s = 1/2$ and $\kappa - 1 \approx 6e^{-\alpha/2} \to 1$, and the absolute-bound route is *provably* closed (Perron $\rho(|C|) > 1$ everywhere): the bridge input this contract demands must control the *phases* of the pair kernel, not its magnitudes — the pair-correlation substance in its sharpest operator clothing (`PRIME.KREIN.CONTRACTOR.01`); still [O].] *[2026-08-08 round-34 note ([verification/v869_cdcore_clarkphase.py] – [verification/v872_damping_compensation.py]):* the wall now reads as *one* two-term inequality at *one* named direction, both terms source-built — the exact split $I - C^*C = T_1 + T_2$ (the indefinite cross-measure Jacobi geometry plus the PSD Christoffel damping, warded at 10^{-12} on all 42 rungs) makes the floor demand $v^{\mathsf{H}} T_2 v > |v^{\mathsf{H}} T_1 v|$ at the soft direction (kz 9: $-0.0328 + 0.0330 = 1.68 \times 10^{-4} = \tau$; $s = \kappa\tau$ with the certified bulk floor at 98–100% of λ_2), and the kernel behind both terms is *named*: the Christoffel–Darboux kernel of the explicit source measure — one Jacobi chain plus arm weights, the controls' chains $O(1)$ different. The pair-correlation substance this contract types is now the deviation of the source Jacobi chain and Christoffel weights from their free counterparts, read at a single named direction; the classical escapes are stop-listed (pole transformations at x_{pole}, finite-rank cross-defect corrections, μ_4-offset Clark kernels); the preregistered next test is `PRIME.PRUEFER.COMPENSATION.01` (see the contractor block); still [O].] *[2026-08-08 round-35 closing note ([verification/v873_pruefer_cotlar_decision.py], the executed preregistration):* the Cotlar decision is in — the round-34 frozen contract was executed under the full provenance chain (committed pre-execution in `526ca3eb`; two documented mechanical fixes with the spec literal SHA byte-identical throughout; the honest FAILs pattern-gated, nothing refit) and returns COTLAR-GROWING by its own rules: the blockwise Cotlar–Stein / phase-cell class on source-only Prüfer phases is *closed* and stop-listed — the battery is flat (S_U 5.55–6.30) but the frozen deep holdouts explode (kz 142: S_U 32.07/S_C 58.78; slopes 0.209/0.408 $\ge$ 0.20, both channels GROWING; kills 2 + 5 fire). The demand narrows: what survives is a *non-stationary* phase bound on the explicit kernel — the predeclared π-resonance cells $d = 7$–8 are the known critical structure (the envelope is bimodal exactly there), and the typed growth law constrains every candidate: the explosion tracks kz-*depth* (α), not window size h (kz 90/116 at $h = 1430/1433$ stay in band; kz 177 at smaller $h = 1219$ explodes) — a uniform certificate must be α-aware. The carry-forward assets: the danger geometry is real (source-only cells capture 0.98–0.61 of $\lambda_{\min}(T_1)$ without the soft eigenvector) and the discrimination stands (Epstein 3927%, scramble 314462%); still [O].] *[2026-08-08 round-36 correction note ([verification/v877_complete_comb_reversal.py]):* the round-35 note's holdout data was comb-truncated (`ATOM_MAX` artifact; complete-comb floors strictly positive) — on complete combs the identically frozen rule returns BOUNDED, the class closure and stop-list entry are amended, the α-not-h law is retired, and the route reopens (`PRIME.PRUEFER.COMPENSATION.02`); the floor demand in embedding coordinates is now the registered target `PRIME.CARLESON.PRIME.01` [O] (the Carleson embedding of $\tilde{\nu}_-$ into $\tilde{\mu}_+$ at degree $h{-}1$ with constant $1 - \tau_h$, machine-exact reduction [verification/v876_carleson_polygon_wave.py]); still [O].]

The character corner and the Hjelmslev tower (`PRIME.CORNER.CHARACTER.01` / `PRIME.HJELMSLEV.CPTOWER.01`). *Registered* 2026-08-07 ([verification/v835_corner_hjelmslev_tower.py], $21/21 + 20/20$, CORNER-IDENTITY-SYMBOLIC + HJELMSLEV-STRUCTURE-ONLY). *Certified/exact:* the corner identity $T_{\mathrm{GL1},X} = V^* e_{\mathrm{GL1}} \pi(P_X) e_{\mathrm{GL1}} V$ as polynomial algebra in free event weights (symbolic toy leg exact; all three deployed windows on two independent code paths); $\hat{c}_j = -1$ exactly for all 136 events, consuming only the glue identity $\sum_m \Theta_m = 240\,\sigma_3$ and the uniform H^* mass; e_{GL1} a rank-one projector, corner compression single-Kraus CP; the chain-ring ladder $\mathbb{Z}[i]/(1+i)^m$, $m = 1..5$, with exact counts $15 \cdot 8^{m-1}/35 \cdot 16^{m-1}/105 \cdot 32^{m-1}$, strictly projective CP channels (certified cb defect identically zero) and level-exact corner coefficients. *Typed (the decision):* identity and state corner are comb-blind (scramble control); the level-m corner motion is the pure register-dilution law 16^{1-m} — no register/geometry lift carries the prime placements; the arithmetic wall relocates into the identification step and stays `PRIME.FLOOR.PAIRCORR.01`.

Register-only tower lifts are stop-listed; the live door is position-dependent Kraus data on the level-m flags (register-side dressings provably cannot carry it: POSITION-KRAUS-TRADEOFF, 13/13, probe-level). *Kill:* any level breaking the exact counts/defect-zero wards, or a register-only lift moving the comb-specific gap. No marker move; no RH statement.

The commutant SOS closure (`PRIME.COMMUTANT.SOS.01`). *Registered* 2026-08-07, closed at registration ([`verification/v836_commutant_sos_closure.py`], 29/29, COMMUTANT-SOS-INFEASIBLE; the declared SPEC v1→v2 witness-triple repair documented, intent/bars/verdict rule unchanged). *Certified/exact:* the five commutant coordinates $\Pi_A(l)$, $l \in \{0,1,4,7,9\}$ (exact Lagrange projections, traces $\{5,2,5,2,1\}$, structure constants $\Pi_a\Pi_b = \delta_{ab}\Pi_a$ on all 25 pairs, $\dim C(\{K,\sigma\}) = 39$ exact); the reduction lemma (abelian conjugation reads only Gram diagonals), collapsing the SDP to a rational LP with the *trivial diagonal rewriting* as unique feasible point (Vandermonde kernel $\{0\}$) — no cross-sector positive transfer; the flag-space Gram lift forced to $G = \mathrm{diag}(1,-1,-1)$ (not PSD) with the rational dual certificate $q(0,1,0) = -1 < 0$ and achievable q-negative directions (including 832/2799 nonneg-cone truncation cuts). *Typed:* FENCE-CLEAN on the algebra layer, FENCE-HIT on the positivity branch — the nontrivial version of the route is pair-correlation substance (`PRIME.FLOOR.PAIRCORR.01`). Route closed; stop-list entry. No marker move; no RH statement.

The corner-closure quantifier (`PRIME.CORNER.OPENDOORS.01` / `PRIME.CORNER.LEVEL2.01` / `PRIME.CORNER.EXPECTATION.01` / `PRIME.CARRIER.POSITION.01`). *Registered* 2026-08-07, closed at registration (stop-list entries; [`verification/v837_corner_closure_quantifier.py`], 19/19+17/17+3, DOORS-CLOSED + LEVEL2-CLOSED; [`verification/v838_corner_expectation_position.py`], 13/13 + 13/13, EXPECTATION-CLOSED + POSITION-CARRIER-TRADEOFF). *Certified/exact:* the 18-class collapse of all 128 characters (product law verified on the full carrier Fourier, exact Fractions); the generalized mass ward (the character corner reads one position-blind carrier mode per event); the level-m duals and polar annihilators ($m \leq 3$; identity characters exactly $\mathrm{Ann}(\langle S_m\rangle)$, Vfac spectra $\{1,0,-1/7\}$); the full subgroup-lattice census (5276 subgroups, 74 259 components; the pinning criterion $\mathrm{Ann}(H) \subseteq \{1,\psi_0\}$ exact); the pinching completeness; the exact tower address (period 4 on rational integers). *Typed (the closure):* identity ⇒ not visible in every cell of every class map; the only position-reading slot is exactly the slot the identity freezes; the deployed-form identity is an *extremal* state condition (EXTREMAL PINNING) — no state-preserving, identity-true carrier map $F(n,u)$ separates self-consistent combs. The structural half kernel-checked (`TfptCarrier/SectorPositiveDescent.lean`; `IdentificationPositivity` the single named hypothesis). *Kill (of the closure):* a cell of any measured class map carrying identity, visibility and placement together, or an identity-true state carrier separating the self-consistent null. The wall stays `PRIME.FLOOR.PAIRCORR.01`. No marker move; no RH statement.

The bridge map, the saturation and the loop (`PRIME.FLOOR.BRIDGEMAP.01` / `PRIME.FLOOR.GUESAT.01` / `PRIME.FLOOR.GUEABLATION.01` / `PRIME.FLOOR.LOOPGAIN.01`). *Registered* 2026-08-07, evidence lines feeding `PRIME.FLOOR.PAIRCORR.01` ([`verification/v839_paircorr_bridge_saturation.py`], 9/9 + 7/7, BRIDGE-FULLY-BEYOND + GUE-SATURATING; [`verification/v840_gue_ablation_loopgain.py`], 10/10 + 8/8, SATURATION-STRUCTURAL + LOOP-SHORT; scholarship correction throughout: Montgomery's $F(\alpha)$ on $|\alpha| < 1$ is RH-conditional, every supply row tagged). *Measured:* the demand ledger in Montgomery $F(\alpha)$-form on all 73 frames; the two-tier gap (unconditional supply misses by 52×; conditional window covers 43–86%; full-GUE rms closes 39/73); the named minimal missing input ($F(\alpha,T) \leq 0.33$–1.7 out to $\alpha \approx 3$); the saturation plateau $R_\infty = 1.11$ inside the frozen band, structural across the Guinand-admissible family and pinned in the *unfolded* coordinate; the calibrated form-factor instrument $\hat{F}(\mathrm{win}) = 0.44 \pm 0.14$ at $T \leq 2 \times 10^4$; the loop-shortfall law $g = 1/(k^2 R^2 c_{\mathrm{sup}})$ with headline $g(k{=}1) = 0.529$ and the ideal-supply bound $1/R^2 = 0.655 < 1$; the Poisson circularity ward. *Typed:* necessary-side consistency and instrument calibration only — the saturation is the strongest form of comb-zero bridge consistency a necessary-side battery can deliver, the loop is *not* a proof architecture, and the measured positivity everywhere is finer-than-statistical information. *Kill:* a Guinand-admissible variant landing off-plateau with positivity intact (the structural reading fails), or the instrument's Poisson control drifting from 1. No marker move; no RH statement.

The relation carrier (`PRIME.RELATION.MULT.01` [O], carrying `PRIME.RELATION.LADDER.01`). *Registered* 2026-08-07, OPEN ([`verification/v841_relation_carrier_ladder.py`], 18/18 + 13/13, RELATION-CARRIER-EXISTS + EXCESS-NONNEGATIVE). *The finding:* the identification carrier *exists* with the relational input — the first structure in the program to pass all four gates (state, identity at truth by degeneration, placement, and the self-consistency null that killed every compression and every single-event carrier). *Certified/exact:* $\mu * 1 = \delta$ and $\Lambda = \mu * \log$ as exact structure in the truncated convolution frame (the relational input *is* the Euler-product datum); the four-comb discriminator (true comb 0 exactly; $\zeta_{\mathbb{Q}(i)}$ and χ_4 blind — Selberg-class-correct; the $h = 2$ Epstein comb $x^2 + 5y^2$ separates at 200.0 on the class-group obstruction sites); the amplitude wiring (L-safety automatic); support and identity wards on all 67 reachable rungs. *Measured:* the identified corner's excess over its comb-blind structural skeleton is positive on all 67 rungs (+2.29 to +3.70) while τ_X decays to 1.7×10^{-5} — the sharpest wall form $\tau_X = \lambda_{\min}(\text{structural}) + \text{EXCESS}$. *Demand (sharpened):* a *uniform* lower bound on the excess margin along the ladder, with its arithmetic entry point declared (`PRIME.FLOOR.PAIRCORR.01` discipline binding — no hidden circularity); interim: amplitude-grade wiring for combs with claimed coefficient data beyond the four frozen test combs. *Kill:* a reachable rung with certified-negative excess at the true comb, or an L-function comb falsely flagged by the amplitude reading. *Close:* the uniform excess bound delivered and typed. Explicitly *not* promised: positivity along the ladder, or any RH-relevant bound. No marker move; no RH statement.

The certified skeleton (`PRIME.RELATION.SKELETON.01` [O]). *Registered* 2026-08-07, OPEN ([`verification/v842_excess_certified_skeleton.py`], 9/9, SKELETON-CERTIFIED). *Certified:* strictly positive interval enclosures of τ_X on *all* 67 reachable rungs ($\alpha = 2.773$–6.304, masses to 3×10^5; widths 5.4×10^{-11}–6.6×10^{-9}, three to five orders below the margin; chol_cert discipline above the deployed float data; certified midpoints reproduce the float ladder at 4.9×10^{-13}); the certified discriminator (the $h = 2$ fake fails its margin certificate on every anchor, routed-corner enclosure strictly negative, exact rational Λ_F routing). *Typed (the scaling law):* certifiability degrades faster than the margin decays (width slope +0.62 vs margin slope −0.52 per α), so the coordinate system has a finite certification horizon — with large headroom: $\alpha^* \approx 9.1$ (mass $\sim 7.5 \times 10^7$), far beyond the deployed ladder end; certifiability is not the binding constraint on any reachable rung. *What stays open (the wall in its sharpest coordinates):* the infinite quantifier over the sequence of strictly-positive certified enclosures — the sharpened demand of the relation route: a *uniform* lower bound on the excess margin, a finer-than-statistical datum by the GUE-side findings; no finite table, certified or not, settles it. *Kill:* a reachable rung whose certified enclosure at the true comb loses strict positivity under the frozen budget discipline, or a certified-positive enclosure for the $h = 2$ fake. *Close:* the uniform excess bound delivered with its entry point declared. Explicitly *not* promised: an unconditional proof. *[2026-08-07 round-29 closing note:* the quantifier is in its final coordinates ([`verification/v843_margin_law_excess_lean.py`]): the demanded object sits on the one scalar series $e_1(m) = (\tau/\tau_{\text{pnt}})\, h^{3/2}$, certified ≥ 4.335 on every reachable rung and doubly derived (the consistency ward); the wall is *self-similar at cell level* — the excess is itself a growing cancellation (CI median 15.6, rising 5.65 → 29.46; 22 287 of 351 122 per-prime aggregates nonpositive — no per-prime positive control); the tower gives scale, not recursion; the compiler connection is null (0/96). The finite-table limit is now *kernel-checked*: `TfptCarrier/ExcessSkeleton.lean` proves `pointwise_pos_not_uniform` (per-rung positivity at every rung does not yield a uniform bound) and the bridge theorem `excess_floor` consumes `UniformMarginBound` as the single named hypothesis — this registration's "no finite table settles it" is a theorem, not a typing. Same-class margin-law re-attacks are stop-listed per the measured self-similarity. Demand, kill and close conditions unchanged; still [O].] *[2026-08-07 round-30 closing note (the three-architecture decision):* the three v5.4 proof-architecture candidates aimed at this registration's demand are measured and typed *closed* at three walls that are one wall ([`verification/v846_schur_spectral_mother.py`], [`verification/v847_wedge_cellcone_transport.py`]): (i) the relational Schur–Gram route dies structurally at the forced Cauchy–Schwarz diagonal price (price/$\tau \sim 3.1 \times 10^4$–$4.5 \times 10^4$ at the density-anchored floor; trace *and* form; the untouched-projection deviation *is* the price), and the price is *geometry-independent* — the unitary spectral mother ($J^2 = I$; $U_d U_m = U_{dm}$ exact) gains 20–33× at symbol grade but never at Gram grade (the modulus bound); harvesting interference is Fejér–Riesz of the total symbol = the positivity itself; (ii) the sign-register wedge lift *exists* as exact algebra (nonnegative wedge weights per frame, where the naive signed Lagrange provably cannot) but the frame-uniform law fails at ∼15% against the 4.4×10^{-5} floor need — the commutant uniformity wall transported; (iii) the completed-cell Lorentz transport breaks the cone at the $n = 2$ cell on 67/67 rungs under both predeclared groupings (endpoint back in-cone = the certified $\tau > 0$; ray cascade to $(5, -3, 4)$ confirmed, Kendall ±1.000), with the inverted Epstein signature kept as its frozen FAIL: the fake stays in-cone — the violations are the true comb's own $\psi(x) - x$ fluctuation; the arithmetic sits *in* the violations. *Stop list (extended):* diagonal relational mothers and manifest-Gram interference harvesting; constant (frame-uniform) wedge weight laws (the named next object is an h-dependent, analytically controlled law); unrenormalized small-n completed-cell transports (the named next object is a renormalized cell map absorbing the $n = 2$ fluctuation). The extraction side is now separated and complete (`PRIME.EXTRACTION.CHAIN.01` below): the wall is exactly hypothesis (H) on a ladder cofinal in the mesh-refinement order. Demand, kill and close conditions unchanged; still [O].] *[2026-08-07 round-31 closing note (the invariance atlas):* the wall this registration names is now proven *invariant* across four axes ([`verification/v850_completion_tax_flow.py`], [`verification/v851_cluster_kernel_field.py`], [`verification/v855_invariance_atlas.py`]): (i) *compressions* — the nuclear-norm tax theorem closes the entire positive-dilation completion class with fixed comb coupling (exact optimum $\geq 4057.6 \times \tau$ on every rung, diverging $\sim h^{0.24}$ against $h^{-3/2}$; zero-gap dual certificates; the measured v846 designs were suboptimal by only 13.7–20.4× — optimality could not have saved them), the finite-dimensional spectral-flow index is a reformulation of the endpoint pivot signs (distance-to-crossing = metric margin / flow velocity, ratio 1.00), and the 2-dim corner cluster expansion is blind to multiplicativity (position-geometric weights); (ii) *geometries* — the tax is basis-invariant ($\leq 1.1 \times 10^{-15}$): the round-30 TV-price geometry-independence is now a corollary; (iii) *criteria* — Weil-window = Li = Nyman–Beurling (BD-constant = $2\lambda_1$ at 4.9×10^{-17}; the NB span *is* the spectral-mother geometry; the Li transfer fails by cone geometry: certified partial data everywhere, one uniform statement missing everywhere); (iv) *input classes* — the sieve toolbox reaches the $\alpha \in 1..2$ band in scope but misses the grade by $205\times$–$1.1 \times 10^4\times$, and the demand at truth exceeds the factor-2 gate on every rung: unclosable by *any* upper-bound input class. *The one object in four languages:* the uniform e_1 bound = the form-factor band $\alpha \in 1..2$ = the bilinear chain fluctuation $d\mathrm{Chain}_2$ = the comb block's soft direction (the minimizer tracking law, 0.0056 rad). *Stop list (extended):* positive-dilation completions with fixed comb coupling, in any basis; finite-dimensional index/spectral-flow reformulations; 2-dim corner cluster expansions; elementary upper-bound supply rows against the factor-2 gate. The graded kernel law (`PRIME.CELLCONE.GRADEDKERNEL.01` below) is the round's positive core; the minimal hypothesis is now (H_{cof}) (`FORM.PRIME.COFINAL.WEIL.01` below — `UniformMarginBound` demoted to an over-strong sufficient lemma). Demand, kill and close conditions unchanged; still [O].] *[2026-08-08 round-32 note (the grade law):* the graveyard this registration accumulated is now *one* kernel-checked homogeneity theorem (`TfptCarrier/GradeNoGo.lean`, `FORM.PRIME.GRADE.NO_GO.01` [F]; numeric mirror [`verification/v859_grade_no_go_elevator.py`]): free positivity lives at *quadratic* grade (Gram squares of linear maps are 2-homogeneous), the deployed target at *linear* grade, and no exact match exists on any scaling ray — with the affine escape priced exactly (the representation error is the sum of two PSD prices) and the background killed too. The measured exchange rate is 10^4–10^5 across four instances: the Schur–Gram diagonal price (3.1–$4.5 \times 10^4\,\tau$), the class optimum ($\geq 4057.6\,\tau$), the covariance variance price (2.5–$4.1 \times 10^5\,\tau$, [`verification/v856_connected_current_lean.py`] — the first design that *evades* the nuclear tax, dying on the grade gap instead), and the wedge uniformity gap. The only licensed grade-1 mechanism is the *rates elevator* (`tangent_psd_on_kernel_null`, kernel-checked), and its deployed instantiation is verified real but the Lévy/conditional-positivity class is *closed on both deployed sides* by signed spectral densities (the comb at exactly 50% negative mass; the arch on the digamma band ending at $\tau^* = 6.27$ where $\mathrm{Re}\,\psi(\frac{1}{4} + i\tau/2) = \log \pi$). The finite-head phase-rescue family is closed by *theorem* ([verification/v860_phase_lever_closure.py]: private-best-phase budget margins ≥ 1.3894 / 0.6296 on all 67 rungs; the D-frontier saturates at $0.73 < 1$). *Stop list (extended):* finite-head phase rescues on the deployed frame family; amplitude-Gram constructions without grade accounting (exact matching of the linear window read by any Gram square, affine backgrounds included); Lévy/conditional-positivity elevators on the deployed sides. The rates door itself stays licensed and typed; demand, kill and close conditions unchanged; still [O].] *[2026-08-08 round-33 note (the wall's operator form):* the demand now has exact operator coordinates ([verification/v861_krein_normalform_lean.py] – [verification/v868_divisor210_audits.py]): the deployed window form is the Krein normal form $Q = \|B_+t\|^2 - \|B_-t\|^2$ with the Douglas pencil identity $1 - \|C\|^2 =$ the τ-margin, the defect has rank *one* ($\lambda_1 =$ the τ-margin exactly — one arithmetic-carrying soft mode), the contractor factors *exactly* as the target-free closed-form source expression $C = W_- F G_+^{-1} F^{\mathsf{H}} W_+$ (blind holdouts, certified defect transfer), the norm statement is the exact reformulation $\|C\| \leq 1 \Leftrightarrow G_- \preceq G_+ \Leftrightarrow K \succeq 0$, and the theorem frame is kernel-checked (`FORM.PRIME.KREIN.DEFECT.01` [F]: `SourceContractor` the named hypothesis, composed to Weil nonnegativity). What the campaign *proves* about the tool space: the absolute half of the classical toolbox is closed (the Perron value $\rho(|C|) = 2.4$–$3.1 > 1$ everywhere — no absolute-value inequality can ever certify: the certification must be *phase-aware*); quadrature does certify $s > 0$ on every reachable rung (the Neumann frontier killed) but the depth law $m^* \sim \sqrt{\mathrm{cond}(G)}$ is *intrinsic* (no source preconditioner waives it — the soft end is a continuum edge of distributed arithmetic structure); the scalar shadow is the pole-port impedance with the closed-form a-term and $\kappa \to 1$; the register is canonical up to *proven* gauge. *Stop list (extended):* source-word searches at all lengths (the monomial closure theorem: every word of any length misses by ≥ 0.87); absolute-value bounds on the weighted frame kernel (the Perron closure); fixed-depth quadrature certificate families without a conditioning bound; chirality deciders through quadratic readouts. The named demand is sharpened to `PRIME.KREIN.CONTRACTOR.01` [O] (below): one phase-aware bound on the closed-form kernel; still [O].] No marker move; no RH statement.

The margin law and the kernel-checked wall (`PRIME.MARGIN.LAW.01` + `FORM.PRIME.EXCESS.SKELETON.01`). *Registered* 2026-08-07 ([verification/v843_margin_law_excess_lean.py], 10 checks with the one preregistered-honest S2.2 FAIL kept and pattern-gated, +4 Lean-mirror checks, MARGIN-RECURSES-THE-WALL; Lean core `TfptCarrier/ExcessSkeleton.lean`, lake build green, no `sorry`, carried in the shipped Lean manifest). *Measured (the law):* $\tau = e_1(\alpha)\, h^{-3/2}\, \tau_{\mathrm{pnt}}(\alpha)$ with the envelope constant doubly derived ($e_1 \in [4.855, 24.209]$ on all 67 rungs, min ≥ 4.335); τ_{pnt} carries the entire α-decay, e_1 the arithmetic; the fit-free h-exponent median -1.341 vs bar edge -1.35 (the one honest FAIL, not refit). *Measured (the negatives, typed):* the tower gives scale, not recursion (dyadic-level ratio defects non-decaying); the excess is a growing cancellation — the wall self-similar at cell level (no per-prime positive control, no Euler-factorized positivity); the compiler connection is null (0/96, Bonferroni-honest). *Kernel-checked [F]:* the two-giants decomposition, the interval-certificate and certified-negative discriminator lemmas, the finite-ladder quantifier, `excess_floor` with `UniformMarginBound` the single named hypothesis (implying the per-rung `IdentificationPositivity` instances; the converse fails), and `pointwise_pos_not_uniform` — no finite certificate table discharges the uniform bound. *Stop list:* same-class margin-law re-attacks (tower-recursion / cell-factorization of the excess). *Demand (unchanged, final coordinates):* a *uniform* lower bound on e_1, with its arithmetic entry point declared (`PRIME.FLOOR.PAIRCORR.01` discipline binding). *Kill:* a reachable rung with $e_1 < 4.335 \times 0.999$, or a kernel-checked discharge of `UniformMarginBound` from structural input alone (which would contradict the typed boundary and force a re-audit). *[2026-08-07 round-31 demotion note (`FORM.PRIME.COFINAL.WEIL.01`):* `UniformMarginBound` is from this round *formally demoted to an over-strong sufficient lemma* in the load-bearing chain — the minimal H theorem (`TfptCarrier/CofinalWeil.lean`, numeric mirror [verification/v849_cfin_unique_cofinal_lean.py]) proves uniform $\subsetneq$ pointwise $\subsetneq$ cofinal kernel-checked in both directions, and the extraction chain consumes exactly the strictly weaker (H_{cof}): PSD at the rungs of one pre-fixed cofinal ladder, no uniform δ. This block's lemmas and `excess_floor` remain kernel-checked and load-bearing as the *sufficient* side; the named wall moves to (H_{cof}). Still [F]/[O] as stated.] No marker move; no RH statement.

The continuum extraction chain (`PRIME.EXTRACTION.CHAIN.01` [O]). *Registered* 2026-08-07, OPEN ([`verification/v848_extraction_chain.py`], 19/19, EXTRACTION-CHAIN-COMPLETE; hypothesis (H) is *never evaluated* — no positivity statement about any actual tower form or the actual Weil functional is made). *Measured:* on the canonical (D, X) ladder, cofinal in the *mesh-refinement* order ($D_j = 2^{-j}$, $j = 4..11$, 68–8704 lags, the full pinned prime-power table) every element of the canonical dense family identifies its independently computed Weil value Q_W = POLE+ARCH+ATOM (exact piecewise-cubic autocorrelations in $\mathbb{Q}$, fifth-point certificates; Suzuki normalization, W1-certified layers) with rates $\sim 4^{-j}$ (median -1.818 per level), a geometric ledger (median ratio 0.302, decay law -1.752), and the X-direction *exactly* stable once the window covers the element (44 cutoff pairs at float zero); the pure-box class carries the *exact* Weil value at every finite level (maximal error 5.3×10^{-14} — the sharpest dyadic-inheritance statement); recovery is constructive and Cauchy (the exact strong-L^2 law in $\mathbb{Q}$); the gated far-spike liminf family has strictly positive margins, while the band directions are typed and *never* gated — wall content under (H), refused by design. Both controls fire (the density-only wrong limit stalls at ratio 1.6×10^6; the scrambled tower inflates the ledger by 5.5×10^5). *The deliverable (the quantifier reduction):* under (H) — finite positivity of the tower forms along a ladder cofinal in that same mesh-refinement order — every ladder value is nonnegative and the limit passage is arithmetic-free: *no* Mosco compactness, *no* uniform $\delta > 0$, *no* diagonal argument enters the implication; per-element convergence is the only analytic input, and the three infinite-level ingredients of `PRIME.MOSCO.SELECTION.01` migrate to the operator-level selection programme, where they stay typed. With the classical legs cited (density; the continuity of Q_W at fixed support — *corrected* in the round-CCCLXV note below, the pure sup-norm C^0 form of this leg being false; Weil 1952, Bombieri 2000, Suzuki arXiv:2606.09096, Iwaniec–Kowalski Thm 5.12), the chain *cofinal finite positivity* ⇒ *Weil positivity* ⇒ (via Weil's criterion) *the target* is theorem-grade modulo the named citations. *What stays open (by design, the whole wall):* hypothesis (H) — equivalently the uniform demand of `PRIME.RELATION.SKELETON.01` restricted to a ladder cofinal in the mesh-refinement order — and *nothing else*. *Kill:* an element of the dense family whose ladder values converge to a value different from the independently computed Q_W, a non-summable ledger on the true tower, or a failure of the X-stabilization on a covered element. *Close:* any discharge of (H), which via this chain yields Weil positivity and, via the criterion, the target. Explicitly *not* promised: an unconditional proof. *[2026-08-07 round-31 note:* step 2 of the chain — the arithmetic-free limit passage under (H) — is now *kernel-checked* (`TfptCarrier/CofinalWeil.lean`: the $\varepsilon/2$ argument, the no-diagonal dense-family form, the assembly; axioms clean), and hypothesis (H) is formalized as `CofinalHypothesis` (H_{cof}), preregistration-shaped (the ladder is data, never mined from measured signs) and proven strictly weaker than `UniformMarginBound` (`FORM.PRIME.COFINAL.WEIL.01` below). The genuinely open item is unchanged: (H_{cof}) itself; still [O].] *[2026-08-13 round-CCCLXV note* ([`verification/v912_form_convergence_theorem.py`], 35/35, `PRIME.FORM.CONVERGENCE.THEOREM.01`): *two of the inputs above change type.* (i) Per-element form convergence — the "only analytic input" of the quantifier reduction — is no longer measured but *proven unconditionally* on the deployed dyadic cellwise-affine family, with the explicit rate $O(D^2 \log(1/D)) = O(2^{-2j} j)$ and explicit constants ($\Theta_0 = 3\log 2 + \pi/2 + \gamma + \log \pi$; log coefficient exactly 7/24 for continuous f); the load-bearing step is the exact defect lemma $k_d - K(dD) = -(D^2/12)G(dD)$, verified with zero deviation in exact rationals, and the only cited arithmetic input is Rosser–Schoenfeld $\psi(x) < 1.03883x$ for support uniformity. (ii) The continuity leg as cited above is *false* in the pure sup norm: the explicit even Lipschitz family $e_n(w) = \frac{1}{n}\min(1, we^{n^2})(1 - w/2)_+$ has $\|e_n\|_\infty \to 0$ while $|A[e_n]|$ grows linearly ($2.57 \to 12.09$). The correct hypothesis — uniform convergence plus an equi-Lipschitz (Dini) condition at the origin, supplied by the admissible even compactly supported BV class — replaces it here and wherever the contract cited it; the *density* leg in that topology stays cited. This is a correction of a wrong cited statement, not a downgrade. What stays open is unchanged: (H_{cof}), still [O], still of RH strength.] No marker move; no RH statement.

The finite-compiler compressions of round 30 (`MESSAGE.LADDER.01`, `DOILY.PASCAL.RANK.01`,

`NORMALFORM.CFIN.01`). *Recorded* 2026-08-07 (evidence lines, not contracts; exact identity grade; no marker moves). The message ladder $M_n = 15 \cdot 2^n = (15, 30, 60, 120, 240)$ with one named structure per rung and $h(E_8) = 30$ derived from the rebuilt root system, and the Doily–Pascal rank theorem $NN^\mathsf{T} = B + 2I$ with rank $N = 10 = A_\Lambda$, $\dim\ker N = 5 = g_{\text{car}}$ and recovery value $2/3$ ([`verification/v844_message_doily_rank.py`], 23/23 + 16/16); and the finite compiler normal form $C_{\text{fin}} = (V, \hbar, q^*, \sigma, \iota)$ assembled as one object with the code-to-matter kill built in and P2 narrowed to the `v799` residual R1′ ([C]), not eliminated ([`verification/v845_cfin_normal_form.py`], 28/28, SPEC v2). Full statements and keyboxes in Paper 1; audit cards in Paper 3. The $(1/3)^6$ typed gap of the six-step decision is re-affirmed with no upgrade.

The minimal H theorem (`FORM.PRIME.COFINAL.WEIL.01` [F]). *Registered* 2026-08-07, FORMAL (`TfptCarrier/CofinalWeil.lean`, 11 declarations, kernel-checked — lake build green at 3415 jobs, no `sorry`, no `native_decide`, axioms propext/Classical.choice/Quot.sound only, verified by `#print axioms` on all eight theorems; numeric mirror [`verification/v849_cfin_unique_cofinal_lean.py`] part B; carried in the shipped Lean manifest at 87 files). *Proven:* the cofinal hypothesis (H_{cof}) — a *pre-fixed* strictly monotone ladder, cofinal in the *mesh-refinement* order in which the form convergence holds and not in the window or cap parameter alone (at a fixed mesh the read is cap-independent, hence eventually constant, and its limit carries the frozen quadrature defect of that mesh), along which the tower matrices are PSD; preregistration-shaped: the ladder is *data* of the structure, supplied before any certificate, never mined from measured signs — and the minimal implication: `limit_nonneg_of_cofinal_seq` (the $\varepsilon/2$ argument at sequence level) and `weil_nonneg_of_cofinal` (per-element form convergence $+(H_{\text{cof}}) \Rightarrow$ the limit functional nonnegative on the *whole* dense family, with *no* diagonal argument — one PSD certificate per rung covers every element simultaneously); the strict hierarchy *uniform* $\subsetneq$ *pointwise* $\subsetneq$ *cofinal*, kernel-checked in both directions (witnesses $1/(m+1)$ — reusing `pointwise_pos_not_uniform` — and ± 1 on the even ladder); and the assembly `cofinal_weil`. *Absent by design:* no uniform margin, no inverse/resolvent, no Mosco coercivity, no limit operator, no diagonal argument. *The formal demotion (the round's status statement):* the load-bearing chain of `PRIME.EXTRACTION.CHAIN.01` consumes *exactly* (H_{cof}); `UniformMarginBound` implies (H_{cof}) and the converse fails kernel-checked — `UniformMarginBound` is from this round an over-strong *sufficient lemma*, no longer the named wall; the wall content has strictly decreased to cofinal PSD on one pre-fixed mesh-refining ladder. *The honest boundary:* (H_{cof}) stays a hypothesis everywhere; nothing proves, gates or evaluates positivity of any actual tower form; no continuum theorem, no RH statement, no arithmetic identification is formalized or claimed.

The graded kernel + tail theorem candidate (`PRIME.CELLCONE.GRADEDKERNEL.01` [O]). *Registered* 2026-08-07, OPEN ([`verification/v851_cluster_kernel_field.py`] part C, 7/7, verdict KERNEL-GRADED). *The candidate statement* (measured on the 67 deployed rungs under the G1/Stieltjes completion; *not* proven, *not* uniform in ε, not an RH claim): for every $\varepsilon > 0$ there exist finite $P_0(\varepsilon)$ and $X_0(\varepsilon)$ such that for every window $X \geq X_0(\varepsilon)$, every state of the completed-cell flow beyond $n = P_0(\varepsilon)$ has relative depth $-\lambda_{\min}/\lambda_{\max} \leq \varepsilon$. *Measured witnesses:* $P_0(10^{-1}) = 0$ on all rungs; $P_0(10^{-2}) \leq 73$, collapsing to $\{2, 3\}$ above $\alpha \approx 3.9$; the skin-depth law $D(X) = 0.754\, X^{-0.548}$ ($R^2 = 0.99$, Kendall -1.00) predicts $X_0(\varepsilon) = (\varepsilon/0.754)^{-1/0.548}$ ($X_0(10^{-3}) \sim 1.8 \times 10^5$); the strict ($\varepsilon \to 0$) census wanders with the horizon ($P_0 \sim X^{1.03}$ at relative position 0.98) — the wandering is vanishing-depth skin, *not* kernel growth; the G2 mechanism typed by contrast (centroid offset 1653×, running imbalance 39× — delayed compensation). *Honest scope:* "resolved" is an in-range statement; the $\varepsilon < 10^{-2}$ rows are extrapolated only through the fitted law. *Kill:* a rung family on which the depth-10^{-2} kernel grows without bound, a measured break of the skin-depth power law, or a G1 state beyond the predicted $P_0(\varepsilon)$ with depth $> \varepsilon$ inside the verified range. *Close:* a proof of the candidate statement for the canonical completed-cell flow — the graded weakening of the cone-transport demand; note it does *not* by itself give (H_{cof}): the graded statement bounds depth, not sign. No marker move; no RH statement.

The code-layer compressions of round 31 (`CFIN.UNIQUE.01`, `ARF.VACUUM.SYNDROME.01`, `OVOID.DECODER.01`, `ARF.BENT.CSS.01`, `PRIME.RELATION.MANGOLDT.01`). *Recorded* 2026-08-07 (evidence lines, not contracts; exact identity grade; no marker moves). The uniqueness theorem: every admissible compiler tuple is isomorphic to C_{fin} (census 14400, one $Sp(4,2) \times S_5$ orbit, $\mathrm{Aut}(C_{\text{fin}}) \cong C_6$; the strict terminal-object reading fails honestly — unique up to *non*-unique isomorphism; P2 stays narrowed to the `v799` residual R1′) ([verification/v849_cfin_unique_cofinal_lean.py], 22/22). The Arf–vacuum syndrome theorem and the ovoid decoder: the two-bit syndrome table 15/20/45/60 on the Hamming code with the local Hecke theorem $12/16 \to 28/{-4}$ at all 15 points, the code pencil with the canonicity lemma and the vacuum bit as the $\mathrm{RM}(2,4)$ parity bit; $\mathrm{SNF}(N) = \mathrm{diag}(1^{10}, 0^5)$ torsion-free with the integral decoder $N^+ = N^\mathsf{T}/4 - J/36$ and the S_6 standard representation; the three-ring correspondence ([verification/v852_arf_vacuum_ovoid.py], 24/24 + 23/23). The bent CSS pair and the commutator theorem: $\hat{s}_q = -4\, s_q$, the $(16,6,2)$ difference set, $[[15,5,3]] \to [[16,4,4]]$ with the vacuum transformation typed, the -4 triple point (Hecke = Gauss = Walsh = a_3); and $\mathcal{L} = -M[D,Z] = -[D, \log Z] = T(\Lambda)$ exact with the four-comb discipline at operator level and the carrier bridge (typed honestly as classical incidence algebra) ([verification/v853_bent_css_mangoldt.py], 22/22 + 30/30). Full statements and keyboxes in Paper 1; audit cards in Paper 3.

The grade barrier as a class closure (`FORM.PRIME.GRADE.NO_GO.01` [F]). *Registered* 2026-08-08, FORMAL (`TfptCarrier/GradeNoGo.lean`, 13 declarations (10 theorems), kernel-checked — lake build green at 3415 jobs, no `sorry`, no `native_decide`, axioms propext/Classical.choice/Quot.sound only, verified by `#print axioms` on all ten theorems; numeric mirror [verification/v859_grade_no_go_elevator.py] part A; carried in the shipped Lean manifest at 87 files). *Proven:* the core no-go `grade_no_go` — a 1-homogeneous map and a 2-homogeneous map that agree on a scaling ray agree with 0 at the base point (evaluate at $t = 1$ and $t = 2$: $2A(m) = 4A(m)$; two lines — a *class* closure, not a construction failure); `gram_two_homogeneous` and `grade_no_go_gram` (Gram squares of linear maps are 2-homogeneous, so exact matching annihilates target *and* factor: grade 1 cannot live inside grade 2); the affine trilemma — `affine_gram_expansion` (the exact split into constant + cross + quadratic components, the cross term the *only* grade-1 slot a Gram square owns), `affine_gram_tax` (the representation error against the pure cross term is *exactly* the sum of two PSD prices: the tax), and `affine_grade_no_go` (exact affine matching forces target, background *and* factor to vanish); and the elevator lemma `tangent_psd_on_kernel_null` — derivatives of PSD families at kernel-null directions are positively signed: *rates, not amplitudes*, the only grade-1 mechanism the no-go does not exclude, proven once, abstractly. *The unifying diagnosis:* the four graveyard instances (the Schur–Gram tax, the covariance grade gap, the wedge uniformity failure, the phase-lever closure) are instances or relatives of this one law; the measured exchange rate is 10^4–10^5. *Honesty notes:* the no-go excludes exact Gram matching on scaling rays and nothing more — denominator representations died separately by measurement (cited); non-Gram mechanisms are outside the class closure; the trilemma exhaustiveness outside the affine family stays an informal doc-level reading, typed. No continuum theorem, no RH statement, no arithmetic identification.

The conditional route to the cofinal hypothesis (`FORM.PRIME.COFINAL_CURRENT.01` [F]). *Registered* 2026-08-08, FORMAL (`TfptCarrier/CofinalCurrent.lean`, 13 declarations (7 theorems), kernel-checked — lake build green at 3415 jobs, no `sorry`, no `native_decide`, axioms propext/Classical.choice/Quot.sound only, verified by `#print axioms` on all seven theorems; numeric mirror [verification/v856_connected_current_lean.py] part C; carried in the shipped Lean manifest at 87 files). *Proven:* the conditional theorem `cofinal_current` — *ConnectedTail* (every tail block PSD) + *PositiveHeadCell* (an open phase cell with a uniform Rayleigh floor) + *PhaseRecurrence* (cofinal visits) $\Rightarrow$ (H_{cof}), with the index sequence *extracted* from the recurrence; the composed corollary `cofinal_current_weil` through the minimal H theorem: the three hypotheses + per-element form convergence force the limit functional nonnegative on the whole dense family — no hidden wall behind the two analytic lemmas; the $\delta = 0$ relaxation (the chain consumes only PSD on the cell; the strict floor is a stronger input than required, typed) and the Rayleigh reading; and the counterexample `connectedTail_not_sufficient` (a PSD-tail split with everywhere-negative head admits no cofinal ladder — the hypotheses are jointly necessary).

Provenance typed, not proven: hypothesis 1 is what the connected-covariance construction delivers structurally (`PRIME.RELATION.CONNECTED_COVARIANCE.01`); hypothesis 2 is what the phase-cell census *measured empty at the deployed frame* (`PRIME.COFINAL.PHASE_CELL.01` — the frame awaits a head with a nonempty cell); hypothesis 3 is the classical Kronecker/Weyl citation (arithmetic input: log-prime independence, the two-line UFD lemma warded in the probe). *The honest boundary:* nothing proves any of the three hypotheses; a continuation contract, not a discharge; no continuum theorem, no RH statement, no arithmetic identification.

The E8 audit compressions of round 32 (`E8.SIMPLEX.FOURIER.01`, `E8.G31.CLOCK_ALPHABET.01`, `FLAVOR.WINDING.QUADRATIC.01`). *Recorded* 2026-08-08 (evidence lines, not contracts; exact identity grade; no marker moves). The simplex-Fourier character theorem: the census $240 = 15 \times 16$ *is* the spectral statement $\hat{r} = (240, -16^{15})$ — the uniform-nonzero channel $P = (J - I)/15$ with spectrum $\{1, (-1/15)^{15}\}$; the `v817` packet numerator equals the code Walsh transform at every level ≤ 16, and $\hat{m}_2(n) = -1/15$ holds as an integer identity at every odd $n \leq 16500$ — the per-prime measurement is the prime restriction of a character theorem, typed spectral-only ([verification/v857_simplex_fourier_winding.py], 26/26; upgraded in round 36 to *all* odd shells, `E8.SHELL.GLOBAL.01` — see the round-36 record below). The winding quadratic: the whole `tfpt_2` winding line is one adjugate quadratic $q_{\mathrm{wind}} = t^2 - g_{\mathrm{car}} t + |\mathbb{Z}_2|$ with the triple lock $s = 6$, $\Delta(6) = 39200 = 2 \cdot 140^2$ and the reality threshold $s^* \approx 2.8250$ typed ([verification/v857_simplex_fourier_winding.py], 20/20). The G31 clock alphabet: $\mathrm{Deg}(G_{31})/4 = \{2, 3, 5, 6\}$ as a normal form with the unique product-46080/sum-64 quadruple, the 607-group kill scan (G_{31} the sole full-battery passer) and the $W(D_5) \times W(A_3)$ fence re-killed by computed centers 4 vs 1 ([verification/v858_g31_clock_alphabet.py], 19/19). Full statements and keyboxes in Papers 1 and 2; audit cards in Paper 3.

The Krein-defect theorem frame (`FORM.PRIME.KREIN.DEFECT.01` [F]). *Registered* 2026-08-08, FORMAL (`TfptCarrier/KreinDefect.lean`, 14 declarations (10 theorems), kernel-checked — lake build green at 3416 jobs, no `sorry`, no `native_decide`, axioms propext/Classical.choice/Quot.sound only, verified by `#print axioms` on all ten theorems; numeric mirror [verification/v861_krein_normalform_lean.py] part C; carried in the shipped Lean manifest at 88 files). *Proven:* the defect representation `defect / formAt_defect` — the quadratic form of $B_+^{\mathsf{H}} B_+ - B_-^{\mathsf{H}} B_-$ *is* $\|B_+ f\|^2 - \|B_- f\|^2$ exactly; the main theorem `defect_psd_of_contraction` — $B_- = C B_+$ with the Loewner certificate $(1 - C^{\mathsf{H}} C) \succeq 0$ forces the defect PSD, via the exact factorization $B_+^{\mathsf{H}}(1 - C^{\mathsf{H}} C) B_+$ and congruence; the converse `douglas_contraction_of_defect_psd` (full-rank case proven, $C := B_- B_+^{-1}$; the singular case deliberately the classical Douglas 1966 citation — the range bookkeeping not formalized, typed); `SourceContractor` — the *named hypothesis*: contractor + factorization + *independent* norm certificate as data a consumer must supply from the source side, with the circularity warning typed (a target-computed C — eigendata or the converse's pseudoinverse — is a reformulation of the positivity, not a proof); and the cofinal composition `krein_cofinal / krein_cofinal_weil` — contractors on a cofinal ladder yield (H_{cof}) (index extraction from the recurrence) and compose through the minimal H theorem to Weil nonnegativity on the whole dense family: no hidden wall behind the contractor certificates. *Provenance typed, not proven:* the contractor hypothesis is exactly what the Krein source-algebra search must deliver; [verification/v866_source_contractor_formula.py] delivers its closed *formula* and proves the absolute half of the toolbox cannot certify it — the certificate demand is the registration below. *The honest boundary:* nothing here proves the contractor exists; no continuum theorem, no RH statement, no arithmetic identification.

The phase-aware contractor certificate (`PRIME.KREIN.CONTRACTOR.01` [O]). *Registered* 2026-08-08, OPEN ([verification/v866_source_contractor_formula.py], 10 + 8/8 with the one frozen-honest FAIL kept and pattern-gated, SYLVESTER12-ANGLES-FAIL + NORM-REFORMULATION; with [verification/v861_krein_normalform_lean.py], [verification/v862_defect_polar_weld.py], [verification/v864_softport_kappa_law.py], [verification/v867_radau_conditioning.py]). *The demand, verbatim:* supply a *target-free* certificate that the explicit closed-form source contractor $C = W_- \, F \, G_+^{-1} F^{\mathsf{H}} \, W_+$ (the factorization warded at 1.4×10^{-14} incl. blind holdouts; density roots and window geometry only) is a contraction, $\|C\| \le 1$, uniformly along the deployed cofinal ladder — equivalently the Loewner comparison $G_- \preceq G_+$ of the two weighted frame Grams, equivalently $K \succeq 0$ (the Douglas equivalence, machine-verified) — with the certificate's constants independent of the measured target signs. *What is proven about the tool space (the registration's teeth):* (i) *no absolute-value bound can succeed* — the Perron value $\rho(|C|)$ is the infimum over all absolute Schur/Hilbert-type tests and it exceeds 1 on every heavy rung ($2.363 \to 3.103$, growing with depth): any valid certificate must use the *phases* of the kernel; (ii) quadrature certifies $s > 0$ pointwise on every reachable rung but the depth grows as $m^* \sim \sqrt{\mathrm{cond}(G)}$ and the conditioning is *intrinsic* — a uniform certificate cannot be a fixed-depth quadrature family; (iii) no source word of *any* length reaches C (the monomial closure theorem) — the certificate cannot be a word bound; (iv) the discharge target is kernel-checked: per-rung `SourceContractor` instances on a cofinal ladder compose through `krein_cofinal_weil` to Weil nonnegativity. *The named candidate shape:* an oscillatory-integral / stationary-phase bound on the entries of the weighted frame kernel, strong enough to beat 1 by the measured $e^{\mathrm{slope}\,\alpha}$ margin (fit-free defect law, slope -2.40). *Kill:* a rung with $\rho(|C|) \le 1$; a proof that every phase-aware bound of the named shape degrades to an absolute bound on the deployed kernel class; a measured break of the factorization on any new rung. *Close:* the certificate itself — it would discharge the `SourceContractor` hypothesis and with it (H_{cof}) for the defect ladder. *[2026-08-08 round-34 note ([verification/v869_cdcore_clarkphase.py] – [verification/v872_damping_compensation.py] + the Prüfer preregistration): the kernel is named* — the closed-form contractor's reproducing kernel K_+ is the Christoffel–Darboux kernel of the explicit source measure $\mu = \sum[2\sin^2(\varphi/2)\, d_+(\varphi)/L]\, \delta_{2\cos(D\tau)}$ (rank-2 displacement collapse with a 10^{13} gap on all rungs incl. blind holdouts; firewalled chain-only reconstruction at $\sim 10^{-13}$): the demand now reads — control *one* source Jacobi chain and the arm weights. *The compensation identity:* $I - C^*C = T_1 + T_2$ exactly ($T_1 = I - U^*U$ indefinite Jacobi cross-geometry; $T_2 = U^*(I - D_-^2)U$ PSD Christoffel damping) with the exact global minimizer equal to the soft direction and the isolated scalar $s = \kappa\tau$ — the demand localized to the two-term inequality at one named direction, both terms source-built. *The closures (stop-list entries):* the Gauss-node unitary core does not exist (co-isometry defect $O(1)$; (H-U) the open hypothesis, $D_-^G \le 0.997041$ the surviving asset — [verification/v870_christoffel_gauss_frame.py]); classical pole transformations at $x_{\mathrm{pole}} = 2\cosh(D/2)$ are dead (typed residue: small-negative-mass Geronimus at 3.5–8.1%, no $\lambda(D)$ law); finite-rank cross-defect corrections are dead (rank grows linearly with h, Pearson $+0.996$ — [verification/v871_pole_uvarov_crossdefect.py]); μ_4-offset Clark kernels are dead (the equal-phase law). *The preregistration (the contract's next test, preregistered pre-execution):* `PRIME.PRUEFER.COMPENSATION.01` — source-only Prüfer phases of the two arm chains, exactly 16 fixed phase cells (boundaries $k\pi/8$, entrywise pairing primary, danger families predeclared), Cotlar–Stein decision rules and all success/kill criteria frozen *before* any deployed-data run, deep holdouts kz 90/116/142/177/243 never used by the CD wave, the decisive analysis locked behind the operational unfreeze flag, 8/8 synthetic freeze wards green; committed byte-exact with round 34: SPEC SHA-256 `4621b89958531d6d3baae9fe762dbd10d23dc22eed77a57aa1c6b179a7440811`, FILE SHA-256 `93208a1b2edebde31388a87192055c9bf3e5a3de29af9fb3e34ddc904825d9fd` (`experiments/tfpt-discovery/pruefer_compensation_probe.py`; any modification invalidates the preregistration). Demand, kill and close conditions unchanged; still [O].] *[2026-08-08 round-35 note ([verification/v873_pruefer_cotlar_decision.py], the executed preregistration):* the contract was executed as committed — the unfreeze flag created *after* the round-34 commit `526ca3eb` (audit line committed), the two mechanical fixes documented verbatim (the $h \le 900$ battery filter the spec's own text mandates; the overflow renormalization, atan2-invariant, validated at $\le 6.6 \times 10^{-4}$ against a 60-digit `mpmath` reference), the spec literal SHA byte-identical throughout, the executed FILE SHA-256 `15d35b26c652ef4144df713cafba7e049e2dcb9bb7a168462cd65735f5ffb4cc` — and it returns COTLAR-GROWING by its own frozen rules: danger geometry real (capture 0.98–0.61 of $\lambda_{\min}(T_1)$, source-only), local dominance and envelope decay FAIL (the envelope bimodal at the predeclared π-resonance cells), the battery flat but the frozen deep holdouts exploding (slopes $0.209/0.408 \ge 0.20$, both channels; kz 142: $S_U\, 32.07/\ S_C\, 58.78$), the contrast bars passing, kills 2 + 5 firing, RUN 3 correctly withheld. The blockwise Cotlar–Stein phase-cell route is closed and stop-listed; the surviving demand is a non-stationary phase bound with the π-resonance as the known critical structure and the α-not-h growth law as the typed constraint (the explosion tracks kz-depth, not window size). The first fully preregistered uniformity test of the program, executed and booked exactly as frozen; still [O].] *[2026-08-08 round-36 CORRECTION note ([verification/v877_complete_comb_reversal.py], the complete-comb re-decision):* the round-35 note above is corrected, dated — the RUN-2 deep-holdout data was *comb-truncated* (`core.build_window` silently caps the atom table at `ATOM_MAX` $= 4 \times 10^5$; kz 142/177/243 demand support to $4.4 \times 10^5/8.0 \times 10^5/1.8 \times 10^6$; `PRIME.DEEPALPHA.SIGN.01` certifies the complete-comb floors strictly positive). The v1 verdict was correct *on the data as measured* and `v873`'s booking stands as the record of the preregistered v1 execution (spec SHA `4621b899...`, executed file SHA `15d35b26...`); but on complete combs the *identically* frozen v1 rule returns BOUNDED in both channels (v2 spec SHA `5fd6bf61...`; kz 142: S_U 32.07 → 5.65, S_C 58.78 → 4.10; slopes 0.006/0.007) and RUN 3 passes (K_1/K_2 h-stable on all five holdouts). *The stop-list entry is amended:* "blockwise Cotlar–Stein on Prüfer phase cells" is removed — the class is *not* closed; the route reopens as `PRIME.PRUEFER.COMPENSATION.02` with the remaining gap typed (the Cotlar constant $B_C \sim 3.9$–4.3 vs the needed 1; the cell-local defects $\sum_r \varepsilon_r \sim 7$–$10$ — the RUN-1 anatomy was in-cap and stands); the α-not-h law is retired (it was the truncation's signature). The π-resonance critical structure and the discrimination stand; the surviving demand is now carried in embedding coordinates by `PRIME.CARLESON.PRIME.01` [O] (registered below). Demand, kill and close conditions otherwise unchanged; still [O].] Existence is *not* claimed; no RH statement.

The 210 canonicity audits of round 33 (`E8.DIVISOR210.CANONICITY.01`). *Recorded* 2026-08-08 (evidence lines, not contracts; no marker moves). The label register $\mathbb{F}_2^4$ *is* the divisor lattice of $210 = 2 \cdot 3 \cdot 5 \cdot 7$ (Walsh–Hadamard carries the lattice Möbius function; the vacuum column is the rank-one Redheffer closure with $\det = \prod_p (1 - 1/p) = 8/35$ exact, [verification/v863_redheffer_colligation.py]); the canonicity guard quantifies the selection ([verification/v868_divisor210_audits.py], $36/36 + 12/12 + 15/15$): the Boolean layer is measured generic (210/210), the two Euler determinants pin $\{2, 3, 5, 7\}$ uniquely, the anchor prime 2 is forced by ramification, and after the crossref exactly two gauge classes remain — the family-cycle chirality, then *proven gauge* by two exact no-go wards (the orientation functional vanishes identically via the Möbius complement $d \mapsto 210/d$; quadratic readouts provably blind by transposition); the $17 = 12+3+1+1$ moving-sector identification honestly buried at space level (principal angles $\sim 90°$); the 6-vs-7 honesty note stands untouched. Full statements and keyboxes in Paper 1; audit cards in Paper 3.

The Prime Carleson embedding contract (`PRIME.CARLESON.PRIME.01` [O]). *Registered* 2026-08-08, round 36 ([verification/v876_carleson_polygon_wave.py]; the ward making the reduction machine-exact at rel $1.5 \times 10^{-13}/1.3 \times 10^{-15}$ on kz 9/13). *The chain:* for the canonical window family, the Krein floor $\tau_h > 0$ is *equivalent* to the Carleson embedding of the negative arm measure into the positive arm measure at polynomial degree $h - 1$ — $\|C_h\| \le 1 \Leftrightarrow G_- \le G_+$ (exact, `PRIME.SOURCECONTRACTOR.NORM.01`) $\Leftrightarrow \int |P|^2 \, d\tilde{\nu}_- \le \int |P|^2 \, d\tilde{\mu}_+$ for all $P \in \mathcal{P}_{h-1} \Leftrightarrow M(\tilde{\mu}_+ - \tilde{\nu}_-) \ge 0$, with the embedding constant $\lambda_{\max}(M_-) = \|C_h\|^2 = 1 - \tau_h$. *The demand:* a proof of the embedding uniformly along a window subfamily cofinal in the mesh-refinement order would discharge `SourceContractor` (`FORM.PRIME.KREIN.DEFECT.01`) and, through `krein_cofinal_weil`, hypothesis (H_{cof}). *The typed constraints a candidate must respect:* the certificate must be phase-aware (the Perron closure of `v866`); the π-resonance is the known critical structure and it is *thick* — not removable by excision (`PRIME.PI.RESONANCE.ANATOMY.01`); the polygon/two-gauge compression is the *wrong* compression of the Loewner order (`PRIME.PHASE.POLYGON.01`); the constructive side is the residual quadrature at full degree (`PRIME.RESIDUAL.QUADRATURE.01`) under the complete-comb constraint (`PRIME.TRUNCATION.AUDIT.01`). Arrow (iv) (a prefix-local scalar reduction) is typed NEEDS PROOF. *Kill/close:* closes if the embedding is proven on a cofinal family, or if a certified negative canonical window family is exhibited (none exists: `PRIME.DEEPALPHA.SIGN.01`). Existence is *not* claimed; no RH statement. *[2026-08-09 round-37* *note ([verification/v879_envelope_partition_closure.py]): the constraint set is extended, dated — the partition class is closed* entirely *(PRIME.PARTITION.CLOSURE.01: Prüfer cells at every tested resolution with no predeclared accounting below* 3.568 *vs the bar* 1*; honest tight wave-packet frames on both natural phase spaces with diagonal mass* $\leq 18\%$ *under any pairing and a growing Schur constant* 4.9–9.2*), so no partition-based accounting can certify* $\|C\| \leq 1$*: the embedding certificate must be* non-decompositional — *eigen-enclosure, Loewner-order or passivity arguments that treat* $M(\tilde{\mu}_+ - \tilde{\nu}_-)$ *whole, all three measured territory. The new carried assets: the exact tridiagonal multiplication identity on the odd frame (CD telescoping at* 1.3×10^{-13}*, rank-2 boundary defect at* $k = 0$*), the derived entrywise bound with h-uniform source constants (*$C_{\text{int}} \in [1.232, 2.328]$ *from* $h = 168$ *to* 1445 *— the route's first analytic uniformity statement), and their source-law persistence as* $h \to \infty$ *as the named open object. Demand, kill and close conditions otherwise unchanged; still [O]; no RH statement.]*

The Carleson wave of round 36 (PRIME.PHASE.POLYGON.01, PRIME.RESIDUAL.QUADRATURE.01, PRIME.PI.RESONANCE.ANATOMY.01). *Recorded* 2026-08-08 (evidence lines with typed honest FAILs; no marker moves; [verification/v876_carleson_polygon_wave.py], $6/8 + 7/8 + 5/5$, the three frozen-honest FAILs pattern-gated). The polygon: Lemma A exact (the difference of the two source canonical systems *is* a signed rank-one phase sum; round-trip 2.4×10^{-12}), but $S_0(m) \geq |S_2(m)|$ breaks on all 47 rungs at $2\theta \approx 1.5\pi$ — the two-gauge compression is the *wrong* compression of the Loewner order, typed with both readings; holonomy null with inverted contrast; the deep-holdout rows retype NUMERIC-VOID per the round-36 truncation audit (battery verdict unaffected). The residual horizon law: the difference Jacobi chain (row-scaled Wheeler + mp escalation past the 53-bit noise wall) realizes the positive representing measure ρ_h at *full degree h* on all 44 floor-positive rungs (k^*/h median 1.000, slope 0.974, $r_h = 0.998$); a single atom sits just beyond the pole port (mass beyond the fold is *required*: the Radau-anchored functional breaks early, genuine); discrimination 7–9×; the three sub-h stalls were exactly the truncated windows — a validated sign detector. The resonance anatomy: the resonant set is *thick* (a constant $\sim$1/5 of the minus nodes; excision gains nothing; the soft direction disjoint) — the wall is phase-distributed at this partition resolution; the α-law addendum confounded by truncation, typed. Full narrative in the prime-front paper; audit card in Paper 3.

The artifact resolution and the Cotlar reversal of round 36 (PRIME.DEEPALPHA.SIGN.01, PRIME.PRUEFER.COMPENSATION.02, PRIME.TRUNCATION.AUDIT.01). *Recorded* 2026-08-08 (the round's headline; [verification/v877_complete_comb_reversal.py], $8/8 + 9/9$). The deep-alpha negative floors at kz 142/177/243 were *comb-truncation artifacts* of the deployed atom table (ATOM_MAX $= 4 \times 10^5$, silent cap; the windows demand support to 1.8×10^6): with the complete canonical comb the certified corner enclosures are strictly positive ($+1.06 \times 10^{-5}/+1.52 \times 10^{-5}/+1.20 \times 10^{-5}$), no negative canonical window exists, the certified ladder's end at kz 121 was the atom-table *data* boundary, and (H_{cof}) is unaffected; the silent truncation is typed an infrastructure bug with the registered follow-up (the builder must fail loudly beyond its table). Under the *identically* frozen v1 bars the complete-comb deep sums collapse into the battery band and both channels return BOUNDED (slopes 0.006/0.007); RUN 3 unlocks and passes (K_1/K_2 h-stable on all five holdouts) — the route reopens, with the remaining gap typed ($B_C \sim 4$ vs 1; local defects 7–10). The truncation audit re-types every affected finding, dated (pruefer v1 confounded → re-decided; residual strengthened; resonance α-law void; polygon holdouts numeric-void; the other waves unaffected). The dated corrections live on the PRIME.KREIN.CONTRACTOR.01 and PRIME.FLOOR.PAIRCORR.01 blocks above; v873's booking stands as the record of the preregistered v1 execution. No RH statement.

The round-36 compiler theorems (FLAVOR.FEEDBACK.NORMALFORM.01, E8.WINDING.DECODER.01, E8.SHELL.GLOBAL.01; Lean: FORM.FLAVOR.FEEDBACK.01 [F], FORM.E8.SHELL.GLOBAL.01 [F]). *Recorded/Registered* 2026-08-08 (exact identity grade; no marker moves; [verification/v874_feedback_decoder_lean.py] 14/14+11/11, [verification/v875_shell_global_lean.py] 15/15). The feedback normal form: $P = [\,2 \cdot \mathbf{1} \mid e_3 \mid Re_3\,]$ conjugates the whole winding line, symbolic in s, to the gain-plus-companion form with $\chi_{R_s} = (t - s - |\mu_4|)\, q_{\text{wind}} - 12t$ and the honest $\mathbb{Z}_2^2$-not-μ_4 typing load-bearing (keybox in Paper 2). The winding decoder: q_{wind} maps the clock alphabet to $(-|\mu_4|, -|\mu_4|, |Z_2|, h(D_5))$, the decoder polynomial $(y+4)^2(y-2)(y-8)$ carries budget coefficients $(2, 48, 32, 256)$, and $|\mu_4|\,q(|Z_2|)/|R(E_8)| = -1/15$ is exactly the Walsh message eigenvalue — typed *audit* theorem, not a functor (keybox in Paper 1). The global shell theorem: $\hat{\Theta}_u/\Theta_L = -1/15$ for *all* odd shells from $\mathrm{Sp}(4, 2)$ transitivity + $(1+i)$ doubling + $\Theta_L = 240\,\sigma_3$; the 16500 census retained as the exact ward (keybox in Paper 1). *The Lean records:* `TfptCarrier/FlavorFeedback.lean` (16 declarations, 11 theorems — the action identities, the inverse-free conjugation $PF_s = R_sP$ over $\mathbb{Z}$ symbolic in s, χ_{R_s} in $\mathbb{Z}[X]$, the SNF determinantal divisors) and `TfptCarrier/GaussianShells.lean` (the `ShellSystem` hypothesis bundle with provenance typed on each field, named *not* proven, + 7 consequence theorems, every statement an exact integer identity, no division) — both kernel-checked: lake build green at 3418 jobs, no `sorry`, no `native_decide`, axioms propext/Classical.choice/Quot.sound on all 18 theorems (`#print axioms`), carried in the shipped Lean manifest at 90 files. *The honest boundaries:* no μ_4 identification from the $\mathbb{Z}_2^2$ quotient; the decoder needs an intertwiner to be causal; the shell theorem's three inputs stay named hypotheses. No RH statement.

The ledger integrity guard and the `FORM.QGEO.04` QA fix (`GATE.LEDGER.01`). *Recorded* 2026-08-08 (a QA/CI note, no new physics; [`verification/v878_ledger_integrity.py`], 10/10). The round-36 ledger audit found the first *primary-key violation* in `status_ledger.csv`: the claim id `FORM.QGEO.03` was double-booked (2026-06-15 the CohomologyGrading formalisation; 2026-06-18 the SeamDeckClosure flat all-orders closure). The fix, dated: the SeamDeckClosure row is renamed `FORM.QGEO.04` with a QA note in its claim text, and every flat-closure reference is updated (this document, the two 2026-06-18 changelog mentions, `v279/v284/v286` + `seam_equivalence.json`, `make_figures`, the website residual chain, registry and `run_all`); the CohomologyGrading row keeps `FORM.QGEO.03`; claim contents unchanged. The guard enforces from this round: globally unique claim ids (the fix pinned), existing supersession targets with an acyclic supersession graph, unique *active* supersession targets modulo a frozen remove-only allowlist (the two legitimate 2026-06 joint closures), and the dependency cross-reference census frozen remove-only (13 SCCs / 265 ids — acyclicity enforced as *no new cycles*; the `audit_baseline` precedent).

The partition closure of round 37 (`PRIME.PARTITION.CLOSURE.01`; dated notes on `PRIME.CARLESON.PRIME.01` and `PRIME.PRUEFER.COMPENSATION.02`). *Recorded* 2026-08-09 (the compact closing round of the reopened Cotlar route; [`verification/v879_envelope_partition_closure.py`], 7 checks with the two frozen-honest FAILs T1.4/C.1 pattern-gated at exit 1 + 6/6; the contract chain `4621b899...` $\to$ `5fd6bf61...` $\to$ `27d9f0a3...` $\to$ `2199e9b1...` machine-verified at run time). The analytic core, saved: multiplication by $\cos\theta$ is *exactly* tridiagonal on the odd frame (mirror-closed boundary, single rank-2 defect at $k = 0$; the CD telescoping identity at 1.3×10^{-13} on every rung incl. all five complete-comb holdouts), and the derived pointwise bound carries *h-uniform source constants* ($C_{\mathrm{int}} \in [1.232, 2.328]$ from $h = 168$ to 1445, slope +0.106; $\sqrt{(G_+^{-1})_{00}} \in [0.661, 0.899]$) — the route's first analytic uniformity statement, at entrywise grade. The structural closure (the honest FAILs are the finding): the cell-envelope decay is *cancellation-carried*, provably not entrywise-derivable (every Prüfer cell contains near-diagonal pairs; the analytic envelopes blow up as h^2); no predeclared accounting below 3.568 (bar 1); $\sum_r \varepsilon_r = 6.9$–$12.9$ α-uniform on clean complete-comb data; and the honest tight wave-packet frames (both natural phase spaces; $\|B\| = \|C^G\|$ exact) leave diagonal mass $\leq 18\%$ under any pairing with a Schur constant growing $4.985 \to 9.222$ (slope +0.295), while the Gabor envelope does decay in phase space (the first partition to show decay) and the discrimination is massive (truth 4.985 vs Epstein 128.7 vs scramble 9006.6). *The consequence:* the cancellation is *atomically global* — no partition-based accounting (cells, packets, any frame decomposition of this family) can certify $\|C\| \leq 1$; the certificate demand of `PRIME.CARLESON.PRIME.01` is retyped *non-decompositional* (eigen-enclosure / Loewner-order / passivity — all three measured territory); wave-packet partitions and entrywise envelope bounds are stop-listed on the `PRIME.PRUEFER.COMPENSATION.02` block. The surviving assets: the tridiagonal identity, the h-uniform derived constants, their source-law persistence (the named open object). Full narrative in the prime-front paper; audit card in Paper 3. No RH statement.

The rounds-38/39 sandbox campaign (the operator geometry and the source law of the wall). *Recorded* 2026-08-09 (a campaign note; 25 frozen sandbox probes, every spec amendment fail-first on record; the promoted verification modules [verification/v880_finite_anchor_theorems.py], [verification/v881_carleson_port_geometry.py] and [verification/v882_port_source_mellin.py] embed their discovery probes byte-exact). The campaign gives the Carleson wall its exact operator geometry (rank-2 Jacobi displacement, gauge-exact Carleson Gram with the classical testing diagonal, exact port Schur reduction — the port *is* the wall — and the preserved IIKS integrable class), its universal source law (the $\sqrt{n/X}$-uniformized prime measure, the PNT edge law $1 - e^{-1/2}$ as the ∼39% port mass, the Mellin–Cauchy port symbol, the closed criticality budget), and the hardness decision (`PRIME.ERRORTERM.SCALE.01` below); the six ledger-only items of the campaign are registered as the blocks that follow. Full narrative in the prime-front paper; audit cards in Paper 3. No marker moves; no RH statement.

The error-term scale decision (`PRIME.ERRORTERM.SCALE.01`, decided). *Recorded* 2026-08-09, DECIDED (sandbox probe `errorterm_demand_curve_probe.py`; the demand-curve measurement behind the round-39 hardness statement). *The finding:* the wall margin equals injected off-critical perturbation *energy* on the τ law exactly (ratios 1.00–1.10 across the tested grid); the perturbation amplitude scales as $A^* \sim \sqrt{\tau}/\delta^2$ under the frozen convention $\Delta\lambda = O(A^2\delta^4)$; and the kill localizes through the port block. *The decision:* the reduction chain of the wall is an *exact reformulation at RH strength* — there is no unconditional slack anywhere in it; the identities are exact, and the one-sidedness is open at exactly that strength. Recorded as a hardness decision, not a contract demand. No marker move; no RH statement.

The port limit contract (`PRIME.PORT.LIMIT.01` [O]). *Registered* 2026-08-09, OPEN (sandbox probes `port_limit_operator_probe.py`, `port_cocycle_window_probe.py`, `weyl_m_criticality_probe.py`). *Measured:* the compressed operator-norm convergence has slope −2.80 carrying the wall exactly; $\lambda_{\max}$ increases to 1 from below; the atom-at-edge Weyl-m class has $\sigma \to -1$ and $w_1 \to 0.72$. *The demand:* the fixed-space limit theorem — the existence and identification of the limiting port operator on a fixed space, with the wall carried through the limit. *Kill:* a measured break of the convergence slope or a rung family on which the compressed operator fails to carry the wall. No marker move; no RH statement.

The scalar port inequality (`PRIME.PORT.SCALAR.01` [O]). *Registered* 2026-08-09, OPEN (sandbox probe `port_scalar_schur_probe.py`). *Measured:* the wall is *one* scalar inequality,

$$\sigma_h \;=\; \big[\,1 - \tilde{\nu}_{m^*} K_h(y^*, y^*)\,\big] - b^{\mathsf{T}}(I - E_{\mathrm{rest}})^{-1} b \;\geq\; 0,$$

exactness 6.2×10^{-11} on 42 rungs, with σ_h tracking τ_h. *The demand:* one-sidedness — a proof that $\sigma_h \geq 0$ along a cofinal window family. *Kill:* a rung on which the scalar reduction fails at certificate grade. No marker move; no RH statement.

The port cocycle window (`PRIME.PORT.COCYCLE.01` [O]). *Registered* 2026-08-09, OPEN (sandbox probe `port_cocycle_window_probe.py`). *Measured:* a fixed 12×12 window carries the wall exactly, with compressed convergence slope −2.80. *The demand:* the closed-form limit of the window cocycle. *Kill:* a depth at which the fixed window stops carrying the wall. No marker move; no RH statement.

The Case sum-rule contract (`PRIME.CASE.SUMRULE.01` [O]). *Registered* 2026-08-09, OPEN (sandbox probes `case_sum_rule_probe.py`, `testing_mnt_law_probe.py`). *Measured:* the deployed Jacobi chains are measured Killip–Simon class. *The demand:* prove a Case C0/C1 sum rule implying the unconditional testing bound $T_h \leq 1$; the target is exactly the closed criticality budget of round 39 (PNT growth compensated by window geometry plus Christoffel). *Kill:* a measured departure of the deployed chains from the Killip–Simon class. *Route update (2026-08-09, round CVIII) — retyped conditional:* the free-tail bridge is numerically delivered (sandbox `freetail_case_bridge_probe.py`, 8/8: the finite Killip–Simon P2 identity exact at 5×10^{-11}; the P2 energy h-uniformly bounded 0.63–0.75 over all 42 rungs), *but* the route is retyped conditional by [verification/v889_route_decisions.py]: PNT-INSUFFICIENT (the smooth reference violates the port margin on every rung) and HOMOTOPY-INDEFINITE-ARITHMETIC (the required input is the signed pair-correlation functional). The demand becomes: state and prove the weighted prime-pair correlation inequality, or accept conditionality. No marker move; no RH statement.

The port Riemann–Hilbert contract (`PRIME.PORT.RIEMANNHILBERT.01` [O]). *Registered* 2026-08-09, OPEN (sandbox probes `port_riemann_hilbert_setup_probe.py`, `mellin_model_operator_probe.py`). *Measured:* the discrete 2×2 IIKS Riemann–Hilbert problem of the dressed port operator is *well-posed* — the two generators are explicit and the resolvent closure holds at 1.2×10^{-12}. *The demand:* nonlinear steepest descent for $h \to \infty$. *The calibration constraint (load-bearing):* the smooth PNT continuum model *violates* the floor at finite depth (τ_{model} $-1.26 \to -0.11$) — the prime discreteness is load-bearing, and the unperturbed problem must keep the lattice. *Execution note (2026-08-24, rounds 244–255):* the finite formulation this demand must consume is now certified — the bordered end-object is a canonical 2×2 RHP $\ltimes$ a one-dimensional budget cocycle with exact evaluation infrastructure ([`verification/v958_bordered_tau_readout_dictionary.py`]); the irreducible-3×3 steepest-descent variant is dropped without replacement, and the admissible target form is an *integrated* statement over the compressed CD readout (no uniform Y_n parametrix) — the campaign continues as `PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01` (the block below). This umbrella stays [O]. *Kill:* a failure of the IIKS well-posedness on a deployed rung. No marker move; no RH statement.

The rounds-40/41 flow corridor (the determinant chain, the parametrix decision, the certified head, and the complete-comb anatomy). *Recorded* 2026-08-09 (a campaign note; nine frozen sandbox probes, every spec amendment fail-first on record; the promoted verification modules [`verification/v883_tau_chain_parametrix.py`], [`verification/v884_certified_head_positivity.py`], [`verification/v885_ladder_flow_anatomy.py`] and [`verification/v886_register_gapcode_null.py`] embed their discovery probes byte-exact). The campaign gives the wall's sign an exact finite determinant chain ending in a candidate tau function, the decided parametrix constraint (the Euler-product comb is globally rigid — every smoothing-based parametrix is dead), the first rigorous positivity statement of the program (exact integer Sylvester certificates on the three head rungs), the complete flow/induction anatomy of the ladder (degree-2 Lax closure; no monotone one-prime induction in any insertion order — positivity is a property of the complete comb), and the register-gapcode disenchantment. The registrations and updates follow as the blocks below. Full narrative in the prime-front paper; audit cards in Paper 3. No marker moves; no RH statement.

The tau contract (`PRIME.PORT.TAU.01` [O]). *Registered* 2026-08-09, OPEN — the finite-h determinant chain delivered ([`verification/v883_tau_chain_parametrix.py`], 5/5, TAU-CHAIN-EXACT). *Exact:* the Schur determinant identities $\sigma_h = \det(I - E)/\det(B)$ and $\det(I - E) = \det(I - R)\det(I - D_P)$ are machine-exact (4.8×10^{-13} in log space) with positive bulk factors, hence $\operatorname{sign}\tau_h = \operatorname{sign}\sigma_h$ = the sign of the wall with $\tau_h := \det(I - D_P)$; the Jacobi variational identity $\frac{d}{ds}\log\det(I - sD_P) = -\operatorname{tr}[(I - sD_P)^{-1}D_P]$ holds on the true family (10^{-6}); the determinant wall indicator is false on Epstein and scramble. *Measured:* the κ/ρ readout — $\text{slope}(\log\sigma) = -2.93$ vs $2\times$ the Mellin fluctuation slope -2.53 ($\sigma \sim \text{fluctuation}^2$ within 16%). *The demand (the symbolic half):* τ_h = the tau function of the discrete 2×2 IIKS Riemann–Hilbert problem of the `v881` generators, for arbitrary h without numerics. *Execution note (2026-08-24, rounds 224–226):* the demand is *executed at the finite-identity level* — Sylvester as a full characteristic identity, the s-dressed Schur family, the confluent diagonal, the IIKS dressing identity with source-explicit dressed generators, the Fredholm/JMU variational identity in two independent times and the relative tau transfer, all machine-exact on the actual builders (TAU-IIKS-EXACT, 18/18; [`verification/v955_tau_iiks_toda_dictionary.py`]); per the note-DLI claim splitting the finite theorem set is carved out as `PRIME.PORT.TAU.FINITE.IIKS.01` [E] (the block below) and the sign question stands alone as `PRIME.PORT.TAU.NOPOLE.COFINAL.01` [O]; this umbrella stays [O] for the fully symbolic arbitrary-h statement. *Kill:* a heavy rung on which the determinant chain fails at certificate grade. No marker move; no RH statement.

The finite IIKS/Toda theorem set (`PRIME.PORT.TAU.FINITE.IIKS.01` [E]). *Registered and delivered* 2026-08-24 (rounds 224–226, notes DL/DLI/DLII;

[verification/v955_tau_iiks_toda_dictionary.py], probes embedded byte-exact, 18/18+15/15+17/17 first pass). *Exact:* the wall determinant is the exact IIKS Fredholm tau function at the finite-identity level (Sylvester characteristic identity on the whole s-grid; the s-dressed Schur family $\det(I - sE) = \det(I - sR)\det(I - s\,D_P(s))$ with the fixed-kernel alias named and blocked; the confluent diagonal $\leq 3.2 \times 10^{-13}$; the IIKS dressing identity $\leq 7.9 \times 10^{-13}$ with the dressed port generators source-explicit in closed form; the Fredholm/JMU variational identity in two independent times; the relative tau transfer exact over $-212.84/-195.50$ log-unit collapses); the h-direction carries an exact *predictive* degree-1 Lax dynamics (rank-1 h-step at 2.9×10^{-16}; blind next-generator and tau-step predictions at 3.4×10^{-16} / 3.2×10^{-11}; zero curvature identically, world-blind; mpmath wards at dps 80/120); and the Toda dictionary holds: $\tau_{w,n} = D_n(\mu - \nu)/D_n(\mu)$, $r_n = h_n(\mu - \nu)/h_n(\mu)$, the Hirota coefficient source-pure from a scaled signed Stieltjes recursion. *Typed honestly:* the sign is WALL-EQUIVALENT ($\hat{\gamma}_n > 0$ for all $n \iff$ quasi-definiteness of the signed defect measure $\mu - \nu \iff$ the wall; MAIN carries all 184 $r_n > 0$, the controls flip at $n = 25/21/27$); the s-time has no fixed degree (CONDITIONING-ONLY) and across rungs there is no common carrier; the general-h statements are classical theorems (Sylvester, Schur, CD confluence, IIKS dressing) instantiated on the actual builders. *Kill:* any of the finite identities failing on a deployed rung. No RH statement.

The tau sign contract (`PRIME.PORT.TAU.NOPOLE.COFINAL.01` [O]). *Registered* 2026-08-24, OPEN (the honest open half of the note-DLI claim splitting). *The demand:* $\tau_h(1) > 0$ cofinally — equivalently, no Malgrange divisor of the s-family up to $s = 1$. *Equivalent coordinates (all machine-exact dictionaries, [verification/v955_tau_iiks_toda_dictionary.py] / [verification/v956_signedmoment_halffilling_duality.py]):* quasi-definiteness of the signed defect measure $\mu - \nu$ through the full free moment window; all Christoffel scalars $r_n > 0$; all Jacobi couplings $\beta_n > 0$ in every one of the four source-pure coordinate paths; the positive orientation of the signed Stokes multiplier of the two-sided midpoint problem. *Measured seeds:* MAIN survives the entire free moment window plus $O(1)$ forced-tail degrees (offsets 0/2/2/3/1) while the controls die at 11–15% of it; $\lambda_{\max}(Q_{N_w}) = 0.999832..0.999999 < 1$. *Kill (of any mechanism candidate):* failing the frozen micro falsifier — predict the forced tail 0/2/2/3/1 *and* the control flips 25/21/27 blind. No marker move; no RH statement.

The parametrix decision (`PRIME.PORT.LATTICE.PARAMETRIX.01`, decided). *Registered and decided* 2026-08-09 ([verification/v883_tau_chain_parametrix.py], 3/3+3/3, PARAMETRIX-MEASURED (FLUCTUATIONS-REQUIRED) + ZONES-MEASURED (BOTH-REQUIRED)). *Measured:* four worlds through the identical pencil pipeline on every heavy rung — true comb $\tau = +1.7 \times 10^{-4}$ to $+6.7 \times 10^{-7}$ (ward); lattice with fully smooth quadrature masses -78 to -1706; lattice with exact local $\pm\frac{1}{2}$-log-unit mass sums -78 to -5537; plain continuum -0.11 to -1.26 (ward) — and the zone refinement kills both ways (edge-only smoothing -3.7 to -22.1; interior-only -77 to -7551). *The decision (a design constraint for every RH analysis of the wall):* the exact Λ-mass-position pairing — the multiplicative Euler-product structure — is load-bearing in both zones; the main/unperturbed object must carry the full von Mangoldt comb; every smoothing-based parametrix is dead; the small parameters live exclusively in the flow/deformation direction (ladder order, isomonodromy/tau, s-deformation). No marker move; no RH statement.

The certified head (`PRIME.PORT.CERTIFIED.HEAD.01`, delivered). *Registered and delivered* 2026-08-09 ([verification/v884_certified_head_positivity.py], 5/5, CERTIFIED-HEAD; SPEC v2 amendments on record). *Proven:* $\sigma_h > 0$ on kz 9 ($h = 184$), kz 12 (151) and kz 13 (168) — the first rigorous positivity statement of the program: independent `mpmath` rebuild of the window lags (dps 60/100; conservative per-entry error $\varepsilon_c = 10\times$ dps-disagreement; cross-ward at 1.1×10^{-14}, which caught a sign bug in the rebuilt Arch near-branch), matrix shift by the total error budget ($\sim 2 \times 10^{-18}$, $\sim$14 orders below the margins), then exact integer Bareiss/LDL on the $Q = 10^{20}$ grid: all pivots positive (Sylvester) $\Rightarrow K \succ 0 \Rightarrow \sigma_h > 0$ via the exact `v866` equivalence; certified floors $\lambda_{\min} \geq 3.633/3.327/3.842 \times 10^{-4}$; the Epstein control through the identical machinery refused at pivot index 10. *Typed honestly:* proven modulo the declared conservative transcendental-evaluation error model (validated-numerics standard); the ball-arithmetic closure and the 42-rung rollout were the named follow-ups — both since delivered ([`verification/v887_certified_ladder_complete.py`], [`verification/v897_certified_interval_ladder.py`]: the informal error model is retired from the ladder). *Kill:* a certified pivot failure on a head rung under the frozen error model. No marker move; no RH statement.

The leading-sign contract (`PRIME.PORT.LEADING.SIGN.01` [O]). *Registered* 2026-08-09, OPEN (the moonshot of the round-40 review). *The demand:* $\sigma_h = \kappa\rho_h + o(\rho_h)$ with a closed-form $\kappa > 0$ via a relative Case/Killip–Simon identity — the one-sidedness of the scalar wall as an asymptotic law. *Measured seeds:* the ρ candidate is the squared lattice fluctuation scale — slope$(\log\sigma) = -2.93$ vs $2\times$ the Mellin fluctuation slope -2.53 (within 16%; [`verification/v883_tau_chain_parametrix.py`]); the parametrix decision forces the h-ladder direction — $D_P(h)$ as sections of one arithmetic limit object, κ from the cocycle increment structure, not from comb deformations. *Kill:* a measured break of the $\sigma \sim$ fluctuation^2 law on the deep ladder. No marker move; no RH statement.

The tail contract (`PRIME.PORT.TAIL.01` [O]). *Registered* 2026-08-09, OPEN (the final composition contract of the port route). *The demand:* $|\sigma_h - \kappa\rho_h| \le (\kappa/2)\rho_h$ for all $h \ge H$ explicit (the tail half), plus the finite head by certified numerics, plus the Lean composition through `krein_cofinal_weil`. *Head status:* delivered for kz 9/12/13 (`PRIME.PORT.CERTIFIED.HEAD.01`, [`verification/v884_certified_head_positivity.py`]); the rollout to all 42 head rungs and the ball-arithmetic closure are the named next steps; the tail half waits on `PRIME.PORT.LEADING.SIGN.01`. *Promotion note (2026-08-09, round CVIII):* the rollout is *complete* — $\sigma_h > 0$ proven on all 42 reachable rungs, promoted in [`verification/v887_certified_ladder_complete.py`] (`PRIME.PORT.CERTIFIED.LADDER.01`; the 40-rung parent rollout stays sandbox, runtime-excluded); the remaining named steps are the ball-arithmetic closure and the Lean composition. *Interval note (2026-08-09, round CXV):* the ball-arithmetic closure is *delivered* — [`verification/v897_certified_interval_ladder.py`] (`PRIME.PORT.BALLLADDER.01`) re-proves all 42 reachable rungs with rigorous interval-arithmetic shifts (15 exact-rational + 27 validated-precision) and retires the informal lag-evaluation error model; the remaining named steps are the Lean composition and the asymptotic tail. *Kill:* a certified negative σ_h on any head rung. No marker move; no RH statement.

The isomonodromy-flow measurement (`PRIME.PORT.ISOFLOW.01`). *Registered* 2026-08-09, measured ([`verification/v885_ladder_flow_anatomy.py`], 8/8, ISOFLOW-MEASURED (FLOW-OPEN)). *Measured:* the gauge-fixed IIKS generator flow over 41 consecutive ladder pairs closes in degree 2 (residual medians 0.244/0.248 with $\{Y^2 f, Y^2 g\}$; degree 1 fails at 0.498/0.457) — isomonodromy-compatible, gauge stable (consecutive cos median 0.943), $\log\tau$ monotone $-58.6 \to -396.6$. *Typed honestly:* the controls close similarly — degree-2 closure is a structural precondition, not a discriminator; the arithmetic stays in the value. *The demand:* the closed-form degree-2 deformation polynomial in a well-conditioned coordinate (the `LAX2` block below). No marker move; no RH statement.

The window-ladder transfer (`PRIME.PORT.LADDER.TRANSFER.01`). *Registered* 2026-08-09, measured ([`verification/v885_ladder_flow_anatomy.py`], 5/5, LADDER-MEASURED (PERTURBATIVE + NON-MONOTONE)). *Measured:* over 40 consecutive steps the first-order transfer is marginally accurate (median 0.458 at bar 0.5) — the step is first-order graspable — but raw h order is not a natural flow (only 52% of steps increase λ; $\|\Delta\|/\mathrm{gap} = 594$–$2026$: the accuracy comes from port concentration, not step smallness). *The typed lesson:* the induction needs a better order than h; the nested one-prime ladder was the named candidate and is decided in the block below. No marker move; no RH statement.

The one-prime ladder decision (`PRIME.PORT.PRIMELADDER.01`, decided). *Registered and decided* 2026-08-09 ([`verification/v885_ladder_flow_anatomy.py`], 13/13, PRIMELADDER-MEASURED). *Measured (exact per-atom lag rows; linearity warded at 0; endpoint warded vs the true floor at 2.5×10^{-9}):* the pure-Arch start is catastrophically subcritical ($\tau(0) = -9.9 \times 10^{13}$ at kz 9; G_+ not even PD at kz 13/26/40); insertion crosses into positivity only at $u^*/U = 0.991$–0.999 in every insertion order; the edge-first ordering reaches stable positivity only at 100% of the atoms (the ∼39% edge mass alone does not carry the wall); per-atom increments 54–64% positive; the deterministic mass scramble drops the endpoint to −474.7 (control fires). *The decision:* positivity is a property of the complete comb — `v877`'s truncation reversal sharpened to atom resolution; no monotone one-prime induction exists in any order; induction must run in a flow coordinate, not in comb content. No marker move; no RH statement.

The degree-2 Lax coefficient flow (`PRIME.PORT.LAX2.01`). *Registered* 2026-08-09, measured ([`verification/v885_ladder_flow_anatomy.py`], 11/11, LAX2-MEASURED (FLOW-NOISY / LAX2-PARTIAL)). *Measured:* the twelve deformation-polynomial coefficients are extracted on all 41 ladder steps, but normalized by the α step the coefficient series behave as noise (median roughness 1.365 vs lawful bar 0.7); the operator-level IIKS prediction misses the bar narrowly (0.350 vs 0.3); the *integrated* tau-transfer identity $\log\det(I + (I - D_k)^{-1}\Delta D)$ holds exactly (warded at 5.6×10^{-11}). *The named lead:* the dictionary columns f, Yf, Y^2f are nearly collinear — the raw coefficient basis is ill-conditioned; a well-conditioned reparametrization may expose a lawful flow the raw basis masks. *The demand:* the flow in a conditioned coordinate; then couple to the `PRIME.PORT.TAU.01` variational identity. *Adjudication note (2026-08-24, round 225):* the demand is adjudicated — **the word "2" was a hypothesis**: with the canonical basis-invariant conditioning the true minimal degree in the closing direction is *one*, and the closing direction is h, not s (LAX1-H-EXACT: rank-1 h-step, blind-predictive degree-1 connection, zero curvature world-blind; the s-time has no fixed degree, CONDITIONING-ONLY, and across rungs there is no common carrier — [`verification/v955_tau_iiks_toda_dictionary.py`], `PRIME.PORT.TAU.FINITE.IIKS.01` [E]). Per the contract's no-go rule, no further Lax cosmetics in the s/across directions. *Kill:* a conditioned basis in which the flow remains noise at the frozen bars (executed: the h-basis closes exactly). No marker move; no RH statement.

The half-filling / free-moment theorem set (`PRIME.FREEMOMENT.JFRACTION.01` [E] + `PRIME.JACOBI.DUAL.REVERSAL.01` [E]) and the positive-prefix contract (`PRIME.FREEMOMENT.POSITIVEPREFIX.01` [O]). *Registered and delivered / registered* 2026-08-24 (rounds 228–231, notes DLIV–DLVII; [`verification/v956_signedmoment_halffilling_duality.py`], probes embedded byte-exact, 14/14 + 17/17 + 17/17 + 13/13 first pass). *Exact (JFRACTION):* the half-filling law $N_w = \lceil \#\mathrm{supp}(\tilde{\mu})/2 \rceil$ on all five windows with the wall dying immediately past the cap (first r-flip at $N_w + 0/2/2/3/1$ — the wall is *maximal*); the complement identity and h-duality at dps 80; the free moment window law (the largest Hankel window inside the S free moments is $n = (S+1)/2 = N_w$; MAIN exhausts the entire free window, the controls die at 11–15%); the inertia theorem; the four-path dictionary (moment prefix = Jacobi chain = Euclidean chain = value chain, exact in rationals; Epstein's first wrong Euclidean pivot exactly at its flip $n = 25$). *Exact (DUAL.REVERSAL):* the full chain reversal ($\alpha^\#_m = \alpha_{S-1-m}$, $\beta^\#_m = \beta_{S-m}$; NO-SOURCE-REVERSAL — not a builder transformation) and the L-gauge midpoint connection theorem ($Y^\#_{N-1+l} = J\, Y_{N-1-l+1}\, W$, $W = \mathrm{antidiag}(1/L, L)$, built from the node polynomial alone; exact in rationals, real w9 at 9.1×10^{-94}, dps 100); the gauge is h-free — the wall's orientation sits exclusively in the h-chain it does not see (SIGNED-STOKES-WALL-EQUIVALENT). *The open contract (POSITIVEPREFIX):* $\beta_n > 0$ through the entire free moment window as a theorem with an identified mechanism — the wall itself in free-moment coordinates; half-filling is the *location*, its positive reachability is the result to be proven; named sub-deliverables: the source-pure scalar normalization beyond the node-log pole remover, and a source-side critical filling t_c as output. *Kill (of any mechanism candidate):* the frozen micro falsifier — a parametrix must predict the forced tail 0/2/2/3/1 and the control flips 25/21/27 blind; a model that only paints the positive side is a fit, not a mechanism. Design rules unchanged: the full von Mangoldt comb stays in the main problem, no PNT smoothing. No marker move beyond the two [E] registrations; no RH statement.

The bordered-RHP / tau-readout theorem set (`PRIME.PORT.RHP.BORDERED.READOUT.01` [E]) and the full-source campaign contract (`PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01` [O]). *Registered and delivered / registered* 2026-08-24 (rounds 244–255, notes DLXXIII–DLXXXV; [`verification/v958_bordered_tau_readout_dictionary.py`], probes embedded byte-exact, 23/23+29/29+24/24 + 20/20 + 22/22 + 26/26 first pass, plus the module-own exact-rational PSD base theorem).

Exact (BORDERED.READOUT): the bordered-RHP dictionary (the Uvarov border column carries F_n as its z^{-1} coefficient; the budget is a CD-kernel functional of the terminal Y_n data alone, $S_{n-1} = \iint K_n \, d\sigma \, d\sigma$; the Uvarov tau step and the T-sourced bordered LAX1 flow, mp-exact through all 184 w9 degrees); the Schlesinger rank-1 insertion ($\det Y_3 = -h_{n-2}F_{n-1}$; IIKS generator rank exactly 2; canonical 2×2 RHP $\ltimes$ one-dimensional budget cocycle carrying the full pivot chain — the irreducible-3×3 campaign dropped without replacement); the centering congruence ($B^\circ = B - \rho_0$, exact in rationals), the σ° dictionary, the dual-norm identity $Q_m = t^T H^{-1} t$ and the tail-kernel compression $T = \iint [K_N - K_8] \, d\sigma^\circ d\sigma^\circ$ (dps 160 through $N = 184$, rel 3.3×10^{-160}); the three exact error formulas ($\delta h_n = (R_1)_{12}$; the central rest identity; the error-kernel sandwich at 1.1×10^{-81}); the contour formula $R_1 = \frac{1}{2\pi i} \oint (R(z) - I) \, dz$ (3.2×10^{-14}); the exact augmented tau quotient $D = B^\circ - Q_7 - T = \tau^{\text{aug}}/\tau$ at full depth (three routes, 10^{-11} / mp 10^{-16}) with the pole classification (rank-1 border poles removable by residue closure; the D-layer essentially singular, $\exp(c/(z-s))$ type); and the PSD base theorem (positive basis $\Rightarrow$ Gram PSD $\Rightarrow$ border readout ≥ 0; bordered PSD = wall + budget by exact Schur complement; the frozen signed witness fires — positive-basis structure alone proves the wrong statement). *Typed honestly (measurements, not identities):* the corner census (CORNER-IMPORTED-ONLY, PSD 0/42 for all three sealed candidates) and the head-heavy budget profile (IRREGULAR-BULK, Gini 0.91); amplification extensive ($\mathfrak{A} \sim N^{1.01}$), annihilation neutral, the fiber fully in the error; the gauge thesis refuted (NO-GLOBAL-GAUGE, MODEL-ORIENTATION-BLIND — BASE-BLOCKED-BY-DRIFTING-LAYER on record); the diagonal pairing thesis refuted (diagonal 19%, wrong sign on scramble; REL-STRUCTURELESS, PAIRING-EXTENSIVE). *The open contract (FULLSOURCE.BASEFIBER, the campaign; the coupled-tau dictionary and telescope certified in [verification/v959_coupledtau_terminal_dictionary.py]; carried at the wave-4 state since 2026-08-25 — surface and orientation certified in [verification/v960_terminal_surface_closure.py] and [verification/v961_midpoint_orientation_dictionary.py], block below):* per the round-256 causal decision the separate treatment of base and fiber is *artificial* — the campaign object is the indivisible full-source pair $(\tau, \tau^{\text{aug}})$ with $D = \tau^{\text{aug}}/\tau > 0$ plus the no-pole statement, on the full von Mangoldt comb; admissible target form stays an *integrated* statement over the compressed CD readout (no uniform Y_n parametrix; the v959 coupled recursion is the one-dimensional budget cocycle of the v958 formulation in closed form). *The precise localization after rounds 256–259:* **the last fiber edge is a positive tau cross-ratio** — the entire fiber positivity is the one terminal inequality $h_{N-1}/F_{N-1}^2 > 7/5$ (equivalently $q_N < 1$; the 5/7 floor typed FLOOR-IMPORTED, derivation open; the measured driver is the rank-perfect F-h covariance +1.000: the ratio is the object) — and **the last base edge is a globally oriented prefix resummation** — the determinant sign is carried by coherent cancellation over the configuration ensemble (a resummation gap, not a selection gap). The two edges are registered as the successor contracts `PRIME.PORT.COUPLEDTAU.TERMINAL_CROSSRATIO.01` [O] and `PRIME.PORT.FULLSOURCE.PREFIX_RESUMMATION.01` [O] (blocks below; wave-4 state: the 42-rung surface of the fiber edge is *certified* in `v960`, the orientation edge has its blind Maslov GO and its named successor in `v961`). *The formalized target form (wave 4):* both edges are corollaries of the one master theorem `augmented_prefix_positive` (`rh/lean/RH/Window.lean`) — $A_{w,n} = \left[\begin{smallmatrix} H_n & u_n \\ u_n^T & B \end{smallmatrix}\right] \succ 0$ for every admissible MAIN window and prefix degree, with `MainWindow` an opaque predicate (the three reviewer counterexamples are permanent Lean guards); modulo this one `sorry` the two ex-edges are Lean-proved, and per the wave-4 measurements the predicate must encode *metric* exactness (sub-gap-exact comb placement), not combinatorial Euler properties. *The north star reformulated (wave 5, [verification/v962_halffilling_pinning_theory.py]):* by the moment counting theorem T1 (the free pivots are exactly $h_0..h_{N_w-1}$), the crossing budget theorem T2 (the crossing count is S_-, world-blind) and the main window reduction T4, the campaign's open center is, in reinstform, **why is the signed prime moment form quasi-definite up to the maximally free order?** — $\forall n < N_w : h_n > 0$, exactly equivalent to $\min C \geq N_w$, and via T4 the base half of the master `sorry` (Lean: `free_window_positivity`, `rh/lean/RH/Window.lean`; carried on the successor contract `PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01` [O], block below). *The canonical form (wave 6, [verification/v963_lstar_reduction_dictionary.py]):* by the r283 A2 chain the north star is exactly **the missing lemma L*** — for every real polynomial p with $\deg p < N_w$:

$\int p^2\,d\nu < \int p^2\,d\mu$, equivalently $\lambda_{\max}(E_{N_w}) < 1$ (the ν-dressed μ-CD kernel is a strict contraction at half-filling depth) — a two-measure moment problem instead of a determinant cascade; registered as `PRIME.LSTAR.SUBORDINATION.01` [O] (block below) with the standalone problem document `rh/problem/lstar_problem.tex` and the Lean statement `lstar_subordination` (the direction L* ⇒ free-window positivity is proved). *The measured state after rounds 250–259 (rounds 250/254/255 experiments-side, SPEC `f9bf299d` / `1083f7a0` / `45875904`):* OUTER-MODEL-FAILS (rates carried — plateau $((b-a)/4)^2 = 0.2500$ exact within 0.040, h-rate slopes ≤ 0.007 dec/degree — pointwise and fiber failed; the naked electrostatic model beats the dressed one 2.2–3.8×); OFFDIAG-EXTENSIVE ($D_{\text{part}} \approx 0.195\,N$; the atom diagonal carries 78.6% of the world gap; the round-254 compactness-inversion headline and the Epstein fiber anomaly are *downgraded on record* by the round-256 firewall — both are base contamination of the indefinite continuation and may not be cited as border/fiber discriminators); COCYCLE-LAW (round 257: the drain is monotone on MAIN through the entire free window and its increment law breaks exactly at the control flips 21/25/27 — base and fiber are coupled in one scalar law; both augmented border variants cross the wall: no bordered spectral no-pole shortcut); and the round-255 one-bit reading is *revised* (round 259: the same short-range swap structure sits on the mains with fully overlapping gap distributions — the base flip stays a threshold event of an extensively accumulated quantity, but the missing information is not one-dimensional). *Kill (of any mechanism candidate; unchanged plus the v959 no-gos):* a base layer must produce the control flips on Epstein/smooth (orientation, not just rate); a fiber main term must separate scramble by ≥ 1 decade before readout and survive the $\times 2$ share window; mass majorants, level crossing, one-swap statistics and k-swap truncations are stop-listed (measured dead in rounds 258/259). No marker move beyond the [E] registrations; no RH statement.

The coupled-tau / terminal-budget theorem set (`PRIME.PORT.RHP.COUPLEDTAU.TERMINAL.01` [E]) and the two terminal-edge contracts (`PRIME.PORT.COUPLEDTAU.TERMINAL_CROSSRATIO.01` [O], `PRIME.PORT.FULLSOURCE.PREFIX_RESUMMATION.01` [O]). *Registered and delivered / registered* 2026-08-25 (rounds 256–259, notes DLXXXVII–DXC; [`verification/v959_coupledtau_terminal_dictionary.py`], probes embedded byte-exact, 21/21 + 26/26 + 27/27 + 19/19 first pass, plus the module-own exact section S0). *Exact (COUPLED-TAU.TERMINAL):* the positive-prefix firewall (MAIN $p_{\max} = 184 = N$ with zero negative modes, the only positive-prefix world; controls INDEFINITE-CONTINUATION at 21/25/27; every prefix kernel numerically PSD at 5.7×10^{-16}, re-proven as exact algebra in-module: H_p PD iff the whole h-prefix is positive, $\det H_p = \prod h_i$, Schur-complement must-fail firing) with the exact 2×2 causal census and its two contamination downgrades on record (the r254 scramble compactness is base contamination, $f_{\text{neg}} = 0.9311$; the Epstein fiber anomaly lives entirely in the indefinite continuation $k \geq 25$); the coupled two-term recursion $\tau_{n+1} = a_n \tau_n$, $\tau_{n+1}^{\text{aug}} = a_n \tau_n^{\text{aug}} + b_n \tau_n$ with $a_n = c_n^2 h_n$, $b_n = -(c_n F_n)^2$, equivalently $D_{n+1} = D_n - F_n^2/h_n$ (anchor rel 4.8×10^{-11}, mp dps 220 through the scramble flip), with the bilinear tau form $\tau_n^{\text{aug}} \tau_{n+1} - \tau_{n+1}^{\text{aug}} \tau_n = (c_n F_n)^2 \tau_n^2$ derived and gated in-module — the base alone carries the sign, the border only nonnegative magnitude, and the cocycle breaks exactly at the control flips; the blind coefficient-field pass of the frozen micro-falsifier (control flips 3/3 and forced tail 0/2/2/3/1 on all five windows; the parametrix half open); and the Schur budget telescope $D_N = B - S_{N-1}$ (mp dps 220 at 2.4×10^{-16}; $B = S_{N-2} + 5/7$ by construction) with the consequence gate margin $> 0 \iff q_N < 1$ and the corner-ray t^* normal form (slope exactly 5/7, root exactly q_N). *Typed* NUMERICAL *(measurements, not laws):* the q census (42/42, flat, min margin 0.0139); the F-h covariance +1.000 (a measured structural fact); the 5/7 surface (FLOOR-IMPORTED, derivation open); the r259 ensemble-coherence findings and the parametrix orientation blindness. *Honest negatives (load-bearing):* the level-crossing no-go (branches cross 5–16 degrees too early; MAIN crosses 45 times with zero true flips), the one-swap no-go (r255's ORIENTATION-LOWDIM(1) revised on record), the mass-majorant no-go (all three sealed majorants 0/42, median ×490, break at $0.97N$). *The open contract (TERMINAL_CROSSRATIO, the fiber edge):* prove $h_{N-1}/F_{N-1}^2 > 7/5$ on the cofinal ladder — equivalently derive the true terminal floor source-pure; the bound must bind the *ratio* directly (no separate F-majorant against an h-minorant), hold against the flat measured margin profile (min 0.0139, no N-decay to spend) and survive the near-terminal zone at $0.97N$;

kill list (stop-listed): the mass-majorant class, compression, constants, band locality, any one-sided bound on the covariant pair. *The open contract (PREFIX_RESUMMATION, the base edge):* the closed analytic form of the coherent ensemble resummation — the determinant as a globally oriented signed partition sum, completing the coefficient-field pass to a parametrix-level pass (predict the control flips and the forced tail blind from a closed model, no minors/slogdets, AST-scanned); *kill list (stop-listed):* two-branch level crossing, one-swap/k-swap statistics, saddle-point dominance, the r255 parity-rule and hysteresis hybrids (0/3 + 0/5 blind). *Wave-4 progress (2026-08-25, block below):* on TERMINAL_CROSSRATIO the 42-rung surface is now *certified* (41 mechanism target-blind + kz15 exact-finite; the cofinal front is the single hypothesis H5 / lemma L2). *Wave-7 progress (2026-08-25, [verification/v964_lstar_coherence_census.py]):* the H5/L2 front has its *first named classical theorem candidate* — the exact van der Corput inequality at $H = \lceil\sqrt{m}\rceil$ (constant-free, arithmetic-free) delivers $\delta' = +0.309 > 0.21$ world-blind and certifies 6/7 exceptions +38/42 rungs; only the kz15 razor blocks 7/7 (exact-finite per r270); registered as the successor contract `PRIME.PORT.L2.VDC_LEMMA.01` [O] (block below) — the remaining input is the chain origin of the P-variance scaling. On PREFIX_RESUMMATION round 261 closed the finite-combinatorial half honestly negative and the named path is the oriented midpoint theorem (Maslov census GO; successor contract `PRIME.PORT.RHP.MIDPOINT.ORIENTED_THEOREM.01` [O], executed as rounds 279–281 and consumed in wave 5, block below). Mincut base 4 / refined 5 unchanged; no RH statement.

The terminal-surface closure and the midpoint-orientation theorem sets (`PRIME.PORT.COUPLEDTAU.SURFACE_CLOSURE.01` [E], `PRIME.PORT.RHP.MIDPOINT.ORIENTATION_DICTIONARY.01` [E]) and the two wave-4 contracts (`PRIME.PORT.RHP.MIDPOINT.ORIENTED_THEOREM.01` [O], `PRIME.PORT.WALL.METRIC_FIREWALL.01` [O]). *Registered and delivered / registered* 2026-08-25 (rounds 260–278, notes DXCII–DCXI; [verification/v960_terminal_surface_closure.py] with 31/31 + 26/26 + 26/26 + 28/28 + 29/29 + 29/29 + 28/28 + 28/28 + 24/24 + 24/24 first-pass gates and [verification/v961_midpoint_orientation_dictionary.py] with 32/32 + 25/25 + 29/29 + 31/31; probes embedded byte-exact and executed in their sealed `-smoke` stage, SPEC SHAs pinned, full records re-verified by `rh/verification/run_rh.py` — since this wave a gate of `bash build.sh audit`). *Exact (SURFACE_CLOSURE):* **the 42-rung surface census $q_N < 1$ is certified on all 42 rungs as a finite fact** — the two-branch theorem (U = the sealed C1b triangle form, $M = \sqrt{5/7}$, Z-identity at 1.8×10^{-14}; cheap branch $g_w \geq 0$ exactly 35/42 including both mains *without cancellation*; exception set exactly $\{kz15, kz20, kz22, kz36, kz38, kz39, kz52\}$), the sealed phase bounds (edge truncation: kz22, the first source-pure exception certification ever; block alternation: +0.26 to +0.41 decades on all 42, six exceptions certified, no detector fires), the kz15 interval certificate (outward-rounded intervals, dps 640, end width 1.5×10^{-92}, margin +0.02680, dps-halving ward — typed an exact finite enclosure, no mechanism), and the universal pair theorem (H1–H4 structural, Lean-proved in `rh/lean/RH/PairBound.lean`; H5 = the margin $|Z_{\text{local}}| + \varepsilon < \sqrt{5/7}$ the single window-dependent hypothesis). *Exact (ORIENTATION_DICTIONARY):* the base Casoratian is the pivot chain itself ($\hat{\pi}_{n+1}q_n - \hat{\pi}_n q_{n+1} = h_n$, $c' = 1$; right solution triple-constructed, the node polynomial is the right Dirichlet boundary), the h-free midpoint form *is* the node polynomial (the r231 sign-blindness as an identity), the augmented telescope $D_{n+1} = B - W_n^{\text{aug}}/W_n^{\text{base}}$ exact against the independent r263 route, and the Maslov census GO (rule R2 = Jacobi interlacing/reality: blind 42/42, mains safe at full depth, controls fire exactly at flip+1, one-way, not h-equivalent). *Typed* NUMERICAL *(measurements, not laws):* the L2 scaling anatomy (BOUND-COARSENESS: the truth margin rises with N; flip condition $\delta' > 0.21$ of available 0.45, address c3), the Euler surgery ladder (PERTURBATION-INSENSITIVE + the firewall map: the wall, not the cancellation rate, is what is arithmetically special), the metric-firewall continuum ($D \sim \theta^b$, support exactness first), the exact position gradients (Hellmann–Feynman class, identity fine type) with the predictive small-primes-loaded u-profile, and the perturbative-only stability law. *Honest negatives (load-bearing):* the eventual-triangle no-go (kz52 itself an exception), three identity routes dead with break loci, the retracted w13 Gram provenance, the KYP memory no-go from structure (exact double obstruction o1/o2 on

46/46 worlds; the admissible Riccati memory target-inverse), and the raw atom-Sturm census refuted as winding quantity (STURM-CHAIN-VERIFIED honestly not awarded). *The open contract (ORIENTED_THEOREM, the orientation edge):* the contraposition form — an independent index obstruction that forbids the R2 interlacing break before half-filling on MAIN; the detection direction (break at flip+1, after the flip) forces exactly this form, and the `MainWindow` arithmetic must enter here as *metric* exactness; *wave-5 state*: rounds 279–281 are executed and consumed ([`verification/v962_halffilling_pinning_theory.py`], block below) — the theorem share is promoted, the candidate generic two-sided obstruction is refuted exactly, and the center is precisified to the budget localization (free-window survival). *The open contract (METRIC_FIREWALL, the firewall lane):* the global form of the wall's metric stability — a wall lemma quantitatively stable under family/assignment permutations at small dose while consuming the metric data exactly; the certified state is graded-continuous with exact gradients but perturbative-only (θ^* 25–170× below the smallest measured dose, L_D window-specific, no uniform lemma candidate); the MAIN positivity itself (H5 / the master theorem) is not part of this demand and stays the open center. *The surface fence (binding):* the surface closure is a census-plus-certificate result, *not* a cofinal theorem — the window quantifier stays. Mincut base 4 / refined 5 unchanged; no RH statement.

The half-filling pinning theory (`PRIME.WALL.HALFFILLING_PINNING_THEORY.01` [E]) and the two wave-5 contracts (`PRIME.PORT.REPRESENTATION.CONTEST.01` [O], `PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01` [O]). *Registered and delivered / registered* 2026-08-25 (rounds 279–281, notes DCXII/DCXIV/DCXV; [`verification/v962_halffilling_pinning_theory.py`] with 32/32 + 29/29 + 28/28 first-pass gates; the three probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs pinned, full records re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 in pure Fractions). *Exact (the four theorems as named gates):* (T1) the **moment counting theorem** — the pivot h_n consumes $m_0..m_{2n}$, the S free moments are $m_0..m_{S-1}$, hence the free pivots are exactly $h_0..h_{N_w-1}$ and h_{N_w} is the first forced one: half-filling is the end of the free moment space, arithmetic-exact for all $S = 2..2000$ with the exact-rational freedom demonstration — "why half-filling" is answered by counting; (T2) the **crossing budget theorem** — $\#\{n < S : h_n < 0\} = S_-$ exactly (Jacobi/Sylvester), world-blind; (T3) the **two-sided parity theorem** — node-sign pattern, gap parity and the census bilanz, h-blind at every degree, with the 87376-case exhaustion; (T4) the **main window reduction** — given T1 and T2 the entire open statement of the wall is $\min C \geq N_w$, exactly equivalent to $\forall n < N_w : h_n > 0$: one placement question, the north star in reinstform. *The refutations (named negative gates, load-bearing):* NO-UNIVERSAL-O1-PINNING (exact; the one-negative toy has offset $N_w - 2$, unbounded in S; only provable ceiling: the pigeonhole $\min C \leq S - S_-$; $C \leq 5$ is a 42-rung measurement), NO-EXTREMALITY (measured + exact witness: three mp-confirmed directions push the w9 crossing 184 → 185; a 4-atom one-negative measure has $\min C = N_w + 1$ exactly), NO-GENERIC-MASLOV-OBSTRUCTION (exact; three rational counterexamples — the two-sided machinery carries no arithmetic), NO-SIMPLE-OFFSET-LAW (measured; six sealed source-pure candidates, max $|\mathrm{sp}| = 0.273 \ll 0.75$, all world-blind). *Lean state:* T1 and T4 *proved* (`moment_counting_free_pivots`, `main_window_reduction`); T2 stated (`crossing_budget`, `sorry` with v962 reference); the fog-free central hole is `free_window_positivity` (`rh/lean/RH/Window.lean`, via T4 the base half of the master `sorry`); the exact one-negative instance is the permanent guard `upper_pinning_not_universal`. *The open contract (REPRESENTATION.CONTEST):* the sealed contest of representations for the localization statement — which coordinate system (dual chain, moment/Weyl, gradient/criticality, Jacobi interlacing, tau/cocycle) carries a MAIN-separating, non-restatement handle on free-window survival; *forbidden (binding):* no derived 5/7, no bound mechanism claim, no asymptotic law, no localization claim from any census, no undeclared h-reading candidates, no world-blind carrier; the v962 refutation set binds the candidate space. *The open contract (FULLSOURCE.QUASIDEFINITENESS, the open center):* $\forall n < N_w : h_n > 0$ on every admissible MAIN window — why is the signed prime moment form quasi-definite up to the maximally free order? *Forbidden (binding, the measured kill lists):* no bookkeeping-only proof (the Lean counterexample guards are permanent), no world-blind mechanism, no Weyl μ/ν perturbation route (range 1.1%), no dual restatement, no variational local-maximum route, no simple offset law, no universal $O(1)$ upper pinning. Both rounds (`representation_contest_probe.py`, `fullsource_quasidefiniteness_probe.py`) were in flight at the wave-5 cut and are *executed and consumed in wave 6* ([`verification/v963_lstar_reduction_dictionary.py`], block below): REPRESENTATION.CONTEST adjudicated CONTEST-ALL-DEAD (the four classical languages closed with exact reasons), FULLSOURCE.QUASIDEFINITENESS adjudicated MECHANISM-PARTIAL(A2) — L* established as the canonical reduction, the open center carried forward on `PRIME.LSTAR.SUBORDINATION.01` [O]. Mincut base 4 / refined 5 unchanged; no RH statement.

The L* reduction dictionary (`PRIME.LSTAR.REDUCTION_DICTIONARY.01` [E]) and the L* subordination contract (`PRIME.LSTAR.SUBORDINATION.01` [O]). *Registered and delivered / registered* 2026-08-25 (rounds 282–285, notes DCXVI/DCXVII/DCXIX–DCXXI, plus the standalone problem document of note DCXX; [`verification/v963_lstar_reduction_dictionary.py`] with 30/30 + 26/26 + 30/30 + 33/33 first-pass gates; the four probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs pinned, full records re-verified by `rh/verification/run_rh.py`; plus the module-own exact section S0 in pure Fractions). *Exact (the reduction as named gates):* the μ-frame congruence $\mathrm{minor}_k(D_\mu - G) = D_k(\tilde{\mu})$ at every $k \le S_+$ (truncation must-fail loud); the frame contraction $h_0..h_{m-1} > 0 \iff \lambda_{\max}(E_m) < 1$ (both truth values); the crossing dictionary $\lambda_{\max}$ crosses 1 exactly at $\min C + 1$ (CAPACITY-CEILING typed where $\min C = S_+$); the pigeonhole ceiling $\min C \le S_+$ re-proved in the frame; capacity-as-counting refuted by the rank-one pair (metric values 1.25×10^{-13} vs 1.25×10^{2} decide); the r285 bookkeeping $\lambda_{\max} = \mathrm{maxdiag} \times (1 + \mathrm{assist})$ with the sign-exact budget equivalence; and the r282 four-language elimination as named negative gates (CONTEST-ALL-DEAD — SOS iff empty negative register, Kasteleyn orientation iff $S_- = 0$ by full 2^S exhaustion, Hamiltonian-PSD = $h > 0$ bookkeeping, dual-pair sync by theorem; the common deep reason: every classical positivity language forces positivity exactly for the positive measure class, and MAIN is signed). *Typed* NUMERICAL *(measurements, not laws):* the one-atom-vs-collective dichotomy ($n_{\mathrm{DIAG}} = 187$ vs crossing 185; controls collective), the sub-classical edge growth ($p = 0.38$, classical coverage 0/42 — the (D) route must be a discrete bound), the ensemble position (MAIN's wall assist the extreme LOW-OUTLIER, pct 0.00), and the first two positively MAIN-separating detectors of the program (assist-at-crossing 0.0195 vs dead 1.69..2.99; random-sign $z = -3.15$ destructive vs dead $+4.95.. + 12.44$). *The open contract (SUBORDINATION, the canonical open center):* for every admissible anchor z and every real polynomial $p \ne 0$ with $\deg p < N_w = (S+1)/2$: $\int p^2\, d\nu_z < \int p^2\, d\mu_z$ — equivalently $\lambda_{\max}(E_{N_w}) < 1$; *state (wave 7, [`verification/v964_lstar_coherence_census.py`]):* 57/57 measured (the 42 plus the 15 extension anchors, N_w up to 1218), **the margin decay resolved harmless** (r286: no counterexample, the $O(1)$ census offset survives beyond the cap, flattening power law $\alpha \approx 3.05$ with driver $c_w \to 1$, and the margin falls *slower* than the local loading speed — no acute falsification threat; the unbounded family stays open); the coherence half has a *named carrier class* (r288: antiphase next-nearest ARCH–ARCH pairs, $z_v = -3.149$, total control reversal, finest alignments below phase resolution); **the diophantine route is excluded** (r289 METRIC-ONLY: the rational twin keeps the full signature identical while every exact log-relation is destroyed; metric coherence threshold $10^{-3}..3 \times 10^{-3}$ of the local gap; Baker ∼7900x too weak *and* unnecessary); *state (wave 8, [`verification/v965_lstar_curvature_arc.py`]):* the profile-functional question is **functionally negative-sealed** — after five sealed class families (local scalars, non-local/low-rank, the curvature two-form in the mixed metric, the metric-reconciled two-form, the hardened partial-free |sp| form) no closed predictive profile functional of the working set exists over the honest bar (r294 F10-FRAGILE + r295 F10-SP-MAJORITY: F10 unpromoted), while the working-set geometry is now *measured* (anisotropic tube, one-degree budget ridge, rank-1 DENS curvature valley, simple h_{184} flip zeros); the successor question is `PRIME.LSTAR.CLOSED_FUNCTIONAL.01` [O] (block below); no round is in flight at the wave-8 cut; the standalone document `rh/problem/` is the external reference; *kill lists (inherited, binding):* the v962 refutation set, the r282 four-language closure, the r283 forbidden list (no triangle/Gershgorin bound — dead at $n = 21 \ll 185$ — no target-inverse quantity, no truncation, no posthoc window), the r284 shielding/Nyquist refutations, the r285 no-fixed-δ/δ'

trend, plus the wave-7 kills (no diophantine/linear-forms route, no plain phase-dispersion separator, no sinc/CD-only sign mechanism), no derived 5/7, no bound mechanism claim, no asymptotic law. Lean: `lstar_subordination` stated (`rh/lean/RH/Window.lean`, documented `sorry`); the direction L* ⇒ free-window positivity is *proved* (`lstar_implies_free_window`). *State (wave 11, [verification/v968_architecture_adjudication.py]):* the formal frame is **restructured** — the master theorem is now *proved from* L* + terminal (`lstar_terminal_implies_master`, r305, with `augmented_prefix_positive` and `free_window_positivity` as corollaries; sorries 9 → 5, the two TRUE holes standing alone: `lstar_subordination` and `terminal_positive_main`), the window source is a real Lean construction (r310/r310b: nodes = $\log p^k$, weights = $\Lambda(p^k)$ derived; source exactness, mass conservation of any folding, the strong stabilization form), and the surrounding architecture is fully adjudicated (Lane A closed as cone language, B not overtaking with the rank-one identities banked, C parked); additional kill entries: no fixed-head route (r307), no constructive PSD-G rule in the planned cone language (r312), no transfer-operator contraction at chain level (r313), no two-regime pointwise majorant under a mid-ladder freeze (r316). *State (wave 12, [verification/v969_forks_and_redteam.py]):* the indefinite fork is adjudicated (r318 P2-MAIN-SPECIFIC: P1 banked *as language* — the index statement is a total restatement of L*, no lever; classical sign regularity dead on MAIN) and the antiphase fingerprint is closed as ALGORITHM-ARTIFACT (r322: a least-norm-proximal Dykstra-basin property, not a psd-solution-set property; the selection geometry is the named residual); the Lean statement class is red-team hardened (the r319 U-findings repaired by the r320 retypes + guards, census 5); **per the binding reviewer adjudication the lane is internally paused** (allocation 0 %; the selection geometry is not pursued further; reopening only under the six documented conditions of the r322 round notes; the external memo candidate: the two-weights sampling formulation $\|J_N\| < 1$), and the hole itself is typed **window-local** (Level B of the wave-12 three-level proof graph — see the red-team block below). Additional kill entries (wave 12): no Pontryagin/Krein index lever, no sign-regularity/variation-diminishing route, no antiphase sign-law theorem without a canonical selection rule. *State (wave 13, [verification/v970_extraction_and_composition.py]):* the pause held — the planned r323 fork was cleanly aborted at the end of its design phase, before any write access (nothing counts as a round); the Lean census is now 5 → 8 (three typed Level-C extraction statements, r326 — none touches this hole; `lstar_subordination` byte-identical). Mincut base 4 / refined 5 unchanged; no RH statement.

The L* coherence census and the vdC L2 lemma contract (`PRIME.LSTAR.COHERENCE_CENSUS.01` [E], `PRIME.PORT.L2.VDC_LEMMA.01` [O]). *Registered and delivered / registered* 2026-08-25 (rounds 286–289, notes DCXXIV/DCXXII/DCXXV/DCXXVI; [`verification/v964_lstar_coherence_census.py`] with 29/29 + 25/25 + 31/31 + 42/42 record gates; the four probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs `0a44ac4e/761d88fa/da46f7ee/91cdc2b1` pinned; plus the module-own exact section — the Fejér window-sum identity, the van der Corput inequality with its Cauchy–Schwarz proof route, the Abel/Erdős–Turán bound, the mutant must-fails, the 2-power resonance count $\binom{8}{2} = 28$, in pure Fractions). *Exact (the vdC theorem):* $|\sum_j P_j|^2 \le \frac{m+H-1}{H}\sum_{|h|<H}(1-\frac{|h|}{H})A(h)$ — constant-free, arithmetic-free, proved module-own; at the frozen window $H = \lceil\sqrt{m}\rceil$ it delivers $\delta' = +0.309 > 0.21$ world-blind (admissible: L2 is the generic half), certifies 6/7 exceptions +38/42 rungs, and only the kz15 razor blocks 7/7 (exact-finite per r270, reserve 0.0268); F1 discrepancy/Erdős–Turán dies loud ($\delta' = -0.21$, 0/7); the autocorrelation map relocates the c3 address: on 39/44 worlds the block sequence *reinforces* — the cancellation is already absorbed into the P_j magnitudes, the frames win through the $\sqrt{m}$ count economy. *Typed* NUMERICAL *(measurements, not laws):* the r286 extension census (15 new anchors $N_w = 942..1218$, all margins mp-sign-safe positive, min $+1.806 \times 10^{-8}$; off census $O(1)$, max +4; flattening power law $\alpha \approx 3.05$; HARMLESS quantified; q_N reconciled; EXTRAP-CALIBRATED 15/15), the r288 carrier map (antiphase next-nearest ARCH–ARCH pairs, $z_v = -3.149$, total control reversal, finest alignments below phase resolution), the r289 twin table and jitter ladder (metric coherence threshold $10^{-3}..3 \times 10^{-3}$ of the local gap). *Honest negatives (load-bearing):* SAMPLING-BLIND (the sinc/CD predictor partial, 0.728/0.878 < 0.90), SOURCE-SEPARATOR-NOT-

FOUND (plain phase dispersion world-blind, ensemble-generic), DIOPHANTINE-EXCLUDED(METRIC-ONLY) — the rational twin (every tent center replaced by a rational $p/q \times \Delta$ at position cost $\leq 2.1 \times 10^{-9}$, every exact log-linear-form relation destroyed) keeps the full signature *identical*: the effect of the diophantine structure is null, Baker territory cleanly excluded (∼7900x too weak in the exponent *and* unnecessary). *The open contract (VDC_LEMMA, the L2 chain statement):* prove the vdC form $|R| \leq \sqrt{\frac{m+H-1}{H} \sum_{|h|<H} (1 - \frac{|h|}{H}) A(h)}$ at $H = \lceil \sqrt{m} \rceil$ as a *chain* statement — derive the P-variance scaling (measured small and large-lag structureless) from the chain/source, source-pure and window-independent, so the bound closes the L2 lemma (H5, address c3, $\delta' > 0.21$ of available 0.45) as a theorem; kz15 needs no new mechanism (exact-finite per r270); *kill list (inherited):* l1 pair hierarchies, state-space/KYP, s-flows, mass splits, H-peek (selection-by-answer, sealed away), fit primitives; world-blind validity is *admissible* here (the declared difference to the L* lane); *state (wave 9,* [`verification/v966_l2_reduction_chain.py`]*, block below):* the remaining input stands **reduced to one inequality** — the four-round chain r297→r300 froze the target level $\sigma^* = -0.516$, proved the exact sign-preserving transfer to the difference measure, settled the ratio half structurally and left NEFF-TARGET (slope(n_{eff}) $\geq +0.908$, measured $+0.963$, margin 0.055; `PRIME.L2.NEFF_TARGET.01` [O]); *state (wave 10,* [`verification/v967_l2_cascade_closure.py`]*, block below):* the reduction route is **documented stopped** — the r303 regress audit retyped the cascade r297→r302 as an exact dictionary around one measured core $S = \sigma^* - \mathrm{sl}_D = +0.0547$ (REGRESS-CONFIRMED) and the r304 stop case fired (LAW-LONGRANGE: the global lag profile is a stable period-4 comb, no k_0); the honest stop state is L2 generic ⇔ anti-concentration of the explicit *dc* block field; return later with new tools. *The successor front on SUBORDINATION:* the firewall reads the tent-split fraction *profile* metrically, not its arithmetic — the open question was the profile functional (rounds 290–295 executed and consumed in wave 8, [`verification/v965_lstar_curvature_arc.py`], block below: the question is functionally negative-sealed over five class families and carried forward as `PRIME.LSTAR.CLOSED_FUNCTIONAL.01` [O]). Mincut base 4 / refined 5 unchanged; no RH statement.

The L* curvature arc and the closed-functional contract (`PRIME.LSTAR.CURVATURE_ARC.01` [E], `PRIME.LSTAR.CLOSED_FUNCTIONAL.01` [O]). *Registered and delivered / registered* 2026-08-26 (rounds 290–295, notes DCXXVII/DCXXIX–DCXXXIII; [`verification/v965_lstar_curvature_arc.py`] with 32/32 + 30/30 + 36/36 + 43/43 + 42/42 + 44/44 full record gates; the six probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs `f953dd71/bb512c17/050821ff/33c44cc6/88c6fd1e/4d7d8095` pinned; plus the module-own exact section — the r292 polarization identity $H(u, v) = [d^2(u+v) - d^2(u-v)]/4$ with its wrong-prefactor must-fail, the r291 budget telescope with the dropped-atom must-fail, the r293 $\alpha = 1$ simple-zero doubling law with the double-zero mutant, the sealed r294/r295 decision bars as exact logic with tipping must-fails, in pure Fractions). *Measured (the geometry of the working set):* a soft-shouldered, strongly anisotropic TUBE in the exact θ_{eq} coordinate (killfrac 0.38/0.62/1.00 at $5 \times 10^{-4}/10^{-3}/2 \times 10^{-3}$; world axes kill 5–50x earlier; the SMOOTH axis a privileged yet gradient-orthogonal killer, $\cos = -3 \times 10^{-5}$); a REAL one-degree ridge ($\min C = 185$ at factors 1..8) whose lift is a first-order budget phenomenon (one threshold in $(1.280, 1.291]$ over all 18 matched-dose cases, $k_{\min} = 9$, one TOP6 retraction; no fixed point, plateau 185; MAIN-specific — EPSTEIN first-order flat); a RANK-1 DENS curvature valley (92.5 %, $\lambda_{\text{top}} = -0.418$; not SMOOTH, not the ridge; EPSTEIN second-order structureless); and SIMPLE h_{184} flip zeros ($\alpha = 1.000 \pm 0.003$; the TOP6 retraction a second simple zero at $f_{\text{ret}} = 7.107$; MSTAR-NO-LAW — no fixed safety distance in any unit). *Honest negatives (load-bearing, the frozen content):* ALL-FUNCTIONALS-BLIND over FIVE sealed class families (local scalars best |sp| 0.263; non-local/low-rank best 0.471 < 0.881; curvature two-form mixed 0.884 < 0.907; the metric-reconciled F10 — home win +0.024 and partial +0.423, but r294 F10-FRAGILE: win 5/5, partial 2/5; the hardened partial-free form — r295 F10-SP-MAJORITY 14/20, margin median +0.028, min −0.079: the sealed three-clause HARDENED bar fails on two clauses); R293-LUCK (the r293 partial +0.423 and the r294 5/5 were both top-of-distribution; the composition-matched subsample gains +0.067 and misses the level clause by 0.004); L2-NOT-DENS + L2-VIA-CONSERVATION (near-tautological, η_0 share 2.9×10^{-30}) — F10 is NOT promoted, the sealed rejections stand verbatim. *The open contract (CLOSED_FUNCTIONAL, the successor question):* exhibit a closed, source-pure functional of the tent-split fraction profile that predicts the survival depth of the working set at HARDENED grade (the sealed r295 three-clause bar: $\geq 18/20$ wins AND margin median $\geq +0.02$ AND no loss beyond 0.02) with a stable partial channel — or prove non-existence in a named class; *kill list (sealed, binding):* the five class families above, MSTAR-NO-LAW, no diophantine/linear-forms route (wave-7 METRIC-ONLY inherited), no plain phase-dispersion separator, no sinc/CD-only sign mechanism, no re-training of the hash-sealed H_{tr} (`3447ed198a56`), no posthoc bar moves; *open fronts (named):* the loss-corpus forensics (K05/K07/K19, margins $-0.041/-0.057/-0.079$), the rank-2 DENS core (top DENS eigenaxis alone +0.855, DENS share 0.989, mechanism-constant across w7/w9/w11: 0.795/0.989/0.834), the substantive "why L2" question, and larger window corpora for the w7/w11 knife edge; *measurement standard (sealed):* PARTIAL-STD = the partial on the full sealed 147-point fresh protocol (r295 pins: median +0.279, IQR $[+0.182, +0.401]$). Lean: no new statement — the wave carries no new formal target form, the hole stays `lstar_subordination`. *State (wave 9, [verification/v966_l2_reduction_chain.py]):* the DENS-identity fork is **closed** — r296 adjudicated DENS-WORLD-BLIND under the sealed rules (the coupling number $\cos(e_{\mathrm{top}}, \nabla\lambda_{\max}) = +0.394$ below the sealed 0.40 bar, noise max 0.331, miss 0.006 honest; the λ_3 control couples *harder*, -0.574: a near-crossing eigenvalue-band property, not top-specific; every arithmetic candidate ≤ 0.38; the moment subspace (0.825) construction-adjacent, holding on both dead controls 0.571/0.603): front (2) — the rank-2 DENS core — is answered negatively at the identity level (a mechanism hint, not an arithmetic identity, not the L*-gradient direction), the lane routed to L2 per the pre-adjudicated reviewer fork; the remaining open fronts are (1), (3) and (4). *State (wave 11, [verification/v968_architecture_adjudication.py]):* **Lane A (the constructive PSD-G rule fixed by the r308 feasibility census) is closed as a cone language** — r311 STRICT-SOURCE-CONE (exact rational in-span Farkas certificates on MINI#0 + SM1#0; the r308 world separation demarcated as the budget sign / a chordal restatement) and r312 COEFFICIENT-SIGN-WALL with the mechanism *named* (block-psd membership without rank-one-SDD membership, obstructed at the budget/border sign; Farkas mass on the border/budget row, t-share 0.583/0.725) — no GO per the reviewer tree, the resources moved to the fiber; the PSD-G rule stays open as a *question*, closed as a *lane*; the closed-functional demand itself is untouched (the sealed r295 bar stands). Mincut base 4 / refined 5 unchanged; no RH statement.

The L2 reduction chain and the NEFF target contract (`PRIME.L2.REDUCTION_CHAIN.01` [E], `PRIME.L2.NEFF_TARGET.01` [O]). *Registered and delivered / registered* 2026-08-26 (rounds 296–300, notes DCXXXV–DCXXXIX; [verification/v966_l2_reduction_chain.py] with $39/39 + 27/27 + 28/28 + 32/32 + 31/31$ full record gates; the five probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs `ffb413c8/e42a76eb/05e831be/f432e944/55218b5d` pinned; plus the module-own exact section — the σ^*-composition arithmetic ($\delta'(\sigma^*(\mathrm{pad})) = \mathrm{pad}$ exactly, the pad-dropped mutant composes to 0), the decomposition identity $S_F = B(\omega, \omega) + B(\Delta, \omega + \beta)$ on the exact positional Fejér block form (toy $3 = 1 + 2$; the $|\Delta|$ and wrong-weight mutants break by exactly 4 and 2/3), the participation identity $D \times n_{\mathrm{eff}} = L_1^2$ with ratio multiplicativity and the machine-checked DIAG↔NEFF equivalence (margin 0.055 exact in both coordinates; signed-sum mutant 8/3), the kernel envelope $F_H(s) \leq \min(H, 1/(Hs))$ exact-rational with equality witnesses on both branches, and the five sealed verdict bars as exact decision logic with tipping mutants, in pure Fractions). *Delivered (the chain):* the r296 DENS fork closed honestly (DENS-WORLD-BLIND, above); the r297 target inequality frozen ($\sigma \leq \sigma^* = -0.516$, measured -0.714, margin 0.198 — provenance missing, not room; the chain exactly orthogonal on the WINDOW measure); the r298 exact sign-preserving transfer (TRANSFER-DOMINANT: the window main term empty, $S_F \approx B(P\Delta, P\Delta)$ — the vdC input IS the Fejér energy of the difference measure); the r299 decay split (LOWPASS 0.93; full-support overlap 42/42 — Δ a pure c-value difference; ET/Abel dead +1.948; the live edge $\mathrm{sl}_D = -0.571 \leq \sigma^*$ with falling ratio -0.168); the r300 diagonal anatomy ($\mathrm{sl}_D = 2\,\mathrm{sl}_{L1} - \mathrm{sl}_{\mathrm{neff}}$ exact; both magnitude routes closed honestly — the $|dw|$ sum-rule attach BREAKS, the fill decay -0.225 is invisible to max ×mass bounds; RATIO-BOUNDED-STRUCTURAL: R_{env} med 1.61 falls at -0.122). *Honest* *negatives (load-bearing, the magnitude catalog):* r297 B1 pair-gap (-0.535 vs needed ≤ -1.178) and B2 mass imbalance ($+0.244$ growth); the r299 ET/Abel wall ($+1.948$, MASS-TARGET missed by 2.46); the r300 chain-norm and max $\times L_1$ closures; no pointwise c-value convergence (cconv 0.86, $+0.045$); the lobe-width heuristic refuted (sp $= -0.47$). *The open contract (NEFF_TARGET, the one remaining inequality):* prove slope$(n_{\text{eff}}) \geq 2\,\text{sl}_{L1} - \sigma^* = +0.908$ for the participation number $n_{\text{eff}} = L_1^2/D$ of the aggregated difference profile (measured $+0.963$, margin 0.055, thin) — exactly equivalent to DIAG-TARGET $\text{sl}_D \leq \sigma^*$, and through the settled chain it closes $\delta' > 0.21$ on the generic half: [NEFF-TARGET] $\Rightarrow \text{sl}_D \leq \sigma^* \Rightarrow \sigma \leq \sigma^* \Rightarrow$ r297 target $\Rightarrow$ v964-S0 vdC $\Rightarrow \delta' > 0.21$ (exceptions: 6 via r287 F2, kz15 exact-finite via r270); *the named bridge (correlational, not causal):* the carrier count grows where the boundary discrepancy falls ($D_{\text{rank}}(\Delta)$ slope -0.117, $\text{sp}(D_{\text{rank}}, n_{\text{eff}}) = -0.81$) — a proof must turn the correlation into a mechanism; *kill list (measured dead, binding):* every max/mean magnitude majorization (r297 B1/B2, r300 chain-norm CS on $|dw|$, r300 max $\times L_1$), the ET/Abel position-bound composition, pointwise c-value convergence, the lobe-width heuristic, H-peek, fit primitives; world-blind validity admissible (L2 is the generic half); D_{rank} statistics typed MEASUREMENT-ONLY (forbidden as proof premises). *State:* round 301 (NEFF-EQUIDISTRIBUTION) was in flight on exactly this question at the wave-9 cut; *wave-10 resolution (rounds 301–304 executed and consumed, [verification/v967_l2_cascade_closure.py], block below):* the demand is **retyped to the documented stop state** — the r303 regress audit shows NEFF/UNIF/ATOM are *one* algebraic slack $S = \sigma^* - \text{sl}_D = +0.0547$ (invariance $\leq 9 \times 10^{-16}$; the $\frac{1}{2}$-conversion refuted), so the rest is no longer a ladder of targets but the stop state: L2 generic $\Leftrightarrow$ anti-concentration of the explicit *dc* block field (core slack $S = +0.0547$ measured); the global-profile mixing route is closed (period-4 comb, LAW-LONGRANGE); the mechanism is two-scale; return later with new tools — *the hard reviewer rule binds: a round counts as progress only if it adds new information, not if it re-expresses the same slack in a new coordinate.* Lean: no new statement — NEFF-TARGET is a measured ladder-slope aggregate, not an exact finite identity, and a universal ladder form would be refutable (the r273 guard pattern); the hole stays `lstar_subordination`. Mincut base 4 / refined 5 unchanged; no RH statement.

The cascade closure and the documented lane stop (`PRIME.L2.CASCADE_CLOSURE.01` [E]). *Registered and delivered* 2026-08-26 (rounds 301–304, notes DCXL/DCXLII–DCXLIV; [verification/v967_l2_cascade_closure.py] with 32/32 + 30/30 + 26/26 + 37/37 full record gates; the four probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs `6f8cc404/36df9424/375e9f2b/2cc5d23f` pinned; plus the module-own exact section — the r303 margin-invariance algebra (all four target margins $m_D = m_{\text{NEFF}} = m_{\text{UNIF}} = m_{\text{ATOM}} = 3/50$ exact on the rational slope set; the halved-need mutant breaks by exactly $\text{sl}_{L1} = 1/5$; the σ decomposition $0.198 = 0.055 + 0.143$ exact; the $\frac{1}{2}$-conjecture refutation $0.099 \neq 0.055$), the r302 coherence identity $(1 + CV^2)\,\text{surv}^2\, n_{\text{eff}}^{\text{atom}} = n_{\text{act}}\,\chi$ on two rational toys (mutant breaks 5/7 and 48/35 exact), the r304 χ-lag decomposition $\chi = 1 + 2\sum_k T_k/Q = D/Q$ on both toys (5/7 and 5/19 exact; factor-1 mutant 1/7), the r304 zero-sum tautology as a gate (mutants 1/2 and 3/4 exact; $NC(1) = 0.556$ exact, $NC(16) = 0.712 < 1$; the period-4 comb certificate with no k_0), the $-/-/+/+$ sign-pattern certificate with the kz18/kz23 Fractions pins, the four sealed verdict bars as exact decision logic with tipping mutants, and the lane-stop composition gate, in pure Fractions). *Delivered (the rejections and retypings ARE the content):* r301 NEFF-SPLIT ($n_{\text{eff}} = n_{\text{act}}/(1 + CV^2)$ exact with the perfect count link $n_{\text{act}} = m$ on 42/42; the Rényi family at one exponent — echt anti-concentration; jackknife-stable; rest relocated onto UNIF-TARGET); r302 UNIF-DERIVED(B1+A2) — the *first* DERIVED of the lane, all hypotheses typed MEASURED (PROFILE-STATIONARY, KS 0.043; exact $1/N$ transient onto $m_2^\infty = 1.973$; $\chi = 0.63$ destructive, falling; rest relocated onto ATOM-TARGET); r303 REGRESS-CONFIRMED + MIXING-INSUFFICIENT (the retyping: all four margins ONE number $S = +0.0547$, invariance $\leq 9 \times 10^{-16}$; the cascade an exact dictionary, not six proof steps; the first causal coordinate — the ρ_1 ladder is monotone and the flip kills the inequality at -0.044 — but the χ level misses by 0.134; $n_{\text{eff}}^{\text{atom}}$ a pure marginal functional, 10^{-15} on all builds); r304 LONGRANGE-STRUCTURE + LAW-LONGRANGE (the reviewer stop case: a stable, world-specific period-4 comb, no $k_0 \leq 8$; the reviewer condition splits — $NC(16) = 0.712 < 1$ holds, summability fails at 1.563; the χ-relevant structure is exactly short-range, the r303 gap at $k \le 3$, lag-2-dominated; lag-8 matching hits the level, breaks the slopes: the mechanism is **two-scale**; the graduated ladder is monotone in χ, not in the margin). *The consequence (sealed map (iii)):* **the global-profile mixing route of the L2 lane is documented closed** — L2 generic $\Leftrightarrow$ anti-concentration of an explicit block field with long-range period-4 structure; return later with new tools. *What stands honestly:* the exact identity dictionary (count, coherence, χ-lag), the two-scale split, $NC < 1$, the $-/-/+/+$ sign pattern (Fractions-exact kz18/kz23), the marginal-functional invariance (10^{-15} on all $2100 + 1008$ builds). *The hard reviewer rule (binding):* a round counts as progress only if it adds new information, not if it re-expresses the same slack in a new coordinate — no further coordinate change counts. Lean: no new statement — the stop carries no new exact target form (measured ladder aggregates; the r273 guard pattern); the hole stays `lstar_subordination`. Mincut base 4 / refined 5 unchanged; no RH statement.

The architecture adjudication and the wave-11 claim split (`PRIME.LSTAR.ARCHITECTURE_ADJUDICATION.01` [E], `PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01` [E], `PRIME.SOURCE.TENSOR.MECHANISM.01` [O], `PRIME.L2.RENYI3.PROVENANCE.01` [O]). *Registered and delivered / registered* 2026-08-26 (rounds 305–316 incl. 310b, notes DCXLV–DCLVIII; [`verification/v968_architecture_adjudication.py`] with $27/27 + 28/28 + 30/30 + 33/33 + 34/34 + 32/32 + 32/32 + 29/29 + 29/29 + 35/35$ full record gates; the TEN probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs `3bb365e1/ec2bb008/d5147850/f8d99877/fac7a8df/6c32f749/6505dd10/841b3196/92d35a3a/5c28b12b` pinned; plus the module-own exact section — the r306 Hill/Lagrange chain with both witnesses and the Rényi bridge with all constants, the r314 opening-flux lemma $F_1 \equiv 0$ with the exact flux telescope, the collision counting identity $3p_1p_2 - 2p_3 = 8(n + 3n(n-1))$ with the exact factor-8 mutant break, an r311-class exact rational Farkas mini-certificate (a PD target in the span with a forced negative generator weight, strictly outside the cone), the TEN sealed verdict bars as exact decision logic with tipping mutants, and the four-level composition gate, in pure Fractions). *Delivered (the reviewer's binding four-level structure):* **Level 1** — real formal theorems (Lean, already green; no Lean change ships: `lstar_terminal_implies_master`, the four closed Inertia theorems, the real PrimeWindow construction, source exactness + mass conservation of any folding + the strong stabilization form, the r309 rank-one update identities, the Hill monotonicity + Rényi-3 $\Rightarrow N_2$ bridge; sorries $9 \to 5$, the two TRUE holes standing alone). **Level 2** — certified finite statements (Rényi-3 on 57 rungs at $C = 1.069$, $A = 2$; the fixed-head census 77/77; the block-Green identity + the strict cone with exact rational Farkas certificates; paired-cone soundness on 20714 steps; fold multiplicity exactly 2; 93 % far-triple cancellation; the signed cubic identity with vanishing boundary, Fractions-exact on the real data; the 21 small-m certificates). **Level 3** — negative architecture decisions verbatim (FIXED-HEAD-DEAD; PAIRED-CONE-NO-INDUCTION + WORLD-BLIND — B does not overtake A; the r308 discriminator = the budget sign / a chordal restatement; COEFFICIENT-SIGN-WALL — Lane A closed as cone language with the mechanism named: block-psd membership without rank-one SDD, obstructed at the budget/border sign; FLOQUET-EXPANDING; TRIPLE-TYPE-MAJORANTS-WRONG; PHI3-ALL-BLIND; TWO-REGIME-DEAD with the anatomy: the obstruction family cuts across the FCIX stratum, the first-5 protocol load-bearing, kz55/kz67 near-one-block). **Level 4** — open mechanisms typed (the constructive PSD-G rule as a question, the signed cubic flux theorem, the quantitative source functional, `crossing_budget`, the definitional source bridge, L*, the terminal). *The claim split (binding — not one TRANSFER title):* `RANKONE.UPDATE.IDENTITY` [E] carries *only* the exact update identities (determinant dictionary, Sherman–Morrison chain, $\pm(\eta \pm e_t)$ border split, local Schur reserve, signed source updates); `TENSOR.MECHANISM` [O] carries the open question whether a source-pure cofinal control arises from them, registered *after* r313/r316 with the documented negatives. *The provenance contract (RENYI3.PROVENANCE):* the pointwise GO stands as a certified finite statement ($\sum q^3 \le 1.069(\log m)^2/m^2$ on 57 rungs, growing reserve; fiber target $n_{\text{eff}} \ge m/(1.034 \log m)$); the provenance is open — trilogy state: the identity exact (r314), the functional blind (r315), the two-regime form dead (r316); the R317 fork verbatim (reviewer adjudication pending): a near-critical family coordinate (TOP1/QMAX source-pure in advance; kz53 needs a second coordinate) *or* a certified exception family (the r287-F2 pattern). *Framing:* the universality-class insight (base fine-metric — METRIC-ONLY, twin invariance; fiber recursive — EPSTEIN holds, SCRAMBLE breaks; shared algebra, not necessarily shared mechanism); the reviewer maturity raise 4 → 5 (formal architecture 9.5) documented as a *reviewer assessment*, not a machine-checked fact. Mincut base 4 / refined 5 unchanged; no round in flight at this cut — R317 waits on the reviewer fork; no RH statement.

The extraction-chain red team, the sliding bound, and the wave-12 adjudication (`PRIME.REDTEAM.EXTRACTION_AUDIT.01` [E], `PRIME.L2.RENYI3.SLIDING_BOUND.01` [O]). *Registered and delivered / registered* 2026-08-27 (rounds 317–322, notes DCLXI–DCLXVI; [`verification/v969_forks_and_redteam.py`] with 38/38 + 25/25 + 39/39 + 25/25 full record gates; the FOUR probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs `04fbe5c0/f2d98683/e68883ad/761b51d4` pinned; plus the module-own exact section S0 — the two-sided concentration bracket $q_{\max}\Phi_{H1} \le \rho_2 \le \Phi_{H1}$ re-proved from scratch with both mutant directions caught, the U1–U3 red-team witnesses in exact arithmetic (the $B = -1$ window against the terminal type; the mesh-level-0 total node collision reduced to three rational facts about e on the certified bracket $685/252 \le e \le 685/252 + 1/35280$, kill $0 < 0$ exact; the adversary $(|F| + 1)^2 < 5/7$ false for every F), the r320 separation/canonical-split certificates (witness mesh level 4 via $2^7 < 3^5 < 2^8$, reduction $1/8 < 2/5$ exact; the level-0 failure), six sealed verdict bars with tipping mutants and the wave-12 composition gate, in pure Fractions; no Lean call — the r319/r320 Lean rounds are consumed as *reports*, their artifacts in `rh/lean/` re-verified by `lake build` and the RH suite). *Delivered (the audit and the repair):* the r319 reconstruction found three kernel-checked *type inconsistencies* three levels above the boxes (U1: the r310b bridge bound neither u nor B — a $B = -1$ window represents the empty spec against `terminal_positive_main`; U2: the mesh-level-0 tolerance admits the total node collision and $p = X - 1$ forces $0 < 0$ against the L* type; U3: the pre-r320 `pair_margin_main` forces its source predicate empty) plus the mesh-vs-anchor cofinality seam — **the false cofinality direction is documented, not solved**; r320 repaired the interface completely (bridge retype with u/B fidelity + separation discipline + `budget_pos`; `SourceExact` for the additional arch/border finding; the canonical extraction with `canonical_split` *proved*; three permanent sorry-free guards; the sorry-free witness; census 5 → 5 with two typed retypes). *The honest chain reading (binding):* “two lemmata ⇒ RH” does not survive the literal reading — the two lemmata give *only window-local master positivity* for the opaque, currently witness-less `MainWindow`. *The three-level proof graph (reviewer adjudication, binding; the two holes are named* ***window-local*** *everywhere from wave 12 on):* Level A — local window mathematics, *proved* ($L^*(w)+T(w) \Rightarrow \mathrm{MasterPositive}(w)$); Level B — the universal window family, the two arithmetic window-local holes; Level C — extraction to Weil/RH, *open* (source-exact realization + admissible transport + convergence/stabilization in the correct directed order + density/continuity). Two binding caveats: the sorry census does *not* measure the Level-C distance (the missing extraction architecture appears in no `sorry`), and `SourceExact` must not stand as a free assumption — the named open Lean target is `CanonicalPrimeWindow` + exactly one construction theorem `sourceExact_buildPrimeWindow` (load-bearing; the extraction-repair fork R325 in flight, not consumed). *The fiber fork (SLIDING_BOUND):* fork (b) sealed WHAC-A-MOLE(kz53, kz83) by contract (the spike family $B = \{kz55, kz67\}$ is real, $C_B = 1.0536$; the F_A top-3 are exactly the near-critical family, but as a *continuum*); fork (a) fired SLIDING-BOUND-GO(G_{SQ}): $\sum_j q_j^3 \le 1.3056\, F_A(w)^2\, (\log m)^2/m^2$ for $m \ge 73$, 0/39 test violations, all four named violators inside at reserves 7.0–9.6 (honest: $C_{\mathrm{impl}} = 7.97$ is 7.5× looser — form, not sharpness; the demand is the two-part F_A/qmax provenance split; the terminal sharpened to the QMAX × M_2 shape route — R324 in flight, not consumed). *The typed reviewer assessment (supersedes the wave-11* 4 → 5 *scalar):* finite dictionaries 9.5/10, formal window-local architecture 9/10, terminal mechanism 6/10, L* mechanism 2/10, source/extraction architecture 2–3/10, complete RH proof path ∼ 3/10, audit quality 9/10 — “stronger as a window-local research program; as a complete RH proof chain currently less far than previously assumed — the removal of a shortcut that did not exist.” Mincut base 4 / refined 5 unchanged; no RH statement. *State (wave 13, [verification/v970_extraction_and_composition.py]):* both in-flight rounds are executed and consumed — the extraction-repair fork R325 fired ELEMENTWISE-STABILIZATION-GO and the named open Lean target is *implemented and proved* in r326 (`sourceExact_buildPrimeWindow`; census 5 → 8 with three named typed Level-C statements — the census caveat partially discharged; see the wave-13 block below); the terminal QMAX × M_2 route composed MEASURED in R324 (PILEUP-GROWS-SUBCRITICAL +0.172 — subcritical also at the r328B chain-honest bar 0.188; m_0^* record $10^{59.6}$, chain-honest 10^{238}; `PRIME.L2.RENYI3.MEASURED_COMPOSITION.01` [O]).

The elementwise extraction repair and the measured terminal composition — the wave-13 block (`PRIME.EXTRACTION.ELEMENTWISE.01` [E], `PRIME.L2.RENYI3.MEASURED_COMPOSITION.01` [O]). *Registered and delivered / registered* 2026-08-27 (rounds 323–327 incl. the r326 Lean round, notes DCLXVII–DCLXXII; [`verification/v970_extraction_and_composition.py`] with 36/36 + 36/36 + 18/18 + 34/34 full record gates; the FOUR probes embedded byte-exact in their sealed `-smoke` stage, SPEC SHAs `9a6696f8/dc36cacb/31277f91/11e4fd40` pinned (r324 imports the embedded r324-pre, r327 the embedded r324-pre and r324); plus the module-own exact section S0 — the r324 interpolation $M_3 \le q_{\max} M_2$ with slack exactly 1/36 and double equality at the flat profile, the F_A identity $F_A B \log m = m\, q_{\max}$ as exact constructional algebra, the r327 group ledger with the two-ancestor bound (both mutants break by exactly 1), the r325 onset formula $\alpha^* = (n_g + 1)D_0/2$ as an exact toy stabilization, the m_0^* log-space solver verbatim (landing $10^{59.6}$; the disclosed pre-freeze linear-space bug rebuilt and caught), the r328B chain-audit gate S0-T5d, six sealed verdict bars with tipping mutants and the wave-13 composition gate, in pure Fractions; no Lean call — the r326 Lean round is consumed as a *report*, the r323 abort as a *note*: the planned L* fork was cleanly aborted before any write access, nothing counts as a round). *Delivered (ELEMENTWISE, the extraction repair):* r325 adjudicated the three reviewer variants for the R319 cofinality seam — ELEMENTWISE-STABILIZATION-GO: on the native v749 class all three channels (comb/arch/pole) of the canonical tower windows stabilize *exactly* at the predicted finite anchor onset and are constant under dyadic mesh refinement — no mesh-cofinal ladder, no transport, H_{cof} **is not needed and is replaced as the target route** (variant B = the construction substrate, variant C = the quantified density step inside the rigorous interpolation envelope, no wall margin anywhere); r326 implemented the architecture (`rh/lean/RH/Elementwise.lean`, merged and green): `sourceExact_buildPrimeWindow` *proved* (`SourceExact` eliminated as a free assumption from the extraction route), the comb stabilization elementwise *proved* (onset predefined from the element's support alone), `weil_nonneg_of_windowlocal` *proved* (the extraction without the ladder: one finite instantiation per element); sorry census 5 → 8 with three named typed Level-C statements (arch/pole transcription classical, source completion definitional). *The honest formulation of the new connection (binding, verbatim):* "two window-local lemmata (over the canonical family) + the elementwise architecture + cited classics (density, v912 continuity, the Weil criterion) ⇒ RH; open links: the arch/pole transcription (classical), the compression bridge, the source completion (definitional)." *Registered (MEASURED_COMPOSITION, the terminal end state):* r324 — after the binding mid-round course correction, with the pre-work banked as r324-pre (FA-BOUNDED-DISTRIBUTIONAL + QMAX-SHARE-OPEN + NOT-COMPOSED; F_A de-black-boxed by the exact identity $F_A B \log m = m\, q_{\max}$) — fired PILEUP-GROWS-SUBCRITICAL ($e_{\mathrm{tot}} = +0.172 < 0.224$, decided): the $O(\log m)$ scale count certifies pointwise ($C_{\mathrm{NSC}} = 2.0258$, 0/39), the per-scale mass is refuted — the near-critical windows are *one heavy source scale*; the composition typed MEASURED: $\sum_j q_j^3 \le 8.941\,(\log m)\, m^{+0.172}/m^2 \Rightarrow N_2 \ge m^{0.888}$ for all $m \ge m_0^* = 10^{59.6}$, the gap $(274, 10^{59.6})$ the disclosed extrapolation hypothesis — no cofinal claim (the independent chain audit r328B, carried as a chain annotation with the record sealed: the bar 0.224 used the atom-level need 0.888, the block-level need is $908/1002 \approx 0.9062$ — the chain-honest bar is 0.188, the verdict survives $0.172 < 0.188$ exactly, the chain-honest m_0^* is 10^{238}, the going-forward bar is 0.188; exact v970 gate S0-T5d); r327 located the spike structurally (CAP-PARTIAL, anatomy SOURCE): on all 65 rungs the heaviest group of the argmax block is exactly *one* β/ω fold pair (kz53 = one bulk/window coincidence at 88.8 %); the Λ-pair cap refuted, **the group count certified** ($C_{NG} = 2.6351$, 0/39, named 4/4 — the third certified

$O(\log m)$ count); *the demand (the coincidence geometry):* where can the single heavy bulk/window coincidence *sit* — a source-geometry question, not group count (certified) or multiplicity (proven); alternatively the MEASURED composition is the honest end state of the route. *Honest boundary:* the window-local positivity premise is proved nowhere; the two true window-local Lean holes are unchanged; L* stays paused; nothing bounds anything beyond the measured rungs. Mincut base 4 / refined 5 unchanged; no RH statement.

The wave-14 consolidation — terminal martingale, cover/growth/K2, the closed L* margin chain, the Borodin duality and the second arithmetic (`PRIME.TERMINAL.DENSITY_MARTINGALE.01` [E], `PRIME.L2.COVER_GROWTH_K2.01` [E], `PRIME.LSTAR.MARGIN_CHAIN.01` [E], `PRIME.LSTAR.DUAL_HOLE.01` [E], `PRIME.WORLD.DIRICHLET_MATCHED_FRAME.01` [E]). *Registered and delivered* 2026-08-27 (rounds 334–357, notes DCLXXXI–DCCIX; reviewer-authorized after the double substance condition r355 + r357; five modules [`verification/v978_terminal_density_martingale.py`] / [`verification/v979_cover_growth_k2.py`] / [`verification/v980_lstar_margin_chain.py`] / [`verification/v981_lstar_borodin_duality.py`] / [`verification/v982_dirichlet_matched_frame.py`]; each re-derives its exact layer from scratch — sympy symbolic + pure Fractions, no probe imports — and re-runs the sealed round verdicts as exact decision logic on the frozen record aggregates with tipping mutants; the eighteen discovery probes stay sealed in `experiments/tfpt-discovery/`, pinned and re-verified by the RH suite). *Delivered (theorem-grade exact layers):* the density martingale from mass conservation alone with the moment dictionary $E[X_\infty^3] = m^2 M_3$, the tilted tower, the exact hand-off and the pair ceiling $\Gamma \le 4$ (v978); the FAB identity, the data-free K1 formula $R_{\text{ALG}} = 4^{1/3} \to 8/5$ and the mesh identity $h - \nu u \in (0, 3/2]$ (v979); the one-line identity, the two-level dressed-scalar theorem, the pinning theorem ($m_2' =$ margin identically under orthonormal top-2 geometry), the rate-equality theorem and the ρ_K identities (v980); the Borodin complementation at half filling with rational conjugator, the equivalence $L^* \Leftrightarrow R > \frac{1}{2} I$, and the reciprocal dual weight $u^\vee \propto c_j(1 - x_j)/|f|$ that makes the r354 anti-correlation *duality algebra by design* (v981); the derived χ-archimedean side with bitwise trivial reduction, exact conductor shift and parity kernel (v982). *Registered (census, honest):* the first certifying three-arm cover ($m_0^* = 10^{22.6}$), the P02 predictor, the growth ceiling $C_{\text{FAB}} = 14.93$ and every m_0^* of the lane are *Frame-A census* (binding r353: the second construction family breaks the ceiling and kills the reserve floor at $m \sim 760$ — the sliding coverage is finite); K2 (≤ 11.87) is the sole cross-family survivor, NU-free over three aspects and 0/126 on the second arithmetic; the L* margin chain is complete and typed (weights dictionary / deficits candidate law / pinning theorem / $\rho_r = \varphi$-difference computable), the duality verdict is DUALITY-REPARAM-ONLY, and the lane is *frozen* as a specialist problem at `rh/problem/lstar_problem.tex` with the one-object question halved (the $\sim 500\times$ smallness of by-design-coupled blocks, localized in local dual two-point data); SECOND-ARITHMETIC-LIVES: the r330 death was a frame artifact, the wall holds on $n = 3$ arithmetics on 126 windows and the matched scramble breaks — the wall needs the arithmetic; the zeta-graded signatures (exact pinning, saturated reserve, φ suppression, counting constants) type the L* margin legislation as a statement about how close zeta's world sits to the wall. *Honest boundary:* no cofinal claim anywhere (the r324 MEASURED composition stays the terminal end state); the equivalence is a reparametrization of the open condition; `PRIME.LSTAR.SUBORDINATION.01` and `PRIME.L2.RENYI3.SLIDING_BOUND.01` stay [O] with wave-14 notes; everything census-side is finite, kein Satz. Mincut base 4 / refined 5 unchanged; no RH statement, no GRH statement.

The kernel-Loewner positivity certificate at $L = 0.3$ (`PRIME.RDAGGER.KERNEL_LOEWNER.01`, Numerical/certified). *Registered and delivered* 2026-09-01 (rounds 494 + 495; module [`verification/v1017_kernel_loewner_positivity.py`]; re-derived, no probe imports). *Delivered:* $Q_W(h) \ge 2.1\times10^{-3}\|h\|_2^2$ on $\operatorname{supp}(h) \subset [-0.3, 0.3]$ ($2L = 0.6 < \log 2$, prime term empty); float64 enclosed floor 2.122×10^{-3} after a $3\times$ Hilbert–Schmidt tail charge — *not* interval arithmetic and *not* [E](repo convention: positivity [E] is Fractions or integer-Sylvester/`mpmath.iv`); G1 calibration vs defining digamma ($\sigma_A(0) = -5.3721834192256654$); G2 Loewner after zero-extension; independent r495 translation identity 6/6 exact over $\mathbb{Q}$ and doubled-c_L false-world budget -2.188. *Honest* *boundary:* r496 `NO_GO`(compact-tail@$L = 0.8$) is the named method boundary — not $\lambda_*(L) \geq 0$ in general; no cofinal claim; no RH statement.

The gapcode disenchantment (`PRIME.DIVISOR210.GAPCODE.01`). *Registered* 2026-08-09, an honest negative/limit row ([`verification/v886_register_gapcode_null.py`], 24/24, GAPCODE-MEASURED; five honest SPEC amendments on record, none loosening a criterion). *Measured:* the error-code reading of the 210-register is literally true on the prime side — the register bits $\{2, 3, 5, 7\}$ are the first four parity checks of the sieve (rate 48/210; 16 syndrome labels; 0 bits of message capacity); the hard floor $g_k \geq w(p_k \bmod 210)$ holds with zero violations in 5.76×10^6 consecutive pairs; $I(\text{state}; \text{gap}) = 1.14$ bits — and the code is the sieve: a zero-parameter wheel–Cramér null explains $R^2 = 0.99839$ of the state profile, the arithmetic excess is 4.02% and shrinking (21.81 $\to$ 4.02% over $X = 10^5$–10^8), no knee at four checks, 210 on the foreign-wheel trend ($z = -0.61\sigma$), all eight anchor Walsh seats below 2.5σ exactly as CRT predicts. *The bound:* everything the register knows about prime gaps, every 4-prime wheel knows; nothing moves with c_3 or φ_0; the register's uniqueness remains the arithmetic forcing $\prod(p-1) = 48$ (`v868`). No marker move; no RH statement.

The rounds-42–49 consolidation (the complete certified ladder, the finite Pfaffian/K_6 signature, the four route decisions, the surviving positivity architecture, and the Möbius kills). *Recorded* 2026-08-09 (a campaign note, round CVIII; fourteen frozen sandbox probes, every spec amendment fail-first on record; the promoted verification modules [`verification/v887_certified_ladder_complete.py`], [`verification/v888_finite_pfaffian_signature.py`], [`verification/v889_route_decisions.py`], [`verification/v890_total_positivity_flow.py`] and [`verification/v891_moebius_firewall_kills.py`] embed their discovery probes byte-exact). The campaign completes the certificate rollout ($\sigma_h > 0$ proven on all 42 reachable rungs), closes the finite Pfaffian/K_6 signature of the compiler (the canonical sign character; the self-hosting counting theorem; the flavor hexagon as the automorphism orbit), decides the four routes of the sum-rule program (the wall dies off-diagonally; PNT is insufficient; the diagonal input is pair-correlation class — the sum-rule contract retyped conditional), measures the surviving positivity architecture (flag positivity – all leading principal minors, Sylvester; typing corrected per the round-51 review, classical total positivity is not claimed; the wall margin as the pole distance of the s-flow; multi-factor inheritance), and honestly kills the Möbius/carrier-invariance route with mechanisms. The registrations and updates follow as the blocks below (plus the dated retypes on the sum-rule and tail contracts above). Full narrative in the prime-front paper and Paper 1; audit cards in Paper 3. No marker moves; no RH statement.

The complete certified ladder (`PRIME.PORT.CERTIFIED.LADDER.01`, delivered). *Registered and delivered* 2026-08-09 ([`verification/v887_certified_ladder_complete.py`], 7/7, CERTIFIED-LADDER-COMPLETE). *Proven:* $\sigma_h > 0$ on *all* 42 reachable rungs of the deployed window ladder — 12 rungs ($h \leq 300$) by exact-rational integer Bareiss/LDL certificates, 30 rungs by validated-precision `mpmath` Cholesky at dps 120/200 (precision doubling; every pivot $> 10^6\times$ the accumulated per-pivot rounding bound at both precisions); 0 refusals, 0 out-of-reach; the module closes the two deepest rungs $h = 859/878$ with floors $\geq 8.352/9.187 \times 10^{-6}$ and pivot-to-bound ratios $\sim 10^{115}/10^{195}$; the Epstein control refused at pivot index 10 by the identical machinery; the 40-rung parent rollout stays sandbox evidence, runtime-excluded. *Typed honestly:* the validated-precision tier is never labeled exact-rational; proven modulo the declared conservative lag-evaluation error model (validated-numerics standard); the ball-arithmetic closure and the Lean composition through `krein_cofinal_weil` remain the named steps of `PRIME.PORT.TAIL.01`; the asymptotic tail is untouched. *Interval note (2026-08-09, round CXV):* the error model is upgraded — [`verification/v897_certified_interval_ladder.py`] (`PRIME.PORT.BALLLADDER.01`, below) re-proves all 42 rungs with *rigorous* interval-arithmetic shifts (15 exact-rational + 27 validated-precision; Epstein refused at pivot 10 by the identical interval machinery); the informal conservative ε_c model is retired from the ladder, and only the Lean composition remains named. *Kill:* a certified pivot failure on any rung under the frozen error model. No marker move; no RH statement.

The $Gr(2,5)$ Plücker reading, the Specht channels, the Hadamard anchor and

the vacuum selection (DOILY.PLUCKER.GR25.01, closed). *Registered and closed* 2026-08-09 ([verification/v888_finite_pfaffian_signature.py], 27/27, S6PLUCKER-CLOSED; exact integer/sympy). *Exact:* the 15 doily lines are the 15 monomials of the five Plücker quadrics of $Gr(2,5)$ — bijection 15/15 with perfect sign match (every quadric reads $(+,-,+)$, the v880 Hodge table); the S_6 duad representation decomposes $15 = 1 + 5 + 9$ with the doily operator acting by the exact scalars $(1, 0, 2/3)$, $2/3 = e_3/(e_2 - e_3)$ at the anchor $(4, 5, 2)$; the q^* zero set is a $(16, 6, 2)$ Hadamard difference set with anchor signature $(2^{e_1}, p_2, e_3, \binom{e_2}{2}) = (16, 6, 2, 10)$; and the vacuum-selection theorem: $\mathrm{Stab}(q^*)$ of order 120 acts faithfully and surjectively as S_5 on the remaining five Arf-1 refinements, orbits $[1, 5, 10] = (\Lambda^0, \Lambda^4, \Lambda^2)$. *Controls:* Arf-0 census $(16, 10, 6)$; a secant breaks the bijection; the wrong scalar breaks the channel law. The physical vacuum-selection mechanism stays [O]. No marker move; no RH statement.

The flavor-hexagon automorphism orbit (CFIN.AUT.FLAVOR.C6.01, closed). *Registered and closed* 2026-08-09 ([verification/v888_finite_pfaffian_signature.py], 19/19, C6FLAVOR-MEASURED + INTERTWINED). *Exact:* $\mathrm{Aut}(C_{\mathrm{fin}}) \cong C_6$ by exhaustive enumeration; the pinned generator satisfies $g^2 = \sigma$ (the deployed family 3-cycle) exactly and $g^3 =$ the W-edge sheet swap; the action on $\Lambda^2 E$ has cycle type $1+3+6$ with orbits equal to the charge shells; the g-orbit of $\{f_1, w_1\}$ is the deployed hexagon transport edge for edge (D_6-equivalent); and $(|Z_2|/N_{\mathrm{fam}})^{|\mathrm{Aut}(C_{\mathrm{fin}})|} = (2/3)^6 = 64/729$ equals the frozen recovery value as an exact Fraction. *Typed honestly:* the exponent reading is a consistent *new reading*, not the unique derivation (the corpus route via $p_2 = |R_+(A_3)|$ stands). *Control:* 0 of 600 symplectic non-automorphisms factor; scrambled transport fails the D_6 match. No marker move; no RH statement.

The fermionic Pfaffian signature (DOILY.PFAFFIAN.WICK.01, closed — the K_6 motif, the sign theorem, the self-hosting counting theorem, and the mod-30 clock). *Registered and closed* 2026-08-09 ([verification/v888_finite_pfaffian_signature.py], 42/42, K6PFAFFIAN-CLOSED; exact integer/sympy, first frozen run). *Exact:* the K_6 motif theorem — the 35 lines of $PG(3, 2)$ are exactly 15 perfect matchings (the doily) + 10 triangles through the vacuum vertex (the secants) + 10 avoiding it (the externals), with $q^*(e) = 0$ iff e touches the vacuum; the sign theorem — the Pfaffian signs match the Hodge table under the canonical character $\chi(i) = (-1)^{i+1}$, triple-identified (vacuum-row Laplace, Λ^4/Λ^5 volume form, C_6 crossing character; a global gauge, not a line-dependent fudge — the flipped-χ control produces exactly the 3 expected mismatches); the self-hosting counting theorem — $\binom{g+1}{2} = g!!$ has $g = 5$ as its only nontrivial solution, $(g-2)!! = 3 = N_{\mathrm{fam}}$; and the mod-30 anchor clock, period exactly $4 = |\mu_4|$, CLOCK-DISTINGUISHED (period-4 moduli exactly $\{5, 10, 15, 30\}$). *Typed honestly:* the counting theorem is [E]; the Wick-compiler premise stays [O] — P2 narrowed to one premise, not eliminated; the operative-Coxeter-30 reading is separately dead (sandbox); no canonical vertex–one-factorization bijection exists (outer-automorphism obstruction, report only). No marker move; no RH statement.

The diagonal separation measurement (PRIME.PORT.DIAGSEP.01). *Registered* 2026-08-09, measured ([verification/v889_route_decisions.py], 8/8, DIAGSEP-MEASURED (DIAG-INTERMEDIATE)). *Measured:* under the exact off-critical injection family the wall dies 100% through off-diagonal coherence — at the kill point the diagonal testing profile is statistically frozen (max $T_m = 0.94$–0.99; the coherence share equals the entire baseline diagonal margin); the separation ratio is 2.0–2.7 on the head rungs and > 10 deep (no diagonal crossing up to $10\times$ the wall's kill amplitude). *The typed reading:* the $O(1)$ diagonal margin does not track the exponentially collapsing wall margin — $T_h \le 1$ is a genuinely weaker finite statement; the RH-scale content stays in the coherence share. *Kill:* a rung on which the diagonal crosses before the wall. No marker move; no RH statement.

The zone split (PRIME.CASE.ZONESPLIT.01, frozen). *Registered and frozen* 2026-08-09 ([verification/v889_route_decisions.py], 12/12 under SPEC v2, ZONESPLIT-MEASURED (ZONE-FROZEN($\theta^* = 0.700$))). *Measured:* outside the critical zone $a > h^{2\theta^*}$ the testing margin $1 - T$ is uniformly positive on all 42 rungs (deep-half infimum $+0.0214 \ge$ the 0.02 bar; cut-stable; $\theta = 2/3$ fails at $+0.0191$); inside, the margin law is $\sim 0.87/\log^2 h$ — the bar the arithmetic input must beat. *Typed honestly:* OUTLIERS-DANGEROUS (on kz 13 the margin minimizer coincides with a top-3 oscillation node — the density defects are not provably measure-zero for the margin). *Control:* scramble drives the outside margin to -1.4×10^4. *Kill:* a rung with negative outside margin at the frozen θ^*. No marker move; no RH statement.

The PNT decider (`PRIME.CASE.PNTGAMMA.01`, decided). *Registered and decided* 2026-08-09 ([`verification/v889_route_decisions.py`], 10/10 under SPEC v2, PNTGAMMA-MEASURED (PNT-INSUFFICIENT)). *Measured:* the four matched worlds (mass × mask in {actual, PNT-smooth}, identical pipeline, exact tent-integral construction self-tested at 6×10^{-12}) show the truth's port correction $\gamma > 0$ at every alias on every rung, *but* the PNT-smooth reference itself has $\min \gamma < 0$ on every rung (-0.96 to -7.8×10^{-3}) — the smooth reference violates the very inequality the diagonal theorem needs; the dominant rescue piece is the mask–mass *interaction* (positive on 4/4 deep rungs); the measured arithmetic deviation $\varepsilon_A = 0.23$–6.2 is 6–970× larger than every gap. *The decision:* the diagonal theorem's arithmetic input is genuinely stronger than PNT on this family — no classical zero-free-region-strength input closes it by smooth comparison. *Control:* scramble flips 6/6 port gammas. No marker move; no RH statement.

The signed-homotopy classification (`PRIME.CASE.SIGNEDHOMOTOPY.01`, decided). *Registered and decided* 2026-08-09 ([`verification/v889_route_decisions.py`], 12/12 under SPEC v2, HOMOTOPY-MEASURED (HOMOTOPY-INDEFINITE-ARITHMETIC); ward metrology only — run 1 ended honestly PIPELINE-BROKEN on an over-sharp deep ward). *Measured:* the signed homotopy identity along the exact PNT-to-truth interpolation is exact calculus (4.3×10^{-14} with exact rational mask-crossing times); 66.3% of the 14047 samples have $J < 0$ (pointwise positivity dead); the fixed-chamber quadratic kernel is concavity-locked negative-semidefinite on both frozen rungs; the polynomial response drains the gain (-17 to -231%, worsening with h). *The classification:* the truth endpoint wins only through the signed *first-order* alignment of the arithmetic residual with the extremal-polynomial weights — the diagonal route's required input is *pair-correlation class*, not any PSD kernel positivity; `PRIME.CASE.SUMRULE.01` is retyped from unconditional candidate to conditional reduction theorem (the dated update above). No marker move; no RH statement.

Flag positivity (Sylvester) and the pole-free s-corridor (`PRIME.PORT.HIROTA.01` [O]). *Registered* 2026-08-09, OPEN — the finite architecture measured ([`verification/v890_total_positivity_flow.py`], 10/10 + 7/7, HIROTA-MEASURED (MINORS-POSITIVE) + SFLOW-MEASURED (CORRIDOR-CLEAR)). *Measured:* every nested leading principal minor $\tau_h^{(k)}$ of $I - C_h$ ($k = 1, \ldots, 12$) is strictly positive on all 37 full-window rungs — the section family is *flag-positive* (Sylvester: all leading principal minors positive; classical total positivity of all minors is not claimed), a strictly stronger finite statement than the wall scalar; the Desnanot–Jacobi identity is warded exact; the s-corridor $s \in [0, 1]$ is pole-free on every rung (the controls pull their pole inside), and by Cauchy interlacing $s_h^* = 1/\lambda_{\max}(C_h)$, so $s_h^* - 1 = \tau_h/(1 - \tau_h)$ — the wall margin *is* the pole distance of the integrable s-flow (ladder slope -2.90). *Typed honestly:* no naive rung-to-rung bilinear closure in raw h (median residual 0.655); the Toda form holds pointwise (0.001) but with drifting coefficients — TODA-VARYING. *The demand:* the multi-factor inheritance question as a total-positivity propagation statement, derived from the IIKS generators. *Kill:* a full-window rung with a non-positive section minor. *Skeleton update* (2026-08-09, round CXI): the propagation question is answered in *bounded-margin* form by [`verification/v892_closure_architecture.py`] and [`verification/v893_relative_margins.py`] (the hermitian congruence margin η and the flag chain; the soft flag is the τ scale after the kz-21 coordinate artifact) — the demand narrows to the $O(1)$ ratio bound (`PRIME.PORT.ETASOURCE.01`). *Dictionary update* (2026-08-24, round 226): the TODA-VARYING finding is resolved into an exact dictionary — the bilinear Toda form holds with the source-pure coefficient $\hat{\gamma}_n/\gamma_n$ from a scaled *signed* Stieltjes recursion, and its positivity is WALL-EQUIVALENT (quasi-definiteness of the signed defect measure $\mu - \nu$; [`verification/v955_tau_iiks_toda_dictionary.py`], `PRIME.PORT.TAU.FINITE.IIKS.01` [E]); no source-pure positive cone was found (HIROTA-CONE-GO not reached — measured and closed via the Cauchy–Schwarz route). No marker move; no RH statement.

The multi-factor determinant anatomy (`PRIME.PORT.TAU.MOEBIUS.01`, decided). *Regis-*

tered and decided 2026-08-09 ([verification/v890_total_positivity_flow.py], 14/14 under SPEC v2, TAUMOEBIUS-MEASURED (BRIDGE-DIFFUSE)). *Measured (the reviewer's key run):* the exact factorization $Q_h = \det(W_{h+1})/\det(W_h) = \prod_i(1+\mu_i)$ is warded at 9.2×10^{-12}; the truth keeps every factor off -1 on all 31 full-window steps over 24 orders of magnitude, while the frame-surviving smooth-mass world crosses factors on 22 of its 28 steps (first at $h\,210 \to 218$) and flips Q thirteen times — the arithmetic detector fires. *The decision:* the one-scalar Möbius bridge is closed — BRIDGE-DIFFUSE (bulk factor margin 0.005 vs the frozen 0.5 bar; the fitted Möbius datum $|\alpha| \sim 10^{-3}$ is determinant-inert); positivity inheritance, if it exists, is a multi-factor/total-positivity statement, not a scalar cocycle inequality. No marker move; no RH statement.

The Möbius kill battery (`PRIME.PORT.MOEBIUS.KILLS.01`, decided). *Registered and decided* 2026-08-09 ([verification/v891_moebius_firewall_kills.py], $10/10+13/13+7/7$, CRFIREWALL-MEASURED (CR-DEAD) + GAUGEFW-MEASURED (GAUGE-MADE) + MOEBIUSCENSUS-MEASURED (UNDECIDED) — the verdicts are kills and are recorded as such). *Measured:* the earlier carrier-invariance finding was a normalization artifact — fit-free gauge-free cross-ratios over 8126 well-conditioned quadruples: pooled median defect 0.128 (bars $10^{-3}/10^{-2}$), anchor triples disagree, the steps do not compose; none of the three source-frozen gauges is raw-invariant (two contrast-degenerate — invariance vacuous; the determinant-phase frame shows real motion); the scalar cocycle does not track the wall (corr = 0.13); the Jacobi-transfer identification is dead ($d_P = 0.997$); and the six-world census shows the arithmetic separating *at the frame* — every control world including the smooth-mass ones loses window subcriticality (truth 41/41 alive, all controls 0), and scramble beats truth on the bare Möbius bars (0.0060/0.0011 vs 0.0218/0.0446): rank-2 window kinematics, not arithmetic dynamics. *The surviving remnant (the named object):* the deep-core 8-node cross-ratio coherence (4.3×10^{-4} on truth vs 2.7×10^{-2} smooth — prime-mass-driven); related sandbox kills recorded (the static Cauchy identification dead; dev/τ grows at slope +1.46 — the Cauchy route can never certify the wall). No marker move; no RH statement.

The rounds-50–53 campaign (the closure architecture of the wall, the bounded-margin skeleton, the diagonal refinement, the collectivity, and the Wick functor arc). *Recorded* 2026-08-09 (a campaign note, round CXI; fourteen frozen sandbox probes, every spec amendment fail-first on record; the promoted verification modules [verification/v892_closure_architecture.py], [verification/v893_relative_margins.py], [verification/v894_diagonal_refinement.py], [verification/v895_collective_comb.py] and [verification/v896_wick_block_functor.py] embed their discovery probes byte-exact). The campaign gives the wall its closure architecture (the exact hermitian congruence with a non-decaying inheritance margin; the flag/pivot chain positive by pure algebra; the honest two-sided Rouché kill; the wall as one fixed 8×8 Schur core), sources both bounded margins and prints the theorem skeleton whose *one* unconditionally open statement is the τ-sign inheritance in bounded-margin form, makes the conditional diagonal contract concrete (the exact band-limited prime kernel, frozen in-probe), measures the collectivity of the arithmetic (the one-sided law; the deep core as the entire multiplicative von Mangoldt comb), and closes the Wick functor arc of the finite compiler at the structure level (the proved scalar obstruction; the block resolution; the seam-diagonal demand). The registrations follow as the blocks below (plus the dated skeleton update on `PRIME.PORT.HIROTA.01` above). Full narrative in the prime-front paper and Paper 1; audit cards in Paper 3. No marker moves; no RH statement.

The hermitian congruence inheritance form (`PRIME.PORT.RELCONG.01`). *Registered* 2026-08-09, measured ([verification/v892_closure_architecture.py], 16/16, RELCONG-MEASURED). *Measured:* $A_{h+1} = A_h^{1/2}(I+H_h)A_h^{1/2}$ exact (6.5×10^{-14}; $H_h = A_h^{-1/2}\Delta_h A_h^{-1/2}$ exactly hermitian); the margin $\eta_h = 1 + \lambda_{\min}(H_h)$ is *non-decaying* (min 0.0050, median 0.29, power slope +0.108 while τ collapses at -2.74) — the inheritance inequality is bounded-margin-shaped, not collapsing; $\rho(W^{-1}\Delta C)$ was a misleading statistic (strongly non-normal: $\|M\|/\rho$ up to 26, $\kappa(V)$ up to 134); the smooth-mass world is CONE-EXITED on all 28 pairs. *Mini-theorem (verified):* base PD + $\eta_h > 0$ for all $h \Rightarrow$ the whole ladder (Sylvester congruence). *Source:* the η scale is explained by `PRIME.PORT.ETASOURCE.01`; the $O(1)$ ratio is the open piece. *Kill:* a truth step with $\eta \leq 0$. No marker move; no RH statement.

The flag-chain induction (`PRIME.PORT.PIVOTFACTOR.01`). *Registered* 2026-08-09, measured ([`verification/v892_closure_architecture.py`], 15/15 under SPEC v2, PIVOTFACTOR-MEASURED). *Measured:* the 12 $\mathrm{LDL}^{\mathsf{T}}$ pivots of $A_h = I - C_h$ factor rung-to-rung as $d_{k,h+1} = d_{k,h}(1+\mu_{k,h})$ with *every* factor positive — all 12 flag quotients $Q_k > 0$ on all 31 truth steps (pivot = minor quotient warded 8.3×10^{-14}); the ladder minimum sits at the soft flag $k = 12$ at 2.06×10^{-14}, sign-decidable at $6.4 \times 10^{9}\times$ the eigenvalue noise floor (Cauchy-interlacing guard); the smooth world violates at the *bulk* flags first ($k = 1$ on 10 steps, $k = 3$ on 9; zero steps violate the soft flag alone); per-prime non-decompositional (leverage to 987×). *The shape:* the shortest induction of the program — base (`v884`/`v887`) + $\min_k Q_{k,h} > 0 \Rightarrow$ the whole ladder by pure algebra; the soft-flag scale is τ itself (`PRIME.PORT.RELFLAG.01`). *Kill:* a truth step with a non-positive flag quotient. No marker move; no RH statement.

The Rouché route (`PRIME.PORT.EULER.ROUCHE.01`, dead). *Registered and closed* 2026-08-09 ([`verification/v892_closure_architecture.py`], 14/14 under SPEC v2, ROUCHE-MEASURED — an honest kill with the mechanism documented). *Measured:* on complex contours enclosing the corridor $s \in [0,1]$, $\sup_\Gamma \|A_h(s)^{-1}\Delta_h(s)\| \geq 1$ on 22/31 truth steps (max 63.4) *including* off-cap — not merely contour-limited: a two-sided contour norm bound is the wrong theorem shape for inheritance on this ladder (consistent with the one-sided avoidance law of `PRIME.PORT.FACTORAVOID.01`); the argument-principle machinery itself verifies exactly (integer zero counts at 5×10^{-15}); the smooth world is corridor-broken on 28/28 but the mechanism is non-discriminating since truth fails the bound too. No marker move; no RH statement.

The 8×8 Schur-core reduction (`PRIME.PORT.DEEPCORE.SCHUR.01`). *Registered* 2026-08-09, measured ([`verification/v892_closure_architecture.py`], 16/16 under SPEC v2, DEEPCORESCHUR-MEASURED). *Measured:* on the full wall operator, the block split at the fixed deep-core aliases $\{2,\ldots,16\}$ gives Haynsworth-exact inertia and the wall *is* the fixed 8×8 Schur core: $\lambda_{\min}(S_h)\, w_{\text{core}}/\tau_h = 1 \pm 8.4 \times 10^{-8}$ with core mass $w_{\text{core}} \geq 0.999937$ on all 41 full-core rungs — the RH-critical object is a fixed 8×8 family, not a growing operator; the absolute exterior margin shrinks ($\lambda_{\min}(R) \sim h^{-2.865}$) but $\lambda_{\min}(R)/\tau$ is trendless at 210–2200 — the reduction is conditional on a *τ-relative* exterior bound only ($c_R = 210$ measured, `PRIME.PORT.RELFLAG.01`); core inheritance holds (0 crossings, η_{core} min 0.0315); the per-prime grain is dead at fixed dimension too. *Kill:* a full-core rung breaking the seven-digit identity. No marker move; no RH statement.

The η-margin source (`PRIME.PORT.ETASOURCE.01`). *Registered* 2026-08-09, explained ([`verification/v893_relative_margins.py`], 20/20 under SPEC v2, ETASOURCE-MEASURED (SCALE-MATCHED)). *Measured:* in the dangerous direction the increment $d_h = |v^{\mathsf{T}}\Delta v|$ and the operator $a_h = v^{\mathsf{T}}Av$ collapse *together* (slopes −3.716 vs −3.456, residual correlation 0.974, 94% shared variance): the non-decay of the inheritance margin η is *explained* — η is scale-free because numerator and denominator are sections of one arithmetic object. *Open:* only the $O(1)$ scatter of the ratio (the fitted τ-laws predict none of it); seat anatomy: the minimizer sits in the core while the binding restricted block is the 4-alias exterior (0.0056 vs core 0.0319). The round-53 sandbox continues with the mechanism decomposition. No marker move; no RH statement.

The relative margin laws and the theorem skeleton (`PRIME.PORT.RELFLAG.01`). *Registered* 2026-08-09, measured ([`verification/v893_relative_margins.py`], 18/18 under SPEC v2, RELFLAG-MEASURED). *Measured:* $d_{12} = 1/(A^{-1})_{12,12}$ exact with the factorization $r_{12} = (\tau'/\tau)(c'/c)$ at 2.1×10^{-16}; the $c = d_{12}/\tau$ ladder is trendless with one outlier (kz 21) diagnosed in the round-53 sandbox as a *coordinate artifact* (the kz-19 mechanism repeats; corrected spread 124 → 15.0 — the soft flag is τ-tied); $\lambda_{\min}(R)/\tau$ trendless with ladder min $c_R = 210$; COMPRESSIONS-DISTINCT (honest: the port bulk margin is an *absolute* floor while the core exterior is τ-tied — different statements at two compressions). *The printed skeleton:* (i) base certified (`v884`/`v887`; interval-rigorous since `v897`); (ii) $\lambda_{\min}(R) \geq 210\,\tau$ trendless; (iii) core $\leftrightarrow$ τ at seven digits; (iv) $\eta > 0$ non-decaying — the *one* unconditionally open statement is the τ-sign inheritance, reorganized from collapsing-margin form into *bounded-margin* form (c, c'/c, c_R, η — all $O(1)$; round-55 sharpening: c'/c sensitivity range $[0.223, 5.258]$ and the exact Christoffel identification of c, [`verification/v899_christoffel_normsquare.py`]). No marker move; no RH statement.

The extracted conditional contract (`PRIME.CASE.PAIRCORR.CONTRACT.01` [O]). *Registered* 2026-08-09, OPEN — the conditional reduction extracted with every object machine-defined ([`verification/v894_diagonal_refinement.py`], $11/11 + 13/13 + 12/12$, PAIRCONTRACT-EXTRACTED (KERNEL-BANDLIMITED) + ENDPOINT-MEASURED (ENDPOINT-PARTIAL) + INNERLINEAR-MEASURED (INNER-EMPTY); the route stays conditional per `v889`). *Measured:* the first-order homotopy gain regroups exactly (4.5×10^{-15}) into the prime-side form with the explicit node kernel $K_{h,m}$; the kernel is band-limited (compact spectral support — the contract hypothesis is one-sided band-limited explicit-formula positivity at the port frequencies, pair-correlation class); the full LaTeX contract statement is frozen in-probe; all 79 (rung, alias) finite margins positive; the endpoint refinement: the global linear collapse fails (142/227) but deep small-a cells pass monotonically; the inner-zone accounting localizes the irreducibly-hard part to a *non-shrinking* band (∼20% of aliases, 38–56% of the mass). *Demand:* state-and-prove the band's weighted pair inequality, or accept conditionality. *Norm-square note (2026-08-09, round CXV):* the kernel's frequency weights are $W_{f,m} \geq 0$ *exactly* and the single top-edge tent defect is an extension-convention artifact killed by the periodic full-weight fold (negative index 0; 79/79 modified margins positive; the boundary term β carried closed-form, max β/margin 0.21) — the contract is now a *norm square* in the frequency weights ([`verification/v899_christoffel_normsquare.py`], `PRIME.CASE.KERNEL.SOS.01` + `PRIME.CASE.EDGEDEFECT.01` below); the arithmetic hypothesis itself stays conditional. *Kill:* a negative finite margin on the frozen census. No marker move; no RH statement.

The one-sided collective law (`PRIME.PORT.FACTORAVOID.01`, decided). *Registered and decided* 2026-08-09 ([`verification/v895_collective_comb.py`], 18/18 under SPEC v2/v3, FACTORAVOID-MEASURED (AVOIDANCE-STRUCTURAL)). *Measured:* the norm-bound theorem shape does not exist — $\rho(W^{-1}\Delta C) \geq 1$ on 17/31 truth steps (max 57.3) but always via the harmless $\mu > +1$ side; the true law is *one-sided*: $\max(-\min \mu) = 0.9950 < 1$ on all 31 steps, which *is* PD inheritance itself; the geometry/atom split is exactly degenerate (the atom cutoff is slaved to the tent-support edge; every leaving step is atom-nil bitwise) — no separable atom channel: the increment is the collective window re-test of the full comb; per-prime leave-one-out is non-decompositional (a leverage census, not a decomposition). No marker move; no RH statement.

The deep core is the port core (`PRIME.PORT.DEEPCORE.01`). *Registered* 2026-08-09, measured ([`verification/v895_collective_comb.py`], 12/12, DEEPCORE-MEASURED). *Measured:* the 8 deepest common nodes of the carrier ladder are the fixed even aliases $\{2, \dots, 16\}$ = the Bessel coordinates $a_m = \pi^2 m^2$ (0.1% at depth; census 39/41 steps; coherence knee $k^* = 10$); the certificate-level cross-ratio coherence (truth 4.3×10^{-4} vs smooth 2.7×10^{-2}) is killed by *all four* surgical modifications — edge-only smoothing, interior-only smoothing, prime-power echoes removed, and the wrong-Λ world: **the signal is the entire multiplicative von Mangoldt comb** — positions, powers, and exact $\log p$ values all load-bearing; no smaller frozen object exists — the global rigidity of `v883` at the finest coordinate. *Kill:* a surgery that preserves the coherence. No marker move; no RH statement.

The Wick functor arc (`SEAM.CFIN.WICKFUNCTOR.01` [O]). *Registered* 2026-08-09, OPEN — structure exact, obstruction proved, block candidate constructed ([`verification/v896_wick_block_functor.py`], 27/27 + 33/33 + 26/26, WICKFUNCTOR-PARTIAL + WICKMIX-MEASURED (WICKMIX-OBSTRUCTED) + WICKBLOCK-MEASURED (BLOCKFUNCTOR-CONSTRUCTED / SEAM-DIAGONAL)). *Exact:* the six compiler roles exist in the deployed seam (D_5 slot pairs + the A_3 boundary block; $\mathfrak{so}(16)$ sector law, q^* edge grading, C_6 intertwining all exact) but the deployed vacuum kernel is channel-diagonal; *the proved scalar obstruction:* the pinned C_6 generator's 2-cycle $\{4, 5\}$ forces a duad zero — every C_6-covariant antisymmetric scalar covariance reaches at most 14/15 duads (a theorem of the deployed action); *the block resolution:* the C_6-covariant cross-block space has dimension 33; the canonical block covariance is exactly covariant, strictly CAR-positive, carries 15/15 blocks and 15/15 block Wick monomials with the canonical χ sign law; *the decisive fact:* the deployed seam covariance has *zero* cross-blocks everywhere (SEAM-DIAGONAL). *Demand:* a physical channel-mixing mechanism (a boundary/transport-twisted state, not the vacuum) realizing the block functor's value level — then the K_6/Pfaffian grammar of v888 would be the measured correlation grammar of the seam. *Candidate note (2026-08-09, round CXV):* the demanded state now has a *constructed candidate* (SEAM.CFIN.KMSMIX.01 below, [verification/v898_kms_schur_mixing.py]); whether the *actual* seam realizes it is untouched. The [O] premise is unmoved. No marker move; no RH statement.

The rounds 54–56 block (2026-08-09; promoted in [verification/v897_certified_interval_ladder.py], [verification/v898_kms_schur_mixing.py], [verification/v899_christoffel_normsquare.py] and [verification/v900_core_update_anatomy.py]) delivers the reviewer's "immediate" list: the interval-rigorous certificate base, the channel-mixing candidate state, the Christoffel/norm-square anatomy of the diagonal contract, and the exact core-update coordinates with their honest negatives.

The certified interval ladder (PRIME.PORT.BALLLADDER.01, delivered). *Registered and delivered* 2026-08-09 ([verification/v897_certified_interval_ladder.py]; probe verdicts BALLHEAD-PROVEN and BALLLADDER-COMPLETE, 10/10, 3960 s). *Proven:* $\sigma_h > 0$ with *rigorous* mpmath.iv interval-arithmetic shifts on all 42 reachable rungs $h = 142$–878 — 15 exact-rational integer Bareiss/Sylvester certificates ($h \leq 300$, floors 4.5×10^{-5}–2.1×10^{-4}) + 27 validated-precision Cholesky certificates (dps 120/200, floors 5.2×10^{-6}–8.4×10^{-5}, pivot-to-bound ratios $\sim 10^{115}$); the node-enclosure lemma, native outward-rounded interval transcendentals, exact midpoint/radius extraction and the rigorous shift $2h\,\mathrm{rad}_{\max}$; **the informal ε_c lag-evaluation error model of v884/v887 is retired from the ladder**; the Epstein control refused at pivot index 10 by the identical interval machinery. Suite economy declared: the promoted module re-certifies all 15 tier-1 + 3 tier-2 rungs verbatim and freezes the full census with both probe SHAs. *Remaining named step:* the Lean composition through krein_cofinal_weil; the asymptotic tail stays with PRIME.PORT.TAIL.01 / PRIME.PORT.LEADING.SIGN.01. *Kill:* a refused pivot on any rung under the interval shifts. No marker move; no RH statement.

The channel-mixing candidate state (SEAM.CFIN.KMSMIX.01 + SEAM.CFIN.SCHURMIX.01). *Registered and constructed* 2026-08-09 ([verification/v898_kms_schur_mixing.py], 30/30, KMSMIX-MEASURED [KMSMIX-FOUND(U=1, T=1/8, BETA=1), SCHURMIX-GENERATES(10/10), RP-THETA-OPEN]). *Constructed:* the C_6-covariant channel-mixing KMS state exists — the first point of the frozen grid passes all block-functor gates (strict CAR, exact covariance with the forced zeros, all 15 blocks at the floor, grading $5/5 + 10/10$, all 15 block Wick monomials with the canonical Pf_4 signs; robust at $\beta = 2$); *exact (sympy):* Schur/Feshbach elimination of the boundary gives $A_{\mathrm{eff}} = \kappa A_{CC} + \frac{t^2 m}{1-m^2} V J_3 V^{\mathsf{T}}$ and generates all 10 carrier duads from the bare diagonal state — on both fronts (RH port, Wick functor) the real correlation lives in the *eliminated boundary*; the frozen 15-signature table is the registered prediction. *Named open check:* RP-THETA-OPEN (no spatial reflection is deployed on the 16-dim space). *Demand:* the physical realization — the [O] premise of SEAM.CFIN.WICKFUNCTOR.01 is unmoved. *Kill:* a physical seam state that violates the frozen signature pattern. No marker move; no RH statement. *Dated note (2026-08-10, rounds 57–59):* RP-THETA-OPEN is now *measured* at family level and the answer is an exclusion — strict reflection positivity and the mixing floor are mutually exclusive on this family (see the seam RP/modular exclusion block below, [verification/v903_seam_rp_exclusion.py]); the [O] premise stays unmoved.

The Christoffel pivot identity and the norm-square contract (PRIME.PORT.CHRISTOFFEL.RATIO.01 + PRIME.CASE.KERNEL.SOS.01 + PRIME.CASE.EDGEDEFECT.01). *Registered and measured* 2026-08-09 ([verification/v899_christoffel_normsquare.py], 19/19 + 15/15 + 19/19, CHRISTRATIO-MEASURED (WINNER-CHRISTOFFEL / EXTSOURCE-DECOUPLED) + KERNELSOS-MEASURED (INDEX-LARGE; RMIN=1; NOT-ABSORBED) + EDGEDEFECT-MEASURED (NORMSQUARE-ACHIEVED; KILL=PERIODIC; CHAIN=CARRIED)). *Exact:* $1/d_{12} = 1 + v_* K_\sigma(y_*)$ — the soft pivot is a deflated-Christoffel evaluation (independent route at 2.0×10^{-11}), winning the boundedness exposure by two orders of magnitude; *the honesty fence, warded exactly:* $\lambda_{\min}(I - G) = \tau$ — the positivity premise is the wall's own PD premise in Christoffel coordinates, a coordinate change, not an independent source; c'/c sensitivity range $[0.223, 5.258]$, $r_{12} > 0$ on 31/31, exterior reserve decoupled (corr 0.454). *The norm-square form:* the pair kernel's frequency weights are $W_{f,m} \geq 0$ exactly; the entire negative content is one top-edge tent (an extension-convention artifact), killed exactly by the periodic full-weight fold (negative index 0; 79/79 modified margins positive; the boundary term carried closed-form, max 0.21) — the conditional contract of `PRIME.CASE.PAIRCORR.CONTRACT.01` is a norm square in the frequency weights; **the arithmetic hypothesis itself stays conditional** (pair-correlation class per `v889`). *Kill:* a negative modified margin on the frozen census. No marker move; no RH statement.

The normalized core update anatomy (`PRIME.PORT.NORMALIZED.CORE.01` + `PRIME.PORT.GRAPH.REGION.01`). *Registered and measured* 2026-08-09 ([`verification/v900_core_update_anatomy.py`], 20/20 + 23/23, NORMCORE-MEASURED (FAMILY-BOUNDED / UPDATE-EXACT / REGION-OPEN) + GRAPHREGION-MEASURED (COUPLING-FREE / GRAPH-REGION-OPEN / SHADOW-LAWLESS)). *Exact:* the normalized deep core $X_h = S_h/\tau_h$ obeys $X' = c\,(X + U)$ at 8.3×10^{-16} with $c = \tau/\tau'$; the family is bounded (diameter 2533) and the input effectively two-dimensional (c + one matrix mode at 97.5% energy). *The registered honest negatives (boundary conditions for any invariant-region theorem):* no linear coupling law $u \sim F(X)$ (both R^2 below the 200-shuffle null q_{95}); the rank-1 truncation fails — the positivity is carried by the 2.5%-energy off-mode; the naive product and graph-tube invariant sets are refused. *The registered direction:* keep the exact $X + U$ with its sign-dependent off-mode structure — no mode may be truncated. *Kill:* an update defect above the ward, or a coupling law beating the null. No marker move; no RH statement.

The seam RP/modular exclusion (`SEAM.STATE.DERIVATION.01` + `SEAM.CFIN.TWISTED.RP.01` + `SEAM.CFIN.RP.DILATION.01` + `SEAM.CFIN.GAP.PENCIL.01`). *Registered and measured* 2026-08-10, rounds 57–59 ([`verification/v903_seam_rp_exclusion.py`], 25/25 + 20/20 + 23/23 + 18/18 on these four claims, SEAMSTATE-MEASURED + TWISTEDRP-MEASURED + DILATION-SPLIT + GAPPENCIL-MEASURED; six frozen probes embedded byte-exact, all fail-first spec amendments on record). *The exclusion:* the odd-sector Gram eigenvalues under both deployable spatial reflections are *exactly* $\pm|a_J|$ of the $\{4, 5\}$ duad (identity 2.2×10^{-16}), so OS positivity $\iff a_J = 0 \iff$ the `v898` mixing floor falls: **reflection positivity and the block-functor mixing floor are mutually exclusive on this family**; strict RP forces $t = 0$; $u \geq t$ is forced (the first non-decorative normalization demand; $u = 0$ is RP-excluded, correcting the round-57 reading); the 2π-KMS locus of the boundary reduction is exactly the point $(t{=}0, \beta{=}2\pi)$ — the geometric temperature kills the mixing instead of pinning it. *The complete twisted census:* 48 candidates → 16 involutive → 6 admissible → 0/6 admit strict RP at any $t > 0$ (TWISTED-RP-EXCLUSIONARY; a measurement on the complete finite candidate set, not a universal no-go). *The dilation split:* the `v898` M2 parent is a θ_{abT}-marginal witness *on* the RP cone boundary whose Schur compression carries the full Pfaffian mixing with the exact rational identity $t^2 \cdot 3m/(1-m^2) = 1/200$; the strict collar route is family-wide obstructed by an exact linear law (30 defect entries, all of magnitude $2t$). *The gap pencil (exact):* $\mathrm{Pf}(A_{\mathrm{dep}}+tA_{\mathrm{int}}) = -(t-1)(3t^2-1)\,q(t)$ with $q = 9t^3+21t^2-t-1$ irreducible; $t_{\mathrm{gap}} = 0.2309488708\ldots$ is the unique root in $(0, 1/4]$ (exact Sturm), a generalized eigenvalue of the fixed integer pencil with kernel dimension exactly 2; the three critical surfaces (1p-gap closure, modular rank-2 ground-state flip $\|\Delta\|_F = 2\sqrt{2}$, hermitized RP boundary $t_c(\beta) \to t_{\mathrm{gap}}$) are projections of one variety; the coupling is load-bearing for the root; *no root is* $1/8$. *Demand (sharpened):* a state construction that carries the mixing *and* a positivity structure must leave this family — the named parent route is non-equilibrium (NESS-PARENT, named, not built). *Dated note (2026-08-10/11, round 60; promoted rounds 60–63):* the NESS-PARENT route is now *built and closed as not-needed* — at finite size stationarity forces zero entropy production exactly, drive buys no reflection positivity, and *equilibrium* witnesses on the $\{J, Z\}$ coupling kernel carry the full 1/200 mixing (see the seam equilibrium wiring block below, [`verification/v908_seam_equilibrium_wiring.py`]); the [O] premise stays unmoved. *Dated note (2026-08-27, `SEAM.STATE.RPMIXING.01`, [`verification/v972_seam_interaction_front.py`]):* the exclusivity is now *typed* as ray- and reflection-relative, not slice-wide — on the full 33-dimensional C_6-covariant mixing slice θ_{abT} kills exactly *one* coordinate (a_J; a 32-dimensional marginal-RP hyperplane remains) and θ_S leaves a 17-dimensional first-order-neutral subspace with strict finite-s witnesses: reflection positivity plus covariance do *not* force the channel-diagonal point. *Dated note (2026-08-28, completeness wave, `SEAM.STATE.RPMIXING.01`, [`verification/v995_kms_subnet_rigidity.py`]):* all 33 v972 mixing directions are killed by KMS *alone* on the finite CAR algebra ($\mathrm{CAR}_8 = M_{256}$);

covariance is provably redundant under full KMS; the unique E-invariant crossed-product state extension is $196608 \to 0$. Type-III/continuum KMS uniqueness stays **[O]**. *Kill:* a strict RP point with $a_J \neq 0$ inside the family. The [O] premise of SEAM.CFIN.WICKFUNCTOR.01 is unmoved; no marker move; no RH statement.

The two dead readings (SEAM.CFIN.MIXING.NORMALIZATION.01 + SEAM.CFIN.MINIMAL.MEDIATOR.01) — registered negatives. *Registered as first-class negatives* 2026-08-10, rounds 57–58 ([verification/v903_seam_rp_exclusion.py], $23/23 + 20/20$ on these two claims, SEAMNORM-MEASURED + MEDIATOR-MEASURED). *Dead reading 1:* '$t = 1/8 = 2\pi c_3/|Z_2|$ is a compiler value' is a *decoration* — the three factors are individually deployed but the combination is free: all 55 scanned t pass all frozen v898 gates (the allowed set is mathematically $(0, \infty)$; $u \in \{0, \ldots, 2\}$ and $\beta \in \{1/2, \ldots, 4\}$ pass too), no frozen functional distinguishes 1/8 (PIN-ABSENT), and the anti-numerology census finds four equally cheap deployed-constant readings; the kept structure fact: the channel mixing is *thermal* (it dies to the ground state, 0/15 cross-blocks at $\beta = 100$). *Dead reading 2:* '$N_{\text{fam}} = 3$ is the minimal boundary-mediation rank' is *refuted* at theorem grade for this finite size — one boundary pair suffices ($\det B_x = \gamma_{ab}\gamma_{cde}$ symbolically for every symplectic-rank-1 mediator; the explicit integer witness passes the full requirement 10/10; seeded census 2000/2000), the deployed mediator has rank 2 *exactly* ($S = 1_5 \otimes 3J$), and what 3 really pins is boundary *purity* = dimension bookkeeping ($\dim(A_3) = 6 = |Z_2|N_{\text{fam}}$), not a mediation theorem. Negative results carry their own ledger rows, not footnotes. *Kill (for the deadness itself):* a frozen v898-gate functional that pins $t = 1/8$, or a physical demand that excludes the rank-1 witness. No marker move; no RH statement.

The rounds 60–63 block (2026-08-10/11; promoted in [verification/v905_bfloor_ideal_certificate.py], [verification/v906_tail_cartography.py], [verification/v907_halfgap_registered_target.py] and [verification/v908_seam_equilibrium_wiring.py]) closes the round-60 seam questions on the compiler side and delivers, on the RH side, the certified B-half surface floor, the tail mechanism map and the registered half-gap target with its first blind holdout (the three RH-side items live in the prime-front document; the seam wiring is registered here). The rounds 64–71 block (2026-08-11/12; promoted in [verification/v909_finite_wall_closure.py], [verification/v910_finite_zero_transfer.py] and [verification/v911_wiring_freedom.py]) closes the *finite* composed wall census and its transfer law on the prime front (PRIME.WALL.FINITE_CLOSURE.01, PRIME.WALL.FINITE_ZERO_TRANSFER.01; both in the prime-front document) and, on the seam side, answers the wiring selector below.

The seam equilibrium wiring (SEAM.CFIN.MARGINAL.STABILITY.01 + SEAM.CFIN.NESS.PARENT.01, closed not-needed). *Registered and measured* 2026-08-10, round 60 (promoted rounds 60–63; [verification/v908_seam_equilibrium_wiring.py], 22/22+25/25, MARGSTAB-MEASURED (INTERIOR-EMPTY / STRATUM-NOT-ISOLATED / NOT-A-CLOSURE-POINT / ESCAPE-NONCOVARIANT / FLOOR-EXCHANGE) + NESSPARENT-MEASURED (PRICE-ZERO / TWO-SEAT-LAW / FINITE-NESS-NOGO / DRIVE-RP-NEUTRAL / NESS-NOT-FORCED); both probes exact-arithmetic, embedded byte-exact). *The 2-cycle side:* the closure-vs-isolated dilemma of the v903 marginal witness is refuted *as a dilemma* — covariance reduces the θ_{abT} seat law exactly to $M_1 = \operatorname{diag}(a_J, -a_J)$ with $\det M_2 = -a_J^2$: strict 2-cycle reflection positivity is **impossible on the whole C_6-covariant class** (interior empty); the witness is neither a limit of covariant strict equilibria (there are none) nor isolated (the whole $s = 0$ family is a marginal stratum; sup(RP-margin × mixing) = 0 exactly); the only strict-enabling escape is the covariance-*breaking* X coordinate at exact linear price 2ε; and the sharpest exact object, the floor exchange: $\mathrm{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$ — the canonical sign gate *is* the negative effective RP margin, with the crossover exactly at the 1/200 floor. *The collar side, no NESS needed:* strict θ_S hermiticity is a *two-seat linear law* of the coupling (exact rank 12, kernel 12 = the $\{J, Z\}$ coordinates of the 24-dim covariant coupling space); the deployed seam wiring is PURE-I (every 2×2 subblock exactly $-I_2$) = a *maximally obstructed* covariant direction, explaining the v903 strict-collar law ('30 entries, magnitude $2t$') exactly — rounds 69–71 add the frame qualifier: the obstruction is maximal *in the deployed θ_S collar frame*, and frame-relative as such (the wiring selector below); the kernel contains *equilibrium* witnesses — V_J (and V_Z) pass strict collar RP in exact arithmetic while their Schur compressions carry the full canonical Pfaffian mixing with the J coordinate exactly $t^2 \cdot 3m/(1-m^2) = 1/200$; at finite size every exactly stationary quasi-free state has $\sigma \equiv 0$ exactly ('NESS with positive entropy production' is category-inapplicable) and drive buys *no* reflection positivity — the minimal entropy production at which the mixing gates open is exactly zero: `SEAM.CFIN.NESS.PARENT.01` **closes as not-needed**. *Demand (the registered successor):* the wiring selector below — **delivered and closed in rounds 69–71** ([verification/v911_wiring_freedom.py]). *Kill:* a covariant strict 2-cycle RP point, or an equilibrium collar witness whose compression breaks the canonical census. The witnesses live on a different covariant coupling direction than the deployed A_{int} wiring — whether the actual seam allows them is untouched; the [O] premise of `SEAM.CFIN.WICKFUNCTOR.01 / v898` is unmoved; no marker move; no RH statement.

The wiring selector (`SEAM.STATE.WIRING.SELECTOR.01`, closed: deployment choice). *Registered* 2026-08-11 (rounds 60–63) as the question: is the deployed PURE-I seam wiring *compiler-forced*? *Closed* 2026-08-12, rounds 69–71 ([verification/v911_wiring_freedom.py], 30/30 + 26/26 + 21/21, verdicts WIRING-DEGENERATE + THETA-CONVENTIONAL + RP-FRAME-COVARIANT; three frozen probes, exact integer/rational arithmetic, embedded byte-exact), and the answer is a *freedom theorem*: the exact Gröbner census of the constraint classes (unit law, CAR/KMS, Pfaffian pencil, orientation propagation, covariance + seat stack) decomposes the real variety certified and leaves exactly **one admissible three-dimensional wiring component modulo the rule gauge** (the rotation cell C_{rot}); the Z/X wirings fall at exactly one class — orientation propagation ($\det u < 0$ identically) — and this exclusion is genuinely rule-gauge-orbit-invariant; PURE-I is an *interior* point of C_{rot}, not a vertex, and the integer gauge element $(\oplus_5 J_2) \oplus I_6$ (preserving the vacuum, the C_6 action, the rule subspace and the constraint ideal) connects it to the pure-J ray: **pure-i is a deployment representative, not a compiler theorem**. The companion probes close the frame side: no compiler-side demand distinguishes the θ_S sheet-swap frame within the 9-dim rule-gauge algebra (five demand classes — μ_4/C_6, CAR/state, pencil/orientation, doily combinatorics, the round-59 census property — all compatible-not-forcing or silent on the angle; THETA-CONVENTIONAL), and the strict-collar RP Grams transform *exactly covariantly* under the rule gauge (symbolic kernel-covariance law on the full torus + exact Gram witnesses): reflection positivity of a (frame, wiring) pair depends only on their *relative angle* on the δ-circle — the same deployed PURE-I parent that fails strict collar RP at θ_S (defect $2t$) *passes it exactly* in the integer-conjugated frame, so the `v903/v908` obstruction is a *frame statement*, not a wiring no-go (RP-FRAME-COVARIANT). The frozen canonical sentence: the compiler determines the wiring *orbit*; strict Hermiticity in the θ_S frame would select pure-J, the same demand in the integer-conjugated frame selects PURE-I; a canonical wiring selection requires an additionally *derived* physical frame. (Forbidden from now on: "PURE-I is compiler-forced" and equivalents without the frame qualifier.) *Honest scope:* the census covers the five enumerated demand classes — a compiler demand outside the list is not excluded; whether the actual seam physics derives a specific collar frame remains outside the formalized rules; the [O] premise of `v898/v903` is unmoved. *Kill (of the closure):* a compiler-side demand outside the censused classes that cuts the δ-circle, or a derived physical collar frame (which would then *select* a wiring via the exact selection map). No marker move; no RH statement.

The Reed–Muller identification (`PRIME.PACKET.RM14.01`). *Registered* 2026-08-06 ([verification/v819_prime_packet_rm14.py], 21/21 incl. the three must-fire controls, RM14-EXACT; Lean companion `TfptCarrier/PacketRM14.lean`, 30 theorems). *Certified:* the 15-label channel row hull is the punctured $\mathrm{RM}(1,4)^* = [15,5,7]$ (enumerator $1 + 15z^7 + 15z^8 + z^{15}$); $C_0 = [15,4,8]$ simplex; χ_{NSR} one codeword (7+8 its two sides, C_2 the affine bit); the self-reproduction cycle closes with exactly 1344 equivalences; the CSS code $[[15,1,3]]$ with dual $[15,10,4]$, logical distances $(7,3)$, triorthogonality 8/4/2. *Typed [H]:* the ζ_8/T-gate phase pointer (`v798`); the TFPT fingerprints ($14 = \dim G_2$, $3 = N_{\mathrm{fam}}$, $4 = |\mu_4|$), audit-only. *Kill:* any census identity failing on the frozen frame. No marker move; no RH statement.

The canonical check rule and the plane frames (`PRIME.KRAUS.RM24.01`). *Registered* 2026-08-06, carrying `PRIME.C2.PLANEFRAMES.01` ([verification/v820_prime_kraus_rm24.py], 16/16 + 14/14,

KRAUS-RM24-RULE-CANONICAL + PLANEFRAMES-EXACT; one honest coset-census fix carried). *The honest double kill:* the literal leg-support claim is dead twice (weights 1/2/3 < distance 4; $Sp(4,2)$ orbits 15+90 vs 45+60 — no equivariant bijection exists). *Certified:* the canonical ω-built rule identifies the rule-A protocol set with the 105 weight-4 checks of $[15,10,4]$ exactly, $Sp(4,2)/\sigma$/deck/KMS-covariant; the six Gram identities entrywise; $A_3 = 12$ / $B_3 = 16$ as per-point plane completion counts ($\sigma_3(3) = 28$, $a_3 = -4$, independent eta-product rebuild); the graded frame intertwiner $V^*V = \sigma_3(n)I$, $V^*\Gamma V = a_nI$ exact for $n = 3, 5, 7$. *Kill:* an equivariance mismatch of the ω rule under the frozen generators. No marker move; no RH statement.

The vacuum completion (`PRIME.VACUUM35.01`). *Registered* 2026-08-06 ([`verification/v821_prime_vacuum35.py`], 22/22 incl. the three must-fire controls, VACUUM35-EXACT). *Certified:* $140 = 105 + 35$; the Grams $22I{+}6J$ / $6I{+}J$ / $28I{+}7J$ entrywise with spectra $\{112, 22^{14}\}$ / $\{21, 6^{14}\}$ / $\{133, 28^{14}\}$; the E_7 completion $112 + 21 = 133$ as a matrix identity (the $E_7 \times A_1$ corpus reading a [C] fingerprint, anchor `v170`); Weitzenböck $H_{105}^T H_{105} = 2B^2 + 14I$ as the canonical label-side left factor (`v758`'s continuum wall explicitly untouched); the $\pm B$ cancellation inside one Gram; the tight frame constant 28 on mean-zero; $\mathrm{RM}(2,4)$ dual-containing. *Typed [H]:* the vacuum/continuum reading — decided negatively by the closure row below. No marker move; no RH statement.

The vacuum-route closure (`PRIME.CONTINUUM.UNSHORTEN.01`, carrying `PRIME.VACUUM.DILATION.01`). *CLOSED — honest double kill* (registered and decided 2026-08-06; [`verification/v822_prime_vacuum_dilation.py`], 17/19 + 12/14 with the four FAILs being the preregistered PSD deciders themselves, controls 3/3 + 3/3, UNSHORTEN-DEAD + DILATION-DEAD; first runs after freeze, no repairs). *The dead reading:* the archimedean/pole continuum as the $\mathrm{RM}(2,4)$ vacuum completion of the 15-label window objects — dead in *both* transcriptions: the one-slot bordering by the `v758` mechanism verbatim and on scale (with the shape theorem: one slot induces only a rank-one pure-J coupling; the demanded $6I{+}J$ needs 35 separate rows), and the corrected target itself — the 35-row Stinespring dilation, shape fixed exactly, $\times 13.1$ distribution gain — still failing both sectors under both frozen normalizations. *The frozen fence (the round's quantitative product):* any continuum re-entry through the RM check geometry is PSD iff its coupling satisfies $c \le c^* = (2/21)\,\lambda_{\text{pencil}}(T_{\text{dep}}, T^2_{\text{cont}})$; measured deficits $\times 3 \times 10^7$ (GL1) / $\times 2 \times 10^2$ (f8) against the mass calibration — the reading dies *with the calibration*, not the geometry: the continuum's completion role is already exhausted inside T_{dep} (the explicit-formula balance), and the path-state construction is not buildable on this gating. The RM integer layer stays certified. No marker move; no RH statement.

The PD-persistence universal form (`PRIME.PD.PERSISTENCE.01`). *OPEN* (registered 2026-08-04): the universal-theorem form of the PD-persistence wall ([`verification/v727_l1_identification.py`], [`verification/v734_s1_canonical.py`]) with the important correction $\lambda_{\min} \ge 0$ instead of $\ge \delta$ — the Ihara lesson (`FTR.IHARA.01`): true RH analogues can have *exact zero modes after rank saturation*, so a uniform margin is stronger than necessary and the wrong instrument. *Target form:* cofinally infinitely many PD windows, established via a state / Gram representation / cone-preserving recursion — *not* via eigenvalue bounds. *The review blocklist, confirmed as permanent kills:* the raw symbol minorant, norm triangles, perturbation theory, position-blind estimates, the direct Sonin transfer ([`verification/v732_strat3_sonin_prolate.py`]) and naive reflection positivity ([`verification/v729_strat2_rp_universality.py`]) stay retired; further zero-matching records do *not* change the quantifier. A research contract, *not* a claim; no marker move; no RH statement. *[2026-08-04 update (the incidence programme): the contract now carries a REGISTERED INDUCTION FORM, a CLOSED route, and a NAMED open object. (i) The induction form: the ramified Schur ladder on the canonical tower ([`verification/v755_simpler_schur_recursion.py`], 13/13, SCHUR-RECURSION-ALIVE) — $S_k \succeq 0$ measured on all 8 ram-odd stages ($n_k = 2^{2k+1}$, $\lambda_{\min}$ flat, long-double-hardened), with basis, Douglas range condition, Albert/Haynsworth step and cofinality all exact: the open proof need is* exactly *the structural Gram certificate $S_k = R_k^T R_k$ per step; the measured boundary data for any candidate R_k are recorded (the label part enters* inverted, *$\rho_k < 0$ with $|\rho_7|$ within factor* 1.34 *of* 49/4 *— cross-term structure, not an added block — and continuum vs* *atom sides are order-*1 *comparable). Honestly: for* ζ *data this is* equivalent *to tower PD — the wall in recursion shape. (ii) The closed route: the review's assembled certificate* [*continuum root*$\oplus$*Kraus*] *is* not mountable *(*[*verification/v758_simpler_certificate.py*]*,* 8/8*,* CERT-CONTINUUM-ROOT-DEAD*, the keystone negative): the arch+pol block is PSD on* no *stage (Levinson break at cell* 47*, exactly between the* $n = 2$ *and* $n = 3$ *cells, ladder-wide), the continuum Schur complement is no Haynsworth object, the additive split fails on every stage (the order-*1 *balance sits in the cross-term), and the rescue cascade is strictly* cell-ordered *— the wall IS the missing left factor. (iii) The named open object: the corrected target shape is the* cell-wise cascade factorization $R = \prod(\textit{cell factors})$ *— rescue order* $n = 2, 3, 4, 5, \ldots$ across *the channels — with the exact building blocks of the incidence programme (*[*verification/v752_projective_hamming_incidence.py*]*,* [*verification/v753_ramified_polarity.py*]*,* [*verification/v754_ramodd_twostep.py*]*,* [*verification/v756_kms_incidence_stinespring.py*]*) as the label-side material. No marker move; no RH statement.]*

The diagonal Gram closure theorem (`PRIME.GRAM.DIAGONAL.01`). *CLOSED* (registered 2026-08-04, the diagonal gram round; *closed 2026-08-05 per its own frozen stakes* — an honest negative adjudication, not a marker upgrade; the full closure cascade is the dated note at the end of this block). The target form, kept verbatim for the record: construct, for the countable dense test family $\{\phi_i\}$ of the canonical (D, X) tower ([`verification/v762_dense_weil_core.py`], 26/26, DENSE-CORE-CANONICAL: frozen enumeration, explicit stage map, admissibility located per class), PSD source matrices G_n *without* any target decomposition, zero data, or positivity test of the target, satisfying

$$\max_{i,j \le n} \left| (G_n)_{ij} - W(\phi_i * \tilde{\phi}_j) \right| \;\le\; 2^{-n};$$

then every fixed m-corner of W is a PSD limit of PSD matrices — Weil positivity on the rational span of the family, and the density step closes by the classical approximation argument. *The round's measured decision state:* Gate 1 positive — the handoff error is boundary/cutoff, not quadrature ([`verification/v759_handoff_fixed_window_resolution.py`], RESOLUTION-BOUNDARY); Gate 2 positive — the anti-alias identities are exact with sharp count $N_{f,\min} = 2M-1$ ([`verification/v760_antialias_exact.py`], ANTIALIAS-CORRECTED) and the paired atom+pole remainder carries the explicit unconditional vanishing bound η_K with *no RH-strength input* ([`verification/v761_atom_pole_abel.py`], combined ABEL-PAIRED-BOUND); Gate 4 positive — the family is canonical ([`verification/v762_dense_weil_core.py`]); Gate 3 *negative on the canonical class* — the battery near-kernel pairing q_f is non-Cauchy at honest maximal depth and survives orthogonalization by structural subordination ([`verification/v763_yosida_handoff.py`], QF-DEAD): *the* q_f *wall* — density and near-kernel avoidance collide on the same function class — is the named wall object; Gate 5 (the assembled diagonal certificate, the planned `v764`) was *not executed* — its precondition (theorem-capable bounds from Gates 2 *and* 3) failed via Gate 3; the red team is green ([`verification/v765_handoff_redteam.py`], REDTEAM-GREEN-COMBINED, with the binding note that certificates must always carry *both* gate families — symbol/PSD and rate/ratio/identification). *Kill criteria that did NOT fire:* no RH-strength input was needed anywhere (the v761 circularity flag and the v763 KILL audit are machine-checked: no $1/\varepsilon$ bound, no PD-margin assumption); no spectral gap was assumed (exact zero modes stay admissible — the Ihara lesson). *The binding stop-list* (ten items, frozen): no atom/prime-layer positivity induction; no separate absolute atom/pole bounds; no 16-dimensional channel square sums; no exact neighboring-window embeddings; no unbounded A^{-1} observables; no Ward-drift signs; no rate fits as proof substitutes; no zero-matching records as progress metric; no positive continuum root as certificate factor; no cellwise PSD demands. A research contract, *not* a claim; no marker move; no RH statement. *[2026-08-05 closure note — the route is CLOSED per its own frozen stakes. The final decision cascade, full tally: Gates 1/2/4 positive from the diagonal gram round (above); Gate 3 DEAD AT THE CORNER LEVEL — the audit pair (*[*verification/v769_qf_contract_necessity.py*]*, combined* QF-GAUGE / GATE3PRIME-DEAD*) first proves the full-ladder Cauchy demand on* q_f *was* stronger *than the contract needs (*q_f *is representation gauge: a deterministic* ε*-net subsequence of density* 0.144 *makes all 14 profiles Cauchy at the parent bars, and the contract's own* E_Y *cor-*

ner objects are blind to q_f jumps, corr $+0.038$*, median* $\rho = 1.13$*), and then the corner-native weakened Gate* $3'$ *dies on its own predeclared* 1.6×10^7 *comb (*$X = 16.5625$*): the corner increments fall to* $X \sim 12.9$ *and then rise sustained (*$+0.10..+0.34$ *per* X *unit at* $\varepsilon = 10^{-3}$*,* $0/56$ *cells) — the* **v764** *precondition is refuted; Gate 5 /* **v764** *therefore was* never executed*. The qf offensive's own line: gauge → corner rise (*[verification/v769_qf_contract_necessity.py]*) → settled-positive levels (*[verification/v770_qf_spectral_bundle.py]*,* QF-SETTLES-POSITIVE*: the band weights plateau at positive per-function constants, R2 family* 0.225*–*0.358*, R1 family* 0.008*–*0.079*; zero-bar depth* $X^* \sim 227$ *= never) → no fixed-d object (*[verification/v772_qf_feshbach_effective.py]*,* FESHBACH-PARTIAL / EDGE-KEEPS-MOVING*: the moving spectral edge — avoided crossing* 0.0039 *at* $M = 1240$ *on the* 10^9 *comb, widening non-regular entry cadence* $\Delta X = 1.625/1.8125/2.625$*, spread* 0.381*) → cocycle domain-only (*[verification/v773_qf_cell_cocycle.py]*,* COCYCLE-DOMAIN-ONLY*: exact identity and zero-breach domain transport,* $0/6$ *convergence cells — structure without limit). What survives as theorem-shaped facts, each typed in its own module row: the exact anti-alias theorem (*[verification/v760_antialias_exact.py]*); the unconditional paired Abel bound (*[verification/v761_atom_pole_abel.py]*); the canonical dense family (*[verification/v762_dense_weil_core.py]*); the exact source-native domain-preserving monotone Möbius/Redheffer cell cocycle through mode entries (*[verification/v773_qf_cell_cocycle.py]*); the Kato bundle frame (*[verification/v770_qf_spectral_bundle.py]*); the Herglotz-certified exact Feshbach reduction (*[verification/v772_qf_feshbach_effective.py]*). NO new variants; the ten-item stop-list stays binding; the sharpened* `PRIME.Z1.OPERATOR.01` *takes over with the measured handover content [2026-08-05: executed as the* **v780** *trilogy — compactness carried, selection the one shared open object, merged target* Z1-COMPACTNESS*].* Lean legacy (2026-08-05): *the exact cell-cocycle core of the closed route is machine-checked as general Mathlib theorems over* $\mathbb{R}$ *—* `TfptCarrier/CellCocycle.lean`*,* 18 *declarations: the exact Banachiewicz/bordered-Schur update identity, PD cell block ⇒ PSD corner correction ⇒ monotone Loewner flow with domain preservation, bounded-monotone entrywise matrix convergence (the theorem* names *the missing analytic ingredient precisely — a uniform Loewner upper bound, which the finite ladder did* not *certify and which is not claimed), and the Redheffer/linear-fractional composition laws over arbitrary commutative rings. No RH statement.]*

The structural upgrade: the canonical nested tower replaces the historical window family (`PRIME.TOWERNEST.01`). The window quantifier of the PD-persistence formulation reduces *exactly* to the canonical nested (D, X) tower ([verification/v749_simpler_tower.py], 8/8, TOWER-NESTED-REDUCES): in X an ascending chain — X-nesting exact to 6.7×10^{-18}, so limit positivity is *one* limit object per chain — and in D dyadic downward inheritance, exact: PD fine *forces* PD coarse; every deployed window is a tower member at deviation 0.0. The window quantifier was an *artifact* of the historical selection: box step functions are dense, so tower-PD for $D \to 0$, $X \to \infty$ is equivalent to the Weil criterion (classical density step). `PRIME.PD.PERSISTENCE.01` is therefore restated with the canonical truncation in place of the historical window family. *Honest limits:* incommensurable root-D chains connect only by approximate interpolation (the residual does not close the PD transfer), and the margins *fall* along the tower — the wall stands, in canonical nested form. *Companion diagnostics of the simplification trilogy:* the gluing identity is an identity-level *off-line detector* — it sees $\delta = 0.01$, sharper than the PD break at $\delta^* = 0.02$, and the envelope slope measures δ — while the rigidity thesis is honestly false (the identity holds unconditionally; [verification/v750_simpler_gluing.py], 5/5, GLUING-DETECTOR); and the component running is deterministic in direction with the exact $\mp u/2$ drift anatomy from $p^2 \equiv 1 \bmod 4$ and a slope-Ward cancellation at 4×10^{-4} — the margin inherits *zero* deterministic drift — but the balance *sign* is zero content (budget $2.9 \times 10^4\times$ the margin): the induction route is RH-circular at the balance level ([verification/v751_ward_monotone.py], 15/15, WARD-MONOTONE-MIXED). The cone-preserving layer recursion is retired by a one-liner: cone-preserving iff layer-positive, and no prime layer is ($c_0 = 0$; [verification/v743_schur_cone_recursion.py], 7/7, SCHUR-CONE-DEAD). No marker move; no RH statement.

The four fixed-size angle instruments (`PRIME.ANGLE.INSTR.01`). The instrument cores of the discovery parts T155 and T157, recomputed on the same declared surface as `PRIME.RITZ.CEIL.01` (the 12 deepest in-cap frame-A windows, $n = 29 \dots 139$, $m = 96 \dots 291 \le 300$; a no-go size ladder $m = 48 \dots 288$; parity and Dirichlet controls at $m = 64, 128, 256$; each statement with at least one mutation control that must fail loudly) — none of the four closes an angle, and none is promoted as anything but a per-instance instrument. *(1)* The 12×12 complement-floor certificate: $\min_{v \perp W} v^\top L_P v \ge \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^\top)M^{1/2})$ with $M = \operatorname{diag}(\mu_{13}^P - \mu_k^P)$ and $G = T_{12}Q_W$ — an identity plus *one* certified Rayleigh bound on the high block, a theorem for every m and every subspace, direction checked: the certificate never exceeds the exact size-m complement bottom (12/12) and agrees to $1 - \text{ratio} \le 1.01 \times 10^{-3}$ against the declared cap 2×10^{-3} on this small surface; exactness controls both ways ($W = \operatorname{span}\{t_1..t_8\} \to \mu_9^P$; the *worthless* configuration $W = \operatorname{span}\{t_2..t_{13}\} \to \mu_1^P$; both to 2.0×10^{-11} — the instrument reports its own worthlessness); a corrupted overlap row moves the certificate by ≥ 0.549, only downward. *(2)* The sine-block confinement: from the certified pencil floor $A \succeq tL_P$ and the certified ladder ceiling $\lambda_1 \le S\mu_1^P$ *alone*, $\|\gamma_H\|^2 \le \lambda_1/(t\,\mu_{17}^P) \le (S/t)/\rho_{17} = 0.0153$–$0.0245 < 1$ on 12/12 — the bottom eigenvector lives 97.5–98.5% inside the first 16 parity sines by a per-instance theorem (measured tail 6.7×10^{-6}–3.0×10^{-5}; the discovery part's measured "98 per cent" replaced by one line); negative control: the pencil ceiling κ in place of the ladder S is vacuous (17.6–$477 > 1$ on 12/12); mutation: the inflated floor κL_P is refuted at $e_1(A)$ by 3.3×10^3–1.0×10^5. *(3)* The Schur-entry identity: $s = \mu_1^P\, t_1^\top A^{-1} t_1 = (B^{-1})_{11}$ exactly (1.1×10^{-11}) and $(B^{-1})_{LL} S_L = I$ to 5.8×10^{-11} with $S_L = B_{LL} - B_{LH} B_{HH}^{-1} B_{HL}$ the *same* fixed-size 16×16 Schur complement the chain already forms, with the literal-entry direction $1/s \le (S_L)_{11}$ on 12/12: $1/s = 4.0460$–5.8962 is an $O(1)$ number carried by one entry, so the R2$''$ loss ceiling $r \le 1/(Ls)$ is a statement about *one* number; negative control: the inversion-free Cauchy–Schwarz ceiling is valid and useless, off by 250–7.27×10^3 (the cancellation in $b^\top B_{HH}^{-1} b$ is nearly complete); mutation: a one-index shift breaks the identity by 1.55–2.15. *(4)* The adaptive Lipschitz ceiling: interval bisection with the *local* Lipschitz constant certifies $\inf f^{\text{arch}}/(4\sin^2(\theta/2)) \ge 0.25$ on 12/12 windows in 519–1201 symbol evaluations (certified floor 0.2505–0.3062, never above the fine-grid minimum; the global constant is documented insufficient — a uniform grid would need up to 61.4× more); mutation: fed a target above the true grid minimum it *refuses* on 12/12. Stress: on the T145 no-go family the Schur floor survives ($t = 0.05$) while the confinement goes vacuous (7.93–413, ×52.0) and $(S_L)_{11}$ explodes ($1185 \to 7.11 \times 10^4$, ×60.0) against the flat 4.05–5.90 real band; parity control exact to 4.9×10^{-14}; the Dirichlet residual hits the predicted $2/\sqrt{N}$ to 7.4×10^{-5}. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D; *nothing about the angles is closed* — the two open terms after T157 (an m-free lower bound on $\hat{g}_1^2$, equivalently an m-free upper bound on the one Schur diagonal entry $1/s$; an m-free R1$''$ domination with a non-shrinking margin — certified per window with a 7.3×10^{-4} margin at the largest window and a shrinking trend) stay open, typed open; every floor t and ceiling S consumed is certified per window and its m-freeness is exactly what remains open; `PRIME.RITZ.CEIL.01`'s ceiling and `PRIME.ODD.SECTOR.IDENT.01`'s ladder stay as stated and nothing is re-opened. Fences: Kac–Murdock–Szegő 1953 / Schur 1917–Haynsworth 1968 / Courant–Fischer 1920–Cauchy 1829 / Cauchy–Schwarz (refuted as a ceiling, wired as a control) / Kantorovich 1948 / Temple 1928–Kato 1949 (floor devices, not used here) / Sylvester 1852–Bunch–Kaufman 1977 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 / Szegő 1915–Widom 1958 (a named address, never a licence) named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v552_angle_instruments.py`]. **[E]** (per-instance identities and certified inequalities)

The exact-form identities (`PRIME.EXACT.FORM.IDENT.01`). The theorem cores of the discovery parts T158 and T159, recomputed on the same declared surface as `PRIME.RITZ.CEIL.01` and `PRIME.ANGLE.INSTR.01` (the 12 deepest in-cap frame-A windows, $n = 29 \dots 139$, $m = 96 \dots 291 \le 300$; no-go ladders $m = 48 \dots 288$ on *both* T145 forms; parity and Dirichlet controls at $m = 64, 128, 256$; each statement with at least one mutation control that must fail loudly) — the algebra of the sixteen-form itself, and it closes neither open term. *(1)* The Thomson dual form and the positive Cholesky ladder: $s = (B^{-1})_{11} = \max_x(2x_1 - x^\top Bx)$ (Maz'ya 1985 / Miclo 1999) with the direction *executed*, not asserted — 8 random trials per window never exceed s; the ladder $g_K = \sum_{j \le K} y_j^2$, $y = L^{-1}e_1$ from one Cholesky of the leading 16×16 block (nested Schur complements): all terms strictly positive, partial sums strictly monotone, $1/s \le 1/g_{16} \le 1/g_1 = B_{11}$ literally on 12/12, tight to 1.0642–1.1522 against the declared cap 1.5; mutation: a 5% corruption of one off-diagonal entry destroys positive definiteness — the Cholesky *refuses* on 12/12. *(2)* The two-dimensionality: $\mathrm{span}\{t_1, A^{-1}L_P t_1\}$ attains s exactly ($g/s = 1$ to 9.0×10^{-13}) for the theorem reason $L_P t_1 = \mu_1^P t_1$ (KMS 1953); negative controls: the two-sine span misses by 1.42–4.8, the corrupted Green column by 255–7.7×10^3. *(3)* The gauge identity and the Toeplitz-minus-Hankel signature: the reflection-odd section annihilates constant lag vectors *entrywise with zero tolerance* ($\text{odd\_toeplitz}(\mathbf{1}) = 0$ exactly), hence $\sum_d w_d = 0$ to 1.1×10^{-14} of $\sup|w|$ — the form is blind to the lag mass of c, exactly where the archimedean and atom halves are h^2-sized and cancel; the signature $A_{r,s} - A_{r+1,s+1} = f(r+s)$ holds to 1.4×10^{-16}; mutations: a one-index Hankel shift breaks the identity by 2.7×10^{-7}–4.2×10^{-5}, and the T145 form has signature defect 7.4×10^{-2} — not a parity section, the identity is not even defined for it. *(4)* The exact lag sum and the two closed scalars: $x^\mathsf{T} B_{LL} x = \sum_d c_d w_d$ ($w_d = T_d - H_d$) against the direct assembly to 2.7×10^{-16} of the absolute scale $\sum_d |c_d w_d|$, with the honest price *measured* (relative to the cancelled total only 1.4×10^{-11} on this small surface — that depth is the h^2 obstruction); $w_0 = \sum_k x_k^2/\mu_k^P$ to 4.2×10^{-16} and $2w_0 - w_1 = \|x\|^2$ to 7.4×10^{-16}; mutations: a 1% lag perturbation and a one-index μ shift both break loudly. *(5)* The Z-matrix law and the Collatz–Wielandt floor: B_{HH}^{arch} is a symmetric Z-matrix *raw* — every off-diagonal entry ≤ 0 with no tolerance on 12/12; the closed sign-based floor $\min_k (B_{HH}^{\mathrm{arch}}\mathbf{1})_k = 0.9093$–$1.1284 \ge 0.25$ with the theorem direction verified per instance against the exact $\lambda_{\min} = 0.9138$–1.1432 (Perron 1907 / Frobenius 1912 / Collatz 1942 / Wielandt 1950), beating the sign-free norm route by 0.69–8.8; the honest limit as content: the atom block obeys no sign law (0.44–0.53, a coin toss), the full block is not a Z-matrix and its row-sum floor is negative ($-48.3 \ldots -0.79$) — admissible and vacuous, R1$''$ stays open; mutation: the checkerboard similarity (an isometry, $\lambda_{\min}$ moves by 0 exactly) destroys the raw law — an entrywise fact, not a spectral one. Stress: the T145 no-go breaks on three axes of this module's structure; on the $c(l) = 1/(1+l)$ reconstruction the ladder theorem survives while $1/g_{16}$ explodes ($1187 \to 7.11 \times 10^4$) — the instrument reports the collapse instead of hiding it; parity: for $A = L_P$ the machinery returns $s = 1$ with every ladder partial sum constant in K; Dirichlet controls exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D, and explicitly *no* statement for all D; the two open terms after T159 (the one explicit pairing inequality for R2$''$ — $\sum_d (\Delta c)_d W_d^1$ with $\|\Delta c^{\mathrm{arch}}\|_1 \le 6$, needing a correlation bound instead of sizes; a third device for the atom off-band block for R1$''$, neither norm nor sign) stay open, typed open; the pairing bound and m-freeness are explicitly *not* claimed. Fences: Maz'ya 1985 / Miclo 1999 / Schur 1917–Haynsworth 1968 / KMS 1953 / Dirichlet 1829 / Abel / Fejér 1915 / Perron 1907–Frobenius 1912 / Collatz 1942–Wielandt 1950 / Gantmacher–Krein 1950 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v553_exact_form_identities.py`]. [E] (per-instance identities and certified inequalities)

The sampling/harmonics identities (`PRIME.SAMPLING.HARM.01`). The closed theorem cores of the discovery parts T160 and T161, recomputed on the same declared surface as `PRIME.EXACT.FORM.IDENT.01` (the 12 deepest in-cap frame-A windows, $n = 29 \ldots 139$, $m = 96 \ldots 291 \le 300$) plus a declared Λ-battery of six log-spaced cut-offs $X = 10^3 \ldots 3.2 \times 10^5$ (finite von-Mangoldt sums only, no matrix anywhere; each statement with at least one mutation control that must fail loudly) — what the sixteen-form *pairs against*, and it closes no open term. *(1)* The sampling identity: the atom half of the pairing equals $-\frac{1}{2}\sum_{n \le X} (2\Lambda(n)/\sqrt{n})\, \widehat{w}(\log n)$ (plus one reflected term below the grid; $\widehat{w}$ the linear interpolant of the lag weights), exact to 5.6×10^{-15} of the absolute atom scale on 12/12 — hence identically a finite combination of sums $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$ at 32 explicit frequencies; mutation: a one-index grid shift breaks by 5.1×10^{-5}–3.1×10^{-3}. *(2)* The moment laws and the total-variation theorem: $m_0 = 0$ (1.0×10^{-16} of $\|w\|_1$), $m_1 = -[S_0^2 + 2\sum_j P_j^2]$ (9.2×10^{-16}) and $m_2 = -[2S_1 - (M-1)S_0]^2$ (3.4×10^{-14}, a perfect square), both strictly negative on every window; U3 as a theorem with *checked* hypotheses ($c^{\mathrm{arch}} \le 0$ and monotone on $d \ge 1$, checked 12/12): $\mathrm{TV}(c^{\mathrm{arch}}) = 3.9817$–$4.8991 \le |c_0| + 2\sup < 6$, closed and window-independent; mutations: a prefix shift, an $(M-2)$ mis-normalisation, and the full atom-including kernel all fail loudly. *(3)* The harmonic-frequency theorem and the PNT main term: $t_j(2\alpha) = 2\pi j$ *exactly* (2.8×10^{-14}) — the 32 frequencies are precisely the Fourier harmonics of the log-window $[0, 2\alpha]$; at those frequencies the partial-summation main term collapses to the closed parameter-free prediction $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$ (de la Vallée Poussin 1896), matching the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12 on the battery; mutations: a wrong Mellin pole worsens the aggregate residual by $\times 1.8$–$\times 95.8$, and a mass-matched *uniform* fake measure misses by 0.23–0.51 — the agreement is a property of the von-Mangoldt measure. *(4)* The head split and the scale-free Bernstein rate: $c_d^{\mathrm{arch}} = \Psi_d + D\widehat{G}(dD, D)$ with $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$ closed, D-free and m-free (the triangle-smeared simple pole; the $\log D$ cancels identically), exact to 3.2×10^{-16} at every lag of every window; $\rho^* = (3 + \sqrt{5})/2 = 2.618034$ is the root of $x^2 - 3x + 1$, returned exactly by the $D \to 0$ limit of the peeled interval and invariant under $\alpha \times 10$ (Bernstein 1912); mutations: dropping Ψ misses by 0.99–1.00, the wrong interval ratio moves the limit to 3.0. *(5)* The off-diagonal fraction bound and the sign refutation: the gauge identity licenses one boundary-free Abel step, the decomposition $Q^{\mathrm{arch}} = \sum_{k,l} a_k a_l \sum_d (\Delta c^{\mathrm{arch}})_d R^1_{kl}(d)$ is exact (2.0×10^{-13}), and the certified per-window inequality is $|\text{off-diagonal}| \le \frac{1}{4}|Q^{\mathrm{arch}}|$ (0.171–0.212 on 12/12) — with the *refuted* sign law wired as the negative control (arch half negative, off-diagonal block positive on 12/12, 176/240 pairs violate the per-pair form): nothing single-signed is promoted; mutation: a one-index R-kernel shift breaks by $\ge 6.8 \times 10^{-5}$. Stress: the head split misses the T145 no-go reconstruction by 2.39 of $\sup|c|$; the Dirichlet cosine-sum identity (including the degenerate branch), the KMS eigenpairs and $\mathrm{odd\_toeplitz}(c^L) = L_P$ (zero tolerance) are exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D; finite Λ-sums are allowed and used, *no* RH statement is made, and the δ triage of T161 (required cancellation depth against RH strength) is a documentation item, *not* a claim of this module; the open terms after T161 (R-A′ the prime-free log-moment $\sum_d \Psi_d w_d$, outside the polynomial ladder; R-B′ the m-free 16×16 Gram fraction bound, Fejér 1915 / Schur 1917; R-C′ *one* split of the pairing with $\delta_{\mathrm{eff}} < \frac{1}{2}$, T161 refuting the arch/atom and arch+smooth/fluctuation splits with numbers; R-D a fifth R1″ device) stay open, typed open. Fences: Dirichlet 1829 / Abel 1826 / Fejér 1915–Schur 1917 / Bernstein 1912 / Chebyshev 1852–Mertens 1874 / de la Vallée Poussin 1896 / Clausen–Lerch / KMS 1953 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v554_sampling_harmonics_identities.py`]. **[E]** (per-instance identities and certified inequalities)

The Pareto/total-variation identities (`PRIME.PARETO.TV.01`). The closed theorem cores of the discovery parts T162 and T163, recomputed on the same declared surface as `PRIME.SAMPLING.HARM.01` (the 12 deepest in-cap frame-A windows, $n = 29 \ldots 139$, $m = 96 \ldots 291 \le 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table in $[100, 3.2 \times 10^5]$ (reproduced to 4.2×10^{-7}; each statement with at least one mutation control that must fail loudly) — the *price structure* of the pairing, and it closes no open term beyond the negative closure that the floor inequality itself is. *(1)* The exchange law and the crossing criterion: $\delta_{\mathrm{bnd}}(x) = \frac{1}{2} + \log(2\kappa\, g_{16}\, \mathrm{TV}(x)/P(x))/\log X$ is an identity at all 612 grid points of the declared Fejér knob (50 geometric σ on $[1, 4096]$ plus ∞; 5.1×10^{-16} relative, admissibility $Q(x) > 0$ everywhere), and $\delta_{\mathrm{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16}\mathrm{TV}$ holds as an exact equivalence at every point — price and demand are two coordinates of *one* object; mutation: the wrong constant (κ for 2κ) moves the law by 6.5×10^{-2}–3.4. *(2)* The total-variation floor — the theorem that carries the T163 verdict: (T1) $w_0(x) = \|a(x)\|^2$ $(4.9 \times 10^{-16}$; orthonormal parity rows); (T2) $\mathrm{TV}(x) \ge |w_0(x)|$ by telescoping with $w_M = 0$; (T3) $x_1 = 1$ forces $\|a\|^2 \ge 1/\mu_1^P$, hence $\mu_1^P \mathrm{TV}(x) = 5.00$–$28.84 \ge 1$ on all 708 trial vectors built (51 knob settings plus 8 random admissible sixteen-vectors per window), the closed floor $1/\mu_1^P = 1/(4\sin^2(\pi/N)) = 943.6$–$8609.6 \sim h^2$;

(T4) every sub-$\frac{1}{2}$ grid point pays $P > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$ (all 54 respect it) — a flat-price sub-$\frac{1}{2}$ demand is impossible in the parity sector on this surface, the inequality that closes R-C‴ negatively; negative control: the *inadmissible* vector $0.01x^*$ ($x_1 \neq 1$) does undercut the floor ($\mu_1^P\mathrm{TV} = 2.5$–2.9×10^{-3}) and is recognised as inadmissible — the floor is a property of the normalisation meeting the smallest parity eigenvalue (Kac–Murdock–Szegő 1953). *(3)* The chain-derived price axis: $\sigma = 1$ gives $x = e_1$ exactly with $P = g_{16}/g_1$ (the $K = 1$ rung of the T158 Cholesky ladder, the T157 route-(0) certificate; 4.1×10^{-16}), $\sigma = \infty$ gives $g_{16}Q(x^*) = 1$ (the $K = 16$ top rung; 1.2×10^{-12}); the ladder carries all 16 terms strictly positive with monotone partial sums, and the knob is monotone in both coordinates on every window — the knob curve *is* the Pareto front; mutation: a corrupted off-diagonal sixteen-block entry makes the Cholesky *refuse* on 12/12. *(4)* The Mellin cell moments and the Abel step: the k-th rung $\sum_d w_d C_d(-(2k + \frac{1}{2}))$ with $C_d(s)$ the closed tent integral (Mellin 1896) agrees with an independent per-cell Gauss–Legendre quadrature to 2.8×10^{-13} on every rung $k = 1 \dots 4$ (shifted tent centres break by $\geq 1.3 \times 10^{-2}$); the boundary-free Abel identities hold at levels $k = 1 \dots 3$ (8.3×10^{-17}; Abel 1826), the dropped-boundary level-1 form agreeing because the gauge identity kills the boundary term exactly; the optimal level is *one* for T162's closed reason ($32\pi/\alpha = 40.6$–$59.4 > 1$, measured level-2/level-1 ratio 30.8–69.1 > 1), and a gauge violation breaks the dropped form by its closed prediction $v_0 c_0 M$ to 1.2×10^{-15} — the positive control. *(5)* The Lerch/Frullani log-moment: $L = \sum_d \Psi_d w_d$ agrees on three independent routes on every window — direct; double-Abel *with* the Wronskian boundary term $-\frac{1}{2}M \log M\, w_{M-1}$ (2.8×10^{-14}; dropping it breaks by 2.2×10^{-4}–1.3×10^{-2}); the Lerch/Frullani integral with the $d = 1$ term peeled closed (1.8×10^{-15}; the peel constant $2\log 2$ to 2.2×10^{-16}, $\Psi_1 = -\log 2$ exact; Frullani 1828 / Lerch 1887 / Clausen 1832). Stress: the monotone no-go staircase — coarsening over $\nu = 4/2/1/0.5$ degrades the arch/atom cancellation monotonically in the median ($1.00 \to 3.49 \to 10.81 \to 24.98$, the $\nu = 4$ column an exact null control); the Dirichlet cosine-sum identity (2.4×10^{-14}, including the degenerate branch), the KMS eigenpairs (1.4×10^{-14}) and odd_toeplitz$(c^L) = L_P$ (zero tolerance) are exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D; the T163 h-exponents ($P_{\mathrm{aff}} \sim h^{3.05}$, $P_{\mathrm{cross}} \sim h^{1.91}$, the TV exponent $\sim h^{2.00}$) are sandbox fits and are *not* consumed — what is consumed is the closed per-window floor; finite Λ-sums are allowed and used, *no* RH statement is made ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required depth is compared against); the open terms after T163 (R-E the successor, both arms prime-free — arm A: a growing $1/s$ ceiling for the downstream T159/T160 chain; arm B: a sector whose lowest eigenvalue does not vanish like h^{-2}; R-B‴ the h-uniform positivity margin; R-D a fifth R1″ device) stay open, typed open. Fences: Fejér 1915 / Mellin 1896 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Legendre / Dirichlet 1829 / KMS 1953 / Frullani 1828–Lerch 1887–Clausen 1832 / Schur 1917–Haynsworth 1968 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v555_pareto_tv_identities.py`]. **[E]** (per-instance identities and certified inequalities)

The gauge/P_{pr} identities (`PRIME.GAUGE.PPR.01`). The closed theorem cores of the discovery parts T164 and T165, recomputed on the same declared surface as `PRIME.PARETO.TV.01` (the 12 deepest in-cap frame-A windows, $n = 29 \dots 139$, $m = 96 \dots 291 \leq 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table in $[100, 3.2 \times 10^5]$ (each statement with at least one mutation control that must fail loudly) — the *gauge structure* of the pairing, and it does not close the one remaining quantifier. *(1)* The gauge theorem (T164): $x_1 = 1$ fixes $a_1 = 1/\sqrt{\mu_1}$, i.e. only the *scale* of a; Q and TV are homogeneous of degree two in a; hence the demand ratio Q/TV of the same a-direction — and with it δ_{bnd} — is the same number in *every* sector, unchanged to 2.9×10^{-11} on 2520 sector-by-trial-vector combinations (5 weight families $\times$ 7 shifts $\times$ 6 Fejér taper widths on 12 windows), with $\mathrm{TV} \geq |w_0|$ in every sector and the pure shift scaling the certified value by exactly $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$ (2.2×10^{-11} on 432 points); the full space is settled in closed form and strictly worse ($\mu_0 = 0$ exactly, the lowest remaining eigenvalue $\le \mu_1^P$ on 12/12); mutation: the wrong transport map (the x-coordinates carried unchanged instead of the a-direction) moves the ratio by 2.0×10^2–5.4×10^3. *(2)* The h^2 conservation law (T164): floor × transfer $= 1/(\mu_1^P + \kappa_s) \cdot (1 + \kappa_s/\mu_1^P) = 1/\mu_1^P$ *identically* at every shift on the declared grid (72 window-by-shift points, 2.2×10^{-16} of $1/\mu_1^P$) — the h^2 is conserved, never removed; it lives in the ratio, not in the floor; mutation: the wrong transfer misses by 0.9995 at the largest shift on every window. *(3)* The power-one spend as an identity (T164): the r-ceiling $r \le U/L$ with $L = \lambda_1(A)/\mu_1^P = 1.1019$–$1.5845$ (computed per instance from the full window) is linear in the entry ceiling, so $d\log r/d\log U = +1$ exactly (within 1.1×10^{-14} on 12/12) — an h^ε ceiling costs $h^{-\varepsilon}$ of r-margin for every $\varepsilon > 0$, no free tolerance in the consumption step (the measured power of the full T156 loss margin stays sandbox); mutation: the quadratic consumption reads slope 2.0000 and is recognised. *(4)* The P_{pr} identity and the predicate equivalence (T165): $P_{\mathrm{pr}} = g_{16} \cdot R \cdot (\mathrm{TV}/(t_1 v)^2)/\mu_1^P$ with no inequality anywhere — four factors, each a named object: the R-E-A quantifier $g_{16} = 0.1559$–0.2227, the demand $R = Q/\mathrm{TV}$, the T163 floor tvf ≥ 1, and the KMS 1953 scale $1/\mu_1^P = 943.6$–$8609.6 \sim h^2$ — exact to 4.3×10^{-16} on all 192 battery vectors (6 knob settings, 6 random sixteen-vectors and 4 random *full-window* vectors per window), gauge-invariant under $v \to tv$ to 1.5×10^{-11} against the declared cancellation horizon; the predicates $R > 2\kappa$ and $P_{\mathrm{pr}} > 2\kappa g_{16}\,\mathrm{tvf}/\mu_1^P$ agree as booleans on all 192 (no search is run — T165's ascent stays sandbox), so the demand half of R2$''$ and the gauge-invariant crossing condition are *one* predicate — R-F is strictly stronger than the R2$''$ demand and its m-uniform form *is* R-E-A — and tvf ≥ 1 pointwise, hence every crossing vector pays $P_{\mathrm{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4$–$142.8$, above the preregistered tolerance $\Theta_{\mathrm{TOL}} = 10$ on 12/12 windows: the R-F conjunction is *empty* on this surface by an identity plus a floor; mutation: the price with $(t_1 v)$ unsquared scales like t and moves by 1.0–1.0×10^4 under the same round trips. *(5)* The Schur-cascade direction and the exponent closure (T164/T165): $s = \mu_1^P t_1^{\mathsf{T}} A^{-1} t_1 = 0.1696$–$0.2472 \ge g_{16} \ge g_1$ window by window ($1/s \le 1/g_{16} \le 1/g_1 = B_{11}$; Schur 1917–Haynsworth 1968), $P_K = \hat{a}\, s = 254.8$–$7.7 \times 10^3 \ge 1$ (Kantorovich 1948); the exponent closure is an identity of the lists — per instance $\log(1/g_1) = \log(g_{16}/g_1) + \log(1/g_{16})$ to 1.6×10^{-15}, the least-squares fits closing to 1.3×10^{-15} by linearity (the exponent *values* are sandbox fits, not consumed — what is promoted is that the whole certification lives in the cascade gain g_{16}/g_1, the object of the one open quantifier); mutations: the g_8 closure breaks by 0.10–0.24, a corrupted sixteen-block makes the Cholesky *refuse* on 12/12. Stress: the monotone no-go staircase ($1.00 \to 3.49 \to 10.81 \to 24.98$, the $\nu = 4$ column an exact null control); the Dirichlet cosine-sum identity (2.9×10^{-14}, including the degenerate branch), the KMS eigenpairs (1.4×10^{-14}) and odd_toeplitz$(c^L) = L_P$ (zero tolerance) are exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D; the T164/T165 h-exponents and the measured anatomy (the free ascent, the η(Cap) profile, the decoupled ν surface with its zone-depth finding, the retired constant $U_{\mathrm{ref}} = 4.90 \to 5.7327$) stay in the diary and the paper; finite Λ-sums are allowed and used, *no* RH statement is made ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required depth is compared against); **the one open object after T165 stays open, typed open**: $\inf_m g_{16}(m) > 0$, equivalently a lower bound on the sixteen-step Schur-cascade gain g_{16}/g_1 uniform in m — a quantifier over a certified flat list; R-B$'''$ (an independent α/h surface) stays open as narrowed by T164. Fences: Schur 1917–Haynsworth 1968 / KMS 1953 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Kantorovich 1948 / Fejér 1915 / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v556_gauge_ppr_identities.py`]. **[E]** (per-instance identities and certified inequalities)

The cascade/vector identities (`PRIME.CASCADE.VECT.01`). The closed theorem cores of the discovery parts T166 and T167, recomputed on the same declared surface as `PRIME.GAUGE.PPR.01` (the 12 deepest in-cap frame-A windows, $n = 29 \ldots 139$, $m = 96 \ldots 291 \le 300$), with the one arithmetic input $\kappa = 0.038821$ *verified* at every jump point of $\psi(x)/x$ on the von-Mangoldt table in $[100, 3.2 \times 10^5]$ (each statement with at least one mutation control that must fail loudly) — the *anatomy of the cascade* and the *exactness of the closed vector at* $K = 2$, and it does not close the one remaining scalar. *(1)* The cascade identities (T166): the prefix property — one

Cholesky of the sixteen-block yields all sixteen rungs, $g_K = e_1^{\mathsf{T}} Q_K^{-1} e_1$ at every K (192 comparisons, 1.1×10^{-12}); the minor form $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$ on the raw arithmetic Gram block $G = TAT^{\mathsf{T}}$ (2.7×10^{-12} on 36 (window, K) pairs, $K \in \{2, 6, 16\}$; Schur 1917–Haynsworth 1968); the regression form $g_K/g_1 = 1/(1 - R_K^2)$ with R_K^2 the squared multiple correlation of mode 1 on modes $2 \ldots K$ (8.8×10^{-13}) — the open object *is* a near-collinearity statement; mutation: the minor with the wrong row/column deleted misses by 0.99–1.45. *(2)* The diagonal invariance (T166): the gain $g_K/g_1 = B_{11}(Q_K^{-1})_{11}$ is unchanged under $B \to DBD$ on 144 random positive diagonals (2.7×10^{-12}) — the cascade does not see the KMS $1/\sqrt{\mu}$ normalisation, the gain is a property of the arithmetic Gram block alone; mutation: a non-diagonal congruence moves the gain by 2.7–3.8. *(3)* The exact $K = 2$ identity (T167): the closed vector $u = (1, -Q_{21}/Q_{22})$ *attains* the constrained minimum, $u^{\mathsf{T}} Q_2 u = 1/g_2 = Q_{11}(1 - r_{12}^2)$ to 1.1×10^{-12}, and $B_{11} g_2 = 1/(1 - r_{12}^2)$ identically to 1.2×10^{-12} (measured $1 - r_{12}^2 = 2.8 \times 10^{-4}$–$5.9 \times 10^{-3}$ on this surface, the h-trend a sandbox fit): at $K = 2$ the fast-null-vector needs no approximation — the third dress collapses onto the second, one scalar built from three closed lag sums; mutation: the wrong-sign vector pays the relative excess $4r_{12}^2/(1 - r_{12}^2) = 6.7 \times 10^2$–$1.5 \times 10^4$. *(4)* The pivot identity and the unification (T167): for $u = x_K^* + \delta$ with $\delta_1 = 0$, $u^{\mathsf{T}} Q_K u = 1/g_K + \delta^{\mathsf{T}} Q_K \delta$ exactly (2.0×10^{-12} on 252 evaluations across three δ scales) — every relative excess is an exact number; the Rayleigh envelope holds with its $\delta_1 = 0$ ceiling attained exactly (1.1×10^{-15}) and Cauchy-interlaced; the sign-aligned entrywise perturbation attains $|x^{*\mathsf{T}} E x^*| = \varepsilon S_K$ exactly (3.8×10^{-16}), so the entry threshold $\rho_{\text{carry}}(1/g_K)/S_K$ is exact — and the unification: $(1/g_K)/S_K$ is the one cancellation ratio of all three dresses, at $K = 2$ literally $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$ (1.2×10^{-12}; the closed factor $1 + 3r_{12}^2 \to 4$ is T166's independently derived quarter): *one* inequality, not three; mutation: $\delta_1 = 0.1$ breaks the pivot identity by exactly the closed cross term $2\delta_1 = 0.2$. *(5)* The trial theorem and the two must-breaks (T166/T167): all 288 random trials with $u_1 = 1$ satisfy $u^{\mathsf{T}} Q_K u \geq 1/g_K$ (min ratio 15.14) — every closed candidate is a certified *lower* bound (Schur 1917); the L_P no-go: the normalised block is the identity (1.6×10^{-15}) and the cascade gain exactly 1 (deviation 0.0) at $m = 64/128/256$; the scramble type-change: uniform random atom positions at the same total mass make B_{11} itself lose positivity on 49/60 declared draws and on a majority of draws on 12/12 windows — at $K = 2$ the collapse under scrambling is a change of *type*, not a factor. Stress: exact Dirichlet/parity controls (2.3×10^{-14} / 1.4×10^{-14} / zero tolerance). **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D; the T166/T167 h-exponents ($1 - r_{12}^2 \sim h^{-2.921}$ on frame A, the certified gain exponent $h^{+2.921}$, the threshold monotonicity in K, the $K = 6$ separation $h^{+1.138}$) and the measured anatomy (the rung profile, the polarisation split, the Kato radius, the headroom) stay in the diary and the paper; finite Λ-sums are allowed and used, *no* RH statement is made ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required depth is compared against); **the one open object after T167 stays open, typed open**: R1, the single scalar — an m-free upper bound on $1 - r_{12}^2 = 1 - Q_{12}^2/(Q_{11}Q_{22}) \leq C\, h^{-3+\varepsilon}$, three closed lag sums and nothing else; the routes T167 closed off (perturbation theory of every order around the KMS bottom mode — the Kato series converges to the *wrong* object; position-blind surrogates) stay sandbox negatives; R-B‴ stays open as narrowed. Fences: Schur 1917–Haynsworth 1968 / KMS 1953 / Rayleigh 1877–Courant–Fischer 1920 / Cauchy 1829 / Kato 1949 (cited as the closed-off route, never used) / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v557_cascade_vector_identities.py`]. [E] (per-instance identities and certified inequalities)

The bilinear/rank identities (`PRIME.BILINEAR.RANK.01`). The closed theorem cores of T168 (contract `LAGRANGE.MINORS`), T169 (contract `TSTAR.RATIO`) and T170 (contract `BILINEAR.SIEVE`) as one deliberately narrow module (27 checks, $\sim$0.2 s), recomputed on the declared small frame-A surface (12 deepest in-cap zones, $m = 96 \ldots 291 \leq 300$; $\kappa = 0.038821$ verified at every jump point of $\psi(x)/x$ on the table) — the *exact shape of the one remaining scalar*, and it does not close that scalar. *(1)* The Lagrange/Wronskian anatomy (T168): the lag-sum identity $\hat{a}_{ij} =$

$t_i^{\mathsf{T}} A_h t_j = \sum_d c_d W_d^{ij}$ with the closed correlation weights (1.6×10^{-15}; the $1/\sqrt{\mu}$ factors cancel, T169-TH3), Euclidean orthogonality $t_1 \cdot t_2 = 0$ (4.4×10^{-16}) — the near-parallelism is created *entirely* by the arithmetic metric — the Wronskian telescope $(\mu_1 - \mu_2) u_r v_r = W_{r-1} - W_r$ with $W_{-1} = W_{h-1} = 0$ (1.9×10^{-17}) and the maximal-minor identity $\frac{1}{2} \sum M^2 = 1/(\mu_1 \mu_2)$ exactly (4.6×10^{-16}; Lagrange 1773); mutation: the wrong eigenvalue pair misses by 1.66–1.67 relative (the closed miss $(\mu_2 - \mu_3)/(\mu_1 - \mu_2) \to 5/3$). *(2)* The exponent account and the Gershgorin certificate (T168/T169): $1 - r_{12}^2 = \nu_1 \nu_2 / (\hat{a}_{11} \hat{a}_{22})$ identically (2.5×10^{-13}; measured $1 - r_{12}^2 = 2.8 \times 10^{-4}$–$5.9 \times 10^{-3}$ on this surface, the h-trend a sandbox fit); $\nu_1 \leq \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$ (Gershgorin 1931, unconditional, closed in the three entries) on 12/12 with loss 1.1885–1.1947; mutation: dropping $|\hat{a}_{12}|$ fails on every window ($\nu_1 - \max \text{diag} = 0.30$–$1.17 > 0$). *(3)* The det-collapse theorem (T169-TH7): on the Rayleigh family $u(t) = (t, -1)$ the candidate $K7 = \sqrt{\hat{a}_{22}/\hat{a}_{11}}$ collapses in closed form, $R(K7) = 2\sqrt{\hat{a}_{11}\hat{a}_{22}} \det \hat{A} / [(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$ (8.7×10^{-13}) — it meets the threshold precisely because it reintroduces $\det \hat{A}$: the loop T167 scalar → T168 factor → T169 candidate closes as an identity; usable form $\nu_2 \leq 2 \det \hat{A} / (\hat{a}_{11} + \hat{a}_{22}) = 2\nu_1\nu_2/(\nu_1 + \nu_2)$, tight to 1.9980–1.9999 < 2 exactly (Rayleigh 1877); mutation: the closed constant $1/\sqrt{2}$ overshoots ν_2 by 17.1–2.3×10^2. *(4)* The exact bilinear von Mangoldt form (T170 TH1–TH3): the exact split $\hat{A} = B - S$ with $S_{ij} = \sum_{n \leq X} (\Lambda(n)/\sqrt{n}) X_n[ij]$ against the direct block (1.7×10^{-14}); the polarisation $\det \hat{A} = \det B - D(B, S) + \det S$ (1.0×10^{-10}); the double sum $\det S = \sum_{n,m \leq X} \Lambda(n)\Lambda(m) K(n, m)/\sqrt{nm}$ with $K = \frac{1}{2} D(X_n, X_m)$ through the *explicit* kernel matrix (7.5×10^{-16}; Cauchy–Binet 1812–1815); the wedge reading to the measured rank-one defect 4.6×10^{-3}–3.1×10^{-2} (honest certificate); mutation: the wrong cross-coefficient breaks by 64–430. *(5)* The rank-3 theorem and the Type II collapse (T170 TH4–TH5): the polarisation matrix on (X_{11}, X_{22}, X_{12}) has eigenvalues $\{-2, -1, +1\}$ exactly — rank 3, signature $(1, 2)$ — so the kernel $K = PMP^{\mathsf{T}}$ has $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$ with nonzero spectrum $= \operatorname{spec}(M\, P^{\mathsf{T}} P)$ (1.0×10^{-15}): the bilinear form *is* the rank-3 polynomial $S_{11}S_{22} - S_{12}^2$ in three linear Λ-sums; Vaughan's identity coefficient-exact for every $U = V \in \{3, 4, 5, 6\}$ (residual 0.0; Vaughan 1977), Type II kernel 99.9%-energy rank $[2, 2, 2, 2]$ against generic $[20, 15, 11, 9]$ (4.5×–10×; Montgomery–Vaughan 1973 cited as the priced route). *(6)* The R4-free chain and the two discriminators (T170-TH6): $1 - r_{12}^2 = \det \hat{A} / (\hat{a}_{11}\hat{a}_{22})$ is an identity (0.0) — no positivity of A_h, the Weil fence never approached; the L_P no-go: $\hat{a}_{12} = 1.1 \times 10^{-17}$ identically, no decay; the scramble discriminator: rank-3 *untouched* on all 60 draws while the value of $1 - r_{12}^2$ moves by a median factor 834.8 (range 69–1.8×10^4) — the arithmetic sits entirely in the joint values of (S_{11}, S_{22}, S_{12}) against B. Stress: exact Dirichlet/parity controls (1.2×10^{-14} / 1.4×10^{-14} / zero tolerance). **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D; the T168–T170 h-exponents ($1 - r_{12}^2 \sim h^{-2.92}$ on the deep frame-A surface, the route-table deltas — large sieve +0.669, MV −0.384, exact split +0.996, Vaughan +0.669 against the measured truth 2.747 — the five-order cancellation and the $h^{-2.711}$ row-angle trend) and the measured anatomy stay in the diary and the paper; finite Λ-sums are allowed and used, *no* RH statement is made ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required depth is compared against); **the one open object after T170 stays open, typed open**: R1, now finally *classified* but not closed — an m-free unconditional certificate that the two rows of $\hat{A}$ become collinear at rate $h^{-3+\varepsilon}$, equally a joint near-degeneracy of the three explicit linear Λ-sums (S_{11}, S_{22}, S_{12}) against the archimedean block — a rate statement the module neither makes nor approaches; the T168/T169/T170 verdicts (MINORS-RESIST / RATIO-RESISTS / SIEVE-RESISTS) are sandbox verdicts. Fences: Lagrange 1773 / Cauchy–Binet 1812–1815 / Gershgorin 1931 / Rayleigh 1877–Courant–Fischer 1920 / Montgomery–Vaughan 1973–Vaughan 1977 (the priced route) / KMS 1953 / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v558_bilinear_rank_identities.py`]. **[E]** (per-instance identities and certified inequalities)

The phase-2 capstone (`PRIME.PHASE2.CAPSTONE.01`). Promotion of discovery T171 (contract `FINAL.MAP`, verdict MAP-COMPLETE) as one module (35 checks, ∼0.3 s): *the capstone verifies* *the map and claims nothing new* — the whole sixteen-link reduction chain of phase 2 as one machine-checked pass, the eight-route no-go battery, and the precision ledger, recomputed on the declared small frame-A surface (12 deepest in-cap zones, $m = 96 \dots 291 \le 300$; $\kappa = 0.038821$ verified at every jump point of $\psi(x)/x$ on the table; the 16×16 low block positive definite on 12/12 — a numerical fact, never routed through the Weil criterion). *(1)* The chain, 13 THEOREM / 3 CERT: T152 Schur two-block (KMS 1.3×10^{-15}; holds at $0.5\lambda_{\min}$, fails at $1.02\lambda_{\min}$ — sharp; Haynsworth 3.8×10^{-15}); T151 Sobolev $b_k \le C_S k$, $C_S = (2\pi + 2)/\sqrt{N}$ (worst ratio 0.4816); T153 Ψ-collapse $\Psi = \sum_d c_d w_d = x^\mathsf{T} G x$ (6.3×10^{-16}; $\sum_d w_d = 0$ to 9.0×10^{-17}); T154 Ritz ceiling (interlacing; $g_{32}/s = 0.9520$–0.9849 from below); T155 12×12 floor [CERT] (holds at $(1 - 10^{-9})\lambda$, fails at $(1 + 10^{-4})\lambda$; $t = 0.2295$–0.3719); T156 $F(P, r)$ [CERT] ($P_K \ge 1$, $r_K \ge 0$, $L(16) = 6.6$–15.6); T158 Thomson/ladder (3.1×10^{-12}; increments strictly positive; 36/36 trials below); T160 sampling (closed Dirichlet weights = correlation weights 1.1×10^{-15}; atom half = sampled Λ-mass 9.9×10^{-15}); T163 TV floor/swap (Abel 1.8×10^{-11}; overshoot $\times 5.2 \times 10^4$–1.1×10^6); T164 gauge/quantifier (invariance 2.2×10^{-11}; $\Psi(x^*) = 1/g_{16}$ to 7.0×10^{-12}; 48/48 perturbed larger); T165 P_{pr} (1.1×10^{-16}; $= 1$ at x^*); T166 cascade $1/(1 - R^2_{16})$ (1.8×10^{-12}; diagonal-invariant); T167 $K = 2$ exactness (6.8×10^{-13}; milder $\times 1.62$–5.20); T169 det collapse R4-*free* ($\le 4.2 \times 10^{-12}$; no positivity of A_h consumed, the Weil fence never approached); T170 rank-3 (J-eigenvalues $\{-2, -1, +1\}$ exact; $\sigma_4/\sigma_1 \le 1.6 \times 10^{-16}$); T170 R1 shape [CERT] (the 2×2 Lagrange identity 3.6×10^{-16}; $\sin\angle = 1.169 \times 10^{-4}$–$2.467 \times 10^{-3} \le 10^{-2}$ per window; rate and quantifier OPEN). *(2)* The no-go battery, 8/8 failing as classified: size budget $\times 2.7 \times 10^3$–1.6×10^4; symbol infimum -47.6–$-21.3 < 0$ on 12/12; sector change an invariance ($\le 4.2 \times 10^{-12}$); perturbation $\times 9.8 \times 10^2$–2.0×10^4 off scale (Kato 1949, the closed-off route); atom band dominates $\times 1.71$–2.83; sieve operator-norm on $\det S$ alone $\times 8.8 \times 10^3$–7.5×10^4 over (by the rank-3 theorem a bound on $\det S$ alone cannot see the three-term cancellation; Montgomery–Vaughan 1973 / Vaughan 1977 the priced route); scramble $\times 679$ median (range 68–2.3×10^6) with rank-3 *untouched* on all 60 draws; L_P gain $= 1$ to 0.0. *(3)* The precision ledger: needed 2.752×10^{-4} at the deepest window ($m = 285$) — finer than the RH yardstick $h^{-0.5}$ by 215× and the best unconditional $h^{-0.996}$ by 13× on this surface; both self-similarity identities (6.8×10^{-13} / 3.1×10^{-13}, the t^* overshoot ≥ 0 everywhere, sign-agnostic). Mutations fail loudly (wrong regression row, wrong Thomson entry, off-by-one swap, wrong polarisation coefficient — the 10^{-2} bar missed by orders). **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in D; the T171 trend fits (pooled sin-angle $h^{-2.33}$, frame-split $\nu = 4$: $h^{-1.86}$ / $\nu = 6$: $h^{-2.87}$, the no-go growth exponents, $\delta = +1.044$ against 3.0, the full-surface precision numbers 2.284×10^{-7} / $1.2 \times 10^5\times$ / $3.1 \times 10^3\times$ at $h = 1444$) are sandbox numbers on T171's two-frame surface and are *not* consumed; **zero new uniform-in-m statements are added — which is exactly why the capstone is promotable**; **the one open phase-2 object stays open, typed open**: R1 — an m-free unconditional certificate that the two rows of $\hat{A}$ become collinear at rate $h^{-3+\varepsilon}$, a joint near-degeneracy of the three explicit linear Λ-sums against the archimedean block; R2/R3 booked (T169), R4 off the path (T170-TH6); the T171 verdict MAP-COMPLETE is a verdict about the *map*, not about R1, and stays a sandbox verdict ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required precision is compared against). Fences: Schur 1917–Haynsworth 1968 / KMS 1953 / Cauchy 1829–Courant–Fischer 1920 / Rayleigh 1877 / Abel 1826 / Dirichlet 1829 / Lagrange 1773 / Cauchy–Binet 1812–1815 / Gershgorin 1931 / Kato 1949 / Montgomery–Vaughan 1973–Vaughan 1977 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Fejér 1915 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v559_phase2_capstone.py`]. **[E]** (per-instance identities and certified inequalities)

The frame/deficit identities (`PRIME.FRAME.DEFICIT.01`). Promotion of the theorem cores of discoveries T172 (contract `FRAME.BEYOND`, verdict PARTIALLY-PORTABLE), T173 (`FRAME.RATE`, DEFICIT-VARIES) and T174 (`CANCEL.IDENTITY`, PARTIAL-CANCEL) as one module (24 checks, ∼0.5 s): *the gauge route of phase 2, closed as algebra* — the companion to the capstone above, certifying what T172/T173/T174 established *about* the map's frame dependence, everything recomputed on a small declared surface (a 4×3 (anchor, h) rectangle of gap-blind frame-B cells, prime-power anchors $n = 41 \ldots 557$ crossed with the declared ladder $h = (160, 226, 320)$, comb complete on every cell, positive definite on 12/12; one non-prime-power instance $n = 210$; one truncated-comb stress instance $n = 1009$; $\kappa = 0.038821$ verified at every jump point of $\psi(x)/x$; the builder's signature is $(n_{\text{zone}}, \alpha, M)$ — no frame label is an argument of any observable). *(1)* The identity transfer + the sieve-horizon discriminator: the KMS/Schur pair (3.2×10^{-16}; quadratic-form pair 5.2×10^{-15}), the lag-sum / correlation identity (1.4×10^{-16} of the larger half; Abel 1826), the R4-free det collapse (at 0.020 of the conditioned bar $64\varepsilon/|1 - r_{12}^2|$) and the additive split (7.6×10^{-16}), re-derived *exactly* on a gap-blind frame-B instance *and* a non-prime-power-anchor instance — the links use no property of the frame rule or the anchor arithmetic, which is why they transfer; the truncated-comb instance turns *indefinite* ($\lambda_{\min} = -9.6 \times 10^4$) against 12/12 positive definite complete-comb cells, and the identity links keep holding there (none uses positivity): indefiniteness sits at the declared sieve horizon, not at a frame or zone. *(2)* The $q = 1 - s$ identity + the calibration: $hD = \alpha$ per cell (8.9×10^{-16}) forces the OLS slopes to obey $q = 1 - s$ to 2.9×10^{-14} (OLS is linear in its response); the gauge-degree account comes out as integers — $G2 = t/\lambda_{\max}(B_{LL})$ and the delivery $1 - r_{12}^2$ both $(d_{\text{amp}}, d_\mu) = (0, 0)$, $G1$ at $d_{\text{amp}} = 1$, $G3$ at $d_\mu = 1$ — so $G2$ is the unique doubly gauge-invariant arithmetic member of T173's family; $G3 = \mu_1^P G2$ identically (2.2×10^{-16}; exponent $h^{-1.9955}$, the offset-2 explained); $(D/\alpha)^3 h^3 = 1$ a tautology (2.2×10^{-16}). *(3)* The blindness theorems + the additive limit: the amplitude $c \to \kappa c$ is a gauge of GAP, of $1 - r_{12}^2$ and of R over twelve decades (worst 0.026 of the per-cell conditioned bar; rebuilt from scratch 0.098); the scalar $\mu_1^P = 4 \sin^2(\pi/(2h + 1))$ — *the entire h^{-2} channel of the demand* — cancels exactly on both sides (0.001 of bar); the delivery is invariant under the full positive diagonal group (48 congruences spanning $e^{\pm 8}$, worst 0.036 of bar) while a non-scalar diagonal moves the demand by 0.23–1.40 (the asymmetry is the anatomy); neither the arch term nor the comb term alone has a positive definite low block on a single cell (0/12 + 0/12 against 12/12 for the sum): no factorisation $\hat{A} = (\text{frame}) \times (\text{arithmetic})$ exists, so T173's P6 is *not* promoted to a theorem — only its multiplicative half. *(4)* The KMS shape band (unconditional, declared $\varepsilon = (K\pi/(2h + 1))^2/6$): $|\sqrt{\hat{S}_k/k^2} - 1| \le \varepsilon_{\text{cert}}$ on 12/12 and the k^2-swap moves $\log \text{GAP}$ by ≤ 0.312 of the certified band $2 \log((1 + \varepsilon)/(1 - \varepsilon))$ (Rayleigh 1877; Schur 1917) while the delivery does not move at all. *(5)* The placement discriminator: the value-multiset scramble destroys the positive low block on 24/24 declared draws and moves $|R|$ by a median factor 10^6. Mutations fail loudly (the $d = 0$ halving dropped, the $N = 2h$ ladder, the off-by-one h, the comb-only rescale, the wrong universal profile). **Named limits as load-bearing content:** per instance on a small declared surface — a finite list with an explicit maximum; nothing uniform in the zone index or in D; **the frame-free deficit is a measurement, not a claim of this row**: T174's $+0.1111 \pm 0.0222$ (5.0σ, 151 anchor clusters, cluster-robust; 0.4σ from T173's $+0.155 \pm 0.102$) stays a sandbox MEASURED number, as do all T172/T173/T174 h-exponents, the density anatomy and the ν/density collinearity; **the one open phase-2 object stays open, typed open**: R1, now stated frame-free — an m-free unconditional certificate that two explicit finite Λ-sum vectors become collinear at the demanded rate; the residual per-anchor scatter ($\chi^2/\text{dof} = 3.02$) and the sparse corner are sandbox items under test in T175 ($\frac{1}{2}$ appears in exactly one role: as the strength of a classical statement a required precision is compared against). Fences: Schur 1917 / KMS 1953 / Dirichlet 1829 / Rayleigh 1877 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Wilkinson 1968–Higham 2002 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [`verification/v560_frame_deficit_identities.py`]. **[E]** (per-instance identities and certified inequalities)

Lemma 10 (G3–G4 — reflection positivity, fixed and integrated)**.** *For each Θ-invariant g in the Calderón ball the physical boundary kernel $\mathcal{K}_g = P_{\text{adm}} \Pi_{\text{phys}} \mathcal{N}_g^{TT} \Pi_{\text{phys}} P_{\text{adm}}$ is reflection positive, $\langle \Theta F, F \rangle_{\mathcal{K}_g} \ge 0$ (G3); and $d\mu_{\text{Cal},\chi}$ is Θ-invariant, supported where G3 holds uniformly, so $\int \langle \Theta F, F \rangle_{\mathcal{K}_g} \, d\mu_{\text{Cal},\chi} \ge 0$ (G4). **G4 is the decisive step:** RP after integrating over dynamical metrics, not merely at fixed background.*

Lemma 11 (G5 — gap dominance [E]). *$2\|V_{\rm metric}\|_{\rm rel} < \Delta = 6\log\frac{3}{2}$, with $\|V_{\rm metric}\|_{\rm rel} \le 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$, hence $2\|V\| = 0.7852 < 2.4328$ and $\Delta_{\rm eff} \ge \Delta - 2\|V\| > 1.647 > 0$. The numerical inequality is trivial ([verification/v29_research_contract_certs.py]); the work is the operator-norm bound (hard).*

Sugawara reading + atomic OS moment ([verification/v171_os_moment_cluster.py]). The G5 budget is the E_8 Sugawara constant: $248c_3^2 = \frac{31}{8\pi^2}$ with $31 = 1 + h^\vee(E_8)$ and $c((E_8)_1) = 248/31 = 8$, so gap dominance reads $|R^+(A_3)|\log(N_{\rm fam}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$ — the A_3 transfer gap beats the E_8 Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*: $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ makes every two-point correlator $C(n) = A + B(64/729)^n + C(1/729)^n$ the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant 729/665, correlation length $\xi = 1/(6\log\frac{3}{2}) = 0.4111$).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full $G_{\rm metric}$). *If the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated, $2\|V_{\rm metric}\| < \Delta = 6\log\frac{3}{2}$ (G5),* then *the admissible IR measure is stable under the ambient completion: a strictly positive effective gap $\Delta_{\rm eff} = \Delta - 2\|V\| > 0$ protects the admissible sector from the (un-built) ambient/deep-UV, so every* physical low-energy *read-out (masses, mixings, α^{-1}, the $R+R^2$ regime) is independent of the open ambient measure. Symbolically*

$$\boxed{\mathrm{RP} \;+\; 2\|V\| < \Delta \;\Longrightarrow\; \textit{admissible IR measure stable under ambient completion}\,.}$$

Thus ($G_{\rm metric}$) is not *a diffuse "full quantum gravity missing" block: it is the sharply delimited* ambient projective limit *(G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on.* ***Typing:*** *the inequality is machine-checked [E]; the operator-norm bound on $\|V_{\rm metric}\|$ is the conditional input.*

Lemma 12 (G6–G7 — projective limit & Ward identities). *A projective system μ_{χ_n} with $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$, uniform moment bounds $\sup_n\int\|g\|^p_{H^s}\,d\mu_{\chi_n} < \infty$, tightness and regulator-independence yields $\mu_{\rm QG} = \lim\mu_\chi$ on $\mathcal{G}/\mathrm{Diff}$ (G6, the hardest step); the limit obeys the BRST Ward identities $\langle s\mathcal{O}\rangle = 0$, whence $\nabla^\mu T_{\mu\nu} = 0$ and the η boundary phase cancels the APS anomalies (G7).*

Gate 2 reduction: G6 is a *boundary* projective limit, not a bulk one ([verification/v76_gmetric_reduction.py])

The exact decoupling margin is $\Delta_{\rm eff} = \Delta - 2\|V\| = 6\log\frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\rm metric}\|_{\rm rel} \le 248\,c_3^2 = \dim(E_8)\,c_3^2 = \frac{31}{8\pi^2}$ $(31 = 2^{g_{\rm car}} - 1)$. **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so *G6* is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\mathrm{RP(seam\text{-}Calder\acute{o}n\ boundary\ kernel)} + \mathrm{tightness} \;\Longrightarrow\; G_{\rm metric}\ \mathrm{closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — now discharged as a [C] redundancy (v369, keybox below). So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) a [C] redundancy, no longer a diffuse "full QG missing". [C]

The measure as one-loop fluctuations around the *parameter-free* saddle ([verification/v365_qg_oneloop_saddle.py], QGAMB.SADDLE.01)

With the saddle now parameter-free (v358/v359), the boundary measure of Tier B organises as a Gaussian one-loop fluctuation determinant whose ingredients are atom-fixed. **Fixed quadratic form [E]:** the fluctuation inverse-propagator stiffness is the saddle coefficient $1/c_3 = 8\pi$ — no free dimensionless dial, so the one-loop problem is *well-posed* (it was not before v358). **Gap-controlled convergence [E]:** the fluctuation operator obeys $M \ge \Delta_{\rm eff} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337), so the Gaussian integral converges mode-by-mode with no zero/negative modes besides the saddle directions; on the OS spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$ the regularised one-loop log-det $\mathrm{Tr}'\log M = 6\log\frac{9}{2}$ is finite. **Conformal direction** [C]**:** the one wrong-sign mode ($c_{\rm conf}(4) = -\frac{3}{2}$) is made convergent by the GHP contour ($\int e^{-|c|\phi^2} = \sqrt{2\pi/3}$) with IDG dressing (v334). **Tightness** [E]**:** $\chi = 1/(1-(2/3)^6) = 729/665 < \infty$ (v330), so the one-loop measure has a projective limit on the admissible sector, with *zero* new dimensionless dials (v364). **Residual:** the full *non-perturbative* projective limit (G6 on the ambient sector) is now discharged as a [C] redundancy (v369, keybox below) — this sharpens C7 from "diffuse full QG" to a one-loop-controlled Gaussian problem around a parameter-free saddle and then to a certification object. [E]/[C]

The holographic route: the ambient measure is boundary-*redundant*, not missing dynamics ([`verification/v369_qgamb_redundancy.py`], `QGAMB.REDUNDANCY.01`)

There is a second, strategically stronger way to dispatch Gate 2 than building the measure: prove that the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the measure `QG.AMB.01` is a non-fundamental *certification* object — the role a coordinate choice plays in general relativity — rather than a missing piece of dynamics. The discriminators that make this *ambient-redundancy* statement well-posed are all in hand. **Trivial superselection** [E]**:** the number of DHR sectors of a chiral net is $|\det \mathrm{Cartan}|$, which is 1 for E_8 (one primary, the vacuum; $\mathrm{DHR}((E_8)_1) = \mathrm{Vec}$) against 4 for the same-c rival $SO(16)_1$ — a holomorphic boundary net has *no* nontrivial superselection charge that a bulk could carry invisibly. **No torus degeneracy** [E]**:** the ground-state degeneracy on T^2 equals the number of anyons $= |\det K| = 1$ for E_8, so there is no topologically protected bulk d.o.f. hidden from the boundary (the genus-1 obstruction of v344, in its redundancy reading). **Bounded reconstruction** [E]**:** the seam transfer deviation subspace contracts at the finite gapped Petz rate $(2/3)^6$ (v221), and the gap-decoupling margin $\Delta_{\rm eff} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) separates the un-built ambient sector from the admissible one. **The statement** [C]**:** with the Bisognano–Wichmann modular reconstruction map (v258/v309/v323), under `SEAM.EQUIV.01` every gauge-invariant bulk observable is boundary-reconstructible, $\mathcal{O}^{\rm bulk}_{\rm phys}/\mathrm{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$, and two ambient measures sharing the same boundary state are physically equivalent. **Residual** [O]**:** the construction consumes exactly two premises — `SEAM.EQUIV.01` (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct `QG.AMB.01` but supplies the *alternative* route, so that Gravity-complete = Boundary-complete + Ambient-redundancy. [E]/[C]/[O]

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]: entanglement equilibrium gives the linearised Einstein equation with G from c_3 ([`verification/v358_grav_entropy_equilibrium.py`])

While Gate 2 (above) reduces the metric-sector *measure*, the classical metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms. **Parameter-free coefficient** [E]**.** Equilibrium gives $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed*: all four c_3-roles are mutually consistent — the Unruh factor $1/(2\pi) = 4c_3$, the EH action $1/(16\pi) = c_3/2$, the Einstein coefficient $8\pi = 1/c_3$, the Bekenstein density $1/4 = 1/|\mu_4|$ — so *no free dimensionless Newton dial* enters. **Thermo = geo coincidence** [E] **(new).** The first law requires the coefficient $2\pi/\eta$ (η the entropy density); the geometry gives $1/c_3 = |\mathbb{Z}_2|\, 2\pi\, \chi(S^2)$ (one-sided Gauss–Bonnet, v58); these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\,\chi(S^2) = 4$ — a TFPT atom identity (v216). So the thermodynamic (first-law) and geometric (Gauss–Bonnet) origins of c_3 are *one*. **The matter side, assembled** [C] **(closing J3).** The ball modular Hamiltonian is the boost (Casini–Huerta–Myers; geometric by Bisognano–Wichmann, v323), $K_B = 2\pi\int_B w\, T_{00}$, $w = (R^2 - r^2)/(2R)$; the 3-ball weight integral is $\int_B w\, d^3x = 4\pi R^4/15$, so $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\,\delta\langle T_{00}\rangle$ — the concrete matter boost flux of TFPT's stress tensor. **The entropy density, atom-fixed** [E]**.** The Bekenstein numerator $1/4 = 1/|\mu_4|$ (v57) and the entanglement normalisation = the central charge $c = g_{\rm car} + N_{\rm fam} = 8$ pin the dimensionless coefficient; the area-law form is established (v59). **Residual:** the full covariant extension is now *closed* (v359, next keybox); what remains is only (i) the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) [O] and (ii) the absolute scale $v_{\rm geo}$ (the Planck-area unit) [O], while the global ambient measure (Gate 2/C7) is now a [C] redundancy (v369) — *no free dimensionless parameter survives in the gravity coupling.*

And that boundary measure is a *rigorously constructed* net ([`verification/v77_e8_conformal_net.py`]). The level-1 central charges $c = \dim\mathfrak{g}/(1+h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \mathrm{rank}\, E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5)+c(A_3) = 5+3 = 8 = c(E_8)$

is a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ *E_8-lattice* VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the $(E_8)_1$ lattice net* (carrier = $(D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\text{eff}} > 0$ supplies the clustering/tightness — so $G6$ is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

The full covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]: fixed-volume equilibrium gives the Einstein tensor with conservation as an output ([`verification/v359_grav_nonlinear_einstein.py`])

The linearised result above (v358) extends to the *full* covariant equation by Jacobson's *fixed-volume* (maximal-vacuum-entanglement) equilibrium — no linearisation assumed. **Einstein tensor, not Ricci [E].** Stationarity of the entanglement entropy at fixed volume forces the divergence-free combination: the 4D trace identity $g^{ab}G_{ab} = (1 - \frac{d}{2})R = -R$ (for $d = 4$) selects $G_{ab} = R_{ab} - \frac{1}{2}R\, g_{ab}$, so

$$\boxed{G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}, \qquad c_3^{-1} = 8\pi}$$

with *both* coefficients TFPT-fixed: the Newton coupling $8\pi = 1/c_3$ (v358) and the cosmological constant Λ from α, $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60). **Conservation is an output [E].** By Lovelock's theorem the only divergence-free, second-order, local symmetric tensor built from the metric is $a\, G_{ab} + b\, g_{ab}$, so $\nabla^a G_{ab} = 0$ forces $\nabla^a T_{ab} = 0$ — matter conservation is *derived*, not assumed. **So Gate B6 — the classical metric-sector field equation — carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] at the *local* level. Residual:** the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) and the single unit v_{geo} stay [O], while the global ambient measure (Gate 2/C7, the constructive boundary projective limit) is now a [C] redundancy (v369) — no free dimensionless gravity dial remains. An *external candidate* for the missing action level is quantified in the next keybox (v473); it does not change this typing. [E]/[C]

The entropic-action bridge: an external candidate for the missing action level, quantified — nothing closes ([`verification/v473_entropic_action_bridge.py`])

The one honest [O] left in the classical sector — "equation of state, not a from-action quantisation" (v359) — now has an explicit *external* candidate: Bianconi's entropic action $S_{\text{B}} = -\operatorname{Tr}_F \ln\big(\widetilde{G}\widetilde{g}^{-1}\big)$, the quantum relative entropy between the spacetime metric and the matter-induced metric (G. Bianconi, *Gravity from entropy*, Phys. Rev. D **111**, 066001 (2025); `arXiv:2408.14391`; her Appendix C identifies $\Delta^{1/2}_{\widetilde{G},\widetilde{g}} = \widetilde{G}\widetilde{g}^{-1}$ with an Araki-type modular operator) — a local covariant action whose variation yields the Einstein equations plus an emergent nonnegative Λ at low coupling: exactly the missing variational layer, *if* the bridge identities hold. What is machine-checked today (`GRAV.ENTROPIC.ACTION.01`): **carrier Hodge count [E].** Her $\Omega^{0,1,2}$ matter fiber, placed on the five-slot carrier $\mathbb{C}^5$ (*not* on 4d spacetime, where the fiber is 1+4+6=11), counts $1{+}5{+}10{=}16{=}2^{g_{\text{car}}-1}{=}\dim S^+_{D_5}$ and Hodge-folds ($\star : \Lambda^1 {\cong} \Lambda^4$) onto $\Lambda^{\text{even}}\mathbb{C}^5$ — the carrier half-spinor's own Pascal row (v2/v44); a vector-space identity *only*. **Coefficient match [E] (the one quantitative pin).** Her weak-coupling Lagrangian $\mathcal{L} = 3\beta R - \alpha \mathcal{L}_{\text{m}} + \cdots$ (the 3 = the three form blocks) matched to the TFPT EH normalisation $1/(16\pi G) = c_3/(2G)$ fixes her free constant *exactly*,

$$\boxed{\beta'_{\text{B}} = \tfrac{c_3}{6} = \tfrac{1}{48\pi}}$$

(the $6 = 3 \times 2 = p_2 = |R^+(A_3)|$ is recorded as a structural echo, not a proof). **Entropy potential = quadratic Λ [E].** Her $\Lambda_{\mathcal{G}} = \frac{1}{2\beta}\operatorname{Tr}_F(\mathcal{G} - I - \ln \mathcal{G})$ is nonnegative and quadratic at the fixed point; with the TFPT displacement $\varepsilon_\star = e^{-\alpha^{-1}}Q$ it reproduces the closed Λ branch $\rho_\Lambda/\bar{M}^4_{\text{Pl}} = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), and the prefactor match yields the *exact* normalisation target $\operatorname{Tr}_F Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\,\delta_{\text{top}}$ — a closed algebraic target for the two-form (pair) block ($\binom{5}{2}{=}10$), [C] until an explicit Q is constructed. **R^2 kill test [E] (pre-registered).** The raw log expansion carries a curvature-squared coefficient of order $(c_3/6)^2$; the TFPT Starobinsky sector needs $\bar{M}^2_{\text{Pl}}/(12M^2_{\text{scal}}) = 1/(12c_3^7)$ (v36) — the exact mismatch is $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$: a direct identification of the raw entropic action with the TFPT R^2 sector is *false* without a mechanism (the scalaron as a light trace mode of her $\mathcal{G}$-field generating c_3^{-9}, or a

KMS-spectral renormalisation of the log action). **The bridge proper [C] (recorded, not proven).** The compression conjecture $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$ — the physical Calderón compression of her relative metric operator equals the TFPT seam modular operator; if proven, S_{B} is a local volume representation of the TFPT relative spectral action $S_{\mathrm{rel},\chi}$ (her log action is the all-scale-integrated heat kernel, by Frullani $-\ln A = \int_0^\infty \frac{dt}{t}(e^{-tA} - e^{-t})$, verified). Open with it: the operator-level Hodge items (Clifford action, Dirac intertwining, leaf involution, Calderón polarisation), and Lorentzian positivity of $\widetilde{G}\widetilde{g}^{-1}$ is *not* automatic (timelike gradients) — TFPT's OS/reflection-positive route is the sturdier construction. **Residual [O] (unchanged).** Nothing closes: the v359 equation-of-state typing stays [O]; adopting S_{B} would close it *only* after the compression conjecture *and* an R^2 mechanism; zero TFPT knobs are added.

The operator level, executed ([`verification/v474_entropic_hodge_carrier.py`]**).** Work packages 1 and 4-algebra of the bridge are now machine-checked on the finite carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ ($\dim 2^{g_{\mathrm{car}}}$=32). **Spinor structure exhibited [E]:** the ten CAR gammas satisfy $\{\Gamma_a,\Gamma_b\}$=$2\delta_{ab}$ exactly, the 45 bilinears span $\mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — S^+ *is* the even exterior algebra with its spinor structure, and Bianconi's Hodge–Dirac operator is Clifford multiplication at symbol level ($(c(\xi))^2$=$-\,|\xi|^2$, symbolic). **The fold is the $5\to\bar{5}$ conjugation [E]:** the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1, so her fiber 1+5+10 becomes the GUT half-spinor 16=1+$\bar{5}$+10 — the SM carrier decomposition, not a dimension match; honest obstruction: $\Omega^{\le 2}$ is *not* Clifford-invariant, so the identification must go through the fold, never through degree truncation. **The Q-target, decided [E]:** the uniform-block ansatz $\operatorname{Tr} Q^2$=$d\,k^2c_3^4$=$32c_3^4$ with integer k admits *exactly* the supports (d,k)=$(2,4),(8,2),(32,1)$ = $\{|\mathbb{Z}_2|, \operatorname{rank} E_8, 2^{g_{\mathrm{car}}}\}$, with the minimal uniform solution q=c_3^2 per Fock mode and the identity $\operatorname{Tr} Q^2/\delta_{\mathrm{top}}$=32/48=2/3; the two-form block 10 (the naive pair-sector reading) and the fiber 16 require irrational multipliers and are *excluded.* The physical choice among the three supports stays [C]; AP2 and the R^2 mechanism stay **[O]**.

The R^2 kill test, executed ([`verification/v475_entropic_scalaron_gate.py`]**).** On a maximally symmetric background the relative curvature operator has eigenvalues *exactly* $\{R;\ R/4\times 4;\ R/6\times 6\}$ (her Appendix-B flattening verified, incl. the 6×6 two-form block), so the exact vacuum action is $\mathcal{L}_{\mathrm{B}}(R) = -[\ln(1-\beta R) + 4\ln(1-\beta R/4) + 6\ln(1-\beta R/6)] = 3\beta R + \frac{17}{24}\beta^2R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) and $\frac{17}{24}$ is the exact tensorial factor behind the pre-registered gap. With the pinned β'=$c_3/6$ *both* mass routes (Starobinsky matching and $f'/3f''$) give the raw entropic scalaron

$$\boxed{m_{\mathrm{raw}}^2 = \frac{4608\pi^2}{17}\,\bar{M}_{\mathrm{Pl}}^2 \;\Rightarrow\; m_{\mathrm{raw}} \approx 51.7\,\bar{M}_{\mathrm{Pl}}}$$

— *trans-Planckian*, not light: the naive reading "the TFPT scalaron is the light trace mode of her $\mathcal{G}$-field" is *dead* [E]; a viable mechanism must supply exactly $m_{\mathrm{raw}}^2/m_{\mathrm{TFPT}}^2 = \frac{72}{17}(8\pi)^9 \approx 1.69\times 10^{13}$ in mass^2. The domain is not the obstruction (first log pole at R=$384\pi^2\bar{M}_{\mathrm{Pl}}^2$, trans-Planckian), and the Lorentzian caveat is now an explicit witness: a timelike gradient gives the Gg^{-1} eigenvalue $1-\alpha v^2 \le 0$ for $v^2 \ge 1/\alpha$ (ln undefined), while spacelike/Euclidean stay positive — the OS route is the sturdier construction. **Net verdict [C]:** of the three bridge interpretations, the entropic action covers the Einstein + Λ level (survives); the light-mode reading is dead; KMS-spectral renormalisation is the *only* surviving route to the R^2 sector. Nothing closes.

AP2, made well-posed ([`verification/v476_entropic_compression_toy.py`]**).** The compression conjecture itself is sharpened on an exactly solvable gapped chain. **The reading is decided:** on a *pure* bulk (the seam situation: pure bulk, thermal boundary) the covariance is a projection, so $f(x)$=$x/(1-x)$ is singular on its spectrum — the literal operator-side reading $P_\Sigma f(C) P_\Sigma$ is *ill-posed*, while the state-side $f(P_\Sigma C P_\Sigma)$ (Peschel) is finite: the identity can only mean *build* Δ_Σ *from the compressed relative metric*, which is exactly Bianconi's own local construction — AP2 retyped, not weakened. **The mismatch is exactly second order [E]:** symbolically, $[C^n]_{AA} - (C_{AA})^n$ carries the cross-cut block twice (first order vanishes identically), so for any analytic f the two readings agree up to $O(\|C_{AB}\|^2)$ — the modular (entanglement) content *is* that quadratic correction. **Gap control:** at the bounded (covariance) level the mismatch obeys $d \le x^2$ and falls with the gap, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates (Δ_{eff}>0, v76/v337). **Residual [O] (unchanged):** the continuum seam statement itself. No marker moves.

The surviving R^2 route, typed as one moment ([`verification/v477_entropic_scaleflow.py`]**).** The Frullani identity upgrades to a *scale-flow representation*: $-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta} - e^{-t}) = 2\int_0^\infty \frac{d\chi}{\chi}(e^{-a/\chi^2} - e^{-1/\chi^2})$ — Bianconi's log action *is* the flat scale-integral of relative heat-kernel actions, while TFPT's $S_{\mathrm{rel},\chi}$ evaluates *one* KMS scale with a weight f: the bridge between the two action principles is a choice of *scale measure*. In the moment dictionary (weight $w(t)\,dt/t$, μ_k its e^{-t}-moments) the coefficients read $3\beta R\,\mu_1 + \frac{17}{24}\beta^2R^2\mu_2$; the flat weight has $\mu_1{=}\mu_2{=}1$ (= the raw v475 coefficients), and demanding $m^2{=}c_3^7\bar{M}_{\rm Pl}^2$ forces *exactly*

$$\boxed{\mu_2/\mu_1^2 = \tfrac{72}{17}(8\pi)^9} \qquad \text{with} \qquad \tfrac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7 \text{ (identically)}$$

— the v475 kill-test number *is* a moment ratio, and the "13 orders of magnitude" are not a bridge mismatch but a scale-measure datum: the flat measure fixes it wrongly, the TFPT KMS measure ($f_0{=}6(4\pi)^2/c_3^7$, v36) fixes it correctly — one KMS moment, two parametrisations, zero new dials. [C] (consistency, not derivation: the measure is TFPT input); the R^2 gate and the v359 typing stay [O].

The two remaining legs, first computable steps ([`verification/v478_entropic_continuum_legs.py`]). *Leg A (AP2 continuum):* on the exactly solvable critical chain (infinite-volume kernel, 50-digit precision) the compressed interval state carries the conformal/BW structure quantitatively — the entanglement entropy follows Calabrese–Cardy with $c_{\rm est} = 1.00089/1.00038/1.00021$ (monotone → the Dirac $c{=}1$; gapped control $\sim 6{\times}10^{-8}$), and its Peschel modular Hamiltonian organises on the CHM weight $\beta(x){=}x(L{-}x)/L$ (corr 0.87→0.99, even-range bands vanishing identically, odd tail 1.8%): the state-side Δ_Σ (the reading *forced* above) flows to the CHM/Bisognano–Wichmann geometric form — *exactly* the object the entanglement-equilibrium derivation consumes (v323/v358); the bridge's modular side and the gravity side meet in the continuum limit (a lattice exhibit, not a continuum proof — AP2 stays [O]; the natural continuum frame for the seam version is the *four-interval* multilocal modular geometry of the four marks, v480). *Leg B (the measure):* the single-scale corollary is exact — $d\mu{=}\delta(t{-}t_0)dt$ gives $\mu_2/\mu_1^2{=}e^{t_0}$, so the one-moment condition forces $t_0{=}\ln\frac{72}{17}{+}9\ln(8\pi){=}30.461$, one dimensionless KMS time in closed form; the near-miss to $h(E_8){=}30$ (difference 0.461) is explicitly *declined* per the no-free-pattern rule (v354/v355) — deriving the measure from the entropic structure stays [O]. [E]/[C]/[O]

The Simple-Current Extension Theorem — the central $G_{\rm net}$ closing statement [E]/[C] ([`verification/v154_simple_current_theorem.py`])

The conditional theorem can now be stated as one clean *simple-current extension*. Let $A = (D_5)_1 \otimes (A_3)_1$ be the carrier net and $L = \langle(1,1)\rangle \subset \mathrm{Disc}(D_5) \oplus \mathrm{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$ the isotropic μ_4 glue line ($q(k(1,1)) = k^2 \in \mathbb{Z}$). Then the local extension $B = A \rtimes L$ has

$$\boxed{[B:A] = |L| = 4 = |\mu_4|, \quad c(B) = 5{+}3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1}$$

so B is *holomorphic* (KLM μ-additivity), and a holomorphic $c{=}8$ chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence $\boxed{B \cong (E_8)_1}$ (the same-c rival $SO(16)_1$, $\mu{=}4$, is excluded). Every algebraic ingredient is machine-checked: the $\mathbb{C}[\mathbb{Z}_4]$ Longo Q-system ([`verification/v125_glue_qsystem.py`]), the $\mathbb{Z}_4$-graded Frobenius E_8 hull ([`verification/v143_graded_frobenius.py`]), the Ramond glue sectors ([`verification/v148_fock_census.py`]). **Consequence:** $G_{\rm net}$ is internally *closed* if the seam–Calderón boundary net is *defined* to be A; it stays the *one* external identification "the RP seam–Calderón net is A" if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not "find a quantum gravity". *[2026-08-05 note, finite-level evidence for the identification ([`verification/v779_gnet_gns_arf.py`]): the inductive system — Haar isometry tower, commuting index-4 expectation squares, geometric state convergence — is exactly coherent; the finite Haag-duality census gives twisted-duality defect* 0 *with excess ratio* $\dim B'/\dim A' = 4 =$ *the local Watatani index; and the* $\mathbb{C}[\mathbb{Z}_4]$ *quasi-basis obstruction law* $2^{-\ell}$ *types the extension as exactly asymptotic at finite level. Typed finite evidence, never the continuum theorem; no marker move.]*

And that one identification reduces to a single shared premise ([`verification/v155_quasifree_boundary.py`]). The identification splits (v110) into (a) "the seam Calderón involution is sheet-odd" — the double-cover deck, the same $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2\cdot 4\pi}$ (structural) — and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the bulk covariance, and the compression of a contraction ($i\Gamma$ with spectrum in $[-1,1]$) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of Γ_d). Hence (b) *follows* from "the seam bulk is a reflection-positive free field" — the *same* premise already load-bearing in the area law ([`verification/v59_area_law_evidence.py`]) and the EH mechanism ([`verification/v150_replica_eh_model.py`]/[`verification/v151_bfk_split.py`]; since exercised on the discretized collar with the seam's own kernel and real replica sheets, v471 — the R3 kernel premise is discharged at the finite level, its residual retyping to the continuum scaling limit shared with `SEAM.EQUIV.01`, v336, plus the v152 anchor). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by G_{net}, the area law and R3, and given it the chain Gaussian bulk ⇒ quasi-free boundary ⇒ one kernel = net (v113) ⇒ carrier $A \Rightarrow (E_8)_1$ is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

The construction, exhibited and verified ([`verification/v156_seam_net_construction.py`]). The "$B \cong (E_8)_1$" step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode e^{ikx} by the harmonic extension $e^{ikx-|k|y}$: $-\partial_n u|_0 = |k|\,u$, so for a free bulk the seam operator is the massless chiral dispersion $\omega_k = |k|$ that defines the free-fermion net on S^1. Sixteen Majoranas give $c = 8 = 5{+}3$; the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ (the $SO(16)$ bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\boxed{\chi = \frac{E_4}{\eta^8} = q^{-1/3}\big(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \cdots\big) = j^{1/3}}$$

(verified by q-series), the 248 at level 1 being $\dim E_8$ — so the net *is* $(E_8)_1$. *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate ([`verification/v157_rigid_fixed_point.py`]). Three facts click together. (i) The DtN/Calderón symbol is *universally* $|k|$: the boundary operator of *any* second-order elliptic bulk is an order-1 ΨDO with principal symbol $|\xi|_g$ (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic c=8 theory is *rigid*: every field has weight $(h, \bar{h}{=}0)$, an exactly-marginal deformation needs $(1,1)$, and with $\bar{h}$=0 *no* $(1,1)$ field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same* $|\mathbb{Z}_2|$ that halves the one-sided S^2 Gauss–Bonnet ($c_3 = \frac{1}{|\mathbb{Z}_2|\,2\pi\,\chi(S^2)} = \frac{1}{8\pi}$) labels the sheet *and* is the Ramond/GSO projection giving μ=1 (248=120_{NS}+128_{R}) — one object, three roles, so the geometry fixing c_3 is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from "prove the seam is Gaussian" to "the gapped ($\Delta{=}6\log\frac{3}{2}{>}0$) boundary flow reaches the holomorphic fixed point" — given which, freeness is *forced* by rigidity + uniqueness ($(E_8)_1$, v83), not fine-tuned. A strictly smaller and simpler residual.

And the destination is a *proven* stable fixed point ([`verification/v158_fixed_point_stable.py`]). An exact operator-dimension count settles it: the chiral Majorana has $h{=}\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty* (bosonic chiral operators have integer h, lowest 1) — nothing relevant flows the free theory away; the h=1 currents are chiral (v157), not $(1,1)$-marginal; and the lowest genuinely-interacting operator (the quartic, $h{=}2{>}1$) is *irrelevant*, as are all higher $2k$-fermi operators ($h{=}k$). So the free chiral c=8 fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ($\Delta{=}6\log\frac{3}{2}$) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise ([`verification/v160_seam_gaussianity_from_pf.py`]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants κ_{2n}=0 (the signed set-partition/Pfaffian inversion of $\langle\psi_1\cdots\psi_{2n}\rangle = \text{Pf}(\Gamma)$ leaves only the degree-2 kernel — the v113 "one kernel = the net" property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if κ_{2n} were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap > 0 suffices; the value $(2/3)^6$ only sets the rate). *The honest residual:* v56's gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — the Schwinger cone the transport must contract has dimension ≫ 3, and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

And that full-cone premise is now *discharged* to a single finite one-particle statement (`[verification/v161_one_particle_reduction.py]`**).** The lever is Bogoliubov second quantisation: a quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dimensional Majorana space H_Σ, and the Fock space grades by correlator degree, so $\Gamma(t)$ acts on the degree-m sector as the exterior power $\Lambda^m(t)$. Three exact consequences follow (all machine-checked): (i) $\mathrm{spec}(\Gamma(t)|_{\deg m})$ is the set of products of m one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap *equals* the one-particle gap**: the sub-leading eigenvalue of $\Gamma(t)$ over *every* degree is $(2/3)^6$ (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is Λ^* of the fixed subspace — the disconnected Wick powers of K_Σ — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation ⇔ one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the $120 = \dim \mathfrak{so}(16)$ weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport's one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8 K_Σ polarisation (v113), sub-leading gap = $(2/3)^6$ (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A_2), now with the infinite-dimensional cone eliminated [E]/[O].

And that finite statement carries no *new* open content: it factors into the two already-open items (`[verification/v162_seam_transport_identification.py]`**).** Take the one-particle symbol apart into its two data, the gap value (β) and the fixed space (α). (β) *is forced*: a μ_4-equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection, $\sum_{k=0}^{3} U^k X U^{-k} = |\mu_4|\,\mathrm{diag}(X)$ for $U = \mathrm{diag}(1, i, -i)$, and the commutant of U is exactly the diagonal algebra — so its spectrum is the cusp weights $\mathrm{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the parabolic residue, v115), and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of the ω_k=$|k|$ free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is* μ_4-equivariant and admissible — which is the R1 "seam clock" identification (the transfer operator as the seam's gapped generator) — together with the A_2 net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into $A_2 + R_1$ with every number forced. The "free-bulk cloud" is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

And both residual identifications are pinned to their floor, so (A) has no independent open gate left (`[verification/v165_premise_a_closure.py]`**).** A_2 (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net *is* $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally $|k|$ (v156/v157). So A_2 re-types [O] → [C] (closed by assembly). R_1 (the seam clock) reduces to `GATE.QGEO`: a μ_4-equivariant flat gapped generator on the carrier is *unique* ($= A_0^\star$, v115/v116), so its gap $(2/3)^6$ is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the μ_4-punctured sphere $\mathbb{P}^1 \setminus \mu_4$ — whose cohomological half is already established (v137, b_1=3=N_{fam}). That is `GATE.QGEO`, an already-open realisation premise, *not* a new gate. *[2026-08-03* ***`GATE.QGEO`*** *note: quantitative progress on the continuum side via the quasi-free CAR route (*`[verification/v715_qgeo_car_continuum.py]`*, 22/22,* ***`QGEO-CAR-RATES-SUMMABLE`****): all 6+1 deck/twist channels are Schatten-norm Cauchy with geometrically summable rates (N up to 3072/6144), the FH renormalization is fit-free lattice-derived (Barnes G, 8.8×10^{-7}), the RP cone is stable, and the majorant lemma (L1)–(L3) is* named *with constants and a classical proof route — named, not proven: the gate itself does* not *move.] [2026-08-04* ***`GATE.QGEO`*** *note: the first proof block of the majorant lemma lands (*`[verification/v745_qgeo_car_l1_sector_lemma.py]`*,* 11/11*,* ***`L1-SECTOR-PROVEN-MODULO-ELEMENTARY`****): (L1) is proven modulo elementary steps for all* 6 *undressed sectors on the fixed grid — the occupied-arc mode sums are* exact *geometric sums (exact summation replaces Euler–Maclaurin), the entire $1/N$ structure is the sublattice embedding offset, the leading term is the exact matrix $(iu/2MN)f_q'(s)$ (ratio $\to$ 1 at 6×10^{-4}), $\gamma = 2$ is* derived, *the Taylor–Lagrange envelope holds with* 39744+108 *inequalities at* 0 *violations, and $r_1 = 1$ is sharp. The residues stay typed: (a) grid-sup [elementary], (b) L2 uniform FH/RH [hard-technical], (c) L3 uniformity [medium] — named, not claimed: the gate itself still* does *not* move.*] [2026-08-05* **GATE.QGEO** *note: residue (a) is* fully closed *in the corrected* $(1+\ln M)$ *form ([verification/v778_qgeo_gridsup_logsum.py],* $19/19 + 19/19$, ***GRIDSUP-SPLIT-CLOSED-LOG-LAW+FOURIER-LOGSUM-CLOSED****): the continuum entrywise supremum, the configuration-S_1 closure (explicit $B_{\rm conf}(\varepsilon)$ via packing + $\zeta(2)/\zeta(3)$) and the refinement remainder (exact* $\csc^3$ *integrals) close elementarily with zero census violations, while the leading part is honestly* log-divergent *under refinement — the literal* ***v715*** *clause "one $B(\varepsilon)$ for* all *grid refinements" is false at $r_1 = 1$ and is repaired to $\|D_N^{(M)}\|_1 \le [B_0(q,\varepsilon) + (J(\varepsilon)/2)\ln(M{+}3)]/N$, certified at theorem level by the grid Fourier log-sum inequality $\|F_q^{(M)}(\varepsilon)\|_1 \le c_0(q,\varepsilon) + (J(\varepsilon)/4)\ln(M{+}3)$, $J = 2|f_q'(\varepsilon)|$, with all constants exact (*162*-case census to $M = 6144$, zero violations; all* 252 *transfer cases verified; Araki/Powers–Størmer summability explicit). Honest: the measured slope is J/π^2 — the bound is valid, not sharp. Residues (b) and (c) stay open: the gate itself still* does *not* move. *2026-08-06 architectural note ([verification/v802_gnet_martingale.py]): the martingale route bypasses (b)/(c) for the limit-*existence *step of the $G_{\rm net}$ continuum strand (summable state-norm defects suffice) and relocates them into the typed normality step — a reallocation, not a proof of (b)/(c).]* By the No-Unit Theorem (v153) the irreducibles stay exactly $\{\pi, v_{\rm geo}\}$ — closing (A) adds none. **Final accounting:** premise (A) = (A_2 assembly, [C]) + (R_1=GATE.QGEO, already open) + the $\{\pi, v_{\rm geo}\}$ irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate [E]/[O].

And the two structural residuals — net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the one irreducible open premise ([verification/v175_net_existence_full_cone.py]). The CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the *complete* Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}$=65536); by the exterior-power spectrum theorem every eigenvalue of $\Gamma(t)$ is a subset product of spec(t), so checking all 2^{16} products shows $\Gamma(t)$ is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity 2^8=256 (the wedges of the rank-8 K_Σ polarisation) and sub-leading eigenvalue exactly the one-particle gap $(2/3)^6$ — so full-cone RP/gap reduce *for every m* (not only $m \le 6$, v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem.* And A_2 is now an *assembled and verified* certificate: the $(E_8)_1$ character $E_4/\eta^8 = 1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), the embedding 248=120+128, and the E_8 Cartan matrix even with det 1 (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the *same* input**: that the actual seam Calderón kernel is this gapped one-particle contraction on $\mathbb{P}^1 \setminus \mu_4$, i.e. QGEO.REALIZE.01, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]— the single irreducible structural premise; net existence and full-cone RP are [E].

The No-Unit Theorem — $v_{\rm geo}$ is an irreducible metrology primitive [E]/[O] ([verification/v153_no_unit_theorem.py])

$v_{\rm geo}$ is not an open gap. Every load-bearing compiler datum — $g_{\rm car}, |\mu_4|, N_{\rm fam}, \mathrm{rank}\, E_8, \alpha^{-1}$, the determinant ladder $(3,4,8,20)$, $c_3 = \frac{1}{8\pi}$, $\varphi_0^{\rm ret}$ — is mass-dimension 0 and *invariant* under a unit rescaling $L \mapsto \lambda L$, whereas a mass / energy / $1/G$ carries nonzero dimension. A map from dimension-0 inputs to a dimension-$d{\neq}0$ output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** — $v_{\rm geo}$ is a *forced input.* The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\rm point} \sim v_{\rm geo}, \qquad \tfrac{1}{G} \sim v_{\rm geo}^2, \qquad m/\mu = e^{3/4} \text{ (a pure ratio)},$$

so the irreducibles stay exactly {one dimensionful unit, π}. *Re-typing:* $v_{\rm geo}$ is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

Sharpened across gravity, and the final tally ([verification/v364_vgeo_sharpen.py]). After the parameter-free gravity (v358/v359/v361) the *gravity* sector adds no new scale either: the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48 c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\rm geo}^2$ and $\Lambda \sim$ (dimensionless) $v_{\rm geo}^4$ both reduce to $v_{\rm geo}$ times atoms. Hence $v_{\rm geo}$ is the *single* dimensionful input of the *complete* theory (matter and gravity), and the final tally is $\{v_{\rm geo}, \pi\}$ with *zero* dimensionless dials —

a $\sim$26 $\to$ 1 reduction against the Standard Model. ($1/G$ stays UV-sensitive (Sakharov, v68), so it is a metrology readout, not a clean prediction — consistent with $v_{\rm geo}$ being the anchor.)

The torsor closure of the interface ([`verification/v725_phys_vgeo_torsor_audit.py`], `VGEO.TORSOR.01`, 21/21). The $v_{\rm geo}$ interface is now *structurally closed* as an $\mathbb{R}_+$ scale torsor, machine-audited on the corpus: (1) the complete export table $O_i = r_i\, v_{\rm geo}^{d_i}$ with machine-derived dimensionless r_i ($\bar M_{\rm Pl}$, $1/G$, Λ, $U_{\rm point}$, m/μ at $d = 1,2,4,1,0$); (2) the dimension matrix has *rank* 1 — exactly one independent scale (conditional [A] on the flavor block via the declared seam=Planck identification, named, not hidden); (3) *all* consistency conditions are homogeneous under $L \mapsto \lambda L$ — the machine form of this theorem; (4) the leave-one-anchor-out cross-calibrations are compatible (G vs ρ_Λ: the same $\bar M_{\rm Pl}$ to 0.11%, v274); (5) the *calibration theorem* in torsor language: one external reference selects the point in the torsor, everything else is prediction. The closure is the *calibration form* — no absolute-scale derivation (impossible per this theorem; that is the content, not the defect); the one remaining dimensionless scale ratio is *named*, $H_{\rm EW} = \ln(\bar M_{\rm Pl}/v_{\rm EW}) = 37.1776$ (naive $c_3/\varphi_0^{\rm ret}$ ladders honestly excluded); the [E]/[O] typing of this keybox is unchanged.

Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure $\mu_{\rm QG}$. Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$ seam action

$$f_R R_{\mu\nu} - \tfrac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu) f_R = \bar M_{\rm Pl}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: the local covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (v358/v359); Theorem G remains open only for the projective-limit ambient* measure $\mu_{\rm QG}$ *(G6, the deepest item). G5 certified; the regime equation closed [`verification/v28_gravity_fR.py`].*

Research Contract — `CELEST.SEAM.01` (the celestial and twistor continuum route)

Reader's note. A compact synthesis of this contract's executed work packages (WP1–WP5d, both WP5d stages, + the WP5e α, β, γ, δ_2, ε_2 and ε_1 stages + the three back-reaction milestones M1–M3) — the narrative arc from the μ_4 clock to the $(E_8)_1$ boundary shadow — is given as a dedicated section in `tfpt_3_e8_audit_bootstrap` ("The celestial and twistor continuum route"); this contract remains the full technical reference (typing fence, kill tests, work-package statements). The bridge from this contract's compiler-side constructions to reconstructed Minkowski physics is *not* part of this contract: it is the separate central contract `WOIT.OS.TWISTOR.01` below — everything in `CELEST.SEAM.01` is preparation for it.

The object diagram (read this first). The route is one chain of objects; each arrow is a separately typed claim. Where mathematics ends and physics begins is visible at a glance: the last two arrows are [O].

Arrow	Status	Claim / modules	Kill test
$(\Sigma, \mu_4, \rho, \Theta) \to \mathbb{P}^1\backslash\mu_4$	[E]	`SEAM.MARKS` chain: four marks, clock, cross-ratio 2, $\tau{=}i$ pillowcase (v168/v214/v216/v180; bit reduction v506/v507/v510/v512)	a fifth mark, a non-order-4 clock, or a marks-preserving root without flag transitivity
$\mathbb{P}^1\backslash\mu_4 \to \mathbb{C}^2/\mathbb{Z}_4$	[E]	`CELEST.WP1.01`/`CELEST.WP2.01`: the glue is the flat $\mathbb{Z}_4$ monodromy on the A_3 ALE, clock = Kähler $U(2)$ phase, clock^2 = deck (v492/v493)	equivariant-sector closure failure (3112/6720 control) or a surviving shape modulus
$\mathbb{C}^2/\mathbb{Z}_4 \to \mathbb{PT}/\Gamma$	[E]/[C]	`CELEST.WP5E.*`: equivariant twistor uplift — anomaly ledger, level-from-flux, axion slot, back-reacted Ω_N, twisted KS measure, a_0 uplift, the δ_1 decision, the measure decision (v505/v509/v511/v513/v514/v515/v516/v517/v518/v520)	the preregistered level kill ("inflow demands $k \neq 1$"); the M1–M3 kills were evaluated by v515/v516/v517 and did not fire; the δ_1 kill *fired on the derived measure* (v518), and the declared/derived measure question is since *decided at probe level* in favour of the declared reading under the typed premises TP-1..TP-4 (v520) — the residual [O] is the constructive w_m derivation
$\mathbb{PT}/\Gamma \to \mathcal{A}_{\rm hol}$	[O]	the *interacting* holomorphic algebra (global BCOV+SDYM quantisation; WP5e proper; target of `WOIT.OS.TWISTOR.01`; CFT-side anchors v497–v504)	anomaly mismatch at the Costello–Li grade
$\mathcal{A}_{\rm hol} \xrightarrow{\rm OS} \mathcal{A}_{\rm Mink}$	[O]	OS reconstruction with the real structure Θ — the `WOIT.OS.TWISTOR.01` target (the free-collar RP/OS witnesses v379/`FORM.SEAM.MMST.01` are the *free* anchor only; the α stage v519 pins Θ itself and free RP on the seam system — kill test (1) discharged at the free level, live on $\mathcal{A}_{\rm hol}$)	the seven kill tests of `WOIT.OS.TWISTOR.01` below

Definition (the group extension, exactly). The shorthand "$\mathbb{PT}/\mathbb{Z}_4$" used in this contract is precise but deserves unpacking, because *two different* order-4 structures are in play and only one of them is quotiented:

- $\Gamma_{\rm ALE} = \langle\text{Deck}\rangle \cong \mathbb{Z}_4 \subset SU(2)$, the deck group $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$: *triholomorphic*, preserves the holomorphic symplectic form Ω; this is the $\mathbb{Z}_4$ that is *quotiented* ($\mathbb{C}^2/\Gamma_{\rm ALE}$ is the A_3 singularity $XY = Z^4$, and $\mathbb{PT}/\Gamma$ means the quotient by $\Gamma_{\rm ALE}$ acting on the twistor space of the ALE) [E] ([`verification/v492_celestial_z4_orbifold.py`]).
- $\rho = \text{Clock} = \text{diag}(i, 1) \in U(2)$: a Kähler, *not* triholomorphic isometry ($\det\rho = i$ rotates Ω by the μ_4 generator — the WP1 S5 computation); it is *not* in $\Gamma_{\rm ALE}$ and is *not* quotiented — it *normalises* $\Gamma_{\rm ALE}$ and survives the quotient as the residual clock symmetry [E] (v492).
- The global object is the extension generated by both: since $\rho^2 = \text{Deck}$ exactly (v492/v506), $\langle\Gamma_{\rm ALE}, \rho\rangle = \langle\rho\rangle$ is *cyclic* — projectively $\mathbb{Z}_4$ (the clock on the sphere) with ρ^2 generating the deck $\mathbb{Z}_2$ there, and on the spin/fermionic level the canonical $\mathbb{Z}_8$ tower ($V^2 \propto U$, $V^4 \propto (-1)^F$, nonsplit; $8 = 2|\mu_4|$) [E] (v506/v507).

The action table, entry by entry ([E] = arithmetically established at the current stratum, [O] = a quantisation question):

Object	Γ_{ALE} (Deck)	ρ (Clock)	**Status / witness**
$K_{\mathbb{PT}}$ (canonical class)	trivial	weight via $\det\rho = i$	[E] (incidence-forced column weights $(+1,+1,-1,-1)$, v514)
Ω (hol. symplectic)	preserved	rotated by i	[E] (v492 S5; back-reacted Ω_N closed-form with $(2\pi i)^2$-integral periods and forced charge 4, v515)
BCOV fields ($O(-2)$ tower)	orbifold sectors	character series $P_0..P_3$	[E] ledger / [O] quantisation (v514)
open-string fields (SDYM E_8)	glue-equivariant sector	sector rotation $j \mapsto j{+}1$	[E] ($D[d] = 60(d{+}1)/64(d{+}1)$, v492) / [O] interacting
boundary states ($(E_8)_1$ shadow)	μ_4 sector split	clock phase on $\lvert s\rangle$ etc.	[E] character/GNS level (v497–v500) / [O] as an actual net

The A_3 role table (no silent identifications). Several rank-3 A_3's and external gauge factors circulate near this contract; the following table states, for each pair, either the connecting functor (with its witness) or an explicit *not identical*:

Object	**Role and relations**
A_3^{family}	the family factor in the lattice $D_5 \oplus A_3$ ($\mathfrak{su}(4)$-flavour; three families from its exponents, v128/v137). Related to A_3^{ALE} by the McKay correspondence *of types* only (both are type A_3); **not identical** as realisations — one is a sublattice of E_8, the other the singularity type of $\mathbb{C}^2/\mathbb{Z}_4$. The bridge that *does* connect them is the Kronheimer–Nakajima quiver (v479: same ADE datum, different functors).
A_3^{ALE}	the ADE type of $\mathbb{C}^2/\mathbb{Z}_4$ ($XY = Z^4$); its resolution carries the three exceptional spheres (v492/v493).
three exceptional spheres	the Coxeter/Picard–Lefschetz structure of the resolved A_3^{ALE}: the clock acts as a Coxeter element of $W(A_3)$ on H_2 (eigenvalues $\{i, -1, -i\}$, v493); they carry the three twisted axion slots (v505) and the lockstep fluxes (v509).
D_5^{carrier}	the carrier factor ($g_{\mathrm{car}} = 5$); glued to A_3^{family} by the μ_4 Lagrangian glue into E_8 (v92/v125); **no** role on the ALE side beyond the zero-mode algebra $\mathfrak{g}_0 = D_5 \oplus A_3$ (v492).
Woit's $SU(3)_{\mathrm{color}}$	*external programme reference only* (Euclidean Twistor Unification: colour as the rank-three quotient bundle on $\mathbb{PT}$). **Not identical — no identification claimed**; in particular $A_3^{\mathrm{family}} \not\cong SU(3)_{\mathrm{color}}$ (an $\mathfrak{su}(4)$ flavour factor is not a colour gauge bundle), and TFPT's colour lives in the SM readout of the carrier, not on $\mathbb{PT}$.
Woit's $SU(2)_{\mathrm{weak}}$	*external programme reference only* (the internal spin factor of the Euclidean twistor construction). **Not identical — no identification claimed**; whether an internal $SU(2)$ of that kind can be realised on $\mathbb{PT}/\Gamma$ is exactly kill test (6) of `WOIT.OS.TWISTOR.01`.

Hypothesis (as corrected by WP1). *The seam is the glue-equivariant $\mathbb{Z}_4$-orbifold sector of the celestial chiral algebra on the A_3 ALE space $\mathbb{C}^2/\mathbb{Z}_4$; the μ_4 clock is the Kähler $U(2)$ phase* $\mathrm{diag}(i,1)$ *whose square is the deck group (spin bridge $\mathbb{Z}_8$, $8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ seam winding integer); and the associated twistorial bulk is the* self-dual *sector of TFPT gravity.* This is the fourth research contract, alongside (U_{wall}), (G_{metric}) and `CONTRACT.F.01`: a numbered chain of work packages with pre-registered kill tests, *not* a claim. The method is the celestial-chiral-algebra (CCA) construction of Bittleston–Homans–Sharma on Eguchi–Hanson space (`arXiv:2305.09451`, JHEP 09 (2023) 008 — the $\mathbb{Z}_2$ case, refined here to $\mathbb{Z}_4$ with a flat connection at infinity in the Kronheimer–Nakajima / heterotic-on-ALE sense), inside the celestial-holography programme (Strominger, PRL **127**, 221601: $w_{1+\infty}$; Costello–Paquette–Sharma, PRL **130**, 061602: Burns holography, the $SO(8)$ WZW_4 model; Bittleston–Sharma–Skinner, CMP **403** (2023): quantizing the non-linear graviton). The gauge-algebra admissibility gate is already passed in the literature: Costello's one-loop axionic-consistency analysis (`arXiv:2111.08879`) admits the exceptional algebras *including* E_8 for SDYM on twistor space (Okubo: no independent quartic Casimir). The spin bridge in the hypothesis is no longer a bookkeeping convention: on the 16-Majorana seam the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$) and the quarter-shift lifts to a canonical $\mathbb{Z}_8$ tower with $V^2 \propto U$ — "$8 = 2|\mu_4|$" fermionically explained ([`verification/v506_seam_clock_rigidity.py`], SEAM.CLOCK.RIGIDITY.01); and the nonsplit class is arrangement-*sensitive*: the edge (silver) arrangement splits ($U_e^2 = +1$, zero roots), so the fermions *measure* the alignment bit as a Fidkowski–Kitaev-type extension class ([verification/v507_seam_bit_origin.py], SEAM.BIT.ORIGIN.01) — and that class is nonsplit *iff* the deck acts freely (all 17 seam-circle involutions, v510), which the covering deck does by topology.

WP1, executed and verified: the E_8 μ_4-glue *is* the flat $\mathbb{Z}_4$ monodromy of the equivariant CCA on the A_3 ALE space — verdict B ([verification/v492_celestial_z4_orbifold.py], CELEST.WP1.01)

Everything in this box is exact arithmetic (sympy, no floats). **Inner grading / flat monodromy [E]:** the v128 glue grading ($240 = 52{+}64{+}60{+}64$, additive on all 6720 root pairs, dims with Cartan $(60, 64, 60, 64)$, $\mathfrak{g}_0 = D_5{\oplus}A_3$ carrier $= 60$) is *inner*: $h = (2, 2, 2, 2, 2; 0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\mathrm{Ad}(\exp(2\pi i h/4))$ — an order-4 torus element, i.e. a *flat* $\mathbb{Z}_4$ *monodromy* in the Kronheimer–Nakajima sense; the A_3-side detector $h' = (0^5; 1, 1, 1, -3)$ reads the *same* class, so the glue charge is the diagonal $(1, 1)$ of the discriminant group $\mathbb{Z}_4{\times}\mathbb{Z}_4$ — the v92/v125 Lagrangian glue. **The spatial side [E]:** $\mathbb{C}^2/\mathbb{Z}_4$ under the deck $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$ *is* the A_3 singularity $XY = Z^4$ (invariant ring exact; Hilbert series $(1{+}t^4)/((1{-}t^2)(1{-}t^4))$; Molien table verified). **The equivariant sector closes [E]:** the glue-equivariant SDYM(E_8) modes ($j + m \equiv 0 \bmod 4$) form a Lie subalgebra with graded dimensions $D[d] = 60(d{+}1)$ (even d) / $64(d{+}1)$ (odd d) — uniform per parity *only* because $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = 60$ and $\dim \mathfrak{g}_1 = \dim \mathfrak{g}_3 = 64$: the v128 dimensions are exactly the consistency condition. The zero modes ($d{=}0$) are $\mathfrak{g}_0$ alone — *the unbroken algebra on the orbifold is the carrier* $D_5{\oplus}A_3{+}$Cartan $= 60$; the asymptotic density is $62/248 = 1/4 = 1/|\mathbb{Z}_4|$ (the orbifold index), and the $\mathbb{Z}_2$ halfway sector reproduces the Eguchi–Hanson towers $120/128 = SO(16)$ adjoint/spinor — μ_4 refines the sheet $\mathbb{Z}_2$ exactly as $4 = 2 \times 2$ (v125). **Character level [E]:** Θ_{E_8} splits into the four glue-coset thetas (shell $1 = (52, 64, 60, 64)$ re-derived from the *lattice*); the four sector characters Θ_{C_j}/η^8 sum *exactly* to $\chi_{(E_8)_1} = 1{+}248q{+}4124q^2{+}34752q^3$ (the v377 series); the sector conformal weights are $h_{D_5} = (0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})$, $h_{A_3} = (0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$ — the v92/v125 discriminant form $(5x^2{+}3y^2)/8 \bmod 1$ — and the glue diagonal has *integer* $h = (0, 1, 1, 1)$: the v125 isotropy *is* the locality of the extension to $(E_8)_1$. **The critical correction [E] (why verdict B, not A):** the A_3 deck $\mathrm{diag}(i, i^{-1})$ induces on the celestial sphere $z = z_1/z_2$ only $z \mapsto -z$ (projective order 2, the sheet flip) — it is *not* the order-4 clock $z \mapsto iz$. The exact bridge: any $SU(2)$ lift of the clock has order 8 (spin group $\mathbb{Z}_8$, which would quotient to A_7), and (spin clock)^2 = the deck generator *exactly*, with $|\mathbb{Z}_8| = 8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ winding integer. The order-4 clock *is* realised, as the $U(2)$ phase $\mathrm{diag}(i, 1)$: projective order 4, fixed points $\{0, \infty\}$, free μ_4 orbit (all v180 clock properties), normalising the deck (so preserving the glue flat connection), with $\det = i$ rotating the holomorphic symplectic form by the μ_4 generator (a Kähler, *not* triholomorphic isometry — it rotates the twistor sphere by $\pi/2$). **Gravity side [E]:** clock-invariance selects from the 3-parameter versal A_3 family $XY = Z^4{+}a_2Z^2{+}a_1Z{+}a_0$ *exactly* the 1-parameter family $XY = Z^4{+}a_0$, whose four branch points are *one* μ_4 orbit with cross-ratio 2 — the seam marks (v179 clock orbit; v168/v214 cross-ratio $2 \Rightarrow j{=}1728$ pillowcase). **Negative controls [E]:** the false spatial action $\mathrm{diag}(i, i)$ is killed three ways ($\det = -1 \notin SL(2, \mathbb{C})$, no CY/ALE resolution; Poisson closure fails in 14/14 nonzero cases; conjugation symmetry broken), the false glue (60/64 blocks swapped) breaks additivity on 3112/6720 root pairs, and of all 24 relabelings exactly $\{\mathrm{id}, j \mapsto -j\} = \mathrm{Aut}(\mathbb{Z}_4)$ are additive — the grading is rigid. **Verdict [C]:** the celestial and glue $\mathbb{Z}_4$ structures match under *one* precisely named extra identification — clock^2 = deck on the sphere (the sheet-$\mathbb{Z}_2$ spin bookkeeping) — so WP1 lands at verdict B (match modulo one named identification), close to but honestly short of A. [E]/[C]

Typing and non-circularity. Admissible *inputs* are the finite seam/glue data already in the suite — the μ_4 Q-system glue (v92/v125), the graded hull (v128), the seam rigidity and pillowcase geometry (v168/v214), the Möbius clock (v180) — plus the WP1 [E] arithmetic ([verification/v492_celestial_z4_orbifold.py]). *Not* admissible as an input: the *continuum existence* of the holomorphic $(E_8)_1$ net on the seam — that is precisely the SEAM.EQUIV.01 target, and assuming it would make the contract circular. *Scope fence:* the twistorial bulk of this contract is the *self-dual* sector only, never full Einstein gravity — the classical field-equation route (entanglement equilibrium, v358/v359) and the entropic-action bridge (GRAV.ENTROPIC.ACTION.01, v473) are separate, non-competing statements about the full equation. SEAM.EQUIV.01 stays [O] throughout; nothing in this contract moves it.

Work packages.

- **WP1 (*done*, verdict B) [E]/[C]:** the structural $\mathbb{Z}_4$ arithmetic of the keybox above, executed and machine-verified as `CELEST.WP1.01` [`verification/v492_celestial_z4_orbifold.py`].
- **WP2 (*done*, verdict B) [E]/[C]:** the clock-invariant deformation theory, executed and machine-verified as `CELEST.WP2.01` [`verification/v493_celestial_wp2_clock_deformation.py`] (sympy exact, 47 checks). Both targets landed, *sharpened*: (i) a_0 is the pure *scale* of the seam configuration — the binary-quartic invariants of $w^2 = Z^4+a_0$ are $I = 12a_0$, $J = 0$ *identically*, so $j = 1728$ and the $\tau=i$ pillowcase shape (v168/v214) is *frozen* for every $a_0 \neq 0$ (no shape modulus survives clock-invariance; the negative controls $a_1 Z \Rightarrow j=0$ and $a_2 \neq 0 \Rightarrow j=1556068/81$ show the test has teeth); (ii) the three resolved spheres ($3 = \operatorname{rank} A_3 = N_{\mathrm{fam}}$) carry *exactly* the three nontrivial μ_4 characters $\{i, -1, -i\}$ under the clock, which acts on H_2 as a *Coxeter element* of $W(A_3)$ (char x^3+x^2+x+1, order $h(A_3)=4=|\mu_4|=N_{\mathrm{fam}}+1$) and *is* the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i} a_0$ gives the 4-cycle); the surviving deformation direction is the weight-1 χ_1-Fourier diagonal $(1, i, i^2)$ — *not* the naive invariant diagonal ($\det(A-1) = -4$: no invariant cycle) — and all three sphere volumes move in lockstep ($|\Pi_j| = \sqrt{2}\,t$). The BHS deformed-algebra pattern transfers from $\mathbb{Z}_2$ to $\mathbb{Z}_4$: the fibre bracket $-4\,\mathrm{Nambu}(XY - Z^4 - a_0)$, anchored at the $a_0=0$ orbifold, closes with corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade (no sector leak — the WP1 equivariant sector deforms consistently), with a_0 in the weight-0 slot of BHS's c^2. Verdict B via three named [C] identifications ($-4\,\mathrm{Nambu}$ as *the* $k=4$ CCA bracket; period = root difference; the seam-scale reading via clock^2 =deck); the full deformed twisted-sector OPEs and the $W(\mu)$-type scaling limit for $k=4$ stay **[O]**. The v216 [O]-residual is *typed*, not moved: given the order-4 clock, the deformation freezes the $(2,2,2,2)$ modulus to the square automatically — a relocation of the same order-4 carrier input, not a new derivation (since v510 that input reads: the square-modulus datum $\tau = i$; its position half is topology; since v512 it reads equivalently: one discrete symmetry-lift bit, flag transitivity $\iff$ $\tau = i$; and since v521/v525/v529 free OS positivity, the whole mark-decorated free-plus-twist state class and the first genuinely interacting FK toy are documented as the *eighth*, *ninth* and *tenth* side-blind tests — the bit is not derivable from free RP/Θ existence, from any Wick-computable twist decoration or from the FK interaction class, and remains genuine discrete input; since v528 it is the twist-class choice, in principle interferometrically readable; and since v534 it carries the first *positive* selection datum — reflection positivity on the interacting mark family keeps exactly the $\delta = \pi/2$ member alive, toy-fenced).
- **WP3 (*done*, verdict B) [E]/[C]:** the Green–Schwarz alignment test, executed and machine-verified as `CELEST.WP3.01` [`verification/v495_celestial_wp3_gs_alignment.py`] (exact Fraction/sympy, 25 checks). The arithmetic side is sharp: the Okubo coefficient $\mathrm{Tr}_{\mathrm{adj}} X^4/(\mathrm{Tr}_{\mathrm{adj}} X^2)^2 = 5/(2(\dim\mathfrak{g}+2))$ is *derived* as a polynomial identity for all 8 algebras on Costello's list (with a sharp $\mathfrak{sl}_4$ negative control: 5/32 vs 3/32), the closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim+2) = h^\vee+6$ holds exactly across the Deligne series, and for E_8 the axion coefficient is *exactly* $\tilde{\lambda} = 6$ (unit-trace) resp. $\lambda_{\mathrm{fund}} = 1/10$ (adjoint-trace), so the GS coupling $\kappa = \lambda_{\mathfrak{g}}/(8\pi\sqrt{3})$ satisfies $(\kappa/c_3)^2 = 12 = |\mu_4| N_{\mathrm{fam}}$ resp. 1/300 — exact anchor rationals, with the anchor decomposition $h^\vee=30=|\mathbb{Z}_2| N_{\mathrm{fam}} g_{\mathrm{car}}$, $\varphi(30)=8=\mathrm{rank}$, $\tilde{\lambda}=|\mathbb{Z}_2| N_{\mathrm{fam}}$, $\lambda_{\mathrm{fund}}=1/(|\mathbb{Z}_2| g_{\mathrm{car}})$ (a byproduct: the printed $\lambda^2_{\mathfrak{so}_8} = 3/2$ in Costello's appendix A is a factor-2 slip; his own weight content gives 3, Okubo-consistent). κ/c_3 itself is *irrational* $(2\sqrt{3})$: only the squared alignment is exact. **The look-elsewhere caveat is part of the result [C]:** the same squared-rational alignment holds for *all eight* algebras on the list (8/8 — the test as such has zero selective power), $\tilde{\lambda}$-integrality passes 2/8 (shared with $\mathfrak{sl}_3$), and the only $\mathfrak{e}_8$-selective single test is $g_{\mathrm{car}}=5 \mid h^\vee$ (1/8); the conjunction that isolates $\mathfrak{e}_8$ is post hoc, not preregistered. *Alignment survives; selectivity does not* — the c_3-connection of the celestial route is convention-level compatibility plus genuine λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence,

and never a derivation of c_3 from celestial data.

- **WP4 (*done*, verdict B(ii)) [E]/[C]:** the type mismatch, executed head-on and machine-verified as `CELEST.WP4.01` [`verification/v496_celestial_wp4_character_shadow.py`] (exact integer/Fraction/sympy, 25 checks). The $(E_8)_1$ character is *not* a conformal block of the celestial $E_8[\mathbb{C}^2]$ S-algebra in its own (jet) grading — the obstruction is localised *three* independent ways: (a) the CP grading gives spin $1-d/2$, unbounded below, and 248 is never a jet-tower dimension ($60(d{+}1)/64(d{+}1)$, $d \le 100$); (b) the cumulative generator count is quadratic ($31s^2{+}92s{+}60$, $31 = k{+}h^\vee$), so the jet Fock grows as $n^{2/3}$ against the character's $n^{1/2}$ (Meinardus poles $s{=}2$ vs $s{=}1$; the ratio f_n/χ_n increases strictly for $n = 1..12$); (c) the level-2 null ideal of $(E_8)_1$ deletes *exactly* $27000 = 30^3 = (h^\vee)^3$ out of $\mathrm{Sym}^2(248) = 1{+}3875{+}27000$ (the character keeps $4124 = 1{+}248{+}3875$) — and has no jet analogue. **But the boundary/period reading holds exactly at the current stratum**: the zero-mode slice is the 60 vacuum-sector currents, the single-line jet slice cycles the four sector counts $(60, 64, 60, 64)$, one full μ_4 period of loop energies sums exactly to 248 (the single-particle level of the $248q$ stage), and the glue-diagonal weights $(0, 1, 1, 1)$ are integers — while the free loop Fock at level 1 counts $897266 \gg 248$, i.e. the rational truncation must be *imposed in a limit.* That is precisely the MMST scaling-limit shape of `SEAM.EQUIV.01` (v336/v449: the rational net exists in the *limit* of the lattice collar, not in the pre-limit algebra): WP4 is hereby *retyped* from "character as conformal block" to "character as boundary/limit shadow", and the constructive limit question (which limit, which topology, convergence) passes to WP5 — where WP5a has since answered the "which limit" part at the counting level (the coefficientwise χ_w limit below), WP5b has constructed the deleting operator, WP5c the limit state whose GNS kernel contains the ideal, WP5d the local-net legs (the α-stage two-interval index chain and the β-stage split + strong-additivity witnesses — all three KLM ingredients of complete rationality on the lattice), and WP5e-$\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$ have anchored *both* faces of the twistor uplift, closed its exchange sub-branch, executed the full-tensor ledger, executed the level-from-flux dial and built the bulk-axion slot (the CFT-side prefactor + level pinning; the equivariant twistor skeleton with the index bridge and the rigid $32\,T_3$ residual; the exchange no-go with certificate; the full-tensor confirmation of the collapse with the one cubic d-symbol channel that opens; the lockstep theorem making "one level" a theorem of clock invariance, with the sector-counter dial pinning $k = 1$; the equivariant $O(-2)$ slot construction with $\tilde{\lambda} = 6$ triply pinned and the first GH/A_3 back-reaction step), and all three quantised back-reaction milestones M1–M3 have since been executed (the closed-form Ω_N with $(2\pi i)^2$-integral lockstep periods and forced source charge $4 = |\mu_4|$, v515; the twisted KS measure on the declared completion reading — per-sector Okubo squares, the $32\,T_3$ cancelled, the $\psi = 64$ slice supplied without the cubic d-channel, v516; the a_0 uplift coupled to the centre count on four scales, v517), leaving the global BCOV quantisation (WP5e proper) as the remaining open piece — the BCOV *measure question* is since decided at probe level: the δ_1 chain itself was decided by v518 (kill under the derived measure, all three testers fail), and v520 has since decided *which* of the two exact measures the BCOV integral produces — single-valuedness of the one-loop lattice factor is *derived* from two independent constructive sources (F-independence of the moduli integral + the Quillen pairing) under the typed premises TP-1..TP-4, forcing the declared v516 completion reading, and v523 has since *derived* the w_m normalisation itself at probe level ($1/\det_j$ = the Atiyah–Bott/zeta-determinant fixed-point factor, computed from three independent sources under the typed premises TP-REG/TP-Q/TP-NUM/TP-CH); the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor (throughout, $\mathbb{PT}/\mathbb{Z}_4$ is the quotient by Γ_{ALE}, normalised by the clock $\langle\rho\rangle$ — the definition block above; the ε_1 slot itself is executed; the quantised milestones M1–M3 preregistered in v514 are all executed via v515/v516/v517 — the v514 fence is fully worked off).

- **WP5 (the summit) — subdivided; WP5a–WP5d (both WP5d stages) plus the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the back-reaction milestones M1–M3**

are landed. The twistorial construction as a *second continuum route* for `SEAM.EQUIV.01` (Costello–Li grade), carrying the WP4 constructive-boundary-limit question. Subdivision WP5a–WP5e:

- **WP5a (*done*) [E]/[C]:** "the null ideal from the limit", executed and machine-verified as `CELEST.WP5A.01` [`verification/v497_celestial_wp5a_null_ideal.py`] (exact integer/Fraction arithmetic, 34 checks). Three results make the WP4 limit reading precise. (i) *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$) *contains* the chiral jet grading as its w=2 member (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n{+}1$ — strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$): the WP4 "boundary-limit shadow" is a precise *coefficientwise* limit $\varepsilon = 1/w \to 0$ with explicit stabilisation threshold $w = n{+}1$, not a slice. (ii) *The null ideal is derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with peeling residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; level-1 integrability ($\langle 2\theta, \theta^\vee\rangle = 4$, $\langle \omega, \theta^\vee\rangle = 2$) leaves the vacuum as the only level-1 primary ([C] Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. (iii) *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3: 34752). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components ($5304{+}1820{+}135{+}1$), $5304 \neq 14^3$ (the $h^{\vee\,3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), and block weights $(0, \frac{1}{2}, 1, 1)$ that can*not* fuse into one local character; only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor. *Honest limit* [O]: the limit does *not* generate the truncation (the loop Fock counts $897266 \gg 248$ at integer level 1) — WP5a fixes the ideal's *size and location* quantitatively and gives the celestial route the same two-step shape as MMST (limit + maximal ideal); the operator content is exactly WP5b–e, of which WP5b–WP5d and the WP5e α, β, γ, δ_2 and ε_2 stages are executed below.
- **WP5b (*done*) [E]/[C]:** "the singular vector as an operator", executed and machine-verified as `CELEST.WP5B.01` [`verification/v498_celestial_wp5b_singular_vector.py`] (exact integer/Fraction arithmetic, 53 checks, deterministic) — *success on the preregistered criterion*: the deleting object exists and is explicit. $|s\rangle = (E^\theta_{-1})^2|0\rangle$ (weight 2θ, level $2 = 8$ quarter units $= q^2$, an *integer* level) is constructed in a machine-built Chevalley/Frenkel–Kac basis (cocycle asymmetry on all 57600 pairs; the $[e_\alpha, e_{-\alpha}]$ sign *forced* by Jacobi, SGN $= -1$; κ derived, $\kappa(\theta^\vee, \theta^\vee) = 2$), and $J^a_1|s\rangle = 0$ is machine-verified for *all* 248 generators with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$, the central-term cancellation — the *only* case that sees the level k); plus $J^a_2|s\rangle = 0$, $E^a_0|s\rangle = 0$, exact weight and Shapovalov norm 0. *Level dial*: the F^θ_1 coefficient is $2(1{-}k)$ — nonzero at k=0 and k=2, zero exactly at k=1: without the central extension the deletion operator does not exist. μ_4 compatibility: glue class $j(\theta) = 1$ (machine-built h-adapted chamber, $\langle \theta, h\rangle = 5$, height $29 = h^\vee{-}1$), clock phase $i^{2j} = -1$, class$(2\theta) = 2$ sheet-even (WP5a-consistent) with the deleting θ-$\mathfrak{sl}_2$ crossing the sheet-odd classes $(1, 3)$; 8 quarters $= q^2$ via the per-period dictionary. The module it generates is *the* ideal: weight 2θ has multiplicity 1 in the level-2 Fock, and the direct $\mathfrak{g}_0$-lowering orbit BFS from $|s\rangle$ reproduces the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ($27000 = (h^\vee)^3$, quotient 4124; [C] Weyl complete reducibility beyond depth 4). Negative controls separate honestly: the level-1 current state and generic level-2 states are *not* singular; at k=2 the *cube* is (the $(E^\theta)^{k+1}$ pattern is generic Kac level mechanics); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$

breaks one-block fusion) — the singular-vector mechanism is level-1 *generic*, the *one-block closure* is the E_8/μ_4-specific part. *Handover to WP5c* (answered below): in the twisted quarter-*slot* moding of the χ_w limit Fock, two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-*period* dictionary (v496), not the per-slot identification, carries $|s\rangle$ to q^2; exactly the GNS/limit-state question (kernel $\supseteq$ ideal) that WP5c answers.

– **WP5c (*done*) [E]/[C]:** "the GNS limit state", executed and machine-verified as `CELEST.WP5C.01` [`verification/v500_celestial_wp5c_gns_limit_state.py`] (exact integer/Fraction arithmetic, 35 checks, deterministic) — *success on the preregistered criterion*: the family ω_w exists and its limit has the null ideal in its GNS kernel. ω_w is the quasi-free state with loop sector = the affine k=1 vacuum n-point functions (via the machine-determined compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$, the unique anti-automorphism sign on all 61504 basis pairs) and radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x-adic); *the WP5a threshold holds at the state level*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N{+}1$, sharp at $w = N$. The core is fully machine: the *complete* exact level-2 Gram — 31124 monomials in 9361 weight blocks — has rank 4124 exactly (the preregistered target), kernel $27000 = V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights), every block PSD, rank table per Weyl orbit $(0, 0, 1, 8, 44)$, level-1 rank 248 positive definite (the current layer survives). μ_4: the clock descends to GNS with level-2 rank split $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 — the WP5a two-routes identity at the *state* level — and $|s\rangle$ *is* the zero vector of $\mathrm{GNS}(\omega_\infty)$, resolving the WP5b twisted-slot tension. The KILL branch is probed and not triggered: a CCR obstruction shows *no* w-uniform state can damp the radial modes (the family formulation is *necessary*), and the wrong constructions are caught (family $E'_w = mw{+}r$ erases the 248 layer, $710955 \neq 248$; no damping keeps $897266 \neq 248$); k-dial and D_8 controls separate honestly (k=2: $\langle s|s\rangle = +4$, no ideal in the kernel; D_8: one block of four, $h = \frac{1}{2}$). [C] quasi-free extension to the full $*$-algebra; Kac positivity beyond level 2.

– **WP5d — both stages (*done*): α [E]/[C], β [E]/[C] (continuum uplift [O]):** "the local net", approached from the lattice. The α stage — the KLM *two-interval index* as the operational Haag-duality discriminator ($(E_8)_1 \Rightarrow \mu$=1, one sector; $SO(16)_1 \Rightarrow \mu$=4) — is executed and machine-verified as `CELEST.WP5D.01` [`verification/v501_celestial_wp5d_two_interval_index.py`] (39 checks; Gaussian lattice machinery ED-validated to 10^{-15}, exact Fractions for the arithmetic). Measured on the 16-layer seam carrier (critical Majorana ring, $c_- = 8$): the fermionic two-interval mutual information is *extensive* (the μ=1 reference; c fit 0.5000, residual $\to 0$ with N), while the sector-summed (orbifold) prescription pays exactly one classical bit — the $\ln 2$ plateau at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at N=512), so $[F : F_{\mathrm{even}}] = 2$ and $\mu_{\mathrm{gauged}} = 4$, the entropic twin of the v490 parity census (two independent lattice witnesses); the orbifold also breaks two-interval complementarity ($S(E) \neq S(E')$, the direct duality-failure witness), with the complementary-pair budget bounded by $\ln 4 = \ln \mu(SO(16)_1)$ as a *double-limit* statement. The condensation arithmetic is anchored at both measured ends: $\det \mathrm{Cartan}(D_5) \cdot \det \mathrm{Cartan}(A_3) = 16$, KLM/Longo–Rehren $16/4^2 = 4/2^2 = 1$, $\Sigma d_i^2 = (4, 4, 1)$, $\theta_v = 1$ exactly at $\nu = 2c_- = 16$ (rivals $\neq 1$) — so $\mu = 1$ after condensation and the preregistered KILL ("μ-offset $\neq 0$ after condensation") is *not* triggered; the ν=1 control shows a non-removable offset ($\theta_v \neq 1$: the discriminator has teeth), the trivial phase shows none, and the wrong sector sum loses the full $\ln 2$. [C] the continuum dictionary "offset = ln index" (KLM; Longo–Xu; Casini–Huerta–Magan–Pontello). **The β stage (*done*) [E]/[C] — split + strong additivity, the two remaining KLM legs** — is executed and machine-verified as `CELEST.WP5DB.01` [`verification/v504_celestial_wp5d_klm_completeness.py`] (37 checks; Gaussian lattice machinery ED-validated, exact GF2/integer algebra for the exact legs;

the $U(1)$/Dirac number-conserving subnet as the preregistered negative control — the current-net analogue where strong additivity famously *fails* in the continuum). *Strong additivity is algebraically exact on the lattice*: with the shared boundary Majorana ("touching = share the point"), $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ *exactly* (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32), while disjoint intervals give exactly *half* (rank 16/32, index 2 — the missing sector is odd $\otimes$ odd, the single $\ln 2$ bit of the α stage, localised at the split point); the $U(1)$/neutral algebras do *not* generate the union even with the shared site (gaps 2/10/52, growing). The entropic witness uses the honest discriminator *bounded vs divergent*: the $\mathbb{Z}_2$/orbifold touching deficit stays $< \ln 2$ with the Ising approach $(\ln 2 - \Delta_2) \sim N^{-p}$, $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (the naive "defect $\to 0$" expectation is *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index, Longo–Xu), while the $U(1)$ control *bursts* $\ln 2$ from $L = 128$ on and grows as $\frac{1}{2} \ln \mathrm{Var}\, Q_A$ (Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$: infinite index). The split property is witnessed quantitatively: the coupling-block singular values fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% over $d = 16..512$, $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$), trace norm summable — and the orbifold *inherits* the same ladder exactly at machine level (P_A flips $C \to -C$, σ_k identical to 8.3×10^{-17}; the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ with singular values $\{\sigma_a \sigma_b\}$ — Longo heredity, [C]). The Pimsner–Popa bound is a one-line *identity*, $E(a) - a/2 = PaP/2$, so $\lambda = \frac{1}{2} = 1/[F : F_{\mathrm{even}}]$ exactly, attained in exact integers ($16384 + 2048$ monomial sweeps, 0 violations; random pencil $\min \lambda = 0.5000000000$); the two-interval group gives $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ exactly, and the index is pinned by *two independent routes* ($e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\mathrm{PP}}$; $e^{2\Delta_\infty} = 4 = 1/\lambda_{E_4} = \mu$); the $U(1)$ control has $\lambda = 1/(m{+}1) \to 0$. *With the α stage, all three KLM ingredients of complete rationality — split, strong additivity, finite μ — are now witnessed on the lattice.* The continuum uplift stays **[O]** and is honestly fenced: the theorem "finite-group orbifolds of completely rational nets are completely rational (hence split + strongly additive)" is *Xu's theorem* (algebraic orbifold CFTs) — cited, *not* claimed; the continuum proof for the concrete seam quotient net on the collar (v336/v449) and for the interacting condensed $(E_8)_1$ net remains WP5e / Costello–Li territory.

– **WP5e (subdivided): α stage (*done*) [E]/[C] — the CFT-side prefactor + level pinning; β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor space, including the honest a_0 refutation; γ stage (*done*) [E]/[C] — the sphere-axion pairing check, a rigid *negative* result with certificate; δ_2 stage (*done*, intermediate verdict, exact) [E]/[C] — the full-tensor ledger: the collapse confirmed, one cubic door opens; ε_2 stage (*done*, verdict B) [E]/[C] — the CPS level-from-flux dial, with "one level" lifted to a theorem; the global BCOV quantisation stays [O].** The α stage is executed and machine-verified as `CELEST.WP5E.ALPHA.01` [`verification/v502_celestial_wp5e_prefactor_level.py`] (exact sympy/Fraction arithmetic, 33 checks). Both exactly computable consistency anchors of the uplift hold. (i) *The $q^{-1/3}$ prefactor is exact μ_4 vacuum-energy bookkeeping*: the clock is *inner* ($h = (2,2,2,2,2;0^4)$) and the A_3 detector $h' = (0^5; 1,1,1,-3)$ both read the glue class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32), so the induced twist on the 8 torus bosons is $\theta = 0^8$ in *every* sector — a *shift* orbifold, not a rotation orbifold — and each sector carries the same $8 \times (-\frac{1}{24}) = -\frac{1}{3} = -c/24$ (Hurwitz-ζ exact, $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$); the sector ground exponents are $-\frac{1}{3} + (0,1,1,1)$ with h_j = the discriminant form, an identity holding via spectral flow ($j^2(h,h)/32$, mod 1) *and exactly* via the $10 + 6 = 16 = 2^{g_{\mathrm{car}}-1}$ Majorana free fermions of the WP5d seam carrier (R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); the four sector characters sum to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$; the rotation reading *fails* (the geometric A_3 deck gives $\frac{3}{16} \neq \frac{3}{8}$ — the WP1 "clock $\neq$ deck" lesson at the Casimir level). (ii) $k = 1$ *is forced three independent ways*: the current condition $h(J) = k = 1$ (the $\mathbb{Z}_4$ extension closes onto the E_8 adjoint $60{+}64{+}60{+}64 = 248$ only at spin 1); the conformal embedding ($D_5{+}A_3 \subset E_8$: $47k^2{+}219k{-}266 = 47(k{-}1)(k{+}266/47) = 0$; $D_8 \subset E_8$: $128k(1{-}k) = 0$); and the central charge $248k/(k{+}30) = 8 \Leftrightarrow 240(k{-}1) = 0$ (= the prefactor itself); plus the WP5b singular-vector dial ($\|(E^{\theta}_{-1})^m 0\|^2 = m!\,k(k{-}1)\cdots(k{-}m{+}1)$, singular vector exactly at grade $k{+}1$; $k = 1$ deletes $V(2\theta)$: $31124 - 27000 = 4124$). *Honest sharpening*: glue-h *integrality* $h(J^j;k) = k\,(0,1,1,1)$ holds for *all* $k = 1..8$ and fixes nothing — the originally conjectured integrality route is dead; the three other dials carry. Controls: $D_8/SO(16)_1$ has the *same* $q^{-1/3}$ prefactor but $h = (0,\frac{1}{2},1,1)$ (the discriminator is glue-h integrality, not the prefactor); $\mathbb{Z}_2$ or μ_4 *rotation* twists break the common prefactor; wrong $k \in \{2,3,4\}$ fails all five dials coherently. **The β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor space** — is executed and machine-verified as `CELEST.WP5E.BETA.01` [`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`] (exact sympy/Fraction arithmetic, 47 checks). The twistor side is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on $\mathbb{C}^2/\mathbb{Z}_4$ with the μ_4-glue monodromy. (i) *The AB ledger is exact*: denominators $\det(1-g_j^{-1}) = (2,4,2)$ with Dedekind sum $5/4 = (|\mathbb{Z}_4|^2{-}1)/12$; equivariant characters $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248,0,-8,0)$ by two independent routes; invariant average $60 =$ the carrier; Frobenius 61568. (ii) *Okubo per sector, the honest sharpening*: only the *invariant* sector is Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}^2_{e_8}$ — the v495 identity re-derived through the glue grading); the twisted sectors carry irreducible T_5/T_3 content, and the AB-weighted sum cancels the D_5 quartic *exactly* while leaving the *rigid* residual $32\,T_3$ (no admissible reweighting fixes it; the graded GS exchange is rank-obstructed in sectors 1–3) — twisted-sector closed strings must carry the rest **[O]**. (iii) *The index bridge* (the centrepiece): $f(m) = \frac{1}{4}\sum_j (i^{jm}{-}1)/\det_j = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ *exactly* — the fixed-point ledger and the McKay/Kronheimer intersection ledger are the same arithmetic; the glue-bundle ch_2 defect is the integer -78 by *both* routes (classes $(-24,-30,-24)$). (iv) *The level dials say $k = 1$ geometrically*: the lattice current count is 240 at $k{=}1$ and exactly 0 at $k = 2,3,4$; the embedding residual $47k^2{+}219k{-}266 = (0,360,814,1362)$; $h(J;k) = k$ is integer for *all* k (integrality alone fixes nothing — honest, exactly as on the CFT side); one scale $\Rightarrow$ one level (v493 periods; since the ε_2 stage a *theorem* of clock invariance, v509). (v) *The honest refutation*: the clock-invariant modulus $a_0 \in O(8)$ fills the BSS *graviton* slot $O(2)$, *not* the axion slot $O(-2)$ (Hamiltonian weights $X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$, typed [C] as a coincidence) — “the theory brings its own GS axion as a_0” is *refuted*; what does match: the three $H^2(\mathrm{ALE})$ classes carry exactly the three twisted-sector Coxeter characters $\{i,-1,-i\}$ (bijection, no invariant class), so the bulk axion must come from the $O(-2)$ tower field itself (construction **[O]**). Controls: $\mathrm{diag}(i,i)$ breaks the ledger four independent ways; $SO(16)$ glue gives characters $(120,0,+120,0)$ and defect $-30 \neq -78$ with a *failing* bulk Okubo (independent quartic Casimir); $k = 2$ dies on the closure dial ($-31/48 \neq -\frac{1}{3}$) while naive integrality dials stay integer. The preregistered kill does *not* fire on the equivariant skeleton. **The γ stage (*done*) [E]/[C] — the sphere-axion pairing check, an honest rigid *negative* result** — is executed and machine-verified as `CELEST.WP5E.GAMMA.01` [`verification/v508_celestial_wp5e_gamma_pairing.py`] (exact sympy/Fraction arithmetic, 27 checks). Question: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i,-1,-i\}$ on $H^2(\mathrm{ALE})$, the β-stage slot bijection) cancel the rigid $32\,T_3$ residual ($A_{\mathrm{fix}} = 9P_1{-}30P_2{-}15P_3{+}32\,T_3$) by Green–Schwarz-type *exchange* with sector-compatible quadratic vertices? *No, with a certificate.* (i) *Vertex-space theorem*: the $W(D_5){\times}W(A_3)$-invariant quadratics on the 8-dim glue Cartan are *exactly* $\mathrm{span}\{s_5, s_3\}$ (dim 2) and the invariant quartics *exactly* $\mathrm{span}\{P_1,P_2,P_3,T_5,T_3\}$ (dim 5) — complete by Weyl nullspace arithmetic over all bidegrees; the *product theorem* is the master kill: the product of any two invariant quadratics has identically zero T_5/T_3 content, so every exchange image — any axion count, any selection rule, any couplings — is annihilated by Φ_{T_3} while $\Phi_{T_3}(A_{\rm fix}) = 32 \neq 0$. (ii) *Selection enumeration*: the strict two-index $\mathbb{Z}_4$ rule collapses (the class bilinears are nonzero only for $j' = -j$, forcing $m = 0$: the sphere axions have *no* invariant bilinear Cartan vertex at all); the generous twist-insertion channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$ give a rank-2 exchange matrix with $\mathrm{rank}([M \,|\, A_{\rm fix}]) = 3$ and annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\rm fix}) = (0, 32, 72)$ — two independent obstructions, the second driven by the side discovery $K^{(0)} = -15\, K^{(2)}$ (the even-sector quadratics are parallel); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$ (the sphere axions are rigidly *silent*), and the sector-0 Okubo mechanism is undisturbed ($\Phi_P(Q^{(0)}) = 0$). (iii) *Naturalness dissolves*: ch_2-natural and Atiyah–Bott-weight couplings both carry certificate $(0, 0)$ against the required $(32, 72)$ — the failure is scale- and coupling-independent (the ansatz space misses the target identically, not by an unnatural factor). Controls: the flipped rule changes the enumeration but not the T_3 kill (Φ_P is rule-dependent, Φ_{T_3} is not); $SO(16)$ has *no* sphere partners (odd McKay classes empty) *and* uncancelled T_5 ($(15, -18, -15, -4, 40)$ — E_8 is doubly special); D_8 has no T_3 structure at all ($16\, T_8 + 12\, s_8^2$). Number fences typed [C] only: $32 = (h, h) + (h', h') = 20 + 12$, $72 = 2\tilde{\lambda}_{e_8}^2$. The β-stage slot bijection is untouched and the preregistered level-kill still does *not* fire — what dies is the exchange *realisation* of the pairing. **The ε_2 stage (*done*, verdict B) [E]/[C] — the CPS level-from-flux dial** — is executed and machine-verified as `CELEST.WP5E.EPS2.01` [`verification/v509_celestial_wp5e_eps2_level_from_flux.py`] (exact sympy/Fraction arithmetic, 28 checks). Target: $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). (i) *The CPS skeleton is replicated exactly*: the S^3 period of the brane back-reaction 3-form is $(2\pi i)^2 N$ (pullback density $-\sin 2\theta$, their eq. 3.33 — large-gauge invariance quantises N); the exceptional-sphere Kähler flux is $2\pi N$ exactly; the boundary level magnitude is $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$), Dynkin normalisation $T(\text{vector}) = 1$ — "level = flux quantum × Dynkin index", *linear* in the flux. (ii) *The pairing matrix is pinned, not postulated*: the β-stage ch_2 ledger + integrality + unimodularity + effectivity force $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by complete enumeration (row candidates 8/6/8; 48 unimodular solutions; with $P \geq 0$ exactly 2: identity + the A_3 diagram flip), so the glue fluxes are $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ (total $188 = 248 - 60$, flip-invariant) and every charged root current carries *exactly one* flux quantum through exactly one sphere. (iii) *The level formula, honest*: "level = total open-string flux" is *killed by the lockstep test itself* ($k_i = (64, 60, 64)$ unequal — F_i is the flavour multiplicity, the brane matter content); what works is $k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)$, anchored non-circularly by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ and the embedding index $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$ ($60 = 2h^\vee$). The ord-4-vs-level-1 tension is *resolved*: the clock order counts *fractional flux sectors*, not the level ($\mu = 16 = \mathrm{ord}^2$; the μ_4 glue diagonal is isotropic and Lagrangian; KLM $16 \to 16/4^2 = 1$, zero residual fractional flux), and a *new dial* pins the value: $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly *one* sector iff $k = 1$, independent of the CPS transplantation. (iv) *The lockstep theorem* (the lifting of the β-stage "one scale $\Rightarrow$ one level" to a *theorem*): clock invariance forces $a_3 = a_2 = a_1 = 0$ sharply, the four branch points are one μ_4 orbit, $|\Pi_j| = \sqrt{2}\, t$ is *equal* on all three spheres, the monodromy acts as the pure phase i and $\det(A - 1) = -4 \neq 0$ (no invariant flux vector) — with the CPS flux-level dictionary, $k_1 = k_2 = k_3$ *is a theorem* of clock invariance; the new falsifier: the clock-forbidden family $Z^4 - Z$ (smooth, disc -27, branch points not a μ_4 orbit) gives *zero* lockstep chains over all 24 orderings. Honest limit: uniform flux doubling keeps the lockstep — the theorem proves *equality*, not the value; the value 1 comes from the current/sector counters and the CPS minimal quantum $N = 1$. Controls separate honestly: $k = 2$ dies on the closure dials (count 0, residual 360, prefactor $-31/48$,

#prim 3) while the lockstep dial honestly does *not* fire; $SO(16)$ glue leaves spheres 1/3 flux-dark with condensation stuck at 4; the A_2 form has $\mu = 12$ (no Lagrangian subgroup exists at all) and Coxeter order $3 \neq 4$. Typed honestly: the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ stays [C] (their geometry is the $\mathbb{C}^2$ blow-up with one (-1)-sphere and $\mathfrak{so}(8)$ at negative level — not the A_3 ALE); the type-I B-model back-reaction on $\mathbb{PT}/\mathbb{Z}_4$ stays **[O]** only in its δ_{1d} derivation face (all three milestones M1–M3 have since been executed, v515/v516/v517; the δ_1 chain itself has since been *decided* by v518 — kill under the derived measure — and the declared/derived measure question has since been decided at probe level in favour of the declared reading under the typed premises TP-1..TP-4, v520; the residual **[O]** is the constructive w_m derivation). **The δ_2 stage (*done*, intermediate verdict, exact) [E]/[C] — the full-tensor ledger: the collapse is real, and one cubic door opens** — is executed and machine-verified as `CELEST.WP5E.DELTA2.01` [`verification/v511_celestial_wp5e_delta2_tensor_ledger.py`] (exact Kostant/Weyl weight arithmetic + sympy/Fraction, 41 checks). The γ-stage collapse was a *Cartan-restricted* statement; δ_2 recomputes the ledger on the full adjoint tensor structure. (i) *Structure*: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ is *semisimple* with no $\mathfrak{u}(1)$ factor (class-0 roots split 40+12, root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$, each irreducible; and the *innerness theorem* decides the collapse wholesale: the grading element h lies in the Cartan *of* $\mathfrak{g}_0$, so every $\mathfrak{g}_0$-invariant tensor carries total sector charge 0 mod 4 — the γ-stage collapse is a *full-tensor* statement *and* holds across arities (all 15 non-neutral trilinear sector triples have Hom = 0: the sphere axions have no invariant partner operator at any arity ≤ 3). (ii) *The Hom tables*: bilinear invariants exist only at $j' = -j$ (dims 2/1/1/1, the Killing blocks); the neutral trilinear multisets carry dims $(3, 2, 2, 1, 1)$, all cross-sector trilinears are pure brackets (antisymmetric in their repeated legs) — the *only* totally symmetric trilinear on all of $\mathfrak{e}_8$ is the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir: $\mathrm{Inv}(S^3\,45) = 0$ protects T_5). (iii) *The new channel*: the quartic projection of the d-symbol exchange is $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$ by two independent routes (Killing propagator $2h^\vee = 60$ from the roots; hand route $|x^2 - (\mathrm{tr}\,x^2/4)\mathbf{1}|^2/60$), and $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the γ-stage master kill (the product theorem) covers *quadratic* vertices only and does not extend to cubic ones: the T_3 direction *is* reachable through the d-symbol channel, while brackets, tadpoles and mixed $d{\times}f$ die exactly and no T_5 analogue exists. (iv) *The new certificate*: $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ has rank 3, augmented rank 4 — the pairing in the charge reading remains *obstructed*, but with the genuinely *weaker* certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\mathrm{fix}}) = 72 - 8 = 64$; and the relaxed reading (charge bookkeeping dropped on the bilinear P-block) becomes *exactly solvable*: $A_{\mathrm{fix}} = -u + 8v + 2w + 1920\,Q_{dd}$ (residual 0) — where the γ stage found $T_5 = T_3 = 0$ persisting even relaxed, the d-channel closes the T_3 gap with small integer coefficients. Number fences typed [C] only: $64 = \dim\mathfrak{g}_1 = 2^6$; $1920 = |W(D_5)| = 8 \cdot 240$ — and the 1920 fence has now been stress-tested and *fails as a derivation claim* ([`verification/v513_celestial_dterm_nonderivation.py`], `CELEST.DTERM.NONDERIV.01`): the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded (11/924 pre-declared catalog expressions hit 1920, vs. 8/924 for the control target 1800) and convention-contingent (only 1 of 5 normalisations produces 1920: Gell-Mann gives 120, $\mathfrak{su}(4)$-Killing 256, plain trace 32, ch_3-normalised 69120); the charge-legal flux pairings give 2048/1800 — they bracket and miss — while the only flux hit is the charge-forbidden $(1, 2)$ pairing; quantisation delivers only lattices ($60\,\mathbb{Z}$, with 1920 the 32nd multiple; strict ch_3 units exclude it); and the twisted cubic index $32i$ clashes in class provenance with the true quartic 32 (classes 1/3 vs. 0/2: same number, disjoint mechanism). The convention-stable **[E]** core is the factorisation $c_d = \Phi_{T_3}(A_{\mathrm{fix}}) \times 2h^\vee(E_8) = 32 \times 60$; the fence stays [C], the physical generation of the 32 stays **[O]** — no marker moves. Controls: the Killing control replicates the γ-stage channels and A_{fix} exactly; $SO(16)/D_8$ has *no* symmetric cubic ($\mathrm{Inv}(S^3\,\mathrm{adj}_{D_8}) = 0$), no odd sectors and no A_3 block — the δ_2 channel cannot even be posed (E_8/μ_4 doubly special); the false $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$ inflates every Hom dimension ($(5,2,2) \neq (2,1,1)$, $\mathrm{Inv}(S^3\mathfrak{g}_0) = 6 \neq 1$) and the false sector content $(16_s, \bar{4})$ kills every pairing. What stays open after δ_2 [O]: the arity ≥ 4 ledger, the *physical* justification of the cubic GS term ($c_d = 32 \times 60$, the convention-stable core; the $1920 = |W(D_5)|$ reading is look-elsewhere-loaded and convention-contingent, v513; ch_2-naturalness for cubic vertices *undecided* — no canonical geometric datum exists for a cubic vertex), and the burden on δ_1 is *sharpened*: it must supply either the $\psi = 64$ slice or the selection-rule relaxation — a burden since *discharged* by the M2 milestone: the declared KS measure supplies the $\psi = 64$ slice exactly (v516), so the cubic d-channel is not needed (c_d free $= 0$) and its physical-justification question stays dissolved — the δ_1 decision (v518) does *not* revive it (no branch fires under the derived measure). **The δ_1 stage (*done*, DECIDED — kill under the derived measure, the $\mathbb{Z}_2$ anchor a method boundary)** [E]/[C] — is executed and machine-verified as `CELEST.WP5E.DELTA1.01` [`verification/v518_celestial_wp5e_delta1_derived_kill.py`] (exact $\mathbb{Q}(\zeta_8)$/phase arithmetic + 30-digit kernels, 30 checks; the $\delta_{1b}/\delta_{1c}/\delta_{1d}$ probe chain consolidated). The exploration stage had left the verdict undecided with four exact sharpenings — (i) the strict holomorphic q^0 reading *refuted at the $\mathbb{Z}_2$/Eguchi–Hanson anchor* (a method boundary, not a kill of the contract); (ii) the forced leading (T_5, T_3) ratio 4:3; (iii) the contact term = the *modular completion* of the Atiyah–Bott data (a Harvey–Moore-type τ-integral, orbit split $15 = 12 + 3$); (iv) massive coset levels $n \geq 2$ mandatory. The τ-integral has now been *evaluated under a derived measure*, and the route is decided. Four exact results: (i) *the 16-component Weil completion closes*: the discriminant module of $D_5 \oplus A_3$ is $\mathbb{Z}_4 \times \mathbb{Z}_4$ with $q = (5x^2 + 3y^2)/8$, Gauss sums $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$ (signature 0 mod 8 = rank E_8), the exact Weil relations hold in $\mathbb{Q}(\zeta_8)$, the diagonal and anti-diagonal Lagrangians are the two E_8 gluings, and the 16-vector theta S-covariance drops the naive-4-rule residual from $2.91 = O(1)$ to $\sim 10^{-39}$; (ii) *the obstruction is a finite μ_4 character, identified*: the dressed gauge blocks $M = G[a,b]\,\eta^{-8}$ carry a multiplier system whose $SL(2,\mathbb{Z})$ relation defects are $(1,1,1)$ exactly on all 15 pairs (a *character* of the orbit stabilisers $\Gamma_1(4)/\Gamma_0(2)^{\pm}$, not a genuine 2-cocycle), on $\Gamma_1(4)$ explicitly $\lambda(\gamma) = i^{2B+C/4}$; (iii) *the cancellation exists and is the twisted fibre block $f_1 f_3$, not the three sphere axions*: $G[a,b] = f_1 f_3$ is an exact identity, the T-fix mechanism is one line of exact phase arithmetic ($(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$), and the exact cancellation table reads none $\to 4$, $f_1 f_2 f_3 \to 4$, $f_2 \to 6$, $f_1 f_3 \to 1$ (both orientations), wrong weights $\to 4$ — the strict solutions are exactly χ_4 (dims $(3,3)$) and χ_{10} (dims $(1,1)$), certified on the dressed functions; (iv) *all three preregistered testers fail* under both derived solutions (T_5 fractions 0.5/0.83 instead of 0; $W_{13} = 0$ instead of $W_2{:}W_{13} = 4{:}3$; spreads 1.05/1.47 against $-A_{\text{fix}}$; no $\psi = -64N$ slice), and the honest (N_1, N_2) orbit rebalance rescues nothing in the positive cone — a genuine KILL on the derived surface. Controls: the $\mathbb{Z}_2$/EH anchor (gauge-only residual order 2, cancelled to 1 by the same fibre dressing), $SO(16)$ (order-4 supply structurally absent), wrong form/wrong signature (break exactly). Fences [C]: the $f_1 f_3$ = KS-weight identification (deck weights $(1,3)$ = the twistor fibre weights forced by the v514 incidence ledger — exact match, *not* a complete BCOV derivation), the kernel-family convention, the discrete χ_k choice (both evaluated, both fail). TENSION, stated honestly (the mandatory fence): the DECLARED completion reading (v516/M2) delivers the $\psi = 64$ slice and cancels $32\,T_3$; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question was which of the two is the true BCOV measure (the declared reading is supported by the δ_1 modular-completion finding but not derived; the derived measure rests on the $f_1 f_3$ identification). Neither result is hidden behind the other. That question is since *decided at probe level* by v520 in favour of the declared reading (single-valuedness derived from

F-independence + the Quillen pairing under the typed premises TP-1..TP-4; the kill of this module is *sharpened*: the column-canonicalised χ_4 member is boundary-consistent yet tester-dead); and the constructive BCOV derivation of the w_m normalisation itself is since *executed* by v523 (verdict ERFOLG; the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor). **The ε_1 stage (*done*) [E]/[C] — the $O(-2)$ bulk-axion slot** — is executed and machine-verified as `CELEST.WP5E.EPS1.01` [`verification/v514_celestial_wp5e_eps1_axion_slot.py`] (exact sympy/Fraction arithmetic, 34 checks, verdict B). Three results, each exact. (i) *The slot is a construction*: on $\mathbb{PT}' = \mathrm{Tot}(O(1)\oplus O(1) \to \mathbb{P}^1)$ the pushforward ledger $\pi_* O(-2)$ gives per spacetime degree d exactly $(d+1)^2$ classes = the exact wave-operator nullspaces, and the *equivariant* bookkeeping under the v492 action $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$ (incidence-forced column weights $(+1, +1, -1, -1)$) closes *block by block* for all $d \le 6$ and all four characters; the character series are $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \dots$, $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \dots$, $P_2 = 6t^2 + 10t^4 + 28t^6 + \dots$, the $d = 0$ slot has multiplicity $(1, 0, 0, 0)$ — *the bulk axion survives the projection* — and the Molien invariant ring is $(1 - t^8)/((1 - t^4)^2(1 - t^2))$: $Z \in O(2)$, $X, Y \in O(4)$, one relation in degree 8 = the a_0 weight (the v492/v493 bridge); the E3 bijection is sharpened per character (twisted minimal content $(2t, 6t^2, 2t)$, no degree-0 mode, = the Coxeter eigenvalues, $\det(A - 1) = -4$), and the graviton control separates a_0 (the $O(+2)$ slot starts invariant only at fibre degree 4 with multiplicity $3 = \{X, Z^2, Y\}$). (ii) $\tilde\lambda = 6$ *is triply pinned*: Okubo $\mathrm{Tr}_{\mathrm{adj}}\, X^4 = 36\langle x, x\rangle^2 = (6\langle x, x\rangle)^2$ on the 240 glue roots ($36 = h^\vee + 6$, $\tilde\lambda = 6 = |\mathbb{Z}_2|N_{\mathrm{fam}}$); the measure chain on $\mathbb{PT}/\mathbb{Z}_4$ cancels the quotient factor $\mu = 1/4$ *exactly* (vertex^2 μ^2, propagator $1/\mu$, anomaly μ: $\tilde\lambda^2_{\mathrm{eff}} = 36$ for arbitrary μ), with the wrong bookings quantified and *excluded* (anomaly-only $\Rightarrow \tilde\lambda = 3$, missing propagator renormalisation $\Rightarrow \tilde\lambda = 12$); and the flux side (minimal CPS quantum $N = 1$, a single exchange channel exactly at $k = 1$, $(\kappa/c_3)^2 = 12$ exact). (iii) *The first honest back-reaction step (Gibbons–Hawking A_3)*: clock-invariant four-centre configurations form exactly two branches (axis^4 = pure resolution or *one* free μ_4 orbit = pure deformation); the free orbit gives $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$, i.e. $XY = Z^4 + a_0$ with $a_0 = -z_0^4$ — the v493 family *re-derived from the geometry*, with the a_0 monodromy = one clock step; the charge-k GH point is exactly $\mathbb{R}^4/\mathbb{Z}_k$ ($k = 4$: the seam boundary $S^3/\mathbb{Z}_4$); the periods are $\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)$ with equal moduli (lockstep, the v509 counter-check from geometry) and the Coxeter clock re-derived ($A^4 = 1$, $\det(A - 1) = -4$, $\Pi \cdot A = i\Pi$); the CPS "$N \log$" analogue carries source charge $4 = |\mu_4|$, with the honest sharpening that the Ricci-flat ALE has asymptotic log coefficient exactly 0 (the CPS log is an exceptional-locus statement; Burns contrast $\det g \ne 1$), and the multipole selection rule $m \equiv 0 \bmod 4$ stores $-a_0$ in the first symmetry-breaking $(4, \pm 4)$ harmonic. Controls: $\mathfrak{so}_8$ ($\tilde\lambda^2 = 12$ irrational; perfect squares in the Deligne series only $\{9, 36\}$), $\mathrm{diag}(i, i)$ (Veronese cone, no hypersurface), $k = 2$ (three channels). [C]: conditional on Costello's flat-$\mathbb{PT}$ matching $\lambda_g^2 E = A$ (the measure chain is classical bookkeeping). [O] (the honest fence): the *quantised* BCOV one-loop coefficient on $\mathbb{PT}/\mathbb{Z}_4$ and the twisted channels ($32\, T_3$) — preregistered as M1–M3 below; all three have since been executed (v515/v516/v517), the δ_1 chain has since been *decided* by v518 (kill under the derived measure), the measure question has since been decided at probe level by v520 for the declared reading, and the w_m normalisation has since been *derived* at probe level by v523; what stays [O] is the global BCOV integral beyond the fibre zero-mode factor. **WP5e proper [O] (the global BCOV quantisation):** the BCOV/Kodaira–Spencer theory on $\mathbb{PT}/\mathbb{Z}_4$ itself (quantisation, non-fixed-point contributions, the twisted closed-string measure). The exchange sub-branch is *closed* by the γ stage, the full-tensor ledger (δ_2), the level-from-flux dial (ε_2) and the bulk-axion slot (ε_1) are *executed*, and δ_1 is *decided* by v518 (kill under the derived measure); the named remaining target, with preregistered success/kill criteria: **(the measure question — the δ_1 successor; *decided at***

***probe level,* ERFOLG-A)** the δ_1 τ-integral had been evaluated on *both* sides: the M2 milestone (v516) *delivers* the coefficient $(9, -30, -15, 0, 32)$-completion exactly on the *declared* measure (every sector the same Okubo square, the $\psi = 64$ slice supplied), while the *derived* chiral measure (v518: blockwise $SL(2, \mathbb{Z})$ covariance solved for, the μ_4 multiplier character cancelled by the twisted fibre block $f_1 f_3$) fails all three preregistered testers — both are exact, and the stated tension was the sharp remaining question: *derive* from the BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ which of the two measures is the true one. That decision is executed and machine-verified as `CELEST.WP5E.MEASURE.01` [`verification/v520_celestial_wp5e_measure_decision.py`] (the δ_{1e}/δ_{1f} consolidation, 39 checks, ERFOLG-A on the preregistered decision layer): the exact invariant subspace of the dressed 16-component Weil system is $\dim_{\mathbb{Q}}\{v : \rho(T)v = v,\ \rho(S)v = v\} = 8 = 4 \times 2$ with *every* kernel vector in the $\mathbb{Q}(\zeta_8)$-span of $\{e_H, e_{H'}\}$ (both theta $= E_4 =$ the unphased trace), and single-valuedness of the physical one-loop lattice factor is *derived* from two independent constructive sources rather than postulated — *(source 1, the F-lemma)* every strict chiral closure at physical weight carries a nontrivial character ($\chi_4(T) = e(1/3)$, $\chi_{10}(T) = e(5/6)$, exact; pointwise covariance of the canonical column-matched member certified at 3.9×10^{-16} with $|G| \geq 59.6$), and an exact change of variables forces $\int_{\gamma F} G\, d\mu = \chi_4(\gamma) \int_F G\, d\mu$, so fundamental-domain independence forces $\int = 0$ for *every* chiral integrand: the v518 route was never a well-defined nonvanishing moduli integral (the kill is *sharpened*); *(source 2, the Quillen pairing)* the doubled hol×antihol transports satisfy $|t\bar{t} - 1| < 8.3 \times 10^{-40}$ on all 15 pairs ($|\chi_4|^2 = 1$ exactly vs one-sided $\chi_4^2 = e(2/3) \neq 1$), the doubled 256-dim system closes *strictly* at trivial character and physical weight (nullspaces $(28, 17) \geq 11$) containing the unphased diagonal *and* the physical columns $w_b \otimes \bar{w}_b$, while the hol⊗hol control is *empty* $(0, 0)$; the forced unphased numerator then reproduces the v516 Okubo squares *levelwise* inside the v518 scaffold (exact 4-design on every level $n \leq 8$, J-weighted spreads $\sim 5 \times 10^{-41}$, $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$ — the J-weighted mirror of ± 64), the declared reading also wins the column-canonicalisation discriminator (the physical AB column is contained *exactly* in the χ_4 family, canonicalising (N_1, N_2), yet the canonical member fails every tester) and the $\mathbb{Z}_2$/EH anchor (all three derived instantiations fail; only the declared reading hits $9\langle x, x\rangle^2 = (1/4) \times 36$ with the scale tooth $c = 1/2 = |\mathbb{Z}_2| h^{A_1}$). Typed premises [C] (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — it must source $\psi = 64$), TP-3 (the v518 kernel convention), TP-4 (the Quillen structure of BCOV F_1). The residual [O] was the constructive BCOV derivation of the w_m normalisation itself — the success branch (“the BCOV derivation lands on the declared completion reading, carrying the forced 4:3 leading ratio”) has fired at probe level, the kill branch (“it lands on the derived chiral measure”) is dead — and that derivation is since *executed* (verdict ERFOLG) as `CELEST.WP5E.WM.01` [`verification/v523_celestial_wp5e_wm_derived.py`]: the $1/\det_j$ weight is COMPUTED from three independent sources — the equivariant mode ledger of the fibre $\mathbb{C}^2$ (closed form $1/((1 - q\, i^j)(1 - q\, i^{-j}))$ exact at every order $n \leq 120$, regular at $q = 1$ with Abel value $(1/2, 1/4, 1/2) = 1/\det_j$, fixed-point splitting at every truncation level for the phased AND the v520-forced unphased trace, so $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ is computed, not declared), the zeta/Quillen route ($\det_\zeta = 4\sin^2(\pi j/4) = \det_j$ exactly with the unique real positive holomorphic section via the $SU(2)$ conjugation pairing, and the δ_{1f} block constant term $1/\det_b$), and the consistency chain (the v516 numbers reproduced one by one: $w = 4h$, $c = 4 = |\mu_4|$, squares $(18, 9, 18)$, total $45\langle x, x\rangle^2$, $\psi \pm 64$) — with the negative controls behaving ($1/\det^2$ and $1/|1 - i^j|$ break the chain quantifiably, the $\mathbb{Z}_2$/EH anchor passes at $c = 2 = |\mathbb{Z}_2|$, the $SO(16)/D_8$ kill fires, $\text{diag}(i, i)$ breaks the zeta $=$ AB identity itself, the $k = 3, 5$ weights wander correctly) and the typed premises TP-REG/TP-Q/TP-NUM/TP-CH as its own mandatory [C] fence; the residual [O] narrows to the *global* BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre zero-mode factor (the resolved-ALE route of v514/v517 not built); **(the cubic GS term — dissolved)** the $\psi = 64$ slice is *delivered* by the declared KS measure (v516); the cubic d-channel is *not needed* (c_d free $= 0$) — its physical-justification question (the honest v513 target $c_d = 32 \times 60$) stays dissolved: the δ_1 decision (v518) does not revive the channel (no branch fires under the derived measure); a ch_2-type naturalness datum for *cubic* vertices still does not exist; **(the back-reaction milestones, preregistered as M1–M3 in v514; all three executed — the v514 fence is fully worked off)** the type-I B-model *back-reaction* on $\mathbb{PT}/\mathbb{Z}_4$, which would upgrade the ε_2 CPS dictionary from [C] to derived: **M1 (*done*, SUCCESS)** [E]/[C] the A_3 Ω_N (the Beltrami/Kodaira–Spencer deformation sourced by the glue currents on the three lockstep spheres; preregistered success = a closed-form Ω-deformation whose $S^3/\mathbb{Z}_4$ periods are $(2\pi i)^2\times$ an integer flux vector with the lockstep phases $(i-1)(1, i, i^2)$; kill = unequal sphere fluxes or a non-integer period — then the CPS transplant *and* the ε_2 level dial die) — executed and machine-verified as `CELEST.WP5E.M1.01` [`verification/v515_celestial_wp5e_m1_omega_n.py`] (exact sympy, 30 checks): the residue form $\omega_2 = dX \wedge dZ/X$ is *derived* (three ambient routes; orbifold-cover factor $4 = |\mu_4|$; clock phase i), the twistor family closes as $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1-\lambda^4)^2$ ($e_1 = e_3 = 0$ identically; seam fibre $\lambda = 0 =$ the v493 family) with the CY-compatible clock lift $(Z, \lambda) \mapsto (iZ, -i\lambda)$, $\Omega \to +\Omega$, and $\Omega_N = \Omega_0 + \sum_p N_p K_p$ is *closed form* (CPS/Bochner–Martinelli kernels on the four centre twistor lines) with every $S^3/\mathbb{Z}_4$ period $(2\pi i)^2$-integral, the seam-fibre phase vector $2\pi i\, t_0(i-1)(1, i, i^2)$ with clock covariance $\Pi \to i\Pi$, the flux vector $N(1, 1, 1, 1)$ forced uniform (clock orbit + K_4 connectivity) and the lens geometry *forcing* the source charge $4 = |\mu_4|$ ($N \in 4\mathbb{Z}$); the 12 conifold nodes sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$); honest fence: on the clock-forbidden $Z^4 - Z$ family the $(2\pi i)^2$ quantisation *also* holds — integrality alone is *not* the discriminator, the discriminator is the lockstep phase structure (0/24 vs 8/24) plus the clock forcing of equal fluxes; neither kill fired; **M2 (*done*, SUCCESS on the declared completion measure — verdict B)** [E]/[C] the twisted KS measure (quantise the $O(-2)$ tower on $\mathbb{PT}/\mathbb{Z}_4$ with the ε_1 mode ledger and compute the one-loop box coefficient per sector; preregistered success = the bulk sector reproduces $\tilde{\lambda}^2 = 36$ *and* the twisted sectors pair with the three sphere axions cancelling the rigid $32\,T_3$ residual; kill = a leftover independent T_3-type quartic in any sector) — executed and machine-verified as `CELEST.WP5E.M2.01` [`verification/v516_celestial_wp5e_m2_twisted_ks_measure.py`] (exact sympy/Fraction, 23 checks): the *declared* completion contact term $\mathrm{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ (the twisted-sector loop contributes the *unphased* sector trace with the g^j zero-mode normalisation) carries the exact completion-weight identity $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4| h_m = -4\,\mathrm{ch}_2(T_m)$ — the three sphere axions pair through their *own* McKay ch_2 charges, *no free scale, no fit*; the locks are parameter-free ($T_5 = 0$ for *any* scale, ratio $h_2{:}h_1 = 4{:}3 =$ the δ_1-forced leading ratio *reproduced*, the T_3 budget forcing $c = 4 = |\mu_4|$ uniquely); every twisted channel becomes the *same* perfect Okubo square $36\langle x, x\rangle^2/\det_j$ ($T_5 = T_3 = 0$ in every sector — the KILL does not fire), the total is $45\langle x, x\rangle^2 = \frac{5}{4} \times 36\langle x, x\rangle^2 =$ Dedekind $\times$ Okubo = the unique quartic-free weighting of the β-stage rigidity theorem; both γ-stage certificates are killed ($\Phi_{T_3}: 32 \to 0$, $\Phi_P: 72 \to 0$) and the δ_2 slice $\psi = 64$ is *supplied exactly* — no cubic d-channel needed (c_d free $= 0$); controls: wrong scale leaves $T_3 = 32 - 8c$, the shuffle breaks T_5 and T_3, $SO(16)$ keeps $T_5 = 20/T_3 = -40$ (the KILL fires there: E_8 doubly special), $\mathrm{diag}(i, i)$ degenerates, the $\mathbb{Z}_2$/EH anchor passes at scale $2 = |\mathbb{Z}_2|$. Mandatory fence [C]: the completion reading (the loop of twisted states carries the *unphased* sector trace with the same zero-mode normalisation) is *declared*, supported by the δ_1 modular-completion finding, *not* derived from the BCOV integral within that module — the $\delta_{1b}/\delta_{1c}/\delta_{1d}$ chain has since been executed and decided by v518 (the *derived* chiral measure fails all three testers), the declared-vs-derived measure question has since been decided at probe level by v520 for the declared reading (typed premises TP-1..TP-4), and the w_m normalisation has since been derived at probe level by v523 (the residual **[O]** narrowing to the global BCOV integral beyond the fibre zero-mode factor); **M3 (*done*, SUCCESS)** **[E]**/**[C]** the a_0 uplift (lift the $(4,\pm4)$ GH multipole to the Beltrami differential and compute the induced Kähler-potential correction; preregistered success = a log-type correction with coefficient tied to the source charge $4 = |\mu_4|$; kill = decoupling from the centre count) — executed and machine-verified as `CELEST.WP5E.M3.01` [`verification/v517_celestial_wp5e_m3_a0_uplift.py`] (exact sympy, 23 checks): the uplift object is the generalized-Legendre-transform kernel $\chi = \log P_4$ — the *log of the M1 family polynomial* — well-typed (the $O(2)$ section is a null coordinate: *any* kernel is harmonic) and matching the V-ledger through the exact residue identity $\partial_x[\text{transform}(\log\eta_p)] = 1/r_p$ (flux -4π per centre, source charge $4 = |\mu_4|$); the log-type correction arrives on *four coupled centre-count scales*: (i) the asymptotic kernel log $\chi = 4\log\eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \ldots$ with coefficient $4 = |\mu_4|$, the seam-fibre first correction *exactly* a_0/η^4 = the $(4,\pm4)$ multipole (exact m-grading, no log $\times$power terms — the ε_1 honest zero preserved); (ii) the GLT tower $p_{4k} = 4(-a_0)^k$ with the $n \equiv 0 \bmod 4$ selection rule; (iii) the exceptional-locus log $\chi(0) = \log a_0 = 4\log t_0 + i(4\varphi_0 + \pi)$ (response 1 at the locus vs power-law at infinity); (iv) the period response $d\log\Pi_j/d\log a_0 = 1/4 = 1/|\mu_4|$ uniformly, integrating to the monodromy i = one Coxeter clock step (the WP2 monodromy reproduced from perturbation theory), with a_0-rigid $\mathbb{Z}_8$ node support and topological $(2\pi i)^2$ fluxes; controls: the $(4,0)$ multipole is clock-invariant, the $\mathbb{Z}_2$/EH analogue reads $2 = |\mathbb{Z}_2|$ on *every* dial, $k = 3, 5$ orbits move the coefficient with the centre count ($O(6)/O(10)$ slots), the forbidden family fails both dials ($p_3 = 3$, $e_4 = 0$) — the KILL (decoupling) does not fire; fence **[C]**: the GLT dictionary ($\eta_p \log\eta_p$ = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); **[O]**: the full nonlinear Kähler potential of the resolved A_3 ALE. Success (unchanged): partition function $= E_4/\eta^8$ *including* the $q^{-1/3}$ prefactor *derived from the twistor side*; kill: an anomaly mismatch at the Costello–Li grade — neither the CFT-side dials of v502, nor the equivariant skeleton of v505, nor the exchange no-go of v508, nor the flux/sector dials of v509, nor the full-tensor ledger of v511, nor the axion-slot construction of v514, nor the M1–M3 back-reaction milestones of v515/v516/v517, nor the δ_1 decision of v518, nor the measure decision of v520 trigger it (all computable dials say $k = 1$, "one level" is now a theorem, the bulk axion survives the projection, the back-reacted Ω_N carries the forced integral lockstep flux, the twisted KS measure lands on the unique quartic-free weighting, the a_0 uplift couples to the centre count, the δ_1 kill is a statement about the derived measure, not about the algebra, and the measure decision selects the declared reading at probe level), but the global quantisation is untested.

Honest status line: WP5a–WP5d (both WP5d stages), the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the M1–M3 back-reaction milestones are exact-arithmetic **[E]** constructions (plus ED-validated lattice witnesses in WP5d) with **[C]** named identifications ("radial decoupling = collar thickness $\to 0$"; Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the quasi-free extension; the KLM/Longo–Xu/CHMP index dictionary; Longo split heredity; Kac generation of the maximal submodule for $k \geq 3$; the WP5e-β dictionaries — $\mathrm{CS}(\rho) \bmod 1$ = discriminant weight, $32 = (h,h)+(h',h')$, $-30 = -h^\vee$, the Ω-contraction; the WP5e-γ number fences $32 = 20+12$, $72 = 2\tilde\lambda^2_{e_8}$; the WP5e-$\delta_2$ number fences $64 = \dim\mathfrak{g}_1$, $1920 = |W(D_5)|$ — the latter typed look-elsewhere-loaded and convention-contingent by v513; the WP5e-ε_2 CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$; the WP5e-ε_1 conditionality on Costello's flat-$\mathbb{PT}$ matching; the M1 global kernel patching, Hitchin small resolution and CPS brane dictionary, v515; the M2 *declared* completion reading of the twisted KS measure, v516; the M3 GLT dictionary, v517); *WP5a–WP5d plus WP5e-$\alpha/\beta/\gamma/\delta_1/\delta_2/\varepsilon_2/\varepsilon_1$ plus M1–M3 are landed — all three KLM ingredients of complete rationality carry lattice witnesses, both faces of the inflow* *are anchored, the exchange sub-branch is closed with a certificate, the collapse is confirmed full-tensorially with one cubic door open, "one level" is now a theorem of clock invariance with the sector-counter dial pinning $k = 1$, the bulk-axion slot is a construction with $\tilde{\lambda} = 6$ triply pinned, the back-reacted Ω_N is closed-form with $(2\pi i)^2$-integral lockstep periods and the source charge $4 = |\mu_4|$ forced, the twisted KS measure (on the declared completion reading) cancels the $32\,T_3$ residual and supplies the $\psi = 64$ slice without the cubic d-channel, the a_0 uplift couples to the centre count on four scales, and the δ_1 chain is decided — kill under the derived measure, tension with the declared reading stated* — WP5e proper (the global BCOV quantisation) is the single remaining open piece, with the continuum uplift of the WP5d lattice witnesses honestly fenced as Xu's theorem (cited, not claimed); the exchange sub-branch is closed by v508, the full-tensor ledger executed by v511, the level-from-flux dial executed by v509, the axion slot executed by v514, the M1–M3 back-reaction milestones executed by v515/v516/v517 (the v514 fence fully worked off), the δ_1 chain decided by v518, the measure question decided at probe level by v520 (the declared completion reading wins: single-valuedness derived from F-independence + the Quillen pairing under TP-1..TP-4; the derived chiral measure was never a well-defined nonvanishing moduli integral), the w_m normalisation itself derived constructively at probe level by v523 (Atiyah–Bott mode ledger + zeta/Quillen determinant + consistency chain, typed premises TP-REG/TP-Q/TP-NUM/TP-CH), and the named remaining target is the global BCOV integral beyond the fibre zero-mode factor (the cubic-GS-term question stays dissolved: v518 does not revive the d-channel) — `SEAM.EQUIV.01` stays **[O]** and nothing in WP5a–WP5e-M3, the δ_1 decision or the measure decision moves it.

Kill tests (pre-registered). **(K1)** the equivariant sector could have *contradicted* the 52+64+60+64 split (uniformity of $D[d]$ needs exactly $\dim \mathfrak{g}_j = (60, 64, 60, 64)$) — *survived by WP1 and WP2*: the false-glue control shows the test has teeth (3112/6720 violations), and WP2 adds the deformation side (a shape modulus surviving clock-invariance, or a_0-corrections leaking between μ_4 sectors, would have fired — neither occurred; the clock-variant controls fail in two independent ways each). **(K2)** if one-loop associativity of the celestial E_8 theory demanded extra fields that TFPT does not possess, the contract dies (Costello's axionic mechanism is the first pass: E_8 is admissible; the TFPT-specific field content is the open half). **(K3)** — *evaluated by WP3* ([`verification/v495_celestial_wp3_gs_alignment.py`]): the trigger ("no exact alignment between the GS coefficient and the c_3 normalisation") does *not* fire — exact alignment exists ($(\kappa/c_3)^2 = 12$ resp. 1/300). But the look-elsewhere battery *demotes the scope of K3 itself*: the alignment format passes 8/8 across Costello's whole list, so surviving K3 certifies compatibility plus λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence — alignment survives, selectivity does not. **(K4)** — *evaluated by WP4* ([`verification/v496_celestial_wp4_character_shadow.py`]): K4 fires only against the *exact*-block reading (the character is provably not a conformal block in the jet grading); it does *not* degrade the contract to "E_8 is an admissible gauge algebra for celestial SDYM", because the sector arithmetic holds exactly — WP4 lands at the boundary-limit reading (verdict B(ii), the `SEAM.EQUIV.01` scaling-limit shape), with the constructive limit handed to WP5 — and made precise there by WP5a ([`verification/v497_celestial_wp5a_null_ideal.py`]: the coefficientwise χ_w limit with threshold $w = n{+}1$ and the derived ideal), with the deleting operator itself constructed by WP5b ([`verification/v498_celestial_wp5b_singular_vector.py`]: $|s\rangle$ explicit, level dial $2(1{-}k)$), the limit state by WP5c ([`verification/v500_celestial_wp5c_gns_limit_state.py`]: the ideal in the GNS kernel, level-2 rank 4124 exactly), the two-interval index chain by WP5d-α ([`verification/v501_celestial_wp5d_two_interval_index.py`]: μ-chain $16 \to 4 \to 1$ anchored at both measured ends), the split + strong-additivity legs by WP5d-β ([`verification/v504_celestial_wp5d_klm_completeness.py`]: all three KLM ingredients of complete rationality witnessed on the lattice; the continuum uplift is Xu's theorem, cited not claimed), the CFT-side anchors of the uplift itself by WP5e-α ([`verification/v502_celestial_wp5e_prefactor_level.py`]: the $q^{-1/3}$ prefactor as exact μ_4 vacuum-energy bookkeeping, $k = 1$ forced three ways), the equivariant twistor skeleton by WP5e-β ([`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`]:

the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact, glue defect -78 two routes, geometric $k = 1$ dials, the rigid $32\,T_3$ residual, and the honest a_0 refutation — the kill branch does not fire on the skeleton), the exchange no-go by WP5e-γ ([`verification/v508_celestial_wp5e_gamma_pairing.py`]: the sphere-axion pairing by quadratic-vertex exchange refuted with the three-functional certificate $(0, 32, 72)$ — the $32\,T_3$ burden moves to the twisted contact terms, δ_1 binary), the full-tensor ledger by WP5e-δ_2 ([`verification/v511_celestial_wp5e_delta2_tensor_ledger.py`]: the collapse confirmed full-tensorially by the innerness theorem, the cubic d-symbol channel opening T_3 with the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$ and the relaxed solution $c_d = 1920\,Q_{dd}$-coefficient — typed by the v513 negative certificate as $c_d = 32 \times 60$ with the $|W(D_5)|$ reading look-elsewhere-loaded), and the level-from-flux dial by WP5e-ε_2 ([`verification/v509_celestial_wp5e_eps2_level_from_flux.py`]: the lockstep theorem — "one level" as a theorem of clock invariance, falsifier $Z^4{-}Z$ at 0/24 — plus the sector-counter dial $\#\mathrm{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$), the bulk-axion slot by WP5e-ε_1 (v514: the equivariant $O(-2)$ slot construction, $\tilde{\lambda} = 6$ triply pinned, the first GH/A_3 back-reaction step), the back-reacted Ω_N by the M1 milestone (v515: closed form, $(2\pi i)^2$-integral lockstep periods, forced source charge $4 = |\mu_4|$), the twisted KS measure by the M2 milestone ([`verification/v516_celestial_wp5e_m2_twisted_ks_measure.py`]: the completion-weight identity $w = 4h = -4\,\mathrm{ch}_2$, per-sector Okubo squares, the $32\,T_3$ cancelled and the $\psi = 64$ slice supplied on the declared completion reading — the M2 kill did not fire) and the a_0 uplift by the M3 milestone ([`verification/v517_celestial_wp5e_m3_a0_uplift.py`]: the log-type correction coupled to the centre count on four scales, monodromy i — the M3 kill did not fire), and the δ_1 chain decided by the consolidated δ_1 module ([`verification/v518_celestial_wp5e_delta1_derived_kill.py`]: the derived chiral measure fails all three preregistered testers — kill under the derived measure, the declared/derived tension stated), and the measure question decided at probe level by the consolidated δ_{1e}/δ_{1f} module (v520: single-valuedness derived from two constructive sources, the declared reading wins under TP-1..TP-4), and the w_m normalisation derived constructively by the WM module ([`verification/v523_celestial_wp5e_wm_derived.py`]: the $1/\det_j$ fixed-point weight computed from the Atiyah–Bott mode ledger, the zeta/Quillen determinant and the consistency chain — the WM kill did not fire), leaving the global BCOV quantisation (WP5e proper, narrowed to the global BCOV integral beyond the fibre zero-mode factor) as the single remaining open piece.

Honest success estimate. The probability of *full* success (WP5e: a second continuum route to `SEAM.EQUIV.01`) remains low — it is constructive holography at the Costello–Li grade. The *partial*-findings expectation is now realised and *exceeded*: WP1–WP4, all five WP5 constructive milestones (both WP5d stages), the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the M1–M3 back-reaction milestones are executed with sharp structural results (verdicts B, B, B, B(ii); WP5a–WP5d passed; WP5e-α pins the CFT side, WP5e-β the equivariant twistor skeleton, WP5e-γ closes the exchange sub-branch with a certificate, WP5e-δ_2 confirms the collapse full-tensorially and opens the one cubic door, WP5e-ε_2 makes "one level" a theorem and pins its value by the sector counter, WP5e-ε_1 builds the bulk-axion slot with $\tilde{\lambda} = 6$ triply pinned, M1 delivers the closed-form back-reacted Ω_N with forced charge $4 = |\mu_4|$, M2 delivers the twisted KS measure on the declared completion reading (the $32\,T_3$ cancelled, the $\psi = 64$ slice supplied, no cubic d-channel), M3 delivers the a_0 uplift coupled to the centre count on four scales, and the δ_1 decision (v518) closes the chain with an honest kill under the derived measure, the measure decision (v520) selects the declared reading at probe level, and the w_m derivation (v523) computes the fixed-point weight from three independent sources; *WP5e proper — the global BCOV quantisation, narrowed to the global BCOV integral beyond the fibre zero-mode factor — is the single remaining open piece*) — the WP1 critical correction (clock $\neq$ deck; clock^2 = deck), the WP2 sharpenings (the clock *is* the Picard–Lefschetz/Coxeter monodromy of the seam deformation; a_0 pure scale with the $\tau{=}i$ shape frozen), the WP3 non-selective alignment (exact λ-arithmetic, honestly typed by the 8/8 look-elsewhere table), the WP4 boundary-limit localisation of the character mismatch (three exactly-located obstructions, one surviving limit structure), the WP5a null-ideal derivation (stabilisation threshold $w = n{+}1$; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), the WP5b singular-vector construction (the deletion operator explicit, case tally 190/57/1, level dial $2(1{-}k)$, one-block closure typed E_8/μ_4-specific), the WP5c limit state (the ideal in the GNS kernel, the full 9361-block Gram at rank 4124, the family proved *necessary* by a CCR obstruction), the WP5d-α two-interval index (the $\ln 2$ plateau at 10^{-15}, the μ-chain $16 \to 4 \to 1$ measured at both ends), the WP5d-β KLM completion (strong additivity exact on the lattice with the shared boundary point; the bounded-vs-divergent entropic discriminator with the Ising $\frac{1}{4}$-exponent approach against the diverging $U(1)$ control; the elliptic-nome split ladder with exact orbifold inheritance; Pimsner–Popa $\lambda = \frac{1}{2}$ and $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ with integer attainment — all three KLM ingredients witnessed, Xu's continuum theorem cited not claimed), the WP5e-α prefactor/level pinning (the $q^{-1/3}$ prefactor as exact μ_4 vacuum bookkeeping; $k = 1$ forced three ways; the glue-integrality route honestly retired), the WP5e-β equivariant ledger (the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact with the -78 defect; the rigid $32\,T_3$ twisted residual; the honest a_0 refutation with the three sphere classes filling the three twisted axion slots), the WP5e-γ exchange no-go (the invariant vertex spaces exactly dim 2/5; the product theorem; the certificate $(0, 32, 72)$; naturalness dissolved scale-independently — an honest negative result that converts the pairing question into the binary δ_1 contact-term test), the WP5e-δ_2 full-tensor ledger (the innerness theorem confirming the collapse across arities; the unique symmetric survivor = the $\mathfrak{su}(4)$ d-symbol with $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ opening the T_3 channel; the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$ and the exactly solvable relaxed reading with $c_d = 32 \times 60$, the $1920 = |W(D_5)|$ fingerprint honestly retired as look-elsewhere-loaded by the v513 negative certificate), and the WP5e-ε_2 level-from-flux dial (the CPS skeleton exact; the pairing matrix pinned by complete enumeration with fluxes $(64, 60, 64)$ and one quantum per current; the naive "level = total flux" killed by the lockstep test itself; the lockstep theorem with its $Z^4 - Z$ falsifier; the sector counter $\#\mathrm{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$), the WP5e-$\varepsilon_1$ axion slot (the equivariant Penrose ledger closing block by block; the $d = 0$ slot surviving the projection; $\tilde{\lambda} = 6$ triply pinned with the wrong bookings 3 and 12 excluded; the GH/A_3 step re-deriving the v493 family and the Coxeter clock from geometry), the M1 back-reacted Ω_N (the residue form derived with cover factor $4 = |\mu_4|$; the closed twistor family with its CY-compatible clock lift; every $S^3/\mathbb{Z}_4$ period $(2\pi i)^2$-integral with the forced uniform flux vector and the lens-forced charge $4 = |\mu_4|$; the honest integrality-is-not-the-discriminator fence), the M2 twisted KS measure (the completion-weight identity $w = 4h = -4\,\mathrm{ch}_2$ with no free scale; every twisted channel the same perfect Okubo square; both γ-stage certificates and the δ_2 $\psi = 64$ slice supplied exactly; the declared completion reading honestly fenced [C], v516) and the M3 a_0 uplift (the GLT kernel $\chi = \log P_4$ with the log coefficient $4 = |\mu_4|$ on four coupled centre-count scales, the period response $1/4$ integrating to the Coxeter monodromy i, v517) and the δ_1 decision (the 16-component Weil completion closing exactly with the E1.5 residual $2.91 \to\sim 10^{-39}$; the μ_4 multiplier obstruction identified as a *character* of the orbit stabilisers by the exact koboundary test; the cancellation achieved by the twisted fibre block $f_1 f_3 = G$ — not the three sphere axions; all three preregistered testers failing under both derived solutions with the honest (N_1, N_2) rebalance rescuing nothing; the declared/derived measure tension stated as the mandatory fence, v518) and the measure decision (the exact invariant subspace = $\mathrm{span}\{e_H, e_{H'}\}$ forcing the unphased E_4 numerator once single-valuedness is derived from F-independence + the Quillen pairing; the consistency circle to v516 closed levelwise; the typed premises TP-1..TP-4 named, v520) — exactly the kind of sharp, falsifiable refinement the contract format is for. Status: `CELEST.SEAM.01` [O] (research contract, WP5e proper the single remaining open milestone); `CELEST.WP1.01` [E]/[C] (executed); `CELEST.WP2.01` [E]/[C] (executed); `CELEST.WP3.01` [E]/[C] (executed); `CELEST.WP4.01` [E]/[C] (executed); `CELEST.WP5A.01` [E]/[C] (executed); `CELEST.WP5B.01` [E]/[C] (executed); `CELEST.WP5C.01` [E]/[C] (executed); `CELEST.WP5D.01` [E]/[C] (executed, α stage); `CELEST.WP5DB.01` [E]/[C] (executed, β stage — the continuum uplift stays [O], Xu's theorem cited); `CELEST.WP5E.ALPHA.01` [E]/[C] (executed, α stage — the CFT-side prefactor + level pinning); `CELEST.WP5E.BETA.01` [E]/[C] (executed, β stage — the equivariant anomaly ledger; the global BCOV quantisation itself stays [O]); `CELEST.WP5E.GAMMA.01` [E]/[C] (executed, γ stage — the exchange no-go with certificate; the δ_2 ledger since executed by v511, the twisted contact terms δ_1 since decided by v518 — kill under the derived measure, tension with v516 stated); `CELEST.WP5E.DELTA2.01` [E]/[C] (executed, δ_2 stage — the full-tensor ledger, intermediate verdict: collapse confirmed by innerness, the cubic d-symbol channel opens T_3 with the weaker certificate $\psi = 64$; the arity ≥ 4 ledger stays [O]; the $\psi = 64$ burden since discharged by M2/v516 — the cubic-GS question stays dissolved, the δ_1 decision (v518) does not revive it); `CELEST.DTERM.NONDERIV.01` [E] (executed, the c_d negative certificate — the $1920 = |W(D_5)|$ fence typed look-elsewhere-loaded and convention-contingent, the convention-stable core $c_d = 32 \times 60$; the fence stays [C], the generation of the 32 stays [O]); `CELEST.WP5E.EPS2.01` [E]/[C] (executed, ε_2 stage — the level-from-flux dial, verdict B: "one level" a theorem, the sector counter pins $k = 1$; the CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ typed [C], the type-I B-model back-reaction milestones M1–M3 all since executed via v515/v516/v517); `CELEST.WP5E.EPS1.01` [E]/[C] (executed, ε_1 stage — the $O(-2)$ bulk-axion slot construction with $\tilde{\lambda} = 6$ triply pinned and the first GH/A_3 back-reaction step; the M1–M3 milestones it preregistered are all since executed; the BCOV-integral derivation stays [O]); `CELEST.WP5E.M1.01` [E]/[C] (executed, back-reaction milestone M1, SUCCESS on the preregistered criterion — the closed-form Ω_N with $(2\pi i)^2$-integral lockstep periods and the lens-forced source charge $4 = |\mu_4|$; honest fence: integrality alone is not the discriminator; M2–M3 since executed via v516/v517); `CELEST.WP5E.M2.01` [E]/[C] (executed, back-reaction milestone M2, SUCCESS on the preregistered criterion *on the declared completion measure* — the completion-weight identity $w = 4h = -4\,\mathrm{ch}_2$ with no free scale, every twisted channel the same perfect Okubo square $36\langle x, x\rangle^2/\det_j$, the $32\,T_3$ cancelled, both γ-stage certificates and the $\psi = 64$ slice supplied exactly, no cubic d-channel needed; the completion reading stays [C]-declared within that module — the δ_1 chain since decided by v518, and the completion reading since *derived at probe level* from two independent constructive sources (F-independence + Quillen pairing) under the typed premises TP-1..TP-4 by v520; the residual [O] is the constructive w_m derivation); `CELEST.WP5E.M3.01` [E]/[C] (executed, back-reaction milestone M3, SUCCESS on the preregistered criterion — the a_0 uplift on the GLT kernel $\chi = \log P_4$ with the log coefficient $4 = |\mu_4|$ coupled to the centre count on four scales, the period response $1/4$ integrating to the Coxeter monodromy i; the GLT dictionary stays [C], the full nonlinear Kähler potential stays [O]); `CELEST.WP5E.DELTA1.01` [E]/[C] (executed, δ_1 stage — DECIDED: kill under the derived measure; the 16-component Weil completion closes exactly, the μ_4 multiplier obstruction is a character (koboundary defects $(1, 1, 1)$, $\lambda(\gamma) = i^{2B+C/4}$ on $\Gamma_1(4)$), the cancellation is the twisted fibre block $f_1 f_3 = G$, and all three preregistered testers fail under both derived solutions; the $f_1 f_3 =$ KS-weight identification and the kernel-family convention stay [C]; the declared-vs-derived BCOV measure question — the stated v516 tension — was the named [O] and is since decided at probe level by v520 for the declared reading, sharpening this kill; the residual [O] is since narrowed by v523 to the global BCOV integral beyond the fibre zero-mode factor); `CELEST.WP5E.MEASURE.01` [E]/[C] (executed, the measure decision — ERFOLG-A on the preregistered decision layer: single-valuedness derived from two independent constructive sources (F-independence + the Quillen pairing), the invariant subspace exactly $\mathrm{span}\{e_H, e_{H'}\} = E_4$, the consistency circle to v516 closed levelwise, the $\mathbb{Z}_2$/EH anchor and the column canonicalisation both selecting the declared reading; typed premises TP-1..TP-4 stay [C]; the constructive BCOV derivation of the w_m normalisation itself — the named residual [O] — is since executed by v523); `CELEST.WP5E.WM.01` [E]/[C] (executed, the constructive w_m derivation — verdict ERFOLG per the frozen preregistration: the $1/\det_j$ fixed-point weight COMPUTED from three independent sources (the equivariant mode ledger with Abel value $(1/2, 1/4, 1/2)$; the zeta/Quillen determinant $4\sin^2(\pi j/4)$ with the unique real positive section; the δ_{1f} block constant term $1/\det_b$), the v516 chain reproduced number by number ($w = 4h$, $c = 4 = |\mu_4|$, squares $(18, 9, 18)$, total $45\langle x, x\rangle^2$, $\psi \pm 64$), all negative controls behaving (the $\mathbb{Z}_2$/EH anchor at $c = 2 = |\mathbb{Z}_2|$, the $SO(16)$ kill, the $\mathrm{diag}(i, i)$ break, the $k = 3, 5$ wandering); typed premises TP-REG/TP-Q/TP-NUM/TP-CH stay [C]; the global BCOV integral beyond the fibre zero-mode factor stays [O]); `SEAM.EQUIV.01` untouched, stays [O].

Research Contract — `WOIT.OS.TWISTOR.01` (the Osterwalder–Schrader twistor bridge)

Typing. **This is the actual bridge from compiler to physics — everything in `CELEST.SEAM.01` is preparation for it.** It is a research contract [O] (ledger row `WOIT.OS.TWISTOR.01`: Open, research contract), *not* a claim; nothing below asserts that the construction exists. The external programme it engages — Woit's Euclidean Twistor Unification (`arXiv:2104.05099`) — is a *named reference frame* for the shape of the target, never a confirmation in either direction (the Woit-bridge paragraph in `tfpt_3` states exactly what is missing).

Input. $X_+ = (\mathbb{PT}/\Gamma)_+$ (the positive-time half of the quotiented projective twistor space; $\Gamma = \Gamma_{\mathrm{ALE}} \cong \mathbb{Z}_4$, normalised by the clock — the definition block of `CELEST.SEAM.01`); $\mathcal{A}_{\mathrm{hol}}$, the *interacting* open+closed twistorial algebra (SDYM(E_8) open strings + BCOV closed strings — the WP5e-proper object, whose free/equivariant skeleton is pinned by v492–v517); ρ, the order-4 clock diag$(i, 1)$; Θ, the anti-linear real structure induced by the seam reflection; and $\mu_{\mathrm{BCOV+SDYM}}$, the interacting functional (the global BCOV+SDYM quantisation on $\mathbb{PT}/\Gamma$). *Precision (i), from the α stage:* "induced by the seam reflection" means the seam-circle *reflection*, not the deck/covering involution — the deck (whose free-ness is topology, v510) furnishes the $(-1)^F$ Kramers class $\Theta_t^2 = (-1)^F$, not the OS Θ; the seam-circle reflection furnishes $\Theta^2_{\mathrm{Fock}} = +1$ ([`verification/v519_woit_theta_rp_free.py`]). *Precision (ii):* on the RP side the μ_4 marks sit at the *bond midpoints* of the 16-Majorana seam circle (the half-offset/NS-natural placement, 4 sites per mark quadrant) — the cut through the sites fails reflection positivity exactly ([`verification/v519_woit_theta_rp_free.py`]). *Precision (iii), from the β_1 stage:* "gauge-invariant subalgebra" at the free level means the GSO/fermion-parity $\mathbb{Z}_2$ — the only θ-compatible part of the μ_4 tower (the free-fermion shadow of the E_8 glue: $(E_8)_1$ = even NS of $SO(16)_1$ + R spinor); the clock is the euclidean rotation itself (*timelike*: all 16 seam mirror axes invert it, none commutes), and its equivariant statement moves to β_2 — OS quotient first, then the clock as reconstructed transfer operator ([`verification/v522_woit_beta1_gso_gauge.py`]).

Target theorem. $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$; the gauge-invariant algebra is reflection-positive; and the Osterwalder–Schrader quotient produces $(\mathcal{H}, \Omega, U(\mathcal{P}_+^{\uparrow}), \mathcal{A}_{\mathrm{Mink}})$ with: a positive Hilbert metric, positive energy, local causality, *one chiral fermion generation without mirror doubling*, the TFPT charge lattice, an internal $SU(3)\times SU(2)\times U(1)$ action, and the $(E_8)_1$ seam net as an *actual* boundary net (not only as a character).

Kill tests (pre-registered, all seven live). **(1)** Θ is incompatible with the clock (no anti-linear Θ with $\Theta^2 = 1$ and $\Theta\rho\Theta = \rho^{-1}$ exists on $\mathcal{A}_{\mathrm{hol}}$). **(2)** Reflection positivity fails after gauge fixing (the gauge-invariant subalgebra of $\mu_{\mathrm{BCOV+SDYM}}$ is not RP). *Named toy threat (no status change):* on the first genuinely interacting 16-Majorana Fidkowski–Kitaev seam toy (`SEAM.INT.FKTOY.01`, [`verification/v529_seam_interacting_toy_fk.py`]) Kill-Test 1 does *not* fire (Θ exists with $\Theta^2 = +1$ and clock-tower inversion), but Kill-Test 2 *fires at toy level* — RP breaks for every $g > 0$ by an interference mechanism, with the *straddle law* (RP fails exactly on quartet-straddled cuts, stays PD on quartet-avoiding ones; 24/24) as a concrete hurdle for the interacting stages (γ and beyond; the free/quotient-level β_3 is since executed, v565). Fence: one toy / one interaction class / [C] flat-band parent — this narrows the route (every $\mathcal{A}_{\mathrm{hol}}$ must protect RP against exactly this mechanism), it does not close it. *The hurdle, executed as a selector (`SEAM.STRADDLE.CONE.01`, [`verification/v534_seam_straddle_cone.py`]; no status change):* on the equivariant interacting mark family (independent couplings on the four mark quartets, both signs, every admissible cut — including the $\pi/2$ straddled clock axes the 24-entry law had not covered) reflection positivity survives for *exactly one* member × sign: $\delta = \pi/2$ with positive coupling, PD on all four cuts over $g \in \{1/32..8\}$ with the minimum eigenvalue lifted *away* from the RP boundary; every asymmetric member dies. The straddle law refines (asymmetric straddling kills, symmetric straddling protects), the literal leading-order cone formalization is dead (the protection is nonperturbative), and the threat paragraph gains its complement: a passing interaction class *exists* and its unique survivor is the alignment bit — the same member that carries the v528 twist-class order parameter. Toy-level evidence for a dynamical origin of the bit, not a derivation; the kill tests stay formally live on

$\mathcal{A}_{\rm hol}$. *The law derived, and the counterweight given its first OS-viable cut geometry (2026-08-27, SEAM.INT.ODDSPLIT.01 + SEAM.INT.OSAVOID.01, [verification/v972_seam_interaction_front.py]; toy-level census, the v529 fence inherited, no status change):* the straddle law refines to the *odd-split law* — the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (`arXiv:1406.1384`, `arXiv:1601.01994`), made decidable, *predicts* RP survival on the centre-straddled (2+2 mirror-split) cuts and measurement confirms it, with the predicted-vs-measured RP confusion matrix perfect over all 35 (member, cut) cells; the μ_4-covariant quartic space (dimension 464) contains *no* JP-certifiable direction on all cuts. And the threat gains a positive counterweight: the quartet-avoiding cut family of the clock-invariant m=4 member generates the *full* even algebra (rank 15, dimension 32768) and carries a *positive* OS/GNS structure at g=1 (minimum eigenvalue $+4.43 \times 10^{-5}$) while the straddled control fails (-6.94×10^{-2}), and the μ_4 clock maps the avoiding pair onto itself — the kill-test-2 counterweight is *narrowed* (an OS-viable cut geometry exists), not removed. **(3)** The reconstruction produces a vector-like mirror generation (chirality is lost in the OS quotient). **(4)** The Penrose transform reaches only free / self-dual states (the interacting extension does not exist at the Costello–Li grade). **(5)** The reconstructed net has the right character but the wrong OPE / fails net equivalence (character-level agreement without operator-algebraic agreement). **(6)** The internal $SU(2)$ remains a spacetime factor (no genuine internal weak factor separates from the Euclidean rotation group). **(7)** The four μ_4 marks are not incidence-compatibly extendable over spacetime (the mark data of the seam cannot be propagated through the incidence relation).

The α stage — executed (*done*; the free/equivariant milestone WOIT.THETA.FREE.01, [verification/v519_woit_theta_rp_free.py]). The real structure *exists*, and free reflection positivity picks the *same* family. Exactly *two* families of anti-linear structures on $\mathbb{C}^2$ normalise the clock: family D ($z \mapsto \mu\bar{z}$, $|\mu| = 1$) satisfies $\Theta\rho\Theta = \rho^{-1}$ *exactly* with $\Theta^2 = +1$ (-1 impossible: $M\bar{M} = \mathrm{diag}(|\mu|^2, 1)$ — Kramers-free), inverts the deck and reflects the seam circle (2 cut points); family A ($z \mapsto \mu/\bar{z}$) centralises the clock projectively and *never* inverts it — it *defines* the euclidean section (Woit's $\rho_{\rm tw}$ side: $\rho_{\rm tw}^2 = -1$, no real points, replicated exactly on $\mathbb{C}^4$), while family D is the OS conjugation ($\sigma_{\rm std}$, real points $\mathbb{RP}^3$). Mark compatibility pins $\mu \in \mu_4$ (a 4-element torsor, 2 clock orbits); the $\mathbb{Z}_8$ spin plane has no phase leaks; in exact Cl(16), $\Theta_{\rm Fock} = U_r \circ K$ has $\Theta_{\rm Fock}^2 = 2^7 > 0$ (normalised $+1$), inverts the Fock clock tower ($V \mapsto 4096\, V^{-1}$) and normalises the deck, while the deck-induced candidate has $\Theta_t^2 = (-1)^F$ — the v510 split/nonsplit dichotomy *is* the $\Theta^2 = +1$ vs $(-1)^F$ dichotomy. Free RP holds on the bond cut with *no* degree truncation (one-particle Gram $(8, 0, 0)$ PD, even deg ≤ 2 sector $(29, 0, 0)$, complete $N = 8$ half-sided algebra PD; twist $\eta = +i$ forced), the site cut and the clock-centralising family fail RP exactly, and the anti-chiral state flips the odd sector to negative definite (the free shadow of kill test 3). **Kill test (1) therefore does *not* fire at the free/equivariant level — and stays formally *live* on the interacting algebra** $\mathcal{A}_{\rm hol}$; kill tests (2)–(7) are untouched. No marker moves: the contract stays [O].

The β_1 stage — executed (*done*, verdict UNDECIDED per the frozen preregistration; the typed milestone WOIT.BETA1.GSO.01, [verification/v522_woit_beta1_gso_gauge.py]). "The clock is time-like, GSO is the gauge datum." The preregistered clock-tower reading of "gauge" (the $\mu_4/\Gamma_{\rm ALE}$ equivariance averaged *before* the OS quotient) is structurally the *wrong* operationalisation: the one-step clock insertion $T_1 = \langle\theta(e_a), \alpha_S(e_b)\rangle$ violates Hermiticity *exactly* (witness entry $-i/(8\sin(5\pi/16))$ where the conjugate transpose demands $+i/(8\sin(5\pi/16))$; the pairing is complex-*symmetric*: 745 matching, 96 anti-matching, 0 violations — "OS-symmetric" is strictly weaker than Hermitian, so "positivity after gauge fixing" is not even well-posed for the clock reading). The mechanism is typed exactly: *all* 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), none commutes — the μ_4 clock *is* Woit's euclidean rotation, and the θ-compatible (gaugeable) part of its $\mathbb{Z}_8$ Fock tower is exactly the 2-torsion $\{1, (-1)^F\}$ = the GSO/fermion-parity $\mathbb{Z}_2$ (Hermiticity census over all powers; α^4 = the 2π rotation of the seam = fermion parity). Under that corrected typing **gauge-fixed RP holds exactly**: the parity-projected bond-cut Gram is $(29, 0, 0)$ PD at $N = 16$ (deg ≤ 2, min eigenvalue 1.78×10^{-6} at 40 digits) and $(8, 0, 0)$ PD on the complete $N = 8$ half algebra (no degree truncation); the site-cut defect *survives* gauge fixing ($(7, 9, 6)$ — the bond placement of precision (ii) is gauge-invariant information); family A stays indefinite ($(17,12,0)$ vs family D's $(29,0,0)$: the α-stage triangulation survives); the Ramond (wrap $+1$) "true $\mathbb{Z}_4$" lift fails state invariance (the NS/$\mathbb{Z}_8$ tower is forced); and the Kramers class stays trivial on the invariant sector ($\Theta_t^2 = \mathrm{id}$ on $\mathcal{A}^G$, Θ restricts with $\Theta^2 = +1$). **Kill test (2)'s free shadow does *not* fire** (no legitimate Hermitian gauge-fixed form has a negative direction), and kill test (1) stays discharged also gauge-invariantly; both stay formally live on $\mathcal{A}_{\mathrm{hol}}$. Transparency (documented in the module): the first frozen run scored 6/13 — the clock-reading checks failed exactly as the mechanism predicts — and the preregistered deg-2 clock-invariant dimension 30 was a combinatorial error (correct orbit census $28+4=32$); the verdict follows the frozen logic (Hermiticity failure $\Rightarrow$ structural-failure branch $\Rightarrow$ UNDECIDED). β_1 *proper* (the clock-equivariant statement) is re-routed through β_2. No marker moves: the contract stays [O].

The β_2 stage — executed (*done*, verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii); the milestone `WOIT.BETA2.OS.01`, [`verification/v524_woit_beta2_os_quotient.py`]). "The OS quotient of the free system made explicit." $\mathcal{H}_{\mathrm{phys}} = \mathcal{A}_+/\mathcal{N}$ is *explicit and nondegenerate* at both levels: at $N=16$ ($\deg \le 2$) the $37{\times}37$ bond-cut OS Gram is exactly Hermitian and parity-block-diagonal with inertia $(37,0,0)$ PD (min eigenvalue 1.7801×10^{-6} at 40 digits) and null space $\{0\}$, $\dim 37 = 29 \oplus 8$; at $N=8$ the *complete* half algebra gives $(16,0,0)$ PD with $\dim 16 = 8\oplus 8 = 4^2$ — compact euclidean time reconstructs a *thermal* (KMS) representation, with the exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$. The euclidean rotation becomes a Klein–Landau *local symmetric semigroup*: $\tau_k = \omega(\theta(a)\,\alpha_k(b))$ is exactly Hermitian on every shrinking domain D_k ($k=1..7$, dims $29,22,16,11,7,4,2$), the chain identities are exact and the vacuum is fixed; the positivity pattern *is* the α-stage site/bond dichotomy (even steps PSD via the exact square identity $T(2j) = A_j^* A_j$, odd steps indefinite with exactly zero one-particle diagonal — the one-step transfer is *not* positive: the free chirality datum, kill-test-3 shadow sharpened). **The clock is implemented on $\mathcal{H}_{\mathrm{phys}}$ as the positive self-adjoint transfer step $T^{N/4}$** ($(11,0,0)$ PD, min 6.7×10^{-5}; compressed spectrum $[0.0195, 1.6414]$ with vacuum eigenvalue exactly 1; spectral projections certified at $\sim 10^{-40}$ — exactly the calculus the non-Hermitian pre-quotient average of v522 could not have), and at $N=8$ the compressed clock spectrum is *exactly* $\{1, \sqrt{2}-1\} = \{1, 1/\delta_S\}$ in both parity sectors (the silver axes of the v519 μ_4 torsor returning as the clock eigenvalue); the reconstructed rotation group $U(s) = e^{isH}$ is unitary with the group law — per precision (iii) the [C]-operationalisation of the contract phrase "the clock acting unitarily". The v522 non-Hermiticity is *resolved*: the census $(745,96,0)$ is reproduced, every anti-matching entry is a wrap overlap and every overlap-free entry matches — the pre-quotient failure was exactly the domain/wrap artifact. The pre-declared KMS deviation arrives with exactly the declared witnesses: *no contraction* on the compact circle (edge-mode expansion $C(1)/C(3) = 1+\sqrt{2} = \delta_S$ exactly; $\det(G-\tau_4) < 0$) — a typed KMS finding, not a β_2 failure (the contract demands no contraction). GSO/Θ: the grading survives; θ_{cut} and the deck candidate do *not* act on $\mathcal{H}_{\mathrm{phys}}$ (support censuses); the *perpendicular* torsor mirror descends anti-unitarily ($\eta_\perp \in \{\pm i\}$) with $\Theta^2_{\mathrm{phys}} = +1$ on *every* sector — the quotient is Kramers-free — and $\theta_{\mathrm{cut}} \circ \theta_\perp = \alpha_{N/2}$ exactly (the torsor closes onto the deck). Controls: the site cut is indefinite ($(3,3,1)/(2,2,4)$ — the contract kill branch fires for the site placement), family A admits no quotient ($(4,4,0)$), the anti-chiral state flips to $(8,8,0)$ (quotient existence itself selects the chiral orientation), the wrong semigroup direction is non-Hermitian exactly off-domain (all mismatches wrap-overlap). **Kill tests (1) and (2) are strengthened and the (3) shadow is sharpened; none fires** — all seven stay formally live on $\mathcal{A}_{\mathrm{hol}}$. Transparency (documented in the module): run 1 scored 18/20 — a sympy simplify failure on a *true* $\pi/16$ product formula ($\sin(7\pi/16) - \sin(5\pi/16) = 2\cos(3\pi/8)\sin(\pi/16)$), fixed by a Laurent-polynomial certificate; no criterion, tolerance or expectation changed. No marker moves: the contract stays [O]; β_3 is executed next (below).

The β_3 stage, executed (`WOIT.BETA3.DUALITY.01`, [`verification/v565_woit_beta3_pt_duality.py`]; verdict SUCCESS per the frozen preregistration; no status change). The $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality is *constructed as a compatibility class* and typed against σ_{std}, with the contract kill branch decided *empty by algebra.* **(i) The class:** a duality is a nondegenerate Hermitian form Φ on $\mathbb{C}^4$ ($\sigma(Z) = Z^\dagger\Phi$, mapping $\mathbb{PT} \to \mathbb{PT}^*$); clock compatibility has the entrywise residual $(\bar{\lambda}_j\lambda_k - 1)\Phi_{jk}$ and forces the four cross-eigenspace entries to zero exactly: two 2×2 Hermitian blocks, $16 \to 8$ real parameters (the deck adds nothing; the wrong clock lift $\mathrm{diag}(i, i, 1, 1)$ breaks the split). **(ii) The dichotomy and the forced $(2,2)$:** pairing against Woit's Euclidean structure ρ_{tw} via $M_{\mathrm{tw}}^\dagger \Phi M_{\mathrm{tw}} = \varepsilon\bar{\Phi}$ splits the class exactly in two. The *Euclidean* branch ($\varepsilon = +1$; $\sigma_{\mathrm{std}} = \mathbf{1}$ lives here) has seam-line norm $a(|z|^2 + 1)$: no null point, no OS cut, ever — exactly the section-defining role Woit assigns it. The *Minkowski* branch ($\varepsilon = -1$) has seam-line norm $a(|z|^2 - 1)$: **the null locus meets the seam line exactly in the seam circle** — the OS cut is *forced*, not imposed — and the second block is exactly minus the conjugate of the first (charpoly mirrored), so the **total signature is $(2,2)$ for every nondegenerate member**: Woit's Minkowski signature is a theorem of clock + ρ_{tw} pairing, not an input. **(iii) The kill is empty:** every compatible duality inverts clock *and* deck exactly (family D); a clock-centralising (family A) duality would need $\rho_4^2 =$ scalar, but $\rho_4^2 = \mathrm{diag}(-1, 1, -1, 1)$ — no solution; and family A independently has no quotient ($(4, 4, 0)$, the v524 control). **(iv) The identification:** every seam-preserving Minkowski member induces *one* projective conjugation $z \to -\bar{z}$ ($\mu = -1$, fixed points $\pm i$), perpendicular to σ_{std}'s $z \to +\bar{z}$; under the declared RP-side placement (v519 precision (ii); exact rational-π arithmetic, offsets $\mp\pi/2$) that member **is** $\theta_\perp$ — the mirror whose quotient descent v524 certified: Θ_{phys} is the shadow of the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality, anti-unitary, Kramers-free ($\Theta^2 = +1$ per sector), time-reversing the semigroup, closing with the σ_{std}-type cut onto the deck ($\theta_{\mathrm{cut}} \circ \theta_\perp = \alpha_{N/2}$; upstairs: the clock conjugation exchanges $\mu = +1 \leftrightarrow \mu = -1$). *Role separation complete:* $\sigma_{\mathrm{std}} \sim$ the OS cut (the metric), the Φ-twisted Minkowski duality $\sim$ the descending mirror. Free/quotient level only; $\mathcal{A}_{\mathrm{hol}}$ untouched; all seven kill tests stay live there; the declared conventions (seam line, seam preservation, placement, torsor relabeling) are cited, not invented. No marker moves: the contract stays [O]; γ is the next milestone.

Thermal follow-on (no status change). The free OS quotient of β_2 is KMS, and its temperature normalisation closes onto the Nariai/BH chain as the *third leg* of c_3 (`SEAM.THERMAL.KMS.01`, [`verification/v526_seam_thermal_kms_nariai_bridge.py`]: β measured $= N$ at both levels; $\beta_{\mathrm{angle}} = 2\pi = 1/(4c_3)$, $T_{\mathrm{seam}} = 4c_3$; modular Hamiltonian closed; class-1 anti-numerology selection; non-compact $\Rightarrow T = 0$). The silver clock eigenvalue is explained and demystified in parallel (`SEAM.CLOCK.SILVER.01`, [`verification/v527_seam_clock_silver_spectrum.py`]: structure SUCCESS, connection KILL). No marker moves.

The β/γ roadmap (named milestones, each with success and kill criteria). (β_1) Θ on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant SDYM(E_8) sector — success: the v519 OS form extends to the gauge-invariant subalgebra with positivity after gauge fixing; kill: kill test (2) fires (RP fails after gauge fixing). *Executed via v522 (above): UNDECIDED under the frozen preregistration — the clock reading is time-like (wrong operationalisation), the gauge datum is the GSO $\mathbb{Z}_2$, gauge-fixed RP holds under that typing, neither kill fires; the clock-equivariant statement moves to β_2.* (β_2) The OS quotient of the *free* system made explicit — success: $(\mathcal{H}, \Omega)$ constructed from the v519 bond-cut form with the clock acting unitarily; kill: the quotient degenerates (zero or indefinite completion). *Executed via v524 (above): SUCCESS, [C]-typed per precision (iii) — $\mathcal{H}_{\mathrm{phys}}$ explicit and PD at both levels, the clock a positive transfer step with spectral calculus (exact $N = 8$ spectrum $\{1, \sqrt{2} - 1\}$) and a reconstructed unitary rotation group; the pre-declared KMS deviation carries exactly the silver witnesses; neither kill fires; β_3 is next.* (β_3) The $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed against σ_{std} — success: Woit's Minkowski conjugation ($\mathbb{PT} \to \mathbb{PT}^*$) identified with the family-D conjugation on the quotient; kill: the duality forces a clock-centralising (family A) structure instead. *Executed via v565 (above): SUCCESS — the kill branch is empty by algebra (every clock-compatible duality is family D; family A would need $\rho_4^2 =$ scalar), the OS cut and the $(2, 2)$ signature are forced on the Minkowski branch, and the unique induced seam member is $\theta_\perp$, the v524 Θ_{phys} carrier.* (γ) The chirality theorem + the mark incidence — success: the odd-sector definiteness flip of v519 upgraded to "one chiral generation without mirrors" on the reconstructed net, and the four μ_4 marks extended incidence-compatibly; kill: kill tests (3)/(6)/(7) fire. *The FREE slice is* *executed via* [verification/v608_gamma_free_slice.py] *(9/9, GAMMA-FREE-SLICE-LANDED): on the explicit v524 net the mirror state's odd sector is* strictly negative definite *(sector-exact flip — every mirror fermion vector has negative OS norm, so no mirror mode survives reconstruction: kill test (3) cannot fire at the free level), the chiral generation is multiplicity-free (clock spectrum exactly* $\{1, \sqrt{2}-1\}$*), and the mark transport exists (*τ_2 *is the mark-A→B transport, Hermitian PD: kill test (7) does not fire at the free level).* γ *itself stays open on* $\mathcal{A}_{\rm hol}$ *(kill tests (3)/(6)/(7) formally live there); kill test (6) is untouched by the free slice. The INTERACTING toy slice is since executed via* [verification/v615_gamma_toy_interacting.py] *(11/11, GAMMA-TOY-LANDED): on the RP-surviving interaction class (the alignment bit* $\delta = \pi/2$, $s = +$, v534*) the mirror state's odd sector is* strictly negative definite *on every admissible cut and every coupling* $g \in \{1/32..8\}$ *(bounded away from zero — kill test (3) cannot fire on the surviving class), the mark transport* α_4 *is Hermitian PD at every* g *(kill test (7) does not fire at the interacting toy level), the odd transfer spectrum stays multiplicity-free (honest finding: strong-coupling near-doubling at* $g = 8$ *without closing), and the RP-dead member has no gamma home — the dynamical bit selection and the chirality data* co-locate *on the same member. Fence: one toy, one interaction class;* γ *proper stays open on* $\mathcal{A}_{\rm hol}$*. Kill test (6) has since received its first (kinematic) slice via* [verification/v616_su2_internal_kinematic.py] *(17/17, SU2-KINEMATIC-SEPARATED): on* $\mathbb{C}^4 = \mathbb{H}^2$ *the internal su(2) (the right quaternion action, with* $\rho_{\rm tw}$ = *right-j verbatim) preserves every twistor fiber, commutes exactly with the spacetime quaternion action and intersects it in* $\{0\}$ *— the internal factor* separates *kinematically, the clock factorizes exactly as left-*$e^{i\pi/4}$∘*right-*$e^{i\pi/4}$*, the ALE deck is purely spacetime-side, and the complex choice breaks the internal* $SU(2)$ *to* $U(1)$ *with the v519 mark torsor as its* μ_4*; the DYNAMICAL half of kill test (6) (a genuine gauge action on the reconstructed net) stays open and formally live on* $\mathcal{A}_{\rm hol}$*.*

Scope fence (explicit non-claims of the celestial/twistor branch). Self-dual is *not* full 4D physics. Until and unless this contract closes, the celestial/twistor branch claims *none* of the following: both helicities (the anti-self-dual completion), generic scattering amplitudes, local matter fields, full Einstein dynamics (beyond the self-dual sector), electroweak symmetry breaking, confinement. The classical field-equation route (entanglement equilibrium, v358/v359) is a separate, non-competing statement about the full equation; `SEAM.EQUIV.01` and its route split (`SEAM.EQUIV.MMST.01`/`SEAM.EQUIV.TWISTOR.01`) are stated in their own rows and are not moved by anything here. Status: `WOIT.OS.TWISTOR.01` [O] (research contract; the α stage is executed — `WOIT.THETA.FREE.01` [E], v519 — with kill test (1) discharged at the free level only; the β_1 stage is executed — `WOIT.BETA1.GSO.01`, v522, verdict UNDECIDED per the frozen preregistration, with kill test (2)'s free shadow not firing, the gauge datum typed as the GSO $\mathbb{Z}_2$ and contract precision (iii) added; the β_2 stage is executed — `WOIT.BETA2.OS.01`, v524, verdict SUCCESS, [C]-typed per precision (iii), with the OS quotient explicit and PD at both levels, the clock a positive transfer step with spectral calculus and a reconstructed unitary rotation group, and the pre-declared KMS deviation carrying exactly the silver witnesses; the β_3 stage is executed — `WOIT.BETA3.DUALITY.01`, v565, verdict SUCCESS, with the kill branch empty by algebra, the OS cut and the $(2, 2)$ signature forced on the Minkowski branch of the duality class, and the unique induced seam member identified with the v524 $\Theta_{\rm phys}$ carrier; no marker move).

Research Contract — `CCC.SEAM.CROSSOVER.01` (the conformal-cyclic crossover on the seam geometry)

Typing. A research contract [O] (ledger row `CCC.SEAM.CROSSOVER.01`: Open, research contract), *not* a claim. It engages Penrose's Conformal Cyclic Cosmology (Tod, `arXiv:1309.7248`; Penrose–Meissner, `arXiv:2503.24263`) as a *named reference frame* for the shape of the target — never a confirmation in either direction. The cyclic reading of `origin_theory` stays [C] regardless of this contract's fate; nothing below asserts that CCC follows from TFPT.

The certified kinematic base [E] ([verification/v957_ccc_crossover_kinematics.py],

CCC.SEAM.KINEMATICS.01; six discovery probes embedded byte-exact, 28+29+18+21+14+7 **gates).** The corpus objects already carry the complete CCC crossover *kinematics*: (i) Penrose's reciprocal gauge $\hat{\Omega}\check{\Omega} = -1$ is the element $\tau = \rho^2\sigma : z \mapsto -1/z$ of the proven μ_4 Möbius stabiliser D_4 — τ reverses the clock (= the $\mathbb{Z}_2$ sheet parity), swaps the aeon poles $\{0, \infty\}$, and the crossover locus $|\hat{\Omega}| = |\check{\Omega}|$ is the unit circle carrying all four marks; (ii) conformal-factor rigidity: on the 2D seam the clock alone leaves exactly *one* constant-curvature modulus and the reciprocal $\mathbb{Z}_2$ kills it; on the 3D crossover the deck alone kills *all* conformal moduli (the $so(4,1)$ deck-commutant is 4-dimensional, all Killing, boost content 0; globally unique on the explicit Obata family) — TFPT's candidate answer to CCC's open unique-conformal-factor problem, with v_{geo} staying pure unit calibration (VGEO.TORSOR.01); (iii) the entropy reset of the cyclic reading is a *reduced channel*: the v221 transport is CPTP with an explicit Stinespring isometry, $\|T^n - P_\star\|_2 = (2/3)^{6n}$ exactly, the global overlap invariant, the microstate record in the complement — "unitarity + rank-1 attractor" is isometry + partial trace, not a contradiction (with the executed sharpening that this deployed Kraus dilation is *entanglement breaking*: it stores the record entirely in the environment and carries no coherent transport in the visible three-state system, quantum capacity 0, [verification/v976_seam_lift_birkhoff.py]); (iv) the kinematic 2D → 3D → 4D uplift: the seam sphere *is* the Hopf base of S^3 (stereographic $= z_1/z_2$ exactly), the crossover $X = S^3/\mathbb{Z}_4$ is the lens boundary of the corpus A_3 ALE (deck free, clock and reciprocal descend, marks = four linked fibre circles in the two sheet orbits), the 4D bridge is the cone $\mathbb{C}^2/\mathbb{Z}_4$ with the inversion as conformal aeon swap fixing the unit lens pointwise, and the Lorentzian shadow $(1/\cos^2 T)(-dT^2 + g_{S^3})$ is Einstein $R_{ab} = 3g_{ab}$ with a spacelike round-S^3 crossover and $\check{\Omega} = -\cos T$ vanishing to first order; (v) the KMS/modular closure at the free level: $T = e^{-H_{\text{mod}}}$ with $\operatorname{spec}(H_{\text{mod}}) = \{0, 6\ln\frac{3}{2}, 6\ln 3\}$ exactly, detailed balance and OS positivity exact, and the v526 normalisation $\beta_{\text{angle}} = 2\pi = 1/(4c_3)$ = the Gibbons–Hawking normalisation on the crossover fibre — one discrete clock tick = one unit of modular time; (vi) the holonomy dictionary: pole strings carry the full μ_4 monodromy, every seam string carries exactly the 2-torsion -1 = the v522 GSO/fermion-parity wrap (typed [C] correspondence); (vii) the selector necessity: the admissible OS reflections of the mark family $M(\delta)$ close to D_4 *iff* $\delta = \pi/2$, and at Majorana operator level (fermionic signs kept) the clock-axis cuts are Θ-real only at the μ_4 configuration — the *uniqueness* half of the v534 straddle selector is thereby *derived* as a Θ-reality necessity (the survival half and the coupling sign stay v534's nonperturbative toy facts, cited); (viii) the frozen Gate-6 kernel: a sharp causal top-hat disc of radius $\theta_{\max} = \eta_{\text{rec}}/(\eta_0 - \eta_{\text{rec}}) \approx 1.16°$, decay rates $\ln(729/64)$ and $\ln 729$ (ratio $\ln 3/\ln\frac{3}{2} = 2.709511$), κ derived from the v526 normalisation up to a declared convention band (relic contrast $\leq 3.3 \times 10^{-4}$ across all of it) — frozen *before* any data contact and feeding the preregistered search `experiments/ccc-crossover-disc` (hypotheses, kill criteria K1–K6, null battery and decision rule frozen there; verdict slot `data_limited`, no data pass).

Demands (the dynamical crossover, in order). (D1) the interacting seam algebra $\mathcal{A}_{\text{hol}}$ realised *on* the lens crossover with reflection positivity protected on the mid-gap cut class while the aeon transfer uses the through-mark class (the two D_4 reflection classes whose product is the clock; the v529 straddle law is exactly this separation at toy level) — this *is* WOIT.OS.TWISTOR.01 stage γ in the crossover geometry, and closing it closes D1; **(D2)** the aeon-transfer channel as a Stinespring isometry at *field* level (the v957 finite-dimensional dilation lifted to the reconstructed net); **(D3)** the cosmology fork *decided*: exactly one active story — Starobinsky ($r = 12/N_\star^2 \in [0.0033, 0.0048]$, within reach of $\sigma(r) \sim 10^{-3}$ experiments), a crossover/gravitational-wave epoch, or a computed hybrid; **(D4)** the preregistered disc search executed per its frozen hypotheses — **EXECUTED 2026-08-24 with a robust null** (first data contact, Planck PR3 SMICA map, integrity SHA-256 pinned in `results.json`, 2 013 312 960 bytes; freeze intact 3/3 hashes, frozen-kernel guards 5/5 *before* data contact; seed 20260824, reruns bit-identical): $\max|\text{SNR}| = 5.119$ on data vs null mean 5.20 over 100 Gaussian ΛCDM sims (matched mask and spectrum), $p_{\text{global}} = 0.673$, $p_{\text{counts}} = 0.446$, BH-$q_{\min} = 0.673$ — the $q < 0.01$ threshold clearly missed; 171 candidates $|\text{SNR}| \geq 4$ with 83 pos / 88 neg (no sign excess); injection recovery 100% ($20 \times 200\,\mu$K discs at 1.16° — the search is well-powered, the null is a real null); kill criteria K1–K6 all NA (they fire only on a resolved relic;

none exists); the replication legs (half-mission, Commander/NILC) are not required for a null per the frozen decision rule and stay available for future candidates; three amendments disclosed (seed pinning of the unseeded synfast preflight, reporting fields, the documented estimator ℓ-weighting) — statistics, thresholds and kill criteria unchanged. **No kill criterion fired; the contract stays open via D1–D3; no type change.**

Kill criteria (frozen). A free fitted conformal factor anywhere; failure of the mid-gap RP protection on the interacting algebra (kill test (2) of the WOIT γ stage); the interacting level breaking the v522 wrap $= (-1)^F$ identification (which would reopen the deck fork the kinematics closed conditionally); a measured r incompatible with the surviving cosmology branch; a resolved crossover relic violating the frozen template class (K1–K6, including the K6 contrast bound that would kill the derived-κ chain); and — typed numerological, forbidden — any identification of the four μ_4 marks with Hawking-point *counts*.

Honest scope. Deck fork: topology searches cannot decide it (lens geodesic $\geq 28\,D_H$ vs matched-circle reach $6.3\,D_H$; 32σ from curvature detectability) — it is decided *conditionally* by the v522 wrap identification (spacetime branch) and stays typed. The executed D4 null leaves the compiler core untouched (the frozen kernel was *output* of the certified kinematics, not input) and asserts nothing about CCC in either direction; the cyclic reading stays [C]. The contract now closes only through D1–D3; the kinematic base [E] stands on its own either way.

Firewall Contract — `DIMENSION.UPLIFT.FIREWALL.01` (no 1+1D theorem closes a 3+1D claim)

Typing. A *standing discipline rule*, in force immediately (ledger row `DIMENSION.UPLIFT.FIREWALL.01`); the uplift theorems it names are separate [O] contracts. It formalizes the core diagnosis of the 2026-08-27 external deep review: *a chiral* 1+1*D seam net does not imply a local, interacting* 3+1*D quantum world.* The corpus's strongest finite results are boundary/kinematic building blocks — the review found several surfaces where boundary theory, continuous relaxation, modular time, Lorentz geometry and physical 4D dynamics were in danger of being silently equated. This firewall separates them by rule.

The rule (two clauses). (F1) No theorem whose hypotheses live in 1+1D (a chiral net, a VOA, a lattice collar, the seam circle) may be cited as *closing* a claim about 3+1D physics. **(F2)** Every contract closure must state the spacetime dimension of its *hypotheses* and of its *conclusion* (dimension typing), with the same discipline as the [E]/[C]/[O]/[X] markers.

Instances audited at registration (honesty census). (i) `SEAM.EQUIV.01` and both its routes are 1+1D in *and* out: the cited scaling-limit framework (Osborne–Stottmeister, `arXiv:2107.13834`) treats 1+1D free lattice fermions, the OS-reconstruction selector (Adamo–Moriwaki–Tanimoto, `arXiv:2407.18222`) treats 2D full VOAs, and lemma L2 of the externalization package below claims convergence to a *chiral* c=8 *CFT* — nothing more. **Even a fully proved `SEAM.EQUIV.01` is a rigorous boundary building block, not the 4D theory.** (ii) The `WOIT.OS.TWISTOR.01` target theorem (Poincaré representation, local causality, one chiral generation, the internal gauge action) *is* the uplift demand; it stays [O] with all seven kill tests live. (iii) The kinematic 2D → 3D → 4D uplift (v957; the discovery probe `experiments/tfpt-discovery/ccc_uplift_4d_probe.py` correctly self-labels as kinematic) supplies a canonical Hopf fibre / lens crossover / 4D cone with an Einstein Lorentzian shadow — *none of which* closes this firewall: a metric uplift is not an interacting local quantum theory. (iv) Anomaly inflow: the non-gappable c_{-}=8 edge is naturally the boundary of a higher-dimensional *invertible/topological* bulk (Witten–Yonekura, `arXiv:1909.08775`; Hsieh–Tachikawa–Yonekura, `arXiv:2003.11550`); the inflow bulk and the *physical* 3+1D spacetime bulk are two different identifications that must never be silently equated.

The realistic route ordering (registered as the programme skeleton). (1) keep the seam as an internal/topological boundary module (all current [E] results survive unchanged as such); **(2)** build a genuine 4D Euclidean lattice theory — TFPT fixes the representations, charges, coupling relations and boundary conditions; the 4D lattice carries spacetime, gauge fields, Higgs and fermions; **(3)** prove the 4D OS package on it (`QFT4D.OS.RECON.01` below); **(4)** prove chirality and mirror decoupling *in the same 4D model* (`CHIRAL4D.NOMIRROR.01` below); **(5)** only then apply Bisognano–Wichmann/CHM and entanglement gravity (the direction discipline of `GRAV.NONCIRCULAR.01` below); **(6)** then prove bulk–seam equivalence, identifying the seam as the topological/anomalous boundary condition of the *constructed* 4D bulk (`SEAM.BULK4D.RECON.01` below). The twistor/celestial alternative (reconstruct the bulk from boundary correlators) is riskier: boundary correlators do not guarantee bulk locality (cf. `arXiv:2602.04223`); it stays a named parallel route (`WOIT.OS.TWISTOR.01`/`CELEST.SEAM.01`), not the trunk.

Status. Firewall in force (F1/F2 bind every future closure); the uplift theorems live in `SEAM.BULK4D.RECON.01`, `QFT4D.OS.RECON.01` and `CHIRAL4D.NOMIRROR.01`, all [O].

Research Contract — `SEAM.BULK4D.RECON.01` (from seam data to a 4D local quantum theory)

Typing. A research contract [O] (ledger row `SEAM.BULK4D.RECON.01`: Open, research contract), *not* a claim. It makes explicit the theorem that was previously only implicit between the `SEAM.EQUIV.01` conclusion (a chiral c=8 boundary object) and the `WOIT.OS.TWISTOR.01` goal statement (a Poincaré-covariant, causal, chiral 4D theory): **the missing seam→bulk reconstruction is a theorem of its own, and no existing corpus result closes any part of it.**

Input. The certified seam data: the gapped μ_4-equivariant collar (v302), the index-4 Q-system (v125/v154), the $(E_8)_1$ character and modular data (v377/v378), reflection positivity (v379), the clock and mark structure (v519–v524), and the kinematic uplift dictionary (v957).

Target theorem (all seven demands; none is currently closed). From seam data alone, construct a 4D family of local algebras $\{\mathcal{A}(O)\}_{O\subset\mathbb{R}^{1,3}}$ (equivalently a family of Schwinger functions) and prove: **(B1)** 4D locality/microlocality (spacelike commutativity; the correct wavefront sets); **(B2)** Poincaré covariance with a positive energy spectrum (the spectrum condition); **(B3)** spin–statistics and CPT; **(B4)** *both* helicities (the anti-self-dual completion — the known hard wall of the twistor route); **(B5)** cluster decomposition; **(B6)** a nontrivial S-matrix (the theory is interacting, not generalized-free); **(B7)** a *unique* seam↔bulk operator dictionary (which boundary operator is which bulk field, uniquely — not only at character level).

Kill tests (pre-registered). **(1)** every 4D extension of the seam state is a generalized free field (B6 fails structurally); **(2)** the inflow bulk of the c_-=8 edge is *provably* only invertible/topological, with no local propagating completion carrying both helicities (B1/B4 fail); **(3)** the dictionary is non-unique at operator level (character-level agreement with OPE mismatch — the same failure mode as kill test (5) of `WOIT.OS.TWISTOR.01`).

Honest relations. `WOIT.OS.TWISTOR.01` (twistor OS quotient) and the route ordering of `DIMENSION.UPLIFT.FIREWALL.01` (constructive 4D lattice first, seam as its boundary condition) are the two candidate proofs; this contract is route-agnostic and states only the target. A Hopf-fiber uplift, a 4D cone, or an Einstein metric on the bridge (v957) does *not* close it. Status: `SEAM.BULK4D.RECON.01` [O].

Spectator obstruction (2026-09-04, [`verification/v1020_spectator_sector_obstruction.py`]). Demand (B7) as worded—a *unique* seam↔bulk operator dictionary from seam data alone—is unsatisfiable: for any decoupled anomaly-free sector C with trivial boundary, $\partial(B\otimes C) = \partial(B)\otimes\mathbf{1} = \partial(B)$, so B and $B\oplus C$ share every seam datum. A finite free-fermion witness ([`verification/v1020_spectator_sector_obstruction.py`]) exhibits this concretely (Chern, edge modes and seam-restricted covariance identical for B and $B\oplus C$; the bulk differs). Uniqueness is re-scoped to `BULK.PRIME_COMPLETION.01` below (prime/seam-generated completions, trivial relative commutant); the status of `SEAM.BULK4D.RECON.01` is unchanged [O].

Research Contract — `BULK.PRIME_COMPLETION.01` (prime/seam-generated completions, no spectator)

Typing. A research contract [O] (ledger row `BULK.PRIME_COMPLETION.01`: Open, research contract), *not* a claim. Registered 2026-09-04 from the r644 audit harvest; it replaces the "unique from seam data alone" reading of `SEAM.BULK4D.RECON.01`/B7 by uniqueness *in the category of seam-generated (prime) completions*. **Dimension typing** (`DIMENSION.UPLIFT.FIREWALL.01` F2): the hypothesis is 2+1D free-fermion, the conclusion is a re-scoped 4D demand.

Demand. (1) an explicit family of seam-derived subalgebras $\mathcal{A}_i$ (the seam CAR algebra, its time evolution, inclusions/intersections); (2) generatedness $\mathcal{A}_{\rm bulk} = \mathrm{vN}(\bigvee_i \mathcal{A}_i)$; (3) *no-spectator*: trivial relative commutant $(\bigvee_i \mathcal{A}_i)' \cap \mathcal{A}_{\rm bulk} = \mathbb{C}\cdot\mathbf{1}$, equivalently no decoupled tensor factor with trivial boundary map; (4) uniqueness *only* within this category; (5) stability of (2)/(3) in the 3+1D limit.

Kill tests (pre-registered). A seam-generated completion with nontrivial relative commutant, or two inequivalent prime completions.

Finite witness ([verification/v1020_spectator_sector_obstruction.py]). In the 2+1D free-fermion strip, stacking a trivial gapped sector C (Chern 0, gap >1) onto the seam model B leaves every seam datum unchanged (Chern $C_B = C_{B\oplus C} = -1$, $C_C = 0$; zero modes 2/0/2 with B-support 1; edge dispersion coincident to 1.4×10^{-15}; seam-restricted covariance identical to 0; edge-profile overlap 1) and changes the bulk (dimension 16 vs 32, $E_0(C) = -25.31$, mid-strip covariance rank 4 vs 8, $\|\Gamma_0 - \Gamma_\alpha\|_{\rm HS}$ 0.752536 vs 0.760202). Krylov generatedness and the commutant test separate B from $B \oplus C$ by exactly $\dim C$ at every momentum; $\Pi_C = 0 \oplus \mathbf{1}$ lies in the commutant (residual 0); selection is unique (B selected, $B \oplus C$ rejected). *Honest:* spectator_dim$(B) = 10$ at $p = 0$ only (no onsite term at $M = 1$), so the criterion is "generated almost everywhere in p", not pointwise. The 2D modular-invariant uniqueness of the holomorphic $(E_8)_1$ net is untouched and is *not* a 4D no-spectator theorem. Status: `BULK.PRIME_COMPLETION.01` [O].

Research Contracts — `DYN.MARKOV.EMBED.01` / `DYN.UNITARY.DILATION.01` (the three meanings of "dynamics")

Typing. Two contracts that split what the corpus has loosely called "dynamics" into its three inequivalent meanings: **(a)** the positive relaxation T^n to a fixed point; **(b)** a continuous dissipative semigroup $e^{t\log T}$; **(c)** reversible Lorentzian/unitary evolution e^{-itH}. Step (a)→(b) is now a machine-checked theorem for the concrete seam kernel ([E] at the finite level); step (b)→(c) is genuinely open and gets its own contract (its *discrete* collision/QCA leg is meanwhile executed exactly, [verification/v984_markov_qca_dilation.py]; the continuous legs remain the contract). Nothing here moves `CONTRACT.F.01` or the four observables.

`DYN.MARKOV.EMBED.01` — the finite kernel half is executed [E]. For the v221 seam transfer kernel $T = J + (2/3)^6\, u_2 u_2^\top + (1/3)^6\, u_3 u_3^\top$ (spectrum exactly $\{1, 64/729, 1/729\}$), the principal logarithm $Q = \log T$ is an *exact classical Markov generator* [verification/v971_markov_embedding_generator.py]: sympy-exact spectral calculus gives $e^Q = T$; every row of Q sums to zero *symbolically*; the three independent off-diagonal entries are exactly $\{\ln(9/8), 2\ln 3, 2\ln 3\} = \{0.117783, 2.197225, 2.197225\}$, all positive (the reviewer's independently verified numbers), with diagonals $-\ln(81/8), -\ln(81/8), -4\ln 3$; the uniformization certificate $Q + (4\ln 3)I \geq 0$ entrywise (diagonals $\ln 8, \ln 8, 0$, exact) proves $e^{tQ} = e^{-4t\ln 3}\, e^{t(Q+4\ln 3\, I)}$ entrywise nonnegative and doubly stochastic for *all* $t \geq 0$ — so T is embeddable and every root $T^{1/n} = e^{Q/n}$ is again a stochastic matrix. The generator spectrum is $\mathrm{spec}(Q) = \{0, -6\ln(3/2), -6\ln 3\} = -\mathrm{spec}(H_{\rm mod})$ (v957: $T = e^{-H_{\rm mod}}$), the gap is the seam gap $\Delta = 6\ln(3/2)$, and Q is symmetric (detailed balance w.r.t. the uniform state — the KMS/OS-positive free level). *Negative control (embeddability is not generic):* the same construction with second eigenvalue $-1/10$ is still doubly stochastic but has $\det = -1/200 < 0$, while every embeddable stochastic matrix has $\det e^Q = e^{\mathrm{tr}\, Q} > 0$ — provably *not* embeddable. The RP/OS spectral positivity of the seam kernel (`CONTRACT.F.01` axiom 3) is exactly what makes (a)→(b) real. *What stays [O] in this contract:* the same embedding statement at *field* level (on the reconstructed net / quasi-local algebra, uniformly in system size), not just on the 3-state kernel.

`DYN.UNITARY.DILATION.01` — the (b)→(c) step, [O]. The semigroup e^{tQ} is *irreversible statistical* dynamics — an H-theorem, not a Schrödinger equation. Fundamental quantum dynamics requires the chain $T = e^{-aH} \xrightarrow{\text{OS}} U(t) = e^{-itH}$ with H positive self-adjoint, and the dilation must be *physical*: **(D1)** a Stinespring/Hamiltonian dilation of the semigroup whose generator is (quasi-)local; **(D2)** size consistency (the dilation is compatible with the thermodynamic limit — no environment that grows pathologically with the system); **(D3)** a Lieb–Robinson bound (finite propagation speed for the dilated dynamics); **(D4)** the OS analytic continuation to a unitary one-parameter group on the reconstructed Hilbert space. *Kill tests:* every dilation of the seam semigroup is provably nonlocal; the Lieb–Robinson velocity diverges in the scaling limit; the OS continuation fails on the interacting algebra. *Relation:* demand D2 of `CCC.SEAM.CROSSOVER.01` (the Stinespring isometry at field level) already presupposes this contract; it is hereby named as the prerequisite.

Executed sharpening — the lift structure of the finite kernel ([E] at matrix level; the contract itself stays [O]). Two modules now pin down *which* dilations are even candidates. **(i)** *The no-go*: the Kraus dilation the corpus actually deploys, $K_{ij} = \sqrt{T_{ij}}\,|i\rangle\langle j|$, is measure-and-prepare — one step kills every coherence (fully symbolic), its Choi matrix is diagonal in the product basis, hence the channel is *entanglement breaking* with quantum capacity 0, and $T > 0$ entrywise gives zero-error capacity $C_0(T) = 0$ exactly ([`verification/v976_seam_lift_birkhoff.py`]); any dilation satisfying D1–D4 must therefore be a genuinely different object. **(ii)** *The hidden fibre*: the complete set of random-permutation (Birkhoff) quantizations of the single step B is a one-parameter line $t \in [0, 1/18]$ whose coordinate is the sign-character Fourier coefficient ($\widehat{w}_t(\text{sgn}) = 6t$); populations are exactly t-free, the one-step coherence spectrum is $\{1,\, 6t,\, (2/3)^{\times 2},\, (1/3)^{\times 2}\}$, and the triangle loop current is an exact eigen-observable $\Psi_t(A) = 6t\,A$ — one step and one current readout measure t without an ancilla. On the contrast qubit the Choi partial transpose has spectrum $\{-3t/2,\, 1/6 + 3t/2,\, 1/3 + 3t/2,\, 1/2 - 3t/2\}$: the deployed sixth-root spectrum sits *exactly* on the entanglement-breaking boundary because $2/3 + 1/3 = 1$ (mutated spectra miss by $\mp 1/12$), while at the full three-level the family stays NPT at every tested iteration, $\lambda_{\min} = -(1/3)(2/3)^n$, t-independent ([`verification/v976_seam_lift_birkhoff.py`]). **(iii)** *The coherent end*: B is *unistochastic* — an exact $U \in SU(3)$ with $|U_{ij}|^2 = B_{ij}$ entrywise, unique up to rephasing and complex conjugation, with Jarlskog invariant $J = \pm 1/27$ exactly; no *fixed* microstep unitary generates the relaxing sequence ($B^2 \neq |U^2|^2$; a Dirichlet recurrence witness returns U^{n^*} arbitrarily close to the identity while B^{n^*} is uniform), so repeated steps *require* environment contact between steps — precisely the D1–D4 demand — while the six-step macro-gate exists as its own reversible $U_6 \neq U_1^6$ ([`verification/v977_transfer_unistochastic_wilson.py`]). The classical kernel thus admits the full lift trichotomy {EB dilation | Birkhoff fibre with hidden circulation t | reversible macro-gate with hidden orientation sgn J}: the same classical data, three inequivalent quantum dynamics. Ledger rows: `TRANSFER.HIDDEN.CIRCULATION.01` and `TRANSFER.COHERENT.WILSON.01` (executed halves [E]; the fibre-point/branch selection, the defect–Birkhoff forcing functor — named `TRANSFER.DEFECT.BIRKHOFF.01` — and the center–transport intertwiner stay [O]). These results fix the contract's boundary conditions; they do not close it.

Executed update (2026-08-28, external master-object review wave 2): the *discrete* leg of D1–D3 is realized constructively — the exact collision-model dilation ([`verification/v984_markov_qca_dilation.py`]; the continuous legs stay [O]). One fresh ancilla qutrit per step, the mod-3 record $C|j,k\rangle = |j, j{+}k\rangle$, and the step unitary $V = C\,(U_B \otimes I)$ (with U_B the v977 lift) give, as *exact symbolic operator identities*: $V^\dagger V = I$; the Stinespring identity $\Phi(\rho) = \mathrm{Tr}_A[V(\rho \otimes |0\rangle\langle 0|)V^\dagger] = \sum_j P_j U_B \rho U_B^\dagger P_j$ (Kraus completeness $\sum_j K_j^\dagger K_j = I$); $\operatorname{diag}\Phi(\operatorname{diag} p) = Bp$ with vanishing off-diagonals; and six fresh ancillas give $\operatorname{diag}\Phi^6 = B^6 p = Tp$ — the v977 macro-gate as the reduced dynamics of an exactly unitary, strictly radius-1 collision band (unitary amplitudes → local environment coupling → dephasing → Markov transfer, in one construction). The must-fail fires (the closed unitary has $|U_B^2|^2 \neq B^2$), and a typed remark *against* the review text: a *reused* mod-3 adder ancilla still reproduces the populations exactly (the record j_1+i stays distinguishable) — fresh ancillas are needed for the coherences, not the diagonal chain. The conjugate branch V^* dilates the same B ($\operatorname{sgn} J = \pm 1/27$ rides through classically invisible; its selection stays [O]). What stays [O]: the (quasi-)local *Hamiltonian* dilation, D2 in the thermodynamic limit, the OS continuation $T = e^{-aH} \to U(t) = e^{-itH}$ (D4), and any 3+1D reading.

Executed update (2026-08-28, promote round): the OS/dilation *ladder* is executed at kernel + free-band + interacting-kernel + quantum-coupled level ([`verification/v987_os_dilation_package.py`]; continuous field-level OS and branch selection stay [O]). The seam single-step $B = P_1 + \frac{2}{3}P_2 + \frac{1}{3}P_3 > 0$ with exact orthogonal projectors; $H = -\log B$ is PSD with gap $\log\frac{3}{2}$; $T(t) = e^{-tH}$ is a stochastic semigroup inside the cone; $U(t) = e^{-itH}$ is a unitary group with Wick anchor $U(-in) = B^n$ for $n = 1..6$. Must-fail: the v971 negative kernel (second eigenvalue $-1/10$) has no real half-step. Size-uniform free band: $T_L = B^{\otimes L}$ has commuting local $H_L = \sum_x h_x$ with gap $\log\frac{3}{2}$ uniform in L; the per-cell reduced channel is B at $L = 1, 2$; Lieb–Robinson radius 1. Interacting congruence: $T_\varepsilon = K^{1/2} B^{\otimes L} K^{1/2}$ is SPD for every tested diagonal coupling (one-sided mutant $KB^{\otimes L}$ is not symmetric); gap plateau ~ 0.18 at $L = 2..4$. Quantum SWAP coupling: $d(\theta) \sim \theta^{1.99}$, CPTP preserved, LR radius 1. Display marker stays [O].

Executed update (2026-08-29, review wave 4): the finite ladder is now *complete* through continuous time ([`verification/v999_weak_collision_hp.py`]; the thermodynamic field limit stays [O]). The deployed strong collision is singular (rank 9/81) and cannot be the exponential of a finite GKSL generator. The typed successor is a fresh-ancilla weak-collision / Hudson–Parthasarathy family: generator extraction Richardson 1.098×10^{-10}; $\operatorname{diag} L_{\mathrm{GKSL}} = Q = \log B$ symbolically; $e^Q = B$ exact; GNS splitting defect 0; locality 1.355×10^{-20}. The finite ladder is now discrete / kernel / field-free / interacting-classical / quantum-coupled / continuous-time. Open = the thermodynamic field limit only. Display marker stays [O].

Precision (binding for all "universal dynamics" prose). The shared relaxation shape of the four `F_transfer` sectors (`FR.DYNAMICS.01`/`DYN.TRANSFER.UNIVERSAL.01`) is a *universal contraction class* — shared gap, shared multiplier algebra — *not* one physical clock: both thermal-time routes are machine-killed (v723/v724) and the frozen external-clock contract fired its K1 kill exactly (v777: no common continuous clock exists for the four jets). Any corpus phrase of the form "one native flow" is to be read in this contraction-class sense; the clocks stay external and sector-specific.

Research Contract — `QFT4D.OS.RECON.01` (the genuine 4D Euclidean lattice package)

Typing. A research contract [O] (ledger row `QFT4D.OS.RECON.01`: Open, research contract), *not* a claim. This is step (3) of the `DIMENSION.UPLIFT.FIREWALL.01` route ordering: the *nonperturbative* 4D existence statement that no perturbative or seam-level result can substitute for.

Input. TFPT fixes the discrete data: the representation content and charge lattice (carrier + glue), the coupling relations at the matching scale, and the boundary conditions (the seam as internal/topological boundary module). The 4D Euclidean lattice carries everything else: spacetime, gauge fields, Higgs, fermions.

Target theorem (the OS package). For the so-specified 4D lattice theory prove: **(O1)** reflection positivity of the lattice measure; **(O2)** Euclidean covariance restoration (hypercubic → full Euclidean group in the limit); **(O3)** clustering; **(O4)** uniform bounds (moment/regularity estimates uniform in lattice spacing and volume); **(O5)** existence of the continuum limit; **(O6)** Osterwalder–Schrader reconstruction of the Wightman/Haag–Kastler theory.

Firewall against the perturbative leg (binding phrasing rule). The existing all-orders result (`QFT4D.EG.ALLORDER.01`, v381) is a *local perturbative construction under the stated Epstein–*

Glaser/BRST hypotheses — that and no more. The weak adiabatic limit for massless particles exists as a cited theorem (Duch, `arXiv:1801.10147`), but its hypotheses must be *verified for the concrete TFPT model*; a seam-transfer gap does not substitute for the infrared structure of massless photons/gluons. Even a fully successful EG/BRST proof yields *none* of: confinement or the hadron spectrum, the Yang–Mills mass gap, asymptotic charged states in QED (the infraparticle problem), the nonperturbative Higgs vacuum, or a controlled full S-matrix. Every corpus phrase of the form "the perturbative 4D leg closes to all orders" is bound to this reading. Status: `QFT4D.OS.RECON.01` **[O]**.

Registered decision (2026-08-28, `QFT4D.LATTICE.FUNDAMENTAL.01`): the lattice-fundamental reading — a typed route/typing, *not* an [E] claim and *not* a marker move. TFPT's *physical* completeness criterion is the finite, local, unitary lattice quantum theory (Hamiltonian route: Gauss-law Hilbert space, hermitian local H, $T = e^{-aH}$ positive by construction, Lieb–Robinson cone). This contract — the exact continuum Osterwalder–Schrader limit — is retyped as *mathematical reinforcement*: a named strengthening programme, no longer the physical bottleneck. Poincaré invariance is an IR fixed-point property with experimentally bounded irrelevant Lorentz-violating operators. Justification anchors ([`verification/v989_lattice_gate_battery.py`]): the Euclidean overlap shortcut is killed at T2 (exact; N_t-exact confirmation: the $|\det D_{\mathrm{ov}}|^2$ Gram is PSD at $N_t = 2$ and indefinite at $N_t = 4$, det-only $\lambda_{\min} \approx -0.249$; Wilson control PSD everywhere — pinned on overlap time nonlocality), and the Hamiltonian route clears T2 by construction. Display: this row stays **[O]** as the math programme; nothing closes.

Amendment (2026-08-29, review wave 4): the reviewer's precision is accepted — "a single finite box is not a world". The fundamental object is the *quasilocal consistent family* $\{H_\Lambda\}$ with the thermodynamic-limit dynamics $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$. Binding phrasing: continuum ($a \to 0$) *optional* (mathematical reinforcement); thermodynamic limit ($|\Lambda| \to \infty$) *and* controlled IR universality *mandatory*. Decision typing UNCHANGED; no **[E]** claim, no marker move.

Executed update (2026-08-30, review wave 7): the 2+1D master-object scaffold is coherent ([`verification/v1008_master_assembly_scaffold.py`]; Decision typing UNCHANGED). Combinatorial $K = 0$ (every oriented plaquette has vanishing flat-x-holonomy exponent); frozen-link $q^2 \sim 4$ with $K(q = 0) = 0$; unique-contractive selector $|R'| < 1$; QWZ wall + DET4 mirror (physical slope ≈ -0.9989, mirror gap ≥ 1.5 through $t = 0.6$); four-block assembly with $\mathbb{Z}_6$ content $(1, 2, 3, 0)$ and clock $[0, 1, 1, 2]$. $L = 3$ Hilbert spaces of dimension 262144 / 129024 stay out of suite. No **[E]** claim, no marker move.

Executed update (2026-08-30, afternoon harvest; [`verification/v1011_3p1d_ladder_battery.py`]; Decision typing UNCHANGED). The 3+1D ladder B6–B9 is complete at scaffold level: the minimal 3+1D object is viable (exact Gauss 32768 checks, first 3+1D confinement witness, wall separation, exact in-face isotropy, Kronecker-level stiffness $g_\star \approx 1.915$); fully coupled link–wall dynamics are demonstrated (exact maximal-tree gauge fixing $524288 \to 4096$, tree-independence, DET survives coupling, back-reaction); the Kronecker limitation of the 2+1D scaffold is removed. No **[E]** claim, no marker move.

Executed update (2026-08-30, late-evening harvest; [`verification/v1013_thermodynamic_dynamics.py`]; Decision typing UNCHANGED). The mandatory-dynamics leg is CLOSED at Hamiltonian-class level. Cited `articles/2026-08-30/tfpt_thermodynamic_dynamics_en.tex`: uniform Lieb–Robinson bound $J = 12/5$, $R = 2$, $z = 4$, $\kappa = 16043/450$, $v_{\mathrm{LR}} = 32086e/225 \sim 387.64$; norm-convergent τ_t; gauge-invariant quasilocal $\mathcal{A}^G$ (Gauss caveat resolved on the invariant subalgebra); existence of ground and β-KMS states. Numeric twin: measured $v = 2.0 \ll$ proved bound; the analytic finite-volume bound dominates nested differences at buffers 8/16/32. Remaining: phase uniqueness, limit gap, IR universality. No **[E]** claim, no marker move.

Research Contract — `CHIRAL4D.NOMIRROR.01` (the chiral 4D lattice matter theorem)

Typing. A research contract [O] (ledger row `CHIRAL4D.NOMIRROR.01`: Open, research contract), *not* a claim. Step (4) of the firewall route ordering: chirality and mirror decoupling must be proven *in the same 4D model* that `QFT4D.OS.RECON.01` constructs — the seam's c_{-}=8 and Chern/inflow structure are a boundary-*anomaly* signal, not yet 4D matter.

Target theorem. An explicit chiral lattice operator (Ginsparg–Wilson/overlap type) for the TFPT representation content with proofs of: **(C1)** a local, gauge-invariant fermion measure; **(C2)** perturbative *and* global anomaly freedom; **(C3)** no Witten $SU(2)$ anomaly; **(C4)** no mirror generation in the physical spectrum (the mirror decouples with a mass that survives the continuum limit); **(C5)** the correct index and charges (the TFPT charge lattice, generation count N_{fam}); **(C6)** a stable continuum limit compatible with the `QFT4D.OS.RECON.01` package.

Calibration (state of the art, honestly). The strongest comparable theorem is Mastropietro's construction (`arXiv:2303.02790`): a *weakly coupled* 4D $U(1)$ chiral gauge theory with a *massive* gauge field. The TFPT target — chiral $SU(3)\times SU(2)\times U(1)$ with three generations, massless nonabelian gauge fields, EWSB and confinement downstream — is far stronger; the gap between the two is the honest measure of this contract's difficulty. *Kill tests:* the mirror fails to decouple (kill test (3) of `WOIT.OS.TWISTOR.01` at lattice level); a Witten anomaly obstructs the carrier $SU(2)$; every gauge-invariant measure for the TFPT content is provably nonlocal. *Consistency note (ledger wins):* the celestial-branch scope fence above honestly lists interacting RP, mirror decoupling, the internal $SU(2)$, both helicities, local matter fields, EWSB and confinement as *not* closed; the contradictory "zero open internal mechanisms" phrase in `docs/OPEN_PROBLEMS.md` is corrected in the same change that registers this contract (that phrase was a certification statement about a *different*, narrower residual matrix and had drifted into an overclaim). Status: `CHIRAL4D.NOMIRROR.01` [O].

Executed update (2026-08-28, promote round): a 1+1D chiral Gauss census is executed on the Hamiltonian route (`[verification/v989_lattice_gate_battery.py]`**; the 3+1D contract stays [O]).** Two-species $\mathbb{Z}_4$ Hamiltonian, factorized 16×256 (the 4096 Kronecker is never materialised): a *single-exponent* Gauss unitary exists iff $q_L\equiv q_R$; the covariant two-exponent Gauss exists for every charge pair (lattice Dirac doubling / Nielsen–Ninomiya as a Gauss statement); $V^4=I$; the mutant $(g_L,g_R)=(1,0)$ on vectorlike content fails. Honest: not the TFPT representation content and not 3+1D. Display marker stays [O].

Executed update (2026-08-28, completeness wave): the SMG mechanism is toy-proved (`[verification/v997_spin2_smg_witnesses.py]`**; the 4D theorem stays [O]).** Fidkowski–Kitaev SMG toy: at $N=8$ Majoranas a unique Spin(7)-singlet ground state (Schmidt rank 1) with volume-independent gap 14 at $L=1,2,3$; the $N=4$ contrast is 2^L-degenerate (must-fail); the physical chiral edge keeps its π/L spectrum exactly. SMG mechanism toy-proved [E]; the 4D Spin(10) theorem stays [O]. Display marker stays [O].

Executed update (2026-08-29, review wave 5): DET16 finite-cluster hit + the Casimir-projector kill (`[verification/v1002_spin10_mirror_projector.py]`**; the 4D theorem stays [O]).** The number-violating determinant vertex supplies an exact rank-one projector on the full 2^{16} cluster: $\Omega^\dagger\Omega=P_{16}$, $\Omega\Omega^\dagger=P_0$, defect 0; P_ϕ hermitian idempotent (2.8×10^{-18}), rank 1; h_{mir} spectrum $\{0^1,1^{65535}\}$, gap exactly 1; all 45 $\mathfrak{so}(10)$ Fock commutators vanish ($\det\rho_{16}=1$); parity commutator 0; $|0\rangle$ and $|F\rangle$ are singlets. Hopping on the 4-mode analogue is stable at $t=0.2$ (gaps 0.90 / 0.86); spectators untouched. NEW candidate `CHIRAL4D.MIRROR.DET16.01` [C]. Independently, the number-preserving Casimir projector route is *excluded*: Weyl-alternant census $|W(D_5)|$=1920 gives $\{N{=}0{:}1,\ N{=}16{:}1\}$ (none at N=8), k=2 half-filling multiplicity 27, k=3:1463 — uniqueness fails at every cluster size. Display marker of this contract stays [O].

Research Contract — `CHIRAL4D.MIRROR.DET16.01` (DET16 rank-one projector, finite cluster)

Typing. A candidate [C] (ledger row `CHIRAL4D.MIRROR.DET16.01`), registered 2026-08-29 from review wave 5. ([`verification/v1002_spin10_mirror_projector.py`].) **Content.** Exact finite-cluster gapping on the full chiral 2^{16} Fock space via the number-violating determinant vertex, as executed above. **Open.** Dynamical gauge fields, domain-wall geometry, and a 4D volume theorem — this is not a 3+1D closure. **Relation.** Does *not* upgrade `CHIRAL4D.NOMIRROR.01`; the Casimir-projector kill is recorded on that parent. Status: Candidate [C].

Executed update (2026-08-30, review wave 7): T1/T2 proved, T3 Michalakis–Zwolak cited-verified, dynamical [N], 1/192 and BDL boxed ([`verification/v1007_det16_stability_battery.py`]; the candidate stays [C]). The 12-page memorandum `articles/2026-08-30/det16_mirror_gap_theorem_en.tex` proves T1/T2 and verifies every Michalakis–Zwolak hypothesis (Local-TQO, $t_* > 0$, perturbed gap at least 1/2). Ordinary hopping is *not* block-diagonal ($\|(1-P)KP\| \neq 0$), so $t_{0,\mathrm{count}} = 1/192$ remains the uncertified counting threshold, not a theorem. Dynamical $\mathbb{Z}_2/\mathbb{Z}_4$ links on the four-mode analogue are uniquely gapped on the compressed grid (full-probe minimum 0.728854416 pinned); second-order Schrieffer–Wolff is $1 - \alpha t^2 + O(t^4)$ with c_{geo}/L bounded. BDL supplies a ground-energy truncation error with unspecified constants, not a volume-uniform gap remainder — boxed. Display marker stays [C].

Executed update (2026-08-31, Monday-morning harvest; [`verification/v1016_state_gap_batteries.py`]; the candidate stays [C]). Fröhlich–Pizzo hypotheses cited-verified for $\mathbb{Z}_2/\mathbb{Z}_4$ DET open chains: volume-uniform spectral gap $\geq \Delta/2$ in the parametric window τ_{FP} ($a = 0.0233$, $a/16 = 0.00146$). The dynamical-link leg is closed at that finite-group level. Remaining: rotor links, wall mixing, 3+1D placement. Display marker stays [C].

Executed update (2026-09-05, round-3 integration): derivative-filtered wall block (`CHIRAL4D.MIRROR.DFILTER.BLOCK.01`; [`verification/v1023_derivative_filtered_wall.py`]; no parent marker move). For

$$H_{\mathrm{df}} = \Delta b^\dagger b + \lambda (b + gD_{\mathrm{GW}}p)^\dagger (b + gD_{\mathrm{GW}}p)$$

each singular-value sector is the exact 2×2 block with determinant $\lambda g^2 \Delta s^2$. It preserves $\ker D_{\mathrm{GW}}$ on the physical summand, the adiabatically mirror-continuated high branch is at least $\Delta + \lambda$, and $\mu_-(s) = \lambda g^2 \Delta (\Delta+\lambda)^{-1} s^2 + O(s^4)$. The finite covariance and SVD-reduction defects are 2.408×10^{-15} and 1.776×10^{-15}. This is only a finite quasi-free block lemma. For $s \neq 0$ the bare b subspace is not invariant and has low-band weight, so no bare-mirror response gap is proved; the low branch has $z = 2$, not a Weyl cone. No canonical DET16/DET32 realization, interacting parent, admissibility/quasilocality theorem, measure/anomaly/index control, universality or 3+1D continuum limit follows. The derivative-filtered witness therefore does not upgrade `CHIRAL4D.MIRROR.DET16.01` [C], `CHIRAL4D.NOMIRROR.01` [O], T4, or TOE.

Firewall Contract — `GRAV.NONCIRCULAR.01` (the Bisognano–Wichmann direction discipline)

Typing. A *standing discipline rule*, in force immediately (ledger row `GRAV.NONCIRCULAR.01`), with an audit census at registration. Bisognano–Wichmann (BW) is a theorem *within* an already-local, Poincaré-covariant QFT with a suitable vacuum: the modular flow of a wedge algebra acts as the boost. It is *not* a device that generates Lorentz spacetime from an arbitrary modular flow — there exist Poincaré-covariant nets violating BW (Mund, `arXiv:hep-th/0101227`, characterizes when BW holds; Morinelli, `arXiv:1703.06831`, exhibits the failure landscape). The only correct proof order is: 4D net + locality + vacuum + spectrum condition ⇒ BW ⇒ geometric modular time. *Never*: KMS/modular spectrum ⇒ Lorentz space ⇒ BW.

The rule. Every use of BW or of the CHM ball modular Hamiltonian in the corpus must *declare* whether 4D Poincaré covariance is an ASSUMPTION (imported; the result is then conditional on `QFT4D.OS.RECON.01`-class existence) or a RESULT (derived intrinsically). The cycle risk this fences: BW/CHM entering the Einstein derivation as input while the 4D locality/Lorentz structure is still supposed to *come out* of the seam.

Audit census at registration (all current BW/CHM usages, annotated).

- ASSUMPTION — the entanglement-equilibrium Einstein derivation (v323/v358/v359; `tfpt_4_frontier` and the (G_{metric}) section above): the CHM ball modular Hamiltonian and the BW boost are *inputs*; the derivation is conditional on a 4D Poincaré-covariant QFT with the right vacuum existing (`QFT4D.OS.RECON.01`) and must never be cited as evidence *for* that existence.
- ASSUMPTION (DECLARED) — `QG.AMB.01` as a [C] redundancy (v369) is everywhere stated "conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann"; correct as declared.
- ASSUMPTION (DIMENSION-TYPED) — the seam KMS normalisation chain (v259/v526; $\beta_{\mathrm{angle}} = 2\pi$): uses only the universal modular/KMS angle of Tomita–Takesaki theory on the *seam* (1+1D/2D objects), not 4D BW; the 2π is a normalisation, not a spacetime derivation.
- RESULT-DIRECTION (OPEN) — the intrinsic-BW programme (v438/v440/v446 and the half-sided modular inclusion route in the (G_{metric}) section): the honest attempt to *derive* geometric modular action intrinsically; explicitly not closed, and the corpus already records the circularity warning ("BW would presuppose the very covariance it should produce", v215).
- EVIDENCE, NOT DERIVATION — the compressed-state flow to the CHM/BW geometric form (v478): a numerical convergence statement in the result direction; it does not establish 4D covariance.
- CONDITIONAL (DECLARED) — `origin_theory`'s "BW downstream of the seam being a chiral net": stated as conditional on net identification; correct as declared.

Status: firewall in force; no marker moves — the census confirms the declarations exist but binds all *future* uses to carry them explicitly.

Research Contract — `OBS.TRANSDUCTION.01` (the transduction principle: eigenvalue, projection, coupling)

Typing. A research contract (ledger row `OBS.TRANSDUCTION.01`); the principle is machine-checked on the minimal model at exploration level, the per-bundle transduction proofs are [O]. It elevates the S15 typing rule to a named principle:

$$\text{observable signature} \;=\; \underbrace{\text{internal eigenvalue}}_{\text{compiler [E]}} \;+\; \underbrace{B\,P_r \neq 0}_{\text{transduction}} \;+\; \underbrace{\text{physical coupling operator}}_{\text{physics}}.$$

Without the middle arrow, neither a non-observation falsifies nor an observation confirms. The minimal model is machine-checked in `experiments/theory-contracts/fo01_transduction_invisibility.py`: a uniform measurement functional sees the seam kernel *exactly not at all* ($\mathbf{1}^{\mathsf{T}}(T^n p - \pi) = 0$ to 10^{-14}; burst-train statistics KS-indistinguishable from a scrambled control), while a character-resolved channel reconstructs the eigenvalue $64/729$ to 10^{-10} — invisibility is a property of the readout B, not of the channel.

Demands. For each empirical bundle (FRB/GW/pulsar nulls; the axion/birefringence family; any future search), *prove or refute* $B\,P_r \neq 0$ for the declared readout before the null or the hit is scored. *Binding application (the birefringence caution):* the ACT+Planck isotropic birefringence being close to $\beta_{\mathrm{TFPT}} = \varphi_0/(4\pi)$ (`arXiv:2608.06480`) is an interesting coincidence, *not* proof, until the shared normalized axion action and the transduction chain are proven — the axion bundle (β, f_a, m_a, $g_{a\gamma\gamma}$,

Ω_a, EDM) must follow from *one* action with *one* proven coupling. Status: `OBS.TRANSDUCTION.01` — principle registered, minimal model checked; every per-bundle middle arrow [O].

Executed update (2026-08-28, promote round): a finite $W[J]$ transduction shadow is executed ([verification/v990_wj_transduction_mechanism.py]; per-bundle proofs stay [O]). On a 1+1D two-site $\mathbb{Z}_6$ Hamiltonian (species Q and e^c), the mixed derivative $\partial^2 W/\partial J\,\partial\theta$ is nonzero on the seam-coupled link ($\approx -9.3\times10^{-3}$) and *exactly* 0 on the decoupled e^c control; a frozen-link companion transduces on the Q-staggered operator and stays 0 on e^c. The middle arrow is thereby a machine-checked finite-level mechanism, not a 4D bundle proof. Display: per-bundle empirical proofs stay [O].

Research Contract — `PRED.JOINTLIKELIHOOD.01` (from hit-counting to overdetermined relations)

Typing. A research contract [O] (ledger row `PRED.JOINTLIKELIHOOD.01`: Open, statistics-construction contract), *not* a claim. The 27+ individually matched values all descend from the same few atoms (φ_0, c_3, g_{car}, μ_4) and are therefore statistically *correlated*: counting hits overstates independent evidence. The evidence is to be reorganized into *overdetermined relation bundles*, each carried by one proof object: **(i)** the flavor invariant sum rule — one spurion/operator proof must yield θ_{12}, θ_{13}, δ_{PMNS}, J_{PMNS}, the mass splittings and the CKM texture *together*; **(ii)** the inflation consistency relation — $r = 3(1-n_s)^2$ and $\alpha_s = -r/6$ (eliminating $N_\star$ as a fitted middleman); **(iii)** the axion/birefringence bundle — β, f_a, m_a, $g_{a\gamma\gamma}$, Ω_a and the EDM bound from the *same* normalized axion action, with the transduction chain of `OBS.TRANSDUCTION.01`; **(iv)** the character-resolved transfer signature (`OBS.TRANSDUCTION.01` as the readout-side principle).

The statistical demand. A joint likelihood over the frozen registry with: the full covariance matrix induced by the shared atoms; the *effective* number of degrees of freedom (not the raw hit count); a prespecified null space and formula-selection rule; and a genuine holdout dataset (future measurements only). *Honest reading of the existing null test:* the "13/13, $< 10^{-30}$" Monte-Carlo statement (v100, `docs/FALSIFICATION.md`) quantifies the look-elsewhere burden *under its declared null model* and is only interpretable as evidence to the extent that the null space, the formula grammar, and all development data were prespecified — which for historically known observables they were not and cannot retroactively be. The 2026-06-09 freeze (v84) genuinely protects *future* measurements; it does not turn historically known data into blind predictions. The wording of `docs/FALSIFICATION.md` is adjusted accordingly in the same change (honest, not weakened: the freeze and the kill tests stand). Status: `PRED.JOINTLIKELIHOOD.01` [O].

Executed update (2026-08-28, promote round): v1 of the joint likelihood is executed ([verification/v992_joint_likelihood_v1.py]; the contract stays [O]). Nine frozen pull-matched scorecard triplets (no `experiments/` import at suite runtime): $\chi^2 = 11.80/\mathrm{dof}\,9$ ($p = 0.225$), $\nu_{\mathrm{eff}} = \mathrm{rank}(J_{\mathrm{atom}}) = 1$, TFPT at the 0th percentile of 200 scrambled decoders, leave-one-bundle-out stable. Holdouts declared not consumed: JUNO Δm^2, DESI Σm_ν, DUNE δ_{CP}, LiteBIRD r. v1 declarations: Σ_{exp} diagonal (NuFIT/Planck covariances unmodelled, flagged); $\Sigma_{\mathrm{bridge}} = \Sigma_{\mathrm{th}} = 0$. Promotion conditions for full closure: off-diagonal Σ, a formula-grammar census, genuine future holdouts. Display marker stays [O].

Research Programme — the TFPT4D master route (one 4D transfer object instead of twenty formulas)

Typing. A programme section (registered 2026-08-27 from the external bird's-eye review), *not* a claim: it organizes the existing 4D contracts and the four new ones below into one route and moves *no* marker. The organizing observation: TFPT is probably not missing twenty independent formulas; it is missing *one* mathematical growth principle that unfolds the discrete structure into a local, unitary, reflection-positive 4D quantum world. The discrete theory is then not spacetime itself but its *genetic code*; the missing bridge must produce spatial locality, reversible quantum dynamics, chiral matter, gauge fields, a controlled continuum limit and concrete observables *simultaneously*.

The master formulation (working hypothesis, unproven). *The physical world is the unique local, reflection-positive and approximately quantum-Markovian 4D completion of the TFPT seam algebra that is anomaly-free and has exactly one relevant dimensionful direction.* "Exactly one dimensionful direction" is the No-Unit theorem already in force: the dimensionless compiler cannot produce an absolute scale, and v_{geo} is the declared calibration torsor (`ANCHOR.VGEO.02`, v153, v725) — the master route inherits this, it does not re-open it.

The seven conditions, mapped onto contracts. **(1)** seam compatibility — the boundary restriction of the 4D theory reproduces the TFPT seam data *exactly*, not only dimensions or group labels (`TFPT4D.LATTICE.ACTION.01` gate T3 below); **(2)** reflection positivity / positive transfer operator (gate T2; `QFT4D.OS.RECON.01` demand O1); **(3)** a local unitary dilation of the observed Markov dynamics, $\Phi(\rho) = \mathrm{Tr}_E[V(\rho \otimes \sigma_E)V^\dagger]$, with size-uniform locality and a non-divergent Lieb–Robinson velocity (`DYN.UNITARY.DILATION.01`); **(4)** chiral anomaly freedom with no mirror sector and the correct index (`CHIRAL4D.NOMIRROR.01`); **(5)** only one relevant scale direction — no hidden tunable dimensionless parameter (kill test in `TFPT4D.LATTICE.ACTION.01`; v_{geo} calibrates the unit only); **(6)** approximate quantum-Markov recovery, $I(A{:}C \mid B) \le C_0\, e^{-\mathrm{dist}(A,C)/\xi}$, so the bulk is quasi-locally reconstructable from overlapping boundary data — the small-CMI/recovery-map link is Fawzi–Renner (arXiv:1410.0664), and local Markov properties of Gibbs states are established in Chen–Rouzé (arXiv:2504.02208) (`SEAM.BULK4D.RECON.01` route input); **(7)** nontriviality — at least one connected four-point function and ultimately a nontrivial S-matrix survive the limit (gate T7; `SEAM.BULK4D.RECON.01` demand B6).

The dynamics correction (discrete → dynamics). The finite transport object already embeds exactly into a continuous Markov semigroup (`DYN.MARKOV.EMBED.01` [E] at kernel level, v971) — but a Markov process is irreversible and cannot be the fundamental dynamics. The fundamental order must be *local unitary amplitudes → decoherence/partial trace → Markov matrix* T. This matches the now machine-checked finding that the classical unistochastic matrix $B_{ij} = |U_{ij}|^2$ keeps probabilities but loses the orientation sign of the Jarlskog invariant $J = \pm 1/27$ ([`verification/v977_transfer_unistochastic_wilson.py`]; the Birkhoff-fibre circulation parameter t is a second, independent classically hidden datum, [`verification/v976_seam_lift_birkhoff.py`]): the fundamental variable must carry a complex holonomy/amplitude, which makes Wilson loops $W_\gamma = \mathrm{Tr} \prod_{e\in\gamma} U_e$ more natural than bare transition probabilities. Discrete Weyl dynamics on unitary cellular automata show such a continuum transition is possible in principle (Białynicki-Birula, arXiv:hep-th/9304070); for TFPT the discrete collision/QCA leg is constructed exactly ([`verification/v984_markov_qca_dilation.py`]), while the continuum/OS legs remain to be constructed (`DYN.UNITARY.DILATION.01`).

Programme ordering (the efficient sequence). **(P1)** the conditional 4D dimension selector — *done at conditional level* (`DIMENSION.SELECTOR.4D.01` [C], [`verification/v975_dimension_selector_4d.py`], next section); **(P2)** the single simple-current/spin-field generator (`SEAM.SIMPLECURRENT.GENERATOR.01` [O]); **(P3)** the explicit finite 4D lattice action with its seven finite gates (`TFPT4D.LATTICE.ACTION.01` [O]); **(P4)** the local unitary dilation with size-uniform Lieb–Robinson bounds (`DYN.UNITARY.DILATION.01` [O]); **(P5)** the continuum/OS package (`QFT4D.OS.RECON.01` [O]); **(P6)** all four transfer bridges from one generating functional (`FTRANSFER.GENERATING.01` [O]); **(P7)** gravitation strictly downstream — modular Hamiltonians → entanglement geometry → effective Einstein dynamics, in the assumption-declared discipline of `GRAV.NONCIRCULAR.01` (Jacobson's entanglement equilibrium, arXiv:1505.04753, *presupposes* a local QFT and can never be cited as producing one); **(P8)** the prespecified joint likelihood with holdout (`PRED.JOINTLIKELIHOOD.01` [O]); **(P9)** all three gauge couplings as unique stationary points of one nonabelian zeta-det/inflow functional (`GAUGE.DETLINE.FIXPOINT.01` [O]; generalizes `ALPHA.QUILLEN.EXACT.01`; honest: $\alpha_s(M_Z)$ is currently an external input); **(P10)** massless transversal spin-2 emergence from the same spectral determinant (`GRAV.SPIN2.EMERGENCE.01` [O]; Einstein equation stays downstream/conditional;

`GRAV.NONCIRCULAR.01` binding); **(P11)** the Schwinger–Keldysh completion (S, ρ_0) of cosmological transfers (`FTRANSFER.SK.RHO0.01` [O]; `FTRANSFER.GENERATING.01` covers equilibrium only; typed candidate $\theta_i = 3\pi/5$, μ_4 orientation picks $k = 0$). The Yang–Mills mass gap is *not* defined away: it remains the open Clay problem; at best the route reduces it to a concrete, structured lattice theory.

Kill criteria (the route is allowed to die early). No local reflection-positive finite 4D model with the correct seam boundary theory — master route dead. The simple-current generator is nonlocal or fails to converge — seam route dead. The chiral measure term stays nonlocal or a global anomaly survives — chiral route dead. Mirror fermions do not decouple — Standard-Model route dead. The Lieb–Robinson velocity or correlation length diverges uncontrolled — no local continuum limit. All connected four-point functions vanish — only a (generalized-)free theory. Extra freely tunable dimensionless parameters are needed — the strong parameter-freeness claim is refuted. The uniform twistor flat direction (the sector-0 Okubo admixture of [`verification/v973_seam_route_narrowing.py`]) shifts *normalized* observables rather than only the vacuum normalization of Z — a genuine undetermined modulus.

Separation (binding). RH and the `PRIME.*` programmes stay separate mathematical programmes: a green RH probe would be neither necessary nor sufficient for the 4D physics, and this architecture yields no RH statement. Status: programme section; every constructive contract in it stays [O]; the only verified piece is the conditional selector below.

Verified Selector — `DIMENSION.SELECTOR.4D.01` (why $d = 4$, conditionally)

Typing. A *conditional* identity [C] (ledger row `DIMENSION.SELECTOR.4D.01`), machine-checked in [`verification/v975_dimension_selector_4d.py`] (16/16, sympy-exact). The μ_4 structure of the compiler must *not* be read as a proof of four spacetime dimensions (`DIMENSION.UPLIFT.FIREWALL.01`); what *is* checkable is a selector theorem under a named axiom set. **Statement.** Under **A1** $d > 2$ (propagating local gauge fields, $d-2 \geq 1$ transverse polarizations), **A2** a dimensionless Yang–Mills coupling, **A3** real self-dual/anti-self-dual 2-form sectors (the Euclidean Hodge star an involutive endomorphism of 2-forms), and **A4** chiral Weyl representations (even d), the dimension $d = 4$ is *unique and minimal* on the exact scan window $d = 2..12$ — and the selector is *overdetermined*: A2 alone pins $\{4\}$ ($[g_{\mathrm{YM}}] = (4-d)/2$ forced symbolically from dimensionless-action power counting) and A3 alone pins $\{4\}$ ($* : \Omega^2 \to \Omega^{d-2}$ is a self-map iff $d = 4$; the explicit 6×6 star on $\Lambda^2(\mathbb{R}^4)$ satisfies $*^2 = I$ with eigenvalues ± 1 at multiplicities 3/3 — the real $\mathfrak{su}(2)_+ \times \mathfrak{su}(2)_-$ split), while A1 & A4 alone give $\{4, 6, 8, 10, 12\}$: chirality does *not* select 4 by itself (honest ablation ledger). Weyl chirality is certified by exact Clifford algebras $d = 2..7$ (even d: γ_* anticommutes with every γ_μ, $\gamma_*^2 = I$, $\operatorname{tr}\gamma_* = 0$; odd d: the full product is central and scalar — no grading). Signature is typed honestly: in Lorentzian signature $*^2 = -1$ on 2-forms in $d = 4$, so the *real* split is a Euclidean/OS-side statement. Mutants are caught (the wrong power counting $[g] = (6-d)/2$ selects $\{6\}$; the “A4 alone selects 4” overclaim is refused). **Honest boundary.** μ_4 and g_{car} enter nowhere — deliberately, because “$\mu_4 \Rightarrow$ 4D” is exactly the forbidden reading. This is a selector, *not* a construction and *not* a compiler derivation: *axiom provenance is the open half* — if A1–A4 cannot themselves be forced from $\{c_3, g_{\mathrm{car}}\}$, the selector remains a consistency statement, not a prediction (the registered kill criterion). Status: `DIMENSION.SELECTOR.4D.01` [C]; every 4D construction contract stays [O].

Research Contract — `SEAM.SIMPLECURRENT.GENERATOR.01` (one generator instead of 128 currents)

Typing. A research contract [O] (ledger row `SEAM.SIMPLECURRENT.GENERATOR.01`), *not* a claim. **Context.** The seam coverage carve ([`verification/v973_seam_route_narrowing.py`]) shows the $\mathfrak{so}(16)_1$ fermion-bilinear half of the seam target is covered by the cited scaling-limit theorem; the named residual is N1 (the μ_4/GSO extension $SO(16)_1 \to (E_8)_1$ as a *net* statement — the 128 spinor currents are not bilinears), N2 (the twist/spin-field sector), N3 (the one-sided chiral edge) and N4 (the holomorphy selector). **Demand (the compression).** Construct *one* normalized simple-current/spin-field intertwiner — the generator of the index-4 Q-system extension (v125/v154) — and prove **(G1)** convergence on the finite-particle core with energy bounds, **(G2)** locality relative to the bilinear net, **(G3)** the correct braiding/monodromy relation, and **(G4)** Q-system closure, so that symmetry and fusion generate the *entire* 128-dimensional extension sector from the single generator — collapsing N1 + N2 (and, via uniqueness of the extension, the N4 selector) onto *one* convergence theorem instead of 128 separate current controls. **Kill tests.** The generator is provably nonlocal relative to the bilinear net; the energy bounds fail on the core; the braiding closes on the wrong extension ($SO(16)_1$ stays unextended); the convergence rate is incompatible with the measured bilinear collar rate ($p = 3.48$ inside the preregistered band, [`verification/v973_seam_route_narrowing.py`]). *Dimension typing (binding):* hypotheses and conclusion live in 1+1D — closing this contract does *not* close `SEAM.BULK4D.RECON.01`. Status: `SEAM.SIMPLECURRENT.GENERATOR.01` [O].

Executed update (2026-08-28, external master-object review wave 2): the generator is *identified* and its lattice-arithmetic shadow is complete (the analytic half G1–G4 stays [O]). [`verification/v983_simple_current_generator.py`] The candidate is the glue vector $\lambda = (\omega_s, \omega_f)$ — the D_5 spinor weight plus the A_3 fundamental weight, in explicit rational coordinates — with $\|\omega_s\|^2 = \frac{5}{4}$, $\|\omega_f\|^2 = \frac{3}{4}$, $\|\lambda\|^2 = 2$ and conformal weight $h_\lambda = \|\lambda\|^2/2 = 1$ (integer $\Rightarrow$ trivial self-monodromy, the bosonic simple-current precondition). Machine-checked exactly: $[\lambda]$ has *order four* in the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ of $D_5 \oplus A_3$; $\lambda \cdot L_0 \subset \mathbb{Z}$ (isotropic glue, the glued lattice is even); $\det = 16/4^2 = 1$ (unimodular rank 8, hence E_8; explicit certificate v1); the coset root census is $[52, 64, 60, 64]$ — with the rank-8 Cartan sector exactly $248 = (45{+}15) + (16, 4) + (10, 6) + (\overline{16}, \bar{4})$ — so the *odd* fusion powers of the one class $[\lambda]$ carry $64 + 64 = 128$, the full missing spinor sector. Kill tests fire: the vector-class glue (norm 7/4) and the mismatched glue (norm 9/4) both break evenness. The holomorphy *shadow* is pinned at the lattice level ($\det 1 \Rightarrow$ one sector, vs $\det D_8 = 4 \Rightarrow$ four sectors for $SO(16)_1$) — the operator-algebraic N4 statement is untouched. Net effect: the contract's search space collapses from "find a generator" to "prove G1–G4 for *this* $\Psi_\lambda = Y(e^{(\omega_s,\omega_f)}, z)$"; no marker moves.

Executed update (2026-08-28, promote round): S3 exact + the quantitative S1/S2 skeleton + the lemma reduction ([`verification/v988_psi_lambda_reduction.py`]; *not* a closure). S3: the crossed-product/tower exchange is exact (naturality; a non-equivariant mutant fails). S1: windowed covariances converge and sectors collapse (measured rates $\sim N^{-1.85}$ on the reduced QWZ collar). S2 quantitative: windowed Hilbert–Schmidt bounded (Shale ~ 0.376) while the global $\mathrm{HS}^2 = 3.605 + 0.203 \log N$ is topological vs. saturation in the trivial phase; matrix-element Cauchy on the phase-fixed band; energy envelope $f_n \le C^{n+1}(n!)^2$ with $C = 1.204$ on the reduced collar (probe provenance $C = 2.86$). The remaining lemma is *reduced* to cited quasi-free theorems (Shale–Stinespring / Araki / Ruijsenaars) plus the MMST identification — reduction, not closure. Honest typing: lattice shadow [E] ([`verification/v983_simple_current_generator.py`]) + quantitative skeleton [E]-measured + reduction-to-cited-class [C] conditional; open rest = the MMST identification. Display marker stays [O].

Executed update (2026-08-28, review wave 3): the $D{+}A$ architecture is an OUTPUT of the rank-8 ADE census ([`verification/v993_minimal_defect_selector.py`]; axiom typing UNCHANGED). The FULL rank-8 ADE pair census (Smith normal form of Cartan, not a textbook lookup) with cyclic $\mathbb{Z}_4$ glue demanded on *both* factors selects (D_5, A_3) uniquely ($D_3 \cong A_3$ aliases collapsed; the raw second hit $D_5 \oplus D_3$ is the same pair). Kill tests fire: (D_4, D_4) and (A_7, A_1) have product 16 but fail cyclicity (Klein $\mathbb{Z}_2 \times \mathbb{Z}_2$ / $\mathbb{Z}_8 \times \mathbb{Z}_2$); the $\mathbb{Z}_3$ control $(E_6, A_2) = \mathbb{Z}_3 \times \mathbb{Z}_3$. So the $D{+}A$ split that v624 still had to *assume* is an OUTPUT of (rank 8 + cyclic $\mathbb{Z}_4$ both sides). Cited-not-proved remain $c_- = 8$ and holomorphic uniqueness of $(E_8)_1$. Lattice shadow of $U^2 = (-1)^F$: $[\lambda]^2 = [v]$ exactly (the order-4 glue class squares to the fermion-parity / D_5-vector class; non-split: vector glue of norm 7/4 not in $2\mathbb{Z}$). Unique finite trace-preserving conditional expectation $E = \frac{1}{4}\sum_r \mathrm{Ad}(u^r)$ onto the clock-fixed algebra (uniqueness needs the module property;

without it the deviation kernel has dimension 36/720). The P1 arithmetic $1/(4 \cdot 2\pi) = c_3$ is the v484/v813 identity, not a derivation; the equidistribution identification stays the open [C] of v813. `AX.P1.01`/`AX.P2.01` stay Axiom. Display marker of this contract stays [O].

Executed update (2026-08-28, completeness wave): 5 of 7 MMST identification criteria executed, and C6 is the written-out hand-off ([`verification/v994_mmst_criteria_battery.py`]**; display stays** [O]**).** C1: measured $c = 8.005$ (16 Majorana copies; one-cut convention $S_{M,1\text{cut}} = S_{\text{scalar}}/4$). C2: conformal edge spectrum $[0.506, 1.516, 2.523, 3.524]$ with degeneracies $[1, 0, 1, 1, 2]$, $v \approx 0.988$. C3 (flagship): μ_4/GSO-projected lattice character sums reproduce the $(E_8)_1$ vacuum character coefficients $[1, 0, 248, 0, 4124, 0, 34752]$ three orders deep. C4: current normalisation $k = 1.000000000000$. C5: sector collapse $\det D_8 = 4 \to \det E_8 = 1$. C6 is now *written out* as the quasi-free convergence theorem in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex` (analytic log coefficient $2/\pi^2 = 0.20264$ vs the v988 measured slope 0.203, relative 0.18%) — this *is* the G1/C6 hand-off, not a re-proof inside the suite. C7 modular invariance stays cited/open. Must-fail: dropping GSO (O alone) misses the 248; $N_{\text{MAJORANA}} = 8$ cannot make 248. Honest typing: 5/7 executed [E]-measured + C6 written-out [C]+ C7 cited [O]; open rest = C7 + net-level G2/G3/G4. Display marker stays [O].

Executed update (2026-08-29, review wave 4): C7 arithmetic shadow now exact ([`verification/v998_seam_modular_closure.py`]**; net-level C7 stays cited).** Two-edge HS remainder bounded by 2.11 on the frozen plateau $N_x \in (16, 24, 32, 48, 64, 96)$. Lattice character θ_{E_8}/η^8 EXACT four orders: t-grid $[1, 0, 248, 0, 4124, 0, 34752]$. $\Theta_{E_8}(\tau) = E_4(\tau)$ as Eisenstein series: q-expansion match to ≥ 8 orders exact (Jacobi identity); $j^{1/3}$ numeric transform at 60 digits with S-weight bookkeeping (τ^4 cancels in the weight-0 character). Net-level C7 modular covariance of the completed extension stays cited/open. Display marker stays [O].

Externalization package (hand-off for outside specialists in conformal nets / subfactors / VOA). In the same six-part format as the `SEAM.EQUIV.01` externalization contract below: **(1) minimal theorem statement.** Let $\mathcal{A}$ be the $\mathfrak{so}(16)_1$ fermion-bilinear net on the seam circle (the half covered by the cited scaling-limit theorem, [`verification/v973_seam_route_narrowing.py`]), and let σ be the simple-current sector of the index-4 μ_4 Q-system (finite side explicit and machine-verified, v125/v154; net-level locality integer $h_s = 16/16 = 1$, v469; the generator is meanwhile *identified* at the lattice level as $\Psi_\lambda = Y(e^{(\omega_s, \omega_f)}, z)$ with $h_\lambda = 1$, [`verification/v983_simple_current_generator.py`]). *Claim:* there exists one normalized intertwiner Ψ_σ (a charged/spin field for σ — the identified Ψ_λ) satisfying G1 convergence on the finite-particle core with polynomial energy bounds, G2 relative locality to $\mathcal{A}$, G3 the μ_4 braiding/monodromy relation, and G4 Q-system closure; then the extension generated by $\{\mathcal{A}, \Psi_\sigma\}$ is the $(E_8)_1$ net and the v973 residual N1 + N2 (and N4 via uniqueness) closes. **(2) typed hypothesis list.** PROVEN-FINITE: the index-4 Q-system (v125/v154), the crossed-product/locality package (v469), the coverage carve with the named N1–N4 residual and the measured collar rate $p = 3.48$ ([`verification/v973_seam_route_narrowing.py`]), the gapped collar model (v367/v368). CITED-CLASSICAL: the Osborne–Stottmeister scaling limit (arXiv:2107.13834); Longo–Rehren / Böckenhauer–Evans simple-current extension theory; KLM μ-index classification. OPEN: C7 modular invariance of the completed μ_4/GSO extension, and G2–G4 as *net* statements (relative locality, braiding, Q-system closure at the continuum net). G1/C6 is no longer a blank: the quasi-free convergence theorem is written out in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex` (analytic log coefficient $2/\pi^2 = 0.20264$ vs measured 0.203) — that document *is* the hand-off. **(3) runnable finite models:** `verification/v125`, `v154`, `v469`, `v973` (embedded probes with pinned SPEC hashes), `v983` (the identified glue vector and its lattice shadow), `v988` (S3 exact + S1/S2 skeleton), `v994` (5/7 MMST identification criteria, including the $(E_8)_1$ vacuum character $[1, 0, 248, 0, 4124, 0, 34752]$). **(4) honest Lean surface:** none for G1–G4; the finite Q-system arithmetic only. **(5) must-fail controls:** the same construction attempted on $SO(16)_1$ *without* the order-4 clock must *not* produce a holomorphic net ($\det K = 4$ rival); a non-order-4 clock must break the marks; the trivial M=3 collar must give no chiral limit. **(6) acceptance tests:** a solution counts iff it (a) proves G1–G4 for one explicit Ψ_σ, or (b) proves that no local normalized generator exists (the registered kill —

then the seam route is dead and the ledger records it as exactly that).

Research Contract — TFPT4D.LATTICE.ACTION.01 (the explicit finite 4D lattice action and its seven finite gates)

Typing. A research contract [O] (ledger row TFPT4D.LATTICE.ACTION.01); route step (2) of the DIMENSION.UPLIFT.FIREWALL.01 ordering made concrete — the single object on which nearly all open 4D questions become simultaneously testable. **Demand.** Exhibit one fully explicit finite family of 4D Euclidean lattice theories

$$\mathcal{Z}_a[J] = \int \mathcal{D}U\,\mathcal{D}\psi\,\mathcal{D}\Phi\; e^{-S_a[U,\psi,\Phi]+J\mathcal{O}}, \qquad S_a = S_{\mathrm{Wilson}}[U] + \bar{\psi}\, D_{\mathrm{GW}}[U,\Phi]\,\psi + S_\Phi[U,\Phi] + S_{\mathrm{seam/top}},$$

where U are the fundamental holonomies (TFPT fixes representation content, charge lattice, coupling relations and boundary conditions), D_{GW} is a Ginsparg–Wilson/overlap operator for the TFPT chiral representation content (the CHIRAL4D.NOMIRROR.01 operator), Φ is the *only* allowed relevant scalar/link-mixing sector (EWSB downstream), and $S_{\mathrm{seam/top}}$ is the boundary/topological module transporting the seam data into the bulk — first at finite lattice spacing a, deliberately *without* continuum claims. **Finite gates** (all machine-checkable at finite size; the promotion criteria for any candidate): **(T1)** exact gauge invariance; **(T2)** a positive transfer matrix (RP at finite a); **(T3)** the seam boundary restriction reproduces the TFPT seam data *exactly*; **(T4)** anomaly bookkeeping including the global $SU(2)$ anomaly; **(T5)** the index/generation count; **(T6)** the mirror spectrum separated; **(T7)** at least one *nonvanishing* connected four-point function. **Kill tests.** No local reflection-positive action with the correct seam restriction exists (master route dead); every candidate is generalized-free (T7 unreachable); a hidden freely tunable dimensionless parameter is required (the strong parameter-freeness claim is refuted); the uniform twistor flat direction shifts *normalized* observables rather than only the vacuum normalization of $\mathcal{Z}$ (a genuine undetermined modulus). **Relation.** The continuum limit is the separate contract QFT4D.OS.RECON.01; the unitary dilation is DYN.UNITARY.DILATION.01; $d = 4$ is typed by DIMENSION.SELECTOR.4D.01 [C]; the generating-functional collapse is FTRANSFER.GENERATING.01. Chiral lattice state of the art is calibrated in CHIRAL4D.NOMIRROR.01 (Lüscher's exactly gauge-invariant abelian construction, arXiv:hep-lat/9811032; the nonabelian anomaly analysis, arXiv:hep-lat/9904009; reflection positivity proven for *free* overlap fermions, Kikukawa–Usui arXiv:1005.3751 — none of this is the interacting chiral Standard Model). Nothing in the corpus closes any gate. Status: TFPT4D.LATTICE.ACTION.01 [O].

Executed update (2026-08-28, promote round): the T1–T7 harness plus a named Euclidean T2 kill and a Hamiltonian-route T2 clearance ([verification/v989_lattice_gate_battery.py]; no gate closed). T4 exact SM anomaly bookkeeping (a lepton-drop mutant fires). T5/T6: overlap index equals flux q on an 8×8 torus; Ginsparg–Wilson circle to machine precision. The Euclidean overlap candidate is killed at T2: $|\det D_{\mathrm{ov}}|^2$ is PSD at $N_t = 2$ and indefinite at $N_t = 4$ (det-only $\lambda_{\min} \approx -0.249$); Wilson fermions (ultralocal in time) stay PSD — the break is pinned on overlap time nonlocality. The Hamiltonian route clears T2 by construction ($H = H^\dagger$ exact $\Rightarrow T = e^{-aH}$ symmetric PD; interior Gauss exact including the seam-sector intertwiner; background-field μ_4 split with $k = 1, 3$ degenerate). Display marker stays [O].

Registered decision (2026-08-28, QFT4D.LATTICE.FUNDAMENTAL.01): the lattice-fundamental reading — a typed route/typing, *not* an [E] claim. The physical completeness criterion is this finite local unitary lattice theory (Hamiltonian route), not the continuum OS limit. That limit (QFT4D.OS.RECON.01) is retyped as mathematical reinforcement; Poincaré invariance is an IR fixed-point property. The display marker of this contract stays [O]; nothing closes. See the decision paragraph at QFT4D.OS.RECON.01.

Amendment (2026-08-29): a single finite box is not a world — the fundamental object is the quasilocal consistent family $\{H_\Lambda\}$ with $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$; continuum optional, thermodynamic limit and controlled IR universality mandatory. Full prose at QFT4D.OS.RECON.01.

Executed update (2026-08-30, review wave 7): 2+1**D scaffold coherent (** [verification/v1008_master_assembly_scaffold.py]**; Decision typing UNCHANGED).** Combinatorial $K = 0$, unique-contractive $|R'| < 1$, QWZ wall + DET4 mirror, $\mathbb{Z}_6$ assembly $(1, 2, 3, 0)$. $L = 3$ spaces out of suite. No [E] claim, no marker move.

Executed update (2026-08-30, afternoon harvest; [verification/v1011_3p1d_ladder_battery.py]**; Decision typing UNCHANGED).** 3+1D minimal object viable; fully coupled link–wall at minimal scale; Kronecker limitation removed. No [E] claim, no marker move.

Executed update (2026-08-30, late-evening harvest; [verification/v1013_thermodynamic_dynamics.py]**; Decision typing UNCHANGED).** Mandatory-dynamics leg closed at class level (Lieb–Robinson + τ_t + $\mathcal{A}^G$ + state existence). Uniqueness / limit gap / IR universality remain. Full prose at `QFT4D.OS.RECON.01`. No [E] claim, no marker move.

Research Contract — `SEAM.DETLINE.UNIFICATION.01` (the determinant-line unification hypothesis)

Typing. A research *hypothesis* contract [O] (ledger row `SEAM.DETLINE.UNIFICATION.01`) — a precise conjecture, *not* a result. **Observation.** Three separately open structures may be projections of *one* geometric object: **(1)** the seam extension is described by a Q-system/simple current (`SEAM.SIMPLECURRENT.GENERATOR.01`); **(2)** the chiral fermion measure lives on a determinant line (`CHIRAL4D.NOMIRROR.01` demand C1); **(3)** anomalies and CP phases appear as holonomies of that determinant line — the *finite* Quillen identity is already two-sided ([verification/v974_alpha_faces_computed.py]: Berry integral = Poincaré–Lelong divisor integral = +1), and the continuum Bismut–Freed identification is the named missing step of `ALPHA.QUILLEN.EXACT.01`. **Demand (the compatibility theorem).**

$$\mathrm{Res}_{\mathrm{seam}} \det D_{4D} \;\cong\; \det D_{\mathrm{seam}}$$

as line bundles *with connection*, and the holonomy of the bulk determinant line equals the discrete TFPT orientation phase (the classically invisible $\operatorname{sgn} J$ orientation bit, now machine-checked at matrix level with the Wilson-plaquette interference readout as its minimal observable, [verification/v977_transfer_unistochastic_wilson.py]; the bulk-determinant identification demanded here stays open). If such a theorem exists, seam extension, chiral anomaly freedom, the generation index and the discrete CP orientation become different projections of the same object. **Kill tests.** The two determinant lines differ by a nontrivial local factor no counterterm removes; the bulk holonomy disagrees with the discrete orientation phase on a computable family. *Honest boundary:* the continuum Quillen/Bismut–Freed identification is exactly the critical missing step; nothing in the corpus proves any part of the 4D side. Status: `SEAM.DETLINE.UNIFICATION.01` [O].

Executed update (2026-08-28, promote round): a finite 2+1**D/**1**D bulk–edge shadow is executed (** [verification/v991_detline_bulk_edge.py]**; continuum Bismut–Freed stays [O]).** On the QWZ collar, bulk twist-torus curvature $2\pi \cdot (+1)$ equals the seam holonomy winding $+1$ (difference 5.6×10^{-16}); the $M = 3$ control is zero on both sides; a conjugation mutant flips both signs coherently. Honest: finite shadow [E], not the 4D Bismut–Freed identification. Display marker stays [O].

Executed update (2026-08-30, late-evening harvest; [verification/v1014_bridge_refinements.py]**; the contract stays [O]).** The finite restriction isomorphism $\mathrm{Res}_\Sigma L_{\mathrm{bulk}} \cong L_\Sigma$ is verified (c_{phase} constancy $3.7{\times}10^{-16}$, conjugation covariance exact, $C = W_{\mathrm{bulk}} = W_{\mathrm{seam}} = +1$ / mirror -1). Orientation fixes only the conjugate-pair branch $(2 \to 1)$; a constant $U(1)$ anchor remains, fixed by the A_0 reference normalization of `ALPHA.QUILLEN.EXACT.01`. Continuum Bismut–Freed stays [O].

Research Contract — `FTRANSFER.GENERATING.01` (four transfer bridges from one generating functional)

Typing. A research contract [O] (ledger row `FTRANSFER.GENERATING.01`); the *constructive successor* named by the `FR.TRANSFER.01` typing guard. The four continuous-transfer interfaces (F_{pole}: source masses → pole, v183; $F_{\mathrm{Boltzmann}}$: CP source → η_B, v169/v184; F_{relic}: $(f_a, m_a, \theta_i, N_{\mathrm{DW}}) \to \Omega_a$, v185; F_{QCD}: $(\alpha_s, b_3, \Lambda_{\mathrm{QCD}}, \mathrm{lattice}) \to m_p/m_e$, v164) are currently four *separate* bridges. **Demand.** If the 4D theory $\mathcal{Z}_a[J]$ of `TFPT4D.LATTICE.ACTION.01` exists with a controlled continuum limit (`QFT4D.OS.RECON.01`), the four bridges must arise from *one* generating functional $W[J] = \log Z[J]$: lepton and quark masses as poles of two-point functions (zeros of $\Gamma^{(2)}$); m_p/m_e from the baryon and electron correlator poles of the *same* theory; η_B from the real-time Schwinger–Keldysh dynamics of the *same* theory (CP violation, B violation, nonequilibrium); the axion relic from the cosmological solution of the *same* normalized action. F_{transfer} is then not another 3×3 matrix; it is the full $W[J]$. **Binding.** The `FR.TRANSFER.01` guard stays in force: the four frontier observables remain [C]/[O]-typed until $W[J]$ exists and each bridge is *derived* from it — this contract permits no marker upgrade; it names the single object whose existence would collapse the four transfer problems into computations within one theory. **Kill test.** The four bridges provably require mutually inconsistent normalizations of the same $W[J]$. Status: `FTRANSFER.GENERATING.01` [O].

Executed update (2026-08-28, promote round): a finite $W[J]$ is executed at 1+1D $\mathbb{Z}_6$ ([`verification/v990_wj_transduction_mechanism.py`]; the `FR.TRANSFER.01` guard stays binding — no upgrade). Content shadow: the real $6Y$ table as a $\mathbb{Z}_6$ centre. Derivative identities: $\partial W/\partial J = \langle O \rangle$ (residual 6.3×10^{-11}), Hessian = Kubo (residual 1×10^{-7}). The four transfer bridges remain un-derived; this is the finite-level mechanism, not $W[J]$ of the 4D theory. Display marker stays [O].

Research Contract — `GAUGE.DETLINE.FIXPOINT.01` (all three gauge couplings from one zeta-det functional)

Typing. A research contract [O] (ledger row `GAUGE.DETLINE.FIXPOINT.01`), registered 2026-08-28 from review wave 3. **Demand.** All three gauge couplings (g_1, g_2, g_3 / equivalently α, $\sin^2\theta_W$, α_s) arise as stationary points of *one* nonabelian zeta-det / inflow functional on the determinant line — the nonabelian upgrade of the $U(1)$ face `ALPHA.QUILLEN.EXACT.01`. **Acceptance.** Unique stationary points, positive Hessian, $\alpha_s(M_Z)$ without measured input, RG from the matching scale reproduces the data, mutants fail. **Honest bound.** $\alpha_s(M_Z)$ is currently an *external input* (the QCD matching / lattice input of F_{QCD}); this contract names the missing first-principles replacement, it does not claim the number is derived today. **Kill tests.** A unique stationary point fails to exist; Hessian not positive; mutants (wrong level, wrong inflow multiplicity) still pass; RG from the functional fails to land on the observed couplings. **Relation.** Generalizes `ALPHA.QUILLEN.EXACT.01`. Status: `GAUGE.DETLINE.FIXPOINT.01` [O].

Executed update (2026-08-28, completeness wave): the naive grammar extension is a structural kill ([`verification/v996_gauge_grammar_kill.py`]; the contract stays [O]). Simplest-extension H0 of the $U(1)$ α-grammar to $SU(2)/SU(3)$ is structurally inapplicable: negative b_2/b_3 flip the stationary-root signs ($SU(2)$ inverse -151.2, $SU(3)$ -115.9); 64 conventions, 0 hits; the $U(1)$ control 137.0359992168 survives every mutant. Exponential detline-curvature convergence to 2π with rate $0.6936 \sim \ln 2$ is a numerical Bismut–Freed *shadow*, not the continuum theorem (canonical note on `ALPHA.QUILLEN.EXACT.01`). Naive census-only extension excluded [E]; genuinely nonabelian Casimir/instanton structure is required — typed. Honest: $\alpha_s(M_Z)$ remains an external input. Display marker stays [O].

Executed update (2026-08-29, review wave 5): fixed-volume mechanism executed, thermodynamic step-scaling divergent (exploration probes `gauge_stiffness_fixpoint_probe.py` + `gauge_stiffness_stepscaling_probe.py`; *not* a new module; the contract

stays [O]). At fixed volume the stiffness map has a unique fixed point with contraction -0.632, exact q^2 and $SU(2)$ Dynkin-ratio 4.0 content laws, Weyl exact. The preregistered thermodynamic step-scaling $L = 2..8$ (certified Lanczos) is *divergent*: $g_\star$ oscillates 0.65..1.69, $|R'| > 1$ at $L = 4, 5, 8$, and $1/L$ vs $1/L^2$ extrapolations are inconsistent. Typed hurdle: the 1+1D toy has no plaquette self-interaction; the real contract needs $\geq$ 2+1D. The v996 H0 grammar kill stands. Display marker stays **[O]**.

Executed update (2026-08-30, review wave 7): the 2+1D plaquette scaffold lifts the 1+1D no-plaquette hurdle at scaffold level ([`verification/v1008_master_assembly_scaffold.py`]; the 4D fixpoint stays [O]). Combinatorial $K = 0$; frozen-link $q^2 \sim 4$ with $K(q = 0) = 0$; unique-contractive selector $|R'| < 1$ (L2 selector 0.578 $\to$ L3 0.936 harvest pins; $L = 3$ Lanczos out of suite). The thermodynamic 4D fixpoint and $\alpha_s(M_Z)$ as an external input remain open. Display marker stays **[O]**.

Executed update (2026-08-30, afternoon harvest; [`verification/v1011_3p1d_ladder_battery.py`]; BINDING constraint recorded, display stays [O]). Holonomy stiffness collapses in the confined phase (coupled K dies above $g \sim 1.6$; charged spinor channels enhance $\sim 2.014\times$ at weak g but do not prevent the collapse at minimal volume). The matching point of the coupling functional *must* be defined in the deconfined/weak regime. Charge-0 control matches the uncharged periodic-face curve to 10^{-10}. Display marker stays **[O]**.

Research Contract — `GRAV.SPIN2.EMERGENCE.01` (massless transversal spin-2 from the same spectral determinant)

Typing. A research contract **[O]** (ledger row `GRAV.SPIN2.EMERGENCE.01`), registered 2026-08-28 from review wave 3. **Demand.** From the *same* spectral determinant that carries the gauge fixpoints, a massless transversal spin-2 pole with two helicities, universal $T_{\mu\nu}$ coupling, diffeomorphism Ward identities, and positive Hessian / reflection-positive transfer. **Relation.** The entanglement-equilibrium Einstein equation (v358/v359) stays a *downstream / conditional* readout — it presupposes a local QFT and can never be cited as producing one; `GRAV.NONCIRCULAR.01` is *binding* (BW/CHM uses remain assumption-declared). **Kill tests.** A massive or longitudinal extra mode; non-universal coupling; failed diffeomorphism Ward; negative Hessian / RP violation on the spin-2 sector. Status: `GRAV.SPIN2.EMERGENCE.01` **[O]**.

Executed update (2026-08-28, completeness wave): quadratic witnesses, with the local truncation's ghost typed ([`verification/v997_spin2_smg_witnesses.py`]; the contract stays [O]). Exact Barnes–Rivers decomposition of the quadratic spectral action: $R{+}R^2$ is *clean* (one massless TT pole, two helicities, scalaron at $M_{\rm scal}$ exact). The local a_4 Weyl^2 truncation *necessarily* carries an opposite-residue spin-2 ghost — typed. The contract's content is pinned on the untruncated form factor $a(\Box) = e^{-\Box/M^2}$. Einstein equation stays downstream/conditional; `GRAV.NONCIRCULAR.01` remains binding. Display marker stays **[O]**.

Executed update (2026-08-29, review wave 4): quadratic Hamiltonian half executed ([`verification/v1000_spin2_sk_witnesses.py`]; the contract stays [O]). Positivity exact; 2 TT helicities at all 5852 nonzero momenta ($N = 6..12$); $\omega = |q|$ with $q_i = 2\sin(k_i/2)$ and axis expansion $\omega = k - k^3/24$ exact. Must-fail: a mass term gaps TT; a wrong-sign kinetic term yields two negative directions. Spectral-action Hessian identification, nonlinear Ward and universal coupling stay **[O]**. Display marker stays **[O]**.

WAVE-6 firewall (2026-08-30): `GRAV.WEINBERG.WITTEN.01` registered [O]; this contract's display stays [O]. The Weinberg–Witten theorem is the binding firewall *before any spin-2 marker move*: the microscopic Hamiltonian family must *not* possess a fundamental Lorentz-covariant local conserved $T_{\mu\nu}$; Lorentz symmetry, diffeomorphism Ward identities and the spin-2 mode must emerge in the *same* IR limit. Cross-link only (this row does not list `GRAV.WEINBERG.WITTEN.01` as a dependency — that would cycle).

Executed update (2026-08-30, evening TOE-gate wave): interacting TT half-step ([verification/v1012_toe_gate_witnesses.py]; the contract stays [O]). Conserved interaction-complete lattice stress $\sim 10^{-15}$; TT spectral positivity exact; sum rules $\sim 10^{-15}$; the dominant positive-Z pole persists under ϕ^4 (gapped, as expected for massive generic matter); Z trend k_1 suppressed 0.958 / k_2 enhanced 1.033; free-limit analytic match. Missing: TFPT content, $k \to 0$ volume scaling, Ward identities. Display marker stays [O].

Executed update (2026-09-05, round-3 integration): gapped-parent/TT boundary (`GRAV.PARENT.GAP.NO_GO.01`; [verification/v1024_parent_internal_boundaries.py]; no parent marker move). A uniform full-parent gap $\Delta > 0$ excludes a massless pole with $\omega(k_{\min}) \to 0$. Away from $k = 0$ the symmetric-traceless target has five pointwise components and the transverse-traceless projector has rank two, but its direction-dependent $k \to 0$ limit prevents an exact strictly finite-range direct TT projector. A viable escape therefore requires a critical physical sequence together with local tensor gauge constraints whose Ward identities generate the TT sector only in the infrared. The finite representation count is not a derived TFPT stress operator, and no massless pole, universal coupling, diffeomorphism Ward identity or gravitational continuum theorem is constructed. `GRAV.SPIN2.EMERGENCE.01`, T7 and TOE remain [O].

Research Contract — `FTRANSFER.SK.RHO0.01` (Schwinger–Keldysh completion: cosmological transfers need (S, ρ_0))

Typing. A research contract [O] (ledger row `FTRANSFER.SK.RHO0.01`), registered 2026-08-28 from review wave 3; mechanism executed 2026-08-29. **Demand.** Cosmological transfers (F_{relic}, $F_{\text{Boltzmann}}$, and any nonequilibrium 4D readout) need the pair (S, ρ_0), not $W[J]$ alone — the Euclidean-cap / KMS initial state. **Typed candidate.** The $\theta_i = 3\pi/5$ saddle $S_{\text{init}} = -\kappa\cos(5\theta - 3\pi)$ as the selection. **Arithmetic.** Stationary points $\theta = (3 + k)\pi/5$; the μ_4 orientation picks $k = 0$. **Relation.** `FTRANSFER.GENERATING.01` covers *equilibrium* correlators of $W[J]$ only; this contract is the Schwinger–Keldysh completion. **Kill tests.** The (S, ρ_0) pair is inconsistent with the $W[J]$ of `TFPT4D.LATTICE.ACTION.01`; the $3\pi/5$ saddle is not selected by μ_4 orientation (a different k is forced); the Euclidean cap fails KMS. Status: `FTRANSFER.SK.RHO0.01` [O].

Executed update (2026-08-29, review wave 4): mechanism executed ([verification/v1000_spin2_sk_witnesses.py]; the contract stays [O]). KMS/FDT on the exact Bohr grid $\sim 10^{-17}$; ρ_0-dependence contrast 3.550×10^{-3} versus static 10^{-16}; saddle 5-fold degenerate with $3\pi/5$ unique in the chosen μ_4 lift [C]. No cosmological solve. Display marker stays [O].

Executed update (2026-08-30, review wave 7): finite uniqueness closed with two typed review corrections ([verification/v1009_rho0_minimizer.py]; the contract stays [O]). The constrained minimizer is uniquely the normalized KMS compression (Klein); Hessian spectrum strictly positive on $[3.76, 12.6]$; centre dimension 1; compression identity $\sim 8\times10^{-17}$. TYPED: `V1000_CONSEQUENCE_MISMATCH` — the compression kills the SK response (RMS $\sim 1.5\times10^{-31}$ versus frozen contrast 3.550×10^{-3}), so the proposed constraint set is too restrictive; and `CENTER_FLATNESS_PREMISE_FALSE` — Umegaki uniqueness holds with a nontrivial centre. The canonical formula needs a weaker admissible set. Display marker stays [O].

Executed update (2026-08-30, evening TOE-gate wave): selection-principle vacuity typed ([verification/v1012_toe_gate_witnesses.py]; the contract stays [O]). Eight unique candidates. DOUBLE DIAGNOSIS: hard Gauss / μ_4, + sectors kill the SK response (RMS $\sim 10^{-31}$); every response-alive set is exactly ρ_{KMS} even *without* μ_4 — the entropic-proximity selection principle is vacuous at finite level. Verdict `NO_SET_WORKS`. BINDING: the canonical state requires a different principle (e.g. Euclidean-cap / orientation-branch). Display marker stays [O].

Executed update (2026-08-31, Monday-morning harvest; [verification/v1016_state_gap_batteries.py]; the contract stays [O]). The R_4 seam-modular cap is the unique finite-level non-entropic selector ($D_{\text{tr}}(\text{KMS}) = 0.1434$, SK response alive 3.038×10^{-3}, μ_4-invariant, conditional

$3\pi/5$ saddle preserved). R_1 vacuous, R_2 response-dead, orientation-free mutant loses saddles. Caveat: the difference lives in the neutral spectator factor — finite proxy. Display marker stays [O].

Executed update (2026-09-05, round-3 integration): parent-internal cap and enlargement boundaries (`FTRANSFER.PARENT.Z5CAP.01`; [`verification/v1024_parent_internal_boundaries.py`]; no parent marker move). The selected internal twelve-dimensional $\mathbb{Z}_5$ cap has a unique subspace ground state and gap 0.197423715893; its positive vacuum-preserving extension retains the checked finite parent gap and a nonzero mixed response. However, the five-cycle is chosen rather than derived, the cap breaks the checked D_4 action (residual 1.094700), and the extension vanishes on a large complement. It therefore does not select a global ρ_0. The exact Weyl-pair divisibility obstruction applies only to the unprojected parent dimension; the physical Gauss-projected dimension has not been counted. Likewise $M_n' \cap M_{nd} \cong I_n \otimes M_d$ is an irreducibility/prime-completion obstruction, not a general dynamical-coupling no-go. `FTRANSFER.SK.RHO0.01`, T8 and TOE remain [O].

Research Contract — `SEAM.MMST.TYPEIII.CHARGED.01` (one scaling-limit theorem)

Typing. A research contract [O] (ledger row `SEAM.MMST.TYPEIII.CHARGED.01`), registered 2026-08-29 from review wave 5. **Demand.** *One* scaling-limit theorem $\Psi_{\lambda,N} \to \Psi_\lambda$ strongly on the finite-energy core + crossed-product continuity, from which $B \cong (E_8)_1$, $\mu = 1$, type III_1, unique rotational KMS, unique conditional expectation, deck equidistribution and modular invariance all follow as corollaries. **Children** (listed here, not listing back): `SEAM.SIMPLECURRENT.GENERATOR.01`, the continuum leg of `SEAM.STATE.RPMIXING.01`, the rigidity premise of `ALPHA.QUILLEN.EXACT.01`, the P1 reading of `AX.P1.01`. **Cited document.** `articles/2026-08-29/holomorphic_kms_extension_en.tex` (13 pp, compiled): unconditional rotational-KMS uniqueness via KLM + Longo–Tanimoto *given* the identification; P1 = theorem-conditional on seam identification + $\beta_{\text{angle}} = 2\pi$. Axiom typing of `AX.P1.01` UNCHANGED. **Kill tests.** The scaling limit fails to exist; uniqueness of rotational KMS fails; type III_1 fails; a competing conditional expectation exists. Status: `SEAM.MMST.TYPEIII.CHARGED.01` [O].

Executed update (2026-08-30, review wave 7): five lemmas in-house; residual TEL-B-EXTERNAL + ALG-EXH boxed ([`verification/v1006_mmst_lemma_battery.py`]; the contract stays [O]). Cited `articles/2026-08-30/mmst_charged_scaling_limit_en.tex` (21 pp) and `articles/2026-08-30/externalization_mmst_handoff_v2_en.html`. In-house: L3 crossed-product; UGF $K_G = \pi^2/4$ (measured plateau ~ 1.023, max measured K 1.029399 at $N = 16$); TEL-a N^{-2} with exact isometries ($A_{\text{TEL}} = 5120(1+\pi/2)$, measured envelope $3.420384 < 8$); integer D5+A3 pairings; $\mathbb{Z}_4$ outerness in the limit + finite-inner correction (256-dim gauge unitary is inner). Residual boxed: TEL-B-EXTERNAL (Hankel C_R; HS remainder sourced from the paper max 2.105079358 / fit 2.107565497 / v998 bound 2.11, not recomputed) + ALG-EXH (two Buchholz–Verch estimates). Display marker stays [O].

Executed update (2026-08-31, Monday-morning harvest; [`verification/v1016_state_gap_batteries.py`]; the contract stays [O]). The third (certified-tail) TEL-B route is BLOCKED: $N^2 D_N$ grows $530 \to 19032$, remainder UV-supported. Strictly smaller external target: the scalar dyadic increment $|\|R_{2N}\| - \|R_N\|| \le A_R/N$ with $A_R < 43$ sufficient for $C_R < 3$ (measured N-increments 0.120–0.123, factor ~ 350 headroom). N-uniform CAR nuclearity PROVED ($\nu_N \le e^{16C_\beta}$; $C_\beta = (67.97, 29.73, 12.15)$ for $\beta = (0.5, 1, 2)$) as ALG specialist compactness input; ALG-EXH projector-identification UNCHANGED. Display marker stays [O].

TEL-B reduction (2026-09-04)

The residual `TEL-B-EXTERNAL` is restated below. The earlier scalar dyadic increment $|\|R_{2N}\| - \|R_N\|| \le A_R/N$ with $A_R < 43$ sufficient for $C_R < 3$ is superseded — A_R was measured 0.1238 (r645) but is no longer the route. Numerically closed ($1.7833 + 0.7863 = 2.57 < 3$; conditional analytic $2.9495 < 3$ with $c_E = 0.30$); round 3 closed C1, C2a and the interval-TV input. Round 4 below now closes CF, DG and the relaxed uniform TEL-B norm, including the finite-grid aliases: $\|R_N\|_{\mathrm{HS}} < 2.995906 < 3$ for all even $N \geq 16$. ALG-EXH and the charged MMST identification remain open. Not a marker move; display stays [O]. Evidence (experiments, not `veri`-citable): `experiments/tfpt-discovery/mmst_telb_trace_decomposition_probe.py`, `mmst_telb_cover_split_probe.py`, `mmst_telb_bound_a_probe.py` (9/9), `mmst_telb_bound_b_probe.py` (17/20, G18–G20 open by design). The historical v998 2.11 value is not promoted to an all-volume theorem; the new load-bearing certificate is [`verification/v1026_telb_round4_closure.py`].

Cover identity and sawtooth split. Corpus convention e^{-ipx}, $\alpha = \frac{1}{4}$, $M = 4N$, $d = x - y$. The four-fold cover identity is

$$(\Gamma_0 - \Gamma_\alpha)(x, y) = (1 - i)\, C_M(d + N) + 2\, C_M(d + 2N) + (1 + i)\, C_M(d + 3N).$$

The remainder splits as $R_N = R_N^{\mathrm{sm}} + M_N$ with $M_N = -(S_N + H_N) \otimes Q$. The scalar sawtooth response has the closed form

$$s_N(x, y) = \frac{-(1 + t^2) + 3it - it^3}{4N(1 - t^2)}, \qquad t = \tan \frac{\pi(x - y)}{4N}.$$

Triangle target: $\|R_N\| \leq \|R_N^{\mathrm{sm}}\| + \|M_N\| < 3$.

Bound (A) — native interval input closed (`SEAM.MMST.TELB.ROUND3.01`; [`verification/v1022_telb_round3_certificates.py`]). The continuous family $F = P - \mathrm{saw} \cdot dP$ has vanishing derivative jump ($\operatorname{rank} dF' = 0$, $\|dF'\| = 3.4{\times}10^{-10}$); the first discontinuity sits in F'' (rank 4, norm 2.000000029). The $2{\times}2$ crossing lemma supplies the removable crossing chart. The all-mode BV bound $|\hat{F}(m)| \leq \mathrm{TV}(F'')/(2\pi|m|)^3$ with 256-bit Arb/Acb arithmetic on 5812 rational cells gives

$$\mathrm{TV}_p(F'') \leq 15.8176959629889243576\ldots < 15.928674, \qquad K_3^{\mathrm{TV}} < 2.517464500835.$$

The cut contribution has exact Hilbert–Schmidt norm 2; the remaining cells use validated eigensystems, Taylor enclosures and an analytic ninth-derivative majorant. Thus the former float64/Richardson constant is retired. Round 4 additionally certifies the infinite Fourier core below 1.783260551 and the finite-grid alias contribution below 0.017333. The actual uniform smooth remainder bound is $\|R_N^{\mathrm{sm}}\|_{\mathrm{HS}} < 1.800594$; the older finite 1.7833 claim was not justified by the limiting core and is withdrawn.

Bound (B) — historical reduction; closed in round 4 below. Relaxed target $\|S_N + H_N\|^2 \leq 1.2167^2/2 = 0.740179$ (measured limit 0.5913). Write $S_N + H_N = A_\infty + \mathrm{Res}_N$ with $\|A_\infty\|^2 = \frac{5}{4} - 8/\pi^2$ exact. Lemma 6 cross term ≤ 0.3813 (all even $N \leq 512$, and 1024/2048/4096). Needed $\sum |\mathrm{Res}_N|^2 \leq 0.276873914$; measured max 0.154983791. Master identity (divided differences via the cotangent partial fraction; weight $\{(m + \alpha)/N\}$ is N-periodic; residual $2.3{\times}10^{-15}$). One-variable reduction

$$N \cdot \mathrm{Res}_N(k, l) = r(u, v) + A \frac{(1 - \lambda)\, \Delta\eta - \lambda\, \Delta\varphi_N}{v - u}$$

(residual $1.16{\times}10^{-14}$), with $u = k/N$, $v = l/N$, $A = \sin^2(\pi\alpha)/\pi^2$, and

$$\begin{aligned}
\eta(k) &= -\psi(k + 1 - \alpha) + \log|k| + \pi \cot(\pi\alpha)\, 1_{k<0}, \\
\varphi_N(k) &= \Phi_N(k) - \log N - \gamma - \Phi_\star(k/N), \\
\Phi_\star(u) &= \log\Big|\frac{u}{\tan \pi u}\Big| - \pi \int_{u-1/2}^{u+1/2} \sigma \cot(\pi\sigma)\, d\sigma - \pi u \cot(\pi\alpha).
\end{aligned}$$

Closed bounds $\pi^2/3 \leq \Phi_\star'' \leq \pi^2 - 4$. The kernel r is singular only on $u = 0$, $v = 0$. The former offset midpoint values were $C_\star = 0.152044949/0.152045069$; the native certificate below replaces them as the load-bearing bound. Cell averages $(\mathcal{K}_N)_{kl} := N \iint_{Q_{kl}} r$ obey $\|\mathcal{K}_N\|^2 \leq C_\star$ exactly

(Cauchy–Schwarz on each cell). Measured $\sqrt{N}\,\|E_N\| = 0.1943\ldots0.2013$ ($N = 16\ldots512$). Required $c_E \le 0.542702$, margin 2.70. Assembled $(\sqrt{0.1525} + 0.2013/4)^2 = 0.1943 < 0.2769$.

C1 and C2a closed (2026-09-05; `SEAM.MMST.TELB.C1C2A.01`; [`verification/v1025_telb_c1_c2a_certificates.py`]). For C1, the exact simplification of the real kernel is integrated over 3888 analytic rectangles at 100-bit precision. Logarithmic charts remove the two axis singularities, a removable seam chart treats the opposite-sign corner, and the omitted axis union is bounded by $3.1570760454\times10^{-7}$. Parseval supplies the exact imaginary contribution 1/12, giving the outward enclosure

$$C_\star \le 0.1520453048\ldots < 0.1525.$$

For C2a, exact cotangent reindexing gives the 3/4-point Peano identity

$$\varphi_N(k+1) - \varphi_N(k) = hf((k+3/4)h) - \int_{kh}^{(k+1)h} f(u)\,du, \qquad h = N^{-1}.$$

The partial-fraction formula $f'(u) = \sum_{m\in\mathbb{Z}\setminus\{0\}}(u-m)^{-2}$ implies $\pi^2/3 \le f' \le \pi^2 - 4$ and makes the increments monotone. Shifted Euler summation with its periodic B_3 remainder reduces the two endpoints to outward constant comparisons in v1025:

$$N\sup_k|\varphi_N(k)| \le 0.508228779140631\ldots < 0.509 < 0.51 \qquad (N \ge 16 \text{ even}).$$

C2b scalar reduction and the former remaining target ([`verification/v1022_telb_round3_certificates.py`]). The analytic one-dimensional reduction, with its final arithmetic enclosed by Arb, gives

$$c_\varepsilon = 0.288547359238391\ldots,$$
$$c_\zeta = 0.110677064900715\ldots,$$
$$c_{\text{potential}} = 0.399224424139106\ldots < 0.400.$$

The executable encloses $1 \le n \le 15$ and checks positive-coefficient rational tail comparisons for $n \ge 16$. The sign-controlled digamma and midpoint inequalities which justify those tails are analytic lemmas of this proof, not an independent formalization inside the executable. The former CF/DG targets were

$$\sqrt{N}\,\|(K_N' - K_N)_{\text{offdiag}}\|_{\text{HS}} \le 0.139, \qquad \sqrt{N}\,\|(\text{Res}_N - K_N)_{\text{diag}}\|_{\text{HS}} \le 0.060.$$

The diagonal is Hilbert–Schmidt orthogonal to the off-diagonal, and these two bounds suffice for the stated $c_E \le 0.542702$ budget. Round 4 proves the stronger constants 0.065697 and 0.042959, respectively, and gives $c_E < 0.466902$. The proof and interval arithmetic are in the round-4 supplement below and v1026; these are no longer open inputs.

ALG2 correction (binding). The former Vandermonde-to-generated-algebra implication is withdrawn. The exact one-mode countermodel has full CAR algebra M_2, while the even/code-generated subalgebra is diagonal; the annihilator stays at distance at least 1 from it. Fourier/Vandermonde inversion therefore supplies only full-CAR compression, not the microscopic generated algebra. ALG2 requires an independent FE-GEN compression-surjectivity theorem or a precise gauge/code-projected microscopic algebra argument. The positive ALG1 estimate is likewise conditional on the generated projection containing the microscopic low-energy sectors. The round-4 sector/frame lemma gives a sufficient conditional interface, not a microscopic FE-GEN proof. ALG-EXH and `SEAM.MMST.TYPEIII.CHARGED.01` remain [O].

Retired routes (do not retry). Absolute-value majorants ($3\sqrt{N}$); global BV majorant; per-cell log counterterms at $\pm\frac{1}{2}$ (nothing to counterterm); shifted-cell majorisation (ratio 0.73); Szegő/Widom split (Fisher–Hartwig $\pi^{-2}\log N$ per summand); trace decomposition (each term diverges $\propto \log N$, cancellation $2/\pi^2 + 2/\pi^2 - 4/\pi^2$).

Research Contract — `TFPT.TOE.COMPLETE.01` (the named AND of eight completeness gates)

Typing. A research contract [O] (ledger row `TFPT.TOE.COMPLETE.01`), registered 2026-08-29 from review wave 4. This row *names* the AND of eight gates; it does *not* close any of them and it does *not* upgrade a marker. Dual rest: the compiler residual $\mathrm{Rest} = v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$ sits beside $\mathrm{Rest}_{\mathrm{TOE}}$; this contract is the named AND, not a closure of either.

The eight gates. **(T1)** structure postulate → P1/P2/compiler (`AX.P1.01`, `AX.P2.01`, `TFPT.IRREDUCIBLE.01`, `DIMENSION.SELECTOR.4D.01`).
(T2) seam = $(E_8)_1$ proven (`SEAM.EQUIV.01`, `SEAM.EQUIV.MMST.01`, `SEAM.EQUIV.TWISTOR.01`, `SEAM.SIMPLECURRENT.GENERATOR.01`,
`SEAM.MMST.TYPEIII.CHARGED.01`, `SEAM.DETLINE.UNIFICATION.01`, `SEAM.STATE.RPMIXING.01`).
(T3) quasilocal local unitary 3+1D Hamiltonian family (`QFT4D.LATTICE.FUNDAMENTAL.01`, `TFPT4D.LATTICE.ACTION.01`, `DYN.UNITARY.DILATION.01`, `DYN.MARKOV.EMBED.01`).
(T4) chiral SM with local measure + uniform mirror gap (`CHIRAL4D.NOMIRROR.01`, `CHIRAL4D.MIRROR.DET16.01`).
(T5) IR Lorentz + confinement + clustering + nontrivial scattering (`QFT4D.OS.RECON.01`, `SEAM.BULK4D.RECON.01`).
(T6) all three gauge couplings + neutrino texture internally fixed (`ALPHA.QUILLEN.EXACT.01`, `GAUGE.DETLINE.FIXPOINT.01`, `FLAV.NUSCALE.01–.06`, `FLAV.NU.TEXTURE.MECHANISM.01`).
(T7) quantum massless spin-2 with universal coupling (`GRAV.SPIN2.EMERGENCE.01`, `GRAV.NONCIRCULAR.01`, `GRAV.WEINBERG.WITTEN.01`).
(T8) unique initial state + one SK functional generating all readouts (`FTRANSFER.SK.RHO0.01`, `FTRANSFER.GENERATING.01`, `OBS.TRANSDUCTION.01`, `PRED.JOINTLIKELIHOOD.01`).

Separation (binding).

$$\mathrm{TOE}_{\mathrm{Complete}} := \mathrm{AND}(\mathrm{T1}, \ldots, \mathrm{T8}), \qquad \mathrm{TOE}_{\mathrm{Validated}} := \text{Complete AND independent holdouts passed.}$$

Every existing 4D / dilation / detline / SK / flavor / gravity contract is a *child of one gate* (listed above); children do not list this row back. Status: `TFPT.TOE.COMPLETE.01` [O].

Executed update (2026-08-30, evening TOE-gate wave): four witnesses in one battery (`[verification/v1012_toe_gate_witnesses.py]`**; the contract stays [O]).** T5 four IR witnesses executed plus the propagating-mode precondition (common-c needs dynamical gauge excitations). T7 interacting stress mechanism executed. T8 selection-principle vacuity typed — a non-entropic selection is required. T6/nu scaffold-level triple bound complete. Display marker stays [O].

Executed update (2026-08-30, late-evening harvest; `[verification/v1013_thermodynamic_dynamics.py]`, `[verification/v1014_bridge_refinements.py]`**; the contract stays [O]).** T3 mandatory-dynamics leg CLOSED at Hamiltonian-class level (Lieb–Robinson + τ_t + $\mathcal{A}^G$ + state existence). Within the finite W-bridge/compiler lane considered here, the remaining identification was routed through the external MMST/Quillen leg (TEL-B/ALG-EXH/P1 + the character-blind determinant-response map + Bismut–Freed/A_0). This historical statement does not classify or close the independent TOE gates T3–T8. Display marker stays [O].

Executed update (2026-08-31, Monday-morning harvest; `[verification/v1015_axiom_core_closure.py]`, `[verification/v1016_state_gap_batteries.py]`**; the contract stays [O]).** T1 structure-postulate route is now fully premise-supported at finite level (both W-bridge premises derived; axiom-core remainder 0 modulo the externalized MMST identification). First viable canonical-state principle (R_4 seam-modular cap). Third-route volume theorem cited-verified for finite-group DET chains. The old phrase "every remaining fundamental gap" is scope-corrected: only the finite compiler/W-bridge lane was reduced to MMST plus the 4D functional; the independent T3–T8 mechanisms were not thereby classified. Display marker stays [O].

Executed update (2026-09-05, T1–T8 integration; upstream round-3 inputs `[verification/v1022_telb_round3_certificates.py]`, `[verification/v1023_derivative_filtered_` `wall.py`], [`verification/v1024_parent_internal_boundaries.py`], [`verification/v1025_telb_c1_c2a_certificates.py`]). With the round-4 and round-7 modules and complete proof supplements below, the current gate matrix is:

- **T1:** the finite axiom-core/W-bridge premises have executable support, but their unconditional structure-postulate provenance and the dimension selector remain open.
- **T2:** the relaxed uniform TEL-B norm is closed below $2.995906 < 3$ at fixed $M = 1$, $N_y = 8$ for all even $N \geq 16$, including C2b and finite-grid aliases. Microscopic one-edge generation, FE-GEN/ALG-EXH and MMST remain open; the sector/frame bridge is conditional and the old Vandermonde implication remains refuted. Round 7 constructs charged corners only after adding an auxiliary local $\mathbb{Z}_4$ flux register. The simple eight-channel quarter twist has $h = 1/4$, not $h_\lambda = 1$; the required microscopic dressing and gauge/code quotient have not been supplied.
- **T3:** dynamics exists only at Hamiltonian-class level. No shared complete 3+1D parent or continuum theorem is constructed.
- **T4:** exact canonical DET excitations and a signed $z = 1$ wall exist under the stated singlet and fixed-classical-background premises. The source's factorized onsite model has gap at least $1 + \sqrt{3}/2$; this bound is not an interacting hopping theorem. An ordinary bond violates the proposed vacuum-relative form bound, not every possible interacting gap argument. No hard bare-mirror gap, chiral quantum-link realization, measure or 4D continuum theorem follows.
- **T5:** finite IR witnesses and the v1020 spectator obstruction make prime completion binding, but OS reconstruction, confinement, clustering and nontrivial scattering remain open.
- **T6:** full Gauss counts remove the blanket projected-dimension obstruction (one site 3 mod 5, cube 0 mod 5 with the specified nine-irrep link truncation and full 192-mode site character), but give no canonical compatible clock family or neutrino mass texture. The symmetric candidate forbids every Majorana bilinear.
- **T7:** the local first-class constraint target now has an explicit zero-mean Weyl/Fock quantization, a distributional continuum limit and a positive covariant local free curvature field of helicities ± 2. Its unstable periodic zero block must still be removed globally. Prescribed conserved sources obey the linear physical-Fourier Ward identities, but dynamical quantum sources and universal nonlinear coupling are not constructed. None of these target-field results supplies a common-parent TFPT embedding.
- **T8:** a symmetric cap on the selected internal 12D pair space has gap one, but the invariant space is two-dimensional; symmetry does not select its ray, and its vacuum-preserving extension selects no global ρ_0 or SK functional.

All T1–T8 gates and `TFPT.TOE.COMPLETE.01` remain [**O**]; no marker moves. A single shared 3+1D parent satisfying T3–T8 is still missing.

Round 4: proved subgates and binding residuals (2026-09-05)

The following results concern the explicitly stated regulators and algebras. They do not identify a single complete physical 3+1D parent. All T1–T8 and `TFPT.TOE.COMPLETE.01` remain [**O**]. The finite regressions are checks of the proofs below, not substitutes for their quantifiers.

T2: uniform relaxed TEL-B theorem. `SEAM.MMST.TELB.NORM.01`. For the frozen $M_{\text{mass}} = 1$, $N_y = 8$ collar, the original remainder obeys

$$\boxed{\|R_N\|_{\text{HS}} < 2.995906 < 3 \quad (N \geq 16 \text{ even}).}$$

The interval inputs are v1022 (TV and potential), v1025 (C1/C2a), and [`verification/v1026_telb_round4_closure.py`] (CF, DG, signed CROSS, validated Fourier core and all-volume aliases). This is not a proof of the historical sharper 2.11 or Bound-B target 0.62. In particular a limiting Fourier-core bound is not a finite-grid remainder bound.

CF: cell covariance and the periodic corner. Put $h = N^{-1}$, $A = (2\pi^2)^{-1}$, $J = N/2$, $I_N = \{-J, \dots, J-1\}$ and let bars denote normalized means over $[kh, (k+1)h]$. Define

$$f(u) = -\log|u| - \pi \mathbf{1}_{u<0}, \qquad \Phi(u) = \log \frac{|u|}{2|\sin \pi u|} - \pi u, \qquad F = f + \Phi,$$

$$c(d) = d^{-1} - \pi \cot \pi d, \quad c(0) = 0, \qquad Q_g(u,v) = \frac{g(v) - g(u)}{v - u}.$$

Partial fractions give $\Phi' = c - \pi$, $L = \sup|\Phi'| = \pi + 2$, $m = \inf \Phi'' = \pi^2/3$, $M = \sup \Phi'' = \pi^2 - 4$ and $D = M - m = 2\pi^2/3 - 4$, with $\int |\Phi'''| = 2D$. The identities used are the standard cotangent and polygamma identities DLMF 4.22 and DLMF 5.15; no asymptotic truncation is needed. The real kernel cancellation is exact:

$$r(u,v) = A\{c(v-u)(F(v) - F(u)) - Q_\Phi(u,v) - i\pi(F(v) - F(u))\}.$$

For independent uniform cell points U, V, set $d_0 = (l-k)h$ and $\delta = V - U - d_0$. Then

$$(\mathcal{K}_N)_{kl} = h\mathbb{E}r(U,V), \quad (\mathcal{K}'_N)_{kl} = hA\left[c(d_0)(\bar{F}_l - \bar{F}_k) - \frac{\bar{\Phi}_l - \bar{\Phi}_k}{d_0} - i\pi(\bar{F}_l - \bar{F}_k)\right].$$

Thus the imaginary part freezes exactly. If $v_k = \operatorname{Var} F(U_k)$, $|a'| \le a_1$, $|a''| \le a_2$, the independent-copy variance identity, $\mathbb{E}\delta = 0$, and $\mathbb{E}\delta^2 = h^2/6$ give

$$|\mathbb{E}[(a(d_0) - a(V-U))(F(V) - F(U))]| \le \frac{a_1 h}{\sqrt{6}}\sqrt{v_k + v_l} + \frac{a_2 h^2}{12}|\bar{F}_l - \bar{F}_k|. \tag{R4.1}$$

For $|g''| \le B$, the two coordinate Lipschitz constants of Q_g are at most $B/2$, so $\operatorname{Var} Q_g \le B^2h^2/24$ and $|\mathbb{E}(\delta Q_g)| \le Bh^2/12$. The two singular log cells have variance one; all remaining log-cell variances are at most $1/(12n^2)$. Therefore

$$(\textstyle\sum_k v_k)^{1/2} \le V(h) := \sqrt{2 + \zeta(2)/6} + L\sqrt{h/12}, \quad \textstyle\sum_{k,l} |\bar{F}_l - \bar{F}_k|^2 \le 2N^2\zeta(2).$$

The last bound follows from $\operatorname{Var} F = \pi^2/6$ and contraction of cell conditional expectation. Split the nonzero lags into central ($|l-k| \le N/2$), middle ($N/2 < |l-k| \le 3N/4$), and far regions. For the first two regions (R4.1) gives

$$C_{\text{central}} \le \frac{Ac'(9/16)hV(h)}{\sqrt{3}} + \frac{Ac''(9/16)h^{3/2}\sqrt{2\zeta(2)}}{12} < 0.027267,$$

$$C_{\text{middle}} \le \frac{Ac'(13/16)h^{3/2}}{\sqrt{6}}\left(\sqrt{H_{J-1}/3} + L/\sqrt{24}\right) + \frac{Ac''(13/16)h^{3/2}\sqrt{2\zeta(2)}}{12} < 0.055905.$$

In the middle count, negative index $-n$ has $n-1$ partners and positive index s has s partners; the singular cells have none. Consequently the log-variance sum is at most $H_{J-1}/3$ and the smooth part at most $L^2/24$. The sequence H_{J-1}/J^3 decreases for $J \ge 8$, by its one-step recurrence, so the displayed envelopes are evaluated at $N = 16$.

For the far seam let $r = N - (l-k)$, $t = 1 - (V-U)$ and $t_0 = rh$. The short periodic continuation satisfies $F(V) - F(U) = -tQ$ with $|F'| \le L_F = 2\pi$ and $|F''| \le M_F = 2\pi^2$. In $c(d) = (1-d)^{-1} + a(d)$ the apparent pole cancels exactly:

$$\mathbb{E}[(t_0^{-1} - t^{-1})(F(V) - F(U))] = -t_0^{-1}\mathbb{E}[(t - t_0)Q].$$

There are r pairs per orientation. With $a_1 = (16/11)^2 + c'(5/16)$, $a_2 = 2(16/11)^3 + c''(5/16)$,

$$C_{\text{far}} \le AM_F h^{3/2}\sqrt{H_{\lfloor N/4 \rfloor}/72} + Aa_1L_Fh^{3/2}/6 + Aa_2L_F(5/16)h^{3/2}/12 < 0.009280.$$

Here $H_{\lfloor N/4\rfloor}/N^3 \le H_4/16^3$ follows by grouping four successive sizes. The remaining smooth quotient contributes $C_\Phi \le AMh\sqrt{\zeta(2)/72} < 0.002810$. The three lag regions are HS-orthogonal; only the last contribution uses the triangle inequality:

$$\sqrt{N}\|(\mathcal{K}'_N - \mathcal{K}_N)_{\mathrm{off}}\|_{\mathrm{HS}} \le \sqrt{C^2_{\mathrm{central}} + C^2_{\mathrm{middle}} + C^2_{\mathrm{far}}} + C_\Phi < 0.065697. \tag{R4.2}$$

DG: original diagonal, not a replacement recurrence. The csc-squared finite product identity and trigamma recurrence for the original remainder give, for $R_k = (\mathrm{Res}_N)_{kk}$,

$$R_{k+1} - R_k = -Ah^2\Phi''((k+3/4)h), \qquad \sum_k R_k = A\pi + A\tau(J),$$

where $T(x) = (x-1)\psi_1(x) + \psi(x)$ and $\tau(J) = T(J+5/4) - T(J+3/4)$. Since $T'(x) = 2\sum_{n\ge0}(n+1)/(n+x)^3 \le 4/J$ for $x \ge J \ge 8$, $0 \le \tau(J) \le 2/J = 4h$. The 3/4-point Peano kernel and $h\sum_k \Phi'(kh) = -\pi - 2h$ imply

$$|NR_k + A\Phi'(kh)| \le Ah(2 + 3D/2). \tag{R4.3}$$

On each diagonal cell the imaginary mean vanishes, $|Q_\Phi - \Phi'(kh)| \le Mh$, and $|c(d)| \le C_s|d|$, $C_s = (\pi^2/3)/(1-h^2)$. The worst logarithmic integral occurs adjacent to zero and is exactly $\mathbb{E}[|V-U|\,|\log|V/U||] \le h/2$; also $\mathbb{E}|V-U|^2 = h^2/6$. Thus

$$\sqrt{N}\|(\mathrm{Res}_N - \mathcal{K}_N)_{\mathrm{diag}}\|_{\mathrm{HS}} \le Ah(2+3D/2) + AMh + AC_s(h/2 + Lh^2/6) < 0.042959. \tag{R4.4}$$

All scalar envelopes increase with $h \le 1/16$. With the independently certified potential contribution 0.399224424139108, orthogonality gives $c_E < 0.466902$, below the required 0.542702.

CROSS: the sufficient one-sided theorem. Let W_∞ be the unitary fractional shift with entries $e^{-i\pi/4}\sin(\pi/4)/[\pi(k-m-1/4)]$, let P restrict to I_N, $w = PW_\infty e_0$, $B = W_\infty P_{(-\infty,-1]}W^*_\infty$, $t_N = \|(1-P)W_\infty e_0\|^2$ and $\mu_N = \sum k|w_k|^2$. Crucially $BW_\infty e_0 = 0$. For the finite unitary overlap W_N and $\rho_m = \{(m+1/4)/N\}$, the source identity is

$$\mathrm{Res}_N = -\operatorname{diag}(k)/N + W_N \operatorname{diag}(\rho_m) W_N^* - PBP.$$

Hence $w^*\mathrm{Res}_N w \ge -\mu_N/N + (1-t_N)/(4N) - t_N$. Pairing signed momenta and using convex reciprocal-square tail integrals yields

$$\mu_N \le 64/675, \qquad Nt_N \le T_* := \frac{4A}{1-(3/32)^2}.$$

For $A_N = |e_0\rangle\langle e_0|/2 - |w\rangle\langle w|$, (R4.3) at $k=0$ and the exact identity $2\,\mathrm{Re}\langle A_N, \mathrm{Res}_N\rangle = (\mathrm{Res}_N)_{00} - 2w^*\mathrm{Res}_N w$ imply

$$N\,2\,\mathrm{Re}\langle A_N, \mathrm{Res}_N\rangle \le A\pi + \tfrac{A}{16}(2+3D/2) - \tfrac{1}{2} + \tfrac{128}{675} + 2(1+1/64)T_* < 0.282637. \tag{R4.5}$$

This is not an absolute-value bound: the negative finite cross terms explicitly falsify the absolute $0.283/N$ mutant. Cell contraction and C1 give

$$\|S_N + H_N\|^2_{\mathrm{HS}} \le \tfrac{5}{4} - \tfrac{8}{\pi^2} + (\sqrt{0.1525} + 0.466902/4)^2 + 0.282637/16 < 0.714386. \tag{R4.6}$$

Validated Fourier core and the missing finite-grid aliases. Use the sigma-x basis $H(p) = \begin{pmatrix} \sin p\, I_8 & B_p \\ B_p^T & -\sin p\, I_8 \end{pmatrix}$, $B_p = (1-\cos p)I_8 - S$, $S_{y,y+1} = 1$. The jump is $dP = E_{8,8} - E_{7,7}$ (zero-based indices), and $F(p) = P(p) - (1/2 - p/(2\pi))dP$ for $0 < p < 2\pi$. At the 512 midpoint nodes, 256-bit Acb validated eigensystems strictly separate eight occupied and eight unoccupied eigenvalues. The outward matrix DFT A_m obeys the exact midpoint alias formula $A_m = \sum_{\ell\in\mathbb{Z}}(-1)^\ell C_{m+512\ell}$.

The v1022 bound $\|C_m\|_{\mathrm{HS}} \le K_3/|m|^3$, $K_3 = 2.517464500835$, controls the entire alias error by $K_3\zeta(3)[(512-m)^{-3} + 512^{-3}]$. The certified rational caps for $m = 1, \ldots, 16$ are, in order,

$$\begin{aligned}
&.872617451262, .119561574970, .036384250090, .013594203794,\\
&.005638853577, .002552800879, .001281003972, .000727535964,\\
&.000465728162, .000325832279, .000241157627, .000184924134,\\
&.000145312983, .000116348415, .000094610814, .000077965240.
\end{aligned}$$

Since $\sum_{m>16} m^{-5} \le 1/(4\cdot 16^4)$, these give $2\sqrt{\sum_{m\ge 1} m\|C_m\|_{\mathrm{HS}}^2} < 1.783260551 < 1.783261$. This is a bound on a dominant Fourier matrix, not yet on R_N^{sm}. Absolute summability gives the exact four-cover expression

$$A_N(d) = \sum_{q\notin 4\mathbb{Z}} (1-i^q) C_{d+qN}, \qquad \|R_N^{\mathrm{sm}}\|_{\mathrm{HS}}^2 = \sum_{d=-(N-1)}^{N-1} (N-|d|)\|A_N(d)\|_{\mathrm{HS}}^2.$$

Choose $q = 1$ for negative $d = -N + m$ and $q = -1$ for positive d; at $d = 0$ choose $q = 1$ once. The resulting dominant norm squared is $4\sum_{m=1}^{N-1} m\|C_m\|^2 + 2N\|C_N\|^2$, bounded by the infinite core. The other aliases have envelope

$$R(t) = \sum_{q\notin 4\mathbb{Z},\ q\ne 1} \frac{|1-i^q|}{|q-1+t|^3}, \qquad 0 \le t \le 1.$$

This nonsingular series is convex. Sum $|q| \le 100$ and add the full tail $4/100^3 + 2/100^2$; its outward bounds at $0, 1/2, 1$ are denoted a, b, c (approximately $2.528674155, 1.333485721, 2.086708706$). Use their actual Arb enclosures, not those abbreviated printed decimals. The piecewise affine interpolant $L(t)$ majorizes $R(t)$. For $f(t) = tL(t)^2$ the composite trapezoid rule on each half interval is exact for the cubic, including the derivative jump. For every even N,

$$\begin{aligned}
T_N &= \frac{2}{N}\left(\sum_{m=1}^{N-1} f(m/N) + \frac{f(1)}{2}\right) = I + D_*/(6N^2),\\
I &= a^2/24 + ab/12 + b^2/3 + bc/4 + 7c^2/24,\\
D_* &= 4b^2 - 2ab - a^2 + 5c^2 - 6bc.
\end{aligned}$$

Thus the alias norm is at most $K_3\sqrt{I + |D_*|/(6\cdot 16^2)}/16^2 \le 0.017332234304270 < 0.017333$ and $\|R_N^{\mathrm{sm}}\|_{\mathrm{HS}} < 1.800594$. Finally $R_N = R_N^{\mathrm{sm}} - (S_N + H_N)\otimes Q$, with $\|Q\|_{\mathrm{HS}} = \sqrt{2}$ from two orthogonal rank-one projectors. Using the unrounded alias enclosure with (R4.6) gives

$$\|R_N\|_{\mathrm{HS}} < 1.783261 + 0.017332234304270 + \sqrt{2(0.714386)} < 2.995906 < 3.$$

The fixed-width full collar still has two gapless edges. One-edge microscopic identification, FE-GEN, ALG-EXH and charged MMST do not follow from this norm theorem.

T3/T4: canonical DET excitations and the signed wall. `CHIRAL4D.MIRROR.SIGNED.CAR.01`. [`verification/v1027_signed_det_car_wall.py`] checks the following algebra and its finite countermodels. The old Hubbard operators $b_i = |\Omega\rangle\langle e_i|$ satisfy $\{b_i, b_j^*\} = \delta_{ij}P_\Omega + |e_j\rangle\langle e_i|$, not CAR, and $b_i^* b_j^* = 0$. For even $n \ge 4$ (in particular $n = 32$), write $F = |\mathrm{full}\rangle\langle 0|$ and define

$$U = I + (2^{-1/2}-1)(P_0 + P_F) + (F - F^*)/\sqrt{2}, \qquad a_i = Uc_iU^*.$$

This is a two-dimensional rotation and identity elsewhere. It is even; under the inherited full-determinant singlet premise it is onsite gauge invariant. It maps $|0\rangle$ to $\Omega = (|0\rangle + |\mathrm{full}\rangle)/\sqrt{2}$, so the a_i satisfy CAR on the entire 2^n-dimensional Fock space and $a_i^*\Omega = e_i$. Products of even onsite rotations preserve cross-site CAR. The actual new parent is additive:

$$N_a = UNU^* = N + \tfrac{n}{2}(P_0 - P_F - F - F^*), \qquad Q \le N_a \le nQ, \quad Q = I - P_\Omega.$$

Thus ΔN_a has the same unique ground state and gap Δ as ΔQ, but is not the old flat parent. The high-degree local rotation is not asserted to be a linear Bogoliubov map or an efficient Fock simulator.

Let $A = A^*$ be a fixed c-number, gauge-covariant quasilocal one-particle operator, $\|A\| \le L$. Define

$$H = p^*Ap + \Delta a^*a + \lambda(a + gAp)^*(a + gAp), \quad M = \Delta + \lambda, \quad \beta = \lambda g^2, \quad \eta = \lambda g, \quad c = \lambda g^2\Delta/M.$$

If $cL < 1$ and $B = L + \beta L^2 < M$, the spectral block

$$h(s) = \begin{pmatrix} s + \beta s^2 & \eta s \\ \eta s & M \end{pmatrix}, \quad \det h(s) = Ms(1 + cs)$$

has precisely the original kernel, no extra zero, high band $\mu_+ \ge M$, and $|\mu_-| \le B$. Expansion gives $\mu_-(s) = s + cs^2 - \eta^2 s^3/M^2 + O(s^4)$, hence $z = 1$ when A is linear. For $(\Delta, \lambda, g, L) = (3, 1, 1/2, 2)$, $M = 4$, $B = 3$, $cL = 3/8$. Filling the negative band and subtracting its energy gives a nonnegative full-Fock Hamiltonian for each fixed classical background; $E_0 \ge -Lr|\Lambda|$ with r physical modes per site. The bare mirror operator nevertheless has low-band weight of order s^2. There is no hard bare-mirror response gap, nor an isolated entire many-body mirror continuum. Operator-valued quantum-link diagonalization does not automatically lift to a canonical Fock transformation (the explicit SWAP mutant fails CAR).

For a free illustration only, $\gamma_i = \tau_1 \otimes \sigma_i$, $\gamma_5 = \tau_3 \otimes I$, $w = \sum_i (1 - \cos k_i) - 1$, $R^2 = 1 + 2\sum_{i<j}(1 - \cos k_i)(1 - \cos k_j) \in [1, 25]$ and $D = I + (w + i\gamma \cdot \sin k)/R$ give $A = \gamma_5 D$, $A^2 = 2 + 2w/R$ and $\|A\| \le 2$, with one Dirac node at zero. The inverse-square-root series has norm ratio 12/13 and exponential length $2/\log(13/12)$. This four-component regulator contains opposite Weyl sectors: it is not the demanded two-component chiral SM construction. The common-parent dynamical gauge, measure, anomaly/index and continuum obligations remain open.

T6/T8: full Gauss counts, internal cap and selection rules. `PARENT.GAUSS.D4.ROUND4.01`. [`verification/v1028_gauss_d4_cap.py`] uses the full exterior character, not a low-occupation truncation. Let W_{16} consist of $Q : (\mathbf{3}, \mathbf{2})_1$, $u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-4}$, $d^c : (\bar{\mathbf{3}}, \mathbf{1})_2$, $L : (\mathbf{1}, \mathbf{2})_{-3}$, $e^c : (\mathbf{1}, \mathbf{1})_6$, $\nu^c : (\mathbf{1}, \mathbf{1})_0$, with charges $6Y$. For color weights $(1, 0), (0, 1), (-1, -1)$ and weak weights ± 1, set

$$P(z) = \prod_{w \in W_{16}} (1 + z^w), \quad h = 1 + (u + u^{-1})v^3, \quad \Delta_W = (1 - x/y)(1 - x^2y)(1 - xy^2)(1 - u^2).$$

There are twelve SM16 copies per site (192 fermion modes), a three-state Higgs space, and seam dimension four. The singlet multiplicity of a Weyl-invariant character is $\mathrm{CT}(f\Delta_W)$, not its zero-weight multiplicity. All representations descend to the $\mathbb{Z}_6$ quotient. In characteristic five, $P^{12} = P^2(z)P^2(z^5)$, so for $A = P^2$,

$$\mathrm{CT}(P^{12}z^r) = \sum_{a+5b+r=0} A_a A_b \pmod 5.$$

This finite sparse calculation has 20771 integer two-copy coefficients (17151 nonzero modulo five); it never enumerates 2^{192} states. The full one-site Gauss dimension is $4\,\mathrm{CT}(P^{12}h\Delta_W) \equiv 3 \pmod 5$. A neutral Majorana bijects the parity sectors, each of which has dimension $4 \pmod 5$. For the nine link irreps $0, Q, u^c, d^c, L, e^c, H, \mathrm{Adj}_3, \mathrm{Adj}_2$, whose squared dimensions sum to 137, the exact transfer matrix is $T_{RS} = \mathrm{CT}(P^{12}h\chi_R\bar{\chi}_S\Delta_W)$. An n-vertex open path has dimension $4^n(T^n)_{00} \pmod 5$. For $n = 1, \ldots, 8$ this is $3, 4, 3, 0, 0, 1, 4, 0$. The full open $2 \times 2 \times 2$ cube within this specified nine-irrep link truncation (eight sites, twelve links) has dimension $0 \pmod 5$. Two pairwise contraction orders give reduced integer 48544377696195; this is not the Hilbert dimension. The bound $9^{12}4^8 < 2^{63}$ precludes contraction overflow. Independent exact raising-operator ranks check 56 singlets without Higgs $(556 - 500)$ and 120 with Higgs $(1376 - 1256)$ for one copy; the two-copy character result with Higgs is 561848. Thus the universal projected non-divisibility obstruction is false. An abstract Weyl pair exists when five divides the dimension, but locality, covariance, canonical choice, energy compatibility and compatibility between volumes are additional demands.

On the selected even 12-dimensional pair space $|k, g\rangle$, use $C = \mathrm{diag}(1, -i, -1, i) \otimes \mathrm{diag}(-i, -1, i)$ and $J|k, g\rangle = |-k \bmod 4, 4 - g\rangle$. Then

$$\psi = (|1, 3\rangle + |2, 2\rangle + |3, 1\rangle)/\sqrt{3}, \qquad h_{\mathrm{sym}} = I - |\psi\rangle\langle\psi|$$

is D_4 invariant, has unique ground vector and gap one, Schmidt rank three, and nonadditive HS norm $\sqrt{8/9}$. This is ordinary D_4 only on that selected even space, not the whole projective fermion lift. The invariant-vector space has dimension two, so symmetry does not select this ray. Extending the cap by zero outside this subspace leaves the old vacuum and does not select the full-parent state. The unitary $X = P_\psi + e^{2\pi i/5}(I - P_\psi)$ has order five but only two eigenvalues; it is not a Weyl clock, and its chosen order is not derived.

For the actual one-fermion phases $u_g = e^{-i\pi g/4}$, invariance of a Majorana bilinear requires $(u_g u_h - 1)M_{gh} = 0$. Since $2 \le g + h \le 6$, $M = 0$ for this symmetric candidate. A Hermitian pair kernel instead has commutant $\mathrm{diag}(a, b, a)$; the two transformation laws differ. The specified squared seam-pair coupling conserves $Q = N_\nu + 2Q_{\mathrm{seam}}$ and cannot generate anomalous pairing in its unique $Q = 0$ vacuum. These symmetries have not been established for every term of an eventual full parent. No physical neutrino texture is obtained.

T7: local constraints and zero-mode removal. `GRAV.SPIN2.LOCAL.CONSTRAINT.01`. [`verification/v1029_local_tensor_constraints.py`] implements a quadratic target, not a TFPT embedding. For six orthonormal symmetric tensor coordinates, let $t = (1, 1, 1, 0, 0, 0)^T$, $B(k)q = q_{ij}k_j$, $v = B^T k$, $s = v - |k|^2 t$. Set

$$K_p = I - tt^T/2, \quad K_q = |k|^2(I - tt^T) - 2B^T B + tv^T + vt^T, \quad H = \tfrac{1}{2}p^T K_p p + \tfrac{1}{2}q^T K_q q,$$

with constraints $C_0 = s^T q = 0$, $C = Bp = 0$. Direct polynomial algebra gives

$$BB^T = (|k|^2 I + kk^T)/2, \quad Bs = 0, \quad K_q B^T = 0, \quad K_p s = B^T k, \quad s^T s = 2|k|^4.$$

The four constraints commute and $\dot{C}_0 = k^T C$, $\dot{C} = 0$. For $k \neq 0$ they are independent first-class constraints, leaving $12 - 2 \cdot 4 = 4$ phase dimensions. Rotation to $k \parallel e_3$ and constraint/gauge reduction leaves the two normalized TT tensors, on which $H = \frac{1}{2}\sum_{a=+,\times}(p_a^2 + |k|^2 q_a^2)$. Local constraints therefore avoid putting the nonlocal TT projector into the Hamiltonian. This is the standard linearized-gravity mechanism, not a novel derivation of general relativity (see Green–Kiriushcheva–Kuzmin).

A finite-stencil regulator puts diagonal components on sites, off-diagonal ij components at offsets $(e_i + e_j)/2$, and vector parameters on links. Use $\kappa_i = 2\sin(ak_i/2)/a$ in every formula. Under a Brillouin shift, the sign flip U_i on vectors induces D_i on symmetric tensors, and $B(U_i\kappa) = U_i B(\kappa) D_i$, $s(U_i\kappa) = D_i s(\kappa)$, $K_q(U_i\kappa) = D_i K_q(\kappa) D_i$, $K_p = D_i K_p D_i$. Thus the half-grid symbol is consistently glued; its only zero is $k = 0$, unlike the eight-corner $\sin k_i$ mutant. At $k = 0$, however, all constraints vanish and $p_0 = rt$ gives $H_0 = -3r^2/4$: the unmodified periodic model is unbounded below. A positive radiative target requires removal of the entire global canonical block (q_0, p_0), an extra global zero-mean prescription, not a local constraint or a TFPT state-selection theorem. Furthermore finite-dimensional exact CCR are impossible by the trace: $[q_d, p_d] = i(I - dP_{d-1})$ for the explicit truncation. Collective low-energy emergence is not excluded, but needs a controlled microscopic sequence. No critical embedding, nonlinear constraints, universal coupling or global gravitational state is established. The distinction between a tensor target and controlled emergence also matters in Gu–Wen; neither cited paper fills the TFPT embedding gap.

ALG2: sector generation and conditional uniform section. SEAM.MMST.ALG2.FRAME.01. [verification/v1030_alg2_sector_frame.py] does not restore the refuted Vandermonde implication. Suppose, explicitly, $H = \bigoplus_r H_r$ and the actual neutral algebra is $\mathcal{A}_0 = \bigoplus_r \operatorname{End} H_r$. Join r, s if $P_s T P_r \neq 0$. Then $W^*(\mathcal{A}_0, T) = \operatorname{End} H$ precisely when this undirected sector graph is connected: the neutral commutant is $\bigoplus_r c_r I_r$, and each nonzero corner forces $c_r = c_s$. The finite bicommutant theorem proves the result. Genuine superselection components must be treated separately, not bridged by forbidden odd fields.

For a fixed D-dimensional low-energy space choose actual local words $Y_{j,N}$, $j = 1, \dots, D^2$, with $\|Y_{j,N}\| \le M$. Let Φ_N have columns $\operatorname{vec}(P_N Y_{j,N} P_N)$. If $\sigma_{\min}(\Phi_\infty) = \sigma > 0$ and $\|\Phi_N - \Phi_\infty\|_2 \le \sigma/2$, then solving $\Phi_N a = \operatorname{vec} X$ gives, for $\|X\| \le 1$,

$$P_N y_N P_N = X, \quad y_N = \sum_j a_j Y_{j,N}, \quad \|a\|_2 \le 2\sqrt{D}/\sigma, \quad \|y_N\| \le 2MD^{3/2}/\sigma.$$

This supplies a uniform section only under the stated microscopic word, norm and noncollapsing-frame hypotheses. Compression does not preserve products ($PABP$ need not equal $PAPBP$), and double-cutoff convergence does not control $|N\rangle\langle 0|$ on a fixed input cutoff. For ALG2 one needs a joint real-linear frame for $L_N(y) = (P_F y P_E, P_F y^* P_E)$ and uniform high-output tails of both x_N, y_N and their adjoints. If the joint error and each pair of tails are at most ε, the desired one-sided sum is at most 3ε. Neutral generation, charged corners, actual local words, their scaling limits, and these tails are not yet proved for the one-edge TFPT microscopic algebra. They, together with the common-parent gates, are the precise remaining research obligations.

Rounds 6 and 7: free quantum curvature, covariant extension, charged disorder, mirror parent, and first-order matter coupling

Exact scope and evidence ledger

This supplement records five written proofs and their machine regressions:

$$69 + 34 + 35 + 36 + 59 = 233$$

passing checks. The Round 6 prerequisite contributes 69 checks (53 exact and 16 floating regressions) [verification/v1031_quantum_tensor_curvature.py]. Round 7 contributes 164 checks: 130 exact, 13 source-contract, and 21 finite or floating regressions [verification/v1032_covariant_curvature.py] [verification/v1033_charged_disorder.py] [verification/v1034_parent_mirror.py] [verification/v1035_matter_coupling.py]. The finite and floating checks are implementation regressions, never substitutes for the quantified arguments below.

The Round 6/Round 7 Lorentz-covariant helicity-± 2 sector is a declared free regulator/Fock construction. It is not an emergent TFPT parent, a nonlinear completion, or a proof of universal matter coupling. The factorized DET result proves the gap of the momentum-cell parent actually specified there; it does not prove a gap for ordinary real-space hopping. The charged construction distinguishes the common quarter disorder $h_{\text{deck}} = 1/4$ from the sourced field $h_\lambda = 1$. The matter construction retains the full Fourier factors $c_0 = -s^T q$, $\boldsymbol{c} = \mathrm{i}Bp$ and the paired $+k/-k$ real Hamiltonian; its sign is not stripped. All physical completeness gates T1–T8 and TFPT.TOE.COMPLETE.01 remain [O]. **NO RH CLAIM.**

Round 6 prerequisite: a free quantum curvature field on the staggered tensor target

The displayed arguments below are the proof-bearing source. The current repository regression is [verification/v1031_quantum_tensor_curvature.py]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Exact scope. This note quantizes only the free, quadratic, zero-mean curvature target whose classical constraint quotient was established at repository commit `4417f1ba8da3d5dc622c942efcbfc46ebaae546d`. It constructs neither a microscopic TFPT parent nor finite-dimensional quantum cells. It proves no nonlinear constraint algebra, state selection mechanism, universal matter coupling, interacting graviton theory, T7 closure, TOE claim, or RH statement. Strict lattice light-cone support remains false. The microcausality theorem below is a distributional statement about the $a \to 0$ limit of this free target.

Assumptions and preregistered obligations. Let a periodic cubic lattice have $V = L^3$ cells and spacing $a > 0$. The six orthonormal symmetric-tensor coordinates are ordered as $(11, 22, 33, 23, 13, 12)$, with the off-diagonal basis tensors normalized by $1/\sqrt{2}$. Component A lives at $x + am_A/2$, where

$$m_A = (000, 000, 000, 011, 101, 110).$$

We use the physical-position Fourier phase $\exp[-\mathrm{i}k \cdot (x + am_A/2)]$. The staggered derivative symbol is

$$\mathrm{i}\kappa_i^{(a)}(k), \qquad \kappa_i^{(a)}(k) = \frac{2}{a} \sin \frac{ak_i}{2}, \qquad r_a(k) = |\kappa^{(a)}(k)|.$$

It vanishes only at the homogeneous momentum in the first Brillouin zone. The complete homogeneous canonical block is removed as an additional zero-mean prescription. It is not declared gauge.

The proof was preregistered against the following obligations.

O1. Prove, for every nonzero real momentum and not from samples, that the curvature commutator operator is $r^4 P_{\mathrm{TT}}$, is positive, has rank two, and is transverse and traceless.

O2. Construct the Weyl quotient, prove the Heisenberg inequality by an explicit factorization, and identify the finite-volume GNS representation and ground state. A finite matrix approximation to the CCR is inadmissible.

O3. Treat k and $-k$, self-conjugate momenta, and half-grid Bloch signs without choosing a globally inconsistent polarization frame.

O4. Prove convergence of local states as $L \to \infty$ at fixed a.

O5. Either prove an $a \to 0$ tempered-distribution limit and its curvature microcausality, or leave that claim open. Finite grids may only be regression tests for the analytic proof.

The input from the classical target is the canonical reduced Hamiltonian and the already-proved staggered identities. In one momentum fibre set

$$Bq = q\kappa, \quad t = (1, 1, 1, 0, 0, 0)^T, \quad v = B^T\kappa, \quad s = v - r^2 t,$$

$$K_p = I - \frac{1}{2}tt^T, \qquad K_q = r^2(I - tt^T) - 2B^T B + tv^T + vt^T. \tag{1}$$

The four local constraints are the scalar $s^T q = 0$ and the three momentum constraints $Bp = 0$. For $r > 0$ they are independent and first class. The curvature variables are

$$E = K_q q, \qquad F = K_q K_p p, \tag{2}$$

and the classical result supplies

$$K_q B^T = 0, \quad K_p s = B^T\kappa, \quad K_q t = s,$$

$$K_q^2 = r^2 K_q + ss^T, \qquad K_q K_p K_q = r^2 K_q + \frac{1}{2}ss^T. \tag{3}$$

All matrices in (1)–(3) are real in the physical-position trivialization.

The exact spin-two commutator projector. For $\kappa \neq 0$, let $P = I_3 - \kappa\kappa^T/r^2$. The orthogonal spin-two projector on symmetric tensors is

$$(P_{\mathrm{TT}}h)_{ij} = P_{ia}P_{jb}h_{ab} - \frac{1}{2}P_{ij}P_{ab}h_{ab}. \tag{4}$$

It has rank two, $P_{\mathrm{TT}}^2 = P_{\mathrm{TT}} = P_{\mathrm{TT}}^T$, and its range is $\{h : h\kappa = 0,\ \mathrm{Tr}\, h = 0\}$.

Theorem 2 (Polynomial curvature projector)**.** *For every $\kappa \neq 0$,*

$$C := K_q K_p K_q = r^4 P_{\mathrm{TT}}. \tag{5}$$

Consequently

$$C^T = C \geq 0, \quad C^2 = r^4 C, \quad \mathrm{Tr}\, C = 2r^4, \quad \mathrm{rank}\, C = 2, \quad CB^T = 0, \quad Ct = 0. \tag{6}$$

Although (4) has inverse powers of r, C itself is a homogeneous degree-four polynomial in κ.

Proof. The TT space is $\mathrm{Ker}\, B \cap t^{\perp}$. If h is TT, (1) gives $K_q h = r^2 h$ and $K_p h = h$, hence $Ch = r^4 h$. Equations (3) give $CB^T = 0$. They also give $Ct = K_q K_p s = K_q B^T \kappa = 0$. The scalar and momentum constraints are independent at $r > 0$, so $\mathrm{Ran}\, B^T + \mathrm{span}\{t\}$ is four-dimensional and is the orthogonal complement of the two-dimensional TT space. Thus C is r^4 on TT and zero on its orthogonal complement, proving (5). The properties in (6) follow immediately. Finally, $C = K_q K_p K_q$ is polynomial; equivalently the two factors of r^{-2} implicit in (4) cancel against r^4. A direct component multiplication of (4) is included in the executable checker. □

This theorem sharpens the off-shell identity in (3): the term $r^2 K_q + ss^T/2$ is exactly the polynomial numerator of the TT projector. No nonlocal TT projector is inserted into the lattice observable.

CCR quotient and finite-volume vacuum. Let $\mathcal{S}_L$ be the real finite-volume smearing space of curvature pairs $X = (f, g)$, with the homogeneous fibre absent and with the same reality/Bloch conditions as the fields. Define

$$\Phi(f, g) = E(f) + F(g), \qquad \sigma_L(X, Y) = \langle f, Ch_g\rangle - \langle h_f, Cg\rangle, \tag{7}$$

where $Y = (h_f, h_g)$ and the brackets denote the real lattice inner product. Canonical quantization means

$$[\Phi(X), \Phi(Y)] = \mathrm{i}\sigma_L(X, Y)I, \qquad W(X)W(Y) = e^{-\mathrm{i}\sigma_L(X,Y)/2}W(X + Y). \tag{8}$$

The null space of (7) is $\mathrm{Ker}\, C \oplus \mathrm{Ker}\, C$. We therefore define the physical Weyl algebra on

$$\mathcal{S}_L^{\mathrm{phys}} = \mathcal{S}_L/(\mathrm{Ker}\, C \oplus \mathrm{Ker}\, C), \tag{9}$$

not on the unreduced six-component smearing space. By (5), (9) retains exactly the two TT canonical pairs in each nonzero real momentum component.

Let $R = (-\Delta_a)^{1/2}$. It has fibre eigenvalue $r_a(k)$ and is strictly positive after removal of the homogeneous block. It commutes with C. Define the real covariance

$$\mu_L(X, Y) = \frac{1}{2}\langle f, CR^{-1}h_f\rangle + \frac{1}{2}\langle g, RCh_g\rangle. \tag{10}$$

Thus, fibre by fibre,

$$\frac{1}{2}\langle\{E, E\}\rangle = \frac{C}{2r}, \qquad \frac{1}{2}\langle\{F, F\}\rangle = \frac{rC}{2}, \qquad \frac{1}{2}\langle\{E, F\}\rangle = 0. \tag{11}$$

The unsymmetrized mixed covariances are $\langle EF\rangle = \mathrm{i}C/2$ and $\langle FE\rangle = -\mathrm{i}C/2$.

Theorem 3 (Proposition: Exact uncertainty factorization)**.** *The covariance (10) satisfies the CCR positivity condition, and it saturates it on every physical oscillator. If $C^{1/2} = r^2 P_{\mathrm{TT}}$ fibrewise, then*

$$\mu_L + \frac{\mathrm{i}}{2}\sigma_L = \frac{1}{2}\begin{pmatrix} CR^{-1} & \mathrm{i}C \\ -\mathrm{i}C & RC \end{pmatrix} = \frac{1}{2}\begin{pmatrix} C^{1/2}R^{-1/2} \\ -\mathrm{i}C^{1/2}R^{1/2} \end{pmatrix}\begin{pmatrix} R^{-1/2}C^{1/2} & \mathrm{i}R^{1/2}C^{1/2} \end{pmatrix} \geq 0. \tag{12}$$

Equivalently,

$$J(f,g) = (-Rg, R^{-1}f), \qquad J^2 = -I, \qquad \mu_L(X,Y) = \frac{1}{2}\sigma_L(X, JY). \tag{13}$$

Hence $\omega_L(W(X)) = \exp[-\mu_L(X,X)/2]$ is a pure regular quasifree state on the physical Weyl algebra.

Proof. All factors in (12) commute in momentum space, and multiplication gives the middle matrix exactly. This proves the uncertainty inequality rather than testing principal minors. On the quotient, J is well defined, compatible with σ_L, and satisfies (13).

For existence and purity, no abstract reconstruction theorem is needed. Choose a real orthonormal spectral basis common to R and P_{TT}. In the αth physical plane represent Q_α by multiplication and $P_\alpha = -\mathrm{i}\,\partial/\partial q_\alpha$ on $L^2(\mathbb{R}, \mathrm{d}q_\alpha)$, and put

$$E_\alpha = r_\alpha^2 Q_\alpha, \qquad F_\alpha = r_\alpha^2 P_\alpha, \qquad \Omega_\alpha(q) = \left(\frac{r_\alpha}{\pi}\right)^{1/4} e^{-r_\alpha q^2/2}. \tag{13a}$$

Direct Gaussian integration gives $\langle Q_\alpha^2\rangle = 1/(2r_\alpha)$, $\langle P_\alpha^2\rangle = r_\alpha/2$, and $\langle\{Q_\alpha, P_\alpha\}\rangle/2 = 0$. It also gives

$$\left\langle \Omega_\alpha, e^{\mathrm{i}(fE_\alpha + gF_\alpha)}\Omega_\alpha \right\rangle = \exp\left[-\frac{1}{4}\left(r_\alpha^3 f^2 + r_\alpha^5 g^2\right)\right], \tag{13b}$$

which is exactly $\exp[-\mu_L(X,X)/2]$ on that plane. The finite tensor product over all physical planes therefore constructs ω_L directly and makes regularity explicit. Multiplications and translations on $L^2(\mathbb{R}^N)$ have only scalars in their joint commutant (use the Fourier transform for the translation part), so this Schrödinger representation is irreducible. Its normalized Gaussian vector state is consequently pure. □

The state is the oscillator ground state, not just a formal Gaussian. In a real spectral basis of R and P_{TT} the reduced Hamiltonian is

$$H_L = \frac{1}{2}\sum_\alpha (p_\alpha^2 + r_\alpha^2 q_\alpha^2), \qquad a_\alpha = \frac{r_\alpha^{1/2} q_\alpha + \mathrm{i} r_\alpha^{-1/2} p_\alpha}{\sqrt{2}}. \tag{14}$$

Its GNS representation is the bosonic Fock representation, with unique vacuum $a_\alpha \Omega_L = 0$. Normal ordering gives $H_L^{\mathrm{N}} = \mathrm{d}\Gamma(R)$; without normal ordering the finite vacuum energy is $\frac{1}{2}\sum_\alpha r_\alpha$. There are exactly $2(V-1)$ real oscillators. Each oscillator requires an infinite-dimensional Hilbert space: no finite $d \times d$ matrices can satisfy $[Q,P] = \mathrm{i}I_d$, because taking the trace would give $0 = \mathrm{i}d$. Thus "finite-volume" never means a finite-dimensional CCR representation.

Here uniqueness means uniqueness of the regular finite-volume oscillator ground state on the zero-mean physical Weyl algebra. It is not a claim that all infinite-volume states or phases are unique. In particular, the omitted $k = 0$ fibre is not present in this algebra, so there is no zero-frequency coherent sector to translate. Restoring that fibre would remove the positive frequency hypothesis used in (14) and would require an additional state choice; the present theorem says nothing about such an extension.

The induced dynamics is

$$E(t) = \cos(Rt)E + R^{-1}\sin(Rt)F, \qquad F(t) = -R\sin(Rt)E + \cos(Rt)F, \tag{15}$$

so (10) is invariant and is a ground state for the positive frequencies.

Reality and Bloch gluing. Write $G_i = \mathrm{diag}((-1)^{(m_A)_i})$. Physical-position Fourier coefficients obey

$$\widehat{q}(k + 2\pi e_i/a) = G_i \widehat{q}(k), \qquad \widehat{q}(-k) = \overline{\widehat{q}(k)}, \tag{16}$$

and similarly for p, E, F. Changing k_i by $2\pi/a$ reverses $\kappa_i^{(a)}$ and gives

$$K_q(k + 2\pi e_i/a) = G_i K_q(k) G_i, \qquad C(k + 2\pi e_i/a) = G_i C(k) G_i. \tag{17}$$

Also $C(-k) = C(k)$. Therefore (7), (10), and (15) descend to the Bloch bundle and respect the real involution. No polarization vectors are chosen, so there is no hidden global-frame obstruction.

For a non-self-conjugate pair $\{k, -k\}$, the two complex TT amplitudes at k determine those at $-k$ and give four real oscillator coordinates. At a self-conjugate boundary momentum, (16) becomes $z = G_n \overline{z}$; the fixed real form of the complex two-dimensional TT space has real dimension two. If N_{sc} is the number of nonzero self-conjugate momenta, then

$$2N_{\mathrm{sc}} + 4\frac{V - 1 - N_{\mathrm{sc}}}{2} = 2(V - 1). \tag{18}$$

For odd L, $N_{\mathrm{sc}} = 0$; for even L, $N_{\mathrm{sc}} = 7$. This establishes the real mode count including Brillouin-zone corners.

Fixed-spacing thermodynamic state. At fixed a, extend the covariance symbols by zero at $k = 0$. The estimates

$$\|C(k)\| = r_a(k)^4, \qquad \|C(k)/(2r_a(k))\| = \tfrac{1}{2} r_a(k)^3, \qquad \|r_a(k)C(k)/2\| = \tfrac{1}{2} r_a(k)^5 \tag{19}$$

show that all symbols are continuous on the compact Brillouin torus. A finite-support smearing has a trigonometric-polynomial Fourier transform. Consequently every entry of (7) and (10) is a Riemann sum of a continuous periodic function. As $L \to \infty$ it converges to the corresponding Brillouin-zone integral.

Theorem 4 (Local thermodynamic state)**.** *For every finite collection of finite-support curvature smearings, the finite-volume Weyl expectations converge as $L \to \infty$ at fixed a. The limits define a translation-invariant quasifree state $\omega_{\infty,a}$ on the inductive local curvature Weyl algebra over $\mathbb{Z}^3$. Its equal-time CCR kernel is the same finite stencil C as in finite volume, while its vacuum covariance generally has infinite spatial range.*

Proof. Riemann-sum convergence and (19) give convergence of σ_L and μ_L. The global summand contributes zero, so deleting it causes no residual $1/V$ term. Pointwise limits of the positive characteristic functionals in (12) remain positive and normalized, and consistency under inclusion of supports gives a state on the inductive algebra. Translation invariance is manifest in the Fourier multiplier. Since C is polynomial in shifts, its CCR kernel is finite range; the fractional powers in (10) need not be. □

The same argument applies to (15) for each fixed time. It does not produce a strict lattice light cone. The analytic tails of the lattice wave propagator found in the classical construction remain present in quantum commutators.

Continuum distributions and microcausality. Take the thermodynamic limit first. For p in $\mathrm{BZ}_a = [-\pi/a, \pi/a]^3$, let

$$\kappa_{a,i}(p) = \frac{2}{a} \sin \frac{ap_i}{2}, \quad \omega_a(p) = |\kappa_a(p)|, \quad C_a(p) = C(\kappa_a(p)). \tag{20}$$

The band-limited interpolation of a lattice kernel is the tempered distribution obtained by integrating its multiplier over BZ_a. Equivalently, lattice fields with $[q_x, p_y] = \mathrm{i}\delta_{xy}$ are scaled as $a^{-3/2}$ distributions, so their smearings are $a^{3/2} \sum_x f(x + am_A/2) E_A(x)$. For Schwartz test functions, Poisson summation shows that direct sampling and the band-limited interpolation have the same limit. Here is the required uniform estimate. With

$$\widehat{f}_{a,A}(p) = a^3 \sum_{x\in a\mathbb{Z}^3} f_A(x + am_A/2)e^{-\mathrm{i}p\cdot(x+am_A/2)},$$

Poisson summation writes $\widehat{f}_{a,A}(p)$ as $\widehat{f}_A(p)$ plus the nonzero aliases $\widehat{f}_A(p + 2\pi n/a)$, each multiplied by a stagger phase of modulus one. For $p \in \mathrm{BZ}_a$ and $n \neq 0$,

$$|p + 2\pi n/a| \geq \pi|n|_\infty/a.$$

Every Schwartz seminorm is finite, so for every $M > 3$,

$$\sup_{p\in\mathrm{BZ}_a} |\widehat{f}_{a,A}(p) - \widehat{f}_A(p)| \leq C_{f,M}a^M \sum_{n\neq 0} |n|_\infty^{-M}. \tag{20a}$$

The largest covariance or commutator multiplier used below is $O(a^{-5})$, and $\mathrm{vol}(\mathrm{BZ}_a) = O(a^{-3})$. The difference between a sampled pairing and its band-limited pairing is therefore $O(a^{M-8})$; choosing any $M > 8$ makes it vanish. For spacetime Schwartz tests, the same estimate holds after spatial Fourier transformation with a coefficient rapidly decreasing in t, so its time integral obeys the same bound. Thus no unproved no-alias assumption enters the continuum limit.

Let $C_0(p) = C(p) = |p|^4 P_{\mathrm{TT}}(p)$ for $p \neq 0$, with its polynomial value $C_0(0) = 0$. Since $|\kappa_a(p)| \leq |p|$ on BZ_a, the covariance multipliers satisfy the a-independent polynomial bounds

$$\|C_a/(2\omega_a)\| \leq \tfrac{1}{2}|p|^3, \qquad \|\omega_a C_a/2\| \leq \tfrac{1}{2}|p|^5. \tag{21}$$

They converge pointwise to $C_0/(2|p|)$ and $|p|C_0/2$. Dominated convergence against the rapidly decreasing Fourier transform of any Schwartz test proves convergence in $\mathcal{S}'(\mathbb{R}^3)$ of the equal-time vacuum covariances. The limiting quasifree functional remains positive by (12).

For the spacetime commutators define the scalar lattice Pauli–Jordan multiplier

$$d_a(t,p) = \frac{\sin(\omega_a(p)t)}{\omega_a(p)}, \qquad d_a(t,0) = t. \tag{22}$$

Equation (15) gives

$$\begin{aligned} [E(t), E(0)] &= -\mathrm{i}C_a d_a, & [E(t), F(0)] &= \mathrm{i}C_a \partial_t d_a, \\ [F(t), E(0)] &= -\mathrm{i}C_a \partial_t d_a, & [F(t), F(0)] &= \mathrm{i}C_a \partial_t^2 d_a. \end{aligned} \tag{23}$$

For spacetime Schwartz tests, $|d_a(t,p)| \leq |t|$, $|\partial_t d_a| \leq 1$, and $|\partial_t^2 d_a| \leq |t||p|^2$. Together with $\|C_a\| \leq |p|^4$, these are integrable polynomial dominators after spatial Fourier transformation. Therefore (23) converges in $\mathcal{S}'(\mathbb{R}^{1+3})$ to the same formulas with

$$d_0(t,p) = \frac{\sin(|p|t)}{|p|}. \tag{24}$$

Theorem 5 (Continuum curvature microcausality)**.** *The limiting free curvature commutators vanish for spacelike-separated test supports. More precisely, all four matrix-valued distributions in (23) have support inside the Minkowski light cone.*

Proof. The inverse spatial Fourier transform of (24) is, up to the Fourier sign convention,

$$D_0(t,x) = \frac{1}{4\pi|x|}\big(\delta(t - |x|) - \delta(t + |x|)\big), \tag{25}$$

so $\operatorname{supp} D_0 \subseteq \{|x| = |t|\}$. The polynomial $C_0(p)$ is a constant-coefficient fourth-order spatial differential operator $C_0(-\mathrm{i}\nabla)$ in position space. The other factors in (23) are time derivatives of D_0. Distributional differentiation does not enlarge support, proving the claim. □

This closes O5 for the two-step limit $L \to \infty$ followed by $a \to 0$. It does not retroactively give strict support to (22) at any $a > 0$, and it does not prove that a TFPT microscopic dynamics has this scaling limit.

Executable certificate, costs, and remaining limits. The repository certificate [`verification/v1031_quantum_tensor_curvature.py`] is standalone and imports no proof text. It is a separate implementation check, not an external independent review. Its roles were fixed before execution:

- exact SymPy polynomial algebra checks (3), (5), (6), and the separate index formula (4);
- exact rational-momentum fibres, characteristic polynomials, TT action, and the factorization (12);
- exact Bloch transformations, boundary real forms, real mode counts, and the finite-matrix CCR trace no-go;
- finite floating-point checks of complete small-volume fibre spectra and two convergence diagnostics. These last checks are regressions only; they do not establish O1–O5.

The universal claims rest on the displayed polynomial identities, positivity factorization, Riemann-sum theorem, dominated-convergence bounds, and support argument. A failed finite regression would reject the implementation; a passed regression cannot replace any of those arguments.

The reported total of 69 checks is evidence-typed as follows: 53 are exact symbolic, rational, integer-counting, or finite-matrix no-go checks; 16 are floating-point regressions. The latter comprise 12 complete small-volume fibre-spectrum/uncertainty/gap checks and four thermodynamic or continuum diagnostics. No floating-point result is used in a theorem proof.

The exact core uses fixed 6×6 matrices of polynomial degree at most eight. Rational checks use four fibres. Boundary checks use the seven even-L self-conjugate corners. The numerical cost is $O(\sum L^3)$ over the listed small boxes plus at most a 96^3 scalar Riemann grid; memory is $O(96^3)$ floating-point numbers. No repository suite is needed for this isolated artifact.

The construction is ultraviolet-distributional. The continuum covariance grows as $|p|^3$ and $|p|^5$, so point fields and coincident products require smearing and, for composite observables, renormalization not supplied here. The un-normal-ordered vacuum energy diverges in thermodynamic/continuum limits. The curvature state is not claimed to have finite correlation range. The Hamiltonian written only in (E, F) contains inverse powers of R even though the CCR and equations are local. None of these limitations alters the finite-volume Fock construction or the stated distributional limits, but each blocks a stronger physical interpretation.

Finally, (E, F) is the spatial curvature/curvature-velocity pair of the linear target. This note does not reconstruct a complete Lorentz-covariant four-dimensional Riemann or Weyl tensor, construct boost generators, or prove that the regulator arises from a covariant microscopic theory.

Round 7: a Lorentz-covariant free curvature field on the Round 6 Hilbert space

The displayed arguments below are the proof-bearing source. The current repository regression is [`verification/v1032_covariant_curvature.py`]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Scope and exact claim boundary

The Round 6 target gave positive quantum electric-curvature variables E, F and a microcausal continuum limit, but did not construct the full four-dimensional Riemann field or its Lorentz boosts. We close that specific free-field gap. A polynomial covariant curvature kernel equals a positive two-polarization kernel on the null cone. It defines a positive Gaussian Wightman field, the vacuum linearized Riemann (equivalently Weyl) tensor, with a strongly continuous positive-energy Poincare representation and helicities $+2, -2$. Its electric part is exactly $E/2$ and its time derivative is $F/2$; it adds no one-particle states. The construction is a free continuum extension of the target, not the missing derivation from the TFPT parent. It proves neither nonlinear gravity, local-net equality with the smaller E, F system, a local lattice realization of every curvature component, nor a complete T7 or TOE. No RH statement is involved.

Conventions and the algebraic bridge. Use $\eta = \mathrm{diag}(-1,1,1,1)$ and a covariant null momentum

$$k_\mu = (-r, p_1, p_2, p_3), \qquad r = |p| > 0, \qquad e^{ik_\mu x^\mu} = e^{-irt + ip\cdot\mathbf{x}}.$$

All tensor components in this note carry lower indices. For a symmetric metric amplitude h, define the linear curvature map

$$(A(k)h)_{\mu\nu\rho\sigma} = \frac{1}{2}\left(k_\sigma k_\nu h_{\mu\rho} + k_\rho k_\mu h_{\nu\sigma} - k_\rho k_\nu h_{\mu\sigma} - k_\sigma k_\mu h_{\nu\rho}\right). \tag{1}$$

It is the Fourier transform of

$$R_{\mu\nu\rho\sigma} = \tfrac{1}{2}(\partial_\rho\partial_\nu h_{\mu\sigma} + \partial_\sigma\partial_\mu h_{\nu\rho} - \partial_\sigma\partial_\nu h_{\mu\rho} - \partial_\rho\partial_\mu h_{\nu\sigma}).$$

No positive covariant quantum metric potential is assumed. Instead introduce the numerical tensor

$$\Pi_{\mu\nu,\alpha\beta} = \tfrac{1}{2}(\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha} - \eta_{\mu\nu}\eta_{\alpha\beta}), \qquad K(k) = A(k)\Pi A(k)^T. \tag{2}$$

In a Euclidean orthonormal basis of symmetric 4×4 matrices, Π has six positive and four negative eigenvalues. It is not a physical Hilbert-space covariance by itself.

Let S embed a spatial symmetric matrix in the ij block of a spacetime matrix, with $h_{0\mu} = 0$. The orthogonal spatial transverse-traceless projector is

$$P_{ij} = \delta_{ij} - p_i p_j / r^2, \tag{3}$$

$$(P_{\mathrm{TT}}h)_{ij} = P_{ia}P_{jb}h_{ab} - \tfrac{1}{2}P_{ij}P_{ab}h_{ab}. \tag{4}$$

It is an orthogonal rank-two projector on $\mathrm{Sym}^2(\mathbb{R}^3)$.

Lemma 13 (Null-cone positive factorization)**.** *For every nonzero null $k = (-r, p)$,*

$$K(k) = A(k)SP_{\mathrm{TT}}(p)S^TA(k)^T = L(k)L(k)^T, \qquad L(k) = A(k)SP_{\mathrm{TT}}(p). \tag{5}$$

Consequently $K(k)$ is positive semidefinite of rank exactly two. It is nevertheless a polynomial of degree four in all four components of k.

Proof. Substitution in (1) gives $A(k)(k\xi^T + \xi k^T) = 0$ for every ξ. The identity (5) can be checked with $p = (0,0,r)$: the two surviving transverse orthonormal matrices are

$$h_+ = \mathrm{diag}(1,-1,0)/\sqrt{2}, \qquad h_\times = (e_1e_2^T + e_2e_1^T)/\sqrt{2},$$

and direct contraction gives $A\Pi A^T = (ASh_+)(ASh_+)^T + (ASh_\times)(ASh_\times)^T$. Spatial rotational covariance and degree-four homogeneity give the result for every nonzero p. The map is injective on the two-dimensional TT space because $(A(k)Sh)_{0i0j} = r^2h_{ij}/2$. This proves rank two. Polynomiality follows directly from (2); no division by r occurs there.

For an independent algebraic certificate, the checker uses ten orthonormal metric components and 21 pair-symmetric bivector components. Write ω as an independent variable and $C(p) = r^4P_{\mathrm{TT}}(p)$, the polynomial Round 6 bracket. It divides each of the 441 entries of

$$A(-\omega,p)(r^4\Pi - SC(p)S^T)A(-\omega,p)^T$$

by $\omega^2 - r^2$ and verifies an identically zero polynomial remainder. The off-shell identity and the identity without the trace-reversal term are separately required to fail. These are algebraic checks, not sampling of momentum directions or a numerical positivity test. □

The same map obeys both pair antisymmetries, pair interchange, the algebraic Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$, and the differential Bianchi identity $k_{[\tau}R_{|\mu\nu|\rho\sigma]} = 0$. On the TT null range the Ricci contraction vanishes, $\eta^{\mu\rho}R_{\mu\nu\rho\sigma} = 0$. For example, insert $h_{0\mu} = 0$, $p_ih_{ij} = 0$, $h_{ii} = 0$, and $k^2 = 0$ into (1); every contraction contains one of those four zero factors. Thus the constructed vacuum curvature is also its linearized Weyl tensor. These identities hold as operator-valued distributional identities after the null quotient below.

Positive field, local commutator, and boosts. Index the independent pair-symmetric bivector entries by I, J. Define the two-point distribution

$$W_{IJ}(x-y) = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3\, 2r}\, e^{-ir(t_x-t_y)+ip\cdot(\mathbf{x}-\mathbf{y})} K_{IJ}(-r,p). \tag{6}$$

The Bianchi identities make one of the 21 algebraic components redundant; retaining it with its null relation avoids an arbitrary choice of twenty coordinates.

Theorem 6 (Free covariant curvature field)**.** *Equation* (6)*, with Wick's rule for higher vacuum correlations, defines a positive free bosonic operator-valued tempered distribution on a symmetric Fock space. Its commutators vanish at spacelike separation. There is a strongly continuous unitary representation of the proper orthochronous Poincare group, with invariant cyclic Fock vacuum, spectrum in the closed forward cone, and covariant rank-four curvature transformation. Its only one-particle helicities are* $+2$ *and* -2*.*

Proof. For Schwartz test tensors f, factorization (5) gives

$$\|f\|_1^2 = \int \frac{d^3p}{(2\pi)^3\, 2r} \widehat{f}(k)^* K(k) \widehat{f}(k) = \int \frac{d^3p}{(2\pi)^3\, 2r} \|L(k)^T \widehat{f}(k)\|^2 \geq 0. \tag{7}$$

Quotient the zero-norm tensors and complete to $\mathcal{H}_1$. Write u_f for the one-particle image of the test tensor. With $a(u)$ antilinear in u, set $R(f) = a^\dagger(u_f) + a(u_{\bar{f}})$ on the finite-particle domain of $\Gamma_s(\mathcal{H}_1)$, using the Fourier convention implicit in (6). This is complex linear in f and obeys $R(f)^* = R(\bar{f})$ on that domain. It gives (6). The usual creation estimate $\|a^\dagger(u)\psi_n\| \leq \sqrt{n+1}\|u\|\|\psi_n\|$ and continuity of (7) make this an operator-valued tempered distribution. Wick positivity follows from this explicit Fock construction, not from assuming Π is positive.

Infrared convergence and temperedness are immediate: $K = O(r^4)$, so $K/(2r) = O(r^3)$ at zero and has only polynomial growth at infinity. Together with the d^3p measure the radial integrand is $O(r^5)$ near zero. Schwartz test functions control every ultraviolet moment.

Let

$$D_0(x) = \int \frac{d^3p}{(2\pi)^3\, 2r} (e^{-irt+ip\cdot\mathbf{x}} - e^{irt-ip\cdot\mathbf{x}})$$

be the scalar massless commutator distribution. Since $K(-k) = K(k)$, the curvature commutator is the constant-coefficient differential operator $K_{IJ}(-i\partial)D_0(x-y)$. The scalar commutator has light-cone support, and differentiation does not enlarge support. This proves locality of the curvature fields, despite the nonlocal spatial TT projector used to exhibit positivity.

All indices in (1)–(2) are tensorial and Π is built only from η. Therefore

$$K(\Lambda k) = D^{(4)}(\Lambda) K(k) D^{(4)}(\Lambda)^T \tag{8}$$

for the covector action of every proper orthochronous Lorentz matrix. The positive-null-cone measure is invariant. The induced contragredient test-tensor transformation, together with translations, consequently preserves (7) and its null space. It gives a unitary one-particle representation. The test-tensor action is continuous in the Schwartz topology, so the norm estimate gives strong continuity on a dense set and then on the completion. Second quantization gives the claimed Fock representation. Translations act by positive-energy null momenta; finite sums and closure keep the many-particle spectrum in the forward cone. The null cone tip has zero one-particle measure. Thus every nonvacuum n-particle wave function has strictly positive total energy almost everywhere, and the only zero-energy vector in this constructed Fock representation is the vacuum. This is not a theorem of uniqueness across all representations or all possible TFPT states.

For helicity, at $k = (-1, 0, 0, 1)$ rotations about the third axis act on $(h_+, h_\times)$ by

$$\begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}.$$

Hence $h_+ \pm ih_\times$ have phases $e^{\mp 2i\theta}$. The translation part of the null little group is trivial on the curvature: put $q = (u^2 + v^2)/2$ and

$$\Lambda(u,v) = \begin{pmatrix} 1+q & -u & -v & q \\ -u & 1 & 0 & -u \\ -v & 0 & 1 & -v \\ -q & u & v & 1-q \end{pmatrix}.$$

It preserves η and k, and sends the transverse covectors $e_1 \mapsto e_1 + uk$, $e_2 \mapsto e_2 + vk$. Each polarization change is therefore $k\xi^T + \xi k^T$, which $A(k)$ annihilates. This proves the two helicities without assuming away the noncompact little-group translations. The checker verifies these identities for symbolic u, v and all rotation angles through a rational parametrization. □

It is the same free target, with a larger covariant field family. The Round 6 continuum construction has, on the physical TT range,

$$E = r^2 q_{\mathrm{TT}}, \qquad F = r^2 p_{\mathrm{TT}}, \qquad \dot{q}_{\mathrm{TT}} = p_{\mathrm{TT}}, \qquad \dot{p}_{\mathrm{TT}} = -r^2 q_{\mathrm{TT}},$$

with $C = r^4 P_{\mathrm{TT}}$. In radiation gauge (1) gives exactly

$$R_{0i0j} = \frac{r^2}{2} q_{ij} = \frac{1}{2} E_{ij}, \qquad \partial_t R_{0i0j} = \frac{1}{2} F_{ij}. \tag{9}$$

Consequently the spectral electric-electric covariance is $C/(8r)$, electric-time-derivative covariance is $iC/8$ at equal time in the order $\langle R\dot{R}\rangle$, and derivative-derivative covariance is $rC/8$. These are precisely one quarter of the Round 6 EE, EF, FF covariances, respectively, including the nonsymmetrized commutator part.

For completeness this matching is more than equality of a few entries. Both one-particle spaces identify with the completion of spatial TT amplitudes in $L^2(d^3p/((2\pi)^3 2r))$. The map from curvature tests is $f \mapsto P_{\mathrm{TT}} S^T A^T \hat{f}$. Electric tests alone have dense image: on every annulus $0 < \epsilon < r < R$ their map is multiplication by the invertible scalar $r^2/2$ followed by P_{TT}. Every smooth compactly supported TT amplitude on such an annulus can be extended to a spacetime Schwartz test in a tubular neighbourhood of the positive null cone. Cutting off at zero and infinity is dense in the one-particle norm. Thus (9) identifies the completed one-particle spaces, and the associated Fock spaces, with no extra physical oscillator or spectator.

There is an important locality qualification. At a fixed time,

$$R_{0ijk} = \tfrac{1}{2}(\partial_k p_{ij} - \partial_j p_{ik}) = \frac{1}{2r^2}(\partial_k F_{ij} - \partial_j F_{ik})$$

in Fourier notation; the spatial components similarly contain two spatial derivatives of $q = E/r^2$. Inverse powers of r^2 cancel in the covariant commutator kernel, but are still present in this reconstruction from E, F. The global Hilbert-space identification does not by itself prove equality of local observable nets, nor that every R component has a bounded-range lattice approximant in the previous E, F algebra. No such claim is used in the positivity, covariance, or locality theorem above.

Validation, provenance, and the remaining obligation. The repository certificate [`verification/v1032_covariant_curvature.py`] is standalone SymPy code. Its 34 exact checks include the all-momentum polynomial identity, gauge and Bianchi identities, null Ricci contraction, positivity factorizations, a nontrivial rational Lorentz boost, the full null little group, and the exact electric and mixed time-derivative covariance normalization. Four rational fibres supplement, but do not replace, the universal proof. No floating-point eigenvalue scan is used. The distributional/Fock completion rests on the written norm, continuity, density and support arguments, not on a finite matrix test pretending to construct an infinite-dimensional field.

The inputs are the declared Round 6 free target and ordinary linearized curvature algebra. No microscopic TFPT Hamiltonian is imported or derived by this checker; only the repository's reporting harness is shared. In particular this note does not manufacture the required shared parent: it does not prove that the microscopic seam, charged matter, mirror sector and gravity jointly flow to this field or to its nonlinear completion. A finite-spacing Lorentz-breaking regulator is not asserted to be exactly boost invariant. The vacuum homogeneous-block choice and the interacting constraint/renormalization problems remain separate.

For historical context, massless-spin field representations and the role of the non-semisimple little group are discussed in Weinberg's primary paper[1]. The result here is an explicit match of that free-field structure to the existing target, not a claim that free helicity-two fields are a new physical theory. T1–T8 retain their existing open status.

Round 7: a microscopic flux-string disorder operator for the frozen QWZ seam

The displayed arguments below are the proof-bearing source. The current repository regression is [`verification/v1033_charged_disorder.py`]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Scope and exact claim boundary

There is a concrete microscopic boundary-condition-changing operator for the $M = 1$, $N_y = 8$ QWZ seam once one local four-level flux register is adjoined. Couple its clock Z directly to the seam hopping and let its shift X raise the flux. Dressing X by a quarter charge rotation on a fermion arc gives a disorder string D_A. It has nonzero $r \to r + 1$ corners, order four, commutes with canonical fermion parity, and its commutator with the actual QWZ Hamiltonian is supported exactly on the opposite endpoint bond. Thus this is not the earlier arbitrary block shift.

It is nevertheless not the TFPT λ field. The source's common quarter twist on eight complex edge channels has minimal charge $q = (1/4)^8$, hence $h_q = 1/4$, while the sourced $\lambda = (\omega_s, \omega_f)$ has $\|\lambda\|^2 = 2$ and $h_\lambda = 1$. Forcing weight one by an integer dressing inside the common scalar phase class requires odd fermion charge. The actual A_3 fundamental makes the missing datum explicit: it is a four-channel quarter disorder plus one endpoint fermion, combined with five half-twisted D_5 channels and a nontrivial cocycle/GSO identification. Those species-resolved couplings and that GSO map are not present in the audited QWZ path. A sharp endpoint insertion also has measured $O(1)$ lattice excitation energy; a smooth whole-circle dressing lowers this to $O(N^{-1})$ but loses sharp support. The result is therefore a positive microscopic deck-disorder construction and a narrow obstruction to identifying it with Ψ_λ, not a no-go for a future species-resolved extension. No RH claim is made.

Frozen source and target. The construction below uses the hopping matrix literally implemented in `verification/v988_psi_lambda_reduction.py`. With

$$T_x = \frac{\sigma_x}{2i} - \frac{\sigma_z}{2}, \qquad T_y = \frac{\sigma_y}{2i} - \frac{\sigma_z}{2},$$

the one-particle QWZ Hamiltonian h_r has mass $M = 1$, open y boundary, and on the bond $(N - 1, y) \to (0, y)$ the forward hopping is $i^r T_x$. Every other longitudinal bond has T_x. The executable v988 campaign constant is $N_y = 6$, but its builder accepts N_y as an argument. This artifact calls that builder explicitly at the charged-limit manuscript value $N_y = 8$. The manuscript writes

$$p_n^{(r)} = \frac{2\pi}{N}\left(n + \frac{r}{4}\right). \tag{1.1}$$

That is its opposite-sign Bloch convention. With the checker convention $\psi_x = e^{ikx}$ and the executable forward seam amplitude i^r, the boundary eigencondition is

$$e^{i(r\pi/2+kN)} = 1, \qquad k_n^{(r)} = \frac{2\pi n - r\pi/2}{N}. \tag{1.2}$$

[1] S. Weinberg, "Feynman Rules for Any Spin. II. Massless Particles", Phys. Rev. 134, B882 (1964), `https://doi.org/10.1103/PhysRev.134.B882`. The accessible abstract was used for context; no inaccessible theorem is imported as a proof.

The checker reconstructs the complete position-space matrix from these Bloch blocks and compares it entrywise with the imported v988 builder. Using the plus sign in this checker convention fails that matrix test. It also states that sixteen Majorana copies give eight complex edge bosons.

The target is not specified merely by the deck phase. The repository fixes

$$\omega_s = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right), \qquad \omega_f = \left(\tfrac{3}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}\right), \qquad \lambda = (\omega_s, \omega_f), \tag{1.3}$$

so

$$\|\omega_s\|^2 = \frac{5}{4}, \quad \|\omega_f\|^2 = \frac{3}{4}, \quad \|\lambda\|^2 = 2, \quad h_\lambda = 1. \tag{1.4}$$

These exact claims occur in two frozen sources:

- `verification/v983_simple_current_generator.py`;
- `articles/2026-08-30/mmst_charged_scaling_limit_en.tex`.

The latter still lists a support-preserving microscopic realization and an independent FE-GEN or gauge/code-projected algebra theorem as open. We do not silently replace (1.3) by an arbitrary representative of a scalar $\mathbb{Z}_4$ phase.

A common microscopic Hilbert tied to the hopping. Let $\mathfrak{h}_N = \mathbb{C}^{2NN_y}$ and let $\mathcal{F}_N = \mathcal{F}(\mathfrak{h}_N)$ be the canonical number-conserving CAR Fock space. Adjoin one four-level link register $\mathcal{K} = \mathbb{C}^4$ at the chosen periodic seam. In the basis $\{|r\rangle : r \in \mathbb{Z}_4\}$ define

$$Z|r\rangle = i^r|r\rangle, \qquad X|r\rangle = |r+1\rangle, \qquad ZXZ^* = iX, \qquad X^4 = Z^4 = \mathbf{1}. \tag{2.1}$$

If $F = \sum_y |0, y\rangle T_x \langle N-1, y|$ is the forward seam hopping and $h_{\text{cut}} = h_0 - F - F^*$, set

$$\widetilde{h} = \mathbf{1}_{\mathcal{K}} \otimes h_{\text{cut}} + Z \otimes F + Z^* \otimes F^* = \bigoplus_{r=0}^{3} h_r. \tag{2.2}$$

Thus the register is not a decorative sector label: its clock occurs in the actual QWZ hopping term. On $\mathcal{K} \otimes \mathcal{F}_N$ take

$$\widetilde{H} = \sum_{r=0}^{3} |r\rangle\langle r| \otimes d\Gamma(h_r). \tag{2.3}$$

The four sectors now live in one canonical Hilbert space. This is an *auxiliary $\mathbb{Z}_4$ flux link*; no claim is made that the present TFPT code derives it.

The sharp line-ended operator. Choose an arc $A = \{0, 1, \ldots, \ell\}$ with $0 \le \ell < N-1$. Let u_A be the one-particle onsite gauge

$$u_A|x, y, a\rangle = \begin{cases} i|x, y, a\rangle, & x \in A, \\ |x, y, a\rangle, & x \notin A, \end{cases} \qquad \Gamma(u_A) = \exp\left(\frac{\pi i}{2} N_A\right). \tag{2.4}$$

Define

$$D_A = X \otimes \Gamma(u_A). \tag{2.5}$$

Theorem 7 (Microscopic QWZ disorder transporter)**.** *For every $r \in \mathbb{Z}_4$,*

$$P_{r+1} D_A P_r = |r+1\rangle\langle r| \otimes \Gamma(u_A) \neq 0, \qquad D_A^4 = \mathbf{1}. \tag{2.6}$$

Writing $\Pi = (-1)^{N_{\text{F}}}$ for canonical total fermion parity,

$$[D_A, \mathbf{1} \otimes \Pi] = 0, \qquad (Z \otimes \mathbf{1}) D_A (Z^* \otimes \mathbf{1}) = i D_A. \tag{2.7}$$

Moreover

$$[\widetilde{H}, D_A] P_r = |r+1\rangle\langle r| \otimes d\Gamma(h_{r+1} - u_A h_r u_A^*)\, \Gamma(u_A), \tag{2.8}$$

and $h_{r+1} - u_A h_r u_A^*$ is supported only on the single longitudinal bond $(\ell, y) \leftrightarrow (\ell+1, y)$, for all $y = 0, \ldots, 7$. Its one-particle rank is $2N_y = 16$ and its norm is $\sqrt{2}$.

Proof. The corner and charge equations follow from (2.1). Since $u_A^4 = \mathbf{1}$ and X acts on a different tensor factor, $D_A^4 = \mathbf{1}$. A number rotation is an even CAR automorphism, proving the parity commutator.

The $(r+1, r)$ one-particle commutator block is $h_{r+1}u_A - u_A h_r = (h_{r+1} - u_A h_r u_A^*)u_A$; second quantization gives (2.8). Conjugation by u_A multiplies a forward hopping $x \to x+1$ by $u_{x+1}u_x^*$. At the original seam this factor is i, changing i^r to i^{r+1}. All bonds internal to A or its complement are unchanged. Only the other boundary of A has factor $-i$. Hence the original seam defect is transported to the far endpoint. Substitution of T_x gives rank 16 and norm $\sqrt{2}$; the companion probe checks the full matrices for all four sectors at $N_y = 8$. □

This is the promised positive result. A matter-only onsite gauge could not have done it: the product of link phases telescopes,

$$\prod_x a_x \frac{g_{x+1}}{g_x} = \prod_x a_x, \tag{2.9}$$

so it preserves the Wilson holonomy. The new register is exactly the extra degree of freedom that changes it. Unlike the Round-6 arbitrary J_r, the operator (2.5) is tied to (2.2), has a specified CAR action, and obeys the endpoint commutator identity (2.8).

Equation (2.7) proves compatibility with canonical fermion parity and with any GSO projector that is a function of that parity. It does *not* prove compatibility with the repository's character-level $O + S$ selector: the audited source supplies no microscopic projector identifying that selector with Π. This boundary is essential.

Why the constructed operator is not Ψ_λ. For a normalized complex chiral fermion, a twist by $e^{2\pi i q}$ has minimal bosonized weight $q^2/2$, with $q \in (-1/2, 1/2]$. The actual common phase i therefore fixes, for the eight complex channels in the source,

$$q_{\text{deck}} = \left(\tfrac{1}{4}, \ldots, \tfrac{1}{4}\right), \qquad \|q_{\text{deck}}\|^2 = \frac{1}{2}, \qquad h_{\text{deck}} = \frac{1}{4}. \tag{3.1}$$

It has order four but nontrivial spin $e^{2\pi i h} = i$. The target (1.3) instead has integer spin and weight one. Norm is invariant under an orthonormal change of current basis, so this is not repaired by relabeling the eight bosons.

The phase by itself does not determine an excited representative. Indeed

$$q' = \left(\tfrac{5}{4}, \tfrac{1}{4}, \ldots, \tfrac{1}{4}\right) \tag{3.2}$$

has the same scalar phase and $\|q'\|^2 = 2$. But $q' - q_{\text{deck}}$ is one unit of fermion charge. More generally, if $q_{\text{deck}} + n$ has norm two, then

$$2\left(\|q_{\text{deck}} + n\|^2 - \|q_{\text{deck}}\|^2\right) = 2\sum_j n_j^2 + \sum_j n_j = 3. \tag{3.3}$$

Modulo two, $\sum_j n_j$ is odd. Every such weight-one dressing therefore changes canonical fermion parity. Selecting (3.2) solely because it has the desired number would be precisely an arbitrary charge-basis shift, not a microscopic derivation of λ.

The $D_5 \oplus A_3$ coordinates show the missing operator content more constructively. The D_5 spinor part is a half disorder on five complex channels and contributes $5/8$. For the A_3 part,

$$\left(\tfrac{3}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}\right) = \left(-\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}\right) + (1, 0, 0, 0). \tag{3.4}$$

Thus it is a four-channel quarter disorder of weight $1/8$ plus one endpoint fermion, giving the traceless A_3 fundamental weight $3/8$. Combined with the five half twists, this gives $h_\lambda = 5/8 + 3/8 = 1$.

Equation (3.4) is the next microscopic specification. Indeed, $e^{2\pi i \omega_s} = (-1)^5$ componentwise and $e^{2\pi i \omega_f} = (-i)^4$ componentwise. A λ operator therefore needs species-resolved seam clocks Z^2 on the five D_5 channels and Z^{-1} on the four A_3 fermions, the endpoint fermion, removal of the diagonal $U(1)$ in the A_3 realization, and the cocycle/GSO rule that makes the total field local and bosonic. The frozen QWZ code has one common scalar i^r phase and none of these four identification maps. The flux register solves the common-Hilbert and locality problem, but not this internal-charge problem. This redundant fermionic realization has nine complex fermions before the A_3 diagonal-$U(1)$ quotient; it is therefore not automatically an operator on the source's eight-channel Fock space. An explicit conformal embedding/isometry is part of the missing microscopic datum.

Energy and support diagnostics. There is a finite-lattice tradeoff that prevents another overclaim. *Portability amendment, 2026-09-06.* The $r = 0$ one-particle operator has an exactly degenerate zero-energy space. Filling the first half of a numerical eigenbasis silently selects a platform-dependent pure vacuum. Let P_- and P_0 denote its negative and zero spectral projectors, let $d = \operatorname{rank} P_0$, and let $m = NN_y - \operatorname{rank} P_-$. We instead declare the basis-independent one-body covariance

$$\overline{C} = P_- + \frac{m}{d} P_0 \quad (d > 0), \qquad \overline{C} = P_- \quad (d = 0).$$

It is the one-body covariance of the uniform fixed-particle mixture over rank-m zero-space fillings, not a uniquely selected pure Slater vacuum. The numerical zero cluster is required to be separated from all nonzero levels; this finite floating-point diagnostic is not an interval certificate. Applying the sharp D_A to that mixture and measuring above the $r = 1$ ground energy gives, for $N = 16, 24, 32, 48, 64$,

$$4.711531, \quad 4.689142, \quad 4.677637, \quad 4.665918, \quad 4.659977. \tag{4.1}$$

The measured cost is consistent with $O(1)$ at fixed $N_y = 8$ and the sharp endpoint commutator. This bare operator is not a low-energy weight-one state.

One can distribute the string by taking

$$u_N(x) = \exp\left[\frac{\pi i}{2}\left(1 - \frac{x}{N}\right)\right], \qquad D_N^{\mathrm{sm}} = X \otimes \Gamma(u_N). \tag{4.2}$$

Then the actual one-particle matrices give

$$N \,\|h_1 u_N - u_N h_0\| \longrightarrow \frac{\pi}{2}, \qquad N\left(\operatorname{tr}\left[h_1 u_N \overline{C} u_N^*\right] - E_0(H_1)\right) \approx 6.904. \tag{4.3}$$

This does not discard other zero-space vacua. For either dressing u, set $A = P_0 u^* h_1 u P_0|_{\operatorname{ran} P_0}$, with eigenvalues $\alpha_1 \le \cdots \le \alpha_d$. Over every rank-$m$ zero-space filling, the exact variational extrema of the one-body excess are

$$\operatorname{tr}(P_- u^* h_1 u) - E_0(H_1) + \sum_{j=1}^{m} \alpha_j, \qquad \operatorname{tr}(P_- u^* h_1 u) - E_0(H_1) + \sum_{j=d-m+1}^{d} \alpha_j.$$

Indeed, in an eigenbasis of A, the diagonal entries of a rank-m orthogonal projector lie in $[0, 1]$ and sum to m; the minimum and maximum select the m smallest and largest eigenvalues. Both bounds are attained. At $N = 64$ the sharp interval is approximately $[4.644352, 4.675602]$, and the smooth N-scaled interval is $[5.333652, 8.474929]$. The previous 8.475 coefficient corresponds to one endpoint of this filling-dependent range; it was not a unique vacuum value. On all five sampled sizes, the sharp intervals lie in $(4, 5)$ and the smooth N-scaled intervals in $(5.2, 8.6)$. These are finite-grid diagnostics, not all-N asymptotic bounds. This is measured $O(N^{-1})$ energy, but u_N occupies the whole circumference and the coefficient contains full-band polarization. It is not a proof of conformal weight. Conversely, the sharp string has exact endpoint support but the $O(1)$ UV cost (4.1). Recovering both sharp local algebras and the weight-one edge primary requires the missing edge projection, renormalized endpoint field, species-resolved charge, and GSO cocycle.

As an independent finite diagnostic, momentum-block diagonalization gives

$$\frac{N}{2\pi}\left(E_0(h_1) - E_0(h_0)\right) \longrightarrow -\frac{3}{16} \tag{4.4}$$

for one two-edge QWZ copy. This is a sector-vacuum/Casimir shift, not the one-edge primary weight and not h_λ. No fit or finite sample in this section is promoted to an all-N theorem.

Executable certificate and claim boundary. The repository certificate [verification/v1033_charged_disorder.py] audits the frozen sources, proves the finite Weyl/parity/charge identities with rational and symbolic arithmetic, imports the executable v988 QWZ builder and compares it entrywise, verifies the endpoint commutator on the actual $M = 1, N_y = 8$ hops, and reports finite energy diagnostics.

Established: after adjoining one explicitly auxiliary local $\mathbb{Z}_4$ flux link, $D_A = X\Gamma(u_A)$ is a CAR-even, order-four, grade-one microscopic QWZ boundary-condition changer with an exact line-endpoint commutator.

Not established: that the present TFPT/GSO construction contains this register; that its $O + S$ selector is canonical parity; or that the common quarter disorder equals the $D_5 \oplus A_3$ field Ψ_λ. The latter identification is false for the minimal common twist because $h_{\text{deck}} = 1/4 \neq 1 = h_\lambda$.

Round 7: the actual DET parent, its exact factorized gap, and the local-defect obstruction

The displayed arguments below are the proof-bearing source. The current repository regression is [verification/v1034_parent_mirror.py]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Scope and exact claim boundary

The source audit separates three operators which had previously been discussed under one "mirror parent" label. The v1002 DET16 onsite term is the flat defect projector $Q = \mathbf{1} - P_\phi$. The v1008 wall scaffold instead tests the four-flavour momentum-cell Hamiltonian

$$h(d) = d(N - 2) + 2Q, \qquad |d| \leq 3/5.$$

The later v1027 identity constructs an additive dressed-number parent N_a; it is not the same operator as Q. For the actual v1008 momentum-factorized Hamiltonian we prove, without a grid scan, the volume-independent many-body and canonical-field threshold

$$\Delta_{\text{fact}} \geq 1 + \frac{\sqrt{3}}{2} = 1.8660254037\ldots.$$

This is the exact infimum of the continuous dispersion envelope; the source's 129-point grid instead attains 1.8660698881694362. It is not a real-space locality theorem. Indeed, for two real-space DET cells, a newly chosen bare flavour-diagonal bond analogue T_{bare} satisfies (for even $n \geq 4$)

$$\|T_{\text{bare}}|\Omega\Omega\rangle\|^2 = \frac{n}{2}|s|^2 > 0, \qquad (Q_x + Q_y)|\Omega\Omega\rangle = 0.$$

Consequently no finite κ can obey $T_{\text{bare}} \geq -\kappa(Q_x + Q_y)$. An explicitly gauge-dressed operator, with the link shift applied as part of every hop, gives the same obstruction inside finite $\mathbb{Z}_2/\mathbb{Z}_4$ Gauss sectors. Thus the desired elementary relative-Q proof is impossible for the unmodified local parent; a genuine small-perturbation theorem or a changed local bond is needed. An explicit coupling-fixed bond repair is given, but is clearly marked as a new, unlanded interaction. No T3, T4, TOE, or RH claim is made.

Source contract and nomenclature. The certificate reads, but does not import, the following landed files at the audited repository tip:

source	exact object used here	status relevant to this note
v1002	$Q = \mathbf{1} - P_\phi$ on 16 complex modes	onsite projector only
v1008	$h(d) = d(N - n/2) + JQ$, $n = 4, J = 2$	momentum-cell scaffold
v1024	positive seam/number-square parent	separate construction
v1027	$N_a = \sum_i a_i^\dagger a_i$, $Q \leq N_a \leq nQ$	algebraic identity

In particular, "DET32" is sometimes used physically for 32 real components, whereas the repository's operator is named DET16 because it has 16 complex fermion modes. We use n for the number of complex modes and retain the repository's DET16 name.

Let

$$|\Omega_\phi\rangle = \frac{|0\rangle + e^{i\phi}|F\rangle}{\sqrt{2}}, \qquad P_\phi = |\Omega_\phi\rangle\langle\Omega_\phi|, \qquad Q = \mathbf{1} - P_\phi. \tag{10}$$

Here $|F\rangle = c_1^\dagger \cdots c_n^\dagger|0\rangle$ and n is even. Both $|0\rangle$ and $|F\rangle$ are assumed to be gauge singlets; this is the inherited full-DET-singlet premise, not a conclusion of this note. The common rotation $e^{-i\phi N/n}$ removes ϕ from (10) and commutes with N and ordinary flavour-diagonal hopping, so every spectrum below is phase independent.

The elementary algebra already confirms the proposed diagnosis:

$$[Q, (-1)^N] = 0, \qquad [Q, N] \neq 0, \qquad \|[Q, N]\|_{\mathrm{F}}^2 = \frac{n^2}{2}. \tag{11}$$

Thus the flat projector preserves parity, not naive mirror number. For the v1008 analogue $n = 4$, the last value is 8; for DET16 it is 128.

Exact theorem for the parent that v1008 actually computes. The v1008 routine `edge_cluster_hamiltonian` is

$$h(d) = d\left(N - \frac{n}{2}\right) + JQ, \qquad n = 4, \qquad J = 2, \qquad |d| \leq t_{\mathrm{src}} = \frac{3}{5}. \tag{12}$$

The scan uses $d = \pm t_{\mathrm{src}} \sin k$. This is a separate Fock cell at every sampled momentum, not a real-space hopping Hamiltonian. In particular, a standard symmetric nearest-neighbour term $s\sum_x (c_x^\dagger c_{x+1} + \mathrm{h.c.})$ has dispersion magnitude $2|s|$ (up to a momentum/phase convention). Matching the v1008 dispersion amplitude would therefore give $|s| = t_{\mathrm{src}}/2 = 3/10$, not 3/5.

Lemma 1 (complete one-cell spectrum). For $1 \leq r \leq n-1$, every occupation-r state has energy

$$E_r(d) = J + d\left(r - \frac{n}{2}\right). \tag{13}$$

On $\mathrm{span}\{|0\rangle, |F\rangle\}$, after removing ϕ, the matrix is

$$\begin{pmatrix} -nd/2 + J/2 & -J/2 \\ -J/2 & nd/2 + J/2 \end{pmatrix}, \tag{14}$$

with eigenvalues

$$E_\pm(d) = \frac{J}{2} \pm \sqrt{\frac{J^2}{4} + \frac{n^2 d^2}{4}}. \tag{15}$$

Proof. P_ϕ vanishes on occupation sectors $1, \ldots, n-1$, which gives (13). Its restriction to the empty/full doublet is the rank-one matrix specified by (10); the displayed characteristic polynomial gives (15). □

For the source values, $E_-(d) \leq 0$, while the least intermediate energy is $2 - 3/5 = 7/5 > 0$. The ground state is therefore unique throughout the whole declared dispersion interval. A canonical c_i or $c_i^\dagger$ maps this ground doublet only to the $r = 1$ or $r = n-1$ sectors, with nonzero amplitude. The exact canonical spectral threshold at $x = |d|$ is consequently

$$g(x) = \frac{J}{2} - \left(\frac{n}{2} - 1\right)x + \sqrt{\frac{J^2}{4} + \frac{n^2 x^2}{4}}. \tag{16}$$

Theorem 1 (source-fixed all-volume factorized gap). Let K be any finite set of momentum cells, with arbitrary dispersions $|d_k| \le 3/5$, and define the graded tensor-product Hamiltonian $H_K = \sum_{k\in K} h(d_k)$. Then H_K has a unique product ground state and

$$\operatorname{gap}(H_K) \ge \inf_{0\le x\le 3/5} g(x) = 1 + \frac{\sqrt{3}}{2}. \tag{17}$$

The same number is the uniform lower edge of canonical one-fermion spectral support.

Proof. For $n = 4, J = 2$, differentiation of (16) gives the unique interior minimum

$$x_* = \frac{J(n-2)}{2n\sqrt{n-1}} = \frac{1}{2\sqrt{3}} < \frac{3}{5}, \tag{18}$$

and substitution gives $g(x_*) = 1 + \sqrt{3}/2$. The other empty/full eigenstate costs at least $J = 2$, and the $r = 2$ sector also costs at least 2, so neither undercuts (17). The even cell Hamiltonians commute on the graded tensor product and energies add. Any state orthogonal to the product ground excites at least one cell, proving the same bound independently of $|K|$. ☐

The analytic infimum 1.8660254037844386... explains the lower envelope of v1008's reported probe minimum; it is not the exact minimum of the finite scan. The source grid $k_j = -\pi + j\pi/64$, $j = 0, \ldots, 128$, attains 1.8660698881694362. The locality limit is equally exact: a projector local in momentum is not a bounded-range real-space interaction. Equation (17) therefore does not prove the missing local, operator-valued TFPT parent theorem.

Why the desired relative-defect bound fails in real space. Put two DET cells on a bond and let

$$D = J(Q_x + Q_y), \qquad T_{\text{bare}} = s\sum_{i=1}^{n}(c_{xi}^\dagger c_{yi} + c_{yi}^\dagger c_{xi}). \tag{19}$$

This is a bare-bond theorem. The numerical choice $s = 3/5$ below is a newly chosen stress analogue, not a real-space coefficient derived from v1008. For even $n \ge 4$, the full-Fock CAR calculation contains only $2n$ nonzero output monomials and does not require construction of the 2^{2n} matrix.

Lemma 2 (minimal kernel incompatibility). For every even $n \ge 4$ and $s \ne 0$,

$$D|\Omega\Omega\rangle = 0, \qquad \langle\Omega\Omega|T_{\text{bare}}|\Omega\Omega\rangle = 0, \tag{20}$$

$$\|T_{\text{bare}}|\Omega\Omega\rangle\|^2 = \frac{n}{2}|s|^2 > 0, \qquad (Q_x + Q_y)T_{\text{bare}}|\Omega\Omega\rangle = 2T_{\text{bare}}|\Omega\Omega\rangle. \tag{21}$$

Proof. Only the $|F, 0\rangle$ and $|0, F\rangle$ components of the product DET vacuum can hop. Each has amplitude $1/2$. The two hopping orientations and n flavours produce $2n$ mutually orthogonal states, each of modulus $|s|/2$, which proves the norm in (21). Both sites have occupation 1 or $n-1$ after the hop, hence both local projectors P_x, P_y annihilate the image. ☐

The restriction $n \ge 4$ is necessary for that counting statement. At $n = 2$ the two orientations land on the same two basis states and add constructively. Direct CAR evaluation gives $\|T_{\text{bare}}|\Omega\Omega\rangle\|^2 = 2|s|^2$, rather than the former formula's $|s|^2$. The image is still nonzero, so the no-go below survives, but the claimed universal norm formula did not.

Theorem 2 (no finite relative-defect constant). There is no finite κ such that

$$T_{\text{bare}} \ge -\kappa(Q_x + Q_y). \tag{22}$$

Proof. If (22) held, the positive operator $B = T_{\rm bare} + \kappa(Q_x + Q_y)$ would have zero expectation in $|\Omega\Omega\rangle$ by (20). Positivity then implies $B|\Omega\Omega\rangle = 0$. Since $(Q_x + Q_y)|\Omega\Omega\rangle = 0$, this would say $T_{\rm bare}|\Omega\Omega\rangle = 0$, contradicting (21). □

For the newly chosen bare-bond analogue $n = 4, J = 2, s = 3/5$,

$$\|T_{\rm bare}|\Omega\Omega\rangle\|^2 = \frac{18}{25}. \tag{23}$$

Let $b = \sqrt{18/25}$ and normalize $|w\rangle = T_{\rm bare}|\Omega\Omega\rangle/b$. Direct CAR evaluation gives $\langle w|T_{\rm bare}|w\rangle = 0$ and $\langle w|D|w\rangle = 2J$. The compression of $D + T_{\rm bare}$ to $\mathrm{span}\{|\Omega\Omega\rangle, |w\rangle\}$ is

$$\begin{pmatrix} 0 & b \\ b & 2J \end{pmatrix}. \tag{24}$$

Thus the variational principle gives the exact strict bound

$$E_0(D + T_{\rm bare}) \le 2 - \frac{\sqrt{118}}{5} = -0.172556\ldots < 0. \tag{25}$$

The old product vacuum is not frustration free once ordinary hopping is on. This does not prove gap closure; it proves that the proposed elementary relative-Q route cannot establish a gap.

Gauge-dressed theorem and actual link action. The bare theorem above makes no Gauss-sector claim. For the $n = 4$ analogue and $q = 2, 4$, now adjoin an oriented link basis $\{|e\rangle : e \in \mathbb{Z}_q\}$ and define

$$U|e\rangle = |e + 1 \bmod q\rangle, \qquad T_U = s\sum_{i=1}^{n}\left(c^\dagger_{xi} U c_{yi} + c^\dagger_{yi} U^\dagger c_{xi}\right). \tag{26}$$

This operator is implementable on matter Fock space tensored with the finite link space; it is fermion-even because each summand is a matter bilinear. Take the endpoint Gauss constraints to be $n_x - e = 0$ and $n_y + e = 0 \pmod q$. The starting vector $|\Omega\Omega\rangle|0\rangle_E$ is physical because its site occupations are 0 or 4. Applying (26), including the link shift rather than assigning a flux afterwards, produces only

$$(n_x, n_y, e) = (3, 1, 3 \bmod q) \quad \text{or} \quad (1, 3, 1 \bmod q),$$

and in both cases $n_x - e = 0$ and $n_y + e = 0 \pmod q$. Hence the nonzero hopping image is a physical flux excitation. The checker applies the CAR and link actions term by term for both $q = 2$ and $q = 4$ and verifies every nonzero output, so Gauss preservation is not a post-hoc flux assignment.

Let $E^2|\pm 1\rangle = |\pm 1\rangle$ and

$$H_0 = J(Q_x + Q_y) + h_E E^2, \qquad h_E \ge 0. \tag{27}$$

Both terms annihilate the physical starting vector, whereas T_U does not. The same positivity-kernel argument therefore rules out any finite relative bound of T_U by this complete positive local parent. Moreover, with $|w\rangle = T_U|\Omega\Omega, 0_E\rangle/b$, the exact Ritz compression is

$$\begin{pmatrix} 0 & b \\ b & 2J + h_E \end{pmatrix}, \qquad E_0(H_0 + T_U) \le \frac{2J + h_E}{2} - \sqrt{\frac{(2J + h_E)^2}{4} + b^2}. \tag{28}$$

Thus the electric cost is part of the excited diagonal entry. For the same newly chosen bond analogue $J = 2, s = 3/5$ and the illustrative choice $h_E = 1$, $b^2 = 18/25$ and (28) is $(25 - \sqrt{697})/10 = -0.140075756489\ldots < 0$.

An exact local repair boundary. The following repair is included to make the result actionable, not to present a new source claim. On a bond let

$$R_{xy} = \mathbf{1} - P_x P_y = Q_x + Q_y - Q_x Q_y. \tag{29}$$

It obeys $0 \le R_{xy} \le Q_x + Q_y$. For the flavour-diagonal hopping K_{xy} with unit coefficient, $\|K_{xy}\| = n$ exactly. Therefore

$$R_{xy} K_{xy} R_{xy} \ge -nR_{xy} \ge -n(Q_x + Q_y). \tag{30}$$

On a graph of maximum degree z, replacing ordinary hopping by its projected version yields

$$J \sum_x Q_x + s \sum_{\langle xy \rangle} R_{xy} K_{xy} R_{xy} \ge (J - zn|s|) \sum_x Q_x. \tag{31}$$

For the newly chosen periodic-1D bond analogue $J = 2, z = 2, n = 4, s = 3/5$, the margin is $-14/5$: projection alone is not enough. This number is not source-derived. Mode-normalizing K/n would give the positive margin $2 - 2(3/5) = 4/5$, but that is another changed normalization.

There is also a coupling-fixed positive repair with no new tunable parameter:

$$B_{xy} = |s| \left[nR_{xy} + \operatorname{sgn}(s) R_{xy} K_{xy} R_{xy} \right] \ge 0. \tag{32}$$

For the newly chosen $s = 3/5$ bond analogue its added coefficient is $|s|n = 12/5$ per bond; neither value is inferred from v1008. The Hamiltonian $J \sum_x Q_x + \sum B_{xy}$ has the common product DET vacuum and is bounded below by $J \sum_x Q_x$, hence has gap at least J. If P_x and K_{xy} are gauge invariant/covariant as assumed, so are R_{xy} and B_{xy}. Equation (32) is nevertheless a new many-body bond counterterm and may alter the desired continuum dynamics. It is an explicit design option, not evidence that TFPT already contains it.

Keeping the ordinary hopping instead requires a real local-stability theorem. The existing repository audit correctly notes that such theorems yield a volume-independent window only under verified tensor-product/local-gap and small-coupling premises, and that their unevaluated constants do not certify the newly chosen $s = 0.6$ bond point. The exact obstruction above explains why those premises cannot be replaced by the wished-for one-line inequality.

Reproducibility and claim boundary. The repository certificate [`verification/v1034_parent_mirror.py`] is read-only. Its JSON separates 29 exact algebra checks, six exact source-contract checks, and one floating-grid regression. The printed tag identifies each class. The checker performs an AST source-contract audit, exact 16×16 onsite algebra, exact symbolic minimization, and a sparse CAR-monomial two-cell calculation. It does not build the 256×256 two-cell matrix; for general n the obstruction uses only $2n$ output monomials when even $n \ge 4$. It also retains the exceptional $n = 2$ collision as a regression and applies the $\mathbb{Z}_2$ and $\mathbb{Z}_4$ link shifts explicitly. The reported decimal radical values are labelled evaluations; the separately labelled 129-point value is a floating reproduction of the actual finite-grid minimum, not an exact trigonometric certificate.

What is finished is precise: v1008's factorized parent has an exact all-volume gap, while ordinary real-space hopping admits no relative bound against the flat local DET defects. What remains open is equally precise: no local operator-valued TFPT Hamiltonian at a source-derived real-space coupling has acquired a proved volume-uniform mirror gap, and no net-chiral T3/T4 construction follows. **NO RH CLAIM.**

Round 7: first-order matter coupling of the staggered tensor target

The displayed arguments below are the proof-bearing source. The current repository regression is [`verification/v1035_matter_coupling.py`]; its finite checks audit the implementation and do not replace the universal steps in the proof.

Result and boundary. Conditional on the declared staggered Fourier dictionary, every nonzero momentum admits an exact affine coupling to a prescribed classical source obeying the two Ward equations. Source energy and momentum must deform the scalar and vector constraints; with those terms, the fibre constraints propagate and the curvature has the classical retarded response displayed below. The executable certificate does not import repository code, but it does instantiate the declared tensor/vector masks and checks their three reciprocal-axis Bloch transitions. Dynamical matter is tested only in collinear one-dimensional sectors and only through first order in the coupling. A free scalar and a fermion with a nearest-neighbour one-particle Hamiltonian pass those tests after their composite lattice vertices are repaired. A naive scalar stress fails, and an onsite ϕ^4 interaction has an explicit homogeneous momentum defect that no periodic stress divergence can remove. The deleted homogeneous gravity block cannot absorb nonzero total matter energy or momentum. No full three-dimensional matter coupling, quantum Kubo calculation, nonlinear constraint algebra, interacting quantum gravity, Lorentz-covariant lattice Riemann tensor, TFPT parent, T7 closure, TOE claim, or RH claim is proved.

Fixed regulator and conventions. We state the spatial regulator of the preceding classical and quantum artifacts as the intended real-space interpretation. The checker below instantiates these declared masks algebraically, but does not import a repository implementation. On a periodic cubic lattice of spacing a, a component with mask $m \in \{0,1\}^3$ lives at $x + am/2$, and

$$D_i^{(m)} f(x) = \frac{1}{a} \begin{cases} f(x + ae_i) - f(x), & m_i = 0, \\ f(x) - f(x - ae_i), & m_i = 1. \end{cases} \tag{1}$$

Thus $D_i^{(m \operatorname{xor} e_i)} = -(D_i^{(m)})^T$ under the periodic lattice sum. Physical-position Fourier phases give symbol $\mathrm{i}\kappa_i$; the factor i is retained below, where

$$\kappa_i = \frac{2}{a} \sin \frac{ak_i}{2}, \qquad r^2 = \kappa^T \kappa. \tag{2}$$

The diagonal tensor components have mask zero, q_{ij} with $i \neq j$ has mask $e_i \operatorname{xor} e_j$, and a vector component j_i has mask e_i. Products below are placed on the indicated cells or links; no unmentioned continuum Leibniz rule is used.

In the six orthonormal symmetric-tensor coordinates $(11, 22, 33, 23, 13, 12)$ define the following real matrices; they do not include the Fourier factor i:

$$(Bh)_i = h_{ij}\kappa_j, \quad t = (1,1,1,0,0,0)^T, \quad v = B^T\kappa, \quad s = v - r^2 t, \tag{3}$$

$$K_p = I - \frac{1}{2} tt^T, \qquad K_q = r^2(I - tt^T) - 2B^T B + tv^T + vt^T. \tag{4}$$

With the common Fourier-volume normalization suppressed, the real canonical Hamiltonian sums the paired complex fibres; the vacuum constraints at each nonzero momentum are

$$\begin{gathered} H_0 = \frac{1}{2} \sum_{k \neq 0} \left[p(-k)^T K_p(k) p(k) + q(-k)^T K_q(k) q(k) \right], \\ c_0 = -s^T q, \qquad \boldsymbol{c} = \mathrm{i} B p. \end{gathered} \tag{5}$$

Relative to the Round-6 real matrices, these are only nonzero sign/phase changes of the vacuum constraints. Their zero surface, rank, and the reduced E, F observables are therefore unchanged. The inherited exact identities are

$$K_q B^T = 0, \quad K_p s = B^T \kappa, \quad K_q t = s, \quad K_q K_p K_q = r^2 K_q + \frac{1}{2} ss^T = r^4 P_{\mathrm{TT}}. \tag{6}$$

Here c_0 is the Fourier transform of $D_i D_j q_{ij} - \Delta \operatorname{tr} q$, and c_i is the transform of $D_j p_{ij}$. Thus $\dot{c}_0 = \mathrm{i}\kappa_i c_i = -\kappa^T B p$ and $\dot{\boldsymbol{c}} = 0$. Dropping i and restoring it only after the algebra gives the wrong scalar sign, as the plane-wave mutant below demonstrates.

The coupling that actually preserves the constraints. Let (ρ, j_i, τ_{ij}) be a symmetric c-number source on the scalar, vector, and tensor staggered cells. Its two Ward equations are

$$\dot{\rho} + D_i j_i = 0, \qquad \dot{j}_i + D_j \tau_{ij} = 0. \tag{7}$$

At $k \neq 0$, add the spatial metric coupling and deform both constraints:

$$H_g(t) = H_0 + g\langle q, \tau(t)\rangle, \qquad \Psi_0 = -s^T q + g\rho, \qquad \boldsymbol{\Psi} = \mathrm{i}Bp - g\boldsymbol{j}. \tag{8}$$

The lapse and shift multipliers multiply Ψ_0 and $\boldsymbol{\Psi}$ in the total Hamiltonian. Thus (8) is not merely the isolated $q_{ij}\tau_{ij}$ vertex: the ρ and j_i couplings are present as constraint terms. Periodic summation by parts fixes all signs in (8).

Theorem 8 (Conditional affine source consistency in the complex Fourier fibre)**.** *Assume that (3)–(6) are the fibre dictionary of the intended staggered placement and that a prescribed c-number source satisfies (7). Then the four constraints in (8) are first class within this fibre algebra and propagate as*

$$\dot{\Psi}_0 = D_i \Psi_i, \qquad \dot{\Psi}_i = 0. \tag{9}$$

If one instead retains the vacuum vector constraint, its time derivative is

$$\dot{c}_i = -g D_j \tau_{ij} = g\dot{j}_i, \tag{10}$$

which is generically nonzero. Consequently "couple q_{ij} to stress" without deforming the constraints is inconsistent already at order g. Moreover, within the linear ansatz

$$H_{\text{int}} = \gamma g\langle q, \tau\rangle, \quad \Psi_0 = c_0 + \alpha g\rho, \quad \boldsymbol{\Psi} = \boldsymbol{c} + \beta g\boldsymbol{j},$$

Ward propagation uniquely fixes $\alpha = \gamma$ and $\beta = -\gamma$. Thus (8), up to the overall normalization of g, is the smallest affine constraint repair of the spatial stress vertex.

Proof. Hamilton's equations give $\dot{q} = K_p p$ and $\dot{p} = -K_q q - g\tau$. At fixed momentum, (7) reads $\dot{\rho} = -\mathrm{i}\kappa^T j$ and $\dot{j} = -\mathrm{i}B\tau$. Therefore

$$\dot{\Psi}_0 = -\kappa^T Bp - \mathrm{i}g\kappa^T j = \mathrm{i}\kappa^T(\mathrm{i}Bp - gj) = \mathrm{i}\kappa^T \boldsymbol{\Psi},$$

$$\dot{\boldsymbol{\Psi}} = -\mathrm{i}gB\tau - g\dot{j} = 0.$$

This is (9) with no stripped-i intermediate convention. Since the sources are c-numbers, their Poisson brackets vanish and the affine shifts do not change the vacuum first-class brackets. Dropping $-g\boldsymbol{j}$ leaves (10). For the general coefficients, the vector residual is $-\mathrm{i}(\gamma+\beta)gB\tau$, while the scalar residual relative to $\mathrm{i}\kappa^T\boldsymbol{\Psi}$ is $-\mathrm{i}(\alpha+\beta)g\kappa^T j$; both vanish only for the stated coefficients. □

The sign is falsifiable on one mode. Take the plane wave $e^{-\mathrm{i}t+\mathrm{i}z}$ with $\kappa = e_3$ and amplitudes $\rho = j_3 = \tau_{33} = g = 1$. Both Ward equations hold. Choose $Bp = -\mathrm{i}e_3$ and $s^T q = 1$, so the repaired constraints vanish. Directly, $\dot{\Psi}_0 = \dot{\boldsymbol{\Psi}} = 0$. If one instead keeps the old stripped-i choice $\Psi_0 = +s^T q + g\rho$ while restoring $\boldsymbol{\Psi} = \mathrm{i}Bp - gj$, the propagation residual is $\dot{\Psi}_0 - \mathrm{i}\kappa^T\boldsymbol{\Psi} = -2\mathrm{i}$.

The source need not be positive, local in time, or generated by matter for this theorem. Fourier reality is explicit: $B(-\kappa) = -B(\kappa)$ while s, K_p, K_q are even. Hence $\Psi_0(-k) = \overline{\Psi_0(k)}$ and $\boldsymbol{\Psi}(-k) = \overline{\boldsymbol{\Psi}(k)}$ whenever the fields at $-k$ are the conjugates of those at k and g is real. For an unordered non-self-conjugate pair, the actual real Hamiltonian contribution is

$$H_{\{k,-k\}} = p(-k)^T K_p\, p(k) + q(-k)^T K_q\, q(k) + g\big[q(-k)^T\tau(k) + q(k)^T\tau(-k)\big],$$

up to the common Fourier-volume normalization. It equals its complex conjugate. To lift the result from the complex fibre calculation to the actual staggered lattice it must also obey the mask placements and Bloch gluing.

That gluing is checked explicitly for the declared regulator. Under the reciprocal shift $k \mapsto k + 2\pi e_a/a$, let $R_V^{(a)}$ change the sign of vector component a, and let $G_T^{(a)}$ change the sign of an off-diagonal tensor component precisely when its mask contains e_a. Diagonal tensor components are unchanged. Then

$$\kappa' = R_V^{(a)}\kappa, \quad B'G_T^{(a)} = R_V^{(a)}B, \quad s' = G_T^{(a)}s, \quad K_q' = G_T^{(a)}K_qG_T^{(a)}.$$

Since K_p commutes with $G_T^{(a)}$, the source gluing $\tau' = G_T^{(a)}\tau$, $j' = R_V^{(a)}j$, and $\rho' = \rho$ gives

$$\mathcal{J}' = G_T^{(a)}\mathcal{J}, \qquad \Psi_0' = \Psi_0, \qquad \mathbf{\Psi}' = R_V^{(a)}\mathbf{\Psi}, \qquad (q')^T\tau' = q^T\tau.$$

Here $\mathcal{J}$ is the response drive defined in (12). The checker verifies all these matrix identities for $a = 1, 2, 3$. Reality is most cleanly imposed on extended-zone labels $(-k, k)$; if $-k$ is folded into the first Brillouin zone by a reciprocal vector, the displayed transition matrices supply the additional mask signs. What remains unverified is that a particular repository implementation uses exactly these declared conventions.

Gauge-invariant retarded curvature response. The local curvature variables remain

$$E = K_q q, \qquad F = K_q K_p p. \tag{11}$$

They commute strongly with the gravitational parts of all four constraints: $K_qB^T = 0$ and $K_qK_ps = 0$. The c-number shifts in (8) have zero bracket with them, so E, F remain gauge invariant. They obey $\dot{E} = F$ and, on $\Psi_0 = 0$,

$$\ddot{E} + r^2E = -g\mathcal{J}, \qquad \mathcal{J} := K_qK_p\tau + \frac{1}{2}s\rho. \tag{12}$$

The retarded solution with vanishing past data is therefore

$$\boxed{\delta E(t, k) = -g\int_{-\infty}^{t} \frac{\sin(r(t-t'))}{r}\,\mathcal{J}(t', k)\,\mathrm{d}t', \qquad \delta F = \partial_t\delta E.} \tag{13}$$

This is the classical Hamiltonian source response in the complex Fourier dictionary, not a spectral-positivity surrogate. Identifying it with a quantum Kubo response requires importing the Round-6 CCR/Fock representation and evaluating the corresponding commutator; that calculation is not executed in this artifact.

Theorem 9 (Proposition: Ward compatibility of the response)**.** *The source in (12) is transverse and has the trace required by the sourced scalar constraint:*

$$B\mathcal{J} = 0, \qquad t^T\mathcal{J} = -(\ddot{\rho} + r^2\rho). \tag{14}$$

Consequently (13), with constraint-compatible initial data, satisfies

$$BE = 0, \qquad t^TE = g\rho, \qquad t^TF = g\dot{\rho}. \tag{15}$$

For a TT step source $\tau(t) = \Theta(t)e_+$ at $\kappa = (0, 0, r)$ and $\rho = j = 0$,

$$\delta E(t) = -g(1 - \cos rt)e_+, \qquad t \geq 0, \tag{16}$$

where $e_+ = (1, -1, 0, 0, 0, 0)^T/\sqrt{2}$.

Proof. Both $BK_q = 0$ and $Bs = BK_qt = 0$, proving transversality. From (6),

$$t^TK_qK_p\tau = s^TK_p\tau = \kappa^TB\tau = -\ddot{\rho}, \qquad t^Ts = -2r^2,$$

where the second Ward equation followed by the time derivative of the first was used. This proves (14). Taking the trace of (12) shows that $t^TE - g\rho$ obeys the homogeneous wave equation and hence vanishes for compatible initial data. Equation (16) follows from $K_qK_pe_+ = r^2e_+$. □

Notice that sourced E is not generally TT: it is transverse, while its trace is fixed by the energy density. Only its radiative part is TT. Calling all of E a free graviton after matter is added would erase the scalar constraint.

Dynamical matter: what is and is not proved. Let matter have canonical variables z and an uncoupled Hamiltonian H_m. If its zeroth-order densities obey (7), then

$$H = H_0 + H_m + g\langle q, \tau^{(0)}(z)\rangle, \qquad \Psi_0 = -s^T q + g\rho^{(0)}(z), \quad \boldsymbol{\Psi} = \mathrm{i}Bp - g\boldsymbol{j}^{(0)}(z) \tag{17}$$

obeys (9) modulo $\mathcal{O}(g^2)$. This is the precise first-order feedback result. It is not exact nonlinear closure. Indeed, even before computing the g-correction to the matter stress,

$$\{\Psi_0(x), \Psi_i(y)\} = -g^2\{\rho^{(0)}(x), j_i^{(0)}(y)\}_m, \tag{18}$$

plus analogous current brackets. These terms are generally nonzero and must be absorbed by second-order constraint and Hamiltonian corrections if a nonlinear theory exists. The checker gives a canonical scalar datum for which the matter bracket on the right of (18) equals 1/2 exactly.

The distinction between a canonical/Noether stress and the metric-response stress matters here, especially for spinors. Their equivalence requires hypotheses and, for spinors, a tetrad formulation; see Leclerc's primary analysis [1]. We therefore test only explicit lattice Ward complexes below and do not rename them a full Hilbert stress tensor.

Free scalar: a local repair, then a sharp ϕ^4 obstruction. It suffices to examine a collinear sector; failure there is failure of a purported universal cubic-lattice identity. Set $a = 1$ on a periodic one-dimensional row and write

$$g_x = \phi_{x+1} - \phi_x, \qquad \Delta\phi_x = g_x - g_{x-1}, \qquad \dot{\phi}_x = \pi_x, \quad \dot{\pi}_x = \Delta\phi_x - m^2\phi_x. \tag{19}$$

The site energy and link flux

$$\rho_x = \frac{1}{2}\pi_x^2 + \frac{1}{2}m^2\phi_x^2 + \frac{1}{4}(g_x^2 + g_{x-1}^2), \qquad j_x = -\frac{1}{2}(\pi_x + \pi_{x+1})g_x \tag{20}$$

obey the exact energy equation

$$\dot{\rho}_x + j_x - j_{x-1} = 0. \tag{21}$$

The continuum-looking averaged-gradient stress

$$\tau_x^{\text{naive}} = \frac{1}{2}\pi_x^2 + \frac{1}{4}(g_x^2 + g_{x-1}^2) - \frac{1}{2}m^2\phi_x^2 \tag{22}$$

does not obey momentum continuity. An exact same-range correction is

$$\boxed{\tau_x^{\text{free}} = \frac{1}{2}\pi_x^2 + \frac{1}{2}g_x g_{x-1} - \frac{1}{2}m^2\phi_x^2 = \tau_x^{\text{naive}} - \frac{1}{4}(\Delta\phi_x)^2.} \tag{23}$$

Direct substitution gives

$$\dot{j}_x + \tau_{x+1}^{\text{free}} - \tau_x^{\text{free}} = 0. \tag{24}$$

The correction in (23) is exactly the finite-difference product-rule term; it vanishes at leading continuum order but is mandatory at finite spacing.

For the rational four-site datum $(\phi_0, \phi_1, \phi_2, \phi_3) = (0, 1, 2, 4)$, the four residuals obtained from (22) are

$$\left(-\frac{25}{4}, \frac{1}{4}, \frac{35}{4}, -\frac{11}{4}\right), \tag{25}$$

so this is an analytic mutant, not a rounding effect.

Now add $V_{\text{int}} = \lambda\phi^4/24$ to ρ_x, the force $-\lambda\phi_x^3/6$, and the continuum-like pressure $-\lambda\phi_x^4/24$ to (23). Energy continuity (21) remains exact, but the momentum residual is

$$R_x := \dot{j}_x + \tau_{x+1} - \tau_x = \frac{\lambda}{24}(\phi_x + \phi_{x+1})(\phi_{x+1} - \phi_x)^3. \tag{26}$$

For the same datum,

$$\sum_{x=0}^{3} R_x = -\frac{17}{2}\lambda \neq 0. \tag{27}$$

Every periodic stress divergence sums to zero. Therefore no redefinition of τ_x alone—local or nonlocal—can repair (26) on this matter phase space while leaving j_x and its dynamics fixed. This is the first precise interacting obstruction, and it appears in constraint propagation at order $g\lambda$, not only in a higher-order gravity bracket.

There is a limited repair for the already-selected nonzero gravity modes. Let P_0 be the spatial mean and let $D\tau = \tau_{x+1} - \tau_x$. Then

$$\delta\tau = -D^{+}(I - P_0)R, \qquad D\delta\tau = -(I - P_0)R, \tag{28}$$

where D^{+} is any zero-mean inverse of D. This makes the nonzero-mode Ward identity exact and is constructed explicitly by cumulative summation in the checker. It is nonlocal, leaves the homogeneous momentum defect P_0R, and does not constitute a local interacting matter coupling. A full repair must instead alter the matter discretization/current or add degrees of freedom that carry the missing lattice momentum. The broader lattice literature likewise defines and renormalizes energy-momentum tensors by enforcing Ward identities, rather than assuming the continuum expression survives unchanged [2, 3].

Free fermion: exact collinear divided-difference vertices. A second collinear test uses a free chiral fermion whose *one-particle Hamiltonian* is nearest-neighbour, with dispersion $\epsilon(p) = \sin p$. This is a finite-spacing Ward test, not a claim of a doubler-free $3 + 1$ dimensional chiral theory. For a bilinear transferring momentum K, set

$$\kappa(K) = 2\sin\frac{K}{2}, \quad \epsilon_- = \sin p, \quad \epsilon_+ = \sin(p+K), \quad \bar\epsilon = \frac{1}{2}(\epsilon_+ + \epsilon_-). \tag{29}$$

The exact divided-difference identity is

$$\epsilon_+ - \epsilon_- = \kappa(K)\cos\left(p + \frac{K}{2}\right). \tag{30}$$

Consequently the energy, current, and stress vertices

$$\rho = \bar\epsilon, \quad j = \cos\left(p + \frac{K}{2}\right)\bar\epsilon, \quad \tau = \cos^2\left(p + \frac{K}{2}\right)\bar\epsilon \tag{31}$$

satisfy both one-particle Ward identities

$$(\epsilon_+ - \epsilon_-)\rho = \kappa j, \qquad (\epsilon_+ - \epsilon_-)j = \kappa\tau. \tag{32}$$

In real space, ρ, j, and τ in (31) have maximal hopping ranges one, two, and three, respectively: only the one-particle Hamiltonian is nearest-neighbour. The average factors are the fermionic analogue of the scalar repair in (23). Replacing j by the continuum vertex ρ fails: at $p = \pi/6$, $K = \pi/3$, one has $(\rho, j, \tau) = (3/4, 3/8, 3/16)$ and the first Ward residual is $-3/8$. Thus both bosonic and fermionic tests reject naive continuum vertices, while both free tests admit explicit finite-spacing corrections.

The homogeneous obstruction. The preceding free tensor construction removed the complete $k = 0$ canonical block to eliminate its negative trace direction. At $k = 0$, $B = s = K_q = 0$, so its vacuum local constraints vanish identically. Equation (8) would reduce to

$$\Psi_0(0) = g\rho(0), \qquad \mathbf{\Psi}(0) = -g\boldsymbol{j}(0). \tag{33}$$

A positive-energy matter state normally has $\rho(0) \neq 0$, and a state may also have $\boldsymbol{j}(0) \neq 0$. For $g \neq 0$, imposing (33) would delete either such component rather than source gravity. The conditional coupling must therefore be read in one of two narrow ways:

(a) couple only $(I - P_0)(\rho, j, \tau)$, equivalently fluctuations around a specified homogeneous background; or

(b) restore and consistently stabilize a homogeneous/global geometry sector.

The first option is sufficient for (9)–(16) at nonzero momentum but does not describe the gravitational response to total energy. The second is an open construction. Simply calling the removed block "gauge" is not a repair.

Quantum meaning and remaining obligation. Equation (13) is an exact classical Green-function solution of the conditional staggered Fourier equations and is gauge invariant and Ward compatible there. A smooth compact-time c-number source is expected to define linear forcing of the already-constructed oscillator Fock space, but the CCR import and Kubo commutator needed to prove that quantum statement are not executed here. Thus the generic prescribed-source fibre algebra is closed; its full staggered real-space and quantum realization remain open checks.

For dynamical scalar or fermion matter, (17) is an interacting Hamiltonian on the tensor–matter tensor product only as a formal first-order deformation. Equations (18), (26), and (33) are three distinct unclosed edges:

- g^2 current brackets require genuinely deformed constraints and gravity self-couplings;
- finite-spacing interacting matter may fail the spatial Ward identity already at order $g\lambda$;
- total energy has no receiving homogeneous gravity mode.

An exact graph energy current such as $J_{x \to y} = \mathrm{i}[h_x, h_y]$ addresses energy continuity only. It is not by itself a symmetric stress satisfying the vector Ward identity, and it does not close the Hamiltonian constraint algebra. Conversely, positivity of a TT stress spectral measure says nothing about these three equations.

Executable certificate and evidence typing. The repository certificate [`verification/v1035_matter_coupling.py`] imports no proof text. It checks the generic prescribed-source polynomial identities with their full complex Fourier signs, a conserved plane-wave witness, the mandatory failure of the old restored-i sign, paired Fourier reality, all three declared Bloch transitions, the classical retarded TT step solution, the two collinear Ward complexes and mutants, the ϕ^4 obstruction and projected repair, the nonzero matter Poisson bracket, and the fermion divided differences. It does not instantiate a full three-dimensional matter complex or execute a quantum Kubo calculation. Its successful run reports 55 exact symbolic/rational checks and four floating-point regressions. The floating checks sample generic fibres and random scalar/fermion data; they establish no universal statement.

The exact cost is fixed-size symbolic algebra plus a four-site matter system; the numerical cost is constant and negligible. The certificate is read-only.

References for the matter-coupling proof

[1] M. Leclerc, "Canonical and gravitational stress-energy tensors," arXiv:gr-qc/0510044 (2006).

[2] A. Buonanno, G. Cella, and G. Curci, "Lattice energy-momentum tensor with Symanzik improved actions," arXiv:hep-lat/9506008 (1995).

[3] L. Giusti and M. Pepe, "Energy-momentum tensor on the lattice: non-perturbative renormalization in Yang–Mills theory," arXiv:1503.07042 (2015).

Research Contract — `FLAV.NUSCALE.06` (pentagon-class misalignment candidate)

Typing. A candidate [C] (ledger row `FLAV.NUSCALE.06`); mechanism = the Q_+-to-flavor operator [O]. No seesaw closure; `FLAV.NUSCALE.05` unmoved. ([`verification/v1001_nu_pentagon_chain.py`].)

Executed content. $U_e = I$ inventory theorem; misalignment $U = U_{v9}R_{13}(\theta, \phi)$; pentagon double hit $\phi = 288° = 4(2\pi/5)$ frozen, all three measured angles $\le 0.56\sigma$ (honest max pull 0.557) at $\theta = 2\pi/35$, the unique LEE survivor of nine pre-declared candidates (LEE 2.7%); v3 chain SHA-16 `a4c28732fa687620` with $\Sigma = 0.0599$ eV, $m_\beta = 9.0$ meV, $m_{\beta\beta} \in [1.5, 3.8]$ meV, $\delta_{\mathrm{CP}} = 287.66°$; v270–$\theta_{23}$ tension 1.85σ TYPED; census null 0/1607. **Kills.** DESI floor on Σ; DUNE δ_{CP} discriminates 287.7 vs 240; JUNO. Status: Candidate [C]; no closure.

Research Contract — `FLAV.NU.TEXTURE.MECHANISM.01` (both structural nulls as binding constraints)

Typing. A research contract [O] (ledger row `FLAV.NU.TEXTURE.MECHANISM.01`), registered 2026-08-29 from review wave 5. ([`verification/v1003_nu_structural_nulls.py`].) **Executed content.** Charged-sector data $h = (1, 1, 1)$, $d = (64, 60, 64)$ (coset min-norms all 2) [E]. Schur-texture NULL [X]: all 8 preregistered (K, B) combinations fail; D4 reflection forces $\lambda_1 = \lambda_3$; hierarchy $(\varepsilon, 2\varepsilon, 3)$ structurally unreachable from D4-even seam data; required $K \sim [3536.998, 1768.499, 1/3]$ matches none of 9 comparators. Orientation-doubling structure theorem [E]: the sgn J odd sector generically lifts the degeneracy (exact rational witness odd_scale $= 1/4$, $\gamma = 1/10$). Scale NULL [X]: 12/12 declared trials fail; required cancellation $1 - |c/a| = 1.885\times10^{-4}$ is not supplied by any frozen constant; η-branches are isospectral. **Consequence (binding).** The mechanism must be D4-odd *and* supply a $\sim 2\times10^{-4}$ even-odd cancellation; pure seam data excluded [X]-typed. `FLAV.NUSCALE.05/.06` unmoved; no seesaw closure. Status: `FLAV.NU.TEXTURE.MECHANISM.01` [O].

Executed update (2026-08-30, evening TOE-gate wave): κ six-fold NULL completes the scaffold bound ([`verification/v1012_toe_gate_witnesses.py`]; the contract stays [O]). Six pre-declared D4-odd candidates, all parity-checked, all NULL (five at 10^{-14}; $K_4 = 3.55\times10^{-3}$ is 18.8× too large). The 1.885×10^{-4} cancellation scale requires the true 4D functional — mechanism scale strictly beyond frozen finite objects. Display marker stays [O].

Executed update (2026-09-05, round-3 integration): parent-internal A_3–D_4 intertwiner (`FLAV.NU.PARENT.INTERTWINER.01`; [`verification/v1024_parent_internal_boundaries.py`]; no parent marker move). Using only the existing four-state seam clock and six ν^c modes, the finite module constructs a positive, gauge-neutral, parity-even, non-additive local term. Its checked gap is 0.25 conditional on the inherited full-parent bound, and the mixed resolvent is 0.104712041885. The selected pair-operator triplet carries ordinary D_4, whereas the full fermionic lift is projective:

$$C_f^8 = 1, \qquad C_f^4 = (-1)^{N_1+N_3} \neq (-1)^F, \qquad J_f C_f J_f = (-1)^F C_f^{-1}.$$

This supplies neither a neutrino mass matrix nor the required 2×10^{-4} cancellation and is not a 4D functional. `FLAV.NU.TEXTURE.MECHANISM.01`, T6 and TOE remain [O].

Research Contract — `SEAM.MILNOR.LOCALRING.01` (canonical identification of the seam collar with the reduced Milnor algebra)

Typing. A research contract [O] (ledger row `SEAM.MILNOR.LOCALRING.01`), registered 2026-08-30 from review wave 6. ([`verification/v1004_milnor_frobenius_gray.py`].) **Demand.** A *canonical* identification between the raw seam collar / E_8 transition bus and the reduced Milnor algebra $\mathbb{F}_2[y, z]/(y^2, z^4)$ preserving *simultaneously* the dual-number action, the Frobenius pairing, the Gray torsor, the deck action and CP. **Acceptance.** The identification must be geometric/mechanical (not basis-chosen) and must produce the rank clock and socle-CP as *outputs*. **Kill tests.** Any second inequivalent identification with the same invariants; failure of μ_4-equivariance. **Note.** Would reduce P2 / alignment-bit / Gauss-code / Brieskorn origin to one local ring. Currently [O] because the Milnor algebra of $x^2 + y^3 + z^5 \cong \mathbb{Q}[y, z]/(y^2, z^4)$ (dimension 8) and the spectral numbers $= \{1, 7, 11, 13, 17, 19, 23, 29\}$ (the E_8 exponents) are *classical* (cite, no claim), and the corpus map is unproven.

Executed finite algebra (review wave 6; the contract stays [O]). New-in-corpus exact [E]-finite: rank clock $(2-a)(4-b)$, table sum $30 = h(E_8)$; $S = pqr \cdot \mu/8$; self-clock iff $\mu = 8$; $(2, 3, 5)$ unique among ordered spherical triples; Galois–Gray action (11 swaps sheets, 7 cycles the seam clock in binary Gray order $0 \to 1 \to 3 \to 2$, $-1 = (a, b) \mapsto (1-a, 3-b)$); CP = socle/Frobenius duality (residue pairing nonzero iff $a + a' = 1$, $b + b' = 3$, $m + m' = 30$); integral dual-number bridge $\mathbb{Z}[y, z]/(y^2, z^4) \otimes \mathbb{F}_2$ free of rank 4 over $D = \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$, $\mathbb{Z}[i]/(2) \cong D$, basis-dependent. Interpretive [C]: $15 = 8_R + 7_{NS}$ as $PG(3, 2) = AG(3, 2) \sqcup PG(2, 2)$. Must-fail: $(2, 3, 7)$ gives $S = 63 \neq 42$; A_8 self-clock passes but the exponent fingerprint fails; wrong socle rank 4. Status: `SEAM.MILNOR.LOCALRING.01` [O].

Executed update (2026-08-30, afternoon harvest; [`verification/v1010_simplicity_bridge_census.py`]; canonical status SHARPENED, display stays [O]). The strong all-structure identification is finitely obstructed: at $\mathbb{F}_2$, clock order 2 versus 4 and CP Jordan rank 2 versus 4 make the four-structure isomorphism set empty; at every upstairs home the same obstruction persists ($\mathbb{Z}/4$: $J^2 = -I$ versus Gray $P^2 \neq -I$, D_8 versus $C_4 \times C_2$; Gaussian $k = 2$ order collapse, $k = 3$ rank/relation clash; affine conjugacy cannot repair the linear parts). Only a coarse D-module+pairing bridge exists (witness constructed; $|\mathrm{Aut}(X)| = 36864$, $|\mathrm{Aut}(Y)| = 8$). The ring-internal identities of v1004 (rank clock, socle CP, self-clock) are unaffected. No marker move.

P2 rehabilitation note (not a third contract; [`verification/v1005_relative_pencil_bridges.py`], corrected by [`verification/v1010_simplicity_bridge_census.py`]). The v1005 circularity flag is corrected as forced provenance: the bit-selector ladder forces $c = 2$ and family $n = 3$ then forces Q column sums $(9, 5, 1)$; C's carrier-looking 5 is inherited from R column 3, not inserted. Unique primitive positive S_3-invariant covector $\varepsilon = (1, 1, 1)$. The winding bridge remains a *choice* that is conditionally unique: spectrum $\{5/2, 1, 1\}$ alone has 1078 solutions; democratic-image + carrier-anchor kernel leave exactly the corpus shift (count 1). Remaining obligation = seam-geometric derivation of those two directional premises. `AX.P2.01` typing UNCHANGED. Hankel $p_n p_{n+2} - p_{n+1}^2 = 2^{n+1}$ and winding spectrum $\{5/2, 1, 1\}$ remain [E]-finite (provenance v106/v119/v4). Klein-four SNF $(2, 2)$ only after $D = 2d$ is interpretive [C].

Executed update (2026-08-30, late-evening harvest; [`verification/v1014_bridge_refinements.py`]; `AX.P2.01` typing UNCHANGED). W-bridge P-anch is DERIVED: $e_1 = R^{-1}(1, 1, 2)^T$ uniquely via the winding/triple lock (not by g_{car} or by the repeated anchor entries). P-dem is PARTIAL: KMS equidistribution lives on the trivial character; the $\mathbb{Z}_4$ average annihilates the nontrivial character space. The missing object is the character-blind determinant-response map into the $v4$ row space — the same MMST/Quillen externalized leg as TEL-B/ALG-EXH/P1.

Executed update (2026-08-31, Monday-morning harvest; [`verification/v1015_axiom_core_closure.py`]; `AX.P2.01` typing UNCHANGED). W-bridge P-dem is now DERIVED: the abelian total-charge insertion is character-blind, $r = (1, 1, 1)$ exactly at collar sizes 12 and 16, democracy residual 0; mutants split $(1, 0, 0)$ and $(5/4, 1, 5/4)$. Together with P-anch (v1014) both W-bridge premises are derived at finite level; the axiom-core remainder is ZERO modulo the externalized MMST identification. Finite derivations are shadows; the continuum identification is the same external leg. Axiom typing UNCHANGED.

Research Contract — GRAV.WEINBERG.WITTEN.01 (Weinberg–Witten firewall before any spin-2 marker move)

Typing. A research contract [O] (ledger row GRAV.WEINBERG.WITTEN.01), registered 2026-08-30 from review wave 6. Child of TOE gate T7; cross-link GRAV.SPIN2.EMERGENCE.01 (listed here, not back — that would cycle). **Demand.** The microscopic Hamiltonian family must *not* possess a fundamental Lorentz-covariant local conserved $T_{\mu\nu}$; Lorentz symmetry, diffeomorphism Ward identities and the spin-2 mode must emerge in the *same* IR limit. **Acceptance.** Positive TT spectral density; exactly two helicities; massless pole; emergent diffeomorphism Ward identity; universal soft coupling; *and* an explicit statement of which Weinberg–Witten premise is evaded. **Kill tests.** A microscopic covariant local $T_{\mu\nu}$ together with a massless composite spin-2 claim. Status: GRAV.WEINBERG.WITTEN.01 [O].

Externalization Contract — SEAM.EQUIV.01 (the seam scaling-limit theorem, packaged for external specialists)

Typing. This section is a *hand-off document*: it packages the one remaining seam obligation as a self-contained problem statement for an outside specialist in conformal nets / vertex operator algebras / algebraic QFT. It *documents*, it does not close: the ledger row SEAM.EQUIV.01 stays [O] (its conditional route SEAM.EQUIV.MMST.01 is closed modulo cited theorems; the unconditional parent is the target below). Nothing in this section moves a marker, and no claim beyond the corpus is made. Everything an external solver needs — hypotheses, runnable finite models, the Lean axiom surface, the negative controls a valid solution must respect, and the exact acceptance criteria — is listed here with its verification module.

(1) Minimal theorem statement. *The scaling limit of the gapped μ_4-equivariant CAR collar exists in the local-net topology; the index-4 simple-current extension functor commutes with this limit; and the Osterwalder–Schrader reconstruction of the limit state is isomorphic to the holomorphic $(E_8)_1$ lattice conformal net.* Explicitly, with $\mathcal{A}_L$ the finite collar net ($D = 16$ Majoranas, gap $\Delta = 6\ln(3/2) > 0$, carrier clock μ_4) and α_{μ_4} the index-4 simple-current extension (the Longo Q-system of v125/v154), the target decomposes into four lemmata:

- **(L1) Uniform energy / nuclearity bounds.** The rescaled collar family satisfies uniform (in L) energy bounds and a nuclearity/compactness condition strong enough to extract a limit state on the quasi-local algebra. [O] — the finite side supplies the uniform gap (v302) and exact quasi-freeness (Wick/Pfaffian closure, v113/v155); N-uniform CAR nuclearity is now proved as compactness input ([verification/v1016_state_gap_batteries.py]: $\nu_N \le e^{16C_\beta}$) but does *not* identify P_{gen}/ALG-EXH; L1 stays [O].
- **(L2) Convergence of local algebras.** The local algebras of the rescaled collar converge (in the sense of the Osborne–Stottmeister scaling-limit framework) to the local algebras of a chiral CFT of central charge $c = 8$. The cited theorem covers free CAR lattice fermions in the range rank $\le c \le D$; the collar sits at $8 \le 8 \le 16$ (v356/v366). [O] as applied *with the μ_4-equivariant structure retained through the limit.* The coverage of the cited theorem is since carved exactly ([verification/v973_seam_route_narrowing.py]): *covered* is the $\mathfrak{so}(16)_1$ fermion-bilinear current algebra (strong convergence, Virasoro and correlator control, polynomial error bounds); *not covered* — the named residual — are N1 the μ_4/GSO extension $SO(16)_1 \to (E_8)_1$ as a net statement, N2 the twist/spin-field sector, N3 the one-sided chiral edge net, and N4 the OS/holomorphy selector; a measured collar convergence rate $p = 3.48$ lies inside the preregistered band (the trivial $M{=}3$ control does not converge). Completeness wave ([verification/v994_mmst_criteria_battery.py]): 5 of 7 MMST identification criteria are now executed at the finite level (measured $c = 8.005$, $(E_8)_1$ vacuum character $[1, 0, 248, 0, 4124, 0, 34752]$, $k = 1$, $\det D_8 = 4 \to \det E_8 = 1$); C6 is the written-out quasi-free theorem, C7 stays cited.

- **(L3) The crossed product commutes with the limit.** The index-4 simple-current extension (the μ_4 Q-system, equivalently the crossed product by the glue $\mathbb{Z}_4$, v469) commutes with the scaling limit: $\lim(\mathcal{A}_L \rtimes \mu_4) \cong (\lim \mathcal{A}_L) \rtimes \mu_4$. The finite Q-system is explicit and machine-verified (v125/v154); the commutation is [O].
- **(L4) Uniqueness of the holomorphic $c = 8$ reconstruction.** The OS reconstruction of the limit is a *holomorphic* (μ-index 1, one primary) $c = 8$ net, hence the $(E_8)_1$ lattice net — the same-central-charge rival $SO(16)_1$ ($\det K = 4$, four primaries) is excluded by the order-4 clock, *not* by $c = 8$ alone (v351/v378). The classification input is cited (KLM/Müger; lattice-net uniqueness); that the *reconstructed* net satisfies its hypotheses is [O].

Dimension typing (per `DIMENSION.UPLIFT.FIREWALL.01`): hypotheses *and* conclusion of L1–L4 live in 1+1D — the target is a chiral c=8 boundary object. No part of this contract, proved or not, closes a 3+1D claim; the seam→bulk uplift is the separate contract `SEAM.BULK4D.RECON.01`.

(2) Complete hypothesis list. Every input an external proof may use, typed. PROVEN-FINITE = machine-checked in the suite (module cited); CITED-CLASSICAL = a published theorem, used as stated, never re-proved; OPEN = the target itself.

Type	**Hypothesis**	**Witness**
PROVEN-FINITE	spectral gap $\Delta = 6\ln(3/2) > 0$ of the collar transfer	v302
PROVEN-FINITE	quasi-free CAR structure (one kernel, Wick/Pfaffian exact; boundary marginal of a Gaussian bulk is Gaussian)	v113/v155/v160
PROVEN-FINITE	MMST applicability arithmetic: $D = 2^{g_{\mathrm{car}}-1} = 16$, $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$, $8 \le 8 \le 16$	v356/v366
PROVEN-FINITE	explicit lattice realisation: p+ip collar, FHS Chern $\lvert C\rvert = 1$ at M=1 (control $C = 0$ at M=3); 16 Majoranas $\Rightarrow c_- = 8$	v367
PROVEN-FINITE	realised non-gappable chiral edge on the strip (bulk–edge, in-gap states, edge-localised)	v368
PROVEN-FINITE	strict locality topologically forbidden (Wannier winding $= \lvert C\rvert = 1 \neq 0$)	v461
PROVEN-FINITE	central charge $c = 8$ from Calabrese–Cardy entanglement scaling (fitted $c = 1.0000$ per mode, 16 Majoranas = 8 modes)	v376
PROVEN-FINITE	the $(E_8)_1$ character: $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$, one primary, 248 level-1 currents	v377
PROVEN-FINITE	torus (genus-1) discriminator: KLM μ-additivity $\mu = 16/4^2 = 1$ for the index-4 glue vs $16/2^2 = 4$ for index-2	v378
PROVEN-FINITE	reflection positivity of the collar measure by an exact positive-mixture Gram identity (symbolic PSD, no tolerance)	v379
PROVEN-FINITE	the index-4 Q-system exists explicitly (Longo axioms on $\mathbb{C}[\mathbb{Z}_4]$); pinned levels $(D_5)_1 \times (A_3)_1$, $c = 5 + 3$	v125/v154
PROVEN-FINITE	the order-4 clock is forced (four Gauss–Bonnet marks, $\lvert\mu_4\rvert = 4 = h(A_3)$); contract typing + import firewall (routes non-circular)	v216/v286
CITED-CLASSICAL	scaling limit of free lattice fermions → chiral CFT, rank $\le c \le D$ (Osborne–Stottmeister, CMP **398** (2023) 219, arXiv:2107.13834, on the MMST OAR framework arXiv:2010.11121)	v336
CITED-CLASSICAL	OS reconstruction / holomorphy selector (Adamo–Moriwaki–Tanimoto 2024)	v336
CITED-CLASSICAL	index-4 extension theory: Longo–Rehren Q-systems; Kawahigashi–Longo–Müger complete rationality; Xu orbifold heredity	v469
CITED-CLASSICAL	gapped free-fermion phases are invertible (Kitaev); nontrivial invertible phase ⇒ non-gappable edge (anomaly inflow)	v301
CITED-CLASSICAL	holomorphic $c = 8$ classification: #primaries = $\lvert\det K\rvert$; E_8 the unique even unimodular rank-8 lattice (Conway–Sloane)	v281
OPEN	**L1–L4 above** — the continuum existence, equivariant convergence, crossed-product commutation and holomorphic reconstruction, assembled	`SEAM.EQUIV.01`

(3) Available finite models (runnable). All models live in the repository and run standalone (Python 3, `mpmath/sympy`; `python verification/run_all.py` runs the whole suite):

- [`verification/v367_seam_s3_lattice.py`] — the collar lattice model (FHS Chern, controls)
- [`verification/v368_seam_s3_inflow.py`] — the strip edge / anomaly-inflow diagnostics
- [`verification/v376_seam_s3_centralcharge.py`] — entanglement central charge
- [`verification/v377_seam_s3_e8character.py`] — the exact $(E_8)_1$ character
- [`verification/v378_seam_s3_modular.py`] — torus degeneracy / modular data
- [`verification/v379_seam_s3_rp.py`] — reflection positivity
- [`verification/v125_glue_qsystem.py`] — the explicit index-4 Q-system
- [`verification/v154_simple_current_theorem.py`] — the simple-current extension theorem
- [`verification/v356_continuum_mmst_applicability.py`] — MMST applicability arithmetic
- [`verification/v366_mmst_seam_collar.py`] — the collar in the MMST class
- [`verification/v286_seam_equivalence_contract.py`] — contract typing and import firewall

The seam-relevant Lean formalisation lives under `experiments/lean4-carrier-rigidity/TfptCarrier/` (shipped as `lean4-carrier-rigidity/`; `lake build` clean).

(4) Lean axiom surface (honest). The Lean layer certifies the *logical composition* and the finite arithmetic — *not* the analysis. In `TfptCarrier/SeamScalingLimit.lean` (`FORM.SEAM.MMST.01`) the kernel proves, by `decide` with no axioms: the range $8 \le 8 \le 16$ (`mmst_range`), chirality $c_- = 8 \neq 0$ (`chiral`) and the holomorphy discriminator $\det K$: 1 vs 4 (`holomorphic_discriminator`). *Axiomatized* (named, cited, not re-proved): `mmst_scaling_limit`, `adamo_holomorphic_e8` and the structural input `collar_gapped`; the carriers `CollarGappedQuasifree`, `ChiralCFT_c8`, `SeamIsE8` are *opaque* `Prop`s. The same pattern holds in `SeamEquivChain.lean` (`kitaev_invertibility`, `mueger_klm_holomorphy`, `conway_sloane_e8` as axioms), `SeamApplicabilityLedger.lean` and `BWKeystone.lean`. No `sorry` in any seam module; `#print axioms` pins the residual to exactly the named cited theorems. **Caveat stated plainly:** no operator algebras, nets, or states are formalised — L1–L4 have *zero* Lean content beyond this composition audit, so the formalisation is thinner than a casual reading of "Lean-verified" would suggest.

(5) Must-fail controls. A submitted proof is accepted only if it *fails* on the following deformations (each already realised as a machine-checked negative control):

- **MF-S1 ($c = 8$ alone must not suffice).** Any argument that would equally conclude "$SO(16)_1$" is invalid: the proof must consume the order-4 clock / $\det K = 1$ discriminator, not just the central charge (the red-team point; v281/v378).
- **MF-S2 (wrong extension index breaks it).** Replacing the index-4 μ_4 glue by the index-2 $\mathbb{Z}_2$ glue must yield $SO(16)_1$ ($\mu = 4$), i.e. the E_8 conclusion must demonstrably fail (v154/v351).
- **MF-S3 (trivial collar).** At the trivial parameter $M = 3$ ($C = 0$, clean gap, no edge) the construction must produce *no* chiral limit (v367/v368).
- **MF-S4 (marks).** A non-order-4 clock, or a mark configuration away from the free μ_4 orbit, must break the $\tau = i$ pillowcase input (v216/v217).
- **MF-S5 (firewall).** No circular import: the proof must not assume the $(E_8)_1$ net on the seam at any intermediate step (the Route-i/Route-ii import firewall, v286).

(6) Acceptance tests. `SEAM.EQUIV.01` counts as solved exactly when: (i) L1–L4 are proved under the hypothesis list of (2), with any additional hypothesis flagged explicitly; (ii) the composed theorem outputs the $(E_8)_1$ net — one primary, 248 level-1 currents, torus degeneracy 1 — as a *consequence*, matching v377/v378; (iii) all five must-fail controls verifiably fail; (iv) no downstream TFPT readout (in particular no α data) enters as an input. Partial results that close a single lemma (e.g. L1 alone) narrow the target and are recorded as such; only the full chain moves the ledger — and any marker move happens in `status_ledger.csv` by the house process, never in this document.

Externalization Contract — `ALPHA.QUILLEN.EXACT.01` (the determinant-line variation theorem, packaged for external specialists)

Typing. The second hand-off document: the one remaining EM obligation, packaged for an outside specialist in determinant lines / η-invariants / index theory. It *documents*, it does not close: `ALPHA.QUILLEN.EXACT.01` stays [O] (external math, type A, v384), and it is a *face* of `SEAM.EQUIV.01` (v382) — not a second independent gate. The *value* α^{-1} is closed [E] regardless (unique root + ablation, v3); open is only "why *this* functional". Nothing here moves a marker.

Relation (2026-08-28). The nonabelian upgrade of this $U(1)$ face is the new contract `GAUGE.DETLINE.FIXPOINT.01` (all three gauge couplings as unique stationary points of one zeta-det/inflow functional; honest: $\alpha_s(M_Z)$ is currently an external input). This row stays the $U(1)$ face; that row is the three-coupling demand.

Relative-determinant note (2026-08-29, review wave 4). The correct object is the *relative* determinant $Q(A) = \log[\det' \Delta_A / \det' \Delta_{A_0}] + 8\, b_1\, c_3^6 \log \varphi(A)$; the BFK constant 2^{-4} is $A/\tau/\alpha$-independent at fixed cut number, so $\delta \log 2^{-4} = 0$ — no physical compensation needed (consistent with `COMPENSATION_INTERNAL` and the reviewer's correction). Absolute normalisation is fixed by the reference A_0.

(1) Minimal theorem statement. Two steps, both over the moduli of flat $U(1)$ connections on the μ_4 seam (the twist torus):

- **(T1) Gap-stable continuum limit of the determinant line.** For the seam operator family $\{D_L\}$ (the collar Bogoliubov–de Gennes/Dirac operators with twisted boundary conditions) converging in the norm-resolvent sense to the continuum seam operator D_{seam}, with the Fermi gap uniformly bounded below over the whole twist torus (so the Dai–Freed section has no zero), the curvature class of the finite determinant line converges to the Quillen curvature of the ζ-regularised determinant line: $c_1(\det D_L) \;\to\; c_1(\det_\zeta D_{\text{seam}})$. The finite side is computed: the FHS curvature of the occupied-frame determinant line integrates to $2\pi \cdot 1$ exactly, size-independently (v472).

- **(T2) Variation = Bismut–Freed curvature = inflow response, with the exact corpus coefficients.** On the continuum line of T1,

$$\delta_\tau\Big(\log \det\nolimits_\zeta \Delta_{U(1)} \;+\; 8\, b_1\, c_3^6 \,\log \varphi_{\text{seam}}\Big) = 0,$$

  i.e. the EM closure $F_{U(1)}(\alpha) = k_0\,\alpha^3 - 2\,c_3^3\alpha^2 - 8\,b_1\,c_3^6 \ln(1/\varphi_{\text{seam}}) = 0$ *is* the stationarity of the exact ζ-regularised Quillen functional, with every coefficient as it stands in the corpus: the Chern level $k_0 = |C| = 1$ (the computed bulk invariant of the S3 collar, normalised by the embedding index $k_Y = \operatorname{tr}(Y^2)/\operatorname{tr}(T_3^2) = 5/3$, v470); the Gauss–Bonnet boundary coefficient $c_3 = 1/(8\pi)$ with heat-kernel powers c_3^3, c_3^6 (v216/v342); the $U(1)_Y$ a_4/index coefficient $b_1 = 41/10$ ($8b_1 = (4/5)\cdot 41$, v434); and the discriminant exponent $-\frac{5}{4} = -q(D_5)$ (v341). The derived stationarity must then have $\alpha^{-1} = 137.0359992168$ as its unique physical root (v3).

The open content splits into two named faces (one target, per v434/v435): the *analytic* face — T1 plus the identification of $\delta \log \det_\zeta$ with the seam Seeley–DeWitt data, i.e. the diagonal ζ-renormalisation at the marks and the multiplicity matching $41 = 10\,b_1$ (v484/v485) — and the *topological* face, the `EM1` lemma: the from-first-principles proof that $\partial_\tau \log Z_{\text{Maxwell}} = k_0\,\alpha^3$ with k_0 the bulk Chern level (v48/v435). *Both named faces now carry computed witnesses*

(2026-08-27, ***ALPHA.QUILLEN.FACES.01*** ***[E]*** *new; [verification/v974_alpha_faces_computed.py];* ***ALPHA.QUILLEN.EXACT.01*** *stays [O]):* the diagonal-ζ (analytic) face is executed as a BFK gluing — the gluing constant is exactly 2^{-N}, the jump matrix $16\,c_3\,\mathrm{circ}(2,-1,0,-1)$ *is* the inverse mark Green matrix, and the v485 scale channel is reproduced via the KMS unit, while against the v484 mark matrix the comparison is channel-wise exact but with the kernels in *swapped* μ_4 channels, so the analytic face is *sharpened* to the channel-swap question; the topological (EM1) face is computed — pumped charge = spectral flow = Středa = Chern = 1 on the v367 collar; and the finite Quillen identity is verified *two-sided* (Berry integral = Poincaré–Lelong divisor integral = +1, the difference an exact form). The remaining open content is typed: the rigidity premise, the channel-swap question, and the continuum Bismut–Freed identification; $\alpha^{-1} = 137.0359992168$ is re-anchored with split residual 6.4×10^{-58}. *Channel-swap update (2026-08-28, external master-object review wave 2; [verification/v985_quillen_channel_swap.py]; the contract stays [O]):* the swap question is *resolved at the finite level* by μ_4-character grading. Exactly: the jump pattern $\mathrm{circ}(2,-1,0,-1)$ has μ_4 eigenvalues $[0,2,4,2]$ (kernel in the $k{=}0$ channel), the mark pattern $\mathrm{circ}(0,1,2,1)$ has $[4,-2,0,-2]$ (kernel in $k{=}2$); the deck swap $r \to r{+}2$ ($P_0 \leftrightarrow P_2$, $P_1 \leftrightarrow P_3$) maps the jump |spectrum| onto the mark pattern *exactly* (the odd swap $r \to r{+}1$ fails — must-fail fires), so after the swap both kernels sit in the same channel and the graded determinant $\log\det'_{\mathrm{gr}} = \sum_r (-1)^r \log\det'(\Delta|_{P_r})$ compares them well-definedly: the graded difference is c_3-*free*, $\log(\ln 2/4)$ exactly, and the channelwise unit ratio is uniform $= 4/\ln 2$ in every matched nonzero channel (the v974 doublet ratio extended to full uniformity). *Completeness-wave notes (2026-08-28):* the finite KMS rigidity premise is now executed ([verification/v995_kms_subnet_rigidity.py]: all 33 mixing directions killed; covariance redundant; $196608 \to 0$); type-III/continuum stays [O]. The finite-collar detline curvature converges exponentially to 2π with rate $\sim 0.6936 \sim \ln 2$ ([verification/v996_gauge_grammar_kill.py]) — a numerical Bismut–Freed shadow, not the continuum theorem. Still [O]: the continuum Bismut–Freed identification, type-III KMS uniqueness, and the 2^{-4}/Quillen-metric compensation.

Executed update (2026-08-30, late-evening harvest; [verification/v1014_bridge_refinements.py]; the contract stays [O]). Finite restriction isomorphism verified; orientation fixes the conjugate branch ($2 \to 1$); the residual constant phase is fixed by the A_0 reference normalization — recorded. Continuum Bismut–Freed stays [O].

(2) Complete hypothesis list.

Type	Hypothesis	Witness
PROVEN-FINITE	the Quillen split at the root: $\alpha^3 - 2c_3^3\alpha^2 = 8b_1c_3^6\ln(1/\varphi_{\text{seam}})$ at $\alpha^{-1} = 137.0359992$, residual $< 10^{-35}$	v341/v382
PROVEN-FINITE	unique root + ablation battery; CODATA deviation $\sim 2.9 \times 10^{-10}$ *as output, never input*	v3/v48
PROVEN-FINITE	b_1 is the $U(1)_Y$ a_4 coefficient: $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$, $(3/5)\cdot\frac{41}{6} = \frac{41}{10}$	v434 (v159/v246)
PROVEN-FINITE	the embedding index $k_Y = (5/6)/(1/2) = 5/3$; level-1 rigidity fixes the $U(1)_Y$ normalisation	v470
PROVEN-FINITE	the π-power ladder $\{0,3,6\}$: the cubic term is the unique metric-independent (topological) term	v435
PROVEN-FINITE	the quadratic (Calderón) term is the Gilkey a_4 gauge-curvature coefficient $\Omega^2/12$	v342
PROVEN-FINITE	bulk Chern invariant $C = 1/0/-1$ at $M = 1/3/-1$ (FHS, Bloch BZ)	v367/v470
PROVEN-FINITE	the finite Quillen object: twist-torus determinant-line curvature $= 2\pi \cdot 1$ exactly (10^{-9}, $L = 4, 6$); twist integer = Bloch integer (Niu–Thouless–Wu); Fermi gap open over the whole torus	v472
PROVEN-FINITE	the contact unit: "c_3 per insertion" = the KMS seam unit, derived for the finite cycle sector; closed-form mark determinant, $41 =$ Y^2-weighted count of the 48 states	v484/v485
PROVEN-FINITE	solvable-model anchors: circle ζ-det variation in closed heat-kernel form; T^4 reaches the a_4 order; measure $(4\pi)^{-2} = 4c_3^2$	v391/v433
CITED-CLASSICAL	Quillen metric/curvature (Quillen 1985); Bismut–Freed curvature (CMP **106** (1986)); Dai–Freed section (JMP **35** (1994))	v472
CITED-CLASSICAL	η/CS reading of $\delta \log\det$ (APS 1975; Alvarez-Gaumé–Della Pietra–Moore)	v470
CITED-CLASSICAL	integer quantisation of the $U(1)$ response (TKNN; Avron–Seiler–Simon); boundary level = bulk response (Callan–Harvey)	v470
CITED-CLASSICAL	heat-kernel coefficients (Gilkey, *Invariance Theory*, Thm 4.8.16); the $\beta = a_4$ theorem (Vassilevich)	v342/v434
CITED-CLASSICAL	the embedding index $k_Y = 5/3$ (Ginsparg 1987)	v470
OPEN	**T1+T2 above**: the gap-stable continuum c_1 identification and the from-first-principles proof that $F_{U(1)}$ is the exact ζ-regularised Quillen functional (incl. the `EM1` lemma)	`ALPHA.QUILLEN.EXACT.01`

(3) Available finite models (runnable).

- [`verification/v472_quillen_detline_moduli.py`] — the finite Quillen object itself (real-space collar on an $L \times L$ torus with twisted boundary conditions; FHS curvature of the occupied-frame determinant line)
- [`verification/v470_alpha_inflow_level.py`] — the Bloch-level computation
- [`verification/v367_seam_s3_lattice.py`] — the underlying collar model
- [`verification/v3_em_alpha.py`] — the fixed point: unique root + ablation battery
- [`verification/v48_em_ward.py`] — the Ward decomposition
- [`verification/v341_alpha_quillen.py`] — the Quillen split and coefficient identities
- [`verification/v382_alpha_quillen_exact.py`] — the named target registration
- [`verification/v391_alpha_quillen_progress.py`] — circle ζ-det solvable model
- [`verification/v433_alpha_quillen_heatkernel.py`] — the T^4 a_4-order anchor
- [`verification/v434_alpha_quillen_betafunction.py`] — b_1 as the a_4 coefficient
- [`verification/v435_alpha_quillen_chernlevel.py`] — the π-power / level partition

- [verification/v484_seam_contact_unit.py] — the contact unit (c_3 per insertion)
- [verification/v485_contact_diagonal_closed.py] — the closed computable side

(4) Lean axiom surface (honest). There is *no* Lean formalisation of this target: no `TfptCarrier` module formalises the determinant line, the ζ-determinant, or the fixed-point functional. The only adjacent Lean content is the `SEAM.EQUIV.01` face (Sec. above — `SeamScalingLimit.lean` and companions, composition-level only) and kernel-checked edge/Chern arithmetic (`SeamEdgeChern.lean`). An external solver should treat the Lean layer as contributing *nothing* to this contract beyond the seam-face audit.

(5) Must-fail controls. A submitted derivation is accepted only if it respects the following (each realised in the suite):

- **MF-A1 (no experimental α).** The functional and its stationarity must be constructed from $(c_3, b_1, k_0, \varphi_{\text{seam}})$ alone; any appearance of the measured α (CODATA or otherwise) at *any* stage of the construction invalidates the derivation — the measured value is compared *after*, never inserted (v3).
- **MF-A2 (ablations destroy the hit).** Ablating the deck/carrier structure — neighbouring integer budgets for 41, dropping the μ_4 factor, or a Chern level $k_0 \neq 1$ — must move the root away from 137.0359992 beyond the corpus tolerance (the v3 ablation battery; v48).
- **MF-A3 (trivial/reversed collar).** The trivial collar $M = 3$ must give $C = 0$ (no level, no fixed point of this form); the orientation flip $M = -1$ must give $C = -1$ (v470/v472).
- **MF-A4 (the metric-independence partition).** Any proposed identification must reproduce the π-power ladder $\{0, 3, 6\}$: a derivation that attaches a $(4\pi)^{-d/2}$ heat-kernel measure to the cubic term (or strips it from the quadratic or transport terms) fails (v435).

(6) Acceptance tests (the external review's three, verbatim, plus one structural):

- **A1 — no experimental α input:** the derivation is checked to contain no use of the measured value anywhere; α^{-1} emerges as output only.
- **A2 — unique physical fixed point:** the *derived* variational equation has exactly one physical root, and it equals 137.0359992168 . . . to the corpus precision.
- **A3 — ablations destroy the hit:** MF-A2/MF-A3 verified to fail, machine-checkably.
- **A4 — structural:** T1 is accepted only with the uniform-gap hypothesis explicit (the Dai–Freed section nonvanishing over the whole moduli torus); T2 modulo T1 (or vice versa) narrows the target and is recorded as progress, but only the pair closes it. Any marker move happens in `status_ledger.csv` by the house process, never in this document.

Externalization Contract — `PRIME.RESIDUE.EXTERNAL.01` (the prime-front terminal residue, packaged for external specialists)

Typing. The third hand-off document: the *terminal open statements* of the prime-front programme (diary and status record: `tfpt_prime_front`), packaged self-containedly for outside specialists in analytic number theory — large sieve, sampling and frame theory, de Branges spaces, explicit-formula methods. It *documents*, it does not close: the contract identifier names this document, not a ledger row; every prime-front ledger row behind it (`PRIME.*`) is an [E]-typed NOT-RH-EVIDENCE identity, finite theorem or honest kill record, and none moves here. **This contract offers open problems. It asserts no progress toward the Riemann Hypothesis in either direction; nothing below is evidence for or against it, and no submitted solution is solicited as such evidence.** Throughout, "λ-uniform" is house shorthand for *uniform over the whole rung ladder* (the infinite quantifier), as opposed to per-rung, finite, or classical-per-census legs; "census" always means the finite root set defined below, never a zero set of ζ.

(1) The finite objects, self-contained. Fix an integer rung $h \geq 4$ and set $a = \frac{1}{2}\log h$, $K = \lceil 1.25\, h \log h \rceil$, modes $\omega_k = k\pi/a$ ($k = 0..K{-}1$), poles $b_k = \omega_k^2$, and the block height $T_z = 2\pi h$. The *source data* are the von Mangoldt atoms $(u_q, w_q) = (\log q,\ \Lambda(q)/\sqrt{q})$ over the prime powers $q \leq h$ — the only arithmetic input anywhere. The *wall matrix* $M_h = M_{\text{pole}} + M_{\text{arch}} - M_{\text{prime}}$ is the $K\times K$ matrix of the Weil-explicit-formula quadratic form in the normalised even trigonometric mode basis $\{\cos(\omega_k u)\}$ on the window $[0, 2a]$: M_{pole} is the rank-one pole block, M_{arch} the archimedean (Γ-term) block, and M_{prime} the *only* atom-dependent block — *linear* in exactly two atom transforms per mode, $P_k = \sum_q w_q \sin(\omega_k u_q)$ off-diagonal and $\tilde{P}_k = \sum_q w_q\,[(a - u_q/2)\cos(\omega_k u_q) - \sin(\omega_k u_q)/(2\omega_k)]$ on the diagonal (the v935 per-mode linearity; the exact assembly is frozen in the builder cited in (6) and rebuildable from the explicit formula for this window class). Let $\tau_h = \lambda_{\min}(M_h)$ and let d (the *wall argmin ray*) be its unit eigenvector; the *source coefficients* c_k are the de-normalised components of d. From these: $A_0 = \sum_k (-1)^k c_k$, $A_2 = \sum_{k\geq 1}(-1)^k c_k b_k$, $y_t = |A_2/A_0|$, $B_1 = \sum_{k\geq 1} b_k = (\pi/a)^2(K{-}1)K(2K{-}1)/6$, $\kappa = B_1/y_t$; the rational function

$$F(y) \;=\; c_0 + \sum_{k\geq 1}(-1)^k c_k\,\frac{y}{y - b_k},$$

whose numerator $N(y)$ (degree $K{-}1$, leading coefficient A_0) is the *census polynomial*; its roots y_j are *the census*. The moment–Laurent normal form is $(y/y_t)\,F(y)/A_0 = \Phi(z) = z - 1 + \sum_{m\geq 1} J_{m+1} z^{-m}$ at $z = y/y_t$ (v924). The *jet mass* at rung h is

$$\delta_h \;=\; \frac{\sum_\gamma \sin^2(a\gamma)\,F(\gamma^2)^2/\gamma^2}{A_0^2 \sum_\gamma \sin^2(a\gamma)/\gamma^2},$$

the sum over verified ordinates $\gamma \in (T_z + 6,\ \gamma_{7000}]$ (caches in (6)); the step-A identity $\delta_h = |J_h|^2_{G_h}$ with the jet vector $J = d/A_0$ and G_h the Gram of $\psi_k(\gamma) = \gamma^2/(\gamma^2 - b_k)$ under the weight $\sin^2(a\gamma)/\gamma^2$ is *exact*, remainder 0 (v939). The three per-rung hypotheses:

- **H1**$(h; c^*)$**:** no census root with $\operatorname{Re} y \geq c^* y_t$, with $c^* \in [1.10, 1.15]$ (grid values, flat over all built rungs). H1 is *certified source-pure* per rung ($h \leq 26$): one exact evaluation of the monotone envelope, $\mathrm{ENVJ}(c^* y_t) < |A_0|$ — no cache, no zeros (v931).
- **H2**(h)**:** the census is complete-real and nonnegative (all $K{-}1$ roots real, ≥ 0). Landed at all 25 built rungs including the degree-116 census at $h = 28$; finite classical per rung, hard $h \leq 13$ (v931).
- **H3**(h)**:** $y_t \leq 0.155\, T_z^4$. Source-pure certificate, certified 26/26 with margin ≥ 1.99, *refutable* (an explicit inflation witness breaks it) and not norm-continuous (v932).

(2) The three minimal open statements. These are the programme's terminal residue in the canonical form (note DII of the discovery record), stated as clean problems about the finite family above.

- **(R1) The triple.** *Prove $\{H1 \wedge H2 \wedge H3\}$-cofinal: for every sufficiently large k there is a rung h in the dyadic block $(2^k, 2^{k+1}]$ at which H1, H2 and H3 hold simultaneously — one rung per block, all three at the same h.* Only this cofinal form is asked; the limsup form is available only modulo the measured quantifier defect $D = 0.0042$ (block-average + pigeonhole is proven exact yet subsidy-empty, v933). Everything downstream of the triple is finite algebra or classical-per-census: the product floor and its assembly are theorems conditional *exactly* on it (v931).
- **(R2) The loop characterization.** *The census-forall-k statement* — the same census schedule quantified over *all* blocks at once — *is identified as RH-equivalent-in-currency by four independent machine-detected proof-route cycles*: census universalization == the RH loop (v928); the pinning-supply route == the same loop's costume (v929); BANDPOS on the unbounded ladder $\iff$ RH (v930); and the de Bruijn–Newman backward-persistence leg == the $[\Lambda \leq 0]$ class (v938). Stated honestly: within the corpus vocabulary this statement

and RH price each other; no proof of equivalence in the literature sense is claimed. The external value is either a genuinely *new independent sign source* for it, or a *proof that the loop is essential* (that census-forall-k is RH-equivalent), which would close the programme's hardness question honestly.

- **(R3) The H-pin.** *The one λ-uniform edge of the counting pair* $\{$*L1, WPD*$\}$ (v937): L1 = TAIL is proven and the H-pin is open; the weak positional dominance WPD($a < \gamma_1^2$) follows from the H-pin for every integer rung $x \geq 3$. The pin splits as Ω-a ⊕ Ω-b ⊕ H123M counting: Ω-a (EPS-LOCK) is the variational quotient $\bar{\varepsilon} \leq \mathrm{poly}(x)\,|A_0|\sqrt{G}$ — numerator trial-priceable, denominator floor == the flagged A_0-triangle loop; Ω-b (SPACING-REMAINDER) is the QSUBGAP kernel. The exact place the floor escapes linearity is machine-exhibited at *two* maps (v937): the eigenvector map ($\lambda_{\min}$ is not additive — $\lambda_{\min}(A+B) = -\sqrt{2} \neq \lambda_{\min}(A) + \lambda_{\min}(B) = -2$ on an exact 2×2 instance) and the census-root map (not affine, $d^2y^*/dw_1^2 \neq 0$ symbolically); the bridge atoms are c-free while the pin floor $m_j = 4|A_0 \sin(a\mu_j)|\,\mathrm{PR}_j$ is c-loaded — the floor lives on the vector side of the mode-level bridge, and the conditioning wall of v935 applies verbatim.

(3) What is proven and citable (complete typed hypothesis list). PROVEN-FINITE = machine-checked in the suite (module cited); CITED-CLASSICAL = a published theorem, used as stated, never re-proved; OPEN = the targets themselves.

Type	Hypothesis	Witness
PROVEN-FINITE	Theorem PF: the product floor $F/A_0 \geq (1 - c^*/z)^{(1+\kappa)/c^*}$, exponent = the trace, conditional *exactly* on {H1 + H2 per rung}; H1 source-pure (one envelope evaluation per rung); witness modus tollens	v931
PROVEN-FINITE	the rate dictionary $a = p/2 - 1$ proven at $p = 3, 4, 5$; TR-CAP proven *and* typed circular-for-TOPROOT; the H3 certificate (26/26, margin ≥ 1.99, refutable)	v932
PROVEN-FINITE	block-average + pigeonhole exact yet subsidy-empty ($D = 0.004183$); the witness boundary a theorem boundary, DK-scoped	v933
PROVEN-FINITE	the unconditional Landau/Gonek pricing: Landau 1912 + Gonek 1985/1993 identified unconditional; Gonek 1984 machine-checked *not an ancestor* (hidden loop flagged, never consumed); envelope closed form sympy-exact, census-depth-insensitive	v934
PROVEN-FINITE	the mode-level Landau bridge: the first linear per-mode classical identity (M_{prime} linear in two atom transforms per mode); the rate dictionary a proven bijection (transports every θ_∞, pins none)	v935
PROVEN-FINITE	the tau-free DK exclusion theorem: $\sin\theta \leq (r_0/\tau)/(\mathrm{FG} - a)$ with τ cancelling, proven $h \leq 13$ at the 10^{-25} bar + the exact required-bar schedule $b^*(h)$; the BA3 scale-gauge theorem	v936
PROVEN-FINITE	WPD's λ-uniform content == the H-pin edge (AND-fire at the declared edges); the restatement licenses; the two linearity-escape exhibits	v937
PROVEN-FINITE	the exact census semigroup: the de Bruijn–Newman flow on the derived census polynomial is e^{-tT}, $T = 4y\,\partial^2 + 2\,\partial$, closed form, trace law at measured deviation 0.0; the factor-4 normalisation lemma; the dictionary threefold-inequivalent	v938
PROVEN-FINITE	step A exact: $\delta_h = \lvert J_h\rvert^2_{G_h}$ with remainder 0, generic; the cluster normal form (basis-change determinant = the spacing product); Montgomery–Vaughan carried at 3.15π with exact margin 1/21	v939
PROVEN-FINITE	the sub-dof composition chain typed exactly (four exact legs; the floor *value* is the measured leg); the complete unconditional lower-class pricing table	v940
PROVEN-FINITE	the Gram is source-free (proven three ways); the ray-to-law map projective-linear and A_0-free ($s_i A_0$ linear in d); the one shared nonlinear link is the wall eigenmap	v941
CITED-CLASSICAL	Landau 1912 (the classical zero-counting/sum form over all nontrivial zeros, unconditional)	v934
CITED-CLASSICAL	Gonek 1985/1993 (unconditional exponential-sum bounds; the RH-conditional Gonek 1984 family is *flagged*, never consumed)	v934
CITED-CLASSICAL	Montgomery–Vaughan 1974 (the generalized Hilbert inequality; killed at linkage 1, carrying at 3.15π)	v939
CITED-CLASSICAL	Ortega-Cerdà–Seip 2002 (Fourier frames, Ann. of Math. **155**, 789–806: the Carleson box condition is equivalent to the *upper* inequality alone — wrong direction for a floor)	v940
CITED-CLASSICAL	Nazarov–Turán 1993 (explicit unconditional lower bound, exponential in the term count — the classical mirror of the measured occupancy law)	v940
CITED-CLASSICAL	Rodgers–Tao 2020 ($\Lambda \geq 0$), consumed as a *cited ceiling* only — its proof consumes pair correlation, which is exactly the flagged loop class	v938
CITED-CLASSICAL	Polymath 15 2019 ($\Lambda \leq 0.22$), consumed as a *cited ceiling* only	v938
OPEN	**R1–R3 above** — the cofinal triple, the loop characterization, the H-pin	this contract

(4) What is measured (honest). Measured legs are carried as measurements, never as theorems: the unscoped sub-block floor holds with margins ≥ 1.405 at all built rungs and the source-only Bertrand-cofinal selector ($\hat{h}_k = 7/13/31/\ldots$) clears the SF2 demand with margins 2.14/3.05

(v939/v940); the alignment law is a frozen measured *triple* — the half-power envelope (α inside the *descriptive* band $(0.38, 0.54)$ at all 14 rungs, never a prediction), the quarter-depth wall lock ($\mu_m = 0.2372 \log_{10} g_{\min}$, R^2 0.9916) and the deep excess (5.64–7.58 dex) (v941); the classical-evaluation face θ_∞ sits in the measured band $[0.0766, 0.0977]$ and is typed OPEN-NONPERTURBATIVE-VARIATIONAL — deeper census improves the data $\sim 1/T$ but provably never the limit (v933/v935); and the full-space refuter ladder is *always cited paired*: the measured first-order minimum sits 30–33 orders above the 10^{-25} bar, while the *proven* schedule is $b^*(h) = 23.8/21.7/15.7/4.4$ orders at $h = 4/5/8/13$ and *none* at $h \geq 16$ (v936). The quartic growth law behind H3 stays MEASURED (v926); the quantifier defect $D = 0.004183$ is a measurement.

(5) The kill atlas (so external solvers do not rebuild corpses). The campaign's obstruction record is machine-checked: an 11-node implication DAG with exactly one open edge and a candidate map complete over the corpus vocabulary — 24 candidate sign sources, 0 pass (`tfpt_prime_front`, kill-atlas section; sandbox, typed editorial). The named dead route classes, each with its one-line death mechanism:

Route class	Death mechanism	Round / witness
pointwise symbol minorants	the decisive eigenvalue is a smooth-direction cancellation invisible to any pointwise symbol minorant	diary, wall-operator arc
ℓ^1 / absolute-value majorants	the target is a five-order cancellation of three $\sim$200-sized pieces; the ℓ^1 majorant overshoots by 10^3–10^6	diary, bilinear arc (T170)
phase-discarding additive source forms	the source-only additive (Weyl-split) route is vacuous by 10.9–54.6 dex: the cancellation *is* the arithmetic	r166, v927
Vieta pinch / power sums	$1/A_0$ appears in every symmetric function — the resummation never decouples from the Connes scale	r172, v932
moment / trace caps	TR-CAP proven yet circular-for-TOPROOT; finite-P moment models undershoot by 1.4–4.5 dex (rate-blind trace-cap vacuity)	r162/172, v926/v932
Riccati / half-gap induction, manifold transport	transport relabels, never reduces: frame legs exact, source legs non-transporting, the level is cheaper than the increment — the infinite quantifier does not fall by induction	r177, v936
de Bruijn–Newman pinch	theta-flow-inert ($\geq 4021\times$ the unconditional window); wall fragility 8–51 orders below the window; the only closing leg is the flagged $[\Lambda \leq 0]$ class	r179, v938
global CBJ frame inequality	the dof kill ($n = 474$ coefficients vs Landau time-bandwidth rank 128), seam coupling 0.994, occupancy collapse	r180, v939
sub-dof scoping	the jet mass leaves the well-conditioned cell at $+0.999$ dex per rung; the scoped margin dies by $\sim 125\times$ at $\hat{h}(B3) = 13$	r181, v940
Carleson embedding as a floor	wrong direction: the box condition is equivalent to the *upper* inequality alone; every unconditional lower class misses for a named reason	r181, v940
naive floor-power composition	overshoots the honest composed floor by 3.41/10.14 dex; the level needs a per-mode lower bound on the known variational-quotient wall	r182, v941

(6) Available finite models (runnable). All models live in the repository and run standalone (Python 3, `mpmath/sympy/numpy`; `python verification/run_all.py` runs the whole suite):

- [`verification/v931_jetmass_floor_theorems.py`] — theorem PF, H1/H2, the assembled floor
- [`verification/v932_toproot_theta_statement.py`] — TOPROOT, the rate dictionary, H3
- [`verification/v933_h3_cofinal_adjudication.py`] — the cofinal adjudication, D, θ_∞
- [`verification/v934_gonek_pricing_unconditional.py`] — the unconditional classical pricing
- [`verification/v935_thetainf_landau_bridge.py`] — the mode-level Landau bridge

- [verification/v936_manifold_invariance_exclusion.py] — the DK exclusion, the refuter ladder
- [verification/v937_l1wpd_closure_reduction.py] — the H-pin split and linearity exhibits
- [verification/v938_dbn_heatflow_census.py] — the exact census semigroup
- [verification/v939_cbj_frame_adjudication.py] — step A, the Gram, the selector
- [verification/v940_cbj_subdof_blockfloor.py] — the sub-dof kill, the lower-class table
- [verification/v941_alignment_law_factorization.py] — the alignment-law factorization

The substrate modules v922–v930 carry the spacing/jet sum rules, the razor enclosure, the moment–Laurent ladder and the block-average/transfer/sigma-floor algebra. The frozen discovery probes (deterministic, SPEC-hashed, in `experiments/tfpt-discovery/`):

- `jetmass_floor_probe.py` (r171), `toproot_theta_probe.py` (r172)
- `h3_cofinal_probe.py` (r173), `gonek_pricing_probe.py` (r174)
- `thetainf_pin_probe.py` (r175), `manifold_invariance_probe.py` (r177)
- `l1wpd_closure_probe.py` (r178), `dbn_heatflow_probe.py` (r179)
- `cbj_frame_probe.py` (r180), `cbj_subdof_probe.py` (r181)
- `alignment_law_probe.py` (r182)

In the same directory: the frozen instrument builder `radius4_an_probe.py` (`build_cell`; worlds MAIN/SMOOTH/SCRARITH/EPSTEIN) and the verified ordinate caches `verified_zeros_big.npy` and `verified_zeros_n7000.npy` (with `_meta.json` sidecars; the big cache holds 2×10^7 verified ordinates and tops out at $T(2 \times 10^7) = 9\,499\,220.48$). The verified height horizon is cited as PT21 ($T_{\mathrm{PT}} = 3\,000\,175\,332\,800$) with HSW22 Cor. 1.2 as the envelope form.

(7) Formalization surface (honest). The only prime-front Lean 4 content is exact finite algebra: the spacing product for generic finite K over an arbitrary field, the jet sum rules, and the cofinal-wall composition/predefinition audits (`TfptCarrier/SpacingProduct.lean`, `JetSumRules.lean`, `CofinalWeil.lean`, `CofinalEnvelope.lean`, `CofinalPredefinition.lean`, `WallCofinalComposition.lean`, `WallCertifiedHead.lean`; `lake build` clean, no `sorry`). **Nothing of R1–R3 is formalised**: H1/H2/H3, the census objects, δ_h, the product floor and the pin have *zero* Lean content — no analysis, no zeros, no operator statements. An external solver should treat the Lean layer as contributing only the finite spacing/jet algebra audit.

(8) Must-fail controls. A submitted solution is accepted only if it respects the following (each already realised as a machine-checked negative control in the probes above):

- **MF-P1 (Epstein).** Any proposed sign source or positivity argument must fail, or demonstrably lose its constant, on the EPSTEIN world (the $x^2 + 5y^2$ lattice atoms: an L-function class with off-line zeros). An argument that goes through unchanged there proves too much.
- **MF-P2 (Scramble).** It must fail on SCRARITH (the golden-ratio permutation of the atom weights on the same support): the law must consume the joint sign-and-magnitude structure of the source, not its histogram.
- **MF-P3 (Smooth).** It must fail on the atom-free SMOOTH world: the conclusion must be arithmetic, not archimedean bookkeeping.
- **MF-P4 (no-zero test).** The proof must consume *no* zero tables: machine ancestry of the delivered chain must be zero-free (the G44-class no-zero gate). The caches of (6) are for *verification* of finite claims, never inputs to a submitted proof.
- **MF-P5 (predefinition test).** Clusters, weights and selectors must be predefined and sealed *before* any sign evaluation (the sealed-pre-ward discipline of r180); no sign-mined selection.

- **MF-P6 (loop firewall).** The proof must not consume the four machine-flagged loop classes: census-forall-k (R2 itself); the A_0-triangle; zero-verification-as-hypothesis (including the RH-conditional Gonek 1984 family); and the RH-conditional second moments (Montgomery pair correlation / Goldston–Montgomery). Consuming any of them re-imports the hypothesis.

(9) Acceptance tests. An external result counts against this contract exactly when it delivers one of:

- **(a) an explicit all-h, all-a bound** — a λ-uniform inequality (over the whole rung ladder, uniform in the window parameter) that implies R1 or the H-pin edge of R3 under the hypothesis list of (3), with any additional hypothesis flagged explicitly and the must-fail controls of (8) verifiably respected;
- **(b) a genuinely new sign source** — one that *separates the worlds and independently orients* (the frozen gate the 24 corpus candidates all fail): it must break on Epstein/Scramble/Smooth while carrying an unconditional orientation on the true source; or
- **(c) a proof that one of R1–R3 is RH-equivalent** — which would close the programme's hardness question honestly and is recorded as exactly that, never as progress toward RH.

Partial results that close a single leg (e.g. the Ω-b kernel alone, or one dyadic block class) narrow the target and are recorded as such. Any marker move happens in `status_ledger.csv` by the house process, never in this document. **No statement about the truth, provability or falsity of the Riemann Hypothesis is made here, and none is implied by the acceptance of a submission.**

Certifiability and recommended order

Building block	Machine	Hardness
W_{wall} enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (*done*)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy–medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (*done*)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order.

$$\boxed{\text{Selector-triangle pairings} \;\to\; v_{\text{geo}} \text{ scale anchor} \;\to\; G_{\text{net}} \text{ inclusion theorem}}$$

The flavor interface (historically U_{wall}) is now reduced to the *selector-triangle pairings* (the dual normal pair (d, n) forcing R columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The v_{geo} scale anchor is the one dimensionful input (same nature as $1/G$). G_{net} — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit. F_{transfer} (source→pole / relic / cosmology) is the remaining downstream interface.

F_{transfer} **is a typed functor, not a bag of open topics.** It is standard physics fed TFPT *source* data, $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, with exactly four interfaces:

Interface	**Map**	**Status**	**Source data → observable**
$F_{\rm pole}$	source→pole masses	[E] / [C]	Koide $53/54 = a^T(R{+}Q)\mathbf{1}/(2\,\mathbf{1}^T Ra)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([verification/v183_koide_f_corner_transfer.py])
$F_{\rm Boltzmann}$	CP source→ η_B	[C]	$\tilde m_1{=}m_3/A_\Lambda$ anchors the washout, but $M_1{=}M_R\varphi_0^4$ only relocates the free input (M_R is the seesaw scale, not a compiler power), so η_B stays a *sharper scenario*, not a closure ([verification/v184_etaB_anchored_boltzmann.py])
$F_{\rm relic}$	$(f_a, m_a, \theta_i, N_{\rm DW}) \to \Omega_a$	[C]	the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — a viable scenario ([verification/v185_axion_relic_solver.py])
$F_{\rm QCD}$	$(\alpha_s, b_3, \Lambda_{\rm QCD}, \text{lattice}) \to m_p/m_e$	[O]	a QCD matching contract ($b_3{=}-7$ a carrier output), *not* a compiler number — the rejected 1836 near-formula is the discipline working

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([verification/v187_ftransfer_laws.py]). Exact *algebraic* sub-parts (the Koide factor, $b_3{=}-7$) may be [E]; the *physical* prediction never is.

The functor contract `CONTRACT.F.01`, four axioms ([verification/v213_ftransfer_functor.py]). Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: **(1) μ_4-deck equivariance** — the discrete kernels are carrier/deck invariants $\{|\mu_4|, N_{\rm fam}, g_{\rm car}, A_\Lambda\}$, and the Koide multiplier $\lambda_2{=}(2/3)^6{=}64/729$ *is* the μ_4-deck transfer eigenvalue (v82/v54); **(2) Plücker preservation** — the dimensionless readouts are Plücker readings ($53{=}a^T(R{+}Q)\mathbf{1}$, $\mathrm{Pl}(F){=}(1,22,30)$; $\|\mathrm{Pl}(K)\|_1{=}11$); **(3) positivity/stochasticity** — $\mathrm{spec}\,T{=}\{1,(2/3)^6,(1/3)^6\}$ is positive (RP/OS), $\kappa_f{\in}(0,1)$; **(4) external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite-T $\chi(T)$ ODE, the lattice C_N) is named and external, never a compiler number. So `CONTRACT.F.01` is the third research contract alongside ($U_{\rm wall}$) and ($G_{\rm metric}$): a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure. *[2026-08-05 dated note — the jet route closes, the fibered refinement stands ([verification/v777_ftransfer_clock_jets.py]): the executed external-clock contract `FTRANSFER.CLOCKS.01` (below) fires its preregistered K1 kill sympy-exactly — the transported Schwarzians are $\{-\Delta^2/2, -\Delta^2/2, 0\}$, $\Delta = 6\log(3/2)$, so* no *common continuous clock exists for the four jets. The surviving architecture is measured: F_transfer is a* fibered *functor — one shared* discrete *PGL_2 base (the anchor cross-ratio of all four channels is $4/3$ exactly, symbolically* and *through the frozen g_{*s} table at dev 6.5×10^{-4}; the deck generator is the unit parabolic σ-translation, $\mathrm{tr}^2 - 4\det = 0$ — the v632 lattice-explicit discrete half) with channel-specific projective connections whose obstruction is the* constant *seam cocycle, values exactly $\{0, \pm\Delta^2/2\}$ (antisymmetry and the cocycle identity exact; $d/d\sigma$ identically zero); the classification matches the predeclared marker — thermal/proper-time clocks $\{\tau,\beta\}$ on the $-\Delta^2/2$ coset, the RG clock $\{\log\mu\}$ on the 0 coset, the relic channel degenerate and honestly unclassified. The functor axioms above stand; the jet/continuous-clock unification demand is* replaced *by the fibered form, with the Schwarzian differences stored as the obstruction cocycle, not fitted away. The one named proof-grade next step — derive the coset assignment from KMS/modular clock typing — is executed one round later and is a* theorem *([verification/v792_ftransfer_kms_coset.py], 18 checks, KMS-COSET-THEOREM, registered as `FTRANSFER.KMSCOSET.01`): $S = -(\ln\lambda)^2/2$ with λ the PGL_2 multiplier of the flow's time-1 holonomy per clock e-fold; the classifier uses no Schwarzian and no derivatives (non-circularity gated on its own source), the blind assignment classifies the four deployed charts exactly as the fibered functor carries them, the unblinding is sympy-exact, and the controls fire (fake KMS caught; affine invariance; the v578 Möbius clock as the positive case) — the fiber classification is derived, not case-inspected. The four observables stay [C]/[O]; no marker move.]*

A preregistered cross-ratio check for the four solvers — designed, and since executed on its discrete half. The design is machine-checked in [`verification/v566_parabolic_anchor_selfcode.py`], the discrete-half execution in [`verification/v571_cr_discrete_corners.py`]. The disc -7 norm line $\alpha_{t+1} = \alpha_t + 2$ (v533) equips the transfer path with a projective invariant: $\mathrm{CR}(\alpha_{-2}, \alpha_{-1}; \alpha_0, \alpha_1) = \frac{4}{3}$ *exactly* — and, honesty first, $\frac{4}{3}$ is the *universal* cross-ratio of any arithmetic progression, so the test content is not the number but the *discipline*: if the four solvers (J, K, C, F) export canonical one-dimensional projective states y_J, y_K, y_C, y_F — declared *before* any evaluation — then Möbius transport along the path forces $\mathrm{CR}(y_J, y_K; y_C, y_F) = \frac{4}{3}$, and the fourth state is determined fit-free by the first three (y_F reconstruction identity). Negative control with teeth: the raw determinant path $(2, 4, 14, 32)$ gives $\mathrm{CR} = \frac{28}{25} \neq \frac{4}{3}$ — the already-known norm sequence cannot pass. *The discrete half is now executed*: the four corners are the in-repo kernels $J, K, C, F = M(1, t) = R + Q\,\mathrm{diag}(1, t, t)$, $t = -2..1$ (v224), and the corpus's one canonical declared solver reading — the Koide anchor response $\ell(M) = a^{\mathsf{T}} M \mathbf{1}$ (v183) — reads $(26, 35, 44, 53)$ on them: $\mathrm{CR} = \frac{4}{3}$ exactly, and the fit-free reconstruction *forces* $y_F = 53$ from $(26, 35, 44)$ alone — the Koide numerator is Möbius-determined by the three upstream corners (a new exact cross-corner identity); honesty theorem stated out loud: ℓ is affine on the path (step $9 = N_{\mathrm{fam}}^2$), and *every* linear functional of the kernel passes — the check separates linear from nonlinear exports (the dominant-eigenvalue export fails at 1.3222, the determinants at $\frac{28}{25}$, both as they must) [E]. Status: the *continuous* half (the external solvers exporting their own states) stays open exactly as this contract fences it; contract markers unchanged [C]. *The frozen export protocol for the continuous half* (design [X], preregistered here): each instance must export, *before* evaluation, the canonical one-dimensional datum of its own native flow — the fixed-point multiplier of its contraction (cf. the native rates of v425: F_pole $(2/3)^6$ exact in-corpus, F_Boltzmann/F_relic/F_QCD external) — as a projective state on the path; only then is $\mathrm{CR} = \frac{4}{3}$ tested, with no renormalisation after the fact. *Executed* ([`verification/v575_cr_continuous_uniflow.py`]): the in-repo native exports form the quadruple $\{(2/3)^6, (2/3)^6, 1, -7\}$ — the repetition is *forced* by the single-flow theorem (v425: F_pole and F_Boltzmann are literally the same seam contraction) — and a repeated quadruple gives $\mathrm{CR} \in \{0, 1, \infty\}$ in every corner assignment (theorem, plus all 24 permutations): the continuous half can never pass *natively*; it closes onto the single-flow reduction, verdict UNIFLOW-DEGENERATE. The four independent external data are the dimensionful anchors (v_{geo}, M_R, C_p, θ_i) — [O] walls, not projective states — so a future continuous test needs genuinely *external* anchor-level exports; the meaningful cross-ratio content was and remains the discrete half. Both halves of the check are now executed [E]/[C]. *The jet-level successor is executed natively too* ([`verification/v578_native_jets.py`]): when point values degenerate, the Möbius-invariant *Schwarzian* of the exported trajectories is the right jet test — and natively, the exact v99 Koide trajectory has $\{q, t\} = -\Delta^2/2$ in wall time (constant, trajectory-independent; F_Boltzmann identical by the single-flow theorem) while being Möbius in its *own* exponential clock $u = e^{\Delta t}$ (Schwarzian 0), and the 1-loop QCD flow is Möbius in its own RG time already (Schwarzian 0): the native quadruple $\{-\Delta^2/2,\ -\Delta^2/2,\ 0,\ \text{degenerate}\}$ shares *no* wall-time jet invariant — each instance is *projective in its own clock*, the obstruction is the clock, and the clocks are anchored by the external data. Verdict MOEBIUS-IN-OWN-CLOCK; a future joint jet test needs external anchor-level jets; markers unchanged [E]/[C].

The discrete→dynamic lens: `F_transfer` is the readout end of the one principle ([`verification/v303_ftransfer_dynamics.py`], `FR.DYNAMICS.01`). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update $m \leftarrow (A+2I)/4\, m \to$ the E_8 marks, gap $6\ln(3/2)$). All four `F_transfer` instances share that *same shape*: [E] **F_pole** (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F'{=}(2/3)^6{=}\lambda_2$, the seam-clock subleading eigenvalue, so $|F'| < 1 \Rightarrow$ a unique attractor at the *seam* rate (v82); [C] **F_Boltzmann** (η_B) is a gapped positive contraction (washout $\kappa_f \in (0, 1)$ toward the unique balance, an H-theorem); [C] **F_relic** (axion) freezes at an adiabatic fixed point (the comoving number relaxes to a constant); [E] **F_QCD** (m_p/m_e) is the 1-loop RG flow with the carrier $b_3{=}-7$, whose Gaussian UV fixed point is the attractor (asymptotic freedom, $\Lambda_{\rm QCD}$ generated). *Honest boundary:* only F_pole has the seam rate $(2/3)^6$ exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So `F_transfer` is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory's simplicity is preserved precisely because the external rates are honestly fenced (v187), not because they reduce to the seam. (Even F_pole's flow time is non-integer, v93: the relaxation is the *continuous* semigroup $e^{t\log F}$ generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.)

The single-flow reduction (`DYN.TRANSFER.UNIVERSAL.01`). The dynamic lens is sharpened to a precise reduction ([`verification/v425_dyn_transfer_universal.py`]): the shared "shape" is literally the *one* native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6\ln(3/2)$) — restricted to each sector, extending the static universal-gap principle (v383) to a dynamic one. **[E]** the v99 Koide generator $\dot q{=}(\Delta/N_{\rm fam})(q{-}2)(q{-}5)$ linearised at the attractor $q{=}2$ has rate $-\Delta$, so the time-1 contraction is $(2/3)^6$ (the recovery eigenvalue, $q{=}5$ unstable); **[E]** the Boltzmann washout is that same contraction, with the CP phase $4\pi/3$ native (v320); **[E]** the QCD rate is $b_3{=}-7$. So the transfer *dynamics* is native; only the *anchors* stay external — the IR pole scale and M_R are $v_{\rm geo}$-class, C_p is the lattice $O(1)$, and $F_{\rm relic}$'s cosmological initial condition θ_i is the one genuine wall. A reduction, *not* internalisation: no new premise, and the firewall (v187) holds — the observable stays **[C]**. *[2026-08-27 precision (binding; `DYN.MARKOV.EMBED.01`, [`verification/v971_markov_embedding_generator.py`]): the "one native flow" is a universal* contraction class *— shared gap and multiplier algebra — not one physical clock; the clocks are external and sector-specific (v723/v724/v777), and the continuous semigroup $e^{t\log T}$ is irreversible statistical dynamics, not unitary evolution — the (b)-vs-(c) firewall of the `DYN.MARKOV.EMBED.01`/`DYN.UNITARY.DILATION.01` contract pair above.]*

Both thermal-time routes are dead: the external clock contract stands (`FTRANSFER.STT.01` / `FTRANSFER.COCYCLE.01`). The radical counter-idea to the external-clock wall — Connes-style thermal time, "the four clocks = four modular flows of one state" — is now machine-killed on *both* checkable routes. **[X]** *Strong thermal time* ([`verification/v723_phys_modular_clock.py`], 11/11, STT-KILLED + INTERNAL-PAIR-CONFIRMED): a modular generator is self-adjoint, hence diagonalizable — but the F_QCD 1-loop clock map $u\mapsto u+2b_0$ is *parabolic* and not diagonalizable (single Jordan block), and the symbolic lemma "exp of a real-diagonalizable matrix is diagonalizable" closes the door: no constant-rescaled modular flow produces the QCD clock; the pole multiplier is confirmed *exactly* $(2/3)^6=64/729$ (hyperbolic, dimensionless ⇒ internal; F_Boltzmann the same element by v425). **[X]** *Conditioned modular flows* ([`verification/v724_phys_t3b_modular_flows.py`], 12/12, T3B-DEAD): with target-free graph projectors, *one* global KMS scale $\Delta=6\log(3/2)$ and the Connes cocycle composition exact to 1.6×10^{-15} along J–K–C–F, the four conditioned flows come out *loxodromic/complex* — the frozen v578/v632 solver classes (hyperbolic / parabolic / degenerate) do *not* arise from conditioned modular flows of one internal mother state (the NEEDS-LINDBLAD escape is honestly refused). *Typed consequence:* the external clock contract is *confirmed* and sharpens to the anchor census {one unit $=\Lambda_{\rm QCD}$ ($v_{\rm geo}$-class by No-Unit v153), one angle θ_i, one lattice $O(1)$ C_p} — T3's clock wall and T1's unit wall are *the same wall*, counted once. No marker moves; the functor typing above is unchanged.

The frozen external-clock contract (`FTRANSFER.CLOCKS.01`) — preregistered 2026-08-04, executed 2026-08-05. Since both thermal-time routes are dead, the clocks *must* be external — and without clocks fixed in advance every jet statement is arbitrarily reparametrizable. The contract therefore freezes them (design byte-frozen 2026-08-04 in `experiments/ftransfer-clocks/hypotheses/ftransfer_clocks_v1.yaml`; typed **[X]**, a search-target design, *not* a claim). *The four frozen clocks, exactly one physical clock per solver:* F_pole = proper time τ (the on-shell/pole-mass phase clock; frozen gauge: one v99 seam unit = one proper-phase e-fold); F_Boltzmann = inverse temperature β (FIRAS-anchored adiabatic chart; one washout seam unit = one β e-fold; the z-linear alternative is recorded as *excluded*); F_QCD = RG time $t=\ln(\mu/M_Z)$ ($\overline{\rm MS}$, 1-loop $b_0=7$ on the frozen window $[m_t, 1\,{\rm TeV}]$; identity gauge); F_relic = cosmological e-folds $\ln a$ (Planck 2018 background; the flow is frozen, the jets *degenerate* — the channel is scored only on kills and dictionary validity). *The preregistered prediction* (FTC.P1): under the frozen affine dictionaries (slopes $-1, -1, +1, -1$ into the common chart $\sigma = \ln(\mu/M_Z)$) *one* constant-coefficient Riccati equation carries all non-degenerate transported jets — equivalently, the transported Schwarzians of F__pole/F__Boltzmann/F__QCD are *equal*; the seam channels transport to $-\Delta^2/2 = -2.959235170076978$ exactly, so the prediction forces the transported QCD Schwarzian onto that value. *Three kills:* K1 NO-COMMON-CONNECTION (symbolic inequality of the transported Schwarzians; on kill the executor must exhibit the forbidden reparametrization per failing pair); K2 FREE-REPARAM-NEEDED (any post-freeze change of a gauge or dictionary; the YAML sha256 is recorded by the executor); K3 SECOND-CLOCK-NEEDED (any channel export needing two time variables). *The honest prior, preregistered:* the *expected* outcome is the K1 kill — affine transport scales a Schwarzian by slope^2 and cannot move 0 (the QCD value) onto $-\Delta^2/2$ (the seam value); a K1 kill would finalize the v578 diagnosis at the *physical*-clock level and close the jet route of `CONTRACT.F.01`; ONE-CLASS would be the surprise. *Executed (2026-08-05, [verification/v777_ftransfer_clock_jets.py], one module from two probes,* 24 + 14 *checks,* 0 *failures):* the K1 kill fired *exactly* as the frozen prior expected — verdict NO-COMMON-CONNECTION. The freeze guards were byte-verified at run time (YAML sha256 `880224f7...`; the two data tables — Saikawa–Shirai 2018 g_{*s}, PDG 2024 $\alpha_s(Q)$ — were curated and SHA-frozen *before* the first run); all four dictionary-validity legs pass (Koide scheme check 571×; washout entropy-chart drift 1.15×10^{-4}; QCD overlay max dev 0.83%; relic onset slope dev 0.0602); the transported Schwarzians are sympy-exactly $\{-\Delta^2/2, -\Delta^2/2, 0\}$ — no tolerance involved — and the K1 duty is discharged: the pass-enabling reparametrization $h = \text{Möbius}(e^{\pm\Delta\sigma})$ is exhibited and is non-affine, exactly the freedom the freeze forbids; K2/K3 stay quiet. The same module carries the surviving architecture (the fibered refinement — see the `CONTRACT.F.01` dated note above). *[2026-08-06 completion (`FTRANSFER.KMSCOSET.01`, [verification/v792_ftransfer_kms_coset.py],* 18 *checks,* KMS-COSET-THEOREM*): the coset assignment of these clock charts is now a theorem from KMS/modular typing —* $S = -(\ln \lambda)^2/2$ *from the time-1* PGL_2 *holonomy multiplier per clock e-fold, with a Schwarzian-free, derivative-free classifier (the detailed-balance multiplier cross-ratio as the discrete KMS signature), blind assignment and exact unblinding; the three-probe chain executor → fibered groupoid → coset theorem is whole.]* Nothing here upgrades `CONTRACT.F.01`: the four observables stay [C]/[O]; no marker move; no RH statement (unrelated surface; discipline note). So the visible residual is one geometric postulate (`QGEO.SYM.01`), one unit gauge (v_{geo}), and this one typed transfer functor — everything else is out-of-sample data.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

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- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.

- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
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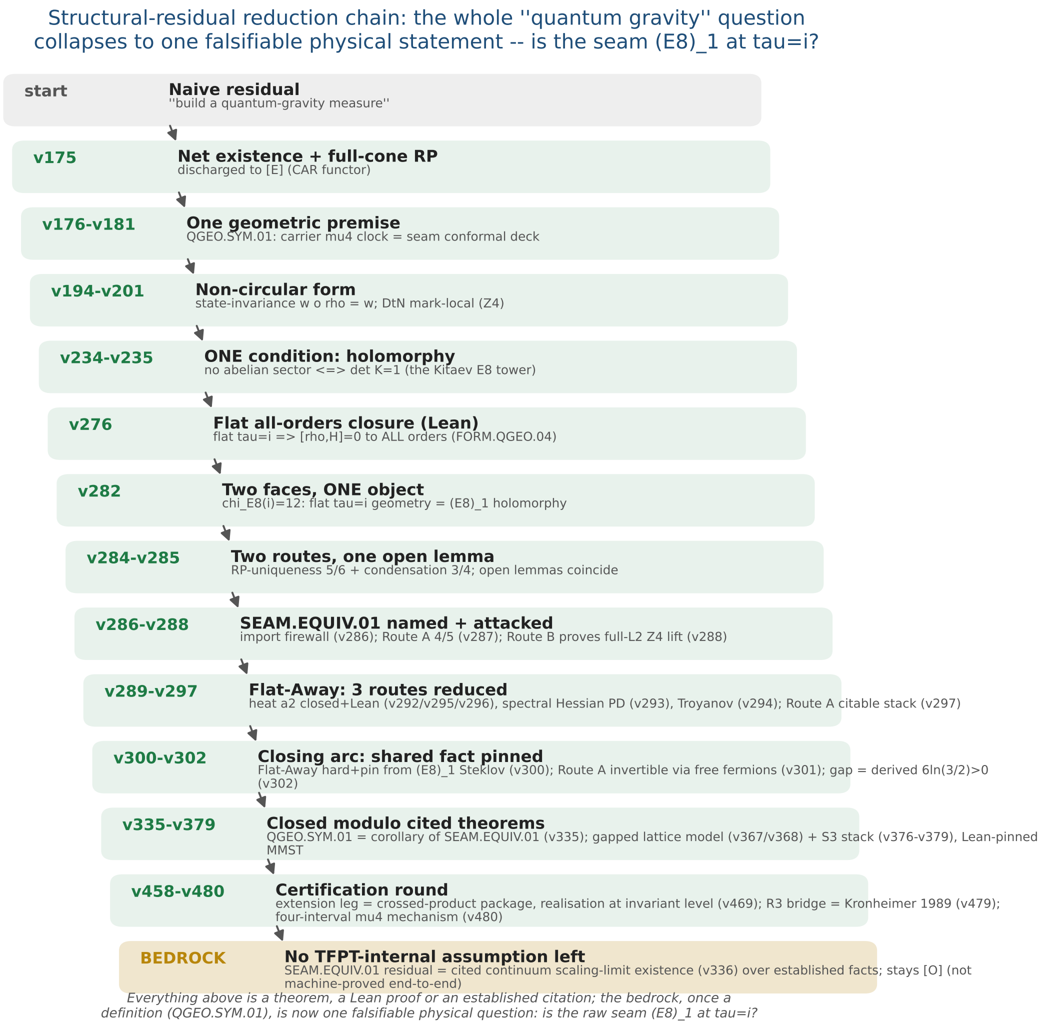

Figure 3: The structural-residual reduction chain (v175 $\to$ the certification round). The naive "build a quantum-gravity measure" shrinks, one machine-checked step at a time, from a vague measure to one falsifiable physical statement — *is the raw seam* $(E_8)_1$ *at* $\tau = i$*?* (`SEAM.EQUIV.01`; its conformal-deck face `QGEO.SYM.01` is a corollary, v335). The closing arc (v276–v302) pins the shared fact and derives the gap $6\ln(3/2) > 0$; the certification round discharges the extension leg to the crossed-product package (v469), the R3 graph$\to$geometry bridge to Kronheimer 1989 (v479) and exhibits the four-interval μ_4 mechanism (v480). Every step above the bedrock is a theorem, a Lean proof or an established citation; the residual — the cited continuum scaling-limit existence (v336) — stays honestly [O].

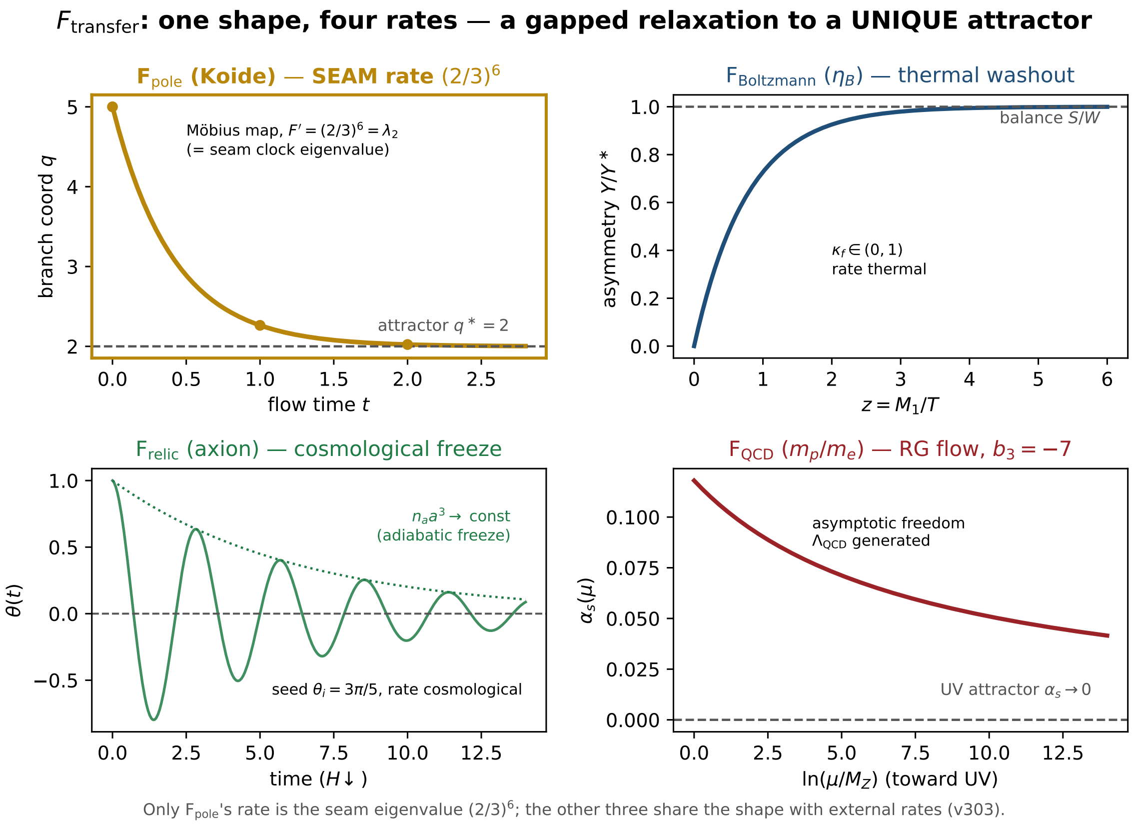

Figure 4: **The discrete→dynamic lens on `F_transfer` ([`verification/v303_ftransfer_dynamics.py`]).** All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update $m \leftarrow (A+2I)/4\,m$ → the E_8 marks. *Only* F_{pole} (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is $F'=(2/3)^6=\lambda_2$, the seam-clock subleading eigenvalue. $\mathrm{F}_{\text{Boltzmann}}$ (η_B, thermal washout), $\mathrm{F}_{\text{relic}}$ (axion, cosmological freeze) and F_{QCD} (m_p/m_e, the RG flow to the Gaussian UV fixed point with the carrier $b_3 = -7$) share the *shape* with *external* rates. So `F_transfer` is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.